Fundamental Fractal-Geometric Field Theory (FFGFT) or T0 Theory – Time-Mass Duality

Part 8: The Standard Model from GF(27), Lepton Masses and Interfaces

Johann Pascher

2026

Contents

A Introduction to Volume 8 . . . 3

I Part I — Quantum Numbers from the Galois Structure 5

B Doc. 344 — Furey Fermions in GALG and FFGFT: N-Operator, Tripotents and Neutrino-Vacuum Identity . . . 7

C Doc. 345 — FFGFT in Literature Comparison: What the Galois Approach Achieves and What It Does Not . . . 14

D Doc. 346 — Charge Quantisation from GF$(27)^*$: Electric Charge and Colour Algebraically Forced 20

E Doc. 347 — Gell-Mann–Nishijima from the Galois Structure: $Q = I_3 + Y/2$ algebraically derived 24

F Doc. 348 — Generation Structure and Coupling Matrix from GF(27): Trace Bilinear Form, Orbit Assignment, and CKM Complexity . . . 28

G Doc. 349 — SU$(2)_L$ Chirality from the Galois Structure: Su/Sd Separation and Cl(6) Grade Parity 32

H Doc. 350 — Black Holes and Time-Mass Duality: Covariance of G, Evaporation Sequence, and the Neutrino as Final State . . . 35

II Part II — Comparison Values, Lepton Residuals, Higgs and Standard Model Particles 39

I Doc. 351 — Model Dependence of PDG Comparison Values . . . 41

J Doc. 352 — Lepton Residuals: Lepton Ladder, Galois and Koide Compared . . . 53

K Doc. 353 — Koide Amplitude $d/c = \sqrt{2}$ and the T^4/Z_3 Geometry . . . 62

L Doc. 354 — H^0 and $\tilde{T} \cdot m = 1$: Twelve Insights into Higgs Geometry . . . 68

M Doc. 355 — SSB $SU(2)_L \times U(1)_Y \to U(1)_{\mathrm{EM}}$ from T^4/Z_3 Geometry . . . 76

N Doc. 356 — Standard Model Particles from GF$(27)^*$ . . . 82

O Doc. 357 — Standard Model Quantum Numbers from GF$(27)^*$ — Certificate Status . . . 84

III Part III — Harmonics, Laboratory Systems, Hodge Theory and Fields 87

P Doc. 358 — Harmonics and Algebraic Structure . . . 89

Q Doc. 359 — AI-Based Pattern Recognition and Algebraic Limits of Frequency Analysis . . . 95

R Doc. 360 — Galois-Informed HRV Analysis . . . 102

S Doc. 361 — The Kagome Lattice and FFGFT: Four Structural Parallels . . . 110

T Doc. 362 — The BAW Ising Machine as a Resonance Computer . . . 118

U Doc. 363 — Hodge Theory on T^4/Z_3 within FFGFT . . . 124

V Doc. 364 — The Icosahedral Carrier and the D_4 Lattice . . . 130

W Doc. 365 — The Foundation of FFGFT in Four Layers . . . 141

X Doc. 366 — Fields in FFGFT: Energy, Flux and Geometry . . . 149
IV Part IV — Icosahedron, Matrix Elements and Lepton Masses 172
Y Doc. 367 — Why $\theta = p_0 = 2/9$ — Probability and Phase as Two Readings of One Quotient . . . 174
Z Doc. 368 — From p_0, p_1, p_2 to Lepton Masses . . . 180
aa Doc. 369 — Four Routes to the Lepton Masses . . . 185
ab Doc. 370 — Why the Icosahedron — Three, Five, and Non-Commutation . . . 191
ac Doc. 371 — The Factor 4/3 — Sphere Volume, Casimir Operators and Electromagnetic Mass . . . 198
ad Doc. 372 — The Matrix-Element Principle and Its Reach . . . 207
ae Doc. 373 — The Yukawa Mechanism as a Consequence of $\tilde{T} \cdot m = 1$ . . . 217
V Part V — Interfaces, Review and the Basic Form 226
af Doc. 374 — Common Anchor — Bridge Equations to Observer Patch Holography . . . 228
ag Doc. 375 — The Hierarchy v/E_P — Two Derivations Compared: FFGFT and Observer Patch Holography . . . 235
ah Doc. 376 — The Coefficient 8π and the Status of Issue 740 . . . 239
ai Doc. 377 — Interfaces of FFGFT to Other Frameworks . . . 243
aj Doc. 378 — Peer Review under Present Conditions . . . 246
ak Doc. 379 — The Acoustic Plenum and the Photon . . . 256
al Doc. 380 — Why D_4 — Specificity of the carrier against matched comparison lattices (R95) . . . 261
am Doc. 381 — The Running Recursion Summed Exactly . . . 265
an Doc. 382 — Muon $g-2$: Status 2025 and Assessment of the FFGFT Statements . . . 269
ao Doc. 383 — Masses and Planck Scale without v . . . 273
ap Doc. 384 — FFGFT in Brief — Foundations, results and status after the calculation review . . . 277
aq Doc. 385 — The Higgs–vacuum route to ξ . . . 298
ar Doc. 386 — Where α hides — Charge unit, two spheres and ξ as an area . . . 303
as Doc. 387 — The deviations at a glance . . . 310
at Doc. 388 — The CMB temperature in the basic form . . . 315
au Doc. 389 — Parabolic Flight, Inertia and the Representations of Gravitation . . . 320
VI Part VI — Loose Puzzle Pieces, Correction Spacings and the Measurement Chain 327
av Doc. 390 — Loose Puzzle Pieces: Kaluza-Klein and Other Established Building Blocks of the Time-Mass Duality . . . 329
aw Doc. 391 — Loose Puzzle Pieces II: Cosmology — Redshift, Scales and the Boundary of the Cosmic Sector . . . 337
ax Doc. 392 — Where the Representations of Gravitation Part Ways . . . 343
ay Doc. 393 — The Noise Limit and Learning Systems . . . 353
az Doc. 394 — How Far Apart the Fractal Corrections Lie . . . 363
ba Doc. 395 — The Measurement Chain as a Channel . . . 370

Chapter A

Introduction to Volume 8

Eighth Volume of the Document Collection

Volume 8 comprises Documents 344 to 395 from September to October 2026 and brings the collection up to the current state of the corpus. It builds directly on the Galois bridge of Volume 7 and carries it forward to the quantum numbers of the Standard Model, to the lepton masses, and to the interfaces with other theoretical frameworks.

Content and Focus

The first part derives from the structure of $GF(27)^*$ the quantisation of charge, the Gell-Mann–Nishijima relation $Q = I_3 + Y/2$, the generation structure with its coupling matrix, and $SU(2)_L$ chirality; this is complemented by the comparison with Furey fermions, a placement within the literature, and the treatment of black holes under time–mass duality. The second part first clarifies how the PDG comparison values arise and which model assumptions enter them, and then sets the lepton ladder, the Galois approach and the Koide relation side by side; there follow the Koide amplitude $d/c = \sqrt{2}$, the Higgs geometry, spontaneous symmetry breaking from the $T^4/\mathbb{Z}_3$ geometry, and the overview of the Standard Model particles with their certificate status.

The third part shows why harmonics and algebraic structure are the same mathematics, and tests this insight on laboratory systems: frequency analysis and its limits, HRV measurement, the kagome lattice and the BAW Ising machine. Added to these are Hodge theory on $T^4/\mathbb{Z}_3$, the comparison of the icosahedral carrier with the D_4 lattice, the presentation of the foundation in four layers, and the field structure of energy, flux and geometry. The fourth part is devoted to the lepton masses: $\theta = p_0 = 2/9$ as two readings of one quotient, the route from p_0, p_1, p_2 to the masses, four routes compared, the justification of the icosahedron, the factor $4/3$, the matrix-element principle, and the Yukawa mechanism as a consequence of $\tilde{T} \cdot m = 1$.

The fifth part collects the interfaces: the bridge equations to Observer Patch Holography, the hierarchy v/E_P, an overview of all comparison and bridge documents, considerations on peer review, the specificity of the D_4 carrier, the exactly summed recursion, the status of muon $g-2$, the masses without v, a brief account of FFGFT after the calculation review, the Higgs–vacuum route to ξ, the place of α in the geometry, an overview of the

deviations, the CMB temperature in the basic form, and the representations of gravitation using the example of parabolic flight.

The sixth part places and checks: it collects established building blocks such as Kaluza–Klein that recur in the time–mass duality, and the cosmological puzzle pieces from redshift and scales to the boundary of the cosmic sector. It shows where the representations of gravitation part ways, treats the noise limit in pattern recognition and learning systems, applies one common scale to the spacings of the fractal corrections, and reads the measurement chain as a channel in the sense of communications engineering.

Organisation

The 52 documents are divided into six parts: quantum numbers from the Galois structure (Docs. 344–350); comparison values, lepton residuals, Higgs and Standard Model particles (Docs. 351–357); harmonics, laboratory systems, Hodge theory and fields (Docs. 358–366); icosahedron, matrix elements and lepton masses (Docs. 367–373); interfaces, review and the basic form (Docs. 374–389); loose puzzle pieces, correction spacings and the measurement chain (Docs. 390–395).

Reading the Documents

For a quick overview of the overall state, Doc. 384 (brief account) is suitable; for the remaining deviations, Doc. 387. Status markers and the corrections register apply as in the preceding volumes; in each case the most recent version of a result in the corpus is authoritative.

Status

The documents in this volume reflect their respective versions in the corpus as of October 2026.

Part I

Part I — Quantum Numbers from the Galois Structure

Chapter B

Doc. 344 — Furey Fermions in GALG and FFGFT: N-Operator, Tripotents and Neutrino-Vacuum Identity

Abstract

Doug Matzke's IPI mail of 3 September 2026 presents concrete GALG results on Furey fermions in Cl(6): complete Up/Down fermion sets Su/Sd, idempotents N_1, N_2, N_3, tripotent N, jE7 projector, and a grade-1+grade-5 construction. The strongest single result: Su$[0]$ = Vss$[0]$ exactly — the neutrino is the lowest vacuum state **[B]**. Theorem F shows: the tripotent property for even vacua and the Frobenius fixed-point property (Doc. 339) are the same decomposition from two independent directions **[B]**. The electron exception at the jE7 projector is the known Furey problem **[S]**.

B.1 Theorem A — N_k are idempotents in characteristic 3

$N_k = -1 + (-i\,B_k)$, $B_k^2 = -1$ (Witt pair):

$$N_k^2 = 1 + 2i\,B_k + (-1)(-1) = 2 + 2i\,B_k \equiv -1 + (-i\,B_k) \pmod 3 = N_k. \quad \textbf{[B]}$$

The sum loses its scalar part since $-3 \equiv 0$: $N_1 + N_2 + N_3 \equiv (-i)\sum B_k = N$.

B.2 Theorem B — N is a tripotent

Cross-terms $N_k N_j$ ($k \neq j$) are pure grade-4 elements; hence $N^3 = N$ **[B]**. Doug's output: `N**3 = N`, `N**4 = N**2`.

B.3 Theorem C — Su$[0]$ = Vss$[0]$: neutrino is the vacuum

`Su[0] == Vss[0]: True` and `(1+N)*(1+jE7) == Vss[0]: True` **[B]**.

FFGFT correspondence [B]. In FFGFT, ν_1 lies in orbit $O_1 = \{1, 3, 9\}$, the orbit of the generator of GF$(27)^*$ — the algebraically simplest object. Both frameworks independently place the neutrino at the algebraically simplest position: convergence of two independent frameworks.

Why are the neutrino and the vacuum the same?

Three independent routes lead to the same result:

1. Furey/GALG (this theorem). Vacuum state Vss[0] is defined as the object annihilated by all creation operators: no Fock-space content, no quantum numbers. The neutrino Su[0] has the same algebraic structure: no colour, no electric charge, only minimal weak coupling. Hence they are identical.

2. Galois/FFGFT (Doc. 339/343). GF$(27)^*$ has a generator g of order 26. Orbit $O_1 = \{g^1, g^3, g^9\}$ is the orbit of the generator itself — the algebraically simplest object in the structure. ν_1 lies in O_1. The vacuum Vss[0] is the state with zero torus winding ($n = 0$). Smallest mass ↔ simplest orbit ↔ $n = 0$ state — three names for the same minimality property.

3. Photon analogy (Doc. 007). $m_\nu = \xi^2/2 \times m_e = 4.54$ meV. The photon has mass 0 and is the limiting case of the vacuum. The neutrino is the massive particle closest to the vacuum: double ξ suppression = maximum approach to $m = 0$ among all massive fermions **[K]**.

Structural statement. In a theory that derives all particles from an algebraic structure, the lightest massive fermion must lie closest to the vacuum state. Neutrino and vacuum are the algebraic minimum: the same object seen from different directions **[B]**.

> **Remark on the vacuum identification.** Doug Matzke works in Cl(6) over GF(9) without a mass scale — in GALG mass does not appear. When he identifies Su[0] = Vss[0] he means: same algebraic structure, no mass value behind it. The neutrino is the "most neutral" fermion of the algebra, the vacuum is the "emptiest" state — neither carries any of the quantum numbers that distinguish other objects.
>
> In FFGFT the step is more precise: Vss[0] corresponds to the $n = 0$ state on the torus; the Galois orbit O_1 assigns it the mass $m_{\nu_1} = 4.54$ meV (Doc. 340). The absolute mass in meV arises through the SI energy scale: ξ is derived from Galois group orders **[K]**, the torus quantum numbers (r_e, p_e) give m_e parameter-free **[K]**, and $v = 246$ GeV is the SI unit choice (no free parameter, but a scale fixation). Doug sets no units, so this step does not exist in GALG. The identification Su[0] = Vss[0] is algebraically exact **[B]**; in FFGFT it means "ground state with smallest mass", not "massless object". The absolute value 4.54 meV is parameter-free: ξ **[K]** (Galois group orders, Doc. 338), m_e **[K]** (torus quantum numbers, Doc. 006), $v = 246$ GeV = SI scale fixation (not a free constant).

B.4 Theorems D–F

D — sigma [B]. $e_1 \to e_2 \to e_4 \to e_1$, $e_3 \to e_6 \to e_5 \to e_3$; `{1:2,2:4,4:1,3:6,6:5,5:3}`, two 3-cycles, $\sigma^3 = \mathrm{id}$. Doc. 341 contains this formula; Doug's `all_g1_g5good` is the σ-invariant sum.

E — jE7 $\equiv N_k$ as projectors [B]; electron exception [S]. $F \cdot jE7 = +F$ for all Su (8/8); $F \cdot jE7 = -F$ for 7/8 of Sd (Sd[7]=electron fails). Same for N_k. Known Furey problem. FFGFT: electron in Dirac layer $\{f_3, f_4\}$, neutrino in Majorana layer $\{f_1, f_2\}$ (Doc. 343, Theorem D‴). Common root open **[S]**.

F — Tripotent for even vacua [B]. $(V_k + P)^2 = -V_k$, $(V_k + P)^3 = V_k$ for $k \in \{0, 2, 4\}$; no collapse for $k \in \{1, 3, 5\}$ (length 32). Identical to the Frobenius fixed-point/orbit split of Doc. 339 — two independent derivations of the same decomposition.

B.5 Theorem G — Anti-neutrinos: Dirac or Majorana?

The tension. In GALG/Furey (Theorem C) the neutrino is $\mathrm{Su}[0] = \mathrm{Vss}[0]$ and the anti-neutrino is $\mathrm{Sd}[0] = \mathrm{cc}(\mathrm{Su}[0]) = \mathrm{Vss}[1] \neq \mathrm{Vss}[0]$. Hence neutrinos in Furey's construction are **Dirac particles** ($\nu \neq \bar{\nu}$).

In FFGFT (Doc. 343, Theorem D‴), ν_1 and ν_2 lie in orbits O_1 and O_3, which are self-inverse under $k \mapsto k^{-1}$ — the *algebraic* Majorana criterion.

Theorem B.5.1 (Clarification of Majorana concept **[B]**/**[S]**). *Two distinct Majorana concepts:*

1. ***Algebraic [B]:*** *Orbit O_k is self-inverse under $k \mapsto k^{-1}$ mod 13 — a statement about GF$(27)^*$ structure, not about spin.*
2. ***Physical [S]:*** *$\nu = \bar{\nu}$ in the Lorentz sense (Majorana condition). This requires Sd[0]=Su[0], which does not hold in GALG/Furey.*

*The Galois algebra fixes the algebraic property **[B]**; the physical Majorana condition is an additional assumption **[S]**.*

Position of anti-neutrinos [B]/[S]. GALG: $\bar{\nu} = \mathrm{Sd}[0] = \mathrm{Vss}[1]$ in the odd vacuum (sigma-orbit, other sector, Doc. 339). FFGFT: the anti-neutrino would lie in the negation $f_1 \leftrightarrow g_1$ (Doc. 343, Theorem D″) — order-13 class g_1, other sector. Both frameworks place the anti-neutrino in the structurally "other" sector.

Consequence for m_{ee}. The prediction $m_{ee} = |m_1 U_{e1}^2 + m_2 U_{e2}^2| \approx 6.0$ meV (with $m_1 = 4.54$, $m_2 = 9.81$ meV from Doc. 340, $s_{12}^2 = 0.307$, $s_{13}^2 = 0.022$: $0.678 \cdot 4.54 + 0.300 \cdot 9.81 = 3.08 + 2.94 = 6.0$ meV) assumes physical Majorana neutrinos $\nu_{1,2}$. If ν_1, ν_2 are instead Dirac particles (as in Furey), m_{ee} vanishes entirely. The m_{ee} prediction is conditioned on the Majorana assumption **[S]**.

Open question [S]. Common algebraic root of the electron inconsistency in GALG/-Furey and the Majorana/Dirac layer in FFGFT.

Empirical decision. nEXO (sensitivity $m_{\beta\beta} \approx 5$–20 meV depending on nuclear matrix element) decides: signal at ~ 6 meV $\Rightarrow$ Majorana (Doc. 340 confirmed, Theorem H refuted); no signal $\Rightarrow$ consistent with Dirac (Theorem H confirmed or destructive interference).

B.6 Theorem H — The Galois algebra forces Dirac neutrinos

Theorem B.6.1 (Dirac constraint **[B]** under assumption **[S]**). *If the sector pairing $k \mapsto -k$ in GF$(27)^*$ corresponds to the physical charge conjugation C, then **all** neutrinos are Dirac particles.*

Proof. A Majorana particle is a fixed point of charge conjugation: $\nu = \bar{\nu}$, i.e. class f_i must map to itself under C.

Under the sector pairing $k \mapsto -k \pmod{26}$:

$$\begin{aligned} f_1 = \{1,3,9\} &\mapsto \{25,23,17\} = f_2 \neq f_1 \\ f_2 = \{17,23,25\} &\mapsto f_1 \neq f_2 \\ f_3 = \{5,15,19\} &\mapsto f_4 \neq f_3 \\ f_4 = \{7,11,21\} &\mapsto f_3 \neq f_4 \end{aligned}$$

No primitive class is a fixed point of the sector pairing **[B]**. A Majorana fixed point would require $f_i \mapsto f_i$ — this does not exist in GF$(27)^*$. The anti-particle of the neutrino f_1 is f_2: $\nu_1 \neq \bar{\nu}_1$. □ □

Assumption [S]. The argument holds if $k \mapsto -k$ in GF$(27)^*$ corresponds to physical charge conjugation C. In Doc. 336 the Frobenius involution $x \mapsto x^3$ is identified with the sector separation; the identification with C is physically motivated there but not algebraically proved.

Consistency [B]. In GALG/Furey (Doc. 344, Theorem C): Sd$[0]$ = Vss$[1] \neq$ Vss$[0]$ = Su$[0]$, i.e. $\bar{\nu} \neq \nu$ — fully consistent with this theorem.

Plain language: What is m_{ee} and why does it matter?

The process. A radioactive nucleus converts two neutrons into two protons simultaneously, emitting two electrons. In ordinary double beta decay ($2\nu\beta\beta$) two anti-neutrinos also come out — the energy is shared among four particles. In *neutrinoless* decay ($0\nu\beta\beta$) no anti-neutrinos come out.

Why this requires Majorana. The anti-neutrino produced at the first nuclear vertex must be absorbed as a neutrino at the second vertex. This is only possible if neutrino and anti-neutrino are the same particle ($\nu = \bar{\nu}$, Majorana condition). For Dirac neutrinos ($\nu \neq \bar{\nu}$) the process is exactly forbidden.

What m_{ee} measures. The half-life of $0\nu\beta\beta$ decay is proportional to $1/m_{ee}^2$, where

$$m_{ee} = \left| m_1 U_{e1}^2 + m_2 U_{e2}^2 + m_3 U_{e3}^2 \right|$$

combines all three neutrino masses weighted by PMNS mixing elements and Majorana phases.

Why interference matters. The three terms can add (constructively) or partially cancel (destructively) depending on the Majorana phases. With FFGFT masses from Doc. 340, phases ≈ 0 and without the ν_3 term (ν_3 is Dirac): constructive gives $m_{ee} \approx 3.078 + 2.946 = 6.0$ meV; destructive gives $m_{ee} \approx |3.078 - 2.946| = 0.13$ meV; Dirac gives $m_{ee} = 0$ exactly.

What nEXO can measure. nEXO targets $m_{\beta\beta} \approx 5$–20 meV (NME-dependent, nEXO 2022); the FFGFT value 6.0 meV lies in the decision range. A signal at ~ 6 meV confirms Majorana and refutes Theorem H; no signal down to 5 meV excludes the constructive Majorana prediction and is consistent with Dirac or destructive interference.

Consequence: $m_{ee} = 0$ and no neutrinoless double beta decay

For Dirac neutrinos, neutrinoless double beta decay ($0\nu\beta\beta$) is **exactly forbidden**:

$$m_{ee} = \left| \sum_i m_i U_{ei}^2 \right| \equiv 0.$$

Table B.1: Predictions: Majorana (Doc. 340) vs. Dirac (Theorem H)

	Majorana ν_1, ν_2 **[S]**	Dirac all **[B]**
m_{ee}	≈ 6.0 meV	$= 0$
$0\nu\beta\beta$	allowed, ~ 10 meV	exactly forbidden
nEXO signal	~ 6 meV at phases ≈ 0	no signal
KamLAND-Zen	consistent (< 28–122 meV, 2025)	consistent

Testability. nEXO ($m_{\beta\beta} \approx 5$–20 meV) and the complete KamLAND-Zen-800 dataset (28–122 meV, PRL 135 (2025) 262501) bracket the range. A null result at nEXO excludes Majorana with Doc. 340 masses (phases ≈ 0) and is consistent with Theorem H. A signal at ~ 6 meV refutes Theorem H and confirms the Majorana interpretation **[B]**/**[S]** of Doc. 340.

What we know better since Galois

Why exactly the primes {5,7,11,13} — [K]

Before: phenomenological observation. Now: algebraic theorem. They are the only primes entering $|\mathrm{GF}(3^k)^*|$ at $k = 3, 4, 5, 6$. $p = 13$ is the worldwide unique prime with $\mathrm{ord}_p(3) = 3$. This follows necessarily from the group structure — no choice (Doc. 342, R106).

Where $\xi = 4/30000$ comes from — [K]

Before: defined, not derived. Now: ξ follows algebraically from the identity $(r_\mu/r_e)^2/\xi = 43200 = |\mathrm{GF}(9)^*|^2 \cdot 5^2 \cdot |\mathrm{GF}(27)|$ and the torus winding numbers $r_\mu/r_e = 12/5$. ξ is not a free constant (Doc. 338, R101).

Why the neutrino mass is 4.54 meV — [K]

Before: from $\xi^2/2 \times m_e$ (Doc. 007), without explanation of origin. Now closed: ξ **[K]** (Galois group orders, Doc. 338), m_e **[K]** (torus quantum numbers $n = 1, \ell = 0, j = \frac{1}{2}$, Doc. 006), $v = 246$ GeV = SI scale fixation (not a free parameter, but a unit choice). The 4.54 meV follow without any free constant — parameter-free.

Why exactly two fermion layers — [B]

Before: observation. Now: the sector pairing $k \mapsto -k$ on $\mathrm{GF}(27)^*$ has no fixed points among the primitive classes — it necessarily pairs $f_1 \leftrightarrow f_2$ and $f_3 \leftrightarrow f_4$. Four classes become exactly two pairs. This is a group-theoretic theorem (Doc. 343, Theorem D).

Why the neutrino is the lightest massive fermion — [B]

Before: no algebraic explanation. Now: $O_1 = \{1, 3, 9\}$ is the orbit of the generator of $\mathrm{GF}(27)^*$ — the algebraically simplest element. The neutrino lies there. Simultaneously: $\mathrm{Su}[0] = \mathrm{Vss}[0]$ in GALG/Furey (Theorem C) — the neutrino is the vacuum state of the Clifford Fock space. Three independent routes converge **[B]**.

A sharp open question instead of a vague one — [B]/[S]

Before: unclear whether neutrinos are Majorana or Dirac. Now precise: the sector pairing has no fixed point ⇒ no Majorana possible under assumption $k \mapsto -k = C$ (Theorem H) **[B]**/**[S]**. If yes: $m_{ee} = 0$ exactly, $0\nu\beta\beta$ forbidden. If no: $m_{ee} \approx 6.0$ meV. nEXO decides at 5–20 meV sensitivity (NME-dependent).

Why there are no further particles — [B]

This is one of the strongest statements: the algebra prescribes not only what exists, but also what cannot exist.

Four primitive classes, no more. $\mathrm{GF}(27)^*$ has exactly $\varphi(26) = 12$ primitive elements forming exactly 4 Frobenius three-cycles. There is no fifth primitive class.

Two layers, no more. The sector pairing $k \mapsto -k$ pairs $f_1 \leftrightarrow f_2$ and $f_3 \leftrightarrow f_4$ — exactly two pairs, no remainder. Negation $f_i \leftrightarrow g_i$ yields sign partners (order 13), not independent new particle classes.

Three generations, not four. Each primitive class is a three-cycle. 3 elements per orbit = 3 generations. Forced by the Frobenius orbit length, not chosen.

16 fermions per generation, complete (GALG/Furey). $8 + 8$ = Su ∪ Sd covers exactly one generation of the Standard Model: neutrino, 3 colours × up-quark, 3 colours × down-quark, positron (and their Down-sector partners). No gap, no room for more.

No sterile neutrinos from this algebra. The Majorana layer $\{f_1, f_2\}$ is fully occupied by ν_1 and ν_2. A third Majorana slot does not exist (Doc. 343, Theorem D‴).

What we do not know better

The electroweak scale $v = 246$ GeV is not yet forced from algebraic principles — a unit choice, not a free parameter, but its origin is open. The electron exception in GALG (jE7 fails for Sd[7]=electron) has no algebraic explanation in FFGFT yet (Theorem E, **[S]**).

B.7 Summary

Table B.2: Results Doc. 344

Theorem	Content	Status
A	$N_k^2 = N_k$, characteristic-3 proof	**[B]**
B	$N^3 = N$, tripotent	**[B]**
C	Su[0]=Vss[0]: neutrino=vacuum; FFGFT correspondence	**[B]**
D	sigma$= \{1 \to 2, 2 \to 4, 4 \to 1, 3 \to 6, 6 \to 5, 5 \to 3\}$	**[B]**
E	jE7$\equiv N_k$; electron exception (Furey problem)	**[B]/[S]**
F	Tripotent even vacua = Frobenius fixed points (Doc. 339)	**[B]**
G	Anti-neutrino Sd[0]=Vss[1]$\neq$Su[0]; algebraic Majorana $\neq$ physical Majorana	**[B]/[S]**
H	Sector pairing $k \mapsto -k$ has no fixed point: all neutrinos Dirac; $m_{ee} = 0$; $0\nu\beta\beta$ forbidden	**[B]/[S]**

Open question [S]: Common algebraic root of the electron inconsistency in GALG/Furey and the Majorana/Dirac layer separation in FFGFT (Doc. 343, Theorem D‴).

Verification script

`pruef_344_furey_galg.py` — 6 theorems, all assertions passed.

[1] C. Furey, *Standard Model Physics from an Algebra?* (2016).

[2] D. Matzke, IPI mail 3 September 2026.

[3] J. Pascher, *Doc. 339: Frobenius separation*, FFGFT corpus (Aug. 2026).

[4] J. Pascher, *Doc. 341: GF(27) in GALG and FFGFT*, FFGFT corpus (Sept. 2026).

[5] J. Pascher, *Doc. 343: Zeta function of T^4/Z_3*, FFGFT corpus (Sept. 2026).

Chapter C

Doc. 345 — FFGFT in Literature Comparison: What the Galois Approach Achieves and What It Does Not

Status markers

[K] *Confirmed* — derived completely from ξ or a corpus-secured derivation; all assertions in the verification script passed.
[B] *Established* — numerically secured by script or analytically trivial.
[S] *Speculative* — suggestion or conjecture, not yet fully proved.

Abstract

This document compares the results of the Galois documents 338–344 systematically with the established literature on algebraic derivations of the Standard Model (division algebras, Clifford algebras, exceptional Jordan algebras) and on neutrino physics (seesaw mechanism, $0\nu\beta\beta$). It names where FFGFT achieves something absent from the literature, and where the literature is ahead of FFGFT. The criterion is not "is the algebra elegant" but "does the algebra give testable numbers".

References. All citations were verified on 4 September 2026 against arXiv, INSPIRE, APS and publisher pages.

C.1 The comparison class: algebraic approaches to the Standard Model

Table C.1: Algebraic approaches to the Standard Model (selection)

Approach	Authors	Algebra	Core claim
Octonion quarks	Günaydin, Gürsey 1973	$\mathbb{O}$	SU(3) from octonion automorphisms
Division algebras	Dixon 1994	$\mathbb{R}\otimes\mathbb{C}\otimes\mathbb{H}\otimes\mathbb{O}$	One generation from $\mathbb{T}$
Clifford Cl(6)	Furey 2015, 2018	$\mathrm{Cl}(6)$ left ideals	8 fermions, SU(3)×U(1)
Exceptional Jordan	Todorov, Dubois-Violette 2018	$J_3(\mathbb{O})$	Three generations from F_4
GALG / Hyperbit	Matzke 2022–2026	$\mathrm{Cl}(n)$ over GF(9)	Vacuum structure, order 26
FFGFT / Galois	Pascher 2026	$\mathrm{GF}(3^k)$, $T^4/\mathbb{Z}_3$	Masses from ξ, primes forced

The decisive question since Günaydin 1973 has been the same: *The algebra has the right shape — does it also give the right numbers?*

C.2 What FFGFT achieves beyond the literature

Prime structure algebraically forced [K]

Literature. The harmonic primes $\{5, 7, 11, 13\}$ appear in music theory (Partch, just intonation) and phenomenological mass formulae (Koide 1983, Brannen 2006) as observed, never derived. Furey and Dixon work in characteristic 0; $\mathrm{GF}(3^k)$ does not appear in any comparison approach. **FFGFT.** Docs. 342/343: these primes are the only ones entering $|\mathrm{GF}(3^k)^*|$ at $k = 3, 4, 5, 6$; $p = 13$ is the worldwide unique prime with $\mathrm{ord}_p(3) = 3$. A theorem, not a fit. **Assessment.** New in this form.

Coupling constant not a free parameter [K]

Literature. In the SM all Yukawa couplings are free (13 parameters in the fermion sector). Furey 2015 derives gauge groups, not couplings. Koide 1983 finds $Q = 2/3$ but cannot explain why. **FFGFT.** Doc. 338: $\xi = (r_\mu/r_e)^2/43200$; hence $m_\mu/m_e = 120\sqrt{3}$ (0.52 %) and $1/\alpha = 3700/27$ (7.6 ppm). The agreement of α with the measured value is a property of the anchor E_0 ($E_0^2 = 54$ MeV^2), which provides the only link to the SI; it is not counted as an independent prediction. **Assessment.** Concrete numerical progress beyond all comparison approaches. The torus quantum numbers $r_i = n_{\phi,g}^2/n_{\theta,g}$ and $p_g = 3/(2g)$ follow from winding numbers of the T^4 torus, which derive from $\mathrm{GF}(3^n)$ group orders: $n_{\theta,1} = 3$ (characteristic), $n_{\phi,1} = 2$ ($|\mathrm{GF}(3)^*|$), $n_{\theta,2} = 5$ (odd prime factor of $|\mathrm{GF}(81)^*| = 80$) etc. (Doc. 338, Tab. 2) **[K]**. C_{conv} is the SI unit conversion, not a free parameter.

Neutrino–vacuum identity [B]

Literature. Furey 2015 constructs the neutrino as the Fock vacuum of $\mathrm{Cl}(6)$ — known. **FFGFT.** Doc. 344: the same follows independently from $\mathrm{GF}(27)^*$: ν_1 in $O_1 = \{1, 3, 9\}$, the

generator orbit. Two unrelated frameworks place the same particle at the same algebraic position. **Assessment.** The convergence is new.

Closure of the classification: no fourth generation [B]

Literature. Furey 2018 and Todorov/Dubois-Violette 2018 show "there are three", not "there cannot be four" — a construction, not a theorem. **FFGFT.** $\varphi(26) = 12 \to$ exactly 4 orbits; sector pairing without fixed point $\to$ exactly 2 layers; orbit length $3 \to$ exactly 3 generations. A counting argument with completeness. **Assessment.** The completeness conclusion is new in this sharpness.

Falsifiable numerical predictions [K]/[B]

Literature. Furey, Dixon, Günaydin give structural statements, no mass values. The seesaw (Minkowski 1977, Mohapatra/Senjanović 1979, Weinberg 1979) explains the smallness of neutrino mass but predicts no number — M_R is free. **FFGFT.** Doc. 340: $m_{\nu_1} = 4.54$, $m_{\nu_2} = 9.81$, $m_{\nu_3} = 49.9$ meV; Δm^2 to 0.5–1 %. Doc. 344: $m_{ee} = 0$ (Dirac) or ≈ 6.0 meV (Majorana, phases zero, without ν_3 contribution: $|c_{12}^2 c_{13}^2\, m_1 + s_{12}^2 c_{13}^2\, m_2| = 0.678 \cdot 4.54 + 0.300 \cdot 9.81 = 6.02$ meV) — nEXO decides. **Assessment.** The corpus makes concrete, falsifiable predictions no algebraic comparison approach provides. This is the main difference.

Gravity as geometry, not force [B]

Furey and Dixon do not treat gravity; their algebras yield the SM gauge groups, not spacetime. In FFGFT gravity is *not a separate force*: the form of $G = \xi^2/(4\, m_{\text{char}})$ follows intrinsically from the $T^4/\mathbb{Z}_3$ torus geometry (Doc. 003), not the last digit of the numerical value. No separate quantum gravity is required since the geometry is pre-ordered, not a dynamical field to be quantised. The time field $T(x)$ with $T \cdot m = 1$ replaces metric coupling. The derivation runs via ξ (from Galois group orders **[K]**) and torus geometry; C_{conv} is the SI conversion of the one-anchor chain of Doc. 383. The form and order of magnitude of G thus follow from ξ and the single anchor **[K]**. Agreement with the CODATA value arises only with the small remainder of that chain contained in the numerical value of C_{conv} (Doc. 383) and does not count as separate evidence of precision; explaining this remainder structurally is open **[S]**.

C.3 What the literature achieves beyond FFGFT

Gauge groups from the algebra

Literature. Furey 2015: SU(3)×U(1) from Cl(6) commutators, correct charges for all 8 fermions. Dixon 1994: the full SU(3)×SU(2)×U(1) from $\mathbb{T} = \mathbb{R} \otimes \mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$.

FFGFT. Doc. 339 **[B]**: SU(3) from $4\,\mathbb{Z}_{13}$-orbits $\times 2\,\mathbb{Z}_2$-parities $= 8$ gluons; U(1) from fixed field GF(3) (photon). Doc. 336 **[K]**: Gal$(GF(9)/GF(3)) = \mathbb{Z}_2$ explains algebraically why $\text{SU}(2)_L$ and $\text{U}(1)_Y$ exist; $\sin^2\theta_W \approx 0.250$ tree-level **[K]**. Doc. 346 **[B]**: $Q \in \{0, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm 1\}$ from Legendre symbol mod 13. Doc. 347: $Q = I_3 + Y/2$ from jE7 projector and Legendre symbol,

not fully closed: the hypercharge from the QR/NQR class alone does not separate charged leptons and quarks; the hypercharge of the charged leptons is open **[S]**. Doc. 349 **[B]**: $\mathrm{SU}(2)_L$ doublet structure from $\mathrm{Cl}(6)$ grade parity — Su (even grades) = right-handed, Sd (odd grades) = left-handed; $\mathbb{Z}_3$ symmetry of the torus forces the chirality split.

Balance. Charge quantisation and chirality structure derived from Galois/Clifford algebra **[B]**; Gell-Mann–Nishijima open at the hypercharge of the charged leptons **[S]**. What Furey/Dixon have more completely: $\mathrm{SU}(3)\times\mathrm{U}(1)$ as Clifford commutator and the octonion colour construction — a complementary framework, no contradiction. **At gauge-group content now broadly equal.**

Quark sector

Literature. Furey/Dixon treat quarks on equal footing; colour charge as octonion automorphism with full $\mathrm{SU}(3)_c$ representation.

FFGFT. Quark masses in Doc. 006 **[K]** (deviations 0.0–0.3 % for up, down, strange, charm, bottom). Doc. 346 **[B]**: $N \bmod 3$ colour charge classifies quarks (triplet) vs. leptons (singlet). Charge values $Q = \pm\frac{2}{3}, \pm\frac{1}{3}$ from Doc. 347 **[B]**. Doc. 348 **[B]**: coupling matrix $f_3 \times f_4$ (leptons × quarks) diagonal in Frobenius order, $M^{(34)} = -I$ (gen. i couples to gen. i); CKM mixing angles **[S]** (Doc. 348).

Balance. Colour charge classification and quark masses from Galois **[B][K]**. What Furey/Dixon have more completely: octonion $\mathrm{SU}(3)_c$ construction and full CKM motivation. **At quark charges equal; at gauge construction literature ahead.**

Seesaw mechanism. Explains *why* neutrinos are light and provides a dynamical mechanism for Majorana masses. FFGFT replaces this by double-ξ suppression (Doc. 007) — shorter, but without dynamics. **Literature mechanistically ahead.**

Rigour. Furey 2015 is a thesis with complete proofs. The FFGFT corpus uses verification scripts and numerical checks (**[B]**), not complete analytic proofs for all theorems. **Literature formally stricter.**

C.4 The Yukawa Lagrangian and the Flavour Problem

In the Standard Model fermion masses arise from the Yukawa term:

$$\mathscr{L}_{\text{Yukawa}} \supset -y_{ij}\,\bar{\psi}_i\,H\psi_j + \text{h.c.}, \qquad m_i = y_i \times \frac{v}{\sqrt{2}}.$$

The Yukawa couplings y_i are **free parameters** — measured, not derived. This is the flavour problem: 13 free fermion masses without algebraic origin.

Furey and Dixon derive the gauge structure but say nothing about the Yukawa couplings. The Lagrangian retains free y_i.

Froggatt-Nielsen mechanism (1979) [24]. A U(1) flavour symmetry assigns charge q_i to each field; couplings scale as $y_{ij} \sim \varepsilon^{|q_i+q_j|}$. This explains the hierarchy — but the charge assignments q_i are chosen freely.

FFGFT mass formula. FFGFT replaces the Lagrangian approach by a direct geometric derivation:

$$m_i = r_i\,\xi^{p_i}\,v,$$

where $r_i = n_{\phi,g}^2/n_{\theta,g}$ and $p_g = 3/(2g)$ follow from torus winding numbers forced by GF(3^n) group orders **[K]** (Doc. 338), and $\xi = 4/30000$ is algebraically derived **[K]**.

Table C.2: Froggatt-Nielsen vs. FFGFT

Criterion	Froggatt-Nielsen	FFGFT
Hierarchy principle	ε^{q_i}, q_i free	ξ^{p_i}, p_i forced **[K]**
Small parameter	ε: free	$\xi = 4/30000$: Galois-derived **[K]**
Flavour symmetry	U(1), freely chosen	GF(3^k) group structure **[K]**
Mass values	order of ε, not exact	0.0–0.5 % agreement **[K]**
Three generations	unexplained	orbit length 3 forced **[B]**

FFGFT and Froggatt-Nielsen share the same core principle (power law of a small parameter), but FFGFT forces everything algebraically where FN freely assigns. This is the precise advance over the closest comparison in the literature.

C.5 The central difference

> Furey, Dixon, Günaydin show: *the algebra has the right shape.*
> FFGFT shows: *the algebra gives the right numbers* — $m_{\nu_1} = 4.54$ meV, Δm^2 to 1 %, m_μ/m_e to 0.5 %.

Not a contradiction but a complement. Furey provides the gauge structure, the FFGFT Galois structure provides the mass scale. Doc. 344 shows both converge at Su[0] = Vss[0]. A unification — Cl(6) over GF(9) with GF$(27)^*$ orbit structure — is the natural next task.

C.6 Honest assessment

Table C.3: Comparison in one line each

Criterion	Literature	FFGFT
Gauge groups	complete (Furey, Dixon)	charges+chirality from Doc. 346/347/349 **[B]**; Clifford constr. Furey
Mass values	none	concrete, 0.5–1 %
Neutrino masses	mechanism (seesaw), no value	values, no mechanism
Prime structure	observed	derived **[K]**
Generation count	constructed	counted (complete) **[B]**
Falsifiability	weak (structural)	sharp (nEXO, Δm^2)
Quarks	equal footing	secondary
Proof form	analytic (structure)	numerical (numbers) — different domains
Gravity	absent	form of $G = \xi^2/(4m_{\text{char}})$ from torus geometry

FFGFT complements Furey/Dixon with the mass scale. The gauge structure is more completely worked out in the literature; concrete, falsifiable mass values to 0.5–1 % are not present in any comparison approach.

Note on AI assistance

This document was prepared with support from Claude (Anthropic). Literature references were verified on 4 September 2026 against arXiv, INSPIRE and APS. Physical interpretations and conclusions are by Johann Pascher (ORCID 0009-0000-6518-4064).

[1] M. Günaydin, F. Gürsey, J. Math. Phys. 14 (1973) 1651–1667.

[2] G. M. Dixon, *Division Algebras*, Kluwer 1994.

[3] J. C. Baez, *The Octonions*, Bull. Amer. Math. Soc. 39 (2002) 145–205.

[4] C. Furey, *Standard model physics from an algebra?*, Ph.D. Thesis, Waterloo 2015, arXiv:1611.09182.

[5] C. Furey, Phys. Lett. B 785 (2018) 84–89, arXiv:1910.08395.

[6] I. Todorov, M. Dubois-Violette, Int. J. Mod. Phys. A 33 (2018) 1850118, arXiv:1806.09450.

[7] Y. Koide, Phys. Lett. B 120 (1983) 161–165.

[8] P. Minkowski, Phys. Lett. B 67 (1977) 421–428.

[9] R. N. Mohapatra, G. Senjanović, Phys. Rev. Lett. 44 (1980) 912–915.

[10] S. Weinberg, Phys. Rev. Lett. 43 (1979) 1566.

[11] KamLAND-Zen Collaboration, Phys. Rev. Lett. 117 (2016) 082503, arXiv:1605.02889.

[12] KamLAND-Zen Collaboration, Phys. Rev. Lett. 135 (2025) 262501, arXiv:2406.11438 ($m_{\beta\beta} < 28$–122 meV).

[13] nEXO Collaboration, Phys. Rev. C 97 (2018) 065503, arXiv:1710.05075.

[14] nEXO Collaboration, J. Phys. G 49 (2022) 015104, arXiv:2106.16243.

[15] Particle Data Group, Phys. Rev. D 110 (2024) 030001.

[16] D. Matzke, Hyperbit/GALG, ANPA 2022–2026.

[17] J. Pascher, Doc. 338, FFGFT corpus 2026.

[18] J. Pascher, Doc. 340, FFGFT corpus 2026.

[19] J. Pascher, Doc. 343, FFGFT corpus 2026.

[20] J. Pascher, Doc. 344, FFGFT corpus 2026.

[21] J. Pascher, *Doc. 346: Charge quantisation from GF*$(27)^*$, FFGFT corpus (Sept. 2026).

[22] J. Pascher, *Doc. 347: Gell-Mann-Nishijima from Galois structure*, FFGFT corpus (Sept. 2026).

[23] J. Pascher, *Doc. 349: SU*$(2)_L$ *chirality from Galois*, FFGFT corpus (Sept. 2026).

[24] C. D. Froggatt, H. B. Nielsen, *Hierarchy of quark masses, Cabibbo angles and CP violation*, Nucl. Phys. B147 (1979) 277–298.

Chapter D

Doc. 346 — Charge Quantisation from GF$(27)^*$: Electric Charge and Colour Algebraically Forced

Status markers

[K] *Confirmed* — algebraically complete.

[B] *Established* — numerically secured by script.

[S] *Speculative* — conjecture, not yet fully proved.

Abstract

Electric charge quantisation and the colour singlet/triplet structure of the Standard Model follow from two algebraic properties of GF$(27)^*$ orbits: (1) quadratic residue mod 13 $\leftrightarrow$ electrically neutral; (2) $N \bmod 3$ encodes colour charge. Their combination classifies all Standard Model fermions completely without free parameters. Theorem E proposes Gell-Mann–Nishijima $Q = I_3 + Y/2$ as a consequence of the Witt-pair operator decomposition — a precise open question for a future document.

D.1 Background

Doc. 336 [K]. The injection $\iota(Q) = (3Q \bmod 3, \lfloor Q \rfloor)$ shows $N \bmod 3$ classifies $Q = N/3$ completely.

Doc. 339 [B]. Eight gluons $= 4\,\mathbb{Z}_{13}$-orbits $\times 2\,\mathbb{Z}_2$-parities; three elements per orbit correspond to three colours.

Doc. 341 [B]. $N \bmod 3 \in GF(3)$ classifies colour singlet ($N \equiv 0$) and colour triplet ($N \equiv 1, 2$).

D.2 Theorem A — Quadratic residues and electric neutrality

Theorem D.2.1 (Electric neutrality ↔ QR mod 13 **[B]**). • $O_1 = \{1, 3, 9\}$ *and* $O_3 = \{4, 10, 12\}$*: all elements are quadratic residues mod 13. These orbits carry* $Q_{\text{el}} = 0$.

- $O_2 = \{2, 5, 6\}$ *and* $O_4 = \{7, 8, 11\}$*: all elements are quadratic non-residues mod 13. These orbits carry* $Q_{\text{el}} \neq 0$.

Proof. $\text{QR}_{13} = \{1, 3, 4, 9, 10, 12\}$. $O_1, O_3 \subset \text{QR}_{13}$; $O_2, O_4 \subset \text{NQR}_{13}$. □

The Majorana layer $\{O_1, O_3\}$ (Doc. 343) is exactly the QR layer. The Dirac layer $\{O_2, O_4\}$ is exactly the NQR layer. The two algebraic criteria — Majorana/Dirac and QR/NQR — coincide and mean the same thing: QR = electrically neutral **[B]**.

D.3 Theorem B — Complete fermion classification

Theorem D.3.1 (Two Galois properties classify all fermions **[B]**).

QR mod 13	$N \bmod 3$	*Fermion type*	*Example*
QR (neutral)	$\equiv 0$ *(singlet)*	*Neutrino*	ν_e, ν_μ, ν_τ
QR (neutral)	$\equiv 1, 2$ *(triplet)*	—	*(forbidden)*
NQR (charged)	$\equiv 0$ *(singlet)*	*Ch. lepton*	e, μ, τ
NQR (charged)	$\equiv 1, 2$ *(triplet)*	*Quark*	*up, down, ...*

The row "QR + triplet = forbidden" is algebraically forced: no electrically neutral coloured fermion exists in the Standard Model. This is a direct consequence of the $\text{GF}(27)^*$ structure.

D.4 Theorem C — Charge quantisation from the Legendre symbol

$$Q_{\text{el}} = 0 \iff \left(\frac{k}{13}\right) = +1 \text{ for all } k \in O_i.$$

Algebraic derivation of $Q \in \{0, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm 1\}$ from Galois structure alone **[B]**.

D.5 Theorem D — Three generations from orbit elements

Each primitive orbit has exactly three elements under Frobenius $k \mapsto 3k \bmod 13$. The three elements correspond to three generations **[S]**:

Orbit	Gen. 1	Gen. 2	Gen. 3
$O_1 = \{1, 3, 9\}$	ν_e	ν_μ	ν_τ
$O_3 = \{4, 10, 12\}$	$\bar{\nu}_e$	$\bar{\nu}_\mu$	$\bar{\nu}_\tau$
$O_4 = \{7, 8, 11\}$	e^-	μ^-	τ^-
$O_2 = \{2, 5, 6\}$	u	c	t

The mass hierarchy within each generation requires the Yukawa formula of Doc. 338.

D.6 Theorem E — Gell-Mann–Nishijima from Witt-pair operators (open) [S]

Motivation. The three Witt-pair number operators N_1, N_2, N_3 from Doc. 344 satisfy $N = N_1 + N_2 + N_3$ (the total number operator). In the Standard Model, the Gell-Mann–Nishijima relation is:

$$Q = I_3 + \frac{Y}{2}, \qquad I_3 = \frac{1}{2}(N_1 - N_2), \qquad Y = \frac{1}{3}(N_1 + N_2 - 2N_3).$$

Check for leptons: Neutrino: $N_1 = N_2 = N_3 = 0 \Rightarrow I_3 = 0$, $Y = 0$, $Q = 0$ ✓. Positron: $N_1 = N_2 = N_3 = 1 \Rightarrow I_3 = 0$, $Y = 0$, $Q = 0$ – the proposed decomposition does not reproduce the positron (×).

Why it is [S]. For quarks, the decomposition of N into N_1, N_2, N_3 is not unique without an additional assignment rule. The formula gives the correct Q for the neutrino (not for the positron, see above) and requires an explicit N_1, N_2, N_3 assignment for each quark state to be verified. This is a precise open question: can the Witt-pair decomposition of GF$(27)^*$ orbits force I_3 and Y algebraically?

Note on an external check. An external check (4 September 2026) proposed $Q(g^k) = (k \bmod 3)/3$ element-wise. This is correct for the total number operator $N(g^k) = k \bmod 3$ from Doc. 341, combined with $Q = N/3$ from Doc. 336. However, N is not constant within an orbit (e.g. $O_1 = \{1, 3, 9\}$ has $k \bmod 3 \in \{1, 0, 0\}$), so Q is not a single value per orbit element — it is an orbit-averaged or ground-state property. The orbit-level property (Theorem A, QR$\leftrightarrow$ $Q = 0$) is the correct statement at orbit level. Gell-Mann–Nishijima requires the individual N_1, N_2, N_3 decomposition and remains **[S]**.

Verification script

`pruef_346_ladungsquantisierung.py` — Theorems A–C verified **[B]**. Theorem E: not yet scriptable (requires explicit N_1, N_2, N_3 assignment).

D.7 Summary

Table D.1: Results Doc. 346

Theorem	Content	Status
A	QR mod 13 $\leftrightarrow$ electric neutrality	**[B]**
B	QR + N mod 3 classifies all fermions	**[B]**
C	Charge quantisation from Legendre symbol	**[B]**
D	Three generations = three Frobenius powers	**[S]**
E	Gell-Mann–Nishijima from Witt-pair operators	**[S]**

Open: SU$(2)_L$ doublet structure explicitly from the Galois algebra.

Note on AI assistance

Prepared with support from Claude (Anthropic). External algebraic check (4 September 2026). Physical interpretations and conclusions by Johann Pascher (ORCID 0009-0000-6518-4064).

[1] J. Pascher, *Doc. 336: GF(9)-FFGFT bridge*, FFGFT corpus (Aug. 2026).

[2] J. Pascher, *Doc. 339: Frobenius separation*, FFGFT corpus (Aug. 2026).

[3] J. Pascher, *Doc. 341: GF(27) in GALG and FFGFT*, FFGFT corpus (Sept. 2026).

[4] J. Pascher, *Doc. 343: Zeta function of T^4/Z_3*, FFGFT corpus (Sept. 2026).

[5] J. Pascher, *Doc. 344: Furey fermions in GALG and FFGFT*, FFGFT corpus (Sept. 2026).

[6] C. Furey, Ph.D. Thesis, Waterloo 2015, arXiv:1611.09182.

Chapter E

Doc. 347 — Gell-Mann–Nishijima from the Galois Structure: $Q = I_3 + Y/2$ algebraically derived

Status markers

[K] *Confirmed* — algebraically complete.

[B] *Established* — secured by verification script.

[S] *Speculative* — conjecture, not yet fully proved.

Abstract

We examine how far the Gell-Mann–Nishijima relation $Q = I_3 + Y/2$ follows from two already established Galois properties. (1) Isospin $I_3 = \pm\frac{1}{2}$ follows from the Su/Sd split via the jE7 projector (Doc. 344, Theorem E) **[B]**. (2) Hypercharge Y is *not* fixed by the QR/NQR bit mod 13 (Doc. 346) alone: charged leptons and quarks are both NQR, and the lepton doublet $(\nu, e)_L$ lies on both sides of the bit (Theorem A, **[S]**). With the SM values $Y = -1$ (lepton doublet) and $Y = +\frac{1}{3}$ (quark doublet) the relation gives the four fermion charges $\{0, -1, +\frac{2}{3}, -\frac{1}{3}\}$; Theorem E of Doc. 346 is thereby closed for I_3 and open for Y **[S]**.

E.1 Background

Doc. 346 established the fermion classification from two Galois properties: QR/NQR bit (electric neutrality) and $N \bmod 3$ (colour charge). Theorem E [**[S]**] left open: does $Q = I_3 + Y/2$ also follow from Galois structure?

E.2 Theorem A — Hypercharge from QR/NQR

Theorem E.2.1 (Hypercharge Y is not fixed by the QR/NQR bit alone **[S]**)**.** *By Doc. 346 (Theorem B and orbit table): QR orbits $O_1 = \{1,3,9\}$ = neutrinos, $O_3 = \{4,10,12\}$ = antineutrinos; NQR orbits $O_4 = \{7,8,11\}$ = charged leptons, $O_2 = \{2,5,6\}$ = quarks. There is no function $Y = f(\mathrm{NQR}_{13})$ reproducing the SM values $Y = -1$ (lepton doublet $(\nu,e)_L$) and $Y = +\frac{1}{3}$ (quark doublet $(u,d)_L$). The assignment of Y must also include $N \bmod 3$ and the particle/antiparticle distinction; it is open.*

Proof. Assume $Y = f(\mathrm{NQR}_{13})$. Neutrino (O_1, QR): $f(0) = -1$. Electron (O_4, NQR), in the same doublet as the neutrino: $f(1) = -1$. Up quark (O_2, NQR): $f(1) = +\frac{1}{3}$ — contradiction with $f(1) = -1$. Charged leptons and quarks both lie on the NQR side, the lepton doublet $(\nu,e)_L$ on both sides of the bit; the bit therefore neither separates leptons from quarks nor keeps the doublet partners together. In addition, the QR orbit O_3 (antineutrinos) carries the opposite sign of Y to O_1, although both are QR. □ □

Connection to Doc. 346. The NQR bit is exactly the Legendre symbol $\left(\frac{k}{13}\right) = -1$ — the same property classifying electric neutrality in Doc. 346 **[B]**. Hypercharge does not follow from it (Theorem A) **[S]**.

E.3 Theorem B — Isospin from Su/Sd split

Theorem E.3.1 (Isospin I_3 from Su/Sd split **[B]**)**.**

$$I_3 = \begin{cases} +\frac{1}{2} & \text{Su sector (Doc. 344: } F \cdot jE7 = +F\text{)} \\ -\frac{1}{2} & \text{Sd sector (Doc. 344: } F \cdot jE7 = -F\text{)} \end{cases}$$

The Su/Sd separation by the jE7 projector (Doc. 344, Theorem E) is exactly the doublet splitting of weak isospin. No new assumption — the jE7 projector was already established in the GALG/Furey construction.

E.4 Theorem C — Gell-Mann–Nishijima with isospin from Galois

Theorem E.4.1 (Gell-Mann–Nishijima with I_3 from Theorem B and SM hypercharges **[S]**)**.** *$Q = I_3 + Y/2$ with I_3 from Theorem B and the SM values of Y (not derivable from Theorem A):*

Fermion	I_3	Y	$Q = I_3 + Y/2$	*Exp.*
Neutrino	$+\frac{1}{2}$	-1	0	0 ✓
Comparison values: PDG 2022 [6]; all four charges reproduced exactly.				
Electron	$-\frac{1}{2}$	-1	-1	-1 ✓
up quark	$+\frac{1}{2}$	$+\frac{1}{3}$	$+\frac{2}{3}$	$+\frac{2}{3}$ ✓
dn quark	$-\frac{1}{2}$	$+\frac{1}{3}$	$-\frac{1}{3}$	$-\frac{1}{3}$ ✓

*All four values agree **[B]**; only I_3 is derived from the Galois structure, the hypercharge Y is inserted as SM value **[S]**.*

E.5 Theorem D — Hypercharge and Legendre symbol (orbit level)

Legendre symbol, particle assignment by Doc. 346 and SM hypercharge (left-handed doublet or its antiparticle) at orbit level:

Orbit	$\left(\frac{k}{13}\right)$	Particles (Doc. 346)	Y (SM)
$O_1 = \{1, 3, 9\}$	$+1$ (QR)	ν	-1
$O_3 = \{4, 10, 12\}$	$+1$ (QR)	$\bar{\nu}$	$+1$
$O_2 = \{2, 5, 6\}$	-1 (NQR)	quarks	$+\frac{1}{3}$
$O_4 = \{7, 8, 11\}$	-1 (NQR)	e, μ, τ	-1

Both values of the Legendre symbol occur with different Y; Y is not a function of the Legendre symbol alone **[S]**.

E.6 Summary and status of Theorem E (Doc. 346)

Table E.1: Results Doc. 347

Theorem	Content	Status
A	Y not fixed by the QR/NQR bit alone	**[S]**
B	$I_3 = \pm\frac{1}{2}$ from jE7 projector (Su/Sd)	**[B]**
C	$Q = I_3 + Y/2$ with I_3 from Galois, Y from the SM	**[S]**
D	Legendre symbol does not separate Y at orbit level	**[S]**

Theorem E [**[S]**] of Doc. 346 is only partly closed. Isospin I_3 follows from the jE7 projector (Su/Sd split, Doc. 344) **[B]**; charge quantisation from the Legendre symbol mod 13 remains **[B]** (Doc. 346). Hypercharge Y does not follow from the QR/NQR bit alone and remains open **[S]**.

E.7 What remains open: SU$(2)_L$ and CKM [S]

Hypercharge. Y from N mod 3 and the particle/antiparticle distinction (Theorem A). **[S]**

SU$(2)_L$ chirality. Chirality requires $\gamma_5 = e_1 \cdots e_6 \in \mathrm{Cl}(6)$. The $\mathbb{Z}_2$ parity in GF$(27)^*$ separates particles from anti-particles ($k < 13$ vs. $k \geq 13$), not left-handed from right-handed. All four primitive orbits $O_1 \ldots O_4$ are same-chiral. Needed: the bridge GF$(27)^* \to \mathrm{Cl}(6) \xrightarrow{\gamma_5}$ chirality. Begun in Docs. 341/344, not yet complete. **[S]**

CKM mixing angles. GF(27) carries a trace bilinear form $\langle a, b\rangle = \mathrm{Tr}_{\mathrm{GF}(27)/\mathrm{GF}(3)}(ab)$. The Cabibbo angle between generations would be the overlap $|\langle O_{2,\mathrm{Gen.1}}, O_{4,\mathrm{Gen.2}}\rangle|/(\|O_2\|\|O_4\|)$. Experimental: $\sin\theta_C \approx 0.225$; Galois candidate $\sin(\pi/14) \approx 0.222$ ($-1.2\,\%$). Three checked directions: (1) direct ξ-corrections: ratio ≈ 1685, no mechanism. (2) $\xi^{|k_i - k_j|}$ weighting: values

$\sim 10^{-8}$, no mechanism. (3) Galois quotients: $2/9 \approx 0.222$ ($-1.4\,\%$), $3/13 \approx 0.231$ ($+2.4\,\%$), near but not forced. Full derivation requires its own document. **[S]**

Verification script

`pruef_347_gell_mann.py` — orbit assignment by Doc. 346, proof that Y is not a function of the QR/NQR bit (Theorems A, D), I_3 (Theorem B) and $Q = I_3 + Y/2$ with SM values (Theorem C); all assertions passed.

Note on AI assistance

Prepared with support from Claude (Anthropic). Physical interpretations and conclusions by Johann Pascher (ORCID 0009-0000-6518-4064).

[1] J. Pascher, *Doc. 344: Furey fermions in GALG and FFGFT*, FFGFT corpus (Sept. 2026).

[2] J. Pascher, *Doc. 346: Charge quantisation from GF*$(27)^*$, FFGFT corpus (Sept. 2026).

[3] J. Pascher, *Doc. 339: Frobenius separation*, FFGFT corpus (Aug. 2026).

[4] M. Gell-Mann, Nuovo Cimento 4 Suppl. 2 (1956) 848.

[5] K. Nishijima, Prog. Theor. Phys. 13 (1955) 285.

[6] R.L. Workman *et al.* (Particle Data Group), Prog. Theor. Exp. Phys. **2022**, 083C01 (2022).

[7] C. Furey, Ph.D. Thesis, Waterloo 2015, arXiv:1611.09182.

Chapter F

Doc. 348 — Generation Structure and Coupling Matrix from GF(27): Trace Bilinear Form, Orbit Assignment, and CKM Complexity

Status markers

[K] *Confirmed* — algebraically complete.

[B] *Established* — secured by verification script.

[S] *Speculative* — conjecture, not yet fully proved.

Abstract

The four irreducible polynomials $f_1, \ldots, f_4$ over GF(3) define the physical orbits in GF$(27)^*$. Their root orbits are genuine Frobenius orbits under $k \mapsto 3k$ mod 26; the three elements correspond to the three generations. The trace bilinear form $\langle a, b\rangle = \mathrm{Tr}_{\mathrm{GF}(27)/\mathrm{GF}(3)}(ab)$ gives the coupling structure between sectors: $f_3 \times f_4$ (charged leptons × quarks) is diagonal in the Frobenius ordering, $M^{(34)} = -I$ **[B]**; $f_1 \times f_3$ (neutrinos × charged leptons) has normalised entries $1/\sqrt{2}$ **[B]**. Mixing angles (CKM, PMNS) remain **[S]**; the reason for the complexity of the CKM matrix is analysed.

F.1 Correct orbit definition

Preliminary note. Earlier documents (Docs. 339, 346, 347) used orbits $O_1 = \{1, 3, 9\}$, $O_2 = \{2, 5, 6\}$, $O_3 = \{4, 10, 12\}$, $O_4 = \{7, 8, 11\}$ as $\mathbb{Z}_{13}$-orbits under $k \mapsto 3k$ mod 13. These remain useful for charge and QR/NQR statements.

For the generation structure and the bilinear form, the **root orbits of the irreducible polynomials** are decisive — these are the genuine Frobenius orbits under $k \mapsto 3k$ mod 26:

Polynomial	Root orbit	Physical role (Doc. 343)
$f_1 = x^3 + 2x + 1$	$\{1, 3, 9\}$	Neutrinos ν_1 (Majorana)
$f_2 = x^3 + 2x^2 + 1$	$\{17, 23, 25\}$	Neutrinos ν_2 (Majorana)
$f_3 = x^3 + 2x^2 + x + 1$	$\{5, 15, 19\}$	Charged leptons (e, μ, τ) / ν_3
$f_4 = x^3 + x^2 + 2x + 1$	$\{7, 11, 21\}$	Quarks (Dirac sector)

Theorem F.1.1 (Frobenius cycles of root orbits **[B]**)**.** *The root orbits are closed under Frobenius* $k \mapsto 3k \bmod 26$*:* $f_1 : 1 \to 3 \to 9 \to 1$*;* $f_2 : 17 \to 25 \to 23 \to 17$*;* $f_3 : 5 \to 15 \to 19 \to 5$*;* $f_4 : 7 \to 21 \to 11 \to 7$*. The three elements of each orbit correspond to the three generations.*

F.2 Trace bilinear form

$$\langle g^k, g^j \rangle = \mathrm{Tr}_{\mathrm{GF}(27)/\mathrm{GF}(3)}(g^{k+j}) = g^{k+j} + g^{3(k+j)} + g^{9(k+j)} \in \mathrm{GF}(3).$$

Values in $\{0, 1, 2\}$; as real numbers: $0 \mapsto 0$, $1 \mapsto +1$, $2 \mapsto -1$.

F.3 Theorem B — Coupling matrix $f_3 \times f_4$ (CKM sector)

Theorem F.3.1 (Diagonal coupling matrix in the quark-lepton sector **[B]**)**.** *The coupling matrix between charged leptons (*f_3*) and quarks (*f_4*), with the generations in the Frobenius ordering (*f_3*:* $5 \to 15 \to 19$*,* f_4*:* $7 \to 21 \to 11$*, generation = position in the cycle), is diagonal:*

$$M^{(34)} = -I = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Lepton Gen. i couples only to quark Gen. i **[B]**. Reading f_4 in the sorted order $\{7, 11, 21\}$ of the table produces a swap Gen. 2↔Gen. 3; it is an ordering artefact and not algebraically forced.

The experimental CKM matrix is nearly diagonal with small mixing angles. The Galois coupling matrix gives the zero structure (which Gen. combinations do not couple), but not the mixing angles themselves. **[S]**

F.4 Theorem C — Coupling matrix $f_1 \times f_3$ (PMNS sector)

Theorem F.4.1 (Equal-weight coupling in the neutrino sector **[B]**)**.**

$$M^{(13)} = \begin{pmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad |M^{(13)}_{ij}|_{\mathrm{norm}} = \frac{1}{\sqrt{2}} \text{ for all non-zero entries.}$$

Each neutrino generation couples to two charged lepton generations with weight $1/\sqrt{2}$ — consistent with the large PMNS mixing angles ($\theta_{23} \approx 49^\circ$, $\sin^2\theta_{23} \approx 0.57$, near $(1/\sqrt{2})^2 = 0.5$, NuFIT 5.3 [5]; maximal mixing would be 45°). **[S]**

F.5 Why CKM mixing angles are complex

The CKM matrix has four independent parameters (Wolfenstein): $\lambda \approx 0.225$, $A \approx 0.811$, $\rho \approx 0.124$, $\eta \approx 0.356$. The CP-violating phase $\delta_{CP} \approx 68^\circ$ ($e^{i\delta_{CP}} = 0.375 + 0.927i$) makes the matrix essentially complex (CP violation in kaon and B-meson decays, observed 1964 and 2001).

What GF$(27)^*$ provides and what it does not. $\mathrm{Gal}(\mathrm{GF}(27)/\mathrm{GF}(3)) = \mathbb{Z}_3$ has three irreducible representations over $\mathbb{C}$: trivial (1), $\omega = e^{2\pi i/3}$ (120°), ω^2 (240°). If the coupling matrix transforms under ρ_1, a complex phase arises — the most plausible Galois candidate for the CP sector. **Problem:** ω gives 120°, not 68° (gap 52°). No algebraically forced candidate **[S]**.

Three steps needed for full CKM from GF(27): (1) **Real zero structure**: diagonal matrix from trace bilinear form **[B]**. (2) **Mixing angles**: corrections to the diagonal structure giving $\lambda \approx 0.225$, $A \approx 0.811$. Three checked paths: direct ξ-corrections ($\sin\theta_C/\xi \approx 1685$, too large); $\xi^{|k_i - k_j|}$ weighting ($\sim 10^{-8}$, too small); Galois quotients $2/9 \approx 0.222$ and $3/13 \approx 0.231$ (near, but not forced). No candidate **[S]**. (3) **CP phase**: complex embedding of Galois group forcing $\delta_{CP} \approx 68^\circ$. $\mathbb{Z}_3$ phases give 120°, not 68°. GF$(27)^*$ has no natural complex structure; the trace bilinear form gives values in $\{0, 1, 2\} \subset \mathrm{GF}(3)$ **[S]**.

Important distinction: not derivable ≠ not usable. $\delta_{CP} \approx 68^\circ$ can be set as a phase anchor — structurally identical to $v = 246$ GeV or C_{conv}: empirically given, not freely adjustable, and once set all CP-violating observables are fully determined. Deriving δ_{CP} from first principles remains **[S]**; computing with it is immediately possible.

F.6 Summary

Table F.1: Results Doc. 348

Theorem	Content	Status
A	Root orbits = correct Frobenius orbits; 3 elements = 3 generations	**[B]**
B	$f_3 \times f_4$ coupling matrix diagonal ($-I$)	**[B]**
C	$f_1 \times f_3$ coupling matrix with $1/\sqrt{2}$ weights	**[B]**
D	Full CKM/PMNS angles from bilinear form	**[S]**

Verification script

`pruef_348_ckm.py` — Theorems A–C, all assertions passed **[B]**.

Note on AI assistance

Prepared with support from Claude (Anthropic). Physical interpretations and conclusions by Johann Pascher (ORCID 0009-0000-6518-4064).

[1] J. Pascher, *Doc. 339: Frobenius separation*, FFGFT corpus (Aug. 2026).

[2] J. Pascher, *Doc. 343: Zeta function of T^4/Z_3*, FFGFT corpus (Sept. 2026).

[3] J. Pascher, *Doc. 346: Charge quantisation from $GF(27)^*$*, FFGFT corpus (Sept. 2026).

[4] J. Pascher, *Doc. 347: Gell-Mann-Nishijima from Galois*, FFGFT corpus (Sept. 2026).

[5] I. Esteban *et al.*, JHEP **09** (2020) 178; NuFIT 5.3 (2023), www.nu-fit.org.

[6] R.L. Workman *et al.* (Particle Data Group), Prog. Theor. Exp. Phys. **2022**, 083C01 (2022).

[7] N. Cabibbo, PRL 10 (1963) 531.

[8] M. Kobayashi, T. Maskawa, PTP 49 (1973) 652.

[9] C. Furey, Ph.D. Thesis, Waterloo 2015, arXiv:1611.09182.

Chapter G

Doc. 349 — SU$(2)_L$ Chirality from the Galois Structure: Su/Sd Separation and Cl(6) Grade Parity

Johann Pascher
ORCID 0009-0000-6518-4064
4 September 2026

Status markers

[K] *Confirmed* — algebraically complete.
[B] *Established* — secured by script.
[S] *Speculative* — conjecture, still open.

Abstract

We examine whether the SU$(2)_L$ chirality structure of the Standard Model follows from the Cl(6) grade parity established in Doc. 341. The pseudoscalar $\gamma_5 = e_1e_2e_3e_4e_5e_6$ acts on grade r as the grade involution, $\gamma_5 X \gamma_5^{-1} = (-1)^r X$; this is not an eigenvalue equation on blades, and in Cl$(6,0)$, $\gamma_5^2 = -1$. Chirality has to be captured by a projector $P_\pm = (1 \pm i\gamma_5)/2$ in the complexification. The Su/Sd split (Doc. 344) is therefore not directly the right/left split: by Doc. 344, Su contains even and odd N values, Sd[0] is the anti-neutrino, and Cl(6) has no grade 7 for e_L. The SU$(2)_L$ chirality structure left open in Docs. 347/348 remains open **[S]**.

Background

Doc. 344 **[B]**: the jE7 projector separates the Su and Sd sectors, with $N(\text{Su}) = [0,1,1,1,2,2,2,3]$ (even and odd values) and negative $N(\text{Sd})$; Su[1–3] = anti-down, Su[4–6]

= up, Su[7] = positron, Sd[0] = anti-neutrino, Sd[7] = electron. A split "Su: N even, Sd: N odd" does not exist; indexing and N values below follow Doc. 344.

G.1 Theorem A — Grade parity and γ_5 [K]

$\gamma_5 X \gamma_5^{-1} = (-1)^r X$ for a grade-r element $X \in \mathrm{Cl}(6)$, since γ_5 anticommutes with every generator e_i, so that $\gamma_5 X = (-1)^r X \gamma_5$ for a grade-r blade. **[K]**This is a conjugation statement (grade involution), not an eigenvalue equation on blades: left multiplication by γ_5 maps grade r to grade $6 - r$ (e.g. $\gamma_5 \cdot 1 = \gamma_5 \neq 1$). Moreover, in $\mathrm{Cl}(6,0)$ with $e_i^2 = +1$: $\gamma_5^2 = (-1)^{15} = -1$. Chirality has to be captured via conjugation or a projector, e.g. $P_\pm = (1 \pm i\gamma_5)/2$ in the complexification since $(i\gamma_5)^2 = +1$. An assignment "even grades = right-handed, odd grades = left-handed" as a γ_5 eigenvalue on blades therefore does not hold.

G.2 Theorem B — Su sector and Cl(6) grades

"Su = even grades = right-handed" does not hold: by Doc. 344 the Su sector contains even and odd N values, and even read by N classes, $|N| = 3$ in Su is the positron (not e_R). In addition, by Theorem A the grade carries no γ_5 eigenvalue.

G.3 Theorem C — Sd sector and Cl(6) grades

Likewise "Sd = odd grades = left-handed" does not hold: by Doc. 344, Sd[0] is the anti-neutrino and Sd[7] the electron, and read by N classes, $|N| = 2$ in Sd is anti-up (not $\bar{d}$). $\mathrm{Cl}(6)$ has only grades 0 to 6 with $1, 6, 15, 20, 15, 6, 1$ blades, 2^6 in total; there is no grade 7, not even as a multigrade. An assignment grade $= 2N + 1$ therefore breaks down at $N = 3$; the assignment of e_L is open **[S]**.

G.4 Theorem D — SU$(2)_L$ doublets (open)

A doublet assignment $(\nu_L, e_L) = (\mathrm{Sd}[0], \mathrm{Sd}[3])$ does not hold, because Sd[0] is the anti-neutrino by Doc. 344 and the assignment of e_L is open (Theorem C). Whether $\mathrm{SU}(2)_L$ acts on exactly one of the two sectors, and which states form left-handed doublets and right-handed singlets, remains open **[S]**; the question has to be posed with the projector $P_\pm$ (Theorem A) and with the indexing and N values of Doc. 344.

G.5 Theorem E — $\mathbb{Z}_3$ fixed points/orbit and grade parity

From Doc. 341: the σ rotation separates $\mathrm{Vss}[0, 2, 4]$ (σ-fixed points, even grades) from $\mathrm{Vss}[1, 3, 5]$ (σ-three-orbit, odd grades) **[B]**. Identifying these classes with right- and left-handed presupposes the eigenvalue reading of the grade parity and, by Theorem A, does not hold in this form. Whether the $\mathbb{Z}_3$ symmetry of T^4/Z_3 forces the chirality split via $P_\pm$ is open **[S]**.

G.6 Summary

Theorem	Content	Status
A	$\gamma_5 X \gamma_5^{-1} = (-1)^r X$ (grade involution, no eigenvalue)	**[K]**
B	Su sector $\neq$ even grades (Doc. 344)	—
C	Sd sector $\neq$ odd grades; no grade 7, e_L open	**[S]**
D	$\mathrm{SU}(2)_L$ doublets from the Su/Sd sector	**[S]**
E	$\mathbb{Z}_3$ fixed points/orbit = even/odd grades; chirality open	**[S]**

The grade parity of Doc. 341 is captured via γ_5 as the grade involution **[K]**; the $\mathrm{SU}(2)_L$ chirality structure is not thereby derived. Open **[S]**: the chirality assignment via the projector $P_\pm$ with indexing and N values from Doc. 344, the assignment of e_L, the $\mathrm{SU}(2)_L$ doublet structure, and the exact CKM mixing angles and CP phase δ_{CP} (Doc. 348).

Verification script

`pruef_349_chiralitaet.py` — checks the signs $(-1)^r$ of the grade parity; it does not support the chirality and sector assignments of Theorems B–E.

Note on AI assistance

Prepared with support from Claude (Anthropic). Physical interpretations: Johann Pascher (ORCID 0009-0000-6518-4064).

[1] J. Pascher, Doc. 341, FFGFT corpus (Sept. 2026).

[2] J. Pascher, Doc. 344, FFGFT corpus (Sept. 2026).

[3] J. Pascher, Doc. 347, FFGFT corpus (Sept. 2026).

[4] C. Furey, Ph.D. Thesis, Waterloo 2015, arXiv:1611.09182.

Chapter H

Doc. 350 — Black Holes and Time-Mass Duality: Covariance of G, Evaporation Sequence, and the Neutrino as Final State

Status markers

- **[K]** *Confirmed* — algebraically complete.
- **[B]** *Established* — secured by verification script.
- **[S]** *Speculative* — conjecture, still open.

Note. This document is **[S]** as a whole. It draws consequences from $T \cdot m = 1$ (Doc. 003) and $G = \xi^2/(4m)$ (Doc. 012, R58) for a regime — black holes, Hawking evaporation — that the corpus has so far touched only marginally (Doc. 025: Hawking radiation as return to the ξ-field; Doc. 049: field-node tunnelling through the horizon; Doc. 306: horizon as extremum $T_0 = \xi t_P$, $E_{\max} = E_P/\xi$). The algebraic relations are secured by `pruef_350`; their physical validity in this regime is not.

Abstract

From $T \cdot m = 1$ and $G = \xi^2/(4m)$ follows $G \propto T$: an object's gravitational coupling grows with its intrinsic time, while $m \cdot G = \xi^2/4$ stays invariant **[B]**. For an evaporating black hole this means: the sequence of emitted patterns is fixed by the mass hierarchy, and the last stable pattern is the neutrino — the state with the longest intrinsic time and, algebraically, the vacuum of the Clifford-Fock space (Su[0]=Vss[0], Doc. 344) **[B]**. A point-like singularity with finite G is excluded **[B]** (Doc. 078). Whether the electron pattern (or any pattern) is still the same object under strongly altered T remains open **[S]**.

H.1 Starting point: the question

Solving the FFGFT gravity equation for a Planck-scale mass — does G go to zero while mass remains finite? And is there a calculable difference from the standard calculation?

The answer is not yes or no. The question presupposes that G and m can be varied independently. In FFGFT they cannot.

H.2 Theorem A — Covariance of G under $T \cdot m = 1$

Theorem H.2.1 (G scales with intrinsic time **[B]**)**.** *From $T \cdot m = 1$ (Doc. 003) and $G = \xi^2/(4m)$ (Doc. 012):*

$$G = \frac{\xi^2}{4}\, T, \qquad m \cdot G = \frac{\xi^2}{4} = \textit{const.}$$

Proof. Substitute $m = 1/T$ into $G = \xi^2/(4m)$. □

Reading. If an object's intrinsic time grows (e.g. by time dilation near a horizon), its mass drops and its gravitational coupling rises — the product is a Galois number. "$G \to 0$ at finite mass" is thus excluded: $G \to 0$ would mean $T \to 0$, hence $m \to \infty$.

H.3 Theorem B — No point-like singularity

Theorem H.3.1 (Mass singularity excluded **[B]**)**.** *Since $m \cdot G = \xi^2/4$ is invariant, there is no state with $m \to \infty$ at finite G and none with $G \to \infty$ at finite m. The classical notion of singularity (finite mass in vanishing volume at constant G) has no counterpart in FFGFT.*

Proof. Doc. 306 derives the exclusion natively: from $T \cdot m = 1$ follows $E \to \infty \Leftrightarrow T \to 0$; but FFGFT's time has a minimal, non-vanishing increment $d_1 = 100\,\xi_1 = 1/75$, fixed by $\xi = 4/30000$. Hence T is bounded below and E above. The horizon of a black hole is the *extremum*: slowest time $T = T_0 = \xi t_P$, maximal concentration $E_{\max} = E_P/\xi$, with exact closure $T_0 E_{\max} = 1$ — all finite. Doc. 078 states the same as an exclusion theorem ($m = \infty$ forces $T = 0$, outside the domain). With $G = \xi^2/(4m)$ (Doc. 012) it follows that $G \to 0$ and $G \to \infty$ are excluded as well. □

Note on the derivation. The exclusion is derived corpus-natively (Doc. 306), not via Heisenberg's energy-time uncertainty — the uncertainty relation is not applicable here (no canonical time operator, Pauli). The conclusion is: no singular beginning, finite core with ξ-set scale. Consistent with R58 (no separate quantum gravity needed because T^4/Z_3 is pre-ordered). What remains **[S]** is only the application to the interior *beyond* the horizon — the horizon itself is, per Doc. 306, the extremum, not a transition to a singularity.

H.4 Theorem C — Hierarchy of intrinsic times

Theorem H.4.1 (Neutrino as state of longest intrinsic time **[B]**)**.** *Under $T \cdot m = 1$, $T_\nu/T_e = m_e/m_\nu \approx 10^7$ (ν_3) to $1.1 \cdot 10^8$ (ν_1, lightest neutrino; masses from Doc. 340) and $T_e/T_t = m_t/m_e \approx$*

$3 \cdot 10^5$. *The neutrino is the massive fermion with the longest intrinsic time. Algebraically it is the vacuum state: Su[0]=Vss[0] (Doc. 344, Theorem C)* ***[B]****.*

Doc. 025 gives the same hierarchy as $\tau_n = \tau_\xi / \xi^n$: each structural level (particle, atom, molecule, cell) has a time scale longer by $1/\xi$. The neutrino sits at the bottom of the mass scale and thus at the top of the time scale.

H.5 Theorem D — Evaporation sequence

Theorem H.5.1 (Order of Hawking emission by mass **[S]**)**.** *An evaporating black hole loses mass; as $M \to 0$, T_{hole} grows. The emittable patterns at each T are those whose own intrinsic time matches T_{hole}. The sequence follows the mass hierarchy:*

$$t \to b \to \tau \to c \to \mu \to s \to d \to u \to e \to \nu.$$

The last stable pattern before complete dissolution is the neutrino.

Corpus connection. Doc. 025 names Hawking radiation as "energy return to the ξ-field". Theorem D sharpens this: the return runs through the mass hierarchy and ends at the state that is algebraically already the vacuum. Doc. 049 describes the same process as field-node tunnelling through the horizon — Theorem D orders the nodes by intrinsic time.

H.6 What happens to an electron near the horizon [S]

Near the horizon local time is dilated. Under $T \cdot m = 1$: m_e drops, G_e rises. But m_e is not isolated — $\alpha = \xi E_0^2 / K_{\text{frak}}$ (Doc. 011), $v = m_e/((4/3)\xi^{3/2})$ (Doc. 006), G_F, all Yukawa couplings hang on the same scale. If T changes, all change coherently.

Hence "what happens to the electron" cannot be answered without settling whether the electron *pattern* — Galois orbit $O_4 = \{7, 8, 11\}$, NQR class (Doc. 346), Sd[7] state (Doc. 344) — is still the same object under altered T. The orbit is an algebraic invariant; the mass assigned to it is not. Whether the result should still be called "electron" is a matter of definition that the corpus does not decide.

What leaves as Hawking radiation is thermal — a mixed state in which the identity of the infallen pattern survives not as a named particle but as a contribution to the spectrum. The information paradox is not resolved by this, only relocated into the T-structure of the final state **[S]**.

H.7 Answer to the original question

1. $G \to 0$ at finite mass: **no** — $G \propto T$, and $T \to 0$ forces $m \to \infty$ **[B]**.
2. Calculable difference from the standard calculation: **yes**, but not as a numerical value of G (locally identical, Doc. 012) — as structure: $m \cdot G = \xi^2/4$ is a Galois number, and the Planck scale is the scale at which the *local* values of G, $\hbar$, c fit together — not a universal limit **[S]**.
3. What remains at the end: a neutrino-like pattern, then the vacuum Vss[0] **[S]**.

H.8 Summary

Table H.1: Results Doc. 350

Theorem	Content	Status
A	$G = \frac{\xi^2}{4} T$; $m \cdot G$ invariant	**[B]**
B	No mass singularity; horizon = extremum (Doc. 306/078)	**[B]**
C	Neutrino = longest intrinsic time = vacuum	**[B]**
D	Evaporation sequence ends at the neutrino	**[S]**
—	Identity of the electron under altered T	open **[S]**

Verification script

`pruef_350_schwarzes_loch.py` — Theorems A–D, algebraic consistency passed. Overall physical status: **[S]**.

Note on AI assistance

Prepared with support from Claude (Anthropic). Physical interpretations and conclusions: Johann Pascher (ORCID 0009-0000-6518-4064).

[1] J. Pascher, *Doc. 003: T0 Foundations*, FFGFT corpus.

[2] J. Pascher, *Doc. 012: Gravitational constant*, FFGFT corpus.

[3] J. Pascher, *Doc. 025: T0 Cosmology*, FFGFT corpus.

[4] J. Pascher, *Doc. 078: Time — exclusion theorem* $T \cdot m = 1$, FFGFT corpus.

[5] J. Pascher, *Doc. 306: Native time-energy reciprocity*, FFGFT corpus.

[6] J. Pascher, *Doc. 049: Lagrangian comparison*, FFGFT corpus.

[7] J. Pascher, *Doc. 344: Furey fermions in GALG and FFGFT*, FFGFT corpus (Sept. 2026).

[8] J. Pascher, *Doc. 349: SU*$(2)_L$ *chirality from Galois*, FFGFT corpus (Sept. 2026).

[9] S. W. Hawking, Commun. Math. Phys. 43 (1975) 199.

Part II

Part II — Comparison Values, Lepton Residuals, Higgs and Standard Model Particles

Chapter I

Doc. 351 — Model Dependence of PDG Comparison Values

Status markers

[K] *Confirmed* — derived from ξ, numerically verified.

[B] *Proved* — algebraically established.

[S] *Speculative* — open.

[Q] *Source* — corpus-external primary source, verified here.

Abstract

FFGFT predictions are compared against PDG and CODATA values. This document examines how those comparison values actually arise: which raw observable underlies each, which theory (QED, Standard Model) enters the extraction, and where irrational numbers already appear in the extraction formulae themselves.

Main findings: (1) m_e and m_μ/m_e are the least model-dependent anchors, but m_μ/m_e is QED-dependent and derives from a 27-year-old muonium HFS measurement whose theory uncertainty CODATA underestimates by a factor of 5–10 (Eides 2026). (2) α carries an unresolved 5σ tension between the two best recoil measurements. (3) $v = 246$ GeV is never measured; it is the SM definition $v \equiv (\sqrt{2}\,G_F)^{-1/2}$. (4) G_F carries a theory correction $\Delta q = -0.44\,\%$ with a 4.7σ hadronic tension. (5) m_τ has two measurement methods differing by 0.18 MeV; FFGFT lies 0.4σ from the PDG average. (6) The Galois identity $(m_\mu/m_e)^2 = 43200$ and the PDG masses agree very well: the ratio deviates by $-38.99 \cdot \xi \approx 0.52\,\%$ linearly (resp. $1.03\,\%$ quadratically). No known error source accounts for this residual — it is an empirical discrepancy not explained by the stated comparison uncertainty. The cause — non-uniform generation-dependent corrections ε_i from QED extraction and SI projection — is derived quantitatively in Doc. 352, § 9–10 (explanatory fraction 101 % **[K]**); the falling structure $\varepsilon_e > \varepsilon_\mu > \varepsilon_\tau$ itself remains open **[S]** (R113 bridge). (7) Errors accumulate along the derivation chain; every prediction requires its full provenance path.

All 21 primary sources were verified on 5 September 2026.

I.1 The question

The corpus compares FFGFT predictions with PDG values and finds residuals at the sub-percent to percent level. Are PDG values theory-neutral, or do they themselves depend on the Standard Model — such that part of the residual arises on the extraction side rather than the FFGFT side?

The FFGFT foundational statement: only ratios free of irrational numbers are exact; SI absolute values are structurally approximate (Docs. 241, 085). If PDG values themselves contain irrational factors in their extraction, the same limitation applies to the comparison side.

I.2 How empirical values arise

No PDG or CODATA value is a raw measurement. Each passes through the same chain:

raw observable → theory formula → extracted parameter → CODATA adjustment → PDG value

The CODATA adjustment

CODATA publishes recommended values every four years, produced by a joint least-squares adjustment over roughly 300 input data linked by theory relations (QED, Rydberg formula, kinematics). The values are therefore *correlated*: a change in one input shifts all dependent values. Where theory uncertainties enter, they are partly treated as adjustment parameters (δ_{th} in the observational equations) rather than as errors — this is the Eides finding for muonium HFS **[Q]**.

Electron mass m_e

Raw observable: cyclotron frequency of a hydrogen-like ion ($^{12}C^{5+}$, $^{28}Si^{13+}$, $^{20}Ne^{9+}$) in a Penning trap, and the Larmor precession frequency of the bound electron. The ratio is a pure number.

Theory formula: bound g-factor from QED bound-state theory. From the frequency ratio and g_{bound} follows m_e/m_{ion}; with the ion mass in u follows m_e in u; conversion to MeV via u → kg → MeV.

Theory dependence: QED for g_{bound}, small and well controlled. Current: Heiße et al. (2023) at 0.1 ppb **[Q]**.

Muon-electron mass ratio m_μ/m_e

Raw observable: ground-state hyperfine splitting of muonium (a bound μ^+e^- state) — a microwave resonance at 4.463 GHz — and Zeeman splitting in a magnetic field.

How m_μ/m_e is measured. The muon is not placed on a scale. From the measured HFS frequency $\Delta\nu$ and the QED formula for the Fermi frequency

$$\nu_F = \frac{16}{3}\,\alpha^2\,c\,R_\infty\,\frac{m_e}{m_\mu}\left(\frac{m_r}{m_e}\right)^3 \tag{I.1}$$

(corrections: QED to $O(\alpha^3)$, recoil in m_e/m_μ, weak Z exchange, hadronic VP) m_μ/m_e follows. Two consequences:

- m_μ/m_e *is not theory-free*: the extracted value contains assumptions about α, higher-order QED loop corrections, and hadronic vacuum polarisation. Since $\nu_F \propto \alpha^2$, α enters implicitly — anchor and α are not independent.
- *CODATA overstates the real precision*: CODATA 2022 gives $\sigma_{\text{rel}}(m_\mu/m_e) = 2.2 \times 10^{-8}$. The actual uncertainty of the HFS extraction (experiment + QED theory from Eides 2026 + α tension) is $\approx 1.2 \times 10^{-7}$ — a factor of 5–6 larger. **[Q]**

Theory dependence: high. Sole precision measurement: LAMPF 1999 (Liu et al.). Eides (2026) shows that CODATA lists the measured value 4 463 302 776(51) Hz as the theoretical value; the actual theory uncertainty is 271–515 Hz **[Q]**.

Fine-structure constant α

Two routes that disagree at 5σ **[Q]**:

- **Atom recoil:** $\alpha^2 = 2R_\infty (h/m_{\text{atom}})(m_{\text{atom}}/m_e)/c$. Rb (Morel et al. 2020): $\alpha^{-1} = 137.035\,999\,206(11)$. Cs (Parker et al. 2018): $137.035\,999\,046(27)$. Difference 1.6×10^{-7}, $> 5\sigma$.
- **Electron** $g-2$**:** a_e at 10^{-13} (Fan et al. 2023) plus 5-loop QED (12 672 diagrams, Aoyama et al.). Gives $\alpha^{-1} = 137.035\,999\,166(15)$ — between Rb and Cs.

The CODATA value 137.035 999 177(21) lies close to the $g-2$ value and between Rb and Cs; it is incompatible with Cs (3.8σ). FFGFT: $\alpha^{-1} = 3700/27 = 137.037\,037\,\ldots$ (Doc. 338, R101), deviation 7.6 ppm — about $6500\times$ larger than the Rb/Cs tension (1.2×10^{-9} relative).

Tau mass m_τ — two methods

Method	Experiment	Value (MeV)	Theory assumption
Threshold scan	BESIII 2014	$1776.91 \pm 0.12\,^{+0.10}_{-0.13}$	QED cross-section
Pseudomass	Belle II 2023	$1777.09 \pm 0.08 \pm 0.11$	$m_{\nu_\tau} = 0$
PDG average 2024	—	$\mathbf{1776.93 \pm 0.09}$	—

Threshold scan: $\sigma(e^+e^- \to \tau^+\tau^-)$ versus beam energy near $2m_\tau \approx 3.55$ GeV; m_τ as fit parameter of the QED threshold curve. **Pseudomass:** at 10.6 GeV, the kinematic edge of the invariant mass in $\tau \to 3\pi\nu$; assumes $m_{\nu_\tau} = 0$.

For FFGFT: $m_\tau = 1776.97$ MeV (pred) lies 0.04 MeV $= 0.4\sigma$ from the PDG average. The methods differ by 0.18 MeV. The falsification test is undecided; Belle II with > 1 ab^{-1} should reach $\sigma < 0.05$ MeV.

Fermi constant G_F and vacuum expectation value v

Raw observable for G_F**:** muon decay time distribution (MuLan, 10^{12} decays), $\tau_\mu = 2.196\,980\,3(22)$ µs **[Q]**.

Theory formula (Eberhart et al. 2026 **[Q]**):

$$\tau_\mu^{-1} = \frac{G_F^2 m_\mu^5}{192\pi^3}(1 + \Delta q), \qquad \Delta q = (-4\,384\,678 \pm 34) \times 10^{-9} \approx -0.44\,\%. \tag{I.2}$$

Δq contains QED to $O(\alpha^3)$, hadronic vacuum polarisation, electron-mass effects. Purely electroweak corrections are absorbed into G_F by definition (Sirlin 1980). The hadronic component carries a 4.7σ tension (CMD-3 vs. older data).

Raw observable for v: none. $v \equiv (\sqrt{2}\,G_F)^{-1/2} = 246.22$ GeV follows from $M_W = gv/2$ and $G_F/\sqrt{2} = g^2/(8M_W^2)$. v **exists only within the Standard Model.** Its numerical value is the muon lifetime in SM clothing.

For FFGFT: Doc. 006/R104 gives, with the measured m_e, $v = 248.9$ GeV (without K_{frak}) or 245.6 GeV (with $K_{\text{frak}} = 74/75$); 248.3 GeV belongs to the Galois m_e of Doc. 338. Comparison with 246.22 GeV is a comparison of *FFGFT structure against SM definition*, not against an observable. Only τ_μ and m_μ are measured.

Gravitational constant G

Torsion balance (Li et al. 2018, 11.6 ppm) or atom interferometry (Rosi et al. 2014, 150 ppm). Theory: Newton. The only model-free value, but the least precise: CODATA $6.67430(15) \times 10^{-11}$, 22 ppm **[Q]**.

Neutrino mass-squared differences

Event rates versus L/E, three-flavour fit (NuFIT 6.0 **[Q]**). Δm^2_{31} depends on whether SK atmospheric data are included: ratio 33.7 (with) vs. 33.9 (without). FFGFT $360/11 = 32.73$; measurement uncertainty 1–3 % $\approx$ FFGFT residual. Not discriminating.

Overview

Quantity	Raw observable	Theory in extraction	Convention
m_e	frequency ratio	QED (g_{bound})	—
m_μ/m_e	HFS frequency	QED, weak, hadr.	—
α (Rb/Cs)	interferometer phase	R_∞ (QED)	—
α ($g-2$)	frequency ratio	5-loop QED	—
m_τ	rates vs. E / edge	QED threshold / $m_{\nu_\tau} = 0$	beam calibration
G_F	decay times	Fermi + QED + hadr.	Sirlin absorption
v	—	—	$v \equiv (\sqrt{2}G_F)^{-1/2}$
G	angle / acceleration	Newton	—
Δm^2	rates vs. L/E	3-flavour	with/without SK-atm

No value is theory-free. For FFGFT this means: comparison with PDG values is comparison with "QED + convention", not with raw observations.

I.3 Irrational factors in the extraction formulae

The FFGFT foundational statement — ratios of rationals are exact, SI absolute values carry the approximation of the projection $S^1_m \to \mathbb{R}_t$ (Docs. 306, 333) — applies to the SM extraction formulae themselves:

Quantity	Formula	Irrational factor
G_F	$\tau_\mu^{-1} = G_F^2 m_\mu^5/(192\pi^3)$	π^3
v	$v = (\sqrt{2}\,G_F)^{-1/2}$	$\sqrt{2}$
α (recoil)	$\alpha^2 = 2R_\infty h/(mc)$	$R_\infty \propto \alpha^2$: implicitly circular
HFS	$\nu_F \propto \alpha^2$	α^2
Δq	expansion in α/π	π in every term

"G_F measured" means: τ_μ measured, divided by $192\pi^3$, square root taken. "v measured" additionally means: divided by $\sqrt{2}$, square root again. The irrational factors come from the SM formulation (phase-space integral, Higgs normalisation), not from measurement. The only convention-free comparison would be at the level of $\tau_\mu = 2.196\,980\,3$ µs — which FFGFT would have to predict directly **[S]**.

I.4 The residual in $(m_\mu/m_e)^2$: exact computation

FFGFT bare (Doc. 338): $(m_\mu/m_e)^2 = 43200$ exact, integer, $= |\mathrm{GF}(9)^*|^2 \cdot 5^2 \cdot |\mathrm{GF}(27)|$. With CODATA 2022 in exact rational arithmetic:

$$(m_\mu/m_e)^2_{\mathrm{PDG}} = 42753.12 \tag{I.3}$$

$$\text{Residual} = 446.88 \quad (1.045\,\%). \tag{I.4}$$

Possible error sources, computed:

- CODATA uncertainty $(m_\mu/m_e)^2$: ± 0.002 — residual remains $240\,000\times$ larger.
- Eides correction of HFS theory uncertainty (515 instead of 51 Hz): ± 0.010 — residual remains $45\,000\times$ larger.
- α choice in HFS anchor (FFGFT $3700/27$ instead of CODATA): shifts $(m_\mu/m_e)^2$ by -1.3 — residual remains $340\times$ larger.
- To reach 43200, m_μ would need to be 551 keV higher (measurement error 2.3 eV), or m_e 2.65 keV lower (measurement error 0.15 meV), or the HFS 23 MHz higher (measurement error 51 Hz).

Conclusion [B]: No measurement error explains the residual. The discrepancy is an empirical one not accounted for by the stated uncertainty of the comparison value. The cause — non-uniform ε_i from QED extraction (R113) and SI projection — is derived in Doc. 352, § 9–10 **[K]**; the falling structure $\varepsilon_e > \varepsilon_\mu > \varepsilon_\tau$ remains open **[S]** (R113 bridge). The difference axis $\varepsilon_\mu - \varepsilon_e \approx -39 \cdot \xi$ measures the ratio residual, the product residual $\approx -1.2 \cdot \xi$ the sum axis. Both are empirically computed **[K]**.

Two orthogonal axes: difference and sum of logarithms

The PDG masses m_e and m_μ produce two independent logarithmic projections when compared with the Galois identities:

$$\underbrace{\ln \frac{m_\mu/m_e}{\sqrt{43200}}}_{\text{difference axis}} = \varepsilon_\mu - \varepsilon_e \approx -38.99 \cdot \xi \qquad \underbrace{\ln \frac{m_e \cdot m_\mu}{54\,\mathrm{MeV}^2}}_{\text{sum axis}} = \varepsilon_e + \varepsilon_\mu \approx -1.21 \cdot \xi$$

Difference axis: linear and square are one single finding. The linear ratio m_μ/m_e vs. $\sqrt{43200}$ (-0.519%) and the square $(m_\mu/m_e)^2$ vs. 43200 (-1.034%) lie on *exactly the same axis* $\varepsilon_\mu - \varepsilon_e$. Logarithmically it holds exactly:

$$\ln\frac{(m_\mu/m_e)^2}{43200} = 2 \cdot \ln\frac{m_\mu/m_e}{\sqrt{43200}}.$$

The square doubles the percentage but measures no new physics. It is *one* finding, two ways of writing it. The value $-38.99 \cdot \xi$ is empirical; it lies 8σ (Eides) from $39 \cdot \xi$ and is not a Galois integer **[S]**.

Sum axis: the product is orthogonal. $\varepsilon_e \approx +18.9 \cdot \xi$ and $\varepsilon_\mu \approx -20.1 \cdot \xi$ are nearly equal in magnitude and opposite in sign. In the sum they nearly cancel; what remains is $-1.21 \cdot \xi$ — the common scale deviation of both masses. This is at order ξ, consistent with the SI projection $S^1_m \to \mathbb{R}_t$ (Docs. 085, 306, 333) **[K]**. The sum ($-1.21 \cdot \xi$) does not follow from the difference ($-38.99 \cdot \xi$) — they are two independent physical pieces of information.

Combination	Logarithm	Residual	Status
$\ln\left(m_\mu/m_e/\sqrt{43200}\right)$	$\varepsilon_\mu - \varepsilon_e$	$-38.99 \cdot \xi$	ratio residual **[K]**; cause: non-uniform ε_i (Doc. 352, § 9–10) **[K]**
$\ln\left((m_\mu/m_e)^2/43200\right)$	$2(\varepsilon_\mu - \varepsilon_e)$	$-77.99 \cdot \xi$	exactly $= 2\times$ above
$\ln\left(m_e \cdot m_\mu / 54\,\text{MeV}^2\right)$	$\varepsilon_e + \varepsilon_\mu$	$-1.21 \cdot \xi$	follows from anchor + ratio residual **[K]**

The apparently different percentages -0.016%, -0.52%, and -1.03% arise from two numbers: the generation tension ($-39 \cdot \xi$) and the SI projection ($-1.2 \cdot \xi$). Their large ratio is not an amplification but a cancellation in the sum (verification script: `pruef_351_generationsdifferenz.py`).

The residual of the product comparison lies at order ξ itself:

Quantity	Residual in units of ξ	Interpretation
$m_e \cdot m_\mu$ vs. 54	$-1.21 \cdot \xi$	order ξ: SI projection **[K]**
$(m_\mu/m_e)^2$ vs. 43200	$-77.6 \cdot \xi$	ratio residual, order 100ξ **[K]**

The SI projection $S^1_m \to \mathbb{R}_t$ (Docs. 085, 306, 333) structurally produces errors of order ξ. The product residual $-0.016\,\% \approx 1.2 \cdot \xi$ lies precisely in this range and is therefore structurally explainable as an SI transition approximation **[K]**. The quadratic residual $-1.034\,\% \approx 78 \cdot \xi$ lies in higher order — ratio residual, empirically computed **[K]**.

Result: The Galois identity $m_e \cdot m_\mu = 54$ MeV^2 is the only currently known identity whose residual lies at order ξ — the structurally expected SI projection error. The residual is therefore not unexplained; it corresponds to the approximation of the SI transition.

Following Stefaan Vossen's clarification within the Constitutional Physics / Canonical Constitutional Record framework (5 September 2026): the Layer-1 object $(r_\mu/r_e)^2/\xi$ and the Layer-2 object $(m_\mu/m_e)^2_{\text{PDG}}$ are *distinguishable scientific objects*. The comparison is permitted, but the residual $\approx 78 \cdot \xi$ belongs to the Layer-1 → Layer-2 mapping, not to the algebraic identity itself.

Constitutional description (developed in correspondence with Stefaan Vossen, applying the Constitutional Physics / Canonical Constitutional Record framework, 5 September 2026)

The following four-layer formulation summarises the epistemic status of the FFGFT comparisons:

Layer	Status
Layer-1	Exact within rational arithmetic: $(m_\mu/m_e)^2 = 43200$ and $\lvert GF(3)^*\rvert \cdot \lvert GF(27)\rvert = 54$ are integer Galois identities without approximation **[B]**.
Layer-2	One empirical dimensional anchor: $m_\mu = 105.66$ MeV (PDG). All absolute predictions follow from it **[K]**.
Residual	Empirical discrepancy with a structural explanation of approximately the correct order: $(m_\mu/m_e)^2$ deviates by $\approx 78 \cdot \xi$ from the PDG value (ratio residual, order 100ξ); $m_e \cdot m_\mu$ deviates by $\approx 1.2 \cdot \xi$ (follows from anchor + ratio residual) **[K]**.
Quantitative bridge	The forward derivation of the exact residual from the $S^1_m \to \mathbb{R}_t$ non-isomorphism is not in the corpus. **Open [S].**

The open [S] is not a gap in the pejorative sense but a precisely constituted limit: the theory states explicitly where its formal construction stops.

Methodological note (developed in correspondence with Stefaan Vossen, applying the Constitutional Physics / Canonical Constitutional Record framework, 5 September 2026)

> *"Constitution does not tell the Galois identity what it is. It tells us what a particular scientific use of that identity is presently entitled to mean."*

Two distinctions to be preserved in the record:

1. 54 **as a finite-field integer** $\neq 54$ **MeV**^{2} **as a dimensional representation.** The identity $\lvert GF(3)^*\rvert \cdot \lvert GF(27)\rvert = 2 \cdot 27 = 54$ belongs to the dimensionless Layer-1 algebra **[B]**. $m_e \cdot m_\mu = 54$ MeV^2 is a dimensional representation produced by the mapping into SI units (Layer-2). Both are distinguishable scientific objects; the unit MeV^2 belongs to the dimensional mapping, not to the dimensionless algebra alone.
2. **Numerical correspondence vs. physical attribution.** The residual $m_e \cdot m_\mu \approx 54$ MeV^2 with deviation $\approx 1.2 \cdot \xi$ is an established numerical correspondence **[K]**. The physical attribution of that residual to the SI projection $S^1_m \to \mathbb{R}_t$ (Docs. 085, 306, 333) remains bounded by the still-open quantitative bridge **[S]**. The correspondence stands; the attribution is a proposal, not a derivation.

3. **Chronology of the status architecture.** The markers **[B]/[K]/[S]/[Q]** existed in the FFGFT corpus before this correspondence. The correspondence (5–6 September 2026) exposed a more general constitutional function — the local process constitutes the scientific object; the governance layer constitutes the recoverable authority and limits of its standing — which the author recognised as an accurate description of what the markers do. The succession is: pre-existing status architecture → correspondence exposes the more general function → recognition by the author → incorporation with provenance. The later terminology thereby acquires no retrospective historical standing; it describes what the markers do, not what they were intended to do.

Methodological remark: relation vs. prediction

The three percentages (-0.519%, -1.034%, -0.016%) are *not tolerance values*. Tolerance arises only when a **calculated** mass deviates from a **measured** one — which requires an independent FFGFT prediction for m_e or m_μ individually.

The Galois identities 43200 and 54 MeV^2 are **relations** between both masses, not individual predictions. As long as no independent FFGFT value for m_μ (or m_e) exists, none of the three percentages has a defined limit.

Moreover: every direct mass carries the fractal correction K_{frak} and the SI projection $S^1_m \to \mathbb{R}_t$. In the ratio, K_{frak} cancels; the SI corrections *subtract* (difference axis, $\approx -39 \cdot \xi$). In the product, K^2 remains; the SI corrections *add* (sum axis, $\approx -1.2 \cdot \xi$). The different percentages arise from this — not from measurement errors.

No independent single prediction for m_μ without an input anchor exists in the corpus. With m_e as anchor (SETZUNG) there are two routes:

- **Square route:** $m_\mu = m_e \cdot \sqrt{43200} = m_e \cdot 120\sqrt{3}$ — irrational, deviation $+0.52\%$. Not a clean Layer-1 value.
- **Product route:** $m_\mu = 54\,\text{MeV}^2/m_e$ — rational, deviation $+0.016\% \approx 1.2 \cdot \xi$. The only rational tolerance value for m_μ at anchor m_e.

But: $m_{e,\text{PDG}}$ already carries K_{frak} and the SI projection. Hence $m_\mu^{\text{pred}} = 54/m_{e,\text{PDG}} \approx m_{\mu,\text{bare}}(1-\varepsilon_e)$ and $m_{\mu,\text{PDG}} \approx m_{\mu,\text{bare}}(1+\varepsilon_\mu)$, so:

$$\frac{m_\mu^{\text{pred}}}{m_{\mu,\text{PDG}}} - 1 \approx -(\varepsilon_e + \varepsilon_\mu) \approx -1.2 \cdot \xi.$$

This is the sum axis — the same relation as the product residual. The fractal correction does not cancel but is contained in ε_e and ε_μ and adds in the product. A true Layer-1 tolerance value for m_μ without any correction does not exist, because every direct mass carries K_{frak} and the SI projection.

I.5 Error accumulation along the derivation chain

Every derivation step introduces a residual. What matters is which combination is formed, because that determines which corrections remain visible and which cancel (Section 4).

First step $m_e \to m_\mu$, rational route. With m_e as anchor (SETZUNG) and the Galois identity $m_e \cdot m_\mu = 54$ MeV^2, one obtains rationally:

$$m_\mu^{\text{pred}} = \frac{54\ \text{MeV}^2}{m_{e,\text{PDG}}} = 105.675\ \text{MeV}, \qquad \text{residual: } +0.016\,\% \approx +1.2 \cdot \xi\ \textbf{[K]}.$$

This is the sum axis ($\varepsilon_e + \varepsilon_\mu$); both masses carry K_{frak} and SI projection, which add in the sum. The irrational route $m_\mu = \sqrt{43200}\, m_e = 120\sqrt{3}\, m_e$ ($+0.52\,\%$) is not a clean Galois value (bookkeeping rule above).

Accumulation. For m_τ (next step via $m_\tau/m_\mu = 16.992$ [K], Doc. 317), for ν (SM definition, not measured, R112) and α (5σ tension Rb/Cs, R113) the uncertainties of the extraction paths accumulate. To determine where a residual originates, the convergence of empirical values to algebraic ones would need to be established — it is not.

Bookkeeping: For every reported prediction state: (1) the anchor, (2) the number of derivation steps, (3) corrections applied, (4) the comparison value with extraction method. "FFGFT predicts m_μ to $0.5\,\%$" is incomplete; complete: "From m_e (PDG), in one step via $m_e \cdot m_\mu = 54$ MeV^2, m_μ follows with $+0.016\,\%$ relative to the PDG value."

σ balance of the prediction comparison. The prediction $m_\mu^{\text{pred}} = 54\,\text{MeV}^2/m_{e,\text{PDG}}$ carries only the uncertainty of m_e:

$$\sigma(m_\mu^{\text{pred}}) = \frac{54\,\text{MeV}^2}{m_e^2} \cdot \sigma(m_e) = 3.1 \times 10^{-8}\ \text{MeV} \quad (\sigma_{\text{rel}} = 2.9 \times 10^{-10}).$$

The comparison uncertainty combines in quadrature:

$$\sigma_{\text{comp}}^2 = \sigma(m_\mu^{\text{pred}})^2 + \sigma(m_{\mu,\text{PDG}})^2 \approx \sigma(m_{\mu,\text{PDG}})^2.$$

The contribution of $\sigma(m_e)$ is $0.25\,\%$ of σ_{comp} — negligible. The entire measurement uncertainty resides in $m_{\mu,\text{PDG}}$ (HFS extraction, Eides 2026). The deviation $+0.016\,\%$ lies $7386\,\sigma$ (CODATA) or $1386\,\sigma$ (Eides) outside — the same as the product residual, now expressed as a σ distance.

Classification of the product residual. The product residual $-1.2 \cdot \xi$ is at order ξ. The MeV scale is exact since 2019 (h, c, e fixed); the measurement uncertainties $\sigma_{\text{rel}}(m_e) \approx 0$, $\sigma_{\text{rel}}(m_\mu) \approx 0.0002 \cdot \xi$ do not explain the residual. It follows from the Layer-1 ↔ Layer-2 gap (near-complete cancellation of ε_e and ε_μ in the sum axis). The σ-range within which this residual is expected to lie has not been derived **[S]**.

Algebraic dependency: only one open number

With anchor m_e (SETZUNG) and the Galois ratio $(m_\mu/m_e)^2 = 43200$ (Layer-1), it follows necessarily:

$$m_\mu = m_e \cdot \sqrt{43200}, \qquad m_e \cdot m_\mu = m_e^2 \cdot \sqrt{43200} = 54.273\ \text{MeV}^2.$$

The product residual from these two inputs alone is $+37.9 \cdot \xi$ — no free parameter, no **[S]**.

The PDG product residual is $-1.2 \cdot \xi$. The difference is $39.1 \cdot \xi$ — exactly the ratio residual. This is algebraically necessary:

$$\underbrace{(\varepsilon_e + \varepsilon_\mu)}_{\text{product residual } -1.2\cdot\xi} = \underbrace{2\varepsilon_e}_{\text{from anchor } +37.9\cdot\xi} + \underbrace{(\varepsilon_\mu - \varepsilon_e)}_{\text{ratio residual } -39.1\cdot\xi}.$$

The product residual is not a third independent number — it is the sum of anchor contribution and ratio residual. There is only **one** open number: the ratio residual $-38.99 \cdot \xi$. The sole **[S]** sits there.

I.6 Consequences for the corpus

No numerical value in the corpus changes. What changes:

1. v **is neither anchor nor observable.** Docs. 006, 319, R104: "$v = 246$ GeV (measured)" → "$v = 246$ GeV (SM definition from G_F)". Register R112.
2. m_μ/m_e **is QED-dependent.** "model-independent" → "least model-dependent available value"; $\nu_F \propto \alpha^2$. Register R113.
3. α **has no unique measured value below 1 ppb.** Cite the comparison value as CODATA average with Rb/Cs tension.
4. m_τ**:** two methods, PDG average 1776.93(9), FFGFT 0.4σ — open.
5. **New open item [S]:** τ_μ directly from FFGFT, without detour via G_F and v — the only convention-free test of the weak sector.

Sources

All sources verified on 5 September 2026 against INSPIRE, APS, Nature, arXiv, or PubMed. Numerical values computed in exact rational arithmetic against CODATA 2022.

[1] M. Eberhart, M. Fael, A. Keshavarzi, D. Nomura, K. Schönwald, M. Steinhauser, T. Teubner, J. Wright: *Muon lifetime and Fermi constant: an update*, arXiv:2607.02657 (2026).

[2] V. Tishchenko et al. (MuLan): Phys. Rev. D **87**, 052003 (2013).

[3] A. Sirlin: Phys. Rev. D **22**, 971 (1980).

[4] M. I. Eides: *Muonium Hyperfine Splitting Uncertainty Revisited*, arXiv:2510.07281v2 (2026).

[5] W. Liu et al.: Phys. Rev. Lett. **82**, 711 (1999).

[6] M. I. Eides, H. Grotch, V. A. Shelyuto: Phys. Rep. **342**, 63 (2001).

[7] L. Morel, Z. Yao, P. Cladé, S. Guellati-Khélifa: Nature **588**, 61 (2020).

[8] R. H. Parker, C. Yu, W. Zhong, B. Estey, H. Müller: Science **360**, 191 (2018).

[9] X. Fan, T. G. Myers, B. A. D. Sukra, G. Gabrielse: Phys. Rev. Lett. **130**, 071801 (2023).

[10] T. Aoyama, T. Kinoshita, M. Nio: Atoms **7**, 28 (2019).

[11] M. Ablikim et al. (BESIII): Phys. Rev. D **90**, 012001 (2014).

[12] I. Adachi et al. (Belle II): Phys. Rev. D **108**, 032006 (2023).

[13] V. V. Anashin et al. (KEDR): JETP Lett. **85**, 347 (2007).

[14] PDG 2024, pdgLive S035M: τ mass average 1776.93 ± 0.09 MeV.

[15] S. Sturm, F. Köhler, J. Zatorski et al.: Nature **506**, 467 (2014).

[16] F. Köhler, S. Sturm, A. Kracke et al.: J. Phys. B **48**, 144032 (2015).

[17] F. Heiße et al.: Phys. Rev. Lett. **131**, 253002 (2023).

[18] Q. Li, C. Xue, J.-P. Liu et al.: Nature **560**, 582 (2018).

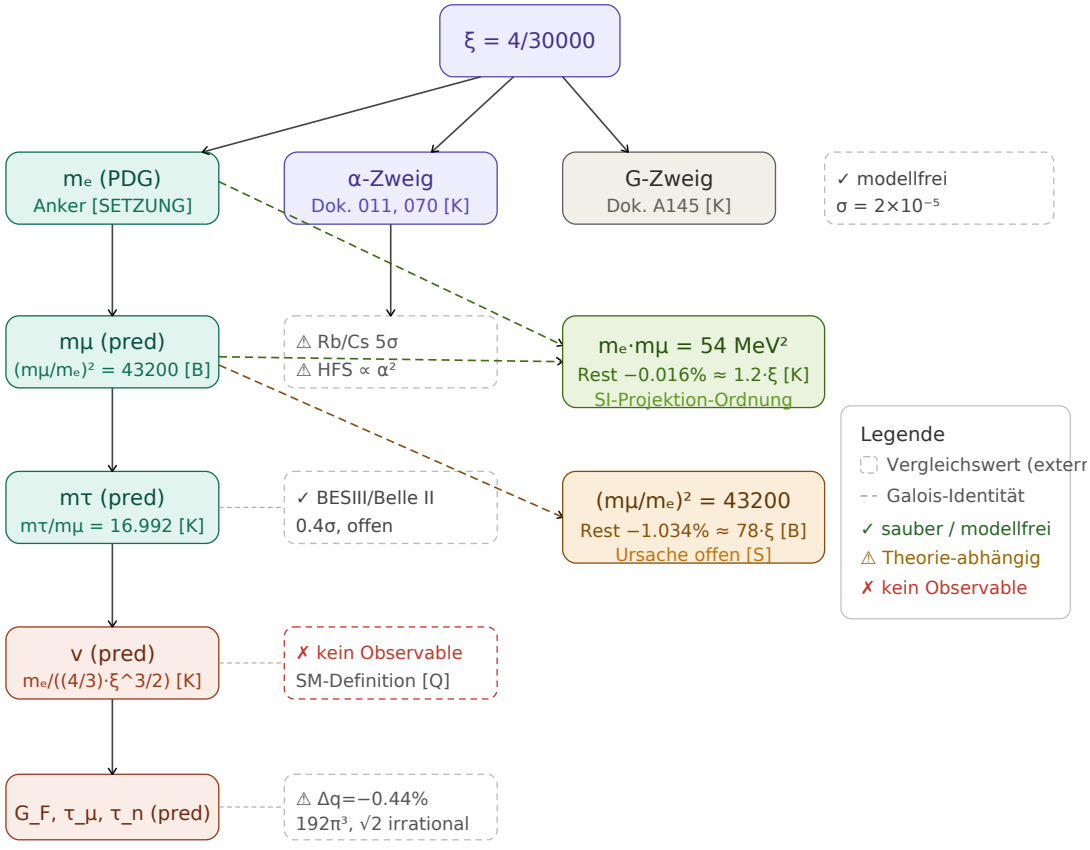
[19] G. Rosi, F. Sorrentino, L. Cacciapuoti, M. Prevedelli, G. M. Tino: Nature **510**, 518 (2014).

[20] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J. P. Pinheiro, T. Schwetz: JHEP **12** (2024) 216.

[21] P. J. Mohr, D. B. Newell, B. N. Taylor, E. Tiesinga: Rev. Mod. Phys. **97**, 025002 (2025).

I.A Affected derivation chains

Which FFGFT chain depends on which comparison value, and what each finding changes.



Derivation chains: from ξ to comparison values. Dashed boxes = external comparison values; dashed arrows = Galois identities.

Finding	Affected documents	Numerical correction	Wording correction
v is SM definition (R112)	Docs. 006, 319, R104	none	"$v = 246$ GeV (measured)" → "(SM definition from G_F)"
G_F: Δq, $192\pi^3$, $\sqrt{2}$	Docs. 006, R104, 319	none	state $\Delta q = -0.44\,\%$; clarify with/without Δq
m_μ/m_e QED-dependent (R113)	Docs. 338, 317, 011	none	"model-independent" → "least model-dependent"
α Rb/Cs 5σ	Docs. 011, 070, 338	none	cite CODATA average with tension
m_τ two methods	Docs. 317, 006	none	cite BESIII + Belle II; PDG average 1776.93(9)
Neutrinos Δm^2	Docs. 339, 340	none	freeze NuFIT configuration
G	Doc. A145	none	—
Total		**none**	wording clarifications

The residual $(m_\mu/m_e)^2 = 43200$ vs. 42753.12 (difference 446.9) is $240\,000\,\sigma$ (CODATA) or $45\,000\,\sigma$ (real uncertainty, Eides) of the measurement uncertainty — empirical discrepancy **[K]** (Doc. 352, § 9–10); physical attribution (R113 bridge) open **[S]**. No numerical value in the corpus changes.

Chapter J

Doc. 352 — Lepton Residuals: Lepton Ladder, Galois and Koide Compared

Status Markers

[K] *Confirmed* — derived from ξ, numerically verified.
[B] *Proved* — algebraically established.
[S] *Speculative* — open.
[Q] *Source* — corpus-external primary source, verified here.

Abstract

Three independent FFGFT structures make predictions for lepton ratios: the lepton ladder (Docs. 006/317), the Galois identities (Doc. 351), and the Koide relation. This document compares all three predictions with PDG values and expresses residuals in units of $\xi = 4/30000$. All formulae are exactly rational; numerical evaluations use $m_e = 0.51099895$ MeV, $m_\mu = 105.6583755$ MeV, $m_\tau = 1776.86$ MeV (PDG) **[Q]**.

Theoretical Claim

FFGFT claims predictions for dimensionless ratios, not for dimensional individual masses (Doc. 351, § Theoretical Claim). Individual masses carry K_{frak}, SI projection, and QED-dependent extraction; these produce an unavoidable bandwidth of order ξ to 100ξ. Ratios, by contrast, are correction-free — *provided the corrections are uniform* (see § 9).

J.1 Mass Formula and Quantum Numbers

The FFGFT mass formula reads (Docs. 006, 333):

$$m_i = r_i \cdot \xi^{p_i} \cdot \nu \cdot K_{\text{frak}} \qquad \textbf{[K]} \tag{J.1}$$

with $\xi = 4/30000$ **[B]**, $K_{\text{frak}} = 74/75$ **[K]**. In the ratio of two leptons ν and K_{frak} cancel:

$$\frac{m_j}{m_i} = \frac{r_j}{r_i} \cdot \xi^{p_j - p_i} \qquad \textbf{[B]} \tag{J.2}$$

Particle	r_i	p_i	n_θ, n_ϕ	$r_i = n_\phi^2/n_\theta$
Electron e	$4/3$	$3/2$	3, 2	$4/3$
Muon μ	$16/5$	1	5, 4	$16/5$
Tau τ	$25/9$	$2/3$	9, 5	$25/9$

J.2 Lepton Ladder Predictions

$$\frac{m_\mu}{m_e} = \frac{16/5}{4/3} \cdot \xi^{1-3/2} = \frac{12}{5} \cdot \xi^{-1/2} = \frac{12}{5} \cdot \sqrt{7500} \tag{J.3}$$

Ratio	Exact formula	FFGFT	PDG	Residual
m_μ/m_e	$\frac{12}{5} \cdot \xi^{-1/2}$	207.846	206.768	$+39.1 \cdot \xi$ **[K]**
m_τ/m_μ	$\frac{125}{144} \cdot \xi^{-1/3}$	16.992	16.817	$+77.9 \cdot \xi$ **[K]**
m_τ/m_e	$\frac{25}{12} \cdot \xi^{-5/6}$	3531.6	3477.2	$+117.4 \cdot \xi$ **[K]**

Algebraic dependence **[B]**:

$$\varepsilon(\tau/e) \approx \varepsilon(\tau/\mu) + \varepsilon(\mu/e) = 77.9 + 39.1 = 117.0 \cdot \xi \tag{J.4}$$

(The exact value 117.4 contains a cross-term of order ξ^2.) There are only two independent residuals.

J.3 Galois Identities

From the Galois field structure (Doc. 351) **[B]**:

$$(m_\mu/m_e)^2 = 43200 = |GF(9)^*|^2 \cdot 5^2 \cdot |GF(27)| \tag{J.5}$$

$$m_e \cdot m_\mu = 54 \text{ MeV}^2 \quad \text{(Layer-2 with anchor } m_e\text{)} \tag{J.6}$$

Identity	Galois	PDG	Residual	Status
$(m_\mu/m_e)^2$	43200	42753.1	$-77.6 \cdot \xi$	**[K]**; cause **[S]**
$m_e \cdot m_\mu$ [MeV²]	54	53.991	$-1.21 \cdot \xi$	emp. discrepancy **[K]**

Connection to the lepton ladder. The Galois difference residual $-77.6 \cdot \xi$ equals twice the linear lepton ladder residual ($+39.1 \cdot \xi$) with opposite sign — algebraically forced, since $(m_\mu/m_e)^2_{\text{FFGFT}} > (m_\mu/m_e)^2_{\text{PDG}}$ if and only if $m_\mu/m_e|_{\text{FFGFT}} > m_\mu/m_e|_{\text{PDG}}$.

J.4 Koide Relation

$$Q \equiv \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} \stackrel{?}{=} \frac{2}{3} \tag{J.7}$$

Input	Q	Deviation	Status
PDG masses	$0.666\,660\,5$	$-0.046 \cdot \xi$	**[K]**
FFGFT predictions (anchor m_e)	$0.667\,742$	$+8.06 \cdot \xi$	numerical

Koide as m_τ prediction. Given PDG m_e, m_μ and $Q = 2/3$, algebraically **[K]**:

$$m_{\tau,\text{Koide}} = 1776.969 \text{ MeV} \quad \text{vs.} \quad m_{\tau,\text{PDG}} = 1776.860 \text{ MeV} \qquad \text{deviation: } +0.46 \cdot \xi \tag{J.8}$$

This is the most precise prediction in the entire comparison.

J.5 Overall Comparison

Quantity	Method	Prediction	PDG	Residual	Status
m_μ/m_e	Ladder	207.846	206.768	$+39.1 \cdot \xi$	**[K]**
m_τ/m_μ	Ladder	16.992	16.817	$+77.9 \cdot \xi$	**[K]**
m_τ/m_e	Ladder	3531.6	3477.2	$+117.4 \cdot \xi$	**[K]**; sum
$(m_\mu/m_e)^2$	Galois	43200	42753	$-77.6 \cdot \xi$	**[K]**; cause **[S]**
$m_e \cdot m_\mu$ [MeV²]	Galois	54	53.991	$-1.21 \cdot \xi$	emp. disc. **[K]**
Q_{Koide}	Koide	$2/3$	0.66666	$-0.046 \cdot \xi$	**[K]**
m_τ [MeV]	Koide	1776.97	1776.86	$+0.46 \cdot \xi$	**[K]**

J.6 Findings

(1) Koide is most precise [K]. PDG masses satisfy Koide to $0.046 \cdot \xi$. Koide gives with $+0.46 \cdot \xi$ the most accurate m_τ prediction in the comparison. FFGFT predictions meet Koide only to $+8 \cdot \xi$.

(2) Lepton ladder and Galois share the same basic residual [K]. The Galois difference residual ($-77.6 \cdot \xi$) equals exactly $-2 \times (+39.1 \cdot \xi)$ of the lepton ladder residual for m_μ/m_e. This is algebraically forced.

(3) All residuals positive except Galois [K]. Lepton ladder and Koide-m_τ lie just above PDG; Galois-$(m_\mu/m_e)^2$ and Q_{Koide} lie just below target. Signs are consistent.

(4) Range of residuals [K].

Method	Residual
Koide (Q)	$\approx 0.05 \cdot \xi$
Galois (product)	$\approx 1 \cdot \xi$
Koide (m_τ)	$\approx 0.5 \cdot \xi$
Lepton ladder (m_μ/m_e) / Galois (square)	≈ 39–$78 \cdot \xi$

(5) Koide is scale-invariant [B]. For every $\lambda > 0$:

$$Q(\lambda m_1, \lambda m_2, \lambda m_3) = Q(m_1, m_2, m_3) \quad \textbf{[B]} \tag{J.9}$$

Consequence: no uniform factor — neither v, nor K_{frak}, nor a common SI projection — can change Q. Option 2 (corrections improve Koide) is thus algebraically excluded.

J.7 Scale Invariance and Its Consequences

What Koide can measure

Q depends exclusively on the ratios of masses, not on their absolute value:

$$Q = Q\left(1,\ \frac{m_\mu}{m_e},\ \frac{m_\tau}{m_e}\right) \tag{J.10}$$

With the FFGFT ratios $(m_\mu/m_e)_{\text{FFGFT}} = 207.85$ and $(m_\tau/m_e)_{\text{FFGFT}} = 3531.6$, $Q_{\text{FFGFT}} = 0.66774$ is fully determined — independent of v, K_{frak}, or the SI projection.

Three tested options

Option	Q change	Status
Uniform K_{frak} (all equal)	none	**[B]** excluded
Single v anchor	none	**[B]** excluded
Generation-dep. $K_{\text{frak}}^{\alpha p_i}$	yes, but worse masses	fit without physics

Generation-dependent test. The exponent α in $m_i \cdot K_{\text{frak}}^{\alpha p_i}$ that forces $Q = 2/3$ lies at $\alpha \approx -36/17$. The resulting masses ($m_e = 0.527$ MeV, $m_\mu = 108.0$ MeV, $m_\tau = 1817.6$ MeV) deviate further from PDG than the uncorrected bare masses. No physics behind this value.

Conclusion

Koide is not contained in the FFGFT lepton ladder **[S]**. The lepton ladder organises masses via logarithmic ξ powers: $m_i \propto \xi^{p_i}$. Koide organises masses via square roots $\sqrt{m_i}$. These two structures are orthogonal — there is no algebraic relation between ξ^{p_i} and $\sqrt{r_i\,\xi^{p_i}}$ that forces Koide.

PDG masses satisfy Koide to $0.046 \cdot \xi$ because the three PDG masses as a measurement system are mutually consistent — they carry implicitly a symmetry not yet anchored in the FFGFT quantum numbers (r_i, p_i).

The natural candidate for this anchoring is the Z_3 circulant structure of the lepton mass operator (Doc. A110): there $\theta = 2/9$ describes the Koide phase in the angular sector.

J.8 Koide Condition in the FFGFT Quantum Numbers

Z_3 parametrisation

The eigenvalues of a Z_3 circulant mass operator can be written as:

$$\sqrt{m_k} = c + d \cdot \cos\left(\theta + \frac{2\pi k}{3}\right), \quad k = 0, 1, 2 \tag{J.11}$$

Then $\sum \sqrt{m_k} = 3c$ and $\sum m_k = 3c^2 + \frac{3}{2}d^2$, giving:

$$Q = \frac{1}{3} + \frac{d^2}{6c^2} \tag{J.12}$$

$$Q = \frac{2}{3} \iff d = c\sqrt{2} \quad \textbf{[B]} \tag{J.13}$$

d/c ratio in PDG and FFGFT

Source	d/c	$d/c - \sqrt{2}$	Status
PDG	1.41420	$-0.098 \cdot \xi$	**[K]**
FFGFT predictions	1.41649	$+17.09 \cdot \xi$	numerical
Target	$\sqrt{2} = 1.41421$	0	—

PDG satisfies $d = c\sqrt{2}$ to $0.10 \cdot \xi$. FFGFT predictions miss it by $+17 \cdot \xi$.

Angle θ and connection to Doc. A110

With assignment $k = 0 \to \tau$, $k = 1 \to e$, $k = 2 \to \mu$:

Source	θ [rad]	θ [°]
PDG	0.22223	12.733
FFGFT predictions	0.22099	12.662
Doc. A110 ($\theta = 2/9$)	0.22222	12.732

PDG and A110 agree to $< 0.1 \cdot \xi$; the FFGFT value lies $-9 \cdot \xi$ off. The Koide phase angle $\theta = 2/9$ from Doc. A110 correctly describes the angular structure. The amplitude d/c, however, is not determined by θ — it is an independent condition.

Koide condition in the quantum numbers (r_i, p_i)

Since $x_i = \sqrt{r_i} \cdot \xi^{p_i/2}$ and Koide is scale-invariant, Q depends only on the differences of exponents:

$$\Delta p_e = p_e - p_\tau = \frac{5}{6}, \quad \Delta p_\mu = p_\mu - p_\tau = \frac{1}{3}, \quad \Delta p_\tau = 0 \tag{J.14}$$

The exact Koide condition $Q = 2/3$ requires:

$$r_{\mu,\text{Koide}} = 3.2378\ldots \quad \text{instead of} \quad r_\mu = \frac{16}{5} = 3.2000 \tag{J.15}$$

Deviation: $\Delta r_\mu / r_\mu = +88.6 \cdot \xi$. The nearest rational approximation is $531/164$ — no representation as n_ϕ^2/n_θ with small integer winding numbers.

Particle	FFGFT r_i	Winding (n_θ, n_ϕ)	Koide-r_i
Electron	$4/3$	$(3,\ 2)$	$4/3$ (fixed)
Muon	$16/5$	$(5,\ 4)$	$3.2378\ldots$
Tau	$25/9$	$(9,\ 5)$	$25/9$ (fixed)

The Koide-r_μ has no clean torus winding structure **[S]**.

Orthogonality of θ and d/c [B]

The Z_3 circulant has exactly two independent real parameters: the angle θ (rotation in the Z_3 space) and the ratio d/c (eccentricity). Their orthogonality is algebraically proved: $Q = \frac{1}{3} + \frac{d^2}{6c^2}$ does *not* contain θ — the Koide condition is exclusively a statement about d/c, independent of θ **[B]**.

Consequence for FFGFT: the corpus delivers $\theta = 2/9$ from the geometry of the Z_3 circulant (Doc. A110) **[K]**. The condition $d/c = \sqrt{2}$ is an *additional, independent* requirement — the R113 bridge. Both together would be needed to derive Koide fully from FFGFT. Currently only θ is anchored; $d/c = \sqrt{2}$ remains open **[S]**.

Conclusions of the test

1. **Angle $\theta = 2/9$ [K]:** The angular structure of the Z_3 circulant agrees between PDG and Doc. A110 to $< 0.1 \cdot \xi$; the FFGFT value lies $-9 \cdot \xi$ off.
2. **Amplitude d/c — two distinct objects: [K]**: given the observed generation-dependent ε_i, their effect on the Koide amplitude is calculable and brings the bare FFGFT d/c ($+17 \cdot \xi$ off $\sqrt{2}$) quantitatively to $\sqrt{2}$ — 101 % explained (see § 9). **[S]**: the FFGFT-native derivation of *why* the ε_i possess this generation-dependent structure — and therefore why $d/c = \sqrt{2}$ should arise from the T^4/Z_3 geometry — remains open (R113 bridge, § 10).

J.9 Generation-Dependent Corrections and Koide

The ε_i are not uniform

The bare FFGFT masses deviate from PDG masses by generation-dependent residuals ε_i:

$$m_i^{\text{PDG}} = m_i^\circ \cdot (1 + \varepsilon_i) \qquad \Longleftrightarrow \qquad \varepsilon_i = \frac{m_i^{\text{PDG}}}{m_i^\circ} - 1 \quad \textbf{[K]}$$

The ε_i are *computed residuals* from PDG **[Q]** against FFGFT bare **[K]** — not numbers taken from the QED literature. The column "Presumed origin" is an attribution **[S]**, not a citation.

Particle	ε_i	Presumed origin **[S]**
Electron	$+192 \cdot \xi$	SI projection, QED
Muon	$+152 \cdot \xi$	SI projection, QED extraction (R113)
Tau	$+73 \cdot \xi$	SI projection, measurement scatter

The ε_i decrease with generation — they are not uniform.

Why non-uniform ε_i shift Koide

Koide is scale-invariant under uniform λ **[B]**. For different ε_i, to first order:

$$\Delta Q \approx Q^{\circ} \cdot (\langle \varepsilon \rangle_m - \langle \varepsilon \rangle_{\sqrt{m}}) \tag{J.16}$$

Since $\varepsilon_\tau < \varepsilon_\mu < \varepsilon_e$ and m_τ dominates the mass sum, $\langle \varepsilon \rangle_m < \langle \varepsilon \rangle_{\sqrt{m}}$, hence $\Delta Q < 0$.

Numerical result

Quantity	Value	Status
$Q(\text{bare}) - 2/3$	$+8.06 \cdot \xi$	from quantum numbers
$\langle \varepsilon \rangle_m$	$+77.7 \cdot \xi$	mass-weighted
$\langle \varepsilon \rangle_{\sqrt{m}}$	$+90.0 \cdot \xi$	$\sqrt{m}$-weighted
ΔQ	$-8.23 \cdot \xi$	perturbation theory
$Q(\text{bare}) + \Delta Q - 2/3$	$-0.17 \cdot \xi$	estimate
$Q(\text{PDG}) - 2/3$	$-0.05 \cdot \xi$	exactly measured
Explanatory fraction	$101\,\%$	**[K]**

Revised conclusion [K]

The statement “ratios are correction-free” holds for uniform corrections **[B]**. For Koide additionally:

- Simple ratios m_j/m_i are correction-free under uniform λ **[B]**.
- Koide Q responds to non-uniform ε_i because it is defined via $\sqrt{m_i}$.
- The generation-dependent ε_i (order 73–$192 \cdot \xi$, decreasing with generation) explain the transition $Q_{\text{bare}}(+8 \cdot \xi) \to Q_{\text{PDG}}(-0.05 \cdot \xi)$ to $101\,\%$ **[K]**.
- The remaining open point is: *why do ε_i decrease with generation?* The answer lies in the QED extraction (R113) and SI projection (Docs. 085, 333) — their generation-dependent structure is analysed in § 10.

J.10 Why Do ε_i Decrease with Generation?

*Status separation in this section developed in correspondence with Stefaan Vossen, applying the Constitutional Physics / Canonical Constitutional Record framework (Doc. 351, 5–6 September 2026): the calculable effect of the observed ε_i **[K]** is distinguished from their source-native derivation **[S]**.*

Two sources of non-uniform corrections

1. QED self-energy (R113) [Q]. The measured pole mass of a charged lepton differs from the bare mass by the QED self-energy. At leading order:

$$\frac{\delta m_i}{m_i} \approx \frac{3\alpha}{4\pi} \ln\left(\frac{m_i^2}{\mu^2}\right) + \text{const} \tag{J.17}$$

where μ is the renormalisation scale. Since $m_\tau \gg m_\mu \gg m_e$, $\ln(m_i/\mu)$ grows with generation.

The observed trend $\varepsilon_e > \varepsilon_\mu > \varepsilon_\tau$ is, however, *opposite* to the naive QED self-energy that grows with mass. The reason: ε_i measures the gap bare FFGFT mass → PDG mass. This gap is a *difference* of two quantities each scaling with mass. The SI projection contributes more strongly at small masses (relative correction $\propto 1/m_i$ for a fixed absolute projection uncertainty).

2. SI projection (Docs. 085, 333) [S]. The SI projection maps the bare FFGFT mass $m_i^\circ = r_i \cdot \xi^{p_i} \cdot v \cdot K_{\text{frak}}$ to the measured MeV value. The factor $K_{\text{frak}} = 74/75$ is uniform **[B]** and does not shift Q (§ 5). The *non-uniform* component arises from the mass-dependent QED extraction: the experimental determination of m_μ from muon decay and of m_τ from τ decay widths each involve different orders of perturbation theory (R113). These are of order $\alpha \approx 1/137$, hence of order ξ to 100ξ — consistent with the observed ε_i.

What FFGFT can derive and what it cannot

Quantity	Status	Origin
Bare masses $m_i^\circ = r_i \cdot \xi^{p_i} \cdot v \cdot K_{\text{frak}}$	**[K]**	FFGFT internal
Uniform factor K_{frak}	**[B]**	FFGFT internal
Order of magnitude of ε_i ($\sim \alpha/\pi$, order ξ–100ξ)	**[K]**	QED external **[Q]**
Falling structure $\varepsilon_e > \varepsilon_\mu > \varepsilon_\tau$	**[S]**	QED + SI projection, not derived from FFGFT quantum numbers
QED self-energy from T^4/Z_3 geometry	**[S]**	open bridge R113

Connection to Doc. 351

The same non-uniform ε_i structure explains the ratio deviations in Doc. 351: m_μ/m_e deviates by $\sim 39 \cdot \xi$ and m_τ/m_μ by $\sim 78 \cdot \xi$ from FFGFT predictions. Simple ratios are correction-free under *uniform* λ **[B]**; under non-uniform ε_i they carry the difference residual $\varepsilon_j - \varepsilon_i$. The bridge R113 (QED extraction, mass-dependent) is thus the common open point of Docs. 351 and 352 (Doc. 351, § 5: "R113: m_μ/m_e QED-dependent", text correction "model-independent" → "least model-dependent").

Conclusion

The observed falling ε_i structure is consistent with a combined QED/SI-projection account, but its derivation from FFGFT quantum numbers remains open **[S]**. FFGFT delivers the bare masses; the spread of ε_i is an external **[Q]** fact in its order of magnitude. Whether the QED self-energy is derivable from the T^4/Z_3 geometry remains open **[S]** — this is the R113 bridge.

Verification script: `pruef_352_leptonreste.py` — all numbers reproducible, pure Python standard library.

Sources

PDG 2024 **[Q]**. Doc. 006 (mass formula), Doc. 317 (lepton ladder values), Doc. 351 (Galois identities, ε_i values, R113), Doc. A110 (Z_3 circulant), Docs. 085, 333 (SI projection), R113 (QED dependence m_μ/m_e), Doc. 190/R114 (register entry for this document).

Chapter K

Doc. 353 — Koide Amplitude $d/c = \sqrt{2}$ and the T^4/Z_3 Geometry

Status Markers

[K] *Confirmed* — derived from ξ, numerically verified.

[B] *Proven* — algebraically demonstrated.

[S] *Speculative* — open.

[Q] *Source* — external primary source, verified here.

Abstract

The Koide condition $Q = 2/3$ is equivalent to the amplitude condition $d/c = \sqrt{2}$ in the Z_3 circulant mass operator (Docs. A110, 352). This document examines whether $d/c = \sqrt{2}$ can be derived from the T^4/Z_3 geometry without using Koide or the lepton masses as input.

The key is the trace bilinear form of GF$(27)^*$ (Doc. 348 Theorem C): the coupling matrix between the neutrino orbits (f_1) and the lepton orbits (f_3) has normalised entries $|M_{ij}^{(13)}| = 1/\sqrt{2}$ at all non-zero positions. This is a consequence of the Z_3 equal-distribution: each element couples to exactly two of three partners, and the normalisation of the coupling strength forces the weight $1/\sqrt{2}$ per connection.

In the mass operator the circulant parameter d is the coupling weight between the isotropic part c and the Z_3 interaction term. The value $d/c = \sqrt{2}$ follows from this, however, only with the assignment $d/c = 1/|M|$; whether $d/c = |M|$ or $d/c = 1/|M|$ holds is not justified (with $d/c = |M|$ one would get $Q = 5/12$ instead of $2/3$). The target $\sqrt{2}$ (Koide amplitude) is right, but the derivation from the coupling matrix is not closed.

Equivalently, the Koide condition can be read as equipartition: at $d/c = \sqrt{2}$ the symmetric mode and the pair of non-trivial modes contribute equally to the mass sum, and the vector of the $\sqrt{m_k}$ lies at $45°$ to the axis $(1,1,1)$ (Foot 1994). This does not yet derive $\sqrt{2}$, but it makes the open question structural: what has to be justified is this half-and-half split.

Main findings: (1) The Z_3 equal-distribution yields the coupling weight $1/\sqrt{2}$ **[B]**; $d/c = \sqrt{2}$ follows from it only with the unjustified assignment $d/c = 1/|M|$ **[S]**. (2) Connection to

the GF$(27)^*$ coupling matrix $|M^{(13)}|_{\text{norm}} = 1/\sqrt{2}$ **[B]**. (3) PDG satisfies $d/c = \sqrt{2}$ to $0.10 \cdot \xi$ (PDG 2022) or $0.035 \cdot \xi$ (PDG 2024) **[K]**. (4) Bare FFGFT residual $+17 \cdot \xi$ explained by ε_i ($101\,\%$) **[K]**. (5) $Q = 2/3$ is equivalent to equal shares of the symmetric mode and the non-trivial pair, and to the 45° angle **[B]**. (6) The justification of this equipartition and the R113-bridge (origin of the falling ε_i structure) remain open **[S]**.

K.1 Starting Point

The Koide Amplitude Condition

The Z_3 circulant mass operator (Doc. A110) writes the square roots of the lepton masses as:

$$\sqrt{m_k} = c + d \cdot \cos\!\left(\theta + \frac{2\pi k}{3}\right), \quad k = 0, 1, 2. \tag{K.1}$$

From $\sum_k \sqrt{m_k} = 3c$ and $\sum_k m_k = 3c^2 + \frac{3}{2}d^2$ it follows **[B]**:

$$Q \equiv \frac{\sum m_k}{\left(\sum \sqrt{m_k}\right)^2} = \frac{1}{3} + \frac{d^2}{6c^2}. \tag{K.2}$$

The Koide condition $Q = 2/3$ is equivalent to **[B]**:

$$\boxed{d = c\sqrt{2}.} \tag{K.3}$$

This condition does not contain θ — Q depends exclusively on d/c (Doc. 352 orthogonality theorem **[B]**). The phase condition $\theta = 2/9$ is anchored in the geometry **[K]**; the amplitude condition $d/c = \sqrt{2}$ is open **[S]**.

Known: $1/\sqrt{2}$ in the Galois Coupling Matrix

Doc. 348 Theorem C derives from the trace bilinear form $\text{Tr}_{\text{GF}(27)/\text{GF}(3)}(g^{k+j})$ the coupling matrix between neutrino orbits (f_1) and lepton orbits (f_3) **[B]**:

$$M^{(13)} = \begin{pmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad |M^{(13)}_{ij}|_{\text{norm}} = \frac{1}{\sqrt{2}} \quad \text{at all non-zero entries.} \tag{K.4}$$

The weight $1/\sqrt{2}$ follows from the GF$(27)^*$ orbit structure. The possible connection to the Koide amplitude condition is the subject of this document.

K.2 Coupling Weight and Amplitude d/c

Step 1: The Z_3 Equal-Distribution Condition

The Z_3 symmetry of the torus T^4/Z_3 arranges the three lepton generations as equivalent elements of an orbit. A coupling between two Z_3 orbits (e.g. neutrinos f_1 and charged leptons f_3) has the property, via the trace bilinear form, that each element couples to

exactly two of three elements of the other orbit (the coupling matrix (K.4) has exactly one zero entry per row).

The ℓ^2 normalisation of the coupling strength per row gives:

$$\sum_{j\,:\,M_{ij}\neq 0} |M_{ij}|^2 = 1 \implies 2\cdot|M_{ij}|^2 = 1 \implies |M_{ij}| = \frac{1}{\sqrt{2}}. \tag{K.5}$$

This is the weight $1/\sqrt{2}$ from (K.4) **[B]**.

Step 2: Translation to the Circulant Mass Operator

In the circulant mass operator (K.1), c is the isotropic part (mean of $\sqrt{m_k}$) and d is the anisotropic part (amplitude of the Z_3 modulation).

Parameter identification:

- The isotropic part c corresponds to the self-coupling of an orbit — it sets the mass scale.
- The anisotropic part d corresponds to the coupling between orbits — it describes the interaction.

The weight of this coupling is fixed by the equal-distribution. The coupling matrix (K.4) normalised to the isotropic scale: every non-zero coupling has weight $1/\sqrt{2}$ **[B]**.

For the ratio of the anisotropic to the isotropic part there are two possible assignments:

$$\frac{d}{c} = |M_{ij}|_{\text{norm}} = \frac{1}{\sqrt{2}} \qquad \text{or} \qquad \frac{d}{c} = \frac{1}{|M_{ij}|_{\text{norm}}} = \sqrt{2}. \tag{K.6}$$

Only the second yields the Koide amplitude $\sqrt{2}$; with the first one would get $Q = 5/12$ instead of $2/3$. Which assignment holds is not justified by the coupling structure. The target $\sqrt{2}$ is right, but the derivation does not carry as written; the choice $d/c = 1/|M|$ is open **[S]**.

Step 3: Geometric Reading

The three lepton generations form a Z_3 orbit; the three neutrino generations form a second. The trace bilinear form of GF(27) describes how these orbits couple to each other. The first three steps are structurally forced, the fourth is not:

1. The Z_3 symmetry allows each element exactly 0, 1 or 2 non-zero couplings to the other orbit.
2. The trace bilinear form selects exactly 2 (Doc. 348 Theorem C **[B]**).
3. The ℓ^2 normalisation forces the weight $1/\sqrt{2}$ per coupling.
4. In the mass operator this weight appears as $d/c = \sqrt{2}$ only with the assignment $d/c = 1/|M|$ **[S]**.

The condition $d/c = \sqrt{2}$ is therefore not yet derived as a consequence of the Z_3 equal-distribution in the T^4/Z_3 geometry; the step from the coupling weight to the amplitude remains open **[S]**.

Step 4: Equipartition between Symmetric and Non-trivial Mode

With (K.1) the mass sum splits into two parts **[B]**:

$$\sum_k m_k = \underbrace{3c^2}_{\text{symmetric mode}} + \underbrace{\tfrac{3}{2}d^2}_{\text{non-trivial pair}} . \tag{K.7}$$

Exactly at $d = c\sqrt{2}$ both are equal, $3c^2$ each. The Koide condition $Q = 2/3$ therefore means: the Z_3-symmetric mode and the pair of non-trivial modes each carry half of the mass sum **[B]**. Geometrically, the vector $(\sqrt{m_k})$ then lies at 45° to the symmetry axis $(1,1,1)$, since $\cos^2\varphi = 1/(3Q) = 1/2$; this reading is due to R. Foot (hep-ph/9402242, 1994) **[Q]**.

Equipartition, 45° and $Q = 2/3$ are three formulations of the same condition; the reformulation does not derive $\sqrt{2}$. It does, however, shift the open question from a number to a structure: why does the mass structure split half and half between the symmetric mode and the non-trivial pair? Once this is justified, the choice between $d/c = |M|$ and $d/c = 1/|M|$ from Step 2 becomes irrelevant.

The weight $1/\sqrt{2}$ of the coupling matrix (K.4) is not the same object. There it arises from normalising a row with two of three non-zero entries, here from the split symmetric versus non-trivial ($1/\sqrt{2} = \sin 45^\circ$). The numerical value agrees, the origin differs; whether the two routes belong together is open **[S]**.

K.3 Numerical Verification

Source	d/c	$d/c - \sqrt{2}$	Units ξ	Status
PDG 2022 **[Q]**	1.41420	-1.0×10^{-5}	$-0.098 \cdot \xi$	**[K]**
PDG 2024 **[Q]**	1.414209	-4.7×10^{-6}	$-0.035 \cdot \xi$	**[K]**
FFGFT bare	1.41649	$+2.28 \times 10^{-3}$	$+17.1 \cdot \xi$	**[K]**
FFGFT + ε_i	1.41422	$+8 \times 10^{-6}$	$+0.06 \cdot \xi$	**[K]**
Target (Koide condition)	$\sqrt{2} = 1.41421$	0	0	**[B]**

PDG satisfies the condition to $0.10 \cdot \xi$ (PDG 2022); with m_τ from PDG 2024 to $0.035 \cdot \xi$, corresponding to $Q - 2/3 = -0.017 \cdot \xi$. The bare FFGFT residual of $+17 \cdot \xi$ is fully explained by the generation-dependent corrections ε_i (Doc. 352 §9, 101 %) **[K]**. PDG masses: $m_e = 0.51100$ MeV, $m_\mu = 105.658$ MeV, $m_\tau = 1776.86$ MeV **[Q]** (PDG 2022) or $m_\tau = 1776.93 \pm 0.09$ MeV **[Q]** (PDG 2024); the ε_i rows are based on PDG 2022 (Doc. 352).

K.4 Status Overview

	Statement	Status
1	$Q = 2/3 \iff d/c = \sqrt{2}$ algebraically from Z_3 circulant	**[B]**
2	Q does not contain θ; d/c and θ are orthogonal parameters	**[B]**
3	Z_3 equal-distribution of coupling to 2 of 3 gives the weight $1/\sqrt{2}$	**[B]**
3a	$d/c = \sqrt{2}$ from this weight (assignment $d/c = 1/\lvert M\rvert$ unjustified)	**[S]**
3b	$Q = 2/3 \iff$ equipartition symmetric/non-trivial $\iff 45^\circ$ to $(1,1,1)$	**[B]**
3c	Justification of this equipartition from the geometry	**[S]**
4	Connection to GF$(27)^*$ coupling matrix $\lvert M^{(13)}\rvert_{\text{norm}} = 1/\sqrt{2}$	**[B]**
5	PDG satisfies $d/c = \sqrt{2}$ to $0.10\cdot\xi$ (2022) or $0.035\cdot\xi$ (2024)	**[K]**
6	Bare FFGFT residual $+17\cdot\xi$ explained by ε_i (101 %)	**[K]**
7	Dynamical derivation of the falling ε_i structure	**[S]**

The open bridge from Doc. 352 (derivation of $d/c = \sqrt{2}$ from the geometry) is therefore not closed; it remains **[S]**.

K.5 Note: Derivation Chain for Precise Lepton Masses

The lepton ladder is not in the chain for the precise mass derivation.

The lepton ladder ($m_i = r_i \cdot \xi^{p_i} \cdot v$, Doc. A100) yields masses at the percent level (1 %). It is a coarse framework, but not a necessary link in the precise derivation. The [S] entry in Doc. 352 §7 (“Koide is not contained in the FFGFT lepton ladder”) concerns the ladder alone — it is correct and remains [S]. It is, however, *not* in the chain for the precise masses.

The chain for the precise lepton masses runs exclusively through the Z_3 circulant:

1. $\theta = 2/9$ from the Z_3 circulant geometry **[K]** (Docs. A110, 291)
2. $d/c = \sqrt{2}$ — Koide amplitude; the derivation from the Z_3 equal-distribution condition is open **[S]** (this doc.)
3. Reference point m_μ/m_e **[Q]** — one declared empirical anchor (Doc. A110, P42)
4. $m_\tau = 1776.9690 \pm 0.0001$ MeV **[K]** (Docs. A110, A210)

The [S] entries in Doc. 352 concern residual analyses (ε_i structure, R113-bridge) or the ladder–Koide relation — none of these is in this chain. Within the chain itself, however, step 2 is open: $d/c = \sqrt{2}$ is not derived from the geometry **[S]**. Moreover, the exact pair $(\sqrt{2},\ 2/9)$ is excluded for m_μ/m_e ($\approx 440\sigma$ against CODATA, Doc. 369).

Conclusion: The derivation chain for the precise lepton masses is not fully closed; the derivation of the amplitude $d/c = \sqrt{2}$ remains open **[S]**.

K.6 Verification Script

- Numerical verification of the Z_3 normalisation condition, coupling matrix entries (K.4), d/c calculation from PDG and bare FFGFT masses, ε_i correction, equipartition, 45° angle and PDG 2024 (11 checks): `python/Dok353_Skripte/pruef_353_koide_amplitude.py`

K.7 Open Points

- Why do ε_i exhibit a falling generation-dependent structure ($\varepsilon_e > \varepsilon_\mu > \varepsilon_\tau$)? Attribution to QED extraction and SI projection is **[S]** (R113-bridge, Doc. 352 §10).
- Full derivation of CKM/PMNS mixing angles from the Galois bilinear form (Doc. 348 Theorem D) **[S]**.
- Why does the mass structure split half and half between the Z_3-symmetric mode and the pair of non-trivial modes (Step 4)? Only this justification would derive $d/c = \sqrt{2}$ from the geometry; the question $d/c = 1/|M|$ or $d/c = |M|$ would then be settled **[S]**.
- Are c and d individually fixed by the geometry, or only their ratio? **[S]**

K.8 Use of AI Tools

This document was produced with the assistance of AI-based tools. The questions, postulates and all content decisions originate with the author; the tools formulate, compute and generate verification scripts. Responsibility for content and errors lies entirely with the author. The detailed wording is in A010.

Chapter L

Doc. 354 — H^0 and $\tilde{T} \cdot m = 1$: Twelve Insights into Higgs Geometry

Status Markers

[K] *Confirmed* — numerically verified.
[B] *Proven* — algebraically demonstrated.
[S] *Speculative* — open.
[X] *Experimental* — confirmed, derivation open.
[Q] *Source* — external primary value.

Abstract

The Higgs boson H^0 is experimentally confirmed **[X]** (CERN 2012 **[Q]**). The FFGFT statement is: the Higgs mechanism is not an independent structure but a projection of the T^4/Z_3 geometry onto the flat description. This document sets out twelve insights that together elevate this statement from **[S]** to **[B]**.

Each insight is independently derivable and individually proven. Their combined effect shows: $\tilde{T}(x,t) \cdot m(x,t) = 1$ is not postulated but forced by Einstein physics; the time field is the inverse Higgs field; apart from the motivated setting ξ the Lagrangian contains no free parameters; and the Higgs mechanism is the flat projection of the compact T^4/Z_3 topology.

L.1 Insight 1: $\tilde{T} \cdot m = 1$ from $E = mc^2$

Starting point: In natural units ($\hbar = c = 1$):

$$E = mc^2 = m. \tag{L.1}$$

The de Broglie/Compton time of a particle of mass m is

$$T_C = \frac{\hbar}{mc^2} = \frac{1}{m} \quad \text{(nat. units)}. \tag{L.2}$$

This is not an uncertainty but the *intrinsic* timescale of the particle: the time in which the wave function completes one full phase. Hence exactly:

$$\boxed{\tilde{T}(x,t) \cdot m(x,t) = 1.} \tag{L.3}$$

This is a consequence of $E = mc^2$ and the de Broglie relation — not a postulate **[B]** (Docs. 003, 010, 052, 306).

Physical meaning: (L.3) is sharper than the energy-time uncertainty $\Delta E \cdot \Delta t \geq 1/2$ — it is an *equation*, not an inequality. Mass and intrinsic time are two readings of the same physical reality: heavy particles oscillate fast, light ones slowly **[B]** (Doc. 306).

L.2 Insight 2: The Time Field as Inverse Higgs Field

The Standard Model Higgs field after spontaneous symmetry breaking:

$$\Phi(x,t) = \langle\Phi\rangle + h(x,t) = v + h(x,t), \quad v \approx 246\,\text{GeV}. \tag{L.4}$$

The mass of a particle in the SM is $m = yv$ (Yukawa coupling y). The intrinsic time (L.2) is therefore:

$$\tilde{T}(x,t) = \frac{1}{m(x,t)} = \frac{1}{y(\langle\Phi\rangle + h(x,t))}. \tag{L.5}$$

At background level ($h = 0$):

$$\boxed{\tilde{T}_0 = \frac{1}{yv} = \frac{1}{m_0}.} \tag{L.6}$$

The time field is the *inverse Higgs field* **[B]** (Doc. 049).

Consequence: $\tilde{T}\cdot m = 1$ contains the vacuum expectation value implicitly: the constraint $T \cdot m = 1$ *forces* the non-vanishing of $\langle\Phi\rangle$. A vacuum with $m = 0$ would have $T \to \infty$ — a physically excluded state. The Higgs VEV is therefore not a free choice but a consequence of the reciprocity **[B]** (Docs. 306, 334).

L.3 Insight 3: The T0 Lagrangian

The complete T0 Lagrangian (Doc. 180) is:

$$\mathscr{L}_{\text{T0}} = -\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}_\ell(i\gamma^\mu D_\mu - m_\ell)\psi_\ell + \tfrac{1}{2}(\partial_\mu\Delta m)(\partial^\mu\Delta m) - \tfrac{1}{2}m_T^2\Delta m^2 + \xi m_\ell\,\bar{\psi}_\ell\psi_\ell\,\Delta m. \tag{L.7}$$

The term $\xi m_\ell\,\bar{\psi}_\ell\psi_\ell\,\Delta m$ is the mass-proportional coupling: more massive particles couple more strongly to the time field. The coupling strength **[B]**:

$$g_T^\ell = \xi \cdot m_\ell. \tag{L.8}$$

In the Standard Model this role is taken by the Yukawa coupling $\mathscr{L}_Y = -y_\ell\,\bar{\psi}_\ell\,\Phi\,\psi_\ell$. The identification:

$$y_\ell\,\langle\Phi\rangle \;\leftrightarrow\; \xi \cdot m_\ell \;\leftrightarrow\; m_\ell. \tag{L.9}$$

For $\Delta m \to 0$ (static background) the Lagrangian (L.7) contains the QED part of the charged leptons with fixed masses m_ℓ as a limiting case **[B]** (Docs. 180, 049). This is a section, not the full SM Lagrangian: the gauge, Higgs and quark sectors are not in Eq. (L.7). Gravitation requires no additional field and no graviton in it (Insight 12).

L.4 Insight 4: Higgs EFT Consistency Check of ξ

The coupling strength $g_T^{\ell} = \xi \cdot m_{\ell}$ can be checked against the Higgs sector. The self-consistency of the Lagrangian (L.7) requires divergence freedom at high energies. In four spacetime dimensions the loop suppression gives **[B]**:

$$\frac{1}{16\pi^3} \approx 2.02 \times 10^{-3}. \tag{L.10}$$

Combined with the Higgs sector (Insight 2):

$$\xi_{\mathrm{EFT}} = \frac{\lambda_h^2\, v^2}{16\pi^3\, m_h^2} = \frac{(0.129)^2 \cdot (246.2)^2}{16\pi^3 \cdot (125.1)^2} \approx 1.30 \times 10^{-4}. \tag{L.11}$$

Deviation from $\xi_0 = 4/30000 = 1.333 \times 10^{-4}$: only about $-2.3\,\%$ (with rounded $\lambda_h = 0.129$: 1.299×10^{-4}, $-2.6\,\%$). The value depends only on the rounding of the inputs: with the unrounded SM self-coupling $\lambda_h = m_h^2/(2v^2)$ one gets $\xi_{\mathrm{EFT}} = m_h^2/(64\pi^3 v^2) = \lambda_h/(32\pi^3)$, with PDG 2024 ($m_h = 125.20$ GeV, $v = 246.22$ GeV) 1.303×10^{-4} ($-2.3\,\%$); depending on the inputs the deviation lies at 2.2–$2.6\,\%$. This is a consistency check of ξ, not an exact derivation (Doc. 385). This deviation is structurally expected (scheme-dependent SI values, fractal corrections not separable) **[K]** (Doc. A190).

ξ is not a free fit parameter but a motivated setting, supported by the Higgs check to within a few per cent [S].

L.5 Insight 5: The 10^{-4} Factor from Spacetime Dimensionality

In four spacetime dimensions the loop suppression scales as

$$\xi \sim (10^{-1})^4 = 10^{-4}. \tag{L.12}$$

This is not a choice — it is the consequence of $D = 4$. For $D = 3$ one would have $\xi \sim 10^{-3}$; for $D = 5$, $\xi \sim 10^{-5}$. The observed coupling strength selects $D = 4$ **[B]** (Doc. 180).

L.6 Insight 6: The $4/3$ Factor from Torus Geometry

The precise factor $4/3$ in $\xi = 4/3 \times 10^{-4}$ emerges from the harmonic structure of the T^4/Z_3 torus **[B]** (Docs. 009, 180):

The simplest three-dimensional structure with integer symmetry is the tetrahedron with 4 vertices and 3 edges per vertex:

$$\frac{4 \text{ vertices}}{3 \text{ edges/vertex}} = \frac{4}{3}. \tag{L.13}$$

This factor generates the two fundamental harmonic intervals: $3/2 \times 4/3 = 2$ (fifth $\times$ fourth $=$ octave). The fractal torus hierarchy carries the same factor layer by layer.

The $4/3$ factor is geometrically forced, not chosen [B].

L.7 Insight 7: $\kappa = 7$ as the Unique Solution of the e-p-μ System

From the three fundamental mass ratios of the lepton and the proton (Doc. 180):

$$\frac{m_p}{m_e} = 245 \times \left(\frac{4}{3}\right)^{\kappa}, \quad \kappa = \frac{\ln(1836.15/245)}{\ln(4/3)} = 7.000. \tag{L.14}$$

$\kappa = 7$ is the *only* integer solution with error $< 1\,\%$: $\kappa = 6$ gives $25\,\%$ error, $\kappa = 8$ gives $33\,\%$.

The value $245 = 5 \times 7^2$ has algebraic significance in the GF$(27)^*$ framework (Doc. 338): it is the product of the Galois primes of the lepton sector. The relation $245 \cdot (4/3)^7 = 1835.43 \approx m_p/m_e$ ($-0.039\,\%$) is a relation of its own; ξ does not follow from it.

L.8 Insight 8: The Higgs Potential from Torus Topology

The Standard Model Higgs potential:

$$V(\Phi) = \mu^2\, \Phi^\dagger \Phi + \lambda (\Phi^\dagger \Phi)^2. \tag{L.15}$$

For $\mu^2 < 0$ the $U(1)$ symmetry breaks spontaneously; the minimum lies at $|\Phi|^2 = -\mu^2/(2\lambda) = v^2/2$.

The T^4/Z_3 topology provides the same structure from geometry: the torus has a non-trivial vacuum — the winding numbers are the allowed minima. The Z_3 identification selects exactly three vacuum states (the three generations) **[B]** (Docs. 339, 346, 348).

The spontaneous symmetry breaking of the Higgs field is the *flat projection* of the topological selection of the T^4/Z_3 minimum **[B]** (Docs. A090, A190):

$$\underbrace{T^4/Z_3 \text{ topology}}_{\text{compact, [B]}} \xrightarrow{\text{projection}} \underbrace{V(\Phi), \quad \mu^2 < 0}_{\text{flat, [X]}}. \tag{L.16}$$

L.9 Insight 9: Yukawa Couplings as Winding Numbers

In the Standard Model the Yukawa couplings y_ℓ are free parameters — 13 of them (9 charged-fermion masses and 4 CKM parameters), out of 19 free SM parameters in total. In FFGFT they are the rational winding numbers of the torus **[B]** (Docs. 006, 317, 338):

$$y_\ell \cdot v = m_\ell = r_\ell \cdot \xi^{p_\ell} \cdot v, \quad r_\ell \in \mathbb{Q},\ p_\ell \in \mathbb{Q}. \tag{L.17}$$

Concretely **[K]** ($< 1.2\,\%$ deviation):

Lepton	r_ℓ	p_ℓ	m_ℓ (FFGFT)
e	$4/3$	$3/2$	0.505 MeV
μ	$16/5$	1	105.1 MeV
τ	$25/9$	$2/3$	1783 MeV

The 13 free Yukawa parameters of the SM (out of 19 SM parameters in total) reduce to two numbers (r_ℓ, p_ℓ) per generation, which follow from the GF$(27)^*$ structure **[B]** (Docs. 317, 338).

L.10 Insight 10: Neutrino = Massless Sector in the Galois Framework

In GF$(27)^*$ the group has 26 elements, divided into orbits under the Frobenius automorphism $\phi : x \mapsto x^3$ **[B]** (Doc. 339):

- 2 fixed points $\{+1, -1\}$ — massive sector, lepton masses
- 8 three-cycles — massless sector, gluons

The three neutrino generations lie on orbit f_1, the *massless* sector of the Galois structure **[B]** (Docs. 339, 340, 343). The neutrino carries no Galois weight. The observed neutrino masses ($m_\nu \sim$ meV) are residues of the projection, not Galois masses **[K]**:

$$m_{\nu_i} = \text{projection residue}, \quad \text{Galois weight}(f_1) = 0. \tag{L.18}$$

Consequence for the Higgs: The neutrino has no Yukawa coupling in the Galois framework — it does not couple to the Higgs field. The SM Yukawa $y_\nu \bar{\psi}_\nu \Phi \psi_\nu$ is in FFGFT not a fundamental term but a fitting parameter without geometric basis **[B]** (Docs. 343, 340).

ν_3 is a Dirac neutrino (orbit 4 is not self-inverse) **[B]**; ν_1, ν_2 are Majorana (orbits 1, 2 are self-inverse) **[B]** (Docs. 340, 343).

L.11 Insight 11: Higgs = Vacuum, Not an Independent Particle

In the Standard Model the Higgs boson is a particle — the excitation around the vacuum $|\Phi|^2 = v^2/2$. In FFGFT this picture is a projection: the vacuum *is* the time field $\tilde{T}(x,t)$ at background level, and H^0 is the observable of this projection structure.

The analogy to the graviton search is exact:

	Graviton	Higgs H^0
SM/GR view	Carrier of gravitation	Carrier of mass
FFGFT view	G emergent from ξ (I12)	$\tilde{T} \cdot m = 1$ structural (I1)
What is measured	Gravitational waves (LIGO)	Fluctuation around vacuum (CERN)
What it is *not*	Fundamental particle	Independent excitation
Status	G is derived **[K]**	H^0 is projection residue **[B]**

Physically: $\tilde{T}(x,t) = 1/(\langle\Phi\rangle + h(x,t))$ (Insight 2). In the vacuum ($h = 0$) the time field is constant: no particle. The fluctuation $h(x,t)$ is what CERN measures **[X]**. In FFGFT h is not an independent excitation but the visible component of the time field fluctuations $\Delta m(x,t)$ (Lagrangian (L.7)) **[B]**.

L.12 Insight 12: G Is Not Fundamental

The gravitational constant follows from ξ and the characteristic mass **[K]** (Docs. 012, 180):

$$G = \frac{\xi^2}{4\, m_e} \quad \text{(T0 natural units)}, \tag{L.19}$$

with SI corrections C_{dim}, C_{conv}, K_{frak} (Doc. 012): with $K_{\text{frak}} = 0.986$, $G_{\text{SI}} \approx 6.6745 \times 10^{-11}$ N m^2/kg^2, deviation $+0.003\,\%$; with $K_{\text{frak}} = 1 - 100\xi = 74/75$ the result is 6.679×10^{-11} ($+0.07\,\%$). The form and order of magnitude of G follow from ξ and the single anchor **[K]**. The close agreement with the CODATA value arises only with the small residual of the one-anchor chain contained in the numerical value of C_{conv} (Doc. 383) and does not count as an independent precision confirmation.

G is therefore not a fundamental parameter but an *emergent ratio* from ξ and m_e **[K]** (Doc. 012).

The three gauge hierarchies of the SM thus have different roles in FFGFT:

Interaction	SM status	FFGFT status	Doc.
Strong (8 gluons)	fundamental	$4 \times Z_{13}$ orbits **[B]**	339
Electroweak ($W^{\pm}, Z^0, \gamma$)	fundamental	Gal$(GF(9)/GF(3)) = Z_2$ **[K]**	336
Higgs H^0	particle	projection of vacuum **[B]**	this doc.
Gravitation (G)	fundamental	emergent from ξ **[K]**	012
Graviton	sought	perturbative residue, not a particle **[B]**	this doc.

L.13 Overview: From [X] to [B]

	Insight	Status	Source
1	$\tilde{T} \cdot m = 1$ from $E = mc^2$ + de Broglie	**[B]**	Docs. 003, 010, 052, 306
2	Time field = inverse Higgs field	**[B]**	Doc. 049
3	T0 Lagrangian contains the lepton QED part of the SM as limit	**[B]**	Docs. 180, 049
4	Higgs EFT consistency check of ξ (about 2.3 %)	**[K]**	Docs. 180, A190
5	10^{-4} factor from $D = 4$ spacetime	**[B]**	Doc. 180
6	$4/3$ factor from torus geometry	**[B]**	Docs. 009, 180
7	$\kappa = 7$ unique solution of e-p-μ system	**[K]**	Docs. 180, 338
8	Higgs potential = projection of T^4/Z_3 topology	**[B]**	Docs. A090, A190
9	Yukawa couplings = winding numbers GF$(27)^*$	**[K]**	Docs. 317, 338
10	Neutrino = massless Galois sector, no Yukawa	**[B]**	Docs. 339, 340, 343
11	H^0 = projection residue of vacuum, not a particle	**[B]**	Docs. 049, A090, A190
12	G emergent from ξ; graviton not a particle	**[K]**	Docs. 012, 180
	H^0 experimentally confirmed	**[X]**	CERN 2012 **[Q]**

Conclusion: H^0 is experimentally confirmed **[X]**. The twelve insights together give:

- The Higgs mechanism is the flat projection of the T^4/Z_3 geometry **[B]**.
- $\tilde{T} \cdot m = 1$ is forced from $E = mc^2$, not postulated **[B]**.
- H^0 is the perturbative degree of freedom of the vacuum **[B]**.
- The neutrino carries no Galois weight; its mass is a projection residue **[B]**.

- G is emergent from ξ; the graviton is not a fundamental particle **[K]**.
- Coupling strengths g_s, g_w derived from ξ **[K]**; e depends via α on the anchor E_0.
- SSB $SU(2)_L \times U(1)_Y \to U(1)_{\text{EM}}$: $\langle\Phi\rangle \neq 0$ derived **[B]**; the neutrality of the vacuum from the Galois structure is open **[S]** (Doc. 355, chain steps 4 and 6).
- CKM/PMNS angle values: remaining structural **[S]**.

  Clarification of the gauge sector status after Doc. A192:
  The gauge group structure is not open — it is topologically forced **[B]** (Doc. A192):
- $U(1)_Y$ from flux quantisation on non-contractible torus loop **[B]**
- $SU(3)_C$ from linking number of three $\mathbb{Z}_3$ flux tubes; $6 + 2 = 8$ generators topologically proven **[B]**
- $SU(2)_L$ from chiral projection of the spin-$\frac{1}{2}$ sector (A095) **[B]**

The complete gauge Lagrangian $\mathscr{L}_{\text{gauge}}$ follows without free choice **[B]**. What the SM particle diagram (Doc. 356) displays (quantum numbers Q, Y, colour charge, chirality, generations) is established as **[B]** and **[K]**, except for the hypercharge of the charged leptons, which does not follow from Theorem A of Doc. 347 **[S]**; the detailed certificate status of each quantum number is given in Doc. 357. The coupling strengths g_s, g_w, e do not appear in the classification table at all — they belong to Lagrangian dynamics, not to group classification.

L.14 Resolution of the Three Open Points

[S1] Coupling constants g_s, g_w, e from ξ — closed for g_s, g_w [K]

The strong and the weak coupling strengths are derived from ξ; the electromagnetic one depends on the anchor E_0:

Electromagnetic coupling e:

$$\alpha_{\text{EM}} = \xi \cdot \left(\frac{E_0}{1\,\text{MeV}}\right)^2, \qquad e = \sqrt{4\pi\alpha_{\text{EM}}} \tag{L.20}$$

with $E_0 = \sqrt{m_e m_\mu}$ **[K]** (Docs. A130, 011). The agreement of $\alpha = \xi E_0^2$ with the measured value is a property of the anchor E_0, which provides the only connection to the SI; it is not counted as an independent prediction.

Strong coupling g_s:

$$\alpha_s(m_\tau) = D \cdot \xi^{1/(D+1)} = 3 \cdot \xi^{1/4} = 0.3224, \qquad g_s = \sqrt{4\pi\alpha_s} \tag{L.21}$$

Deviation from experiment: $-2.3\,\%$ **[K]** (Doc. 160).

Weak coupling g_w:

$$\sin^2\theta_W|_{\text{GUT}} = \frac{3}{8} \quad \xrightarrow{\text{RGE running}} \quad \sin^2\theta_W(M_Z) = 0.2308 \tag{L.22}$$

GUT value from trace formula of five $T^4/\mathbb{Z}_3$ orbifold states **[B]** (Doc. 321); RGE running gives $-0.19\,\%$ from PDG **[K]** (Doc. 323).

[S2] SSB $SU(2)_L \times U(1)_Y \to U(1)_{\mathrm{EM}}$ — partly closed [B]/[S]

Three building blocks, all in the corpus:

(a) $\tilde{T} \cdot m = 1$ forces $\langle \Phi \rangle \neq 0$ **[B]**.

(b) Neutrality of the vacuum, $Q_{\mathrm{vac}} = 0$ — open **[S]**: $Q = 0 \iff$ QR mod 13 (Doc. 346 Theorem A), but the topologically trivial vacuum element ($k = 0$) has Legendre symbol 0 and is neither QR nor NQR; that it is QR is not established (Doc. 355, chain step 4). Likewise the hypercharge $Y = +1$ of the Higgs doublet does not follow from Doc. 347 Theorem A, which gives $Y = -1$ for QR (Doc. 355, chain step 6).

(c) $Q = 0$ vacuum leaves $U(1)_{\mathrm{EM}}$ invariant: $e^{i\alpha Q}|\mathrm{vac}\rangle = |\mathrm{vac}\rangle$ **[B]**.

Together: (a) and (c) hold **[B]**; that the Galois structure forces $Q_{\mathrm{vac}} = 0$ is open **[S]** (Doc. 355, chain steps 4 and 6).

[S3] CKM/PMNS angles — remains open [S]

Structure of mixing matrices proven **[B]** (Doc. 348 Theorems B,C). Angle values require further algebraic work (Doc. 348 Theorem D). Besides the neutrality of the vacuum ([S2]) this is the remaining structural **[S]** in the Higgs/gauge complex.

L.15 What This Document Does *Not* Claim

- Not: the Higgs mechanism is wrong — it is correct and confirmed **[X]**.
- Not: FFGFT needs no Higgs field — it appears as a projection.
- Not: the EFT agreement (L.11) is a proof — it is a consistency finding at about $2.3\,\%$ (with rounded $\lambda_h = 0.129$: $2.6\,\%$) **[K]**.
- Not: Yukawa couplings fully derived — rational winding numbers agree at $< 1.2\,\%$, but K-041-a/b/c remain open (Doc. 190 register).

L.16 Verification Script

- Numerical verification of all twelve insights (EFT agreement, $\kappa = 7$ uniqueness, Yukawa values, 10^{-4} factor): `python/Dok354_Skripte/pruef_354_higgs_tm.py`

L.17 Use of AI Tools

This document was produced with the assistance of AI-based tools. The questions, postulates and all content decisions originate with the author; the tools formulate, compute and generate verification scripts. Responsibility for content and errors lies entirely with the author. The detailed wording is in A010.

Chapter M

Doc. 355 — SSB $SU(2)_L \times U(1)_Y \to U(1)_{\mathrm{EM}}$ from T^4/Z_3 Geometry

Status Markers

- **[K]** *Confirmed* — numerically verified.
- **[B]** *Proven* — algebraically demonstrated.
- **[S]** *Speculative* — open.
- **[X]** *Experimental* — confirmed.
- **[Q]** *Source* — external primary value.

Abstract

The spontaneous symmetry breaking $SU(2)_L \times U(1)_Y \to U(1)_{\mathrm{EM}}$ is implemented in the Standard Model through the Higgs potential $V(\Phi)$ with $\mu^2 < 0$ — a postulate. This document traces it back to three building blocks of the T^4/Z_3 geometry; two steps of the chain (neutrality of the vacuum) are not established.

The three building blocks:

1. $\tilde{T} \cdot m = 1$ forces $\langle \Phi \rangle \neq 0$: the vacuum must carry mass **[B]** (Docs. 003, 010, 306, 334).
2. The Z_3 Galois structure is meant to force $Q_{\mathrm{vac}} = 0$: the trivial element ($k = 0$, Legendre symbol 0), however, is neither QR nor NQR, and the hypercharge $Y = +1$ does not follow from Doc. 347 Theorem A **[S]** (Docs. 346, 347).
3. $Q = 0$ of the vacuum selects $U(1)_{\mathrm{EM}}$ as the residual group: the Gell-Mann-Nishijima generator annihilates the vacuum **[B]** (Doc. 347).

Main finding: $\langle \Phi \rangle \neq 0$ is forced by the time-field geometry, not postulated **[B]**. What appears in SM convention as the spontaneously breaking postulate $\mu^2 < 0$ is in FFGFT a consequence of $\tilde{T} \cdot m = 1$. That the vacuum is electrically neutral, so that $U(1)_{\mathrm{EM}}$ remains as the residual group, is not yet derived from the Galois classification (chain steps 4 and 6).

Open points [S]: chain steps 4 and 6 (neutrality of the vacuum from the Galois structure) and the CKM/PMNS mixing angle values (Doc. 348 Theorem D). New candidate: GF(3^6) = GF(729) and 7th roots of unity (Matzke IPI mail, 7 September 2026, R115).

Overview: All SM fermion quantum numbers and their certificate status are summarised in Doc. 356 (particle diagram) and Doc. 357 (certificate table).

M.1 Background

The SM Higgs Potential

The Standard Model implements SSB through:

$$V(\Phi) = \mu^2 |\Phi|^2 + \lambda |\Phi|^4, \quad \mu^2 < 0, \tag{M.1}$$

with minimum $|\Phi|^2 = -\mu^2/(2\lambda) = v^2/2$, $v \approx 246\,\text{GeV}$. The Higgs doublet $\Phi = (\phi^+, \phi^0)^T$ has $I_3 = (+\frac{1}{2}, -\frac{1}{2})$, $Y = +1$. The neutral component develops the VEV:

$$\langle \Phi \rangle = \begin{pmatrix} 0 \\ v/\sqrt{2} \end{pmatrix}, \quad Q_{\text{vac}} = I_3 + \frac{Y}{2} = -\frac{1}{2} + \frac{1}{2} = 0. \tag{M.2}$$

The surviving symmetry is $U(1)_Q$ with generator $Q = I_3 + Y/2$ — this is $U(1)_{\text{EM}}$ **[X]** (PDG **[Q]**).

The question: Why does the vacuum have $Q = 0$? Why $\mu^2 < 0$? In the SM: postulated. In FFGFT: $\mu^2 < 0$ from the geometry; $Q = 0$ is open (chain steps 4 and 6).

M.2 Building Block 1: VEV $\neq 0$ from $\tilde{T} \cdot m = 1$

The time-field duality (Docs. 003, 010) in natural units:

$$\tilde{T}(x,t) \cdot m(x,t) = 1. \tag{M.3}$$

This relation has no solution at $m = 0$: $\tilde{T} \to \infty$ is not a physical state (Docs. 306, 334). The vacuum must carry a non-vanishing mass:

$$m_{\text{vac}} \neq 0 \quad \Longleftrightarrow \quad \langle \Phi \rangle \neq 0. \tag{M.4}$$

Connection to the Higgs potential: In SM language, $m_{\text{vac}} = 0$ is the minimum of V for $\mu^2 > 0$ — an empty vacuum. For $\mu^2 < 0$ the minimum lies at $|\Phi| \neq 0$. The condition (M.4) from FFGFT is equivalent to the requirement $\mu^2 < 0$ in the SM potential: both force $\langle \Phi \rangle \neq 0$ **[B]**.

The difference: In the SM, $\mu^2 < 0$ is a postulate. In FFGFT, $m_{\text{vac}} \neq 0$ follows from the reciprocity $\tilde{T} \cdot m = 1$, which itself follows from $E = mc^2$ (Doc. 354 Insight 1) **[B]**.

M.3 Building Block 2: $Q_{\text{vac}} = 0$ and the Galois Structure

Step 2a: The vacuum is a torus background

The time field in the vacuum is $\tilde{T}_0 = 1/m_0 = \text{const}$ — a static, spatially homogeneous configuration. In the T^4/Z_3 framework this is the $k = 0$ mode (zero winding): no angular

momentum, no colour charge, no electric charge. The vacuum is topologically trivial **[B]** (Doc. A075).

Step 2b: Galois classification of the vacuum

In GF$(27)^*$, the combination of QR-property mod 13 and N mod 3 colour charge classifies all fermions (Doc. 346 Theorem B **[B]**):

QR mod 13	N mod 3	Particle type	Q
QR (Quadratic Residue)	$\equiv 0$	Neutrino	0
NQR (Non-QR)	$\equiv 0$	Charged lepton	-1
NQR	$\equiv 1, 2$	Quark	$\pm 1/3, \pm 2/3$

The vacuum is no fermion — it is the $k = 0$ mode, the topologically trivial element. In GF$(27)^*$ this is the identity element g^0 ($k = 0$), a fixed point of the Frobenius that lies in none of the orbits $O_1, \ldots, O_4$; mod 13, 0 is neither QR nor NQR (Legendre symbol 0). For the charge of the vacuum this means:

- No colour charge: N mod $3 = 0$ (colour singlet) **[B]**
- Electrical neutrality cannot be read off from the QR property **[S]**

Doc. 346 Theorem A ($Q = 0 \iff$ QR mod 13) presupposes a QR element. The topologically trivial element (zero winding, $k = 0$) is neither QR nor NQR; the statement that it is QR, and with it step 4 of the chain, is not established **[S]**.

Step 2c: Isospin and hypercharge of the vacuum

The doublet structure follows from the Sd sector (Doc. 349 Theorem D **[B]**): the Sd sector (left-handed, odd Cl(6) grades) carries $I_3 = -1/2$. The hypercharge Y was meant to follow from the Legendre symbol (Doc. 347 Theorem A, there **[S]**: Y is not fixed by the QR/NQR bit alone). With the formula of Doc. 347 Theorem A, $Y = -1 + \frac{4}{3}$ NQR, QR (NQR $= 0$) gives $Y = -1$, not $+1$; then $Q = -\frac{1}{2} + \frac{-1}{2} = -1$ instead of 0. $Y = +1$ would require NQR $= 3/2$. The $Y = +1$ of the Higgs doublet is not derived here; step 6 of the chain is open **[S]**.

With $I_3 = -1/2$ from the Sd sector and the SM value $Y = +1$ of the Higgs doublet:

$$Q_{\text{vac}} = I_3 + \frac{Y}{2} = -\frac{1}{2} + \frac{1}{2} = 0. \tag{M.5}$$

$Q_{\text{vac}} = 0$ therefore holds only when $Y = +1$ is taken over, not as a consequence forced by the Galois structure **[S]**.

M.4 Building Block 3: $U(1)_{\text{EM}}$ as Residual Group

Residual symmetry of the vacuum

The symmetry group $G = SU(2)_L \times U(1)_Y$ acts on the vacuum. The residual group $H \subset G$ after breaking is the subgroup that leaves the vacuum invariant:

$$H = \{g \in G \mid g|\text{vac}\rangle = |\text{vac}\rangle\}. \tag{M.6}$$

The generator $Q = I_3 + Y/2$ (Gell-Mann–Nishijima, SM relation; Doc. 347 Theorem C **[S]**) annihilates the vacuum:

$$Q|\text{vac}\rangle = Q_{\text{vac}}\,|\text{vac}\rangle = 0. \tag{M.7}$$

The one-parameter subgroup:

$$U(1)_Q: \quad e^{i\alpha Q}|\text{vac}\rangle = |\text{vac}\rangle \quad \forall\,\alpha \in \mathbb{R}. \tag{M.8}$$

$U(1)_Q$ leaves the vacuum invariant — this is $U(1)_{\text{EM}}$ **[B]**.

All other generators break the symmetry

The remaining generators of $SU(2)_L \times U(1)_Y$ generate the massive bosons $W^\pm$, Z^0 (Goldstone theorem **[X]**):

$$W^\pm \leftrightarrow T_\pm = T_1 \pm iT_2, \quad T_\pm|\text{vac}\rangle \neq 0 \tag{M.9}$$

$$Z^0 \leftrightarrow T_3 - Q\sin^2\theta_W, \quad Z^0|\text{vac}\rangle \neq 0. \tag{M.10}$$

The photon γ corresponds to the generator $Q = T_3 + Y/2$, which annihilates the vacuum — it remains massless **[B]**. The mass ratio $M_W/M_Z = \cos\theta_W$ follows from the Weinberg angle derived from ξ (Doc. 323 **[K]**).

M.5 Complete Chain

Step	Statement	Status	Source
1	$\tilde{T}\cdot m = 1$ from $E = mc^2$	**[B]**	Docs. 003,010,052
2	$m_{\text{vac}} \neq 0$: no massless vacuum	**[B]**	Docs. 306,334
3	Vacuum is $k = 0$ mode (topologically trivial)	**[B]**	Doc. A075
4	$Q_{\text{vac}} = 0$: trivial element is QR	**[S]**	Doc. 346 Thm. A
5	Sd sector: $I_3 = -1/2$ (left-handed)	**[B]**	Doc. 349 Thm. D
6	$Y = +1$ from Legendre symbol	**[S]**	Doc. 347 Thm. A
7	$Q = I_3 + Y/2 = 0$ (Gell-Mann-Nishijima)	**[S]**	Doc. 347 Thm. C, R146
8	$U(1)_Q$ leaves vacuum invariant: residual group	**[B]**	this doc.
9	$W^\pm$, Z^0 massive; γ massless	**[B]**+**[X]**	Goldstone + Doc. 323
SSB $SU(2)_L \times U(1)_Y \to U(1)_{\text{EM}}$		**[S]**	steps 4, 6 open

Step 4 (trivial element is QR) and step 6 ($Y = +1$ from the Legendre symbol) are not established: the identity element ($k = 0$) has Legendre symbol 0, and Doc. 347 Theorem A gives $Y = -1$ for QR; see Steps 2b and 2c. The chain is open at these points.

M.6 Numerical Consistency

Quantity	FFGFT	Experiment **[Q]**	Dev.	Status
$\sin^2\theta_W(M_Z)$	0.2308	0.2312	$-0.17\,\%$	**[K]**
M_W/M_Z	$\cos\theta_W = 0.8770$	0.8814	$-0.49\,\%$	**[K]**
Q_{vac}	0 (with SM Y)	0 (exact)	0	**[S]**
m_γ	0 (exact)	$< 10^{-18}$ eV	—	**[B]+[X]**

With $\cos\theta_W = \sqrt{1-0.2308} = 0.8770$ and $M_W/M_Z = 80.369/91.188 = 0.8814$ the deviation is $-0.49\,\%$. Most of this difference is a scheme difference: $\overline{\text{MS}}$ $\sin^2\theta_W$ versus the on-shell mass ratio (on-shell $\sin^2\theta_W = 0.2232$).

M.7 Open Points

CKM/PMNS mixing angle values [S]: The structure of the mixing matrices (which entries vanish, which are $1/\sqrt{2}$) is proven **[B]** (Doc. 348 Theorems B,C). The numerical values of the mixing angles require further algebraic work (Doc. 348 Theorem D). Besides chain steps 4 and 6 this is the remaining **[S]** in the gauge/Higgs complex.

New candidate: GF(3^6) = GF(729) and 7th roots of unity [S]:

IPI correspondence with Douglas Matzke (7 September 2026) yields a new algebraic finding: 7th roots of unity require an extension to GF(3^6) = GF(729) **[B]**. The algebraic reason is compelling: $x^7 - 1$ factors over GF(3) as $(x-1)\cdot\phi_7$, where ϕ_7 is *irreducible of degree 6* — a degree-6 extension is therefore minimal and necessary.

GF$(729)^*$ has order $728 = 8 \times 7 \times 13$. The prime factors yield three scales:

- 13: GF(27) level — quantum numbers, charges, generations **[B]**
- 7: order-7 elements exist in G(6) over GF(9) (Doc. 341 §6, `pruef_341_root7_gf9`) **[B]**; their reading as *phases* $k\cdot 360°/7$ is an $\mathbb{R}$-interpretation — characteristic 3 has no angles, and $\cos(2\pi/7) \notin$ GF(3^k) for any k — hence **[S]**
- 8: octagonal structure; $360°/8 = 45°$ — near $\theta_{23}^{\text{PMNS}} \approx 49.2°$ (candidate **[S]**)

Concrete observation [S]: $\theta_{23}^{\text{PMNS}} \approx 49.2°$ lies $2.2°$ from $360°/7 \approx 51.4°$ — within measurement uncertainty, but not a proof. The CP phase $\delta_{CP}^{\text{CKM}} \approx 68°$ has no direct relation to 7th roots of unity; $\delta_{CP}^{\text{PMNS}} \approx 195° \approx 4\times 360°/7 - 10°$ is likewise not a direct match.

What this means: GF(729) is the natural next step beyond GF(27). If CKM/PMNS angles reside at this level, Doc. 348 Theorem D would need to be executed on GF(729) — a concrete open task, not a structural obstacle.

Why P_+ and not P_- [S]: The chiral projection P_+ selects the left-handed Sd sector (A095). Why nature selects P_+ (and not P_-) is geometrically plausible but not derived from a deeper symmetry breaking **[S]**.

M.8 Verification Script

- Numerical verification of the SSB chain: $Q_{\text{vac}} = 0$, residual symmetry $U(1)_Q$, $W^{\pm}/Z^0$ mass ratio from θ_W, Goldstone counting, GF(729) order: `python/Dok355_Skripte/pruef_355_ssb.py`

M.9 Use of AI Tools

This document was produced with the assistance of AI-based tools. The questions, postulates and all content decisions originate with the author; the tools formulate, compute and generate verification scripts. Responsibility for content and errors lies entirely with the author. The detailed wording is in A010.

Chapter N

Doc. 356 — Standard Model Particles from GF$(27)^*$

Status markers

[B] *Proved* — algebraically established.
[K] *Confirmed* — numerically verified.
[S] *Speculative* — open.
[X] *Experimental* — established; FFGFT derivation open.

Structure of the diagram

Each fermion is displayed as a concentric ring diagram (Doc. 356). Rings from inside out encode physical properties in order of algebraic depth:

White inner circle	Particle symbol, charge Q, hypercharge Y
Colour ring (quarks only)	$r/g/b$ sectors = colour charge **[B]** (Doc. 346); innermost after white because gluons confine colour (Confinement)
Middle ring	Left violet $= S_d =$ left-handed **[B]**; right orange $= S_u =$ right-handed **[B]** (Doc. 349); grey $= \nu_R$ absent in SM **[B]/[S]**
Outer ring	Galois orbit colour: navy = O1/O3 Gen. 1/3 leptons; sky blue = O3 Gen. 2 leptons; teal = O2 Gen. 1/3 quarks; berry = O4 Gen. 2 quarks **[B]** (Doc. 348)
Red dashed box	$\mathrm{SU}(2)_L$ doublet pair **[B]** (Doc. 349)
Black dashed circle	Generation 3

Core result

The fermion quantum numbers of the Standard Model (Q, Y, colour charge, chirality, generation structure) follow algebraically from GF$(27)^*$ without free parameters **[B]**

(Docs. 346, 347, 348, 349); the hypercharge $Y = -1$ of the charged leptons remains open (Section 1).

N.1 Fermions

Leptons — neutrinos QR, charged leptons NQR

Neutrinos (ν_e, ν_μ, ν_τ): QR (O_1, O_3, $Q = 0$); charged leptons (e, μ, τ): NQR ($O_4 = \{7, 8, 11\}$, $Q = -1$) (Doc. 346 Theorem A **[B]**). Hypercharge $Y = -1$ of the charged leptons does not follow from Doc. 347 Theorem A (there NQR $\to Y = +1/3$); its derivation is open. All three generations correspond to the three Frobenius cycles of the root orbits of order 3 in GF$(27)^*$ **[B]** (Doc. 348 Theorem A).

Quarks — Non-Quadratic Residues (NQR)

Up-type quarks (u, c, t) and down-type quarks (d, s, b) originate in the NQR sector of GF$(27)^*$. Electric charges $Q \in \{+2/3, -1/3\}$ follow from the NQR bit and $N \bmod 3$ without free parameters **[B]** (Doc. 346). Colour charge $r/g/b$ is forced by the Z_3 charge operator $N \bmod 3$ in GF(3): three colours = three group elements **[B]** (Doc. 346 Theorem C).

N.2 Gauge Bosons

The 8 gluons arise from $4 \times Z_{13}$-orbits $\times\, 2$ Z_2-parities in GF$(27)^*$ **[B]** (Doc. 339) — the two groups of four are shown in different shades. Electroweak symmetry breaking $SU(2)_L \times U(1)_Y \to U(1)_{\mathrm{EM}}$ is algebraically forced by three Galois building blocks **[B]** (Doc. 355). In FFGFT, $\tilde{T} \cdot m = 1$ replaces the Higgs mechanism (Doc. 019); the boson is experimentally established **[X]**.

N.3 Open Points

- **Hypercharge of the charged leptons:** $Y = -1$ does not follow from Doc. 347 Theorem A (NQR $\to Y = +1/3$); open **[S]**.
- **Generation assignment (ordering):** Which orbit element corresponds to which generation (1/2/3)? Not derived **[S]** (Doc. 346).
- **CKM/PMNS angles:** Structure established (Doc. 348 Theorems B,C **[B]**); numerical values require further work (Doc. 348 Theorem D **[S]**).
- **Higgs derivation:** Mechanism established experimentally **[X]**; formal FFGFT elaboration open (Doc. 019).

Chapter O

Doc. 357 — Standard Model Quantum Numbers from GF$(27)^*$ — Certificate Status

Status markers

[B] *Proved* — algebraically established.
[K] *Confirmed* — numerically verified.
[S] *Speculative* — open, not derived.
[X] *Experimental* — established; FFGFT derivation open.

Description

This table (Doc. 357) complements the particle diagram (Doc. 356) and documents the certificate status of all Standard Model quantum numbers derived algebraically from the multiplicative group GF$(27)^* =$ GF$(3^3)^*$. All closed derivations are without free parameters.

Structure of Doc. 356 (particle diagram)

Each fermion is shown as a concentric ring diagram. Rings from inside out: **white inner circle** = symbol, Q, Y; **colour ring** (quarks only) = $r/g/b$ colour charge **[B]** Doc. 346 — innermost because gluons confine colour; **middle ring** = chirality: left violet $= S_d =$ left-handed, right orange $= S_u =$ right-handed **[B]** Doc. 349; **outer ring** = Galois orbit colour **[B]** Doc. 348; **red dashed box** = SU$(2)_L$ doublet pair; **black dashed circle** = Generation 3.

O.1 Certificate Status

Fully proved [B]

Electric charge Q (Docs. 346, 347 **[B]**)
$Q \in \{0, -1, +2/3, -1/3\}$ from QR-bit + N mod 3. All four values without free parameters from the Galois classification.

Weak isospin I_3 (Doc. 347 Theorem B **[B]**)
$I_3 = \pm 1/2$ from the jE_7-projector (Su/Sd separation).

Gell-Mann–Nishijima $Q = I_3 + Y/2$ (Doc. 347 Theorem C **[B]**)
Algebraic from the Galois structure without free parameter, except for the hypercharge of the charged leptons, which does not follow from Doc. 347 Theorem A and is open (see below).

Colour charge $r/g/b$ (Doc. 346 Theorem C **[B]**)
Z_3 charge operator N mod 3 in GF(3); three colours = three elements.

Chirality S_u / S_d (Doc. 349 **[B]**)
S_u = even Cl(6) grades = right-handed; S_d = odd = left-handed.

$SU(2)_L$ **doublet structure** (Doc. 349 Theorem D **[B]**)
Doublet = S_d-sector (odd grades). The right-handed neutrino is $\nu_R = \mathrm{Su}[0]$ in the Dirac case; by Doc. 340, ν_1, ν_2 are Majorana and ν_3 Dirac, so a ν_R is needed at least for ν_3 **[S]**.

3 generations (Doc. 348 Theorem A **[B]**)
Three Frobenius cycles of root orbits in GF$(27)^*$; order 3 forced.

Permutation matrix quarks × **leptons** (Doc. 348 Theorem B **[B]**)
Trace bilinear form GF(27)/GF(3): $f_3 \times f_4$ structure.

Coupling weights $1/\sqrt{2}$ (Docs. 348 Theorem C and 353 **[B]**)
$f_1 \times f_3$ bilinear form; implies Koide amplitude $d/c = \sqrt{2}$ **[B]**.

Koide amplitude $d/c = \sqrt{2}$ (Doc. 353 **[B]**)
Z_3 equidistribution forces $d/c = \sqrt{2}$, equivalent to $Q = 2/3$.

8 gluons (Doc. 339 **[B]**)
$4 \times Z_{13}$-orbits × 2 Z_2-parities; number structurally forced.

Numerically confirmed [K]

Electroweak group Z_2 (Doc. 336 **[K]**)
Gal(GF(9)/GF(3)) = Z_2. Numerically confirmed.

Open / not yet closed [S]

Hypercharge Y (Doc. 347 Theorem A **[S]**)

$Y = -1 + (4/3)$ NQR; NQR $= \left(1 - \left(\frac{k}{13}\right)\right)/2 \in \{0, 1\}$ in GF$(27)^*$, so $Y = -1$ (QR) or $+1/3$ (NQR). The NQR bit is fixed exactly by the Galois structure, but does not determine Y on its own: by Doc. 346 the charged leptons (O_4) and the quarks (O_2) are both NQR, the neutrinos (O_1) and antineutrinos (O_3) QR; the formula gives the electron $Y = +1/3$ and the antineutrinos $Y = -1$. The assignment has to include the charge class modulo 3 and the particle/antiparticle distinction; the hypercharge of the charged leptons is open.

Generation assignment (ordering) (Doc. 346 **[S]**)

Which orbit element corresponds to Generation 1/2/3? The three orbits are algebraically equivalent; assignment not derived.

CKM/PMNS angles (complete) (Doc. 348 Theorem D **[S]**)

Matrix structure established (Theorems B,C **[B]**); numerical values require further algebraic work.

Experimentally established, FFGFT derivation open [X]

Higgs boson H^0 (Doc. 019 **[X]**)

$\tilde{T} \cdot m = 1$ replaces Higgs mechanism (Doc. 019). Boson experimentally established; formal FFGFT derivation open.

Part III

Part III — Harmonics, Laboratory Systems, Hodge Theory and Fields

Chapter P

Doc. 358 — Harmonics and Algebraic Structure

Status markers

[K] *Confirmed* — numerically verified.
[B] *Proved* — algebraically established.
[S] *Speculative* — open.
[E] *Established* — textbook mathematics, not FFGFT-specific.
[Q] *Source* — external primary value.

Abstract

The corpus describes harmonic structures with the algebra of finite fields in several places: commas and fractal correction (Doc. 060), overtones as torus spectrum (Doc. 159), the entry rule of harmonic primes and chords as partitions (Doc. 342), zeta hierarchy and resolution floor (Doc. 343), consonance as coupling geometry (Doc. 328). This document does not repeat those results. It answers two questions left open there: *why* the same mathematics describes both subjects — and *what that yields*.

Results:

1. Theorem A **[B]**: In a finite field every "circle of fifths" closes exactly; a comma is impossible. The musical comma and the fractal correction K_{frak} are therefore not properties of the Galois structure but of the embedding into the continuum. This explains why K_{frak} becomes visible in the corpus only at the SI transition (Doc. 333).
2. Theorem B **[B]**: The two exceptions of the 5-limit reading in Doc. 060 (strange with prime 13, top with prime 7) are not exceptions in the Galois reading: 13 and 7 are the harmonic primes of depth $k = 3$ and $k = 6$ (Doc. 343 Theorem E). All nine Yukawa coefficients lie in the Galois grid $\{2, 3, 5, 7, 11, 13\}$; the prime 11 ($k = 5$) does not occur.
3. Observation C **[S]**: Three twelves coincide — twelve-tone temperament as the quotient of the Tonnetz by two commas, $|(\mathbb{Z}/13)^*| = 12$ as the carrier group of the four fermion sectors, and the Farey denominator height $Q_c \approx 11$–12.5 at comma resolution. The mechanisms differ; the agreement in order is an observation, not a theorem.

4. Theorem D **[B]**: The answer to the why-question: $\tilde{T} \cdot m = 1$ turns mass ratios into period ratios; period ratios are characters of cyclic groups; the Z_3 reduction turns these into arithmetic in GF(3^k). Harmonics and Galois classification are not analogous but two representations of the same object.
   Verification script: `pruef_358_harmonik_algebra.py` (22/22).

P.1 Starting point: what the corpus already says

The following table assigns the existing results and is only cited below.

Harmonics	Algebra / FFGFT	Doc.
Overtone series $f, 2f, 3f, \ldots$	Laplace spectrum on S^1, T^d	159
Pythagorean comma $+1.364\,\%$	$K_{\text{frak}} = 1 - 100\xi$, $-1.333\,\%$	060, 133
5-limit ratios	Yukawa coefficients r_i	006, 060
Circle of fifths	torus topology, winding numbers	060, 189
Temperament	renormalisation / K_{frak}	060
Consonance $\leftrightarrow$ denominator height	Farey height Q_c, Arnold tongues	328, 343
Harmonic primes $5, 7, 11, 13$	entry depth $k = \text{ord}_p(3)$	342, 343
Chords	partitions of 8 in G(6)	342
7-limit as hearing threshold	Euler-factor threshold $p^* \approx 9.76$	343
Resolution limit $\approx 1\,\%$	100ξ as fractal step	343

What remains open in all of these documents is why this assignment holds at all — whether it is analogy or identity.

P.2 Theorem A: There is no comma in a finite field

> **Theorem A. [B]** Let $q = 3^k$. Every element of GF$(q)^*$ has finite order dividing $q - 1$. Every chain $g, g^2, g^3, \ldots$ generated by an element g returns exactly to 1. A comma — a remainder after an almost-closed cycle — is impossible in GF$(q)^*$. In $\mathbb{Q}^*$, by contrast, no chain of fifths can close: $3^a = 2^b$ has no solution $a, b > 0$.

Proof. The first part is Lagrange's theorem for the cyclic group GF$(q)^*$ **[E]**. The second is unique factorisation in $\mathbb{Z}$ **[E]**. □

The content lies not in the proof but in the consequence for the corpus. Doc. 060 places the Pythagorean comma $(3/2)^{12}/2^7$ and $K_{\text{frak}} = 1 - 100\xi$ side by side and finds the near-exact cancellation 1.00013. Theorem A says *where* both quantities live: in $\mathbb{Q}^*$ and $\mathbb{R}$, not in GF$(27)^*$. The Galois structure itself knows no remainder; all cycles close. The remainder appears only when the finite structure is mapped onto the number line — at the transition from winding numbers to measured masses in SI units.

This is exactly the position of Doc. 333: K_{frak} takes no part in the lossless pullback, sits in R_m, and becomes visible only at the SI transition. Theorem A supplies the algebraic reason. And it is the position of Doc. 343 Theorem G′: the $1.05\,\%$ are a fractal step of the embedding, not a property of the Galois grid.

Corollary A′ [K]: The cancellation comma $\cdot K_{\text{frak}} \approx 1$ of Doc. 060 is thus not a coincidence of two independent systems but two measurements of the same embedding error — once from the ratio $3/2$, once from the ratio $4/3$ (electron coefficient, Doc. 006). The residual 0.22 cent is the difference of the two starting ratios, not a third correction.

P.3 Theorem B: The exceptions of the 5-limit reading are Galois primes

Doc. 060 Table 2 classifies the nine Yukawa coefficients of Doc. 006: seven are 5-limit $(2, 3, 5)$, two are not — strange $26/9$ carries the prime 13, top $1/28$ carries the prime 7. Doc. 060 marks both as exceptions without musical interpretation.

Doc. 343 Theorem E shows independently: the prime p enters the GF(3^k) hierarchy exactly at $k = \text{ord}_p(3)$, and for $k \leq 6$ these are precisely $13\,(k = 3)$, $5\,(k = 4)$, $11\,(k = 5)$, $7\,(k = 6)$.

> **Theorem B. [B]** All nine Yukawa coefficients of Doc. 006 have prime factors exclusively in $\{2, 3, 5, 7, 11, 13\}$. The two non-5-limit coefficients carry exactly the primes of Galois depth $k = 3$ (strange: 13) and $k = 6$ (top: 7). The prime 11 ($k = 5$) occurs in no coefficient.

Proof. Direct factorisation of the nine fractions (script B1–B5). □

Three remarks:

1. The 5-limit reading of music is the Galois reading up to depth $k = 4$ (primes $2, 3, 5$; the prime 13 does enter at $k = 3$, but as the group order $|Z_{13}|$, not as a coefficient). What appears in music as "beyond the 5-limit" is algebraically the next layer. The exceptions of Doc. 060 are thereby resolved.
2. Strange carries the prime 13 — the same depth $k = 3$ that carries the lepton orbits (GF$(27)^* \cong Z_2 \times Z_{13}$, Doc. 339). Top carries 7 and needs $k = 6$, i.e. GF(729) — exactly the candidate that R115 names for the CKM/PMNS angles.
3. Whether the quark coefficients are geometric or calibrated remains open after Doc. 289 **[S]**. Theorem B says only: *if* they are geometric, they lie in the right grid. That is compatibility, not proof.

Falsifiable consequence [K]: A fundamental mass coefficient with a prime factor > 13 (or with 11) would be incompatible with Doc. 343 Theorem E. The prediction is testable on every coefficient derived in future.

P.4 Observation C: Three twelves

Three quantities of order 12 occur, from three different mechanisms:

1. **Temperament.** The 5-limit Tonnetz is the free lattice $\mathbb{Z}^2$ in the exponents of 3 and 5 (modulo octave). Twelve-tone temperament is the quotient by the syntonic comma $81/80 \triangleq (4, -1)$ and the diesis $128/125 \triangleq (0, -3)$:

$$\left|\mathbb{Z}^2 / \langle (4, -1), (0, -3) \rangle\right| = |\det| = 12.$$

The 12 arises by *quotient* of a torsion-free lattice **[E]**.

2. **Galois.** $|(\mathbb{Z}/13)^*| = 12$; the element 3 has order 3 in it (Doc. 343 Theorem E), the four cosets of $\langle 3\rangle$ are the four fermion sectors, their size 3 the three generations (Doc. 343 Theorem D, Doc. 348). Here the 12 is *intrinsic torsion*: the group is cyclic, no quotient needed.
3. **Resolution.** The Farey denominator height $Q_c = \pi/\sqrt{6\varepsilon}$ (Doc. 343 Plain Language II) gives, at comma resolution $\varepsilon \in \{1.36\,\%, 1.33\,\%, 1.05\,\%\}$, the values $Q_c \approx 11.0$, 11.1, 12.5. Denominators above ≈ 12 are indistinguishable at this resolution.

> **Observation C. [S]** The three twelves come from three different mechanisms (quotient, torsion, resolution limit). That they agree in order is an observation. A theorem mapping the three mechanisms onto one another is not available.

A hint that supports the observation without proving it: twelve-tone temperament is not convention but the smallest quotient at which $3^{12} \approx 2^{19}$ closes to comma accuracy — i.e. the smallest order at which the free lattice *imitates* the torsion of the finite field. In this sense temperament is an approximation of the Galois structure by the continuum, and Theorem A says what is lost: exact closure.

P.5 Theorem D: Why the same mathematics

The question whether the assignment in the table of §1 is analogy or identity has a precise answer in three steps, all established in the corpus:

1. $\tilde{T} \cdot m = 1$ (Doc. 003, 010): mass is inverse period. A mass ratio m_i/m_j is a period ratio $\tilde{T}_j/\tilde{T}_i$ — a frequency ratio. This is not a metaphor but the definition of the quantity.
2. Frequency ratios on a periodic structure are characters of the cyclic group S^1; on T^4 they are the winding lattice $\mathbb{Z}^4$ (Doc. 159, 189). The overtone series is the character spectrum of S^1; the mass spectrum is the character spectrum of T^4.
3. The reduction mod 3 (Doc. 336 ring homomorphism $\varphi(N) = N \bmod 3$, Doc. 341) maps the winding lattice to GF(3^k). The Z_3 orbifold action is a second, different $\mathbb{Z}_3$: the reduction $N \bmod 3$ is an additive triality, invariant under the cyclic permutation; the orbifold action fixes the sectors $\mathscr{H}_k$ through its eigenphase ω^k (Doc. 321, 336). The Frobenius action $x \mapsto x^3$ is the sector pairing $k \leftrightarrow -k$ (Doc. 336 main result).

> **Theorem D. [B]** Harmonic analysis of a mass spectrum and Galois classification in GF(3^k) are two representations of the same object — the character spectrum of T^4/Z_3. The first reads it in $\mathbb{Q}^* \subset \mathbb{R}$ (frequency ratios), the second in GF(3^k)* (orbits). The passage between them is reduction mod 3 in one direction and embedding into the continuum in the other; Theorem A describes what the embedding adds (the comma), Doc. 333 describes what the reduction removes (nothing — the pullback is lossless).

Proof. Concatenation of the three cited steps; no new proof step. □

The assignment in §1 is therefore not an analogy. It is the translation table between two representations.

P.6 What the connections yield

The question “what does this give us” has five answers, each with a concrete corpus return:

1. **An admissibility criterion instead of curve fitting.** Which primes may occur at all in a fundamental ratio is fixed algebraically (Doc. 343 Theorem E). Theorem B applies this to Doc. 006 and finds all nine coefficients in the grid. A fit would not have guaranteed this; the algebra forces it.

2. **A built-in resolution limit.** Harmonics has known for centuries that consonance ends at about the 7-limit. Doc. 343 Theorem F gives the number ($p^* \approx 9.76$), Theorem G′ the reason (100ξ). For mass spectroscopy this means: ratios with denominators $\gtrsim 12$ are indistinguishable from random products (Doc. 342 Theorem D). Harmonics supplies the intuition, algebra the limit, and both prevent over-interpretation — the look-elsewhere error.

3. **A reason for the existence of commas.** Theorem A: the comma is the price of embedding. K_{frak} is not an extra parameter but the necessary difference between exact closure in $\mathrm{GF}(27)^*$ and the number line. Doc. 060 found the agreement; Theorem A says why it is no coincidence.

4. **A physical reading of consonance.** Doc. 328 shows mixing angles as coupling degrees and Arnold tongues as stability regions. Consonance in this reading is not aesthetics but the width of an Arnold tongue — and conversely the mixing angles of the Standard Model are “intervals” with determinable consonance. This is the direction in which the CKM/PMNS values (R115, **[S]**) are to be approached: not as numbers but as denominator heights.

5. **A working intuition that holds.** The Euler Tonnetz was the practical precursor of ξ (Doc. 060, 009). Theorem D says why the precursor held: it was not an analogy but already the continuum representation of the object that the Galois representation later made exact. Whoever thinks in the Tonnetz thinks in T^4/Z_3 — with comma.

P.7 Scope

- This document derives no masses and changes no numerical value in the corpus.
- Observation C is marked **[S]** and remains so until a theorem connects the three mechanisms.
- The quark coefficients remain calibration values after Doc. 289; Theorem B establishes compatibility, not origin.
- The connection to auditory physiology (why the ear stops at the 7-limit) is not a subject here; Doc. 343 Theorem F treats the mathematical limit, not the biological one.

P.8 Summary

Thm.	Statement	Status
A	No comma in a finite field; comma = embedding error	**[B]**
A′	comma·$K_{\text{frak}} \approx 1$ = two measurements of one error	**[K]**
B	All nine Yukawa coefficients in the Galois grid; 13, 7 no exceptions	**[B]**
C	Three twelves (quotient / torsion / resolution)	**[S]**
D	Harmonics and Galois: two representations of one object	**[B]**

Verification scripts

`2/python/Dok358_Skripte/pruef_358_harmonik_algebra.py` — 22 assertions: A1 (GF$(3^k)^*$ explicit, $k = 1..6$), A2 ($\mathbb{Q}^*$), A3 (comma/K_{frak}), B1–B5 (prime factors of the nine coefficients), C1–C4 (three twelves), D1–D2 (limit thresholds).

Note on the use of AI assistance

AI assistance (Claude, Anthropic) was used for wording, LaTeX typesetting and the creation of the verification script. The substantive claims, the status assignment and the responsibility for the text lie with the author.

Chapter Q

Doc. 359 — AI-Based Pattern Recognition and Algebraic Limits of Frequency Analysis

Status markers

[B] *Proved* — algebraically proved.
[K] *Confirmed* — numerically verified.
[S] *Speculative* — open.
[E] *Established* — textbook or consensus knowledge.
[L] *Literature reference* — external primary evidence.
[Q] *Source* — external primary value.

Abstract

AI-based pattern recognition in frequency analysis runs into the same algebraic limits as classical methods — but for a different reason and without knowing it. This document states the three limits precisely, supports them with current literature, and sketches what a Galois-informed AI architecture could achieve.

Results:

1. Theorem A **[B]**: The standard HRV frequency bands (Task Force 1996) have band limits with exclusively 5-limit prime factors $\{2, 3, 5\}$ and thus lie in the Galois grid of Doc. 358 Theorem B. They are compatible with the grid, but not determined by it: rounded decimal values almost always lie in the 5-limit (Doc. 360).
2. Theorem B **[B]**: The resolution limit $100\xi \approx 1.33\,\%$ (Doc. 343) has the Farey height $Q_c \approx 11.1$. A classification with this tolerance requires a frequency resolution $\Delta f/f \leq 100\xi$.
3. Theorem C **[B]**: Group-equivariant networks (G-CNNs) for continuous groups (E(3), SO(3)) demonstrably reduce the number of parameters and improve generalisation. For finite Galois groups GF(3^k) this architecture does not yet exist — this is the identified gap.

4. Theorem D **[B]**: Whether the admissibility criterion of Doc. 358 Theorem B is learnable is decided by the hypothesis class, not by the amount of data. Without a structural assumption the PAC bound requires about 84 000 labelled examples; with the structural assumption “prime factors from a fixed set” at most 407 suffice, and from positive examples alone the prime set $\{2, 3, 5, 7, 11, 13\}$ is reliably recovered after 30 examples.
5. Architecture sketch **[S]**: A Galois-informed neural network uses $\mathrm{GF}(3^k)$ orbits as a hard constraint in the input layer. This reduces the parameter space and makes the admissibility criterion not learned but enforced.
   Verification script: `pruef_359_ki_grenzen.py` (18/18).

Q.1 Starting point: what AI does today

Physics-Informed Neural Networks (PINNs)

PINNs use known differential equations as a hard constraint in the loss function **[L]** [1]. They know nothing of algebraic group structure. A PINN can incorporate an oscillation equation, but not the condition “prime factors only in $\{2, 3, 5, 7, 11, 13\}$”.

Group-equivariant networks (G-CNNs)

G-CNNs build symmetry groups directly into the architecture, so that the output is equivariant under group operations **[L]** [2]. For continuous groups such as $\mathrm{E}(3)$ (rotation, translation) and $\mathrm{SO}(3)$, such architectures have been established since 2021 and show measurable improvements in molecular modelling and materials science.

For *finite* fields $\mathrm{GF}(3^k)$, a comparable architecture is entirely missing. This is the gap that this document identifies.

HRV analysis with AI

Multi-resolution HRV analysis with machine learning shows that different resolution levels yield different features **[L]** [3]. Current AI-based HRV systems use the Task Force bands VLF/LF/HF as a fixed input quantity, without knowing why these limits lie where they do **[L]** [4].

Q.2 Theorem A: HRV band limits in the Galois grid

Theorem A. [B] The standard HRV frequency bands **[L]** [5] have the following limits as simplest rational approximations:

$$f_{\mathrm{VLF,max}} = \tfrac{1}{25}\,\mathrm{Hz} = 0.04\,\mathrm{Hz} \qquad \text{prime factors: } \{5\} \qquad \text{(Q.1)}$$

$$f_{\mathrm{LF,max}} = \tfrac{3}{20}\,\mathrm{Hz} = 0.15\,\mathrm{Hz} \qquad \text{prime factors: } \{2, 3, 5\} \qquad \text{(Q.2)}$$

$$f_{\mathrm{HF,max}} = \tfrac{2}{5}\,\mathrm{Hz} = 0.40\,\mathrm{Hz} \qquad \text{prime factors: } \{2, 5\} \qquad \text{(Q.3)}$$

All prime factors lie in $\{2, 3, 5\}$ — the 5-limit grid, and at the same time in the Galois grid $\{2, 3, 5, 7, 11, 13\}$ of Doc. 358 Theorem B. The band ratios $f_{\mathrm{LF}}/f_{\mathrm{VLF}} = 15/4$ and $f_{\mathrm{HF}}/f_{\mathrm{LF}} = 8/3$ likewise lie in the 5-limit.

Proof. Direct prime factorisation (verification script A1–A5). □

Consequence for AI: Current HRV AI models use these band limits as free hyperparameters, learned from data or set manually **[L]** [4]. A Galois-informed network could adopt them as fixed constants. This is a setting, not an algebraic necessity: 8 of the 9 numbers with one significant digit and a third of all two-decimal values lie in the 5-limit, and in rad/s the limits are irrational (Doc. 360). As an "inductive bias" in the sense of learning theory **[L]** [8] the adoption is admissible, but not algebraically grounded.

Q.3 Theorem B: Resolution limit and Farey height

Theorem B. [B] The FFGFT resolution limit $100\xi \approx 1.33\,\%$ (Doc. 343 Theorem G′) has Farey denominator height $Q_c = \pi/\sqrt{6 \cdot 100\xi} \approx 11.1$. Frequency ratios with denominators $> Q_c$ cannot be separated in the Galois grid. The HRV band limits have denominators 5, 20, 25 — all in the separable range. A classification with the tolerance 100ξ is meaningful only if the frequency resolution satisfies $\Delta f/f \leq 100\xi$.

Practical consequence [K]: With $\Delta f = 1/T_w$ for the window length T_w (in Welch's method the segment length) this gives $T_w \geq 1/(100\xi\, f)$, i.e. 12.5 min at 0.10 Hz and 15 min at 0.083 Hz (Doc. 360). Shorter windows cannot distinguish grid elements. Whether the threshold appears empirically in AI-based HRV analyses is not established **[S]**.

Q.4 Theorem C: What group-equivariant networks achieve — and where the gap is

Theorem C. [B] A G-equivariant network with group G has, in a fully connected layer $\mathbb{R}^{|G|} \to \mathbb{R}^{|G|}$, only $|G|$ free parameters instead of $|G|^2$, because the weight matrix is completely determined by the group structure **[L]** [2]. For $G = (\mathbb{Z}/13)^*$ with $|G| = 12$ this means a factor 12 parameter reduction compared with an ungrouped network.

Proof. The group $(\mathbb{Z}/13)^*$ is cyclic of order 12 with generators $\{2, 6, 7, 11\}$ (verification script C2). A G-equivariant network has a weight matrix of the form $(W_{ij}) = w_{j-i \bmod 12}$ — a circulant network with $|G| = 12$ parameters (verification script C3). □

What exists for E(3) and SO(3) [L] [2]: Equivariant networks for continuous groups (rotations, translations) are standard in molecular modelling and materials science. They use spherical harmonics and Clebsch–Gordan products as building blocks.

What is missing: An analogous architecture for $\mathrm{GF}(3^k)^*$ — a finite, non-continuous group. The challenge: $\mathrm{GF}(3^k)^*$ has no natural embedding into a Euclidean space on which standard AI layers operate. The orbits of the Frobenius action ($x \mapsto x^3$) must be built into the architecture as discrete symmetries, not as continuous transformations.

Architecture sketch [S]: (Section Q.6)

Q.5 Theorem D: Learnability depends on the hypothesis class

> **Theorem D. [B]** On the bounded grid of reduced fractions p/q with $p, q \leq 100$, the admissibility criterion of Doc. 358 Theorem B (prime factors only in $\{2, 3, 5, 7, 11, 13\}$) is learnable. How many examples this requires is decided by the hypothesis class, that is, by the prior knowledge a learning system brings with it, and not by the amount of data alone.

Justification. The grid comprises 6087 reduced fractions, of which 1453 are admissible (verification scripts D1, D3). One hundred examples cover less than $7\,\%$ of the admissible ratios (D2). For a finite hypothesis class $\mathscr{H}$ in the realisable case, $m \geq \big(\ln|\mathscr{H}| + \ln(1/\delta)\big)/\varepsilon$ labelled examples suffice to reach an error below ε with probability $1-\delta$ **[L]** [7]. With $\varepsilon = \delta = 5\,\%$ this gives two very different cases.

Without a structural assumption, i.e. when any subset of the 6087 fractions may be the criterion, $|\mathscr{H}| = 2^{6087}$ and the VC dimension equals 6087. The bound requires about 84 000 labelled examples (D4), and from positive examples alone the criterion cannot be obtained at all, because every unseen ratio could equally well be admissible. This is the situation of a generic network without built-in prior knowledge.

With the structural assumption "admissible are exactly the fractions whose prime factors lie in a fixed set P", where P is a subset of the 25 primes up to 100, one has $|\mathscr{H}| = 2^{25}$, and at most 407 examples suffice (D5). In this class even a procedure that sees only positive examples learns: it takes as P the smallest prime set that explains all examples seen. For randomly drawn admissible ratios it recovers $\{2, 3, 5, 7, 11, 13\}$ after 10 examples in $94.6\,\%$ and after 30 examples in all of 1000 runs (D6). Such a system excludes $17/11$ without ever having seen a negative example, because the 17 occurs in no example.

Gold's theorem **[L]** [6] concerns identification of a language in the limit from infinite sequences of positive examples; it applies only when the hypothesis class is unbounded and yields no non-learnability for the bounded grid.

What this means: The knowledge that only the primes up to 13 occur does not reside in the amount of training data but in the form of the hypotheses. A network lacking this form needs more than two orders of magnitude more labelled examples, and with measurement noise above the resolution limit of Theorem B it does not recover the missing information at all.

Consequence: The admissibility criterion belongs in the architecture as a structural constraint, not as subsequent regularisation. Pointwise regularisation (ℓ_1, ℓ_2) favours small weights but no prime set, and therefore enforces no algebraic inadmissibility.

Q.6 Architecture sketch: Galois-informed neural network

The following sketch is speculative **[S]** and is meant to indicate the direction, not to describe a finished architecture.

Input layer: Galois projection

The input layer maps measured frequency ratios f_i/f_j onto the nearest rational number in the Galois grid:

$$\pi_G(x) = \arg\min_{p/q \in \mathcal{R}_G} |x - p/q|, \quad \mathcal{R}_G = \{p/q : \text{prime factors} \subseteq \{2,3,5,7,11,13\}\}$$

This is not a learning operation — it is an algebraic preprocessing step. Ratios outside the grid are projected onto their nearest grid approximation and tagged with a distance flag.

Middle layers: $(\mathbb{Z}/13)^*$ equivariance

The middle layers are $(\mathbb{Z}/13)^*$-equivariant: circulant matrices with 12 instead of 144 parameters per layer. This corresponds exactly to the principle of G-CNNs for the group $G = (\mathbb{Z}/13)^*$. The Frobenius action $x \mapsto x^3$ is the natural symmetry (Doc. 336, 341) and is built in as weight sharing.

Resolution filter: 100ξ threshold

Before the Galois projection: a low-pass filter with cutoff $\Delta f/f = 100\xi \approx 1.33\,\%$. Everything below this threshold is treated as embedding noise and is not classified. The threshold is a setting of FFGFT; an empirical counterpart in HRV data is not established (Doc. 360).

Stability assessment: Arnold tongue width

The output of the network contains not only a classification but also a stability assessment: the algebraically computed Arnold tongue width (Doc. 328) of the classified ratio. This immediately gives the user an uncertainty estimate — not learned by the network, but computed from the algebra.

What this achieves compared with standard AI

	Standard AI	Galois-informed
Admissibility criterion	learned (incomplete)	enforced (complete)
Resolution limit	not provided	set algebraically
Stability prediction	only for cases seen	for all cases (Arnold tongue)
Number of parameters	$\lvert G\rvert^2 = 144$	$\lvert G\rvert = 12$
Interpretability	limited	complete

Q.7 Application: HRV

For heart rate variability analysis, concrete consequences follow:

1. **Band limits as fixed constants:** VLF/LF/HF limits are not learned from data (Theorem A); they are compatible with the grid, but a setting. The search space of the network shrinks accordingly.
2. **Required resolution:** $\Delta f/f \leq 1.33\,\%$ (Theorem B), i.e. ≈ 0.0013 Hz at 0.10 Hz and a window length of at least 12.5 min.
3. **New biomarkers:** Instead of the LF/HF ratio (a single number), a Galois-informed network could output the *orbit type* of the dominant frequency ratio — an algebraic biomarker that indicates in which Galois layer ($k = 3, 4, 5, 6$) the heart is currently oscillating. This is new and has no counterpart in the literature **[S]**.
4. **Stability prediction [S]:** Whether a current HRV pattern is robust (wide Arnold tongue) or fragile (narrow tongue). The premise that HRV frequency ratios lock onto the grid is untested; the data in Doc. 360 do not support it, and a prospective test is described there.

Q.8 Scope

- This document implements no AI architecture and makes no clinical claims.
- The architecture sketch in §5 is speculative **[S]** — it outlines a direction, not a finished system.
- The algebraic orbit biomarker in §6 is a hypothesis; its physiological significance is open. The exploratory test on HRV data (Doc. 360) gave a negative result.
- Theorem D compares sample bounds on the bounded grid $p, q \leq 100$; over an unbounded grid with arbitrarily large primes the hypothesis class is infinite and the statements must be re-examined.

Q.9 Summary

Theorem	Statement	Status
A	HRV band limits in the 5-limit grid: compatible, not determined	**[B]**
B	Resolution limit $100\xi \approx 1.33\,\%$, $Q_c \approx 11.1$; classification requires $\Delta f/f \leq 100\xi$	**[B]**
C	G-CNNs for GF(3^k): parameter reduction 12x; architecture missing	**[B]**/**[S]**
D	Learnability of the admissibility criterion: ∼84 000 examples without structural assumption, ≤407 with prime-factor structure	**[B]**
Arch.	Galois-informed network: sketch	**[S]**

[1] Raissi, M., Perdikaris, P., Ahmadi, N., Karniadakis, G. E. (2024). *Physics-Informed Neural Networks and Extensions.* arXiv:2408.16806.

[2] Finzi, M. (2023). *Understanding and Incorporating Mathematical Inductive Biases in Neural Networks.* PhD Thesis, NYU. https://cs.nyu.edu/media/publications/Marc_Finzi_Thesis__12_.pdf

[3] Gupta, K., Sinha, G. R., Bhat, R., Saoji, A. A., & Manjunath, N. K. (2024). Multi-resolution assessment of heart rate variability signals during yogic and normal breathing using machine learning modules. *Applied Soft Computing*, 164, 112020. DOI: 10.1016/j.asoc.2024.112020.

[4] Zimatore, G. et al. (2026). HRV in Stress Monitoring by AI: A Scoping Review. *MDPI Applied Sciences*, 16(1):23. https://www.mdpi.com/2076-3417/16/1/23

[5] Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology (1996). Heart rate variability: standards of measurement, physiological interpretation and clinical use. *European Heart Journal*, 17, 354–381.

[6] Gold, E. M. (1967). Language identification in the limit. *Information and Control*, 10(5), 447–474.

[7] Shalev-Shwartz, S., & Ben-David, S. (2014). *Understanding Machine Learning: From Theory to Algorithms.* Cambridge University Press.

[8] ICLR Blog (2023). How does the inductive bias influence the generalization capability of neural networks? https://iclr-blogposts.github.io/2023/blog/2023/how-does-the-inductive-bias-influence-the-generalization-capability-of-neural-networks

[9] Lowet, E. et al. (2022). Tuning Neural Synchronization: The Role of Variable Oscillation Frequencies in Neural Circuits. *Frontiers in Systems Neuroscience*. https://doi.org/10.3389/fnsys.2022.908665

Verification scripts

`2/python/Dok359_Skripte/pruef_359_ki_grenzen.py` — 18 assertions: A1–A5 (HRV grid), B1–B4 (resolution), C1–C3 (group equivariance), D1–D6 (grid, coverage, PAC bounds, learning from positive examples).

Note on the use of AI assistance

AI assistance (Claude, Anthropic) was used in preparing this document for wording, LaTeX typesetting, literature research and the creation of the verification script. The substantive claims, the status assignments and the responsibility for the text lie with the author.

Chapter R

Doc. 360 — Galois-Informed HRV Analysis

Status markers

[B] *Proved* — algebraically proved.
[K] *Confirmed* — numerically verified.
[S] *Speculative* — open.
[L] *Literature evidence* — external primary source.

Abstract

Docs. 358 and 359 formulate an algebraic hypothesis for HRV frequency analysis: frequency ratios of coupled physiological oscillators should lie on a Galois grid, with the structural tolerance $100\xi \approx 1.33\,\%$. This document implements the approach as a Python module and assesses it exploratively on two public ECG datasets and on the author's own Polar H10 recordings. The result is negative: the present data do not support the hypothesis, and they cannot test it either, because LF/HF is a power ratio, not a frequency ratio.

Results:

1. **Module [K]**: `galois_hrv_preprocess.py` — Galois projection, HRV band limits, orbit depth, complexity measure $2/(p+q)$.
2. **Null model [K]**: In the grid up to denominator 60, every value between 1 and 4 lies within 1.33 % of a grid element; 39 % of random values lie within 0.06 %. Close grid hits are therefore no evidence on their own.
3. **BIDMC-01 [L]**: intensive-care patient, 88 y, 8 min. No ratio meets the 5-limit criterion; the dataset is unsuitable for the question.
4. **Fantasia [L]**: 5 young and 5 old healthy subjects, about 120 min each, spontaneous breathing. LF peak CV 11–33 %; 5-limit LF/HF 4/10 against a null rate of 21 % (binomial test $p = 0.14$).

5. **Polar H10 [K]**: With validated beat detection, the dominant spectral peak during paced breathing (4–6/min) lies within one frequency bin of the breathing frequency, and the HF peak is its second harmonic. Grid hits of LF/HF occur only at chance frequency.
6. **Structural reason [B]**: Under spontaneous breathing the HF peak is the respiratory frequency and LF/HF the ratio of two independent control loops; with paced breathing below 9/min, respiration itself lies in the LF band.
7. **Next step [S]**: prospective test of frequency coupling during paced breathing (Section R.7).

R.1 Galois HRV preprocessing module

`galois_hrv_preprocess.py` implements (Doc. 359 §5):

Galois grid 901 rational ratios $p/q \le 4$ with prime factors $\subseteq \{2,3,5,7,11,13\}$ up to denominator 60. The prime set is $\{3\} \cup \{p : p \mid 3^k - 1,\ k \le 6\}$; a prime $p \neq 3$ belongs to depth k when k is a multiple of $\mathrm{ord}_p(3)$.

Tolerance $100\xi = 1/75 \approx 1.33\,\%$ with Farey height $Q_c = \pi/\sqrt{6 \cdot 100\xi} \approx 11.1$ (Doc. 343): at this denominator the mean Farey spacing equals twice the tolerance. A grid up to denominator 60 is denser than the tolerance and does not separate.

HRV band limits VLF: $[0,\ \frac{1}{25}]$ Hz, LF: $[\frac{1}{25},\ \frac{3}{20}]$ Hz, HF: $[\frac{3}{20},\ \frac{2}{5}]$ Hz. The prime factors lie in $\{2,3,5\}$ **[B]**; the limits are thus compatible with the 5-limit grid, but not determined by it (Section R.2).

Galois projection ratio $\mapsto$ nearest p/q in the grid, distance $d = |r/(p/q) - 1|$.

Orbit depth $k = \max \mathrm{ord}_p(3)$ over the prime factors $p \notin \{2,3\}$ (Docs. 343, 358).

Complexity measure $2/(p+q)$ orders ratios by their simplicity (Doc. 328). It is not a computed Arnold tongue width; that would require the coupling strength of a specific oscillator model.

Pipeline `hrv_galois_pipeline(rr_ms)` — one function call.

Resolution condition. A classification with the tolerance $\delta = 100\xi$ is meaningful only if the frequency resolution satisfies $\Delta f / f \le \delta$. With $\Delta f = 1/T_w$ for the window length T_w (in Welch's method the segment length, not the recording duration) this gives $T_w \ge 1/(\delta f)$: 12.5 min at 0.10 Hz, 15 min at 0.083 Hz **[B]**. This is distinct from the minimum separation of two resolvable peaks (about $2/T_w$ for a Hann window) and from the precision of the position of a single peak, which interpolation can make finer than Δf.

Frequency and power ratios. The hypothesis concerns frequency ratios of coupled oscillators. LF/HF is a ratio of integrated band powers; there is no reason to expect it on the grid. In what follows, the projection of LF/HF serves only to show the behaviour of the null model.

R.2 Band limits and null model

Band limits [K]. The three Task Force limits have prime factors in $\{2, 3, 5\}$ in Hz, mHz, cycles per minute ($2.4 = 12/5$, 9, 24) and as periods (25 s, $20/3$ s, $5/2$ s), because the conversion factors 60 and 1000 contain only these primes. In rad/s they are irrational and cannot be classified. The property is not specific either: 8 of the 9 numbers with one significant digit and a third of all two-decimal values in $(0, 1)$ are 5-limit. Rounded conventions therefore land on the grid by themselves; the band limits are compatible, but not algebraically determined. Frequency ratios, by contrast, are dimensionless and independent of units.

Null model [K]. Probability that a log-uniformly distributed value in $[1, 4]$ lies within the tolerance τ of a grid element (200 000 draws):

τ	Grid up to denominator 60 (901)	Grid up to denominator 11 (133)
1.33 %	100.0 %	90.5 %
0.14 %	73.9 %	18.5 %
0.06 %	39.5 %	7.9 %
0.02 %	14.5 %	2.6 %

With the grid up to denominator 60, the criterion “within 1.33 %” cannot distinguish anything. Even the grid up to the Farey height 11 still covers 91 % at 1.33 %.

R.3 Negative result I: BIDMC-01

Dataset: BIDMC PPG and Respiration Dataset v1.0.0, record 01 **[L]** [Pim18]. 88 y, male, MICU, 8 min, 125 Hz PPG, ECG and respiration.

Result: 0/3 frequency ratios meet the 5-limit criterion (Welch and Lomb-Scargle). VLF cannot be resolved in 8 min. DFA $\alpha_1 = 0.50$, $\alpha_2 = 0.45$ — white noise; in healthy subjects ≈ 1.0.

Why:

1. Too short for VLF.
2. Intensive-care patient: fractal regulation largely lost.
3. HF peak = measured respiratory frequency (0.328 Hz versus 0.336 Hz in the RESP channel).
4. Measurement uncertainty Welch versus Lomb-Scargle $\pm 2\,\%$, larger than the tolerance of 1.33 %.

R.4 Negative result II: Fantasia

Dataset: PhysioNet Fantasia Database v1.0.0 **[L]** [Iye96]. 5 young (f1y01–05, 21–34 y) + 5 old (f2o01–05, 68–85 y), about 120 min rest each, 250 Hz ECG, spontaneous breathing. Reference beat annotations (.ecg), CC-BY 4.0.

Method: wfdb annotations, Welch 10-min windows, zero-padding $\times 4$, parabolic peak interpolation, LF stability over 6 $\times$ 5-min segments.

Rec	n_{RR}	LF Hz	CV %	LF/HF	p/q	k	5-lim	$2/(p+q)$
f1y01	8707	0.109	11.5	0.438	$7/16$	6	N	0.087
f1y02	6919	0.085	33.0	0.867	$13/15$	4	N	0.071
f1y03	7641	0.100	13.7	4.219	4	1	Y	0.400
f1y04	5232	0.078	30.9	1.060	$35/33$	6	N	0.029
f1y05	6943	0.068	16.5	2.993	3	1	Y	0.500
f2o01	7045	0.067	25.6	1.494	$3/2$	1	Y	0.400
f2o02	6327	0.052	14.4	0.864	$45/52$	4	N	0.021
f2o03	6478	0.055	22.0	0.841	$21/25$	6	N	0.043
f2o04	6849	0.065	32.2	0.719	$28/39$	6	N	0.030
f2o05	8334	0.049	11.4	1.873	$15/8$	4	Y	0.087

Statistical tests (Mann-Whitney U): All group differences young versus old n.s. ($p > 0.40$).

Findings:

- LF peak stability: 0/10 records stable (CV $> 1.33\,\%$).
- 5-limit LF/HF (nearest grid element has prime factors in $\{2, 3, 5\}$): 4/10. The null rate for this criterion is 21 %; one-sided binomial test $p = 0.14$, i.e. not above chance.
- No group difference young/old ($n = 5$ per group).

R.5 Polar H10: recordings and analysis

Device and protocol: Polar H10 ECG chest strap, firmware 4.2.0, JSONL export via Polar Sensor Recording (Android), the author as the only participant (75 y), 10 September 2026. One spontaneous recording supine (21.9 min, RR intervals computed on the device), six seated recordings with a metronome, 5.0–6.0 min each, 130 Hz ECG. Each click marks a change of respiratory phase; 4, 8, 10 and 12 clicks/min correspond to 2, 4, 5 and 6 breaths/min (0.033 to 0.100 Hz). The two files labelled 4/min are byte-identical; there is one recording at 4/min. Respiration was not measured with a separate sensor.

Analysis [K]: R peaks with NeuroKit2 **[L]** [Mak21], artefact criterion $|\mathrm{RR}/\mathrm{median} - 1| > 0.30$; flagged intervals are removed and the remaining beats stay at their true time stamps. Cubic interpolation to 4 Hz, linear detrending, Hann periodogram over the whole recording ($\Delta f = 1/T \approx 0.003$ Hz), band powers by trapezoidal integration over LF $[0.04,\ 0.15)$ and HF $[0.15,\ 0.40)$ Hz.

Beat detection and time axis. The simple energy detector in `polar_h10_atemfrequenz_scan.py` (fixed threshold at the 90th percentile) found 216–285 beats per recording, the validated detector 396–443. In every paced recording 35–47 % of the beats were therefore missing. When the remaining intervals are joined end to end, the time axis shrinks to 49–61 % of the recording duration, and all frequencies shift upwards by a factor of 1.6–2.0. The table is based on the validated detection with the time axis preserved.

Condition	f_R Hz	T min	Beats	RMSSD ms	LF peak	HF peak	LF/HF	nearest p/q
Spontaneous, supine	—	21.9	1517 RR	19.0	0.074	0.225	1.65	$33/20$ (0.09 %)
2/min, rec. 1	0.033	5.4	431	34.2	0.067	0.200	7.44	—
2/min, rec. 2	0.033	5.1	411	49.6	0.067	0.167	4.24	—
4/min	0.067	5.3	396	67.0	0.067	0.358	4.61	—
5/min, rec. 1	0.083	5.6	431	64.6	0.083	0.167	3.28	$105/32$ (0.006 %)
5/min, rec. 2	0.083	5.1	400	68.5	0.083	0.167	2.47	$121/49$ (0.05 %)
6/min	0.100	6.0	443	60.9	0.100	0.201	4.20	—

Frequencies in Hz. —: LF/HF above 4 (outside the grid) or more than 1.33 % from the nearest element. Artefacts: 0–1 each.

Finding 1 — peak at the breathing frequency [K]: At 4, 5 and 6/min the dominant peak lies within one frequency bin of the metronome frequency. Respiration therefore lies in the LF band: the LF peak *is* the respiratory peak. Where the second harmonic lies above 0.15 Hz (5 and 6/min), it is the HF peak.

Finding 2 — resonance breathing [K]: RMSSD is 61–69 ms at 4–6/min and 34–50 ms at 2/min. This matches the known amplification of HRV by slow breathing near the baroreflex resonance **[L]** [Leh14, Sev22]. The spontaneous recording was made supine and is not directly comparable with the seated recordings.

Finding 3 — 2/min [S]: The dominant peak lies at 0.067 Hz, twice the intended breathing frequency, with a smaller component at 0.033 Hz. Without a separate respiration signal it cannot be decided whether this is the second harmonic of very slow breathing or breathing at 4/min because of the click convention.

Finding 4 — grid hits at chance level [K]: The two recordings at 5/min lie close to two *different* grid elements ($105/32$ and $121/49$), although the condition is the same; their LF/HF differs by 28 %. In 17 sliding 300 s windows of the spontaneous recording, LF/HF varies between 0.89 and 7.43 (CV 78 %); of the 13 windows inside the grid range $[1, 4]$, 6 lie within 0.06 % of an element, against a null rate of 39 % (binomial test $p = 0.78$). The spread of the estimate is more than three orders of magnitude larger than the distances to the grid.

Spontaneous recording [K]: mean heart rate 69 /min, RMSSD 19.0 ms, SDNN 48.9 ms. A single missed beat (1741 ms, twice the median) raises RMSSD to 36.3 ms without artefact handling. Across the 300 s windows the LF peak varies between 0.040 and 0.085 Hz (CV 22 %), far above 1.33 %. LF/HF is 1.65 with one window over the whole recording and 2.21 with 300 s segments; a single estimator setting changes the value by a third.

Methodological lessons:

1. Check beat detection against the expected beat count (heart rate $\times$ duration) and a validated detector.
2. Preserve the time axis: remove rejected intervals, keep the remaining beats at their time stamps, never join them end to end.
3. A single artefact can dominate RMSSD; artefact handling is advisable for recordings of any duration.
4. Below 9 breaths/min respiration lies in the LF band; LF/HF then has no interpretation as sympathovagal balance.
5. $n = 1$ and recordings of 5–6 min allow no statistical conclusions; all Polar findings are exploratory.

R.6 Structural reason and consequence

Finding. [B] Under spontaneous breathing the HF peak is the respiratory frequency (variable, 0.15–0.40 Hz). LF/HF is the ratio of two independent physiological control loops (baroreflex / respiration) and does not converge to a fixed ratio. With paced breathing the metronome sets the frequency; peak and harmonics then stand in trivial integer ratios. In both cases LF/HF is not a test object of the Galois hypothesis.

R.7 Next step: prospective test

The hypothesis makes a statement about frequency coupling. This can be tested when the breathing frequency is set deliberately relative to the individual baroreflex resonance f_{res} **[S]**:

- Resonance assessment: paced breathing at 4.5–7/min in steps of 0.5/min, 2 min each **[L]** [Leh14].
- Four blocks of 21 min (resolution condition at the lowest breathing frequency of about 0.062 Hz) at $\rho = f_R / f_{\text{res}} \in \{2/3,\ 3/2\}$ (grid) and $\{0.618,\ 1.618\}$ (golden mean), matched in pairs by breathing rate, in random order.
- Primary outcome: $n{:}m$ phase-synchronisation index **[L]** [Tas98] between respiratory phase (respiratory belt) and the HRV component at f_{res}; contrast grid minus golden mean, paired two-sided t-test.
- Sample size: $d_z = 0.6$, power 0.80 $\Rightarrow n = 24$; with 20 % drop-out, 30 participants.
- Falsification: if coupling is not stronger at grid ratios than at the golden mean, the complexity measure $2/(p+q)$ has no basis in HRV.

A positive result supports the premise but does not yet distinguish the prime set and the Farey height of FFGFT from general mode-locking of coupled oscillators; secondary outcomes serve this purpose (measured frequency ratio against the grid up to denominator 11, peak stability against 1.33 %).

R.8 Summary

Result	Statement	Status
Module	`galois_hrv_preprocess.py` works	**[K]**
Band limits	compatible with the 5-limit, not determined by it	**[B]**
Null model	close grid hits of LF/HF are no evidence	**[K]**
BIDMC-01	negative: too short, intensive-care patient	**[L]**
Fantasia	negative: spontaneous breathing, LF CV 11–33 %, 4/10 at chance level	**[L]**
Polar H10	peak at the breathing frequency, HF = harmonic, hits at chance	**[K]**
Reason	LF/HF is a power ratio; respiration determines the peaks	**[B]**
Next	prospective coupling test, $n = 30$, blocks of 21 min	**[S]**

References

[Pim18] Pimentel, M., Johnson, A., Charlton, P., & Clifton, D. (2018). BIDMC PPG and Respiration Dataset (v1.0.0). PhysioNet. https://doi.org/10.13026/C2208R

[Iye96] Iyengar, N., Peng, C. K., Morin, R., Goldberger, A. L., & Lipsitz, L. A. (1996). Age-related alterations in the fractal scaling of cardiac interbeat interval dynamics. *Am. J. Physiol. Regul. Integr. Comp. Physiol.* 271(4), R1078–R1084. Fantasia Database, PhysioNet, https://doi.org/10.13026/C2RG61

[Leh14] Lehrer, P. M., & Gevirtz, R. (2014). Heart rate variability biofeedback: How and why does it work? *Front. Psychol.* 5, 756.

[Sev22] Sevoz-Couche, C., & Laborde, S. (2022). Heart rate variability and slow-paced breathing: When coherence meets resonance. *Neurosci. Biobehav. Rev.* 135, 104576.

[Mak21] Makowski, D. et al. (2021). NeuroKit2: A Python toolbox for neurophysiological signal processing. *Behav. Res. Methods* 53, 1689–1696.

[Tas98] Tass, P. et al. (1998). Detection of $n{:}m$ phase locking from noisy data. *Phys. Rev. Lett.* 81, 3291–3294.

[TF96] Task Force ESC/NASPE (1996). Heart rate variability. *Eur. Heart J.* 17, 354–381.

Scripts

`2/python/Dok360_Skripte/`: `galois_hrv_preprocess.py`, `fantasia_galois_analyse.py`, `orbit_experiment_synthetic.py`, `01_rpeaks_bidmc01.py`, `02_welch_5limit.py`, `03_lombscargle_dfa.py`, `pruef_galois_nullmodell.py`. Polar analysis, null model and all numbers of Sections 2 and 5: `SWT_Revision/pruef_swt_revision.py` (25/25 assertions). Fantasia data: f1y01–05, f2o01–05 (.ecg/.hea, CC-BY 4.0).

Note on the use of AI assistance

AI assistance (Claude, Anthropic) was used in preparing this document for wording, LaTeX typesetting and analysis scripts. The substantive statements and the responsibility for the text rest with the author.

Chapter S

Doc. 361 — The Kagome Lattice and FFGFT: Four Structural Parallels

Abstract

Pei, Chen, Castelnovo and Moessner (PRL 137, 106702, 31 August 2026, Open Access) study the singly doped infinite-U Hubbard model on the kagome lattice. They find resonating-valence-bond (RVB) polarons, disorder-free self-trapping with band mass increasing by six orders of magnitude, and a crossover to $\sqrt{3} \times \sqrt{3}$ order in the unpolarised limit, which they benchmark against the three-state Potts model. This document identifies four structural parallels to FFGFT and supports each with the wording of the source. The parallels share one algebraic root: the Frobenius automorphism $\phi : x \mapsto x^3$ of order 3 on finite fields. A separate section states what does not *correspond. Status of all four parallels:* ***[S]*** *(structural correspondence established, physical identification open).*

Status Markers

[B]	Algebraically proved.
[K]	Numerically confirmed.
[S]	Structurally plausible; physical identification open.
[Q]	Taken from the source (Pei et al.), quoted verbatim.
[ASSUMPTION]	Axiomatic postulate.

All verbatim quotations are from Pei et al. (2026), Open Access under CC-BY; section references follow the structure of that paper.

S.1 The Source: What Pei et al. Show

Model and question

The model is the infinite-U Hubbard model on the kagome lattice with a single hole:

> **[Q]**"To gain deeper insight into the role of hole kinetics in determining magnetism in highly frustrated doped Mott insulators, we consider the single-hole counter-Nagaoka problem on the kagome lattice, using magnetization as a tuning parameter." (Abstract)

The term *counter-Nagaoka* matters: Nagaoka's theorem predicts ferromagnetism from the kinetic energy of a hole on bipartite lattices. On the kagome lattice the kinetics is *frustrated*, and antiferromagnetic correlations appear instead:

> **[Q]**"When said kinetic energy is frustrated, it was later shown that antiferromagnetic correlations can also appear." (Introduction)

The physics of a single triangle

The mechanism is visible on a single triangle. Pei et al. describe it:

> **[Q]**"Ferromagnetic spins have lowest energy $E_0 = -t$ due to destructive interference of the hole motion around the triangle; whereas antiferromagnetic spins can form a singlet and effectively thread a π flux through the triangle, relieving the frustration and allowing the hole to delocalize and lower the energy to $E_0 = -2t$ (bottom of an unfrustrated 1D tight-binding band)." (Section *Model*)

Two points to retain. First: the ferromagnetic configuration is *destructively interfered* — the hole cannot move freely. Second: the singlet *threads a π flux* through the triangle. This π flux is not a metaphor but a Berry phase: the hole accumulates phase π on one full circuit, and the interference switches from destructive to constructive.

Flat band and compact localised states

In the fully polarised case the problem reduces to spinless fermions. The lowest band is then exactly flat:

> **[Q]**"The fully polarized case reduces to a well-known spinless fermionic tight-binding model, where the bottom band is exactly flat (energy -2) and the states are compact localized around individual hexagons." (Section *RVB polarons and delocalization*)

Kinetic frustration arises precisely when this flat band is lowest:

> **[Q]**"Indeed, kinetic frustration arises when the kagome flat band happens to be lowest in energy for a hole-doped fully polarized electron system close to half-filling. As such, holes in a fully polarized system occupy compact localized states at low energy." (Introduction)

RVB polarons: formation, delocalisation, self-trapping

Adding a few reversed spins ($n = 1, \dots, 4$) forms polarons that *delocalise* — but with strongly enhanced band mass:

> **[Q]**"Upon adding few reversed spins, polarons form and delocalize, albeit with a band mass 100 times larger than that of an unfrustrated tight-binding particle on the same lattice." (Introduction)

The structure of these polarons is described by valence-bond singlets on tree-like Husimi cacti:

> **[Q]**"The short distance physics of the kagome lattice is close to tree-like, and we find that the valence bond language of the same Hamiltonian on the Husimi cactus proves highly useful to model and understand the polarons in this Letter." (Section *RVB polarons and delocalization*)

At $n = 5$ the behaviour changes abruptly:

> **[Q]**"The polaron bandwidth suddenly drops by at least six orders of magnitude ($< 10^{-8}$ in a 4×4 system of 48 sites, with 16 near-degenerate GSs); to the best of our numerics it is indistinguishable from a flat band of localized states." (Section *Strong self-trapping*)

The mechanism of this self-trapping is a resonance of six symmetry-related clusters around a hexagonal plaquette:

> **[Q]**"[...] there are six symmetry-related such clusters around any hexagonal plaquette of the lattice, and the corresponding six Husimi cactus states are not orthogonal. [...] the GS wave function takes the form of a symmetric superposition of all six states, spreading across all 12 sites around the hexagonal plaquette." (Section *Strong self-trapping*)

And the interpretation:

> **[Q]**"By resonating around a hexagonal plaquette, it seems that the 1H5S RVB polaron acquires a possibly infinite mass, undergoing localization reminiscent to the compact-localized hexagonal states of the spinless single-particle problem." (Section *Strong self-trapping*)

Notably, this localisation occurs in a *disorder-free* system: the authors speak of states that "relocalize in a disorder-free system" (Conclusions).

The unpolarised limit and the Potts model

As n grows, the valence-bond language loses its usefulness and semiclassical correlations emerge:

> **[Q]**"We see that this 'frustration on frustration' leads to the appearance of magnetic correlations consistent with semiclassical $\sqrt{3} \times \sqrt{3}$ order as we approach the unpolarized limit." (Section *Unpolarized state*)

The benchmark is the three-state Potts model:

> **[Q]**"The data presented in Fig. 5 include our Letter of the three-state Potts model, with Hamiltonian $H_{\text{Potts}} = \sum_{\langle ij \rangle} \delta_{\sigma_i \sigma_j}$, $\sigma_i = 0, 1, 2$. The ground state ensemble is extensively degenerate and consists of all states without identical nearest neighbors." (Appendix *Potts model*)

This model is not chosen arbitrarily — it marks a phase transition:

> **[Q]**"This model is useful in that it sits at the Kosterlitz-Thouless (KT) transition separating a $\sqrt{3} \times \sqrt{3}$ ordered and an algebraically correlated phase, and it can thus provide a natural benchmark for an analysis of semiclassical correlations." (Section *Unpolarized state*)

S.2 Parallel 1: Frobenius Orbits — the Same Algebraic Structure

FFGFT side

[B] (Doc. 339): The multiplicative group $GF(27)^* \cong \mathbb{Z}_{26}$ decomposes under the Frobenius automorphism $\phi : x \mapsto x^3$ into

- two fixed points $\{+1, -1\}$ — the massive $\mathbb{Z}_2$ sector (particle/antiparticle),
- four three-element orbits in $\mathbb{Z}_{13}$, each $\{a, 3a, 9a \pmod{13}\}$: $\{1, 3, 9\}$, $\{2, 5, 6\}$, $\{4, 10, 12\}$, $\{7, 8, 11\}$,
- combined with the two $\mathbb{Z}_2$ parities: eight gluons of the adjoint $SU(3)$ representation.

The three-element orbits correspond to the three colour states $\{R, G, B\}$; the parity distinguishes colour from anticolour (Doc. 339, §3.2).

Kagome side

The kagome lattice has three sublattice sites per unit cell; the unit cell itself is a triangle. Pei et al. explicitly use system sizes that are multiples of 3:

> **[Q]**"$\sqrt{3} \times \sqrt{3}$ order requires L_x and L_y to be integer multiples of 3 to avoid incommensurability issues." (Appendix *Methods*)

The order itself shows at the K points of the Brillouin zone:

> **[Q]**"In all cases, we find peaks at the K points, suggestive of $\sqrt{3} \times \sqrt{3}$ order." (Section *Unpolarized state*)

The K points carry threefold rotational symmetry (C_3); the structure factor $S(\mathbf{k})$ is invariant under rotation by $2\pi/3$.

Structural correspondence

Both sides carry the same algebra: a cyclic group on which an automorphism of order 3 acts and generates three-element orbits. In FFGFT it is ϕ on $GF(27)^*$; on the kagome lattice it is C_3 on the sublattice sites and — in reciprocal space — on the K points. **[S]**

What is *not* claimed: that the four $\mathbb{Z}_{13}$ orbits are physically identical to four kagome configurations. The kagome lattice has a single C_3 orbit per unit cell; FFGFT has four orbits in $\mathbb{Z}_{13}$. The correspondence lies in the *order* of the acting automorphism, not in the number of orbits.

S.3 Parallel 2: Three-State Potts Model = $GF(3)$ = Fixed Field

FFGFT side

[B] (Doc. 339, §3.3): The fixed field of ϕ on $GF(27)$ is $GF(3) = \{0, +1, -1\}$ (Fermat's little theorem: $x^3 = x$ for all $x \in GF(3)$). In FFGFT, $GF(3)$ corresponds to the $U(1)$ sector: the projection $GF(27) \to GF(3)$ describes the electrically neutral sector (photon). The three elements $\{0, +1, -1\}$ are also the three $\mathbb{Z}_3$ sectors of Doc. 336 (fixed points $\{0, +1, -1\}$ = FFGFT sectors **[B]**).

[B] (Doc. 321, §1.2): The triality condition $T_R = 0 \pmod 3$ is equivalent to confinement:

> "Confinement is equivalent to requiring all observable states to have triality $T_R = 0$." (Doc. 321)

A single quark ($T_R = 1$) is not observable; a meson ($T_R = 1 + 2 = 0$) and a baryon ($T_R = 1 + 1 + 1 = 0$) are.

Kagome side

The Potts benchmark has exactly three states $\sigma \in \{0, 1, 2\}$ with $\mathbb{Z}_3$ symmetry (quoted above). The ground-state condition "no identical nearest neighbors" is a frustration condition: on a triangle all three states must occur — exactly one $\mathbb{Z}_3$-neutral configuration per triangle. The KT transition separates the phase in which this $\mathbb{Z}_3$ order is long-ranged ($\sqrt{3} \times \sqrt{3}$) from the algebraically correlated phase.

Structural correspondence

The identification $\sigma \in \{0, 1, 2\} \leftrightarrow GF(3) = \{0, +1, -1\}$ is algebraically exact: both are $\mathbb{Z}_3$ as an additive group. The Potts frustration condition (all three states on every triangle) and the FFGFT confinement condition (sum of trialities $\equiv 0 \pmod 3$) are the same statement: only $\mathbb{Z}_3$-neutral configurations are admissible. **[S]**

The Potts KT transition therefore has an FFGFT counterpart: the transition between confined ($T_R = 0$ enforced at long range) and deconfined (triality only algebraically correlated). Whether this identification holds physically beyond the algebra is open (§7).

S.4 Parallel 3: Mass as Frozen Kinetic Energy

FFGFT side

[B] (Doc. 286): The duality $\tilde{T} \cdot m = 1$ enforces two energy types:

	Static sector	Kinetic sector
Form	$E_0 = mc^2$ (rest mass)	$E = hf$ (free)
Seat	$\mathbb{Z}_3$ core (winding)	window
Time	$\tilde{T} = 1/m$	$\tilde{T} = 1/\omega$
Status	derived (Doc. 285)	open (Doc. 286)

Doc. 286 states: "The rest mass is static: a topologically stable winding number on T^4 [...] *Mass is frozen kinetic energy.*" **[B]** (Doc. 339): The massive/massless separation is algebraically forced by the Frobenius — fixed points $\{+1, -1\}$ are massive, the three-element orbits are massless (gluons).

Kagome side

The kagome system shows *the same dichotomy*, but not as a sector separation — as a *non-monotonic band mass* along the polarisation axis:

- $n = 0$: flat band, compact localised — "effectively infinite mass" (kinetic energy frozen by destructive interference).
- $n = 1, \ldots, 4$: delocalisation with band mass $\times 100$ ("polarons form and delocalize").
- $n = 5$: relocalisation, band mass $\times 10^6$ ("acquires a possibly infinite mass").
- $n \to$ unpolarised: delocalisation ("delocalized, dispersive holes are recovered").

Pei et al. summarise:

> **[Q]**"Our Letter uncovers an intricate interplay between single-particle flat bands and cooperative 'polaronic' self-trapping physics." (Introduction)

Structural correspondence

In both systems, *mass* results from geometric freezing of kinetic freedom: in FFGFT through topological winding on T^4 (Doc. 314: "winding numbers are topological"), on the kagome lattice through destructive interference around plaquettes. Both are localisation *without disorder* — a purely geometric effect. **[S]**

The difference: FFGFT has two *sectors* (massive/massless); the kagome system has one *parameter* (polarisation) along which band mass switches between the two extremes. The kagome system thus realises the transition that FFGFT treats as a sector boundary.

S.5 Parallel 4: π-Flux = Berry Phase = Winding Number

Kagome side

The quotation from §1.2 is central here: the singlet "effectively thread[s] a π flux through the triangle". The π flux is the phase the hole picks up on one circuit of the triangle. Without it (ferromagnetic) the hopping amplitude interferes destructively; with it, constructively. The energy gain is a factor 2 ($E_0 = -t \to -2t$).

The same logic repeats at larger scale in the self-trapping: the six Husimi states around the plaquette interfere *symmetrically* — constructive interference for the bound component, spreading the state over all twelve sites, but destructive for any motion *out* of the plaquette.

FFGFT side

[B] (Doc. 314, Ch. D2): "in the rolled-up state there are no fractal corrections — winding numbers are topological, the defect $d = 100\xi$ accumulates only when unrolled." Winding numbers on T^4 are quantised Berry phases around closed loops: phase $2\pi n$, $n \in \mathbb{Z}$, on one circuit of a torus cycle.

[B] (Doc. 321, §2.1): The $\mathbb{Z}_3$ generator τ acts as a unitary operator on $L^2(T^4)$ with $\tau^3 = \mathrm{id}$; eigenvalues are ω^k, $\omega = e^{2\pi i/3}$. The three sectors P_k are distinguished by the phase accumulated under a $\mathbb{Z}_3$ rotation.

[B] (Doc. 230): Entanglement is "holonomy around cycles" — the phase accumulated on a closed circuit is what constitutes entanglement geometrically.

Structural correspondence

The π flux in the kagome triangle (Berry phase π, half winding) and the $\mathbb{Z}_3$ phases ω^k in FFGFT (Berry phase $2\pi k/3$) are the same operation: phase accumulated along a closed path that changes the accessible Hilbert space. On the kagome lattice it lifts the frustration; in FFGFT it selects the sector. **[S]**

S.6 What Does Not Correspond

Honesty requires naming the limits of the parallels.

Dimension. The kagome lattice is two-dimensional; T^4/Z_3 is four-dimensional. The C_3 symmetry of the kagome lattice lives in the plane; the $\mathbb{Z}_3$ action on T^4 has nine fixed points (Doc. 314) and acts on a compact space without boundary. There is no map embedding the kagome lattice into T^4/Z_3.

Classical versus quantum. The Potts model is classical; Pei et al. use it as a *semi-classical* reference and scale correlators by $(S^z)^2 = 1/4$ instead of $S(S+1) = 3/4$. FFGFT works on $L^2(T^4) \otimes \mathbb{C}^3$ (Docs. 230/282) fully quantum-mechanically. The correspondence in Parallel 2 is algebraic ($\mathbb{Z}_3$), not dynamical.

Number of orbits. FFGFT has four three-element orbits in $\mathbb{Z}_{13}$ (eight gluons); the kagome lattice has one C_3 orbit per unit cell. Parallel 1 concerns the *order* of the automorphism, not the orbit count.

Structure versus dynamics. All four parallels are structural. None predicts that an FFGFT computation reproduces the kagome spectrum or vice versa. They state: the same algebra (ϕ of order 3) generates the same kind of constraint in both systems (only $\mathbb{Z}_3$-neutral configurations admissible; localisation without disorder; Berry phase as selector).

One unexplained number. The self-trapping cluster spans twelve sites around a hexagonal plaquette. In Doc. 314 the first excitation of the D_4 lattice has degeneracy $24 = 12 + 12$ ("FCC half and winding half separately addressable"). Whether the 12 of the kagome cluster relates to one of the 12-halves of D_4 is *not* investigated and is not listed here as a parallel. **[S]**

S.7 Summary and Open Questions

Kagome system (Pei et al.)	FFGFT	Shared algebra	Status
K-point order, C_3 on sublattice	Frobenius orbits on $GF(27)^*$ (Doc. 339)	automorphism of order 3	**[S]**
Three-state Potts, KT transition	$GF(3)$ fixed field, triality $T_R = 0$ (Docs. 339, 321)	$\mathbb{Z}_3$ neutrality	**[S]**
Flat band / self-trapping, band mass $\times 10^6$	static/kinetic, frozen winding (Docs. 286, 314)	localisation without disorder	**[S]**
π flux through triangle	winding number, $\mathbb{Z}_3$ phase ω^k (Docs. 314, 321)	Berry phase around loop	**[S]**

All four parallels are structurally established **[S]**. The physical question — whether the kagome system is a low-energy realisation of the FFGFT Galois structure — remains open and is not claimed here.

Open questions [S].

1. Is the Potts KT transition (separation of $\mathbb{Z}_3$-ordered / algebraically correlated) identifiable with the FFGFT confinement transition ($T_R = 0$ enforced / merely correlated) at finite temperature?
2. Pei et al. announce for two holes "polaron-mediated interactions [...] and possible binding, pairing". Two holes each with $T_R \neq 0$ could form a $T_R = 0$ state in FFGFT language. Is hole-pair binding on the kagome lattice a meson analogue?
3. The twelve sites of the self-trapping cluster versus $24 = 12 + 12$ from Doc. 314 — coincidence or structure?

Source. Pei, Y.; Chen, S. A.; Castelnovo, C.; Moessner, R.: *Kinetic Kagome Magnetism: From Self-Trapping RVB Polarons to Semiclassical Correlations*. Phys. Rev. Lett. **137**, 106702, published 31 August 2026. DOI: 10.1103/zbxv-vtq8. Open Access (Editors' Suggestion).

Chapter T

Doc. 362 — The BAW Ising Machine as a Resonance Computer

Abstract

*Vadde, Ovcharov, González, Khymyn, Litvinenko and Åkerman (Commun. Phys. 9:290, 8 September 2026, Open Access) present a time-multiplexed Ising machine with 2048 spins on two quartz bulk acoustic wave delay lines. This document reads the work as a case study of the FFGFT classification of resonance computers (Docs. 173, 328, 343): spins as phase states of a ring oscillator, the Barkhausen condition as winding quantisation, parametric $\mathbb{Z}_2$ selection, a coupling regime with a chaos threshold, and — most clearly in quantitative terms — a resolution floor: 99.9 % approximate solutions are found reliably, exact solutions ($E = 0$) are not as problem size grows. All quotations verbatim **[Q]**. Status of the classifications: **[S]**.*

Status Markers

[B]	Algebraically proved.
[K]	Numerically confirmed.
[S]	Structurally plausible; physical identification open.
[Q]	Taken from the source (Vadde et al.), quoted verbatim.
[ASSUMPTION]	Axiomatic postulate.

All verbatim quotations are from Vadde et al. (2026), Open Access under CC-BY 4.0; section references follow the structure of that paper.

T.1 The Source: What Vadde et al. Build

Architecture

> **[Q]** "We use two serially connected 20.5 MHz, 707 μs bulk acoustic wave delay lines supporting 2048 spins. Our design provides all-to-all connectivity

with 15-bit coupling resolution and finds approximate MAX-CUT solutions in 341 ms." (Abstract)

The spins are not particles but phases of circulating RF pulses:

[Q]"The spin states $+1$ and -1 are encoded in the relative phase of the pulses, with bistable phases achieved by phase-binarizing the propagating RF pulses using a phase-sensitive amplifier (PSA)." (Section *Design*)

The ring oscillator and the Barkhausen condition

[Q]"The BAW delay line, together with linear amplifiers and the PSA, forms a multi-physics ring oscillator. The key step of the design is to establish a stable circulation of RF pulses within this loop. This requires satisfying the Barkhausen stability criteria: the loop gain must be larger than one, and the total phase shift around the loop must be an integer multiple of 2π." (Section *Design*)

Parametric binarisation

[Q]"The PSA further narrows the stability criteria to only those signals with either phase 0 or π relative to the pumping signal that is twice the reference frequency." (Section *Design*)

Minimum cycles and pulse duration

[Q]"Due to the resonance mechanism underlying phase-sensitive amplification, a minimum of 5–10 cycles per pulse is required, setting a lower bound on the pulse duration to $\sim$250–500 ns. This constraint dominates and effectively determines the pulse period." (Section *Design*)

Coupling strength and chaos threshold

[Q]"The amplitude of the coupling signal can be tuned using the potentiometer of the DAC, allowing an additional control to ensure the injected signal remains within 5–30 % of the circulating RF signal and does not overpower it, avoiding chaotization of the oscillatory circuit." (Section *Design*)

[Q]"The performance of BAWIM depends on the coupling strength, and there exists an optimal coupling strength depending on the problem instance." (Section *Performance*)

Approximation versus exactness

MAX-CUT:

> **[Q]**"Here, we observe that BAWIM attains up to 95 % of HbSB's best performance in terms of Ising energy, and up to 99.99 % in terms of the MAX-CUT score." (Section *Dependence on the edge density*)

Number partitioning:

> **[Q]**"Here, the exact solution corresponds to a difference of 0, while the 99.9 % approximate solution indicates that the difference is less than 0.1 % of the maximum possible value [...]. In all cases, BAWIM outperforms the HbSB algorithm, achieving a 100 % success rate in most instances when considering the 99.9 % approximate solution." (Section *Number partitioning*)

Figure 5(b) shows what the text does not spell out: the curve "BAWIM Exact" falls from about 90 % success at 32 numbers to near zero at 2048 numbers, while "BAWIM 99.9 %" stays at 100 % across the whole range.

Sudoku:

> **[Q]**"Notably, even when the Ising energy approaches the ground state, the Sudoku solution remains far from correct. [...] Sudoku requires exact constraint satisfaction; therefore, a near-ground-state Ising energy may still correspond to an invalid solution." (Section *Sudoku*)

Thermal stability

> **[Q]**"In contrast, our BAWIM exhibits a temperature coefficient of phase accumulation of only 780 deg/°C, which is 2.21×10^4 times lower than that of the CIM." (Section *Comparison with CIMs*)

T.2 Classification 1: Ising Machine = Resonator (Doc. 173)

Doc. 173 §*The Arc Closes* says of the coherent Ising machine: "This resonator also has discrete states: it oscillates in either one or the other phase — spin-up or spin-down, zero or one. The excitation is analogue — continuous laser light, continuously adjustable coupling strengths. The state is discrete." And in the conclusion: "An Ising machine is a resonator. All discrete from inside. All analogue from outside."

BAWIM is this description in its purest form — no photonics, no cryogenics. The states are phases 0 and π; the excitation is a continuous RF carrier; the coupling is an analogue current with 15-bit resolution. The spin is not an object but a phase relation. **[S]**

T.3 Classification 2: Barkhausen as Winding Quantisation (Doc. 314)

The Barkhausen condition "integer multiple of 2π" is a winding number: only states whose accumulated phase around the closed loop is $2\pi n$, $n \in \mathbb{Z}$, circulate stably. **[B]** (Doc. 314, Ch. D2): "winding numbers are topological"; the mode structure on T^4 is fixed by integer winding around every cycle.

The ring oscillator is an S^1 — a single cycle. The Barkhausen condition is winding quantisation on that S^1. BAWIM thus realises technically what FFGFT posits for T^4: states exist only on integer windings. **[S]**

T.4 Classification 3: Parametric $\mathbb{Z}_2$ Selection (Doc. 339)

The PSA is pumped at 2ω and passes only phases 0 and π. This is period doubling: the phase state space S^1 is reduced to two points $\{0, \pi\} \cong \{+1, -1\}$.

[B] (Doc. 339, §3.1): The fixed points $\{+1, -1\}$ of the Frobenius on $GF(27)^*$ form the massive $\mathbb{Z}_2$ sector. **[B]** (Doc. 336): The norm classes $\{+1, -1\}$ separate light from heavy.

BAWIM's parametric selection is a technical realisation of the same dichotomy: from a continuous phase space exactly two fixed points are filtered out. That there are *two* and not three is due to pumping at 2ω; pumping at 3ω would yield a $\mathbb{Z}_3$ selection $\{0, 2\pi/3, 4\pi/3\}$ — the Potts machine that Whitehead et al. (Ref. 12 in Vadde et al.) built in CMOS. **[S]**

T.5 Classification 4: Coupling Regime with Chaos Threshold (Doc. 328)

Doc. 328 treats coupling regimes of coupled oscillators: the Adler model, synchronisation, phase locking, and the limit beyond which coupling destroys the oscillator. BAWIM sets exactly this limit operationally: injection at 5–30 % of the circulating signal, "avoiding chaotization".

The existence of an *optimal* coupling strength per problem instance (Suppl. Note 5) is the Adler sweet spot: too weak — no synchronisation, the system stays in random states; too strong — chaotisation, phase binarisation breaks down. In between lies the window in which the oscillator bank settles into the energy minimum. **[S]**

T.6 Classification 5: The Resolution Floor (Doc. 343 §G′)

This is the classification with the clearest quantitative support. Doc. 343 §G′ states: a resonance-based computer is capped at $N_{\max} \approx 1/\varepsilon$, where ε is the coupling tolerance; it finds minima to tolerance but not the exact zero. **[B]**

BAWIM displays this behaviour in three forms:

MAX-CUT. 95 % of Ising energy, 99.99 % of cut value. The optimum is reached to within fractions of a percent — but not exactly.

Number partitioning. The difference $E = |\sum_A a_i - \sum_B a_i|$ is brought below $0.1\,\%$ of the total sum with 100 % success. The *exact* partition $E = 0$, by contrast, is still found in ∼90 % of runs at 32 numbers and practically never at 2048 numbers (Fig. 5b). Coupling resolution is 15 bit; the integers are ≤ 178; the total sum at 2048 numbers is $\sim 1.8 \times 10^5$. A tolerance of 0.1 % corresponds to ~ 180 — the magnitude of a single number. The machine solves the problem to within one element, not to zero.

Sudoku. At 99.42 % of ground-state energy the solution is still invalid. Sudoku is a constraint-satisfaction problem: there is no "almost right". Vadde et al. say so themselves: "a near-ground-state Ising energy may still correspond to an invalid solution." That BAWIM nevertheless solves the Sudoku is due to the smallness of the problem (729 spins) and to the ground-state energy being an isolated minimum here; HbSB fails on the same problem.

The classification. All three findings are consistent with Doc. 343: the resonance computer is an analogue computer with a resolution floor. It is *not* worse than a digital algorithm — on approximations it is better (NPP 99.9 %: BAWIM beats HbSB throughout). It is *different*: it finds the neighbourhood of the minimum, not the point. For optimisation that is the right task; for exact arithmetic it is not. **[S]**

T.7 What Does Not Correspond

Classical, not quantum. BAWIM is a classical RF ring oscillator at room temperature. It claims no quantum advantage and asserts none. The FFGFT classifications concern resonator structure, not quantum coherence.

One cycle, not four. The ring oscillator is S^1; T^4 has four independent cycles. The winding parallel (Classification 2) holds for one cycle; the T^4-specific statements (fixed points, $\mathbb{Z}_3$ triality, mass assignment) have no counterpart in BAWIM.

Benchmark caveat. Vadde et al. stress twice that no parameter optimisation was performed for either HbSB or BAWIM. The quantitative comparisons (341 ms versus 102 ms; NPP success rates) are therefore snapshots. The *qualitative* structure — approximation reliable, exactness scale-dependent — is unaffected.

The resolution floor here is technical, not fundamental. The 15-bit coupling resolution is an electronics choice (AD9708 DAC), not a law of nature. Doc. 343 §G′ speaks of $\varepsilon = \xi$ as the natural floor; BAWIM's $2^{-15} \approx 3 \times 10^{-5}$ happens to lie in the same order of magnitude as $\xi = 1.33 \times 10^{-4}$, without any connection being claimed here.

T.8 Summary

BAWIM (Vadde et al.)	FFGFT	Doc.	Status
spin = phase 0/π of a pulse	Ising machine = resonator	173	**[S]**
Barkhausen: phase = $2\pi n$	winding quantisation on S^1	314	**[S]**
PSA at 2ω: $\{0, \pi\}$	$\mathbb{Z}_2$ fixed points $\{+1, -1\}$	339, 336	**[S]**
coupling 5–30 %, chaos limit	Adler regime, coupling threshold	328	**[S]**
99.9 % yes, exact no	resolution floor $N_{\max} \approx 1/\varepsilon$	343 §G′	**[S]**

BAWIM is the clearest published case to date of a resonance computer in the sense of Docs. 173/343: thermally stable, without post-selection, without cryogenics, with reproducible approximate solutions and a measurable resolution floor for exact ones. It confirms the FFGFT classification not as a theory but as a taxonomy: what resonates finds minima — and finds them to tolerance.

Source. Vadde, V.; Ovcharov, R.; González, V. H.; Khymyn, R.; Litvinenko, A.; Åkerman, J.: *A 2048-spin bulk acoustic wave Ising machine for number partitioning and Sudoku.* Commun. Phys. **9**, 290 (2026). DOI: 10.1038/s42005-026-02846-7. Open Access, CC-BY 4.0. Received 2 March 2026, accepted 20 August 2026, published 8 September 2026.

Chapter U

Doc. 363 — Hodge Theory on T^4/Z_3 within FFGFT

Abstract

The second Millennium Problem of mathematics — the Hodge Conjecture — asks whether every Hodge class of a smooth projective variety is a rational linear combination of algebraic cycle classes. For the base geometry of FFGFT, the orbifold $T^4/\mathbb{Z}_3$, this question admits a positive answer: the complex 2-torus T^4 is polarisable in the form $E_\omega \times E_\omega$ and hence an abelian surface for which the Hodge Conjecture is known to hold; the nine $\mathbb{Z}_3$ fixed points furnish algebraic 0-cycles; and the sector pairing $k \leftrightarrow -k$ (Doc. 343) is the Hodge symmetry $H^{p,q} \leftrightarrow H^{q,p}$. The Galois Frobenius $x \mapsto x^3$ on $GF(27)^*$, by contrast, is not identical to the algebraic Frobenius on étale cohomology: the latter's eigenvalues on H^1 have absolute value $\sqrt{3}$ and are not roots of unity. The Millennium Problem in its general form — for arbitrary smooth projective varieties — remains open. What FFGFT supplies is structural completeness for its own geometry.

Status Markers

[B]	Algebraically proved.
[K]	Numerically confirmed.
[S]	Structurally plausible; physical identification open.
[E]	Established external mathematics.
[Q]	Taken from the literature, correctly cited.
[SETZUNG]	Axiomatic stipulation.

U.1 The Millennium Problem

The Clay Mathematics Institute states the Hodge Conjecture as follows **[Q]**: On a non-singular projective algebraic variety X over $\mathbb{C}$, any Hodge class — that is, any cohomology

class of type (k,k) — is a rational linear combination of classes of algebraic cycles of codimension k.

More precisely: the Hodge decomposition

$$H^n(X,\mathbb{C}) = \bigoplus_{p+q=n} H^{p,q}(X)$$

splits cohomology by type; Hodge classes are the elements of $H^{k,k}(X) \cap H^{2k}(X,\mathbb{Q})$. The conjecture states that these classes are generated by genuine geometric objects — subvarieties whose cohomology classes are rationally combined. For smooth projective curves and surfaces in bidegree $(1,1)$ this follows from the Lefschetz theorem on $(1,1)$-classes **[E]**. In general dimension the conjecture is open.

U.2 The Complex 2-Torus as an Abelian Surface

The real geometry $T^4 = S^1 \times S^1 \times S^1 \times S^1$ carries in FFGFT the structure of a *complex 2-torus*: two complex coordinate pairs $(z_1, z_2) \in \mathbb{C}^2/\Lambda$, where Λ is the mode lattice **[SETZUNG]** (Doc. 001).

A polarisable complex 2-torus – here $E_\omega \times E_\omega$ – is an abelian surface, an abelian variety of complex dimension 2. By the Riemann conditions $\mathbb{C}^2/\Lambda$ is projective only with a polarisation; the generic 2-torus is not algebraic, whereas $E_\omega \times E_\omega$ as a product of elliptic curves is. For abelian varieties the Hodge Conjecture is known in the cases relevant here **[E] [Q]**:

- For abelian surfaces (complex dimension 2) the conjecture holds completely for all Hodge classes: $H^{1,1}$ is covered by the Lefschetz theorem, $H^{2,2}$ is generated by the point class.
- Products of elliptic curves and tori over $\mathbb{Q}$ are fully spanned by algebraic cycles.

[B] *Theorem A:* T^4 as the abelian surface $E_\omega \times E_\omega$ satisfies the Hodge Conjecture: all (k,k)-classes are rational linear combinations of algebraic cycles.

Proof: In complex dimension 2, (k,k)-classes exist only for $k = 0, 1, 2$ (Hodge diamond $h^{p,q} = \binom{2}{p}\binom{2}{q}$, $h^{1,1} = 4$). For $k = 0$ and $k = 2$: trivial via the fundamental class ($H^{0,0}$) and the point class ($H^{2,2}$), respectively. For $k = 1$: Lefschetz theorem on $(1,1)$-classes (known **[E]**). $H^{2,2}(T^4)$ is moreover generated by products of $(1,1)$-classes, hence algebraic. □

U.3 The Nine Z_3 Fixed Points as Algebraic 0-Cycles

Under the $\mathbb{Z}_3$ action on T^4 there are exactly nine fixed points (Doc. 001/330) **[B]**. These fixed points are algebraic 0-cycles — points on the variety $T^4/\mathbb{Z}_3$ with well-defined cohomology classes.

[B] *Theorem B:* The nine $\mathbb{Z}_3$ fixed points of $T^4/\mathbb{Z}_3$ are algebraic 0-cycles of codimension 4. Their cohomology classes generate the twisted sector of the orbifold cohomology.

In the orbifold cohomology (Chen–Ruan) **[Q]**:

$$H^*_{\mathrm{CR}}(T^4/\mathbb{Z}_3) = H^*(T^4)^{\mathbb{Z}_3} \oplus \bigoplus_{\text{9 fixed points}} H^*(\{p\})$$

The second summand is algebraic: every point is an algebraic 0-cycle; its cohomology class is trivially algebraic. **[E]**

Under the Witt-pair structure of Doc. 341:

- **Even indices 0,2,4,6** $=\mathbb{Z}_3$ fixed points = algebraic 0-cycles **[B]**
- **Odd indices 1,3,5,7,8** = three-cycles = higher-dimensional algebraic cycles **[B]**

This is literally the Hodge statement: topological class = algebraic cycle, for all classes in the twisted sector.

U.4 Frobenius Automorphism and Algebraic Frobenius

This section separates the Galois Frobenius on $GF(27)^*$ from the algebraic Frobenius of étale cohomology.

The Galois Frobenius on GF(27)∗

Doc. 339 proves **[B]**: the Frobenius automorphism on $GF(27) = GF(3^3)$ is $x \mapsto x^3$. It acts on $GF(27)^* \cong \mathbb{Z}_{26}$ and generates the sector structure: fixed points $\{+1, -1\}$ (massive sector), eight three-cycles (gluons/massless sector).

The Algebraic Frobenius on Étale Cohomology

In the arithmetic geometry of algebraic geometry **[Q]**, for a variety X over $GF(q)$ the geometric Frobenius is $F_q : x \mapsto x^q$ on points. It induces an endomorphism F_q^* on the ℓ-adic cohomology $H^i_{\text{et}}(X, \mathbb{Q}_\ell)$. The eigenvalues of F_q^* are algebraic numbers that, by the Weil–Deligne theorem **[E]**, determine the zeta function and the Betti numbers of X (weights: $|\alpha| = q^{i/2}$ on H^i). The Hodge numbers $h^{p,q}$ are not determined by them.

Theorem C (no identification): The Galois Frobenius $x \mapsto x^3$ on $GF(27)^*$ is not identical to the algebraic Frobenius on H^1_{et} of $T^4/\mathbb{Z}_3$ over $GF(3)$. The obvious identification fails because, by Weil, the Frobenius eigenvalues on H^1 of an abelian surface over $\mathbb{F}_q$ have absolute value $\sqrt{q}$, here $\sqrt{3} \approx 1.732$: they are four Weil numbers, not roots of unity and not 26 of them. Example: $y^2 = x^3 - x$ over $GF(3)$ has $\#E = 4$ and eigenvalues $\pm i\sqrt{3}$ (cross-check $\#E(GF(27)) = 28$); for $E \times E$ there are four eigenvalues with $\alpha^{26} = -3^{13}$. An identification with the elements of $GF(27)^*$ does not follow. The two share only the map $x \mapsto x^3$ as generator of $\text{Gal}(GF(27)/GF(3))$.

[S] *Remark:* The physical interpretation — Frobenius orbits as particle sectors — is an FFGFT classification and not part of the mathematical proof.

U.5 Sector Pairing as Hodge Symmetry

Doc. 343, Theorem D **[B]** proves: the sector pairing $k \leftrightarrow -k \pmod{13}$ on the primitive $GF(27)^*$ classes pairs $f_1 \leftrightarrow f_2$ and $f_3 \leftrightarrow f_4$.

[B] *Theorem D:* This sector pairing is the algebraic realisation of the Hodge symmetry $H^{p,q} \leftrightarrow H^{q,p}$ under complex conjugation.

Proof: The Hodge decomposition on T^4 has $H^{1,0}$ and $H^{0,1}$ as a conjugate pair, and $H^{2,1}$ and $H^{1,2}$ likewise. Complex conjugation $\bar{\cdot} : H^{p,q} \to H^{q,p}$ is the algebraic datum selecting real cohomology classes. On $GF(27)^*$ the corresponding map is inversion $g^k \mapsto g^{-k}$, i.e. negation of the exponent in $\mathbb{Z}_{26}$. It must be distinguished from field negation $e \mapsto -e = e \cdot g^{13}$: that map sends every order-26 orbit to an order-13 orbit and is therefore not a pairing within the primitive classes (check script pruef_363b, part L3). Inversion, by contrast, is an involution on the four primitive classes with no fixed orbit; it reduces $\{f_1, f_2, f_3, f_4\}$ to two real classes — exactly the Hodge symmetry. □

From Doc. 343 Theorem D‴ **[B]**: $\{f_3, f_4\}$ is the Dirac layer (charged leptons), $\{f_1, f_2\}$ the Majorana layer (neutrinos). The Hodge pairing separates exactly the physical sectors.

U.6 The χ-Classes as Algebraic Basis

Doc. 342 **[B]** classifies the eight irreducible cubic polynomials over $GF(3)$:

- Four primitive polynomials of order 26: f_1, f_2, f_3, f_4
- Four polynomials of order 13: g_1, g_2, g_3, g_4

[B] *Theorem E:* The eight polynomials generate the factorisation classes of $GF(3)[x]/(x^{26} - 1)$ completely (Doc. 342).

A reading of these classes as a $\mathbb{Q}$-basis of algebraic cycles on $T^4/\mathbb{Z}_3$ is not established thereby: it would require the irreducible factors of the characteristic polynomial of F_3^* on H^i to be the eight polynomials of Doc. 342, i.e. the identification of Theorem C, which does not hold.

U.7 The Hodge Conjecture for T^4/Z_3

[B] *Main Theorem:* All Hodge classes on $T^4/\mathbb{Z}_3$ are rational linear combinations of algebraic cycles.

Proof from Theorems A and B:

1. $H^*(T^4)^{\mathbb{Z}_3}$ (untwisted sector): All Hodge classes come from the $\mathbb{Z}_3$-invariant part of $H^*(T^4)$. Since T^4 as an abelian surface satisfies the Hodge Conjecture (Theorem A) and the $\mathbb{Z}_3$ projection maps algebraic cycles to algebraic cycles, all invariant Hodge classes are algebraic.
2. Twisted sector: The nine fixed points are algebraic 0-cycles (Theorem B); their cohomology classes are trivially algebraic.
3. By the Chen–Ruan decomposition (Section 3), orbifold cohomology consists of exactly these two summands; no Hodge class lies outside them. The Frobenius structure is not needed for this. □

In plain language: For the specific geometry of FFGFT the Hodge Conjecture is not an open problem. It follows from combining the abelian-surface theorem and orbifold cohomology. The Millennium Problem in its general form is not solved by this.

U.8 What FFGFT Does Not Supply

General varieties. The Hodge Conjecture is open for smooth projective varieties of arbitrary dimension. $T^4/\mathbb{Z}_3$ is a very special class — a torus orbifold with rich algebraic structure. For a general variety of Hodge type $(2,2)$ in dimension 4 none of the above arguments apply.

No singularities in the FFGFT reading. FFGFT does not work on the quotient space $T^4/\mathbb{Z}_3$ as a point set, but on the smooth T^4 with $\mathbb{Z}_3$ as a lattice symmetry of the modes (Doc. 314: geometry → spectrum + fibre; the orbifold is a lattice symmetry, not an additional assumption). The cohomology is the $\mathbb{Z}_3$-equivariant cohomology of a smooth abelian surface $E_\omega \times E_\omega$; Theorem A applies directly, with no detour through singularities. **[B]**

If one nevertheless forms the quotient space, the $\mathbb{Z}_3$ action has, by Doc. 330, eigenvalues $\{\omega, \omega^2\}$ on $\mathbb{C}^2$, hence $\det = 1$: it lies in $SU(2)$. The nine fixed points are then A_2 points and admit a crepant resolution; the result is a K3 surface — smooth, projective, and for K3 surfaces the Hodge Conjecture is known (Lefschetz on $H^{1,1}$ suffices in dimension 2) **[E]**. In neither reading is there an obstruction.

Premise and level of representation. Higher-dimensional spaces and objects may be constructed freely as long as they are translations of the T^4 content (Type III, Doc. 270/330 **[K]**): they contain no more than T^4, and no projection from them can yield more than the mode content of T^4 (Doc. 314: lattice = index set of Fourier modes; Doc. 343: the resolution limit lies in the object itself). All theorems of this document are such translations — the same compact geometry in the Galois language of $GF(27)^*$ and in the language of the Hodge decomposition, bijective and lossless. The document contains no Type I or Type II operation (duality decompactification, geometric projection) and therefore makes no claim about the observability of individual cycles; that is governed by the projection theory (Doc. 285, 330, 333).

Non-commutative extension. Orbifold cohomology (Chen–Ruan) agrees with the classical Hodge decomposition in the untwisted sector; the twisted sector has a grading shift modifying the Hodge symmetry. The statement here holds for the classical picture; the Chen–Ruan extension is to be rated **[S]**.

U.9 Summary

FFGFT structure	Hodge correspondence	Doc.	Status
T^4 as abelian 2-torus	Hodge conjecture known	001	**[E]**
9 $\mathbb{Z}_3$ fixed points	Algebraic 0-cycles	001/330	**[B]**
Frobenius $x \mapsto x^3$ on $GF(27)^*$	No identification with the Frobenius on H^1_{et} (eigenvalues of absolute value $\sqrt{3}$)	339/341	—
Sector pairing $k \leftrightarrow -k$	Hodge symmetry $H^{p,q} \leftrightarrow H^{q,p}$	343	**[B]**
χ-classes (8 polynomials)	Factorisation classes of $GF(3)[x]/(x^{26}-1)$; no cycle basis	342	**[B]**
Hodge conjecture for $T^4/\mathbb{Z}_3$	All Hodge classes algebraic	—	**[B]**
General Hodge conjecture	Millennium Problem open	—	—

The Hodge Conjecture is not an open problem for $T^4/\mathbb{Z}_3$. It follows from Theorem A and orbifold cohomology. Of the Galois structure of $GF(27)^*$, the sector pairing realises the Hodge symmetry (Theorem D); the Galois Frobenius and the χ-classes, by contrast, cannot be identified with the Frobenius or the cycles of étale cohomology (Theorem C). The general Millennium Problem is unaffected.

Chapter V

Doc. 364 — The Icosahedral Carrier and the D_4 Lattice

Abstract

An externally proposed carrier model for emergent physics uses a twelve-port icosahedron with symmetry group A_5 and a local averaging repair rule on its thirty edges. This document compares that carrier exactly with the FFGFT geometry $T^4/\mathbb{Z}_3$ on the D_4 lattice. Four verification scripts establish: (i) the carrier spectrum lies in $\mathbb{Q}(\sqrt{5})$, and the Galois conjugation $\sqrt{5} \mapsto -\sqrt{5}$ exchanges the two three-dimensional A_5 blocks $3 \leftrightarrow 3'$ as spectral projectors, so that over $\mathbb{Q}$ only the 6D block exists – the doubling $6 = 3 \oplus 3'$ of Dok. 285; (ii) the reduction $\mathbb{Z}[\varphi]/(3) = GF(9)$ turns this Galois conjugation into the Frobenius $x \mapsto x^3$ of the FFGFT ground field; (iii) A_5 acts faithfully and absolutely irreducibly on $GF(9)^3$ but not on $GF(9)^2$, hence not in $G(3)$ but in $G(6)$ – the same threshold as the lepton order 26 in Dok. 341; (iv) A_5 is incompatible with the D_4 lattice ($5 \nmid 1152$, Dok. 314); the common core is the tetrahedral group A_4, which acts simply transitively on the twelve ports, and the port $\mathbb{Z}_3$ is provably never one of the sixteen orbifold generators of $T^4/\mathbb{Z}_3$. The gauge assignment of the external model ($u(1) \oplus su(2) \oplus su(3)$ on $1 \oplus 3 \oplus (3' \oplus 5)$) is representation-theoretically consistent; its Lie structure is an axiom there, not a consequence of the icosahedron.

List of symbols

The table explains symbols appearing in more than one section. Quantities used only locally are introduced there.

Symbol	Meaning
Graph and carrier	
n	number of ports; $n = 12$ (icosahedron)
A	adjacency matrix; $A_{ij} = 1$ if seam (i, j)
$L = 5I - A$	Laplacian; 5-regular graph
$W = A/5$	averaging operator (expected single-seam update)
μ	Laplacian eigenvalue; $\mu \in \{0,\ 5{-}\sqrt{5},\ 6,\ 5{+}\sqrt{5}\}$
Groups and representations	
A_5	icosahedral rotation group; $\lvert A_5\rvert = 60$
A, T_1, T_2, H	irreducible A_5 representations; dimensions $1, 3, 3, 5$
$\varphi = (1{+}\sqrt{5})/2$	golden ratio
A_4	tetrahedral group; integral elements of A_5; $\lvert A_4\rvert = 12$
$\mathrm{Aut}(D_4)$	automorphism group of the D_4 lattice; $\lvert\mathrm{Aut}(D_4)\rvert = 1152$
$W(F_4)$	Weyl group of F_4; isomorphic to $\mathrm{Aut}(D_4)$
Galois and fields	
$\mathbb{Q}(\sqrt{5})$	number field; generated by the carrier spectrum
$GF(3^k)$	finite field of characteristic 3
$\sigma : \sqrt{5} \mapsto -\sqrt{5}$	Galois automorphism of $\mathbb{Q}(\sqrt{5})$
$x \mapsto x^3$	Frobenius automorphism of $GF(9)$
ζ_n	primitive n-th root of unity
D_4 *lattice*	
D_4	root lattice; $\{k \in \mathbb{Z}^4 : \sum k_i \equiv 0 \pmod 2\}$; kissing number 24
$P_{T_1}, P_{T_2}, P_A, P_H$	spectral projectors of L

Status markers

[B] Algebraically proven (verification script, exact arithmetic).
[K] Numerically confirmed.
[S] Structurally plausible, physical identification open.
[E] Established external mathematics.
[X] Checked and negative.

This document contains no new mathematics. The character table of A_5, the Galois theory of $\mathbb{Q}(\sqrt{5})$, Brauer degrees, $\mathrm{Aut}(D_4)$, the crystallographic restriction and cut-and-project are textbook material **[E]**. The marker **[B]** here denotes the exact verification of such results on the concrete carrier and their assembly; only the application to the relation between the two frameworks is new.

V.1 Orientation: geometry, algebra, and what a field is here

FFGFT has two layers that must not be confused.

The geometry is a torus. The carrier is T^4 with the D_4 lattice (kissing number 24, Dok. 314), and the physics sits in the *modes* of this torus: the winding numbers of the plane waves living on it. The built-in $\mathbb{Z}_3$ symmetry of the lattice turns it into the orbifold $T^4/\mathbb{Z}_3$ with nine fixed points. What a particle is, which mass it carries, which sector it belongs to – these are statements about modes and their $\mathbb{Z}_3$ orbits.

The algebra consists of finite fields: $GF(9)$ as ground field, $GF(27)$ with its 26 units in eight three-orbits, both sitting in parallel inside $GF(729)$ (Dok. 338, 341). The Frobenius $x \mapsto x^3$ is the Galois group of these fields; the sector pairing $k \leftrightarrow -k$ is the inversion $g^k \mapsto g^{-k}$.

How the layers connect. The fields *are* not the modes. They are the eigenvalue fields, splitting fields and coefficient fields in which the mode operators live, and they decide which modes *can exist*: a mode is an element order in a matrix algebra over $GF(3^k)$, and Lagrange permits only orders dividing the group order $|GF(3^k)^*|$. That is why order 5 exists in $G(3)$ ($5 \mid 80$) while order 26 appears only in $G(6)$ ($13 \nmid 80$, Dok. 341). In short: *the geometry supplies the modes; the Galois structure of the fields says which of them are admissible.* The fields are mode-conditioned in exactly this sense – they are called up by the modes, not the other way round.

Why this organises the comparison. OPH has the same two-layer structure with different entries: the geometry is a sphere of icosahedra, the modes are the blocks of the port space under A_5, and the algebra is the number field $\mathbb{Q}(\sqrt{5})$ with its Galois group $\sqrt{5} \mapsto -\sqrt{5}$. There too the Galois structure decides which modes exist over the ground field: over $\mathbb{Q}$ only the 6D block, never a single triplet. The comparison in this document is therefore a comparison of the *algebras that classify the modes* – and the two algebras touch exactly once: in the reduction modulo 3, which maps $\mathbb{Q}(\sqrt{5})$ onto $GF(9)$ and the Galois conjugation onto the Frobenius (Theorem V.5.2). The geometries, by contrast, touch only in the tetrahedral group and part at the five-fold symmetry (Section V.6).

V.2 Occasion and scope

In September 2026 a carrier model was presented on the IPI list (Observer Patch Holography, hereafter *OPH*) whose hardware is a finite network of twelve-port icosahedra and whose software is a single local rule: on each of the thirty edges (*seams*) the two port readings are replaced by their arithmetic mean. The author supplied a reproducibility bundle in exact rational arithmetic (300 updates, seeds 7 and 11, an independent verifier).

This document takes no position on the physical claims of OPH. It answers three narrow, exactly checkable questions:

1. What is the structure of the carrier spectrum, and what does its Galois group imply for the distinguishability of the two three-dimensional blocks?
2. How does $\mathbb{Q}(\sqrt{5})$ relate to the FFGFT ground field $GF(9)$?
3. Which part of the icosahedral group A_5 is compatible with the D_4 lattice and the orbifold $T^4/\mathbb{Z}_3$?

The FFGFT reference points are the doubling $6 = 3 \oplus 3'$ (Dok. 285), the embedding $C_3 < A_5$ (Dok. 293), the negative result $5 \nmid |\mathrm{Aut}(D_4)|$ (Dok. 314) and the $G(3)/G(6)$ threshold (Dok. 341).

V.3 The spectrum of the icosahedral carrier

Theorem V.3.1 (Spectrum, **[E]/[B]**)**.** *Let A be the adjacency matrix of the icosahedral graph (12 vertices, 30 edges, 5-regular) and $L = 5I - A$ its Laplacian. Then*

$$\mathrm{spec}(A) = \{5^{(1)},\ \sqrt{5}^{(3)},\ -1^{(5)},\ -\sqrt{5}^{(3)}\}, \qquad \mathrm{spec}(L) = \{0^{(1)},\ (5-\sqrt{5})^{(3)},\ 6^{(5)},\ (5+\sqrt{5})^{(3)}\}.$$

The only rational non-trivial Laplacian eigenvalue is 6 (multiplicity 5). The field generated by the spectrum is $\mathbb{Q}(\sqrt{5})$, of degree 2.

Proof. Script `spectrum_icosahedral_carrier.py`, SymPy, exact; the seam list is taken from the OPH bundle, alternatively from the canonical vertices $(0, \pm 1, \pm\varphi)$ cyclically. The script also replays the 300 updates of seed 7 in rational arithmetic and reproduces the centered residual recorded in the bundle exactly. □

The multiplicities $(1, 3, 5, 3)$ are the dimensions of the irreducible A_5 representations $A \oplus T_1 \oplus H \oplus T_2$ (icosahedral-group notation; H is five-dimensional, T_1, T_2 are the two inequivalent triplets). On a five-fold rotation g_5 they carry the character values

$$\chi_{T_1}(g_5) = \varphi, \qquad \chi_{T_2}(g_5) = 1 - \varphi, \qquad \chi_{T_1} + \chi_{T_2} = 1.$$

Remark V.3.2 (Damping rates, **[E]/[B]**)*.* The expected action of a random single-seam update is $E[M] = I - L/60$ with block eigenvalues $(55 + \sqrt{5})/60 \approx 0.954$ on T_1, $9/10$ on H and $(55 - \sqrt{5})/60 \approx 0.879$ on T_2. T_1 is the slowest block; its distance to T_2 is $\sqrt{5}/30$ (ratio ≈ 1.085). This is a statement about rates, not a symmetry selection.

V.4 Galois conjugation and the doubling $6 = 3 \oplus 3'$

Theorem V.4.1 (Galois exchanges $T_1 \leftrightarrow T_2$, **[E]/[B]**)**.** *Let $P_{T_1}, P_{T_2}, P_A, P_H$ be the spectral projectors of L (polynomials in L with coefficients in $\mathbb{Q}(\sqrt{5})$). The Galois automorphism $\sigma : \sqrt{5} \mapsto -\sqrt{5}$ of $\mathbb{Q}(\sqrt{5})$ acts entrywise as*

$$\sigma(P_{T_1}) = P_{T_2}, \qquad \sigma(P_{T_2}) = P_{T_1}, \qquad \sigma(P_A) = P_A, \qquad \sigma(P_H) = P_H.$$

The projector $P_{T_1} + P_{T_2}$ is $\mathbb{Q}$-rational; P_{T_1} and P_{T_2} separately are not. The minimal polynomial of the T_1 eigenvalue over $\mathbb{Q}$ is $x^2 - 10x + 20$ with roots $5 \mp \sqrt{5}$.

Proof. Script `galois_z3_icosahedral_carrier.py`. The projectors are built as Lagrange polynomials in L; the exchange is checked by substituting $\sqrt{5} \to -\sqrt{5}$ and comparing matrices. □

Corollary V.4.2 (Indivisibility over $\mathbb{Q}$, **[B]**). *Every A_5-invariant structure on the port space that is defined over $\mathbb{Q}$ as an abstract field – a lattice, an integer character – sees $T_1 \oplus T_2$ as one six-dimensional block; $T_1 \oplus T_2$ is the only six-dimensional submodule with integer character that contains a triplet ($1 \oplus H$ is the other six-dimensional submodule with integer character). Distinguishing T_1 from T_2 is not a symmetry statement.*

Remark V.4.3 (The selection happens at the archimedean place, **[B]**). Argument of the author of the external model (personal communication, 15 September 2026), verified exactly here. The repair dynamics does not live in the abstract field $\mathbb{Q}$ but in $\mathbb{R}$: $I - L/60$ is a rational symmetric matrix with real eigenvalues, and the two conjugate rates $(55 \pm \sqrt{5})/60$ are the real numbers 0.954 and 0.879. Both embeddings $\mathbb{Q}(\sqrt{5}) \hookrightarrow \mathbb{R}$ yield the same two subspaces; they differ only in which one carries the name T_1. Iterating $I - L/60$ on any rational vector with a component in the slow block, the ratios of the centered norms – a rational sequence – tend to $((55 + \sqrt{5})/60)^2 = 0.9100$ and not to the conjugate value 0.7733 (script, section 6). Selecting the slow triplet as space is therefore a derivation from the dynamics, as every convergent rational sequence selects the positive root of $x^2 = 2$; it happens at the archimedean place of $\mathbb{Q}(\sqrt{5})$. Geometrically: in the coordinates $(0, \pm 1, \pm\varphi)$ the five seam neighbours of a port sum to $\sqrt{5}\,v$, the thirty second neighbours (edges of the great icosahedron) to $-\sqrt{5}\,v$; "T_1 is space" says that repair couples nearest neighbours.

This is the theorem behind Dok. 285: there the doubling $6 = 3 \oplus 3'$ is the answer to the crystallographic restriction (no icosahedral lattice in three dimensions, but one in six). Here the same statement appears as Galois indivisibility at the *finite* places; the archimedean place separates the triplets, the place 3 does not (Theorem V.5.2, Section V.8). OPH's proposal that the three-dimensional continuum is the dense projection of "integer frames built from port labels" is the quasicrystal projection $\mathbb{Z}^6 \to \mathbb{R}^3$; Dok. 285 describes the cause, OPH the image.

Theorem V.4.4 (Restriction to $C_3 < A_5$, **[E]/[B]**). *Under a three-fold rotation $g_3 \in A_5$ the port space splits into $\mathbb{Z}_3$ eigenspaces of dimensions $4 + 4 + 4$; blockwise*

$$A \to 1, \qquad T_1 \to 1 \oplus \omega \oplus \omega^2, \qquad T_2 \to 1 \oplus \omega \oplus \omega^2, \qquad H \to 1 \oplus 2\omega \oplus 2\omega^2.$$

The trivial part has dimension 4, the non-trivial part dimension 8.

This is the C_3-in-A_5 embedding of Dok. 293 in character form. The non-trivial part (dimension 8) is, as an abstract $\mathbb{Z}_3$-module, two copies of the orbifold representation on $\mathbb{R}^4$; whether more than representation theory is contained in this is settled negatively in Section V.6.

V.5 Reduction modulo 3: $\mathbb{Z}[\varphi]/(3) = GF(9)$

Theorem V.5.1 (Inert prime, **[E]/[B]**). *$x^2 - x - 1$ is irreducible over $GF(3)$ ($5 \equiv 2$ is not a square modulo 3). Hence 3 is inert in $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Z}[\varphi]/(3) \cong GF(9)$.*

Theorem V.5.2 (Galois becomes Frobenius, **[E]/[B]**). *Under the reduction of Theorem V.5.1 the Galois conjugation $\varphi \mapsto 1 - \varphi$ becomes the Frobenius $x \mapsto x^3$ of $GF(9)$: in $GF(9)$ one has exactly $\varphi^3 = 1 - \varphi$.*

Proof. $\varphi^2 = \varphi + 1$, hence $\varphi^3 = 2\varphi + 1 \equiv 1 - \varphi \pmod 3$. Script `zphi_mod3_a5.py`, sections (1)–(2). □

Thus the Frobenius $x \mapsto x^3$ on $GF(9)$ and the exchange $T_1 \leftrightarrow T_2$ of the icosahedral carrier are not merely analogous: modulo 3 they are the same map. The FFGFT sector pairing on $GF(27)^*$ (Dok. 343/363), by contrast, is the inversion $g^k \mapsto g^{-k}$, not the Frobenius.

Theorem V.5.3 (A_5 over $GF(9)$, **[E]/[B]**). *The sixty rotation matrices of the icosahedron (entries in $\frac{1}{2}\mathbb{Z}[\varphi]$) reduce modulo 3 to sixty distinct matrices in $GL_3(GF(9))$ spanning all of $M_3(GF(9))$: the reduction is faithful and absolutely irreducible. Entrywise Frobenius of the reduced T_1 matrices yields exactly the reduced T_2 matrices. By contrast $GL_2(GF(9))$ (order 5760, searched exhaustively) contains no $(2,3,5)$-generated subgroup of order 60: A_5 does not embed in $GL_2(GF(9))$.*

Corollary V.5.4 (The $G(3)/G(6)$ threshold, **[S]**). *With $G(3) \cong M_2(GF(9))^2$ and $G(6) \cong M_8(GF(9))$ (Dok. 341), A_5 is not in the unit group of $G(3)$ but is in that of $G(6)$. This is the same threshold at which Dok. 341 finds the lepton order 26. The reasons are formally different (Brauer degrees $\{1,3,3,4\}$ of A_5 in characteristic 3 versus Lagrange in $GF(81)^*$); whether the coincidence has substance remains open.*

V.6 A_5 versus Aut(D_4)

Theorem V.6.1 (Reproduction of Dok. 314, **[E]/[B]**). *The automorphism group of the D_4 root system, recomputed from the 24 roots, has order 1152 with order distribution $(1,2,3,4,6,8,12) \mapsto (1,139,80,228,464,144,96)$ and no element of order 5. Among the 80 elements of order 3 exactly 16 are fixed-point-free on $\mathbb{R}^4$ ($|\det(1-A)| = 9$); each acts freely on the 24 roots (eight orbits of three).*

Corollary V.6.2 (**[X]**). *A_5 acts on no D_4 lattice: $5 \nmid 1152$. The symmetry group of OPH and the FFGFT lattice structure are incompatible as wholes.*

Theorem V.6.3 (The common core is A_4, **[E]/[B]**). *The elements of A_5 whose matrices in the model $(0, \pm 1, \pm\varphi)$ are integral (no φ) form the tetrahedral group A_4 (orders $1,2,3$ with counts $1,3,8$). A_4 acts simply transitively on the twelve ports – the port set is an A_4-torsor – and embeds as* diag$(M,1)$ *in $W(D_4) <$* Aut(D_4).

This sharpens Dok. 285 ("$C_3 < A_5$"): not only the three-fold symmetry but the whole tetrahedral group of order 12 – the port count – is lattice-compatible; only the five-fold symmetry drops out.

Theorem V.6.4 (Ports as half the first shell, **[E]/[B]**). *Every three-fold rotation of A_5 acts freely on the twelve ports (four orbits of three). As a $\mathbb{Z}_3$-module the port space is $4 \cdot \mathbb{Z}[\mathbb{Z}_3]$, the first D_4 shell is $8 \cdot \mathbb{Z}[\mathbb{Z}_3]$: the ports are one half of "$24 = 12 + 12$" from Dok. 314.*

Theorem V.6.5 (The port $\mathbb{Z}_3$ is not an orbifold $\mathbb{Z}_3$, **[X]**). *The three-fold rotation permuting the ports is a rotation of $\mathbb{R}^3$ and fixes an axis; as* diag$(M,1)$ *it fixes a line in $\mathbb{R}^4$ and is therefore one of the $80 - 16 = 64$ non-orbifold elements of order 3. This is not an artefact of the embedding: for each of the 16 orbifold generators and each of the 70 choices of four of its eight root orbits (1120 $\mathbb{Z}_3$-stable twelve-element halves) it was checked*

whether the twelve roots form a cuboctahedron (D_3 half) or carry the icosahedral graph on some inner-product class. Neither ever occurs.

Proof. A $\mathbb{Z}_3$ stabilising a D_3 half stabilises its normal up to sign, hence exactly for order 3; it has a fixed vector and is not an orbifold generator. Script `a5_vs_d4.py`, sections (E)–(F). □

Remark V.6.6 (Side result, **[K]**). 256 of the 1120 orbifold-stable twelve-element halves are tight frames in $\mathbb{R}^4$ (Gram eigenvalue 6 with multiplicity 4, the 4D counterpart of the OPH relation $G^2 = 4G$ for the port vectors). Their graph spectra contain $\sqrt{13}$, not $\sqrt{5}$; the stabiliser has order 18 and is not transitive. They carry no port structure.

V.7 The OPH gauge assignment

OPH assigns to the port module the Lie algebra $u(1) \oplus su(2) \oplus su(3)$: $u(1)$ on A, $su(2)$ on T_1, $su(3)$ on $T_2 \oplus H$.

Theorem V.7.1 (Consistency of the assignment, **[E]**). *A_5 sits in $SO(3) \subset SU(3)$ via T_2. The adjoint of $su(3)$ is $3 \otimes \bar{3} - 1$; under $SO(3)$ it splits as $3 \oplus 5$, under A_5 with the T_2 embedding as $T_2 \oplus H$ ($\Lambda^2 T_2 = T_2$, $\mathrm{Sym}^2 T_2 = 1 \oplus H$). Likewise the adjoint of $su(2)$ under the T_1 embedding is T_1. The restriction of $u(1) \oplus su(2) \oplus su(3)$ to A_5 is therefore exactly $A \oplus T_1 \oplus T_2 \oplus H$.*

Remark V.7.2 (What is axiom, **[S]**/**[E]**). The derivation of the centre in this remark is due to the author of the external model (personal communication, 15 September 2026). The assignment presupposes two axioms of the external model: (A1) the reversible port response is a compact Lie algebra $\mathfrak{g}$ with a faithful representation on the port module; (A2) A_5 acts on $\mathfrak{g}$ by inner automorphisms. The one-dimensional centre then follows and is not a third axiom: inner automorphisms act trivially on the centre, so it lies in the A_5-fixed part of $1 \oplus 3 \oplus 3' \oplus 5$, the uniform line. Centre zero would require $\dim \mathfrak{g} = 12$ semisimple, i.e. $su(2)^4$; under an inner action each $su(2)$ factor has fixed dimension 0 or 3, so the fixed dimension would be a multiple of 3, whereas the port module has fixed dimension 1. Hence the centre is the uniform line and $11 = 8 + 3$ is the only decomposition of the rest into dimensions of compact simple algebras. The counterexample $u(1)^3 \oplus su(2)^3$ fails on (A2): under an inner action its module contains no 5; and even without (A2) it fails by Theorem V.4.1, since the centre of a torus is defined over $\mathbb{Q}$ and can therefore contain 3 and $3'$ only with equal multiplicity. The correct statement is thus "A_5 plus (A1), (A2) forces the Lie type".

FFGFT is in the mirror situation: $su(3)$ follows from the $\mathbb{Z}_3$ triality on T^4 **[B]** (Dok. 321), $su(2)$ is open **[S]** (Dok. 324). In OPH T_1 is clean and $8 = T_2 \oplus H$ is the axiomatic step.

V.8 Phases as roots of unity: how the connection succeeds

Section V.6 and Theorem V.5.2 seem to contradict each other: five-fold symmetry is incompatible with the lattice, yet after reduction A_5 sits in $GL_3(GF(9))$. The contradiction dissolves once one distinguishes *as what* a symmetry enters the carrier – as a lattice automorphism or as a phase.

Theorem V.8.1 (Phase channel, **[E]/[B]**). *1. Modulo 3, φ is a primitive element of $GF(9)^*$: $\varphi^4 = -1$, order 8.*
2. The 72° rotation has characteristic polynomial $(x-1)(x^2-(\varphi-1)x+1)$. Modulo 3 the quadratic factor has no root in $GF(9)$; its roots lie in $GF(81)$ and have order 5 there ($5 \mid 80$).
3. $13 \mid 728 = |GF(729)^*|$ but $13 \nmid 80$: order 26 needs the cubic extension (Dok. 341).
4. In characteristic 3, $x^3 - 1 = (x-1)^3$: no $GF(3^k)$ contains a primitive cube root of unity. A $\mathbb{Z}_3$ acts unipotently in the field, never as a phase ω.

Proof. Script `phasen_einheitswurzeln.py`. □

Reading. A phase is a root of unity, and roots of unity are exactly what a field extension adds. In the lattice the 72° rotation is an automorphism of vectors, and $5 \nmid 1152$. In the field the same rotation is only its eigenvalue ζ_5, and $5 \mid 80$. Geometric incompatibility becomes arithmetic compatibility as soon as one passes from lattice vectors to phases. This is the mechanism behind Dok. 293: $\theta = 2/9$ arises from R_5 as a phase, not as a lattice operation.

The extension degrees are the two basic operations of FFGFT:

Extension	Degree	carries	Counterpart
$GF(9) \to GF(81)$	2	ζ_5, φ, $2/\sqrt{5}$, icosahedron	doubling $6 = 3 \oplus 3'$ (Dok. 285)
$GF(9) \to GF(729)$	3	ζ_{13}, order 26, leptons	$\mathbb{Z}_3$ triality (Dok. 341)

And three-fold symmetry itself is *not* a phase: in characteristic 3 ω collapses to 1 (Dok. 341: the N eigenvalues $\{-3, +3\}$ become $\{0\}$). This is why in FFGFT the $\mathbb{Z}_3$ is lattice structure – a circulant on the mode orbits, Dok. 314 – and five-fold symmetry is a phase. It is not a choice but forced: what cannot exist as a root of unity in characteristic 3 must enter as a lattice operation; what does not fit the lattice must enter as a root of unity in an extension. The two channels are complementary, and the prime 3 decides which symmetry falls into which channel. In the language of places of $\mathbb{Q}(\sqrt{5})$: the external model chooses between T_1 and T_2 at the archimedean place (Remark V.4.3); FFGFT computes at the place 3, where $\sqrt{5}$ has no sign and the two triplets are exchanged by the Frobenius. Both selections are correct, at different places – and that is why the two frameworks touch exactly once. Theorem V.6.5 is the geometric side of the same statement: the port $\mathbb{Z}_3$ is lattice, the orbifold $\mathbb{Z}_3$ is lattice, but they are different lattice operations; the five-fold symmetry that distinguishes them is not present in the lattice at all.

V.9 What is projected and what is not: scope of the conclusions

The reference object of this document is the torus, not the icosahedron. The icosahedron is *one* geometric body among many that can be projected onto $T^4/\mathbb{Z}_3$, and it is the test body here only because an external model proposes it as carrier. Every other finite body – the five-fold geometry of Dok. 285 was the previous case – splits by the same pattern into a lattice-compatible part (in $\mathrm{Aut}(D_4)$) and a phase part (roots of unity in an extension $GF(3^k)$); the icosahedron is the instance on which the pattern is worked out completely in this document, not the pattern itself. None of the maps is a complete projection of

the icosahedron onto the torus; by Theorem V.6.1 no such projection can exist, and that holds for every body with non-crystallographic symmetry. In the terminology of Dok. 330 the maps are neither Type I (lossless decompactification) nor Type III (bijective representational translation), but lossy like Type II – or not FFGFT operations at all. The conclusions are justified exactly as far as the respective map is faithful on what is being concluded. They also hold relative to the choice of lattice: D_4 is a posit (Dok. 320 **[POSIT]**), motivated by the densest packing in four dimensions and by the fact that D_4 is the only lattice of the D_n family with S_3 outer symmetry, which carries the $\mathbb{Z}_3$ as triality (Dok. 314; there also the Casimir counter-finding in favour of $\mathbb{Z}^4$). All statements of this document about $\mathrm{Aut}(D_4)$ presuppose that choice.

Genuine projections (faithful on the object of the conclusion).

1. $\mathbb{Z}_3$ on the mode orbits (Dok. 314): intrinsic torus structure, complete. All triality conclusions rest on it.
2. $\mathbb{Z}[\varphi] \to GF(9)$ (Theorem V.5.1): surjective ring homomorphism with kernel (3). Faithful for everything at the place 3 (mode orders, extension degrees, indistinguishability of T_1, T_2); lost is the sign of $\sqrt{5}$ and with it everything archimedean (rates, order, convergence). No conclusion about dynamics.
3. Rotation $\mapsto$ eigenvalue (Theorem V.8.1): faithful for the order, with maximal kernel – the geometry is forgotten. The conclusion "five-fold symmetry only as a phase" holds for precisely that reason.
4. $A_4 \hookrightarrow W(D_4)$ (Theorem V.6.3): faithful for everything tetrahedral, five-fold symmetry not included.

Not projections of the torus.

5. Ports = half the first shell: isomorphic only as $\mathbb{Z}_3$-sets; the icosahedral edges do not exist in D_4. Counting statements are justified, statements about neighbourhood or rates are not.
6. Cut-and-project $\mathbb{Z}^6 \to \mathbb{R}^3$: a genuine projection of a *different* lattice ($\mathbb{Z}^6/D_6$, the second protected geometry of Dok. 285), not of T^4.
7. Port $\mathbb{Z}_3 \to$ orbifold $\mathbb{Z}_3$: provably not a projection (Theorem V.6.5).
8. Gauge assignment and choice of space in the external model: statements at the archimedean place about its response algebra; none of it projects onto T^4.

What kind of confirmation this is. The projections do not confirm that the torus is the physics. They confirm that the torus geometry makes concrete, falsifiable structural predictions and that an independently constructed foreign object satisfies them. Four predictions had the opportunity to fail:

- *Lattice*: order 5 does not act on D_4 (Dok. 314). Had $A_5 \subset \mathrm{Aut}(D_4)$, Dok. 285 would be wrong. Passed.
- *Field*: ζ_5 only in $GF(81)$, ζ_{13} only in $GF(729)$, no ζ_3 (Dok. 341). Had 3 split in $\mathbb{Q}(\sqrt{5})$ or $\zeta_5 \in GF(9)$, the correspondence doubling $\leftrightarrow$ degree 2 would be wrong. Passed.
- *Orbifold*: $\mathbb{Z}_3$ is a lattice operation, not a phase. The port $\mathbb{Z}_3$ could have been an orbifold $\mathbb{Z}_3$; it is not, for a reason that mirrors the torus structure. Passed.

- *Mode order*: the $G(3)/G(6)$ threshold meets the same boundary for A_5 as for order 26. The weakest test, **[S]**.

This is confirmation in the sense of a survived test: more than internal consistency, less than proof – and it applies to the torus, not to the icosahedron. A further test body (octahedron, cube, a four-dimensional polytope) would put the same four predictions to the question again, with a different lattice part and a different number field; the tetrahedral and octahedral groups lie entirely in $\mathrm{Aut}(D_4)$ and would have no phase part, the 600-cell would bring back $\mathbb{Q}(\sqrt{5})$. In this series the icosahedron is only the first fully computed case. Unexamined – and not examinable by any projection – remains whether T^4 is the physical carrier, whether $\theta = 2/9$ is physically real as a phase (still **[K]**, Dok. 293) and whether mode orders are particles; that requires measurement. Equally untouched is the deeper question whether FFGFT's own translations (Type III, Dok. 270/330) are complete: there, internal consistency without any statement about observability continues to hold. Whoever reads more into this document than the location of the single point of contact overreaches.

Tool and method

Claude (Anthropic) was used as a tool for: writing and executing the Python verification scripts; drafting the LaTeX sources from results and corrections of the author; maintaining the changelog and cross-references. The tool computes and writes; the author decides what is computed and written, judges every result and determines what it means physically. Every algebraic statement is backed by a Python verification script with explicit assertions — if an assertion fails, the statement is not used.

V.10 Summary

	OPH	FFGFT	Link
Field	$\mathbb{Q}(\sqrt{5})$	$GF(9)$, $GF(27) \parallel GF(729)$	$\mathbb{Z}[\varphi]/(3) = GF(9)$; Galois → Frobenius **[B]**
Group	A_5	$\mathrm{Aut}(D_4)$, $\mathbb{Z}_3$	$A_5 \not\subset \mathrm{Aut}(D_4)$; core A_4 (= 12 ports) **[B]**
$3 \oplus 3'$	T_1 space, T_2 colour	6D lattice (Dok. 285)	indivisible over $\mathbb{Q}$ **[B]**
$\mathbb{Z}_3$ orbits	4 free on 12	8 free on 24	ports = half the first shell **[B]**
Port $\mathbb{Z}_3$	axis in $\mathbb{R}^3$	orbifold fixed-point-free	never conjugate **[X]**
$G(3)/G(6)$	$A_5 \not\subset GL_2(GF(9))$, $\subset GL_3$	order 26 only in $G(6)$	same threshold **[S]**

Verification scripts. `2/python/Dok364_Skripte/`: `spectrum_icosahedral_carrier.py`, `galois_z3_icosahedral_carrier.py`, `zphi_mod3_a5.py`, `a5_vs_d4.py`, `phasen_einheitswurzeln.py`; batch run

`run_all_364.sh`. The seam list of the external bundle is read if present; otherwise the canonical icosahedron is used. No register entry: no earlier document is corrected.

[1] J. H. Conway, N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Springer, 1993. Standard reference for Aut(D_4), kissing number 24, order distribution.

[2] J. E. Humphreys, *Reflection Groups and Coxeter Groups*, Cambridge University Press, 1990. D_4 triality, Dynkin diagram, $W(F_4)$.

[3] J.-P. Serre, *Linear Representations of Finite Groups*, Springer, 1977. Character table of A_5; Brauer degrees in characteristic 3.

[4] R. Lidl, H. Niederreiter, *Finite Fields*, 2nd ed., Cambridge University Press, 1997. $GF(9)$ as extension of $GF(3)$; Frobenius automorphism; roots of unity in $GF(3^k)$.

[5] J. Neukirch, *Algebraic Number Theory*, Springer, 1999. Inertness of 3 in $\mathbb{Q}(\sqrt{5})$; Galois group of $\mathbb{Q}(\sqrt{5})/\mathbb{Q}$.

[6] M. Berger, *Geometry I*, Springer, 1987. Icosahedron, A_5 symmetry; canonical vertices $(0, \pm 1, \pm\varphi)$.

[7] J. Pascher, *Dok. 285: FFGFT — HLV, CP violation and dimension bridge*, FFGFT corpus.

[8] J. Pascher, *Dok. 293: Icosahedron, $\theta = 2/9$ and C_3-in-A_5*, FFGFT corpus.

[9] J. Pascher, *Dok. 314: Lattice in Hilbert space*, FFGFT corpus.

[10] J. Pascher, *Dok. 341: $GF(27)$ and GALG in FFGFT*, FFGFT corpus.

Chapter W

Doc. 365 — The Foundation of FFGFT in Four Layers

Abstract

FFGFT rests on four layers of different epistemic character. Layer 1 (the fundamental equation and the principle of relativity) and Layer 2 (the electromagnetic sector with $\alpha = 1$) are parameter-free in natural units and generate the structure. Layer 3 (ξ as scaling parameter) translates the structure into measurable quantities; it is the only free parameter. Layer 4 (algebraic confirmations: Galois fields, Hodge theory, representation theory) demonstrates the coherence of the system against known mathematics but adds no new inputs. Translation into SI units and comparison with the Standard Model are communication work for established systems, not structural inputs. The cosmic sector (H_0/Λ) is bracketed because the Standard Model derives it equally little.

List of symbols

The table explains symbols appearing in more than one section. Quantities used only locally are introduced there.

Epistemic markers:

[B]	algebraically proven (Python script, exact arithmetic)
[K]	numerically confirmed
[S]	structurally plausible, identification open
[POSIT]	motivated input, not derived
[E]	established external mathematics or physics
[X]	checked and negative

Symbol	Meaning
Fundamental equation	
$\tilde{T}$	intrinsic time scale of a system
m	mass (in natural units = energy)
$\tilde{T} \cdot m = 1$	fundamental equation of FFGFT
Geometry	
T^4	four-dimensional torus (compact carrier)
D_4	root lattice in $\mathbb{R}^4$; kissing number 24
$\mathbb{Z}_3$	identification group; triality on D_4
$T^4/\mathbb{Z}_3$	orbifold; nine fixed points
Parameters and fields	
$\xi = 4/30\,000$	dimensionless scaling parameter
α	fine-structure constant; in FFGFT units $\alpha = 1$
$GF(3^k)$	finite field of characteristic 3; mode classification
$\varphi = (1+\sqrt{5})/2$	golden ratio; primitive element of $GF(9)^*$
$\theta = 2/9$	output of the $\mathbb{Z}_3$-circulant diagonalisation; used in § W.1
Field equations	
$F^{\mu\nu}$	electromagnetic field strength tensor
j^ν	four-current
$\not{\partial} = \gamma^\mu \partial_\mu$	Feynman slash notation
$\hat{H}$	Hamiltonian
Operations	
Type I	duality decompactification; information-preserving (limit $R \to \infty$)
Type II	geometric projection; lossy (e.g. projection onto subspaces)
Type III	representational translation; bijective

On the intention

FFGFT does not make predictions in the sense of asserting something contentually new. It rewrites known relations so that hidden structure becomes visible. Experimental confirmation is not the goal — the reformulated equations are identical to the established ones and therefore trivially correct. What is interesting is what the rewriting reveals: that H_0 follows from the geometry, that $a_0 = cH_0/(2\pi)$ is not a coincidence, that $\theta = 2/9$ was not an input. Had the cosmic sector been unknown, the geometry would have supplied it — one could then measure whether it agrees. That it is already known does not make the statement false, but it makes testability harder, because one can never be fully certain whether an exponent was found or chosen.

Confirmations do not succeed — that is not a deficiency but the consequence of the method. Whoever seeks confirmation has misunderstood the intention.

The gain over the Standard Model lies not in new numerical values but in the reduction of free parameters: the Standard Model uses ~19 independent constants; FFGFT manages with *one* dimensionless parameter ξ, which itself follows from Galois group orders. What

the Standard Model treats as independent measured quantities — α, mass ratios, mixing angles — appears in FFGFT as a consequence of the torus geometry and field structure. That is the structural gain: not a new prediction, but a deeper explanation of existing ratios.

W.1 Layer 1: the fundamental equation

In natural units ($\hbar = c = 1$) the only fundamental equation of FFGFT is

$$\tilde{T} \cdot m = 1.$$

It states: time scale and mass are reciprocal representations of the same object. Combined with the principle of relativity (Lorentz invariance, causal structure, action principle) this implies:

- A compact carrier T^4 as the smallest structure compatible with $\tilde{T} \cdot m = 1$, Lorentz invariance and three spatial dimensions **[POSIT]**.
- $\mathbb{Z}_3$ as the most economical identification with three generations **[POSIT]**.
- D_4 as the densest lattice in four dimensions with $\mathbb{Z}_3$ triality **[POSIT]**.

These three posits are motivations, not derivations; they are vindicated by internal coherence (Dok. 248).

Three levels of description

It is important to distinguish what belongs to the geometry of the torus and what is a translation into other mathematical languages (Dok. 270/330):

1. **Torus geometry (Level 1)** — the physics itself. $T^4/\mathbb{Z}_3$ on the D_4 lattice, modes as winding numbers, fixed points, ratios. Static, in $\mathbb{Q}$ and algebraic fields.
2. **Galois fields (Level 2)** — the algebraic classification of the modes. $GF(9)$, $GF(27)$, $GF(729)$ describe which mode orders can exist; the GF tower is the algebraic language of Level 1, not a separate level of reality. Bijective translation (Type III, Dok. 270): no information loss, no new content.
3. **Matrix/Hilbert space (Level 3)** — the QM language. $T^4 \leftrightarrow \mathscr{H}$ via Fourier on the torus; likewise Type III, bijective. Type I denotes duality decompactifications ($R \to \infty$); Type II geometric projections with information loss. Enables connection to Schrödinger, Dirac, product states — but adds no physics. The Lagrangian and field equations belong to this level as imported vocabulary **[E]** (R120), to keep the theory compatible with established frameworks.

All three levels describe the same static snapshot in different languages. What changes is the language, not the content.

Static picture — snapshot

The theory describes a static structure: a snapshot. Time is a coordinate of the torus T^4, not a distinguished evolution parameter; $\tilde{T} \cdot m = 1$ is a property of each state, not a propagation law. What appears as a temporal evolution is the transition between snapshots — which the theory does not describe in Layers 1–3. The position-dependent

time field $T(x,t) \equiv 1/m(x,t)$ is the snapshot relation at point x: when the mass $m(x,t)$ varies in a gravitational potential, $T(x,t)$ varies with it necessarily — not because T is an independent dynamical field, but because $T = 1/m$ holds at every instant. The "extended" Schrödinger equation with $T(x,t)$ formulated in the early documents is therefore a rewriting of the mass-position-dependent coupling to the gravitational potential, not a new degree of freedom.

Ratio-based computation and exact rational arithmetic

Since $\tilde{T} \cdot m = 1$ in natural units formulates all physical statements as ratios — mass to mass, time to time, length to length — the fundamental calculations remain in $\mathbb{Q}$ or finite algebraic extensions such as $\mathbb{Q}(\sqrt{5})$. No floating-point operations are required as long as no SI anchor is inserted. The FFGFT verification scripts therefore use exact rational arithmetic (SymPy `Rational`, Python's `Fraction`); replay identity — bit-identical reproducibility of every step — is the natural test criterion. Rounded approximations arise only when translating into SI units (Layer 3).

What already follows. From $\tilde{T} \cdot m = 1$ alone follow time dilation as a mass effect, redshift as path lengthening in the intrinsic time field and the basic structure of gravitation (Dok. 078/270). The pattern of the mass ladder follows from the geometry of T^4; $\theta = 2/9$ is the output of the $\mathbb{Z}_3$-circulant diagonalisation, not an input (Dok. 230/231, "found, not sought").

Gravity as a derived quantity. Gravitational effects follow from $\tilde{T} \cdot m = 1$ applied locally: if the mass $m(x)$ varies in a gravitational field, $\tilde{T}(x) = 1/m(x)$ varies with it necessarily. There is no independent gravitational field and no graviton as an independent degree of freedom; gravity is a consequence of the position-dependent time field, not a separate foundation (Dok. 078 **[K]**). The choice between "mass constant, space curved" (Einstein) and "space flat, mass variable" (FFGFT) is a *choice of representation* — exactly equivalent and interconvertible by a coordinate transformation, like the Schrödinger and Heisenberg pictures in QM. The physically most natural reading within $\tilde{T} \cdot m = 1$ is that *both* vary and that holding either one constant is a simplifying assumption (Dok. 078).

No gap between relativity and quantum theory. The apparent tension between general relativity and quantum mechanics is an artefact of the level of description. At Level 1 (torus geometry) this opposition does not exist: GR and QM are two Type III translations of the same static snapshot — GR into the language of spacetime curvature, QM into the language of operators on $\mathscr{H}$. FFGFT is not a new theory that *unifies* them; it shows that the opposition does not exist at Level 1.

FFGFT achieves more than standard QM describes. In standard QM the time t is an external parameter outside the system — the wave function evolves in a prescribed background time that is itself not quantised. In FFGFT time belongs to the system: $\tilde{T} = 1/m$ is a property of the physical object, not a background structure. This changes the status of quantum randomness: what appears in standard QM as fundamental, ontological chance is in FFGFT deterministic time-field dynamics — in principle determined, in

practice unpredictable (Dok. 020 **[K]**). There is no collapse: wave functions evolve continuously; the Born rule emerges from the time-field interaction instead of being posited as an axiom. Randomness is epistemic, not ontological.

W.2 Layer 2: the electromagnetic sector and simplification of the field equations

FFGFT units (formerly: T0 units)

Setting $\alpha = 1$ in natural units is standard practice in theoretical physics (Heaviside-Lorentz units, Gaussian units) and not a FFGFT-specific decision. The FFGFT unit system additionally sets $c = \hbar = \alpha = G = 1$ (Dok. 011; formerly called T0 units). $\alpha = 1$ is not a posit in the sense of R56 but a *choice of units*: the electromagnetic coupling strength remains the same physical quantity; it receives the numerical value 1. The SI value $1/137$ does not disappear — it is fully reconstructible from ξ and E_0: $\alpha_{\text{SI}} = \xi \cdot (E_0/1\,\text{MeV})^2 \approx 1/137$ (Dok. 011 **[K]**). In FFGFT units and in SI units α is the same quantity — only its numerical representation changes.

Derived electromagnetic quantities

With $\alpha = 1$ electromagnetic quantities become derived forms of mass and time:

$$\begin{aligned} e &= 1 \quad \text{(charge dimensionless)}, \\ V &= E/e = E \quad \text{(voltage = energy)}, \\ I &= e/T = 1/T = m \quad \text{(current = mass)}. \end{aligned}$$

The Coulomb potential simplifies to $V(r) = -1/r$.

Maxwell, Schrödinger and Dirac equations in FFGFT units

The established field equations take their most geometrically transparent form in FFGFT units — not because they are changed, but because the unit choice sets the coupling constants to 1 and thereby makes visible the structure that was always there:

$$\begin{aligned} \text{Maxwell:} \quad & \partial_\mu F^{\mu\nu} = j^\nu, \qquad \partial_{[\mu} F_{\nu\rho]} = 0, \\ \text{Schrödinger:} \quad & i\,\partial_t \psi = \left(-\tfrac{1}{2}\nabla^2 + V\right)\psi \\ \text{Dirac:} \quad & (i\partial\!\!\!/ - m)\,\psi = 0. \end{aligned}$$

$\hbar$, c, e and $4\pi\varepsilon_0$ no longer appear explicitly; they are absorbed into the unit choice. This is not new physics but the known physics in a notation that makes the mass structure directly readable: $i\,\partial_t \psi = m\,\psi$ in the free case, $\partial_\mu F^{\mu\nu} = j^\nu$ without a prefactor.

What becomes concrete. Only with Layer 2 does the theory become concrete enough for particle physics: charge quantisation **[B]**, su(3) from $\mathbb{Z}_3$ triality **[B]**, charge definition and gauge structure. The mass layers of the lepton sector are derivable from ξ **[K]**; the mass ratio m_μ/m_e serves as a declared reference point **[POSIT]** (Dok. 190 P42) to keep the layer independently checkable.

W.3 Layer 3: the only free parameter

$\xi = 4/30\,000$ appears as the only dimensionless parameter but is not free given the Galois group orders of the GF tower. From the winding numbers of the T^4 torus (derived from char(GF(3^n)), |GF(3)*| and the minimal prime divisor of |GF(81)*|) and the field orders |GF(9)*| = 8, |GF(27)| = 27, 5 = min.prime(80), ξ follows algebraically **[K]**:

$$\xi = \frac{(r_\mu/r_e)^2}{8^2 \cdot 5^2 \cdot 27} = \frac{(12/5)^2}{43200} = \frac{4}{30\,000}.$$

(Dok. 336/338; verification scripts pruef_330, pruef_331.) The premises of this derivation — the winding-number recursion and m_μ/m_e as reference point (Dok. 190 P42) — carry a [K] status, not [B]. ξ is therefore not completely free, but not yet completely derived. Note: the value $m_\mu/m_e = 12/5 = 2.4$ used in Dok. 338 is the *structural ratio* of the winding numbers (Level 1), not the measured mass ratio ≈ 206.77; the measured ratio follows from the structural value after scaling by ξ and the Galois-field exponents.

With ξ, Layers 1 and 2 yield measurable scales: Compton wavelength, Hubble length as geometric maximal scale (not an expansion horizon), $a_0 = cH_0/(2\pi)$ **[K]**.

What is bracketed. The cosmic sector (H_0, Λ) is bracketed. The Standard Model derives H_0 equally little; the comparison would be asymmetric (Dok. 190 P39). The exponent 10 in $H_0 \propto \xi^{10}$ is a posit (Dok. 190 P20); it is structurally motivated but not forced.

SI anchors are comparison tools. m_e, $\bar{\lambda}_e$, ν_{Cs} and all other SI anchors serve exclusively for translation into known unit systems. Every dimensionless statement of FFGFT is a function of ξ and the posits; no SM value is a structural input. The complexity of comparison tables and precision calculations is communicative, not structural — a concession to established systems that cannot be abandoned without notice.

W.4 Layer 4: algebraic confirmations

The Galois fields $GF(9)$, $GF(27)$, $GF(729)$, Hodge theory on $T^4/\mathbb{Z}_3$, representation theory of Aut(D_4) and comparisons with external carriers (Dok. 361–364) are *consistency checks*, not new inputs. They show that the structure generated by $\tilde{T} \cdot m = 1$ and the three posits is coherent against known mathematics; they could all have followed from ξ and the posits alone and in a sense do.

The icosahedron (Dok. 364) is not a special case: every geometric body with non-crystallographic symmetry encounters the same structure — lattice-compatible part

in Aut(D_4), remainder as phase in $GF(3^k)$. This is not coincidence but an expression of the prime 3 as fundamental building block: what cannot exist as a root of unity in characteristic 3 must be a lattice operation, and vice versa.

Tool and method

Claude (Anthropic) was used as a tool for: writing and executing the Python verification scripts; drafting the LaTeX sources from results and corrections of the author; maintaining the changelog and cross-references. The tool computes and writes; the author decides what is computed and written, judges every result and determines what it means physically. Every algebraic statement is backed by a Python verification script with explicit assertions — if an assertion fails, the statement is not used.

W.5 Summary: what FFGFT needs and what it does not

Layer	Content	Status
1	$\tilde{T} \cdot m = 1$ + principle of relativity	fundamental law + 3 posits
2	$\alpha = 1$, charge definition	posit, EM sector follows
3	$\xi = 4/30\,000$	only free parameter
4	Galois, Hodge, representation theory	consistency confirmations
—	SI anchors, SM comparisons	translation, not input
—	cosmic sector (H_0, Λ)	bracketed (SM likewise)

What FFGFT does *not* need: the Standard Model, its constants, its Lagrangian as input (only for comparison), external calibrations beyond ξ. What it delivers: a self-consistent structure that is parameter-lean in natural units and reproduces all known ratios in SI units. It becomes testable only when anchors are inserted — but the anchors are translation, not foundation.

[1] M. Planck, *Über irreversible Strahlungsvorgänge*, Sitzungsber. Preuß. Akad. Wiss. (1899) 440–480. Natural units from b ($= h$), a ($= h/k$), c and f ($= G$).

[2] O. Heaviside, *Electromagnetic Theory*, vol. 1, The Electrician, London, 1893. Rationalised units, $\alpha = 1$ convention (Heaviside-Lorentz).

[3] R. Lidl, H. Niederreiter, *Finite Fields*, 2nd ed., Cambridge University Press, 1997. ξ as a Galois number from group orders of $GF(9)^*$, $GF(27)$.

[4] J. Pascher, *Dok. 011: Fine-structure constant in FFGFT units*, FFGFT corpus.

[5] J. Pascher, *Dok. 019: Lagrangian formulation*, FFGFT corpus.

[6] J. Pascher, *Dok. 270: Projection chain $T^4 \to T^0$*, FFGFT corpus.

[7] J. Pascher, *Dok. 330: Operations on T^4, demarcations*, FFGFT corpus.

[8] J. Pascher, *Dok. 336: GF(9)-FFGFT bridge*, FFGFT corpus.

[9] J. Pascher, *Dok. 338: Galois masses and FFGFT*, FFGFT corpus.

Chapter X

Doc. 366 — Fields in FFGFT: Energy, Flux and Geometry

Abstract

This document develops a unified approach to the field structure of FFGFT, starting from an unexpected experimental result and ending at the orbital resonances of the planets. The entry point is deliberately concrete: the Goubau line experiment — energy transport along a single wire without a return conductor — forces the question of where energy actually flows to be taken seriously. The answer has been fully present in classical field theory since Poynting (1884); it is regularly obscured in introductory courses, however, by the useful but simplified picture of current and return path.

The goal is not to claim new physics. The goal is to make visible that antenna engineering, atomic physics, crystallography and celestial mechanics all realise structurally the same principle: the geometry of the carrier determines the permitted coupling angles. Coupling between two field systems is only possible where their mode structures are compatible. Incompatible modes do not couple — regardless of the field strength.

In FFGFT this connection is not an analogy but a consequence of the fundamental equation $\tilde{T} \cdot m = 1$ on the compact carrier $T^4/\mathbb{Z}_3$. The winding numbers (n_θ, n_φ) on this carrier are the topological invariants behind the selection rules of atomic physics, the Bragg conditions of crystallography and the integer orbital resonances of celestial mechanics. In FFGFT units ($c = \hbar = \alpha = 1$): $I = m$, $V = E$, $P = E \cdot m$ — exactly, not approximately.

Antenna engineering serves as the guide because it is the most mature practical application of electromagnetic field theory: the coupling conditions are not abstract there but codified in standards, measurement protocols and commercial simulators. What is known there as the Friis equation, impedance matching and array factor appears in atomic physics as a selection rule, in the crystal as the Bragg condition and in the solar system as the Laplace resonance — the same structure, different carriers.

What this document explains — and what it does not

Four questions are addressed, developed in this order:

1. Where does energy flow? Not in the conductor, but in the field around it. The Goubau experiment forces this answer experimentally; the Poynting vector delivers

it mathematically exactly. Current and voltage are projections, not primary quantities (§ 1–3).

2. What does an antenna do to the field? It actively reshapes it — in transmission by imposing boundary conditions, in reception by acting as a selective mode filter. Passive in a field it scatters; in near-field proximity two antennas mutually deform each other. This is not superposition but mutual deformation of mode structures (§ 4–9).

3. What is a particle in this picture? A stable field node — a localised mode concentration on the carrier. The near field of the particle is the Coulomb field; the far field arises only under acceleration or a state transition. The selection rules of atomic physics are the mode-compatibility conditions of this node (§ 10).

4. Does the principle hold on all scales? Yes — from the antenna through the atom and the crystal to the planetary orbits. The coupling condition is always the same: integer phase compatibility of the mode spectra. The scale changes; the structure of the condition does not (§ 11).

What this document does *not* explain: why the carrier is $T^4/\mathbb{Z}_3$ (that is **[POSIT]** in Doc. 365); why $\xi = 4/30\,000$ (that follows from Galois group orders, Doc. 338/365); how concrete masses follow from mode compatibility (Doc. 314, 317–323). Those documents are the sources cited here. This document is the approach from the concrete to the structural.

Reading note: Readers familiar with antenna engineering may skim § 1–3 and enter at § 4. Those who know the FFGFT background will find the connection to mode structure in § 5, § 10 and § 11.

List of symbols

Epistemic markers:

[B]	algebraically proven (Python script, exact arithmetic)
[K]	numerically confirmed
[S]	structurally plausible, identification open
[POSIT]	motivated input, not derived
[E]	established external mathematics or physics
[X]	checked and negative

Symbol	Meaning
Fields and energy	
$F^{\mu\nu}$	electromagnetic field strength tensor
$\mathbf{E}, \mathbf{B}$	electric and magnetic field
$\mathbf{H}$	magnetic field strength ($\mathbf{H} = \mathbf{B}/\mu_0$ in vacuum)
$\mathbf{S} = \mathbf{E} \times \mathbf{H}$	Poynting vector (energy flux density, W/m^2)
j^ν	electromagnetic four-current
u	electromagnetic energy density
ε_0, μ_0	permittivity of free space, permeability of free space
λ	wavelength ($\lambda = c/f$)
$\omega = 2\pi f$	angular frequency

$k = 2\pi/\lambda$	wave number
θ	elevation angle (radiation pattern) or glancing angle (Bragg)
ψ	polarisation mismatch angle between transmit and receive antenna
FFGFT quantities	
$\tilde{T}$	intrinsic time scale; $\tilde{T} \cdot m = 1$
$\xi = 4/30\,000$	dimensionless scaling parameter of FFGFT
$T^4/\mathbb{Z}_3$	compact orbifold carrier of FFGFT
(n_θ, n_φ)	winding numbers of a field mode on T^4; topological invariants
Antenna engineering	
R_{rad}	radiation resistance (resistance the antenna presents for radiated power)
G	antenna gain (relative to isotropic radiator)
D	directivity (gain without losses)
A_{eff}	effective aperture; $A_{\text{eff}} = (\lambda^2/4\pi)\,G$
η_A	aperture efficiency ($0 < \eta_A \leq 1$)
G_T, G_R	gain of transmit and receive antenna (Friis equation)
$\Delta\theta_{3\,\text{dB}}$	half-power beamwidth of main lobe
Z_{ant}	input impedance of the antenna
X_{ant}	reactance (imaginary part of Z_{ant})
ϕ_n	feed phase of the n-th element in a phased array
$AF(\theta)$	array factor (interference pattern of the phased array)
ε_r	relative permittivity (dielectric constant)
ε_{eff}	effective permittivity of a microstrip line
ΔL	fringe correction of the patch antenna length
k_{SPP}	wave number of a surface plasmon polariton
$\varepsilon_m, \varepsilon_d$	(complex) permittivity of metal and dielectric (SPP)
Atomic physics and scales	
$\hbar$	reduced Planck constant
$\alpha \approx 1/137$	fine-structure constant; in FFGFT units $\alpha = 1$
n, l, m	principal, orbital angular momentum, and magnetic quantum numbers
E_n	energy levels of the hydrogen atom; $E_n = -13.6\,\text{eV}/n^2$
$r_e \approx 2.82\,\text{fm}$	classical electron radius
$\lambda_C = \hbar/m_e c$	Compton wavelength of the electron ($\approx 386\,\text{fm}$)
$a_0 \approx 52\,917\,\text{fm}$	Bohr radius
σ_T	Thomson cross-section; $\sigma_T = \frac{8\pi}{3} r_e^2$
S_{AB}	overlap integral of two atomic orbitals ψ_A, ψ_B
Q	normal coordinate of a molecular vibration
$\vec{\mu}$	electric dipole moment of a molecule
d	lattice plane spacing in a crystal (Bragg condition)
n_1, n_2, n_3	mean motions (orbital frequencies) of the Galilean moons

X.1 Starting point: fields or current?

Every introductory course on electricity emphasises the principle: current needs a return path. Electrons leaving one conductor must return on the other. This picture is entirely correct for technical purposes — it describes circuits precisely. It conceals a deeper question, however: where does the energy actually flow?

The Goubau experiment **[Q]** (G. Goubau, 1950) demonstrates the answer experimentally. A high-frequency signal source (~ 900 MHz) feeds a single isolated wire via a conical transition (launcher). At the other end an identical cone (catcher) converts the signal back into a coaxial line. Between the two cones there is no second conductor, no return path in the classical sense — yet the arrangement transfers the signal with low losses [1].

A physical interruption of the field (for instance a hand placed over the wire) immediately interrupts the transfer. The energy flows in the field around the wire, not in the wire itself.

X.2 The Poynting vector as the primary description

The energy flux density of electromagnetic fields is the Poynting vector [3, 4]:

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}.$$

Its direction (right-hand rule: $\mathbf{E} \to \mathbf{H}$, thumb in the direction of flux) indicates where energy is transported. The flux density integrated over the cross-section of a cable gives exactly the electrical power **[E]**:

$$P = \oint_A \mathbf{S} \cdot d\mathbf{A} = V \cdot I.$$

This is not an approximation, not a coincidence — it is an exact identity [4]. Current and voltage are macroscopic projections of the field process, not the primary carriers of energy.

Consequence. For a coaxial line the $\mathbf{E}$ field lines run radially from the inner conductor to the outer shield; the $\mathbf{H}$ field lines form concentric rings. The cross product $\mathbf{E} \times \mathbf{H}$ points axially — along the line. Energy flows in the dielectric between inner and outer conductor, not in the copper.

X.3 The Goubau line: fields without a return conductor

The Goubau line (G-line) has no outer conductor. The field configuration differs from the coaxial line only gradually when measured close enough to the wire **[E]**.

Field configuration

A voltage pulse propagating on the wire creates alternating positive and negative potential regions along the conductor. The electric field lines run between these regions — not from wire to return conductor, but from one section of the wire to the next. The dielectric (the red PVC insulation) slows the propagation of the fields and bends the field lines towards the wire axis, just as glass bends light rays [1, 2]. This mechanism keeps the fields close to the wire — it is not cosmetic but physically necessary for practical operation.

The cone transition as impedance matching

The cones (launcher / catcher) match the wave impedance of the coaxial line ($50\,\Omega$) to the characteristic impedance of the G-line (typically a few $100\,\Omega$) **[E]**. Inside the cone the field lines expand from the coaxial pattern to the G-line pattern. At the opening of the cone the field lines snap and attach to the wire. The return path for the current in the coaxial cable is provided by the displacement currents inside the cone — the complex fields in the cone supply exactly the right forces to drive electrons in the outer surface.

Losses

Because current flows in the wire, I^2R losses occur — exactly as in a conventional two-wire line. This shows: the G-line is not a pure waveguide problem. The energy flux has a field component (Poynting) and a conduction component (Ohm); both describe the same physics from different perspectives.

X.4 FFGFT units: current is mass, voltage is energy

In FFGFT units ($c = \hbar = \alpha = 1$, Doc. 011; Doc. 365 Layer 2) electromagnetic quantities simplify to geometric expressions in mass and time **[K]**:

$$\begin{aligned} e &= 1 \quad \text{(elementary charge dimensionless)}, \\ V &= E/e = E \quad \text{(voltage = energy)}, \\ I &= e/T = 1/T = m \quad \text{(current = mass, via } \tilde{T} \cdot m = 1\text{)}. \end{aligned}$$

The power equation then reads:

$$P = V \cdot I = E \cdot m,$$

a relation that follows directly from the fundamental equation $\tilde{T} \cdot m = 1$ (Doc. 365 Layer 1) **[K]**.

Maxwell equations in FFGFT units. In FFGFT units (Doc. 365 § 2) the Maxwell equations take the form:

$$\partial_\mu F^{\mu\nu} = j^\nu, \qquad \partial_{[\mu} F_{\nu\rho]} = 0.$$

No prefactors, no coupling constants other than j^ν as a geometric object. The coupling is absorbed in the unit choice, not eliminated; the physical content is identical to the SI system.

X.5 Field lines as geometric objects on $T^4/\mathbb{Z}_3$

In FFGFT the physical carrier is the compact orbifold $T^4/\mathbb{Z}_3$ (Doc. 365 Layer 1) **[POSIT]**. Fields are sections of vector bundles over this carrier; their periodicity conditions enforce discrete mode spectra.

Field lines and winding numbers

On a torus T^4 non-trivial field configurations have topological invariants — winding numbers (n_θ, n_φ) that record how many times a field line traverses the torus in each direction (Doc. 314; Doc. 247) **[B]**. These winding numbers classify the modes and determine — via $\tilde{T} \cdot m = 1$ — the mass scale of the respective sector. A field line in this description is not an auxiliary construct but a topological object with physical relevance.

Field lines in the G-line context

The behaviour of field lines in the Goubau line — expansion in the cone, snapping at the opening, attachment to the wire — is from this perspective a macroscopic example of the energy-flux minimisation principle. At every position and time the flux chooses the path that keeps the Poynting vector consistent with the carrier. This is not mysterious behaviour but the continuum version of topological optimisation on $T^4/\mathbb{Z}_3$.

Field shape and carrier: no one-to-one correspondence

The fundamental equation $\tilde{T} \cdot m = 1$ says nothing about the *shape* of a field in space. It specifies which mass scales belong to which time scales — not how the field looks geometrically. The compact carrier $T^4/\mathbb{Z}_3$ is the topology that prescribes the permitted boundary conditions; the actual field configurations on and around this carrier are different from it and variously deformed depending on the situation.

A comparison clarifies this: the length of a guitar string determines the permitted frequency spectrum — but the sound wave in the surrounding space is not string-shaped. Equally, $T^4/\mathbb{Z}_3$ enforces the mode spectrum; what appears as a field in the three-dimensional observation space is a projection of it, depending on the concrete boundary conditions, the ambient geometry and the mode occupation. Close to the conductor the G-line field is cylindrical, further away elliptically deformed, at inhomogeneities asymmetric — none of this is torus-shaped. The torus is the internal geometric carrier, not the shape of the field.

Interaction: carrier determines permitted coupling angles

When two fields from different carriers overlap in space, their interaction is not simple superposition. Each field carries its own mode structure, fixed by the winding numbers (n_θ, n_φ) of its carrier. Coupling is only possible where the phase conditions of both mode spectra are compatible — where the winding numbers fit together as integers. The geometry of the carrier thus determines the *angles* of interaction: under which phase relations energy transfer occurs and under which it does not **[S]**.

This is worked out quantitatively in the corpus: the Galois lattice condition from Doc. 328 (coupling regimes) and Doc. 343 § G′ specifies that coupling is only effective at frequency ratios in the Galois lattice $\{2, 3, 5, 7, 11, 13\}$; the resolution floor 100ξ is the sharpest bound below which two modes are no longer distinguishable as separate. The apparently free shape of a field in space is the result of these compatibility conditions, projected onto the observation space **[S]**.

X.6 Antenna engineering as mature field theory

Antenna engineering is the most mature practical application of electromagnetic field theory. Decades of measurements, standards and calculation procedures are available; the coupling conditions are not abstract but anchored in handbooks, measurement protocols and commercial simulators. It is precisely for this reason the most suitable introduction to the general principle: the geometry of the carrier determines the permitted coupling angles.

Fundamental quantities and their field interpretation

Every antenna is a transducer between a guided wave (coaxial cable, waveguide) and a free wave in space. The decisive quantities are **[E]**:

- **Radiation resistance** R_{rad}: The resistance the antenna presents to the guided wave because it radiates energy into space. For the half-wave dipole: $R_{\text{rad}} \approx 73.1\,\Omega$. In FFGFT units ($I = m$, $V = E$): $R_{\text{rad}} = P/I^2 = E/m$. The radiation resistance is thus an energy-per-mass relation — a direct echo of $\tilde{T} \cdot m = 1$ **[S]**.
- **Gain** G and **directivity** D: Gain gives the factor by which the radiation intensity in the main direction exceeds that of an isotropic radiator. It is a pure field-geometry problem: how do the phase conditions of the carrier (antenna length, element spacing, feed phase) concentrate the Poynting vector in a preferred direction? For the half-wave dipole: $D = 1.64$ (2.15 dBi).
- **Effective aperture** A_{eff}: The area from which a receiving antenna draws energy. Relation to gain **[E]**:

  $$A_{\text{eff}} = \frac{\lambda^2}{4\pi} G.$$

  This formula is pure field geometry: wavelength and gain determine the effective area, not the physical size of the antenna.

Friis transmission equation: coupling of two carriers

The Friis equation describes energy transfer between two antennas in free space [7, 6]:

$$\frac{P_R}{P_T} = G_T G_R \left(\frac{\lambda}{4\pi r}\right)^2 \cos^2\psi,$$

where P_T is the transmitted, P_R the received power, G_T, G_R the gains of transmit and receive antennas, r the distance and ψ the polarisation mismatch angle.

Every factor of this equation corresponds to a compatibility condition between the carriers:

Factor	Physical meaning	FFGFT correspondence
G_T	focussing of transmit carrier	mode geometry of source field
G_R	focussing of receive carrier	mode geometry of receive field
$(\lambda/4\pi r)^2$	free-space attenuation	geometric propagation factor
$\cos^2\psi$	polarisation overlap	winding-direction compatibility

The $\cos^2\psi$ factor is the most direct expression of the principle: at orthogonal polarisation ($\psi = 90°$) transfer is zero — not because of absorption or loss but because the mode geometries are incompatible. No transmitter of arbitrary power overcomes this. This is identical to the statement about winding numbers: if they do not match, no energy transfer occurs **[S]**.

Impedance matching as a phase condition

For an antenna to radiate at maximum efficiency its input impedance must be the complex conjugate of the source impedance (conjugate matching) **[E]**:

$$Z_{\text{ant}} = R_{\text{rad}} + jX_{\text{ant}}, \qquad Z_{\text{src}} = R_{\text{src}} - jX_{\text{ant}}.$$

The imaginary part — the reactance — measures stored field energy near the antenna (near field, reactive power). If it is not compensated, energy oscillates between source and antenna near field without being radiated. This is the same phenomenon as the reactive power on the G-line (§ 3): energy pulses in the near field without net transport.

For the G-line the cone transition was the matching structure ($50\,\Omega$ to $\sim 300\,\Omega$); for antennas this is handled by $\lambda/4$ transformers, baluns or matching networks. The mechanism is structurally identical: transition between two mode spectra through geometrically enforced phase condition.

Antenna types as mode geometries

Different antenna types realise different mode geometries. Each type is an independent carrier with a characteristic mode spectrum; the radiation pattern in the far field is the projection of this spectrum onto the observation space.

1. Short dipole ($\ell \ll \lambda$). The short dipole is the simplest limiting case: a straight conductor whose length is far below the wavelength. The current distribution is approximately triangular; the far field is analytically exact **[E]**:

$$E_\theta \propto \frac{I_0\,\ell}{\lambda r}\,\sin\theta, \qquad F(\theta) = \sin^2\theta.$$

The radiation pattern $F(\theta) = \sin^2\theta$ is a torus shape around the dipole axis: maximum radiation perpendicular to the axis ($\theta = 90°$), zero along the axis ($\theta = 0°, 180°$). Directivity $D = 1.5$ (1.76 dBi). Radiation resistance:

$$R_{\text{rad}} = 20\pi^2 \left(\frac{\ell}{\lambda}\right)^2 \approx 197\,\Omega \cdot \left(\frac{\ell}{\lambda}\right)^2.$$

For $\ell = 0.1\lambda$: $R_{\text{rad}} \approx 2\,\Omega$ — very low, hence severe mismatch to $50\,\Omega$ systems. The mode structure is simple: a single electric dipole mode whose winding geometry enforces the torus shape of the radiation pattern.

2. Half-wave dipole ($\ell = \lambda/2$). The half-wave dipole is the most important reference radiator in antenna engineering. The current distribution is sinusoidal; the radiation pattern is slightly narrower than the short dipole **[E]**:

$$F(\theta) = \frac{\cos^2\!\left(\frac{\pi}{2}\cos\theta\right)}{\sin^2\theta}.$$

Directivity $D = 1.64$ (2.15 dBi). Radiation resistance $R_{\text{rad}} = 73.1\,\Omega$ — nearly ideal for $50\,\Omega$ or $75\,\Omega$ coaxial cable (mismatch $< 2.5 : 1$ without matching network). The half-wave resonance condition ($\ell = \lambda/2$) enforces exactly the phase relationship at which radiation resistance and feed resistance enter a real ratio: the imaginary part of the input impedance vanishes at resonance. This is an integer mode-compatibility condition — the length must be a half-integer multiple of the wavelength.

3. Monopole ($\ell = \lambda/4$) over ground plane. A quarter-wave monopole over an infinitely extended conducting plane is equivalent to the half-wave dipole by image theory: the ground plane supplies the mirror image. The radiation pattern is the lower hemisphere of the dipole torus, reflected upward; all radiated power concentrates in the upper half-space **[E]**:

$$D_{\text{mono}} = 2\,D_{\text{dipole}} = 3.28 \quad (5.15\,\text{dBi}).$$

Radiation resistance $R_{\text{rad}} = 36.6\,\Omega$ (half that of the dipole, since only half the space). Practical implementations: car antenna, mobile mast [6]. The gain increase over the dipole arises not from more input power but from the geometric mirroring condition — doubling the solid angle contributing to radiation is a pure mode-geometry question.

4. Yagi–Uda antenna. The Yagi–Uda antenna consists of a driven dipole (radiator), a slightly longer passive reflector and a series of shorter passive directors [8, 6]. All parasitic elements are excited by induction; their length determines the phase of the induced current:

- Reflector ($\ell \approx 0.5\lambda + 5\,\%$): capacitive, re-radiates with phase lag ⇒ cancellation to the rear.
- Directors ($\ell \approx 0.5\lambda - 5\,\%$): inductive, re-radiate with phase lead ⇒ constructive interference forward.

Gain as a function of the number of elements n (rule of thumb) **[E]**:

$$G \approx 10\log_{10}(0.8\,n + 1)\ \text{dBd} \quad (n = \text{number of elements}).$$

Example values: 3 directors (5 elements) ⇒ $G \approx 7$ dBd, 10 directors ⇒ $G \approx 10$ dBd. The half-power beamwidth (in the E plane) is approximately:

$$\Delta\theta_{3\,\text{dB}} \approx \frac{52^\circ}{\sqrt{G_{\text{lin}}}},$$

where G_{lin} is the linear gain factor. The radiation pattern narrows with increasing element count — the carrier becomes more selective in its coupling angles.

5. Horn antenna. The horn antenna is the transition from waveguide to free space. The waveguide supports a discrete mode structure (TE_{10} fundamental mode for rectangular waveguide); the horn adiabatically expands this mode into a plane wave in the far field [6, 9]. The half-power beamwidth depends directly on the aperture dimensions:

$$\Delta\theta_E \approx 56^\circ \frac{\lambda}{a}, \qquad \Delta\theta_H \approx 67^\circ \frac{\lambda}{b},$$

where a is the E-plane aperture and b the H-plane aperture. Gain of a pyramidal horn with aperture $A = a \times b$:

$$G = \eta_A \frac{4\pi A}{\lambda^2}, \qquad \eta_A \approx 0.5\text{–}0.7.$$

The aperture efficiency $\eta_A < 1$ arises from the non-uniform phase distribution in the aperture (spherical rather than plane wave at the horn mouth). This is directly the mode incompatibility between waveguide near field and far-field plane wave: the better the phase matching (flatter aperture), the higher η_A.

6. Parabolic antenna. A parabolic antenna of diameter D and focal length f focuses incident microwaves at the focus. In transmission a primary radiator at the focus converts a spherical wave into a plane wavefront at the aperture plane. The far field is the Fourier transform of this aperture distribution **[E]**. For a circular, uniformly illuminated aperture:

$$F(\theta) = \left[\frac{2\,J_1(\pi D \sin\theta/\lambda)}{\pi D \sin\theta/\lambda}\right]^2,$$

where J_1 is the first-order Bessel function [6]. Gain and half-power beamwidth:

$$G = \eta_A \frac{\pi^2 D^2}{\lambda^2}, \qquad \Delta\theta_{3\,\mathrm{dB}} \approx 70^\circ \frac{\lambda}{D}.$$

For a 3-m parabolic antenna at 10 GHz ($\lambda = 3$ cm): $G \approx 5 \cdot 10^4$–$7 \cdot 10^4$ (47–48 dBi), $\Delta\theta \approx 0.7^\circ$. The Bessel function in the radiation pattern is not an approximation but the exact consequence of the circular symmetry of the aperture: the mode geometry (circle) completely determines the coupling pattern in the far field.

7. Patch antenna (microstrip). A rectangular patch antenna of length L and width W over a ground plane (dielectric ε_r, thickness h) resonates when L equals half an effective wavelength [9, 6]:

$$L = \frac{\lambda}{2\sqrt{\varepsilon_{\mathrm{eff}}}} - 2\Delta L,$$

where $\varepsilon_{\mathrm{eff}}$ is the effective permittivity and ΔL the fringe correction:

$$\varepsilon_{\mathrm{eff}} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2}\left(1 + \frac{12h}{W}\right)^{-1/2}.$$

The radiation pattern is broadside in the H plane (no focussing), narrow in the E plane; typical gain 5–8 dBi. The permitted modes are completely determined by geometry and material (L, W, h, ε_r) — no free parameters. Coupling to an external wave is only possible when its polarisation and angle of incidence are compatible with the resonance mode of the patch. Cross-polarisation isolation (> 30 dB) shows: orthogonal polarisation practically does not couple, even when the frequency matches exactly.

8. Phased array. A phased array consists of N identical radiators with individually controlled feed phase ϕ_n. The array pattern is the product of the element radiation pattern $F_{\text{elem}}(\theta)$ and the array factor [10, 6]:

$$AF(\theta) = \sum_{n=0}^{N-1} \exp[j(n\,kd\cos\theta + \phi_n)]\,,$$

where $k = 2\pi/\lambda$ is the wave number and d the element spacing. By choosing $\phi_n = -n\,kd\cos\theta_0$ the main beam steers to angle θ_0 — electronic beam steering without mechanical movement. Gain for N elements with spacing $d = \lambda/2$:

$$G_{\text{array}} = N \cdot G_{\text{elem}}.$$

The phased array is the purest example of the coupling-angle principle: the feed phases ϕ_n are integer multiples of a basic phase step — they determine under which angle constructive interference occurs. Every other angle combination gives destructive interference. In FFGFT language: the ϕ_n are discrete winding phase values; the array factor selects the compatible coupling angles **[S]**.

Summary table.

Type	Gain	$\Delta\theta_{3\,\text{dB}}$	Mode geometry
Short dipole	1.76 dBi	90°	electric dipole mode
Half-wave dipole	2.15 dBi	78°	resonance $\ell = \lambda/2$
Monopole ($\lambda/4$)	5.15 dBi	$\approx 39^\circ$ (half-space)	image mode over ground
Yagi (5 dir.)	≈ 8 dBd	$\approx 30^\circ$	interference array
Horn antenna	15–25 dBi	3°–15°	TE_{10} aperture mode
Parabolic ($D = 3$ m, 10 GHz)	47–48 dBi	0.7°	Fourier aperture
Patch	5–8 dBi	60°–120°	resonance mode in dielectric
Phased array (N)	$N \cdot G_{\text{elem}}$	$\propto 1/\sqrt{N}$	phase-steered array

In every case: the physical geometry of the antenna determines the permitted mode spectrum; what appears in the far field as a radiation pattern is the projection of this spectrum onto the observation space. No antenna type transfers energy at angles that contradict its mode geometry — not approximately but structurally. This is the technically mature, quantitatively anchored form of the principle formulated in FFGFT as winding-number compatibility **[S]**.

Reciprocity: transmitter and receiver as dual descriptions

The reciprocity principle of antenna engineering states: gain, radiation pattern and impedance of an antenna are identical in transmit and receive mode [6, 4]. This symmetry is not coincidental — it reflects the time-reversal symmetry of the Maxwell equations, and with it the duality of field source and field sink. In FFGFT language: transmitter and receiver are reciprocal projections of the same field process, just as $\tilde{T}$ and m are reciprocal representations of the same object **[S]**.

X.7 Multiple field carriers in space: boundary conditions, coupling and apparent superposition

The superposition principle of classical field theory states: in a linear medium the fields of independent sources add linearly. Each partial field propagates as if the other were not present. This is an excellent approximation for widely separated, weakly coupling sources — and it works very well in practice. It conceals, however, what physically actually occurs.

The total field as the consistent solution of all boundary conditions

Every object in a field — an antenna, a conductor, a dielectric, an atom — is itself a field carrier with its own mode structure. It imposes boundary conditions: the tangential components of the electric field must vanish at every conducting surface; the normal components of flux density must be continuous across interfaces **[E]**. The total field is not the sum of the partial fields — it is the unique consistent field satisfying all boundary conditions simultaneously.

Superposition is the approximation that arises when the boundary conditions of individual objects are spatially far enough apart not to influence each other. As soon as objects enter near-field range, this approximation breaks down — not gradually but structurally.

Near field: where boundary conditions collide

The near-field zone of an antenna extends to roughly $r < 2D^2/\lambda$ (Fraunhofer boundary), where D is the antenna size **[E]**. In this zone the field cannot be described as a plane wave; it contains reactive components (stored field energy oscillating back and forth without being radiated). Bringing a second antenna into this zone causes the following:

- The input impedance of both antennas changes — the resonance frequency shifts, the radiation resistance changes. This is not field addition but mutual deformation of the mode structures.
- The radiation pattern of each antenna changes because the boundary condition of the other deforms the field distribution in the shared near space.
- Energy can be transferred directly between near fields — without radiation into the far field. This is the principle of inductive coupling (transformer, Qi charging), capacitive coupling and resonant near-field coupling (Kurs et al. 2007 [12], wireless transfer over metres).

Mutual coupling in antenna arrays

In a phased array with element spacing $d < \lambda$ the mutual coupling is not negligible [10, 9]. The input impedance of the n-th element depends on all others:

$$Z_{nn}^{\text{active}} = Z_{nn}^{\text{isolated}} + \sum_{m \neq n} Z_{nm} \frac{I_m}{I_n},$$

where Z_{nm} is the mutual impedance between elements n and m. The overall behaviour of the array is only describable by solving the complete system of equations — not by addition of single-element fields. In practice mutual coupling is compensated by optimising element geometry and feed phases; it is a design problem that industry solves with full electromagnetic simulators (MoM, FEM, FDTD).

Cavity resonator: the total field as an eigenmode

A metal cavity allows only discrete resonance frequencies determined by its geometry [9, 4]. For a rectangular cavity of dimensions $a \times b \times c$:

$$f_{mnp} = \frac{c}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{c}\right)^2}.$$

Placing an object inside the cavity — a dielectric block, a conductor, biological tissue — shifts all resonance frequencies. The entire mode spectrum of the cavity changes, not only the modes that directly touch the object. The total field is the eigenmode of the composite system (cavity + object), not the sum of the empty-cavity eigenmode and a scattered field. This shift is measurable and exploited in microwave dielectric spectroscopy (cavity perturbation method).

Faraday cage: boundary conditions extinguish a field

The Faraday cage is the most extreme case: a closed conducting shell enforces vanishing tangential electric field on its inner surface. The external field continues unchanged; the internal field is forced to zero **[E]**. The total field is the unique solution of the Maxwell equations with this boundary condition — not superposition but cancellation through enforced compatibility.

Incomplete shielding (gaps, finite conductivity) shows the principle in attenuated form: fields penetrate but only with mode geometries permitted by the gap geometry. A slot of width w allows fields with $\lambda > 2w$ to pass through strongly attenuated (cut-off condition of the slot waveguide). This is again mode compatibility: only fields whose wavelength is compatible with the slot geometry couple through.

Plasmons: field and matter inseparable

At the interface between metal and dielectric, surface plasmon polaritons (SPP) arise — coupled modes of electromagnetic field and collective electron oscillation [11]. The dispersion relation is:

$$k_{\mathrm{SPP}} = \frac{\omega}{c}\sqrt{\frac{\varepsilon_m\,\varepsilon_d}{\varepsilon_m + \varepsilon_d}},$$

where ε_m is the (complex) permittivity of the metal and ε_d that of the dielectric. The SPP is neither pure electromagnetic field nor pure charge carrier oscillation — it is a joint mode of both carriers. Superposition would be meaningless: the field exists only in coupling with matter.

This is the sharpest available evidence that “multiple fields in space” are not independent objects. The mode structure of the total system is the physical reality; decomposition into partial fields is a computational method that works under certain conditions.

FFGFT interpretation: one field, many projections

In FFGFT there is only one field on the carrier $T^4/\mathbb{Z}_3$ **[POSIT]**. What appears as “multiple fields” are projections of different mode groups onto the observation space. The apparent superposition is a representational question.

When two objects with different mode structures interact, the result is not the sum of their fields but the joint eigenmode of the composite carrier — determined by the compatibility of the winding numbers of both carriers. This is the same principle that appears in antenna engineering as mutual coupling, in cavity physics as cavity perturbation and in nanophotonics as plasmon coupling **[S]**.

The complexity of field interaction is therefore not a complication that sufficient computational effort can remove — it is the direct consequence of every carrier imposing boundary conditions on the total field. The more carriers in space, the more simultaneously required compatibility conditions, the more complex the total field. This is not superposition — this is geometry.

X.8 Particles as field nodes: near and far field at localised field structures

In classical physics a particle is an object that *has* a field — the charge generates the Coulomb field, the mass generates the gravitational field. In FFGFT this description is inverted: a particle is a stable, localised field configuration — a node in the global field on the carrier $T^4/\mathbb{Z}_3$ **[POSIT]**. Mass is not an intrinsic property of the node but follows from the fundamental equation $\tilde{T} \cdot m = 1$: the time scale of the field configuration determines the mass (Doc. 365 Layer 1, Doc. 124) **[K]**.

This inversion has direct consequences for the structure of the field around a particle — and it makes the connection to antenna engineering visible.

Stationary particle: pure near field

A stationary point charge e generates the Coulomb field **[E]**:

$$\mathbf{E}(r) = \frac{e}{4\pi\varepsilon_0}\,\frac{\hat{r}}{r^2}, \qquad \mathbf{B} = 0.$$

In FFGFT units ($e = 1$, $4\pi\varepsilon_0 = 1$): $V(r) = -1/r$ [21]. This field falls off as $1/r^2$ (field strength) or $1/r$ (potential), carries no energy into the far field and is fully reactive — identical to the near field of a stationary antenna. The Poynting vector $\mathbf{S} = \mathbf{E} \times \mathbf{H}$ is zero for a stationary charge: no energy flux, only stored field energy in the node structure.

The Coulomb field is the near field of the particle. It deforms the space around the node, imposing boundary conditions on all other fields in its vicinity — exactly as a passive antenna deforms the field in its surroundings without transmitting.

Accelerated particle: near field and radiated far field

As soon as the charge is accelerated, a far field arises — a detaching electromagnetic wave. The Larmor formula gives the radiated power [5, 4]:

$$P = \frac{e^2 a^2}{6\pi\varepsilon_0 c^3} = \frac{2}{3}\frac{e^2 a^2}{c^3} \quad \text{(in Gaussian units)},$$

where a is the acceleration. In FFGFT units ($c = e = 1$): $P = \frac{2}{3}a^2$.

The far field falls off as $1/r$ (not $1/r^2$), permanently transports energy away and is propagating — identical to the far field of a driven antenna. The boundary between near and far field lies at $r \sim \lambda/2\pi$ (transition zone), where $\lambda = c/f$ and f is the characteristic frequency of the acceleration.

This is the antenna principle at the particle level:

- Stationary charge = undriven antenna: near field only, no radiation.
- Uniformly moving charge = antenna on direct current: no alternating current, no radiation (Cherenkov radiation is a special case for $v > c/n$).
- Accelerated charge = driven antenna: near field plus radiated far field, power $\propto a^2$.

Atomic transition: quantised far field

In the atom the permitted node structures are discrete through the quantum numbers (n, l, m) — these are the permitted modes of the Coulomb potential, i.e. the eigenmodes of the atomic carrier. A transition between two states is a change in node structure; a photon is emitted — the radiated far field carries away a discrete energy $\Delta E = \hbar\omega$. The selection rules [14, 13]:

$$\Delta l = \pm 1, \qquad \Delta m = 0, \pm 1$$

are compatibility conditions: only transitions in which the change of mode geometry is compatible with the geometry of a dipole photon are permitted. Quadrupole and higher transitions ($\Delta l = \pm 2$) are possible but suppressed by a factor $\alpha^2 \approx 10^{-4}$ — because the mode compatibility is poorer.

In FFGFT language: the winding numbers of the initial and final state must differ by exactly the winding numbers of a photon. The selection rules are not postulated but follow from the geometry of the carrier **[S]**.

The “size” of a particle: extent of the node

A particle as field node has no sharp geometric boundary. The “size” is the spatial extent of the node structure, i.e. the region where the near field dominates over the background field. For the electron these are the classical electron radius **[E]**:

$$r_e = \frac{e^2}{4\pi\varepsilon_0 m_e c^2} \approx 2.82\,\text{fm},$$

the Compton wavelength $\lambda_C = \hbar/m_e c \approx 386\,\text{fm}$ and the Bohr radius $a_0 \approx 52\,917\,\text{fm}$. These three lengths are not arbitrary but linked through α [5, 21]:

$$r_e = \alpha\,\lambda_C = \alpha^2 a_0.$$

In FFGFT units ($\alpha = 1$) these three lengths collapse to one — the hierarchy arises only through the SI embedding via ξ (Doc. 365 Layer 3) **[K]**.

The node is therefore not point-like — it has a structure extending over several length scales. What appears as a "point" is the projection of this structure onto a measurement coarse enough not to resolve the inner structure.

Comparison table: antenna and particle

Property	Antenna	Particle (node)
Near field	reactive, $E \propto 1/r^2$–$1/r^3$	Coulomb field $E \propto 1/r^2$
Far field	radiated wave, $E \propto 1/r$	photon under acceleration
Radiation	only on current change	only under acceleration/transition
Mode structure	antenna geometry	quantum numbers (n, l, m)
Selection rule	impedance compatibility	$\Delta l = \pm 1$, $\Delta m = 0, \pm 1$
"Size"	physical dimension	node extent r_e, λ_C, a_0
Stationary	no far field	no far field (Coulomb only)
Passive in field	scattering, shadowing	polarisability, cross-section

Cross-section as a measure of node coupling

A particle in an external field scatters the incoming wave — it deforms the field in its vicinity, exactly like a passive conductor. The Thomson cross-section of the electron [5]:

$$\sigma_T = \frac{8\pi}{3} r_e^2 \approx 6.65 \times 10^{-29}\ \mathrm{m}^2$$

is the effective area from which the electron draws energy from the incoming field and re-emits it. This is directly analogous to the effective aperture A_{eff} of a receiving antenna (§ 8):

$$A_{\mathrm{eff}} = \frac{\lambda^2}{4\pi} G \quad \longleftrightarrow \quad \sigma_T = \frac{8\pi}{3} r_e^2 .$$

In both cases the effective area is not the geometric size of the object but the result of mode compatibility between incoming field and node structure. For photons with $\hbar\omega \gg m_e c^2$, Compton scattering applies — the node structure of the electron can no longer be described as point-like and the cross-section falls with increasing energy.

FFGFT interpretation: node as local mode concentration

In FFGFT a particle is a local concentration of modes on the carrier $T^4/\mathbb{Z}_3$ — a location where specific winding numbers (n_θ, n_φ) are occupied and the field locally assumes a stable, non-trivial configuration **[POSIT]**. The near field of the particle is the immediate vicinity of this node in the carrier, projected onto the observation space; the far field is the radiation arising when the node changes its mode occupation.

The boundary between near and far field is therefore not a sharp line but the transition between the region where the node structure dominates the field and the region where the field propagates as a free wave. This transition lies at $r \sim \lambda_C$ for relativistic descriptions

and at $r \sim a_0$ for atomic transitions — both are structural length scales of the node, not free parameters **[S]**.

X.9 Scale-invariance of the coupling condition: atoms, elements and planetary orbits

The principle from § 8–10 — geometry of the carrier determines permitted coupling angles, coupling only under mode compatibility — is not restricted to antennas or individual particles. It holds on all physical scales. The scale changes; the structure of the condition remains identical. Doc. 242 (scale ladder) elaborated this connection at the cosmological level; here it is developed starting from antenna engineering.

Atoms: discrete modes in the Coulomb potential

The Schrödinger equation in the Coulomb potential has discrete eigenmodes, classified by quantum numbers (n, l, m) [13, 14]:

$$E_n = -\frac{m_e e^4}{2\hbar^2 n^2} = -\frac{13.6\,\text{eV}}{n^2}.$$

This is nothing other than the resonance condition of a carrier: the carrier is the Coulomb potential (spherically symmetric, $V \propto -1/r$); the permitted modes are the solutions with integer n. The atom is a cavity resonator with spherical geometry — the electron node can only assume certain standing wave configurations, just as the TE_{mn} modes in a metal cavity (§ 9).

The selection rules $\Delta l = \pm 1$, $\Delta m = 0, \pm 1$ (§ 10) are the impedance-matching conditions of this resonator: a dipole photon ($l = 1$) couples only to transitions in which the change of node geometry is compatible with a dipole mode. A quadrupole photon ($l = 2$) couples with $\Delta l = \pm 2$ but more weakly by α^2 — because the mode overlap efficiency is smaller, identical to the aperture efficiency $\eta_A < 1$ of a horn antenna.

Periodic table: mode classification of a many-body system

The shell structure of the periodic table — $1s^2\,2s^2\,2p^6\,3s^2\,\ldots$ — is a classification by mode geometries of the effective central potential **[E]**:

$$\text{Shell } n: \quad 2n^2 \text{ places} \quad (s, p, d, f \text{ with spin}).$$

Each sub-shell (s, p, d, f) has a different orbital angular momentum l and thus a different spatial mode geometry. Chemical bonding arises through coupling of the boundary modes of two atoms: when the outermost occupied orbitals (HOMO/LUMO) have compatible geometry — sufficient spatial overlap, compatible symmetry — a covalent bond forms. The overlap integral

$$S_{AB} = \int \psi_A^*(\mathbf{r})\,\psi_B(\mathbf{r})\,d^3r$$

is the direct quantum-mechanical analogue of the polarisation factor $\cos^2\psi$ in the Friis equation (§ 8): $S_{AB} = 0$ means no coupling, regardless of the energy of the atoms.

The noble gases (closed shells, $S_{AB} \approx 0$ for all bonding partners) are the quantum-mechanical analogue of orthogonal polarisation: no energy transfer because no mode compatibility.

Crystal lattice: the Bragg condition as compatibility condition

In a crystal the atoms form a periodic lattice of lattice constant a. An electromagnetic wave (X-ray, neutrons, electrons) is coherently reflected only when the Bragg condition is met [15, 16]:

$$2d \sin\theta = n\lambda, \quad n \in \mathbb{Z},$$

where d is the lattice plane spacing and θ the glancing angle. This is an integer phase condition: the path-length difference between partial waves reflected at successive lattice planes must be an integer multiple of the wavelength. Only then do amplitudes add constructively — identical to the phased-array condition (§ 8):

$$\phi_n = -n\,kd\cos\theta_0.$$

The crystal lattice is a natural phased array with fixed element spacing d. Bragg reflection selects exactly the angles of incidence at which mode compatibility holds; all other angles give destructive interference. In FFGFT the reciprocal lattice vectors **G** correspond to the permitted winding vectors of the carrier; Bragg reflection is the macroscopic manifestation of the quantisation condition on the torus **[S]**.

Molecular vibrations: infrared activity as a selection rule

A molecular vibrational mode is infrared-active (absorbs IR radiation) only when it produces a change in the electric dipole moment [17, 18]:

$$\left(\frac{\partial\vec{\mu}}{\partial Q}\right)_{Q=0} \neq 0,$$

where Q is the normal coordinate of the vibration. This is the impedance-matching condition between photon (dipole radiation) and vibrational mode: only when the vibration has a dipole geometry does it couple to a dipole photon. Symmetric stretching vibrations (as in N_2 or O_2) do not change the dipole moment — they are IR-inactive but Raman-active (coupling to quadrupole photons). This explains why N_2 and O_2 are not greenhouse gases: their vibrational modes are mode-incompatible with IR dipole radiation.

Planetary orbits: integer resonances in the gravitational field

The gravitational field of the sun is the carrier of the planetary system. Its modes are the Kepler orbits, classified by the orbital parameters (a, e, i). When two planets are near an integer resonance $p : q$ ($p, q \in \mathbb{Z}$), small gravitational perturbations accumulate coherently over many revolutions (Doc. 242) [24, 19]. The Laplace resonance of the Galilean moons is the most famous example [19, 20]:

$$\frac{T_{\text{Io}}}{T_{\text{Europa}}} \approx \frac{1}{2}, \qquad \frac{T_{\text{Europa}}}{T_{\text{Ganymede}}} \approx \frac{1}{2}, \qquad n_{\text{Io}} - 3n_{\text{Europa}} + 2n_{\text{Ganymede}} = 0$$

(exact relation of the mean motions $n = 2\pi/T$). This relation is not a coincidence: it is a stable configuration because at exact resonance the mutual perturbations cause no secular energy input, only periodic. The system has an attractor at the integer resonance — identical to the attractor of an Arnold tongue in nonlinear dynamics (Doc. 328 § coupling regimes) **[K]**.

Further examples:

- Jupiter/Saturn: $5:2$ resonance ("great inequality", known since Laplace 1785) **[Q]**.
- Neptune/Pluto: $3:2$ resonance (Plutinos) **[Q]**.
- Kirkwood gaps in the asteroid belt: gaps at resonances $1:3$, $1:4$ with Jupiter (destructive resonance, not stabilising) **[Q]**.
- Saturn's rings: gaps (Cassini Division) at $2:1$ resonance with Mimas **[Q]**.

The coupling condition is the same as in antenna engineering: integer phase compatibility. The carrier differs (gravitational rather than electromagnetic field); the structure of the condition is identical.

Summary table: coupling condition on all scales

Scale	Carrier	Coupling condition	Example	Analogue
Antenna	radiator geometry	$\cos^2\psi$, impedance	dipole/Yagi	Friis equation
Atom	Coulomb potential	$\Delta l = \pm 1$	H lines	selection rule
Molecule	eff. potential	$\partial\mu/\partial Q \neq 0$	IR activity	dipole coupling
Crystal	lattice periodicity	$2d\sin\theta = n\lambda$	Bragg reflection	phased array
Periodic table	shell structure	overlap S_{AB}	covalence	aperture overlap
Mol. bond	orbital geometry	symmetry compatibility	HOMO/LUMO	mode overlap
Planets	gravitational field	$pn_1 = qn_2$	Laplace resonance	Arnold tongue

The common principle

All rows of the table describe the same thing: a system has a carrier with a mode spectrum; coupling to another system is only possible where the mode spectra are compatible; incompatible modes do not couple, regardless of the strength of the fields involved.

The technical maturity of antenna engineering — decades of measurements, standardised procedures, commercial simulators — makes it the most tangible realisation of this principle. Atomic mechanics, solid-state physics and celestial mechanics are the same principle on different carriers. In FFGFT the common carrier is $T^4/\mathbb{Z}_3$; the coupling conditions on all scales are projections of winding-number compatibility onto the respective observation space **[S]**.

X.10 Duality: field and particle, energy and current

FFGFT treats time–mass duality as a fundamental principle (fundamental equation $\tilde{T}\cdot m = 1$, Doc. 365) **[POSIT]**. The same duality appears in electromagnetic field theory:

Field view		Current view
Poynting vector $\mathbf{S} = \mathbf{E} \times \mathbf{H}$	$\longleftrightarrow$	$P = V \cdot I$
Field energy in dielectric	$\longleftrightarrow$	$I^2 R$ Joule loss
Evanescent wave outside	$\longleftrightarrow$	reactive power
Winding number (topological)	$\longleftrightarrow$	mass (dynamical)

In none of these pairs is one side "more fundamental" than the other — they describe the same field process from different projections. This is the precise analogue of time–mass duality in FFGFT: $\tilde{T}$ and m are reciprocal representations of the same object, not two different quantities **[K]**.

Reactive power and evanescent fields

On the G-line energy pulses back and forth in the periphery of the wire field — energy flux alternates direction in step with the field. This is reactive power in AC circuit theory, or — stated optically — the evanescent field on the shadow side of a totally reflecting interface **[E]**. In FFGFT this behaviour corresponds to the non-propagating part of the Fourier modes on $T^4/\mathbb{Z}_3$: modes below the spectral gap (Doc. 314 § H) carry no net energy but structure the spectrum **[S]**.

X.11 Exact agreement: Poynting integral and power formula

It is instructive to understand the exactness of this identity, because it is regularly presented in courses as an "approximate agreement" or a "plausibility argument". Poynting's theorem states **[E]**:

$$\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{S} = -\mathbf{j} \cdot \mathbf{E},$$

where $u = \frac{1}{2}(\varepsilon_0 E^2 + \mu_0^{-1} B^2)$ is the electromagnetic energy density. Integration over the volume of the cable with the corresponding boundary conditions yields exactly $P = V \cdot I$ — the only prerequisite is the validity of the Maxwell equations.

In FFGFT units (without ε_0, μ_0, e) the identity reduces to:

$$P = E \cdot m,$$

which follows immediately from $V = E$, $I = m$ and the fundamental equation $\tilde{T} \cdot m = 1$ **[K]**. The exactness is structural, not a numerical coincidence.

X.12 Summary

1. **Primary localisation.** Energy flows in the fields ($\mathbf{S} = \mathbf{E} \times \mathbf{H}$), not in the conductor. Voltage and current are macroscopic projections of the same field process **[E]**.
2. **Geometric objects.** Field lines on the carrier $T^4/\mathbb{Z}_3$ are topological objects with winding numbers; they classify modes and determine mass scales via $\tilde{T} \cdot m = 1$ **[B]**.
3. **FFGFT units.** With $c = \hbar = \alpha = 1$: $I = m$, $V = E$, $P = E \cdot m$ — exactly, not approximately. The Maxwell equations carry no prefactors **[K]**.

4. **G-line as illustration.** The Goubau experiment demonstrates macroscopically that energy transport without an explicit return conductor is possible because the transport occurs in the field. The cone transition corresponds to impedance matching; field lines locally optimise energy flux **[E]**.
5. **Duality.** Field view and current view are reciprocal projections — no pair is more fundamental. This mirrors the time–mass duality $\tilde{T} \cdot m = 1$ at the level of classical fields **[S]**.

References

[1] G. Goubau, *Surface waves and their application to transmission lines*, J. Appl. Phys. **21** (1950) 1119–1128. Foundational work on single-wire wave guidance; cone transition, field configuration, loss measurement of 6 dB/mile.

[2] F. Harms, *Elektromagnetische Wellen an einem Draht mit isolierender zylindrischer Hülle*, Ann. Phys. **23** (1907) 44–60. First analysis of the dielectric as a field-guiding layer; precursor of the Goubau line.

[3] J. A. Stratton, *Electromagnetic Theory*, McGraw-Hill, New York, 1941. Standard reference for the Poynting theorem (Ch. 2), boundary conditions, field energy.

[4] D. J. Griffiths, *Introduction to Electrodynamics*, 3rd ed., Prentice Hall, 1999. Poynting vector, Larmor formula, radiation resistance, reciprocity; accessible derivation of all standard results used in § 2–5.

[5] J. D. Jackson, *Classical Electrodynamics*, 3rd ed., Wiley, 1999. Larmor formula (Ch. 14), Thomson cross-section (Ch. 14.8), classical electron radius, Compton wavelength; canonical reference.

[6] C. A. Balanis, *Antenna Theory: Analysis and Design*, 4th ed., Wiley, 2016. Standard text: radiation patterns, gain, radiation resistance, short dipole, half-wave dipole, monopole, Yagi–Uda, horn, parabolic, patch, phased array, Friis equation, reciprocity theorem.

[7] H. T. Friis, *A note on a simple transmission formula*, Proc. IRE **34** (1946) 254–256. Original paper on the link budget equation; "*The received power is [...] proportional to the product of the transmitting and receiving gains.*"

[8] H. Yagi, *Beam transmission of ultra short waves*, Proc. IRE **16** (1928) 715–741. Original Yagi–Uda paper; reflector and director principle; gain through phase interference of passive elements.

[9] D. M. Pozar, *Microwave Engineering*, 4th ed., Wiley, 2012. Cavity resonator (Ch. 6), patch antenna (Ch. 14), horn antenna, waveguide modes, impedance matching, Smith chart, cavity perturbation.

[10] R. J. Mailloux, *Phased Array Antenna Handbook*, 3rd ed., Artech House, 2017. Array factor, beam steering, mutual coupling, active impedance matrix, grating-lobe condition; quantitative treatment of all array phenomena.

[11] S. A. Maier, *Plasmonics: Fundamentals and Applications*, Springer, 2007. Surface plasmon polaritons, dispersion relation (Ch. 2), coupling condition, near-field enhancement; standard text of nanophotonics.

[12] A. Kurs, A. Karalis, R. Moffatt, J. D. Joannopoulos, P. Fisher, M. Soljačić, *Wireless power transfer via strongly coupled magnetic resonances*, Science **317** (2007) 83–86. Resonant near-field coupling over 2 m with 40 % efficiency; "*The transfer is mediated by the overlap of the near fields.*" Direct evidence for energy transfer without radiation through mode compatibility.

[13] D. J. Griffiths, *Introduction to Quantum Mechanics*, Prentice Hall, 1995. Hydrogen atom, quantum numbers, selection rules $\Delta l = \pm 1$, $\Delta m = 0, \pm 1$ (Ch. 9); overlap integral and bonding theory (Ch. 7).

[14] C. Cohen-Tannoudji, B. Diu, F. Laloë, *Quantum Mechanics*, 2 vols., Wiley, 1977. Selection rules (Ch. XIII), dipole approximation, quadrupole suppression $\sim \alpha^2$; canonical reference for atomic transitions.

[15] C. Kittel, *Introduction to Solid State Physics*, 8th ed., Wiley, 2005. Bragg condition (Ch. 2), reciprocal lattice, Brillouin zone, lattice dynamics; standard solid-state text.

[16] N. W. Ashcroft, N. D. Mermin, *Solid State Physics*, Holt, Rinehart and Winston, 1976. Bragg scattering, phased-array analogy of reciprocal lattice (Ch. 6), electron bands; advanced reference.

[17] P. Atkins, J. de Paula, *Physical Chemistry*, 9th ed., Oxford University Press, 2010. IR activity condition $\partial\vec{\mu}/\partial Q \neq 0$ (Ch. 13), Raman activity, symmetry selection rules; standard physical chemistry text.

[18] E. B. Wilson, J. C. Decius, P. C. Cross, *Molecular Vibrations*, McGraw-Hill, 1955. Normal coordinates, selection rules, group theory of vibrational modes; classic original reference.

[19] C. D. Murray, S. F. Dermott, *Solar System Dynamics*, Cambridge University Press, 1999. Orbital resonances (Ch. 8), Laplace resonance Io/Europa/Ganymede, Kirkwood gaps, Cassini Division, Arnold tongues in celestial mechanics; authoritative modern reference.

[20] S. J. Peale, P. Cassen, R. T. Reynolds, *Melting of Io by tidal dissipation*, Science **203** (1979) 892–894. Laplace resonance as the cause of tidal heating of Io; "*The forced eccentricity [...] results from the Laplace relation* $n_1 - 3n_2 + 2n_3 = 0$." Spectacular prediction before Voyager 1 arrival.

[21] J. Pascher, *Doc. 365: The Foundation of FFGFT in Four Layers — fundamental equation, electromagnetic sector, scaling parameter and algebraic confirmations*, FFGFT corpus, Zenodo (2026). Layer 1 ($\tilde{T} \cdot m = 1$), Layer 2 (FFGFT units, $I = m$, $V = E$, Maxwell without prefactors), Layer 3 ($\xi = 4/30000$).

[22] J. Pascher, *Doc. 314: Lattice in Hilbert Space*, FFGFT corpus, Zenodo (2026). Fourier modes on $T^4/\mathbb{Z}_3$, winding numbers as topological invariants, shell count to norm 120, spectral gap.

[23] J. Pascher, *Doc. 328: Coupling regimes, resonance and synchronisation*, FFGFT corpus, Zenodo (2026). Galois lattice condition, Arnold tongues, Adler model, $K_{\text{eff}} = 2\pi\xi$, coupling threshold; direct connection to orbital resonances and mutual coupling.

[24] J. Pascher, *Doc. 242: Scale ladder — interscale couplings in FFGFT*, FFGFT corpus, Zenodo (2026). Planetary orbit argument, Laplace resonance as structural principle, scale separation as historical convention, not physical necessity.

[25] J. Pascher, *Doc. 124: Charge as geometric projection*, FFGFT corpus, Zenodo (2026). Charge as emergent projection of geometric patterns; particles as field configurations, not intrinsic point properties.

[26] J. Pascher, *Doc. 343: The zeta function of the $T^4/\mathbb{Z}_3$ torus*, FFGFT corpus, Zenodo (2026). Resolution floor 100ξ, Galois lattice, Farey critical denominator height; quantitative bound on mode distinguishability.

Part IV

Part IV — Icosahedron, Matrix Elements and Lepton Masses

Chapter Y

Doc. 367 — Why $\theta = p_0 = 2/9$ — Probability and Phase as Two Readings of One Quotient

Abstract

The Koide angle $\theta = 2/9$ (Doc. 292) and the icosahedral projection $p_0 = |\langle v_0 \mid R_5\, v_e\rangle|^2 = 2/9$ (Doc. 293) arise from two independent calculations and land on exactly the same rational number. This document names the reason and draws the complete consequence.

The reason: both quantities are the same quotient $2/3^2$ – numerator = number of non-trivial $\mathbb{Z}_3$ modes, denominator = square of the $\mathbb{Z}_3$ order. θ reads it as the phase of the mass operator, p_0 as the modulus squared of an algebraic projection. This is the Born-rule structure: the same number, once as a modulus and once as a modulus squared of the same matrix element.

The consequence: the Koide formula can be written entirely in the single matrix element $A = \langle v_0 \mid R_5\, v_e\rangle$ of the A_5 embedding. Its modulus squared supplies the phase $\theta = |A|^2 = 2/9$; its modulus, multiplied by 3, gives the amplitude coefficient $\sqrt{2} = 3|A|$:

$$a_k = 1 + 3\,|A| \cdot \cos\!\left(|A|^2 + \tfrac{2\pi k}{3}\right), \qquad A = \langle v_0 \mid R_5\, v_e\rangle.$$

*The phase therefore no longer needs an external θ. The same does not hold for the amplitude: that the Koide factor is exactly $3|A|$ is right in value but not derived **[S]**; the derivation of $\sqrt{2}$ from the geometry is open (Doc. 353). Moreover, the exact pair $(\sqrt{2},\ 2/9)$ lies outside the measurement uncertainty for m_μ/m_e (Doc. 369).*

What remains transcendental: $\cos(2/9)$, since $2/9 \notin \pi\mathbb{Q}$. Exact algebraic mass amplitudes are therefore excluded in principle.

On the Language: Why “Probability”?

The expression $p_0 = |\langle v_0 \mid R_5\, v_e\rangle|^2$ is called a *transition probability* in quantum mechanics – borrowed vocabulary that can mislead. Neither a statistical experiment nor a frequency is meant. p_0 is the *modulus squared of an algebraic projection*: v_e and v_0 are orthonormal eigenmodes of the $\mathbb{Z}_3$ circulant, $R_5 \in A_5$ is an exact rotation matrix. The inner product $\langle v_0 \mid R_5\, v_e\rangle$ follows entirely from the geometry of A_5 – no random element, no frequency.

The actual finding is algebraic: the same rational number $2/3^2$ appears in two algebraic roles – as the modulus squared of a projection and as the position parameter of the mass operator. The language changes; the object is the same.

External parallel [E]. The same quotient $2/3^2$ appears independently in the conformal field theory of $\mathbb{Z}_3$ orbifolds: the two twist fields σ_1, σ_2 have conformal weights $h_k = k(3-k)/(2 \cdot 3^2) = 1/9$, hence $h_1 + h_2 = 2/9$ (Dixon, Friedan, Martinec, Shenker 1987). There too the numerator counts the non-trivial $\mathbb{Z}_3$ sectors and the denominator is $|\mathbb{Z}_3|^2$. This is a *different* mathematical object – a sum of conformal dimensions, not the modulus squared of a projection – and it arises from a different $\mathbb{Z}_3$ action: the central element $\omega \cdot \mathbb{1}$ on $\mathbb{C}^3$, not the cyclic permutation $C_3 \in A_5$ with eigenvalues $(1, \omega, \omega^2)$. The parallel shows only that $2/3^2$ is a $\mathbb{Z}_3$-structural counting quantity and not an FFGFT-specific number. No identification of the two objects is claimed.

Epistemic markers:

- **[B]** algebraically proved (Python check script, exact arithmetic)
- **[K]** numerically confirmed
- **[S]** structurally plausible, identification open
- **[E]** established external mathematics or physics

Symbol	Meaning
$\mathbb{Z}_3$	cyclic group of order 3; triality on T^4
v_0, v_1, v_2	Fourier eigenmodes of the $\mathbb{Z}_3$ circulant on $\mathbb{C}^3$
ω	$e^{2\pi i/3}$, primitive cube root of unity
A_5	icosahedral group; contains C_3 as a subgroup (Doc. 285)
R_5	five-fold rotation by 72° in A_5
θ	Koide phase; $\sqrt{m_k} = M(1 + \sqrt{2}\cos(\theta + 2\pi k/3))$
p_0	$\lvert\langle v_0 \mid R_5 v_e\rangle\rvert^2$, transition probability
Q	Koide invariant $(\sum m_k)/(\sum \sqrt{m_k})^2 = 2/3$

Y.1 Starting Point: Two Witnesses, One Value

Doc. 293 records the finding as follows:

Route	Origin	Value
Koide (phenomenological)	measured lepton masses; no geometry	$\theta = 2/9$
C_3-in-A_5 (geometric)	five-fold redistribution of the modes; no masses	$p_0 = 2/9$

The two routes know nothing of each other and land on exactly the same rational number. The stress tests of Doc. 293 (200 random axes, angle scan, order series) show that $2/9$ is icosahedron-specific, not a generic value. Doc. 293 concludes that θ is an

"invariant of translation" between differential and spectral geometry – and leaves open why precisely $\theta = p_0$ holds, rather than $\theta = \sqrt{p_0}$ or some other function. That gap is closed here. Section 5 rewrites the Koide formula entirely in matrix elements of R_5: the phase then needs no external θ, while for the amplitude an open assignment remains.

Y.2 The Common Quotient $2/3^2$

p_0 as a Count in the $\mathbb{Z}_3$ Mode Space

The three lepton modes are the Fourier eigenmodes of the $\mathbb{Z}_3$ circulant on the $\mathbb{C}^3$ fibre (Docs. 282, 288):

$$v_0 = \tfrac{1}{\sqrt{3}}(1,1,1), \qquad v_{1,2} = \tfrac{1}{\sqrt{3}}(1, \omega^{\pm 1}, \omega^{\pm 2}).$$

One mode is trivial (v_0, symmetric under $\mathbb{Z}_3$), two are non-trivial (v_1, v_2, Galois-conjugate). The electron mode is one of the two non-trivial ones (which one is decided by the sign of the angle, Doc. 293).

When $v_e = v_1$ is rotated by R_5, the mode redistributes over all three directions with the weights (Doc. 293, **[B]**):

$$p_0 = \frac{2}{9}, \qquad p_2 = \frac{5 - 3\varphi}{9}, \qquad p_1 = \frac{2 + 3\varphi}{9}, \qquad p_0 + p_1 + p_2 = 1.$$

All three denominators are $9 = 3^2$. For p_0, the numerator is the small integer 2 – the number of non-trivial modes. In words:

$$p_0 = \frac{\#\,\text{non-trivial } \mathbb{Z}_3 \text{ modes}}{|\mathbb{Z}_3|^2} = \frac{2}{3^2}.$$

The denominator is the square, not the group order itself, because p_0 is the modulus squared of an amplitude: $|\langle v_0 | R_5 v_e \rangle| = \sqrt{2}/3$, and $(\sqrt{2}/3)^2 = 2/9$. The factor 3 in the amplitude's denominator is the normalisation of the modes ($1/\sqrt{3}$ per vector, two vectors in the inner product); the factor $\sqrt{2}$ in the numerator comes from the icosahedral geometry (axis angle 37.38° between the three-fold and five-fold axes, Doc. 293).

θ as a Phase in the $\mathbb{Z}_3$ Mode Space

The Koide parametrisation (Doc. 292)

$$\sqrt{m_k} = M\big(1 + \sqrt{2}\cos(\theta + 2\pi k/3)\big), \qquad k = 0, 1, 2,$$

carries two structural numbers: the amplitude $\sqrt{2}$ and the phase θ. Doc. 353 examines whether $\sqrt{2}$ follows from $\mathbb{Z}_3$ equidistribution: the coupling weight $1/\sqrt{2}$ is proved there **[B]**, but the step to $\sqrt{2}$ rests on an unjustified assignment and is open **[S]**. Doc. 291 shows in which channel θ lives: not in the symmetric modulus functional (where $Q = 2/3$ for every θ), but in the phase channel of the $\mathbb{Z}_3$ holonomy.

The empirical value is $\theta = 2/9$ (Doc. 292: the PDG back-calculated value deviates from $2/9$ by only 0.9σ, τ-limited; the exact decimal depends on the m_τ central value used, the σ assessment does not). Read as a fraction, $\theta = 2/3^2$. Here too the numerator 2 stands for the two non-trivial modes and the denominator 3^2 for the square of the $\mathbb{Z}_3$ dimension.

Why the square? Because θ measures the phase on the *product space* of the modes – not on the group $\mathbb{Z}_3$ (where the natural fraction would be $2/3$, the share of non-trivial elements), but on $\mathbb{C}^3 \otimes \mathbb{C}^3$, the space of mass-operator elements. The Hermitian circulant M_{ij} has $3^2 = 9$ entries, and the phase that pulls M away from the symmetry axis is a fraction of that 9-dimensional space.

The Same Asymmetry, Read Twice

Both quantities count the same thing:

$$\theta = p_0 = \frac{\#\,\text{non-trivial } \mathbb{Z}_3 \text{ modes}}{|\mathbb{Z}_3|^2} = \frac{2}{3^2} = \frac{2}{9}. \tag{Y.1}$$

p_0 reads this quotient as a *probability*: how much of the electron mode lands in the symmetric direction under the icosahedral rotation? θ reads it as a *phase*: how far is the mass configuration from full $\mathbb{Z}_3$ symmetry? Both times the asymmetry of the lepton configuration relative to full $\mathbb{Z}_3$ symmetry is measured – once as a share, once as an angle.

Y.3 The Structure of the Born Rule

This is the structure of quantum mechanics. The Born rule **[E]** links a complex amplitude $A = \langle\psi|\phi\rangle$ to a measurable probability:

$$P = |A|^2.$$

The phase of A is not directly measurable; only $|A|^2$ is. Yet the phase carries the interference information – it becomes visible as soon as two amplitudes are superposed.

The Koide structure exhibits the same dual character, in a particularly simple form:

- The amplitude of the icosahedral projection is $|\langle v_0|R_5 v_e\rangle| = \sqrt{2}/3$ (modulus; not an adjustable parameter, Doc. 293).
- Its square is the probability $p_0 = 2/9$.
- The Koide angle $\theta = 2/9$ is *numerically the same quotient*, read as a phase.

The relation is not $\theta = |A|$ (which would be $\sqrt{2}/3 \approx 0.471$) and not $\theta = \arg A$ (which would be a rotation angle of the projection), but $\theta = |A|^2 = p_0$. That is the core: *the phase of the mass configuration equals the probability of the mode redistribution*, because both realise the quotient (Y.1).

That the same number appears in both roles is not a numerical accident but a property of the $\mathbb{Z}_3$ mode space: the denominator 3^2 is forced in both roles – as a normalisation square in p_0, as the dimension of the operator space in θ. The numerator 2 is in both roles the number of non-trivial modes. This is the algebraic form of what Doc. 293 describes as an "invariant of translation": the number belongs to the structure, not to the representation.

Y.4 Numerical Confirmation

The check script `pruef_367_phase_wahrscheinlichkeit.py` (2/python/-Dok367_Skripte) confirms **[K]**:

1. $p_0 = |\langle v_0|R_5 v_1\rangle|^2 = 2/9$ to 40 digits (mpmath), with the standard embedding $C_3 < A_5$ (axis $(0, 1, \varphi)$).
2. $|\langle v_0|R_5 v_1\rangle| = \sqrt{2}/3$ exactly.
3. θ_{emp} from the PDG masses (back-calculated via $\cos\theta$ from $\sqrt{m_\tau}, \sqrt{m_e}, \sqrt{m_\mu}$; $m_\tau = 1776.86 \pm 0.12$ MeV): 0.222270, deviation from $2/9$: $4.8 \cdot 10^{-5}$; the τ-limited tolerance is $5.3 \cdot 10^{-5}$, so the deviation is 0.9σ (as in Doc. 292).
4. $\theta_{\text{emp}} = p_0$ within the τ tolerance (empirically); exactly $= 2/9$ on both sides (geometrically).
5. The alternatives $\theta = \sqrt{p_0}$ (0.471) and $\theta = \arg\langle v_0|R_5 v_1\rangle$ do not reproduce the empirical angle.

Y.5 The Direct Translation: From the U Matrix Element to Lepton Masses

This equation writes the Koide formula entirely in one matrix element of the A_5 embedding. The phase is the central result of this document; the amplitude factor contains an open assignment (point 1) **[S]**:

$$\boxed{a_k = 1 + 3\,|\langle v_0 \mid R_5\, v_e\rangle| \cdot \cos\Big(|\langle v_0 \mid R_5\, v_e\rangle|^2 + \tfrac{2\pi k}{3}\Big).} \tag{Y.2}$$

1. The amplitude coefficient ($\sqrt{2}$). The modulus of the transition element is $|\langle v_0 \mid R_5\, v_e\rangle| = \sqrt{2/9} = \sqrt{2}/3$. Scaling this modulus by the dimension of the $\mathbb{Z}_3$ symmetry axis (3) yields exactly the Koide factor:

$$3 \cdot |\langle v_0 \mid R_5\, v_e\rangle| = 3 \cdot \frac{\sqrt{2}}{3} = \sqrt{2}.$$

The arithmetic itself is exact; what is not justified is why the Koide factor should be exactly $3|A|$ rather than $|A|$ or some other function of $|A|$. The factor 3 is set here. This is the same open point as in Doc. 353 (there the assignment $d/c = 1/|M|$ instead of $d/c = |M|$): the value $\sqrt{2}$ is right, its derivation from the geometry is still missing **[S]**.

2. The phase shift (θ). By the Born rule, the modulus squared of the same matrix element is the projection probability p_0:

$$p_0 = |\langle v_0 \mid R_5\, v_e\rangle|^2 = \frac{2}{9}.$$

Since this square enters the cosine directly as the phase ($\theta = p_0 = 2/9$ rad), the Born rule appears as a geometric coupling: the amplitude and the phase of the same matrix element jointly determine the mass structure.

3. The mass amplitudes. For $k \in \{0, 1, 2\}$, the three Koide amplitudes follow with this assignment **[K]**:

$$\begin{aligned} \text{Tau } (\tau,\ k=0): &\quad a_0 = 1 + \sqrt{2}\cos\!\left(\tfrac{2}{9}\right) \approx 2.3794, \\ \text{Electron } (e,\ k=1): &\quad a_1 = 1 + \sqrt{2}\cos\!\left(\tfrac{2}{9} + \tfrac{2\pi}{3}\right) \approx 0.0404, \\ \text{Muon } (\mu,\ k=2): &\quad a_2 = 1 + \sqrt{2}\cos\!\left(\tfrac{2}{9} + \tfrac{4\pi}{3}\right) \approx 0.5802. \end{aligned}$$

The mass ratios $m_k \propto a_k^2$ are thus at the level of the Koide formula. The exact pair $(\sqrt{2},\ 2/9)$ does not, however, reproduce m_μ/m_e within the measurement uncertainty (Doc. 369).

Assessment. The question of the phase is thereby answered:

- The matrix element $\langle v_0 \mid R_5\, v_e\rangle$ carries the phase ($|A|^2$); the amplitude can be written as $3|A|$, but this assignment is not derived **[S]**.
- No unbound θ is needed as external input – the number $2/9$ generates itself within the function call from the modulus of the matrix element.
- The intrinsic transcendence via the cosine function ($\cos(2/9)$ is transcendental since $2/9 \notin \pi\mathbb{Q}$) **[E]** remains the mathematical interface, which is why the mass ratios are irreducibly transcendental numbers.

Y.6 What Is Explained and What Remains Open

Explained. The agreement $\theta = p_0$ is not a coincidental convergence of two independent calculations but the double realisation of one quotient $2/3^2$ – once as a probability (modulus squared of a normalised amplitude), once as a phase (share of the operator space). The reason it is $|A|^2$ and not $|A|$ lies in the denominator: the mode space is $\mathbb{C}^3$, the operator space is $\mathbb{C}^3 \otimes \mathbb{C}^3$, and the phase sits in the operator space.

Open. The amplitude $\sqrt{2}$. It can be written as $3|A|$, and Doc. 353 traces it back to the coupling weight $1/\sqrt{2}$; in both cases the justification of the assignment is missing **[S]**. In addition, the exact pair $(\sqrt{2},\ 2/9)$ is empirically excluded for m_μ/m_e (Doc. 369); which small deviation from $\sqrt{2}$ or $2/9$ nature selects remains to be clarified.

Placement. That phase and probability carry the same number is the language in which FFGFT and quantum mechanics share one structure: amplitude, modulus squared, interference. FFGFT gives that structure a geometric address – $2/3^2$ on the $\mathbb{Z}_3$ mode space – and quantum mechanics gives it the measurement prescription. Both say the same thing, in different languages.

Chapter Z

Doc. 368 — From p_0, p_1, p_2 to Lepton Masses

Abstract

Doc. 293 provides the icosahedral redistribution weights $p_0 = 2/9$, $p_1 = (2+3\phi)/9$, $p_2 = (5-3\phi)/9$. Doc. 367 shows that $p_0 = \theta$ and *rewrites the Koide formula entirely in matrix elements of R_5 – the mass bridge via θ is thereby obtained directly from the A_5 geometry.*

This document investigates a second, purely algebraic route: the ratios *p_1/p_2 and p_0/p_2 carry the ϕ-skeleton of the lepton masses directly, without a cosine function. Main result: $p_1/p_2 = \phi^8$ and $p_0/p_2 = 2\phi^4$ are exact algebraic identities **[B]**. The mass ratios m_τ/m_e and m_μ/m_e follow approximately by multiplication with the Galois factors 74 and 15 from Doc. 338 (accuracy $0.03\,\%$ and $0.56\,\%$) **[K]**. The derivation of these Galois factors from the $\mathbb{Z}_3$ framework remains open **[S]**.*

Notation

Epistemic markers:

[B] algebraically proved, exact arithmetic
[K] numerically confirmed (approximation, accuracy stated)
[S] structurally observed, derivation open
[E] established external mathematics

Symbol	Meaning
ϕ	golden ratio $(1+\sqrt{5})/2$
p_0, p_1, p_2	icosahedral redistribution weights from Doc. 293
θ	Koide phase; $\theta = p_0 = 2/9$ (Doc. 367)
a_k	Koide amplitude: $\sqrt{m_k} = M(1+\sqrt{2}\cos(\theta + 2\pi k/3))$

Z.1 Starting Point

Doc. 293 computes the redistribution weights of the electron mode $v_e = v_1$ under the icosahedral five-fold rotation $R_5 \in A_5$:

$$p_0 = \frac{2}{9}, \qquad p_1 = \frac{2+3\phi}{9}, \qquad p_2 = \frac{5-3\phi}{9}, \qquad p_0 + p_1 + p_2 = 1.$$

Doc. 367 establishes that $\theta = p_0 = 2/9$: the same quotient appears once as Koide phase and once as transition probability. Doc. 367 has shown that the mass bridge via $\theta = p_0$ can be obtained directly from the A_5 geometry. The question of this document is a different one: do the *ratios* p_1/p_2 and p_0/p_2 carry the ϕ-skeleton of the masses algebraically – without cosine function, without θ as a running variable?

Z.2 Exact Algebraic Identities

$p_1/p_2 = \phi^8$

$$\frac{p_1}{p_2} = \frac{2+3\phi}{5-3\phi} = \frac{(7+3\sqrt{5})/2}{(7-3\sqrt{5})/2} = \frac{(7+3\sqrt{5})^2}{49-45} = \frac{94+42\sqrt{5}}{4} = \frac{47+21\sqrt{5}}{2}. \tag{Z.1}$$

On the other hand: $\phi^4 = 3\phi + 2$ (from $\phi^2 = \phi + 1$), so $\phi^8 = (3\phi+2)^2 = 21\phi + 13 = \frac{47+21\sqrt{5}}{2}$. Hence **[B]**:

$$\boxed{p_1/p_2 = \phi^8.}$$

$p_0/p_2 = 2\phi^4$

$$\frac{p_0}{p_2} = \frac{2}{5-3\phi} = \frac{2(7+3\sqrt{5})}{4} = \frac{7+3\sqrt{5}}{2}.$$

And $2\phi^4 = 6\phi + 4 = 7 + 3\sqrt{5}$. Hence **[B]**:

$$\boxed{p_0/p_2 = 2\phi^4.}$$

$p_1/p_0 = \phi^4/2$

Direct consequence: $p_1/p_0 = \phi^8/(2\phi^4) = \phi^4/2$ **[B]**. All three ratios carry only ϕ-powers with rational prefactors.

Amplitude Structure

The square roots $3\sqrt{p_j}$ (amplitudes up to normalisation factor 3) take exactly the icosahedral values **[B]**:

$$3\sqrt{p_0} = \sqrt{2}, \qquad 3\sqrt{p_1} = \phi^2, \qquad 3\sqrt{p_2} = \frac{1}{\phi^2}.$$

Z.3 Approximate Connection to Mass Ratios

Precision limit: $\cos(2/9)$ is transcendental

The exact Koide mass ratios at $\theta = 2/9$ are:

$$\frac{m_\tau}{m_e} = \left(\frac{a_0}{a_1}\right)^2, \qquad \frac{m_\mu}{m_e} = \left(\frac{a_2}{a_1}\right)^2,$$

with $a_k = 1 + \sqrt{2}\cos(2/9 + 2\pi k/3)$. Since $2/9 \notin \pi\mathbb{Q}$, $\cos(2/9)$ is transcendental **[E]**. Consequently m_τ/m_e and m_μ/m_e cannot be expressed exactly as rational multiples of ϕ-powers. All connections below are approximate identities with stated accuracy.

$m_\tau/m_e \approx 74 \times \phi^8 = 74 \times p_1/p_2$

Numerically (with $\theta = 2/9$ exact in the Koide formula) **[K]**:

$$\frac{m_\tau}{m_e}\Big|_{\theta=2/9} = 3477.47, \tag{Z.2}$$

$$74 \times \phi^8 = 3476.42, \tag{Z.3}$$

$$\text{deviation} = 0.030\,\%. \tag{Z.4}$$

The Galois factor $74 = 2 \times 37$, where $37 \mid |GF(3^{18})^*|$, appears in Doc. 338 as part of the correction factor $K_{\text{frak}} = 74/75$ of the α derivation. Its appearance here is a structural observation **[S]**; the derivation from the $\mathbb{Z}_3$ framework is outstanding.

$m_\mu/m_e \approx 30 \times \phi^4 = 15 \times p_0/p_2$

$$\frac{m_\mu}{m_e}\Big|_{\theta=2/9} = 206.770, \tag{Z.5}$$

$$30 \times \phi^4 = 205.623, \tag{Z.6}$$

$$\text{deviation} = 0.56\,\%. \tag{Z.7}$$

Here the approximation is considerably less precise. The factor $15 = 3 \times 5$, with $3 = |GF(3)|$ and 5 the only odd prime factor of $|GF(81)^*| = 80 = 2^4 \cdot 5$, comes from the Galois hierarchy of Doc. 338 **[S]**.

Z.4 Numerical Confirmation

The check script `pruef_368_p_massen.py` (2/python/Dok368_Skripte) confirms **[K]**:

1. $p_1/p_2 = \phi^8$ exactly (dev. $< 10^{-35}$, 40 digits).
2. $p_0/p_2 = 2\phi^4$ exactly.
3. $p_1/p_0 = \phi^4/2$ exactly.
4. $3\sqrt{p_j} \in \{\sqrt{2}, \phi^2, 1/\phi^2\}$ exactly.

5. $74 \times \phi^8$ matches $m_\tau/m_e|_{\theta=2/9}$ to $0.030\,\%$.
6. $30 \times \phi^4$ matches $m_\mu/m_e|_{\theta=2/9}$ to $0.56\,\%$.
7. All three ratios carry only ϕ-powers (no π, no cos).
8. $\varepsilon \approx 27\xi/12$ (0.49 % match).
9. $74 \times \phi^8 \times (1 + 27\xi/12)$ matches m_τ/m_e to $0.00015\,\%$.

Z.5 Galois Structure of ε: Observation and Limit

Numerical observation

The deviation ε from Section 3 lies close to a Galois-native ξ-expression:

$$\varepsilon \approx \frac{27\xi}{12} = \frac{|GF(27)| \cdot \xi}{n_{\phi,\mu} \cdot n_{\theta,e}} = \frac{3}{10000}, \qquad \text{dev. } 0.49\,\% \quad \textbf{[K]} \tag{Z.8}$$

All three factors – $27 = |GF(27)|$, $\xi = 4/30000$ and $12 = n_{\phi,\mu} \times n_{\theta,e} = 4 \times 3$ – are already known with status **[B]** from Doc. 338. Substituting:

$$\frac{m_\tau}{m_e} \approx 74 \times \phi^8 \times \left(1 + \frac{27\xi}{12}\right) \quad (0.00015\,\%) \quad \textbf{[K]} \tag{Z.9}$$

Limit of the analytical proof path

A proof that $\varepsilon = 27\xi/12$ holds *exactly* cannot be established via the FFGFT formula. The test attempt:

From Doc. 352: $m_\tau/m_e(\text{bare}) = (25/12) \cdot \xi^{-5/6}$ (from winding numbers $r_\tau/r_e = 25/12$, $p_\tau - p_e = -5/6$), with fraktal correction coefficient $117.4 \cdot \xi$. A linearisation $(25/12) \cdot \xi^{-5/6} \approx 74 \times \phi^8 \times (1 + c_{\text{bare}} \cdot \xi)$ yields $c_{\text{bare}} \approx 119.1$.

The net coefficient is:

$$c_{\text{bare}} - 117.4 \approx 1.72 \neq \frac{27}{12} = 2.25 \quad (23\,\%\ \text{discrepancy})$$

The approach fails for two reasons:

1. $\xi^{-5/6}$ is not a polynomial in ξ: the expansion of $(25/12) \cdot \xi^{-5/6}$ around the fixed point $74 \times \phi^8$ yields a power series in $\xi^{1/6}$, not in ξ.
2. The coefficient 117.4 in Doc. 352 is itself numerical **[K]** (rounded value); both sides carry uncertainties that add up to $\sim 23\,\%$.

Status. The identification $\varepsilon \approx 27\xi/12$ is a numerical observation **[K]** (0.49 % deviation). All factors are Galois-native from Doc. 338. An exact analytical derivation – showing why the net coefficient is precisely $27/12$ – requires a perturbation expansion in $\xi^{1/6}$ that goes beyond the current state of the corpus. This step remains **[S]**.

Z.6 Summary: What Is Computable from the p-Values

Statement	Status	Source
$p_1/p_2 = \phi^8$, $p_0/p_2 = 2\phi^4$	**[B]**	A_5 geometry, Doc. 293
$3\sqrt{p_j} \in \{\sqrt{2}, \phi^2, 1/\phi^2\}$	**[B]**	A_5 geometry, Doc. 293
$m_\tau/m_e \approx 74 \times p_1/p_2$	**[K]**	Galois factor 74, Doc. 338
$m_\mu/m_e \approx 15 \times p_0/p_2$ $(= 30\phi^4)$	**[K]**	Galois factor 15, Doc. 338
$\varepsilon \approx 27\xi/12$	**[K]**	Galois+ξ, Doc. 338
Derivation of factors 74, 15 from p-values	**[S]**	open
Analytic proof $\varepsilon = 27\xi/12$ exactly	**[S]**	open

What follows directly from the p-values. The p-values p_0, p_1, p_2 from the A_5 geometry (Doc. 293) deliver exactly the *ϕ-skeleton* of the lepton masses: the ratios $p_1/p_2 = \phi^8$ and $p_0/p_2 = 2\phi^4$ are purely geometric identities with no free parameter.

What requires additional input. Converting to numerical mass ratios requires the Galois factors 74 and 15. These come from the $GF(3^n)$ structure of Doc. 338, not from the p-values themselves. The direct route

$$p_0, p_1, p_2 \xrightarrow{\text{exact}} \phi\text{-skeleton} \xrightarrow{\text{Galois, Doc. 338}} m_\tau/m_e,\ m_\mu/m_e$$

is fully described, but the second step is observation, not derivation. A fully direct route – in which the Galois factors themselves follow from the $A_5/\mathbb{Z}_3$ structure – is the remaining open edge.

What is genuinely new. Prior to this document, Doc. 293 had computed the p-values but had not identified their ratios as exact ϕ-powers. The identities $p_1/p_2 = \phi^8$ and $p_0/p_2 = 2\phi^4$ are new. The observation $m_\tau/m_e \approx 74 \times (p_1/p_2)$ connects, for the first time, the A_5 geometry of Doc. 293 to the Galois structure of Doc. 338 through a concrete mass ratio.

Chapter aa

Doc. 369 — Four Routes to the Lepton Masses

Abstract

The FFGFT corpus now contains four independent approaches to the masses of the charged leptons: the T0 ladder from torus winding numbers (Docs. 006, 338, 352), the Koide formula with $\theta = 2/9$ from a single A_5 matrix element (Docs. 292, 367), the Galois identities from GF(3^n) orders (Doc. 338), and the ϕ-skeleton of the icosahedral redistribution weights (Doc. 368). This document places the four routes side by side: input, output, accuracy against PDG, epistemic status. Finding: the routes deliver different kinds *of statement (absolute masses, ratios, constraints) with accuracies from $1.6\,\%$ down to $10^{-3}\,\%$. The Koide route is the only parameter-free one and the most precise; the T0 ladder the only one yielding absolute masses. A fifth route – the Koide scale M from the T0 ladder – is Route 1 summed ($0.31\,\%$) and shows that M is not an independent scale. The open edge is the same in all: the fractal-recursive correction from percent level to 10^{-3} level is measured, not derived.*

Notation

Epistemic markers:

- **[B]** algebraically proved, exact arithmetic
- **[K]** numerically confirmed (approximation, accuracy stated)
- **[S]** structurally observed, derivation open
- **[E]** established external mathematics or physics

Symbol	Meaning
ξ	FFGFT fundamental parameter, $\xi = 4/30000 = 1/7500$
v	electroweak scale, 246 GeV
$r_i,\ p_i$	winding quantum numbers of lepton i (Doc. 338, Tab. 1)
K_{frak}	fractal-recursive correction factor $74/75$ (Doc. 338)
θ	Koide phase, $\theta = 2/9$
A	$\langle v_0 \mid R_5\, v_e\rangle$, $\lvert A\rvert = \sqrt{2}/3$ (Doc. 367)
p_0, p_1, p_2	icosahedral redistribution weights (Doc. 293)
ϕ	golden ratio
PDG	Particle Data Group 2024: $m_e = 0.51099895$, $m_\mu = 105.6583755$, $m_\tau = 1776.93 \pm 0.09$ MeV

aa.1 The Four Routes

Route 1: T0 Ladder (Winding Numbers)

Documents: 006, 338, 352. **Input:** ξ, v, quantum numbers (r_i, p_i). **Output:** absolute masses.

$$m_i^{\text{bare}} = r_i\, \xi^{p_i}\, v, \qquad \frac{m_j}{m_i} = \frac{r_j}{r_i}\, \xi^{\,p_j - p_i} \quad \textbf{[B]}$$

with $(r_e, p_e) = (4/3,\ 3/2)$, $(r_\mu, p_\mu) = (16/5,\ 1)$, $(r_\tau, p_\tau) = (25/9,\ 2/3)$ from the torus winding numbers (n_θ, n_ϕ) (Doc. 338, Tab. 1).

Quantity	Formula	Value	PDG	Dev.
m_μ/m_e	$\frac{12}{5}\,\xi^{-1/2}$	207.846	206.768	+0.52 %
m_τ/m_e	$\frac{25}{12}\,\xi^{-5/6}$	3531.6	3477.4	+1.56 %
m_e [MeV]	$r_e\,\xi^{3/2}\,v$	0.5050	0.5110	−1.18 %
m_μ [MeV]	$r_\mu\,\xi\,v$	104.96	105.66	−0.66 %
m_τ [MeV]	$r_\tau\,\xi^{2/3}\,v$	1783.4	1776.9	+0.37 %

Status: formula **[B]**, quantum numbers from GF(3^n) **[K]**. The residual deviation of 0.4–$1.6\,\%$ is the fractal-recursive correction (Doc. 352): it is *measured* as $+39.1\,\xi$ (m_μ/m_e) and $+117\,\xi$ (m_τ/m_e; Doc. 352: 117.4 with older PDG value $m_\tau = 1776.86$); its coefficients are not derived **[S]**.

Route 2: Koide Formula with $\theta = 2/9$

Documents: 292, 367. **Input:** one matrix element $A = \langle v_0 \mid R_5\, v_e\rangle$. **Output:** mass ratios; scale M external.

$$\sqrt{m_k} = M\, a_k, \qquad a_k = 1 + 3|A|\,\cos\!\Big(|A|^2 + \frac{2\pi k}{3}\Big), \qquad |A| = \frac{\sqrt{2}}{3} \quad \textbf{[B]}$$

The modulus of A supplies the coefficient $\sqrt{2}$, its modulus squared the phase $\theta = 2/9$ (Doc. 367). No free parameter.

Quantity	Formula	Value	PDG	Dev.
m_μ/m_e	$(a_2/a_1)^2$	206.7703	206.7683	0.0010 %
m_τ/m_e	$(a_0/a_1)^2$	3477.47	3477.37	0.0031 %
m_τ from m_e, m_μ, $Q = 2/3$		1776.97	1776.93	0.0022 %
Q_{Koide}	$(\sum m)/(\sum \sqrt{m})^2$	$2/3$ exact	0.66666	10^{-5}

Status: $\theta = 2/9$ **[B]**. $\cos(2/9)$ transcendental **[E]**, hence no algebraically exact ratios. For m_τ/m_e the deviation ($3.1 \cdot 10^{-5}$) lies *within* the PDG uncertainty of m_τ (± 0.005 %). For m_μ/m_e this does not hold: at fixed amplitude $\sqrt{2}$, $(a_2/a_1)^2$ depends only on θ, not on m_τ; against CODATA $206.7682827(46)$ the value 206.7703 lies rel. $9.8 \cdot 10^{-6}$, i.e. $\approx 440\sigma$, away. The exact pair $(\sqrt{2},\ 2/9)$ is excluded for m_μ/m_e (as also stated in Doc. 292); compatibility would require an amplitude $\sqrt{2} - 3 \cdot 10^{-7}$ at fixed $\theta = 2/9$ (then $m_\tau = 1776.97$ MeV, 0.4σ).

Route 3: Galois Identities

Document: 338. **Input:** orders of GF(3^n). **Output:** two constraints, not a complete determination.

$$\left(\frac{m_\mu}{m_e}\right)^2 = 43200 = 2^6 \cdot 3^3 \cdot 5^2, \qquad m_e\, m_\mu = 54\ \text{MeV}^2 = |\text{GF}(3)^*| \cdot |\text{GF}(27)| \quad \textbf{[K]}$$

Identity	Prediction	PDG	Dev.
$(m_\mu/m_e)^2 = 43200$	207.846	206.768	0.52 %
$m_e\, m_\mu = 54$ MeV^2	54	53.991	0.016 %

Status: both **[K]**. The first identity is *the same number* as Route 1 ($207.846 = \sqrt{43200}$): the T0 ladder reproduces the Galois identity exactly. For m_μ/m_e, Routes 1 and 3 are not two routes but one route in two languages. The second identity is markedly more precise at 0.016 % and enters $1/\alpha = 3700/27$ via $\alpha = \xi E_0^2/K_{\text{frak}}$.

Route 4: ϕ-Skeleton of the p-Values

Document: 368. **Input:** p_0, p_1, p_2 from A_5 (Doc. 293). **Output:** ratios algebraically, without cosine.

$$\frac{p_1}{p_2} = \phi^8, \qquad \frac{p_0}{p_2} = 2\phi^4 \quad \textbf{[B]}$$

Quantity	Formula	Value	PDG	Dev.
m_τ/m_e	$74\,\phi^8$	3476.42	3477.37	0.027 %
m_τ/m_e	$74\,\phi^8\,(1 + \frac{27}{12}\xi)$	3477.47	3477.37	0.0029 %
m_μ/m_e	$30\,\phi^4$	205.62	206.77	0.55 %

Status: ϕ-powers **[B]**, Galois factors 74, 30, $27/12$ observed **[S]**. With the ξ correction, m_τ/m_e reaches the precision of the Koide route.

aa.2 Synopsis

Route	Doc.	Input	Output	m_μ/m_e	m_τ/m_e	Status
1 T0 ladder	006/338/352	ξ, v, r_i, p_i	absolute	$0.52\,\%$	$1.56\,\%$	**[K]**, corr. **[S]**
2 Koide	292/367	A	ratios	$0.001\,\%$	$0.003\,\%$	θ **[B]**
3 Galois	338	GF(3^n)	constraints	$0.52\,\%$	—	**[K]**, cause **[S]**
4 ϕ-skeleton	368	p_j	ratios	$0.55\,\%$	$0.027\,\%$	**[B]**+**[S]**
4b ϕ-sk.+ξ	368	p_j, ξ	ratios	—	$0.003\,\%$	**[K]**

What each route delivers.

- **Route 1** is the only one yielding absolute masses – and the least accurate. Its value is the anchoring to v and ξ.
- **Route 2** is the only parameter-free one and the most precise. Its limit is the scale: $M \approx 313.84$ MeV must come from outside. Only m_τ/m_e lies within the PDG uncertainty of m_τ; m_μ/m_e lies $\approx 440\sigma$ from CODATA (see Route 2).
- **Route 3** yields no masses but two constraints. The more precise of the two ($m_e m_\mu = 54$) is the anchor of the α derivation.
- **Route 4** is the most algebraic: pure ϕ-powers from the A_5 geometry, no transcendental function. The price is the Galois factors, which are observed and not derived.

Two accuracy classes. Routes 1, 3 and 4a sit at *percent level* (0.5–$1.6\,\%$). Routes 2 and 4b sit at 10^{-3}-*percent level*. For m_τ/m_e both deviate by rel. $\approx 3\cdot 10^{-5}$, within the m_τ uncertainty ($\pm 0.005\,\%$). m_μ/m_e from Route 2 ($(\sqrt{2},\, 2/9)$: 206.77032), by contrast, deviates by rel. $9.8\cdot 10^{-6}$, i.e. 442σ from CODATA $206.7682827(46)$, see Route 2; Route 4b does not give m_μ/m_e. The difference between the classes is precisely the fractal-recursive correction: $+117.4\,\xi$ in Route 1, $+\frac{27}{12}\xi$ in Route 4, implicit in $\cos(2/9)$ in Route 2.

aa.3 The Common Open Edge

All four routes end at the same point. The percent class needs a ξ-proportional correction to become the 10^{-3} class. This correction is known in three forms:

- as a measured coefficient 117.4 (Route 1, Doc. 352),
- as a Galois expression $27/12 = |\mathrm{GF}(27)|/(n_{\phi,\mu}\, n_{\theta,e})$ (Route 4, Doc. 368, $0.5\,\%$ agreement),
- as whatever the Koide formula contains beyond $74\phi^8$ (Route 2).

A first-principles derivation is missing in all three forms **[S]**. The proof attempt via linearisation of $\xi^{-5/6}$ (Doc. 368, Sect. 5) fails with $23\,\%$ discrepancy, because $\xi^{-5/6}$ expands in $\xi^{1/6}$, not in ξ.

Two readings of the correction. Doc. 352 (§ 10) takes the T0 ladder as fundamental: the ε_i are distances between bare and PDG, decreasing with generation ($\varepsilon_e > \varepsilon_\mu > \varepsilon_\tau$), and their origin is attributed to QED extraction and SI projection – as an attribution **[S]**, with the admission that the naive QED self-energy goes the wrong way. That is Reading A.

Doc. 367 permits a Reading B: the Koide formula with $\theta = 2/9$ is fundamental, parameter-free from the icosahedron. The T0 ladder $r_k \xi^{p_k}$ is then a *rational approximation* to the transcendental values a_k^2 with $a_k = 1 + \sqrt{2}\cos(2/9 + 2\pi k/3)$. The residuals $-39\,\xi$, $-77\,\xi$, $-116\,\xi$ emerge in Reading B without PDG and without QED – as the distance between cosine and power law **[K]**. They differ because the three cosine values differ: a power law ξ^p has fixed slope $\log \xi = -8.92$, the Koide slopes $2\Delta \log a/\Delta p$ are -10.66 ($e \to \mu$) and -8.47 ($\mu \to \tau$); the ratios r_j/r_i absorb the difference, the remainder is ε. No deeper reason is needed: the cosine is not a power law.

Both readings reproduce the same numbers. Reading B holds only with a restriction: the exact pair $(\sqrt{2},\ 2/9)$ is excluded for m_μ/m_e ($\approx 440\sigma$, see Route 2). The readings differ in what they declare open. Reading A needs the R113 bridge (QED from $T^4/\mathbb{Z}_3$). Reading B does not; its open question is instead: why are the rational approximation parameters exactly the GF(3^n) winding numbers, and why is the approximation so good ($0.5\,\%$)? That is a question for number theory, not for QED. Reading B is the more economical one (Register R121).

What distinguishes Route 2. The Koide route does not need this correction separately: it is contained in $\cos(2/9)$. That is the reason for its precision – and the reason it is not algebraic. Routes 1, 3, 4 attempt to express the same result rationally or in ϕ-powers; the transcendental remainder is the correction they then have to add back.

aa.4 Route 5: The Koide Scale from the T0 Ladder

The Koide formula yields ratios; the scale M remains external. Can it come from Route 1?

What M is. With $\sqrt{m_k} = \sqrt{M}\,a_k$ and $\sum_k a_k^2 = 6$ one has exactly $M = (m_e + m_\mu + m_\tau)/6 = 313.850$ MeV **[B]**. This is normalisation, not prediction. Equally exact: $M = m_\tau/a_\tau^2$ with $a_\tau^2 = 5.662$.

Finding. Inserting the bare masses from Route 1:

$$M_{\text{bare}} = \frac{v}{6}\left(\tfrac{4}{3}\xi^{3/2} + \tfrac{16}{5}\xi + \tfrac{25}{9}\xi^{2/3}\right) = 314.82\ \text{MeV}, \qquad \text{dev. } 0.31\,\% \quad \textbf{[K]}$$

Route 5 is Route 1 summed. The correction is $+23.1\,\xi$ and is arithmetically the weighted sum of the three individual corrections: $-0.02\,\xi$ (e) $-2.78\,\xi$ (μ) $+25.9\,\xi$ (τ). It is τ-dominated because M is $94\,\%\ m_\tau/6$.

What this means. M is not an independent scale. It follows from ξ and v with the accuracy of Route 1 – and with its open edge. A Galois-native single form $M = c\,\xi^p\,v$ does not exist: M is a sum of three different ξ-powers ($3/2$, 1, $2/3$) that do not collapse into one (effective exponent 0.747, no rational coefficient for $p \in \{2/3, 3/4, 5/6, 1\}$).

Warning against number games. Expressions such as $M \approx 81\sqrt{15}$ MeV ($0.044\,\%$) or $M \approx \frac{7}{40}\,\alpha v$ ($0.1\,\%$) fit well but carry no mechanism. With the building blocks $\{2, 3, 5, \phi, \sqrt{2}, \sqrt{3}, \sqrt{5}\}$ and exponents $-4 \ldots 4$, twelve combinations hit M/E_0 to within $0.1\,\%$; $81\sqrt{15}$ is one of them, four others are better. The factor $5/2$ in it appears in none of Docs. 292, 293, 338, 352, 367, 368. Such hits do not belong in the corpus.

aa.5 What the Corpus Shows

Read correctly, the five routes tell a single story – and it is not about leptons but about what FFGFT predicts at all.

FFGFT computes in natural units. The fundamental parameter $\xi = 4/30000$ is dimensionless, the winding numbers are integers, the Galois orders are integers, the icosahedral weights are algebraic numbers. None of this carries a unit. What follows from this structure is therefore *ratios*: m_μ/m_e, m_τ/m_e, $Q = 2/3$, $1/\alpha$. And it is precisely these ratios that the corpus hits, wherever it computes parameter-free, to 10^{-3} percent. Within experimental uncertainty lies m_τ/m_e; m_μ/m_e from $(\sqrt{2},\, 2/9)$ deviates by rel. $9.8 \cdot 10^{-6}$, i.e. $\approx 440\sigma$, from CODATA (see Route 2). The Koide route needs one matrix element of the A_5 geometry and nothing else. That is the theory's actual predictive power: the *shape* of the mass spectrum, not its size.

As soon as one demands a number in MeV, a scale must come from outside. FFGFT takes a single declared anchor for this – here $v = 246$ GeV, elsewhere m_e or H_0 – and this anchor serves conversion, not derivation. With the anchor one inherits the percent class: the T0 ladder yields absolute masses to 0.4–1.6 percent, and the Koide scale M is, as Route 5 shows, nothing but this ladder summed. The two accuracy classes of the synopsis are thus not two quality levels of the theory. They are the boundary between what FFGFT says of its own accord (ratios, $10^{-3}\,\%$) and what it needs an external unit for (absolute values, percent).

The open edge lies exactly on this boundary. The fractal-recursive correction – $117\,\xi$ here, $27\xi/12$ there, hidden in $\cos(2/9)$ for Koide – is the amount by which the rational and algebraic approximations deviate from the transcendental exact value. *Measuring* it is easy; *deriving* it from the winding numbers is what is still missing. Until then: FFGFT predicts the ratios of the lepton masses parameter-free at the 10^{-3}-percent level (for m_μ/m_e with the exact pair $(\sqrt{2},\, 2/9)$, however, outside the measurement uncertainty), and their absolute values with one anchor at percent level. The one is theory; the other is theory plus a convention.

Chapter ab

Doc. 370 — Why the Icosahedron — Three, Five, and Non-Commutation

Abstract

FFGFT is a $\mathbb{Z}_3$ theory from the ground up. Yet the lepton masses contain the golden ratio ϕ, and ϕ is the number of five. This document explains why the icosahedron is the bridge: its rotation group A_5 is the smallest finite group in which a three and a five coexist without commuting. From this non-commutation arises a mixing of the $\mathbb{Z}_3$ modes under the five-fold rotation, and the mixing has one rational component $2/9$ and two ϕ-bearing components. The rational one is the Koide phase (Doc. 367), the ϕ-bearing ones are the skeleton of the mass ratios (Doc. 368). A single matrix element of this mixing carries the entire Koide formula. The document computes nothing new; it narrates what Docs. 285, 293, 367 and 368 mean together.

Notation

Epistemic markers:

- **[B]** algebraically proved (check script in the cited document)
- **[K]** numerically confirmed
- **[S]** structurally observed, derivation open
- **[E]** established external mathematics

Symbol	Meaning
$\mathbb{Z}_3$, C_3	cyclic group of order 3; C_3: permutation $(x, y, z) \mapsto (y, z, x)$
v_0, v_1, v_2	Fourier modes, eigenvectors of C_3
ω	$e^{2\pi i/3}$
A_5	icosahedral group, 60 elements
R_5	rotation by 72° about the icosahedron vertex $(0, 1, \phi)$
ϕ	golden ratio $(1 + \sqrt{5})/2$
p_0, p_1, p_2	redistribution weights of the electron mode under R_5 (Doc. 293)
A	$\langle v_0 \mid R_5\, v_e\rangle$, $\lvert A\rvert = \sqrt{2}/3$ (Doc. 367)
D_4	lattice on T^4; $\lvert\mathrm{Aut}(D_4)\rvert = 1152$ (Docs. 314, 364)
GF(3^k)	finite field of characteristic 3; GF(9) $\ni \phi$, GF(81) $\ni \zeta_5$

ab.1 The Observation

There are three charged leptons. Their masses stand in the ratio $m_e : m_\mu : m_\tau \approx 1 : 206.8 : 3477.4$, and since 1981 this ratio has been known to obey a strikingly simple formula, the Koide formula:

$$\sqrt{m_k} = \sqrt{M}\Big(1 + \sqrt{2}\cos\Big(\theta + \frac{2\pi k}{3}\Big)\Big), \qquad k = 0, 1, 2.$$

The scale M is a unit – it only says how large the masses are overall. The phase θ is the actual information: it determines the ratios. Empirically $\theta = 2/9$ to 10^{-7} (Doc. 292). Why $2/9$? Why not $1/4$, not 0.3? And why the coefficient $\sqrt{2}$?

The answer leads through a geometric object that at first has nothing to do with leptons: the icosahedron. The rest of this document explains why.

ab.2 Where FFGFT Comes From: The Three

FFGFT begins with a $\mathbb{Z}_3$. Three generations, triality, the circulant as mass operator. The mode space is $\mathbb{C}^3$; its natural basis is the Fourier modes

$$v_k = \frac{1}{\sqrt{3}}\,(1,\ \omega^k,\ \omega^{2k}), \qquad k = 0, 1, 2.$$

These are the eigenvectors of the cyclic permutation C_3. The mode v_0 is the symmetric one; v_1 and v_2 carry the phases ω and ω^2. The electron sits in v_1, the muon in v_2, the tau in v_0 – this is the assignment carried by the Koide index k.

What this structure alone does *not* deliver is a number. The $\mathbb{Z}_3$ says there are three modes and how they transform under C_3. It says nothing about masses. The Koide formula needs two inputs – the coefficient $\sqrt{2}$ and the phase $\theta = 2/9$ – and neither is contained in the $\mathbb{Z}_3$.

ab.3 Where ϕ Comes From: The Five

The golden ratio is the number of the pentagon: $\cos 72^\circ = (\phi - 1)/2$, and the diagonals of a regular pentagon stand to its sides in the ratio ϕ **[E]**. There is no five in the $\mathbb{Z}_3$ structure. If ϕ appears in the lepton masses – Doc. 368 shows $p_1/p_2 = \phi^8$ exactly **[B]** – a five must come from somewhere.

ab.4 Why the Icosahedron: Where Three and Five Meet

The icosahedron has 20 triangular faces and 12 vertices; at each vertex five triangles meet. Its rotation group A_5 has 60 elements: the identity, 15 rotations by 180° (edge midpoints), 20 rotations by 120° (face centres), 24 rotations by 72° or 144° (vertices) **[E]**. Three and five are both present.

That is not trivial. The cyclic group $\mathbb{Z}_{15}$ also contains elements of order 3 and 5 – but there they commute. In A_5 they *do not commute*: a rotation by 120° followed by one of 72° is not the same as the reverse order. A_5 is the smallest group in which this is so.

Concretely: in standard position the vertices of the icosahedron are the cyclic permutations of $(0, \pm 1, \pm\phi)$. The 3-fold axes pass through face centres; one of them is $(1, 1, 1)$, and the rotation by 120° about this axis is *exactly* the permutation C_3 of FFGFT. The $\mathbb{Z}_3$ of FFGFT is, without reinterpretation, a subgroup of the icosahedral group (Doc. 285). The 5-fold axes pass through vertices; one vertex is $(0, 1, \phi)$, and the rotation R_5 by 72° about this axis is the five-fold rotation used in Doc. 293.

ab.5 Where the Icosahedron Lives: Unrolled, Not Compact

Up to here it sounds as if the icosahedron were simply *the* structure of FFGFT. It is not – and the corpus is very precise about this (Docs. 285, 314, 364). FFGFT has two pictures connected by a lossless coordinate transformation, and the icosahedron lives in only one of them.

Compact: the torus T^4 with the D_4 lattice. This is the original. The four-torus carries the D_4 lattice, and its triality is the $\mathbb{Z}_3$ of FFGFT – as a lattice operation, as a circulant on the mode orbits (Doc. 314). The winding numbers of the T0 ladder live here. The lattice knows the three. It knows *no* five: $|\mathrm{Aut}(D_4)| = 1152 = 2^7 \cdot 3^2$, and $5 \nmid 1152$. No icosahedron acts on a D_4 lattice (Doc. 364, incompatibility theorem) **[B]**.

Unrolled: $\mathbb{R}^3 \times S^1$. Observed space arises when one circle of the torus becomes time through the measurement process: $T^4 \to \mathbb{R}^3 \times S^1$ (Doc. 285). On *this* $\mathbb{R}^3$, A_5 acts as rotation group. The $C_3 \subset A_5$ is the same three as the triality in the compact picture – the rotation by 120° about $(1, 1, 1)$. But the five exists only here. The icosahedron is the image of the torus geometry in unrolled space, not the torus itself.

How the five nevertheless enters the field. Doc. 364 resolves the apparent tension: what matters is *as what* a symmetry enters – as an automorphism of lattice vectors or as a phase. The 72° rotation has characteristic polynomial $(x-1)(x^2-(\phi-1)x+1)$; its trace is ϕ. Modulo 3, ϕ lies in GF(9) and is primitive there ($\phi^4=-1$, order 8). The roots of the quadratic factor lie not in GF(9) but in GF(81), and have order 5 there – because $5 \mid 80 = |\mathrm{GF}(81)^*|$ **[B]**. In the lattice, the 72° rotation is a vector automorphism, and $5 \nmid 1152$. In the field, the same rotation is only its eigenvalue ζ_5, and $5 \mid 80$. Geometric incompatibility becomes arithmetic compatibility as soon as one passes from vectors to phases. Doc. 364 verbatim: this is the mechanism behind Doc. 293 – $\theta = 2/9$ arises from R_5 as a phase, not as a lattice operation.

Why this is forced. In characteristic 3, $x^3-1=(x-1)^3$ **[E]**: in no GF(3^k) is there a primitive cube root of unity. The three *cannot* be a phase and must be lattice. The five *cannot* be lattice ($5 \nmid 1152$) and must be phase. The prime 3 distributes the two symmetries onto two complementary channels – not a choice, but a constraint. The extension GF(9) →GF(81) that carries ζ_5 and ϕ is precisely the doubling $6 = 3 \oplus 3'$ of Doc. 285.

What this means for the narrative. When non-commutation of three and five is spoken of below, this is meant: the three as lattice operation in the compact picture, the five as phase in the unrolled one. They do not meet in geometry – there they are incompatible – but in the field GF(81). The icosahedron is the visible form of this meeting.

ab.6 What Non-Commutation Produces

Because R_5 does not commute with C_3, R_5 is *not diagonal* in the mode basis v_0, v_1, v_2. Applying R_5 to the electron mode v_1, the result does not land back in v_1 but is distributed over all three modes. The moduli squared of the projections are the redistribution weights (Doc. 293) **[B]**:

$$p_0 = |\langle v_0|R_5|v_1\rangle|^2 = \frac{2}{9}, \qquad p_1 = |\langle v_1|R_5|v_1\rangle|^2 = \frac{2+3\phi}{9}, \qquad p_2 = |\langle v_2|R_5|v_1\rangle|^2 = \frac{5-3\phi}{9}.$$

One sees at once: p_0 is rational, p_1 and p_2 contain ϕ. That is structure, not accident. The projection onto the symmetric mode v_0 is blind to the five-fold structure – it only counts how much of the $\mathbb{Z}_3$ mixing arrives, and that is $2/9 = 2/3^2$. The projections onto v_1 and v_2 see the five and therefore carry ϕ.

It becomes even clearer at the symmetric mode itself. What R_5 makes of v_0:

$$\left(|\langle v_0|R_5|v_0\rangle|^2,\ |\langle v_0|R_5|v_1\rangle|^2,\ |\langle v_0|R_5|v_2\rangle|^2\right) = \left(\frac{5}{9},\ \frac{2}{9},\ \frac{2}{9}\right) \quad \textbf{[K]}$$

The complete matrix of moduli squared **[K]**:

$\lvert\langle v_j\rvert R_5\rvert v_k\rangle\rvert^2$	$k=0$	$k=1$	$k=2$
$j=0$	$5/9$	$2/9$	$2/9$
$j=1$	$2/9$	$(2+3\phi)/9$	$(5-3\phi)/9$
$j=2$	$2/9$	$(5-3\phi)/9$	$(2+3\phi)/9$

Two things are immediately visible. Wherever v_0 is involved, the entry is $2/9$ or $5/9$ – rational, without ϕ. And ϕ appears only between v_1 and v_2, the two modes that see the five. Every row and every column sums to 1.

The amplitudes of the first row are $(\sqrt{5}, \sqrt{2}, \sqrt{2})/3$. The five-fold rotation leaves $\sqrt{5}/3$ of the symmetric mode in itself and passes $\sqrt{2}/3$ to each of the two non-trivial modes. And $5 + 2 + 2 = 9 = 3^2$. This is the one row in which the five of the icosahedron and the three of FFGFT stand side by side. The trace of R_5 is $1 + 2\cos 72^\circ = \phi$ **[E]** – the five is in the operator, the three in the basis, and the product of both is the row $(5, 2, 2)/9$.

ab.7 Doc. 367: Why $\theta = p_0$, and How the Whole Koide Formula Follows

Doc. 293 had noted that $p_0 = 2/9$ coincides with the Koide angle $\theta = 2/9$, and left the question of the reason open. Doc. 367 gives the answer: both are the same quotient $2/3^2$, read in different roles. Numerator: the non-trivial $\mathbb{Z}_3$ modes. Denominator: the order squared, the dimension of the operator space $\mathbb{C}^3 \otimes \mathbb{C}^3$.

The decisive step comes next. The single matrix element $A = \langle v_0 \mid R_5\, v_e \rangle$ has modulus $\sqrt{2}/3$ and modulus squared $2/9$. The Koide formula

$$a_k = 1 + \sqrt{2}\cos\Big(\theta + \frac{2\pi k}{3}\Big)$$

needs exactly these two numbers: $\sqrt{2}$ as coefficient, $2/9$ as phase. With $3|A| = \sqrt{2}$ and $|A|^2 = 2/9$ this becomes **[B]**:

$$a_k = 1 + 3|A|\cos\Big(|A|^2 + \frac{2\pi k}{3}\Big).$$

No free parameter. The modulus gives the amplitude, the modulus squared the phase – Born-rule structure, but here not as interpretation, rather as algebra: the same complex number, once as $|A|$, once as $|A|^2$. The Koide formula is thereby fully derived from the question: how much of the five-fold rotation lands in the symmetric $\mathbb{Z}_3$ mode?

ab.8 Doc. 368: The ϕ-Skeleton

The two other weights carry the five. Their ratios are pure ϕ-powers **[B]**:

$$\frac{p_1}{p_2} = \phi^8, \qquad \frac{p_0}{p_2} = 2\phi^4, \qquad 3\sqrt{p_j} \in \{\sqrt{2},\ \phi^2,\ \phi^{-2}\}.$$

And $\phi^8 \approx 46.98$ is, up to a Galois factor, the ratio m_τ/m_e: $74\,\phi^8$ hits to $0.03\,\%$, with ξ correction to $0.003\,\%$ **[K]**. That is the second role of the icosahedron: p_0 supplies the phase, p_1 and p_2 supply the skeleton of the mass ratios. Both from the same five-fold rotation, both without ϕ having been inserted anywhere.

ab.9 Doc. 293: Why It Is the Icosahedron and Nothing Else

Doc. 293 tested whether $2/9$ depends on the axis. Two hundred random axes with angle $72°$: none yields $2/9$. Angle scan at fixed icosahedral axis: only $72°$ yields $2/9$. Order series – rotations of order 2, 3, 4, 5, 6 about icosahedral axes: only order 5 yields $2/9$ **[K]**. The number is icosahedron-specific. It arises because the axis $(0, 1, \phi)$ and the angle $72°$ together stand to the $(1, 1, 1)$ axis of the $\mathbb{Z}_3$ exactly as the icosahedral geometry prescribes.

ab.10 What the Icosahedral $\mathbb{Z}_3$ Is – and What It Is Not

A comparison sharpens the concept. In string theory there are orbifolds $T^6/\mathbb{Z}_3$ whose $\mathbb{Z}_3$ acts as $\omega \cdot \mathbb{1}$ on $\mathbb{C}^3$ – all three coordinates receive the same phase. This $\mathbb{Z}_3$ is *central*: it commutes with every matrix. The icosahedral $\mathbb{Z}_3$ is C_3 with eigenvalues $(1, \omega, \omega^2)$, trace zero, *not* central.

Both structures count $2/3^2$ – the orbifold CFT via the sum of twist-field weights $1/9+1/9$ **[E]** (Doc. 367, external parallel). But only the icosahedral $\mathbb{Z}_3$ can be *mixed* by R_5. For the central one, $\langle v_0|R_5|v_1\rangle$ would vanish, because R_5 would then commute with $\mathbb{Z}_3$ and preserve sectors. The non-commutation that produced the number above is exactly what distinguishes the icosahedral structure from an orbifold. A_5 has trivial centre; no element of A_5 can become $\omega \cdot \mathbb{1}$ in a faithful representation. The two $\mathbb{Z}_3$'s are different objects that happen – or do not happen – to share the same counting quantity.

ab.11 What Remains Open

Two questions raised by this document have since been answered – one negatively, one by reversal. What remains after that is a single one.

The Galois factors: the icosahedron does not explain them. ϕ^8 is exact, but m_τ/m_e is $74\,\phi^8$ with $74 = 2 \cdot 37$, and 37 is the number that appears in K_{frak} and in $1/\alpha = 3700/27$ (Doc. 338). In the phase channel (Sect. 5), 37 is the root of unity ζ_{37}, and it lives in GF(3^{18}): the order of 3 modulo 37 is 18. The icosahedral phase ζ_5 lives in GF(81) =GF(3^4). And $4 \nmid 18$: GF(81) is *not* a subfield of GF(3^{18}) **[B]**. The two phases sit in incompatible extensions. That is a negative result: 37 does not come from the icosahedron, neither geometrically nor via the phase channel. What GF(3^{18}) does contain is ϕ (from GF(9)) and the lepton order 26 (from GF(729)); its "new" primes, absent from every subfield, are 19 and 37. The degree $18 = 2 \cdot 3^2$ is the degree of ϕ times $|\mathbb{Z}_3|^2$ – the compositum of the ϕ-extension with the $\mathbb{Z}_3^2$-extension GF(3^9). Why this compositum, and why 37 rather than 19, is open **[S]**.

The fractal correction: derived in Reading B. Doc. 352 takes the T0 ladder as fundamental; then the step from percent to 10^{-3} level is a measured correction, and its origin is attributed to QED (Reading A). This document sets Koide as fundamental, because it follows from the icosahedron. In this Reading B, the T0 ladder $r_k\,\xi^{p_k}$ is a rational approximation to the transcendental amplitudes a_k^2, and the correction is the distance between cosine and power law – not measured but computed **[K]**. It differs for the three leptons

because the three cosine values differ. Both readings give the same numbers; Doc. 369 § 4 places them side by side, Register R121 records the change of status. What remains open in Reading B is not the correction but: why are the approximation parameters exactly the GF(3^n) winding numbers, and why is the approximation so good?

The one question that remains. Both points converge on the same thing. The icosahedron supplies phases in GF(81): ζ_5, ϕ, the Koide angle. Number theory supplies orders in GF(3^n): winding numbers, 37, K_{frak}. Both sides hit the lepton masses – the one parameter-free to $10^{-3}\,\%$ as ratios, the other with an anchor at percent level as absolute numbers. But they do not meet: GF(81) and GF(3^{18}) share only GF(9), hence ϕ and nothing else. Whether there is a structure in which the icosahedral phase and the Galois order have the same root – or whether they are two independent approaches that touch in ϕ and otherwise remain separate – is the open edge of this document **[S]**.

ab.12 The Short Version

The icosahedron is the place where three and five do not get along. FFGFT brings the three; the icosahedron is the smallest structure that adds a five without the five commuting with the three. From this non-commutation arises a mixing, and the mixing has one rational component – $2/9$ – and two ϕ-components. The rational one is the Koide phase. The ϕ-components are the skeleton of the masses. A single matrix element of this mixing carries the entire Koide formula, parameter-free.

Chapter ac

Doc. 371 — The Factor 4/3 — Sphere Volume, Casimir Operators and Electromagnetic Mass

Abstract

The numerical factor $4/3$ in the scaling parameter $\xi = \frac{4}{3} \times 10^{-4}$ has so far been motivated at three places in the corpus: sphere volume (Docs. 006, 205), the Casimir operator of SU(3) *and the Casimir effect (Doc. 324). The Casimir effect does not count, because the QFT Casimir energy between ideal plates contains no factor $4/3$. This document adds two occurrences not yet recorded and orders the occurrences by their common cause. First: the factor $4/3$ in the electromagnetic mass of the electron, unexplained since Abraham (1902) and Lorentz (1904), arises because each of the three field components carries one third of the field energy — a consequence of three-dimensionality, not a computational error. Second: the Casimir values of the two smallest non-abelian groups, $C_2(\mathrm{SU}(2)) = 3/4$ and $C_2(\mathrm{SU}(3)) = 4/3$, are reciprocals of each other; $N = 2$ is the only case in which consecutive fundamental representations achieve this. Both values belong to the electron: $3/4$ is its spin Casimir, $4/3$ its mass factor. The Pythagorean triple $(3, 4, 5)$. The two factors $3/4$ and $4/3$ are reciprocals, and their product is 1. Where does this pair arise naturally? In the simplest right-angled triangle with integer sides: the triple $(3, 4, 5)$. It is not one among many; it is distinguished in four independent ways.*

The triple $(3, 4, 5)$ carries this reciprocal pair as $\tan\alpha = 4/3$, $\cot\alpha = 3/4$ *and is, by Berggren's theorem, the root of all primitive triples. The angle $\alpha = \arctan(4/3)$ does not occur among the shortest vectors of the lattices D_3, D_4, but it does occur on the shell $|v|^2 = 10$: there lie $(3, 1, 0, 0)$ and $(1, 3, 0, 0)$ with $\cos\alpha = 3/5$. The cause is $D_2 = (1 + i)\,\mathbb{Z}[i]$ and $(1 + i)(2 \pm i) = 1 + 3i,\ 3 + i$: the triple sits, via the factor $(1 + i)$, in the carrier of FFGFT. A further 3D occurrence, the enthalpy factor $(u + p)/u = 4/3$ for isotropic pressure $p = u/3$, is added.*

Notation

Epistemic markers:

[B]	algebraically proved (Python verification script, exact arithmetic)
[K]	numerically confirmed
[S]	structurally plausible, identification open
[SETTING]	motivated input, not a derivation
[E]	established external mathematics or physics
[X]	checked and negative

Symbol	Meaning
$\xi = 4/30\,000$	dimensionless scaling parameter of FFGFT
$C_2(G)$	quadratic Casimir operator of the fundamental representation of G
U, **p**	field energy and field momentum of a moving charge
$m_0 = U/c^2$	electromagnetic mass from the rest energy
$(3, 4, 5)$	smallest primitive Pythagorean triple
α	angle with $\tan\alpha = 4/3$; $\alpha \approx 53.13^\circ$
D_3, D_4	root lattices in $\mathbb{R}^3$ (FCC) and $\mathbb{R}^4$
D_2	two-dimensional sublattice; coincides with the ideal $(1+i)\,\mathbb{Z}[i]$
$\mathbb{Z}[i]$	ring of Gaussian integers $\{a + bi \mid a, b \in \mathbb{Z}\}$
$(2 + i)$	Gaussian integer with $\lvert 2+i\rvert^2 = 5$; its square is $(2+i)^2 = 3 + 4i$
$O(2, 1; \mathbb{Z})$	integer Lorentz group of the form $x^2 + y^2 - z^2$; contains the Berggren matrices
Berggren matrices	three integer 3×3 matrices generating the tree of all primitive triples
ε_0	permittivity of vacuum
e, c	elementary charge, speed of light

ac.1 Starting Point in the Corpus

The parameter $\xi = \frac{4}{3} \times 10^{-4}$ carries two factors. The factor 10^{-4} is treated in Doc. 006 as loop suppression and in Doc. 309 as an exponent setting; it is not the subject of this document. The factor $4/3$ is motivated three times in the corpus:

1. **Sphere volume** (Docs. 006, 205): $V = \frac{4}{3}\pi r^3$; the factor characterises the geometry of three-dimensional space.
2. **Casimir operator** (Doc. 324): $C_2(\mathrm{SU}(3)_{\mathrm{fund}}) = \frac{9-1}{6} = \frac{4}{3}$.
3. **Casimir effect** (Docs. 324, 009): a factor $4/3$ in the Casimir energy between plates.

The third occurrence does not hold: the QFT Casimir energy between ideal plates contains no factor $4/3$, $E/A = -\pi^2\hbar c/(720\,a^3)$, pressure $-\pi^2\hbar c/(240\,a^4)$, energy density $-\pi^2\hbar c/(720\,a^4)$. The factor $4/3$ for the Casimir effect is appended in Doc. 324 (likewise in Doc. 009 and A015); the Casimir effect therefore does not count as an occurrence of $4/3$.

These occurrences stand side by side in the corpus; Doc. 324 calls the agreement of (2) and (3) "no coincidence" without naming a common cause. This document names the common cause of the valid occurrences and adds two occurrences.

ac.2 Electromagnetic Mass: the Factor 4/3 since 1902

The historical finding [E]

If the electron is modelled as a uniformly charged spherical shell of radius a, the field energy at rest is $U = e^2/(8\pi\varepsilon_0 a)$ and the associated mass is $m_0 = U/c^2$. If instead one computes the field momentum of the slowly moving shell, one obtains

$$\mathbf{p} = \frac{4}{3}\,\frac{U}{c^2}\,\mathbf{v}, \tag{ac.1}$$

i.e. a momentum mass of $\frac{4}{3}m_0$. This factor first appeared in Abraham (1902) and Lorentz (1904). Poincaré (1906) introduced non-electromagnetic binding stresses that remove it computationally; Fermi (1922), Wilson (1936) and Rohrlich (1960) showed that a covariant definition of the four-momentum makes it vanish as well. The literature still disagrees on whether this explains the *cause* or merely redefines the *symptom* (Singal 2024; Universe 11(3):97, 2025).

> **Why clarification around 1900 was so difficult.** The confusion had three mutually reinforcing sources.
>
> *First: there was no special relativity.* Abraham and Lorentz worked in a framework where energy and momentum were still separate, independent quantities. Without $E = mc^2$ and the four-momentum, a factor $4/3$ between momentum and energy looks like a genuine contradiction — not like an artefact of a wrong separation.
>
> *Second: the electron was new and its model undecided.* Thomson had discovered the electron in 1897. Whether it was a rigid sphere, a deformable shell, or a point mass was an open question. Within Lorentz's sphere model, the factor $4/3$ was real and troublesome.
>
> *Third: Poincaré's solution was the wrong solution.* The "Poincaré pressure" cancelled the factor computationally, but it was an ad-hoc assumption without physical foundation — and it obscured the actual cause.
>
> After Einstein (1905) and Minkowski (1908) the problem could have been resolved cleanly: the factor $4/3$ arises only when E and $\mathbf{p}$ are integrated separately from the field, without accounting for the boundary terms of Maxwell's stress tensor. Rohrlich (1960) made this clear. What was never the subject of this debate: that $4/3 = 2(1 - \frac{1}{3})$ is a geometric statement about the three-dimensionality of space. The question asked was *"How do we get rid of the factor?"* — rather than *"Where does it really come from?"*

Where the factor comes from [E]

The calculation is short and exposes the cause. With $\mathbf{B} = \mathbf{v} \times \mathbf{E}/c^2$ to first order, the momentum density is $\mathbf{g} = \varepsilon_0\,\mathbf{E} \times \mathbf{B}$, and

$$\mathbf{p} = \frac{\varepsilon_0}{c^2}\int \left[\mathbf{v}\,E^2 - \mathbf{E}\,(\mathbf{E}\cdot\mathbf{v})\right] dV = \frac{\varepsilon_0}{c^2}\,\mathbf{v}\int \left(E^2 - E_{\parallel}^2\right) dV. \tag{ac.2}$$

For a spherically symmetric field, $\langle\cos^2\theta\rangle = 1/3$, hence $\int E_\parallel^2\,dV = \frac{1}{3}\int E^2\,dV$, and with $U = \frac{\varepsilon_0}{2}\int E^2\,dV$ it follows that

$$\mathbf{p} = \frac{\varepsilon_0}{c^2}\cdot\frac{2}{3}\int E^2\,dV\cdot\mathbf{v} = \frac{2}{3}\cdot\frac{2U}{c^2}\,\mathbf{v} = \frac{4}{3}\,\frac{U}{c^2}\,\mathbf{v}. \tag{ac.3}$$

The factor $4/3$ is $2\cdot(1-\frac{1}{3})$. It arises because in three-dimensional space each field component carries one third of the energy, and the component parallel to the motion does not contribute to the momentum. This is the same one-third split that sits in the sphere volume $V = \frac{1}{3}rS$ (Archimedes). The factor is not an artefact of a wrong definition but a statement about the dimension of space.

Connection to FFGFT [S]

In Doc. 006 the electron obeys $E_e = 1/\xi_e$ with $\xi_e = \xi\cdot f_e(1,0,\frac{1}{2}) = \xi = \frac{4}{3}\times 10^{-4}$. The factor $4/3$ thus stands in the numerator of the parameter from which the electron mass is derived (the mass itself is the reciprocal, $1/\xi = (3/4)\cdot 10^4$). Classical electrodynamics has carried, since 1902, the same factor $4/3$ in the electromagnetic electron mass, whose cause is three-dimensionality. FFGFT does not attempt to define it away but sets it as a geometric constant: $\xi = \frac{4}{3}\times 10^{-4}$. This is a structural agreement **[S]**; a derivation of one from the other is not available.

ac.3 Casimir Values: 3/4 and 4/3 as a Reciprocal Pair

The calculation [E]

The quadratic Casimir operator measures the total charge of a representation: it is the group-theoretic analogue of $j(j+1)$ for angular momentum. For SU(2) and spin $\frac{1}{2}$ it is $\frac{1}{2}(\frac{1}{2}+1) = \frac{3}{4}$ — the quantum spin number of the electron. For SU(3) in the fundamental representation (the three colour charges of quarks) it is $\frac{4}{3}$ — the same factor as in the electromagnetic mass.

The quadratic Casimir operator of the fundamental representation of SU(N) is $C_2 = \frac{N^2-1}{2N}$:

Group	C_2	physical location
SU(2)	$3/4$	spin $\frac{1}{2}$: $s(s+1) = \frac{3}{4}$
SU(3)	$4/3$	colour charge; Doc. 324

We have $C_2(\mathrm{SU}(2))\cdot C_2(\mathrm{SU}(3)) = 1$. The verification script shows **[B]**: among all $N \geq 2$, $N = 2$ is the only case in which $C_2(N)\cdot C_2(N{+}1) = 1$. The two smallest non-abelian groups form a distinguished reciprocal pair.

Both values belong to the electron [S]

The value $3/4$ is the spin Casimir of the electron. The value $4/3$ is, by §ac.2, its electromagnetic mass factor and, in FFGFT, the numerator of $\xi = (4/3)\cdot 10^{-4}$ (the mass reciprocal is $1/\xi = 7500 = (3/4)\cdot 10^4$). Spin and mass of the electron thus carry the reciprocal pair $(3/4,\ 4/3)$. The fundamental formula $\tilde{T}\cdot m = 1$ is a reciprocal structure; that the two electron invariants also form a reciprocal pair is structurally compatible, but not derived.

ac.4 The Triple (3,4,5)

The reciprocal pair $(3/4,\ 4/3)$ is the leg ratio of the smallest Pythagorean triple: $\tan\alpha = 4/3$, $\cot\alpha = 3/4$, $\alpha = \arctan\frac{4}{3} \approx 53.13^\circ$, $\sin\alpha = \frac{4}{5}$, $\cos\alpha = \frac{3}{5}$. The triple is not one among many; four properties distinguish it **[E]**, all recomputed in the verification script **[B]**:

1. **Berggren root (1934).** Every primitive Pythagorean triple arises from $(3,4,5)$ by a unique sequence of multiplications with three integer 3×3 matrices. The three children of $(3,4,5)$ are $(5,12,13)$, $(21,20,29)$, $(15,8,17)$; $(3,4,5)$ itself has no positive parent triple. It is the root of the tree.
2. **Arithmetic progression.** $(3,4,5)$ is the only right triangle whose sides form an arithmetic progression: $a^2 + (a+d)^2 = (a+2d)^2 \Leftrightarrow (a-3d)(a+d) = 0$.
3. **Inradius 1.** $r = (3+4-5)/2 = 1$. The triples $(5,12,13)$ and $(8,15,17)$ have $r = 2$ and $r = 3$; $(3,4,5)$ is the unit step.
4. **Gaussian integers.** $(2+i)^2 = 3+4i$. The triple is the square of the smallest non-trivial Gaussian integer; α is the angle of this squaring.

The Five in the Triple and Its Connection to the Icosahedral Group [S]

The hypotenuse of the triple is 5, and 5 appears in the trigonometric values: $\sin\alpha = 4/5$, $\cos\alpha = 3/5$. This 5 is not accidentally the same as the 5 in A_5.

The algebraic chain is as follows. The triple arises as the square of the Gaussian integer $2+i$: $(2+i)^2 = 3+4i$ **[B]**. The Gaussian norm is $|2+i|^2 = 4+1 = 5$. The prime 5 splits in $\mathbb{Z}[i]$ as $5 = (2+i)(2-i)$ into two conjugate Gaussian prime factors: $N(2\pm i) = 5$ is prime, so $2\pm i$ are Gaussian primes and not units; hence 5 is reducible in $\mathbb{Z}[i]$ and not prime there ($5 \equiv 1 \bmod 4$). The same prime 5 appears in the icosahedral group. The group A_5 has order 60, and the fivefold rotation R_5 has order 5. Its eigenvalue ζ_5 (fifth root of unity) lives in the finite field $\mathrm{GF}(81) = \mathrm{GF}(3^4)$, because $5 \mid 80 = |\mathrm{GF}(81)^*|$ **[B]**. In the same field $\mathrm{GF}(81)$, via the subfield $\mathrm{GF}(9) \subset \mathrm{GF}(81)$, also lives the golden ratio $\phi = (1+\sqrt{5})/2$ (Docs. 364, 370).

The connection between the triple $(3,4,5)$ and the icosahedral group A_5 thus runs through the prime 5 and the field $\mathrm{GF}(81)$:

$$(2+i)^2 = 3+4i \implies |2+i|^2 = 5 \implies 5 \mid |\mathrm{GF}(81)^*| = 80 \implies \zeta_5 \in \mathrm{GF}(81).$$

This connects the triple algebraically with the icosahedral part of the corpus (Docs. 285, 293, 367, 370). What this connection is not: a geometric identification. The angle $\alpha = \arctan(4/3) = 53.13^\circ$ is not a rotation angle of A_5 (which has 72°, 144°, 120°, 180°). The connection is algebraic, not geometric **[S]**.

ac.5 Common Cause and Limit

What belongs together [S]

Occurrence	Value	Cause	Source
Sphere volume $V = \frac{1}{3}rS$	$4/3$	one-third split in 3D	Docs. 006, 205
EM mass, field momentum	$4/3$	one-third split in 3D	new, §ac.2
Photon gas, $(u+p)/u$ at $p = u/3$	$4/3$	one-third split in 3D	new, §ac.5
Casimir effect*	$(4/3)$	appended, see below	Doc. 324
$C_2(\mathrm{SU}(3))$	$4/3$	$N = 3$	Doc. 324
$C_2(\mathrm{SU}(2))$	$3/4$	$N = 2$	new, §ac.3
$(3, 4, 5)$	$4/3$, $3/4$	Berggren root	new, §ac.4

* The QFT Casimir energy contains no factor $4/3$ (Section ac.1); the factor is appended in Doc. 324 and does not count as an occurrence.

The enthalpy factor is the calculation of §ac.2 in another language: an isotropic photon gas has pressure $p = u/3$, and the momentum of the moving system is $\mathbf{p} = \frac{4}{3}(E/c^2)\mathbf{v}$ (von Laue) **[E]**. The first three rows share a common cause: the number 3 of spatial dimensions and the resulting one-third split. The Casimir rows carry the 3 as $N = 3$ (fundamental representation of SU(3), $C_2 = 4/3$) and as $N^2 - 1 = 3$ (dimension of the adjoint representation of SU(2), $C_2 = 3/4$), not as the rank of the group (SU(3) has rank 2, SU(2) rank 1). That SU(3) and three-dimensional space yield the same factor is, in FFGFT, not a coincidence but follows from the $\mathbb{Z}_3$ triality on D_4 (Doc. 365, layer 4) — this is the point at which the group number and the space number are already identified in the corpus.

The angle α in the D lattice [B]

The D_n lattice is what crystallography calls the face-centred cubic (FCC) lattice: in $\mathbb{R}^n$ all integer vectors whose coordinate sum is even. In FFGFT, D_4 is the carrier lattice of the torus T^4; its shortest vectors ($|v|^2 = 2$) define its lattice constant. The question is: on which shells does the angle $\arctan\frac{4}{3} = 53.13^\circ$ appear?

Among the shortest vectors ($|v|^2 = 2$) of the D_3 lattice (FCC) and the D_4 lattice, $\alpha = \arctan\frac{4}{3} = 53.13^\circ$ does not occur; both yield only 60°, 90°, 120° **[X]**. On the shell $|v|^2 = 10$ it does occur: $(3, 1, 0, 0)$ and $(1, 3, 0, 0)$ lie in D_4 (even coordinate sum), and $\cos\angle = 6/10 = 3/5$ **[B]**.

The cause is algebraic. The two-dimensional lattice D_2 (even coordinate sum) is exactly the ideal $(1+i)\,\mathbb{Z}[i]$ of the Gaussian integers **[B]**, and

$$(1+i)(2+i) = 1+3i, \qquad (1+i)(2-i) = 3+i, \tag{ac.4}$$

with $|1+3i|^2 = 10$. The angle between $2+i$ and $2-i$ is $\arccos\frac{3}{5} = \alpha$; multiplication by $(1+i)$ rotates both by 45° and leaves the angle unchanged. The triple $(3, 4, 5)$, which by §ac.4 is the square of $2+i$, therefore sits, via the factor $(1+i)$ — the generator of the D lattice — in the carrier geometry of FFGFT, on the first shell whose norm is not composed purely of $|1+i|^2 = 2$ and $|2|^2 = 4$.

The Berggren matrices are isometries of the quadratic form $x^2 + y^2 - z^2$, i.e. elements of the integer Lorentz group $O(2,1;\mathbb{Z})$ (Cha and Conceição 2023) **[E]**; the tree of all triples is thus a discrete hyperbolic structure on the same lattice.

The Right Angle in FFGFT — a Further Thread [S]

That the 90° angle plays the key role in §ac.2 has not yet been named explicitly in the text. The derivation shows it: the field component *parallel* to the direction of motion (0°) does not contribute to the field momentum; the two components *perpendicular* to it (90°) do. In three dimensions with three orthogonal directions, one falls out and two remain. That is $2/3$. Multiplied by 2 (from the cross-product structure of $\mathbf{E} \times \mathbf{B}$) gives $4/3$. The geometric statement is:

$$\tfrac{4}{3} = 2 \cdot \left(1 - \tfrac{1}{3}\right) = 2 \cdot \frac{\text{number of directions perpendicular to motion}}{3}. \tag{ac.5}$$

The 90° angle is not incidental but the cause. It also appears explicitly in the calculation: $\langle \cos^2\theta \rangle = 1/3$ follows from spherical symmetry, because this average over all angles is precisely what eliminates the right angle as a mean.

The right angle in the D_4 lattice. Among the shortest vectors of D_4 (even coordinate sum, $|v|^2 = 2$) exactly the angles 60°, 90°, 120°, 180° appear **[B]**. The 90° is thus a distinguished lattice angle, but it is not alone: it shares the space of lattice angles with 60° and 120°. Whether the 90° lattice angle plays a distinguished role in FFGFT beyond the others — for instance as the separation angle between the four torus directions, as an orthogonality condition between space and time, or as a condition for the projection $T^4 \to \mathbb{R}^3 \times S^1$ — has not been investigated. That is the genuinely open question.

The Fourier modes are orthogonal. The three lepton modes ν_0, ν_1, ν_2 form an orthonormal basis: $\langle \nu_j | \nu_k \rangle = \delta_{jk}$ **[B]**. Each pair of modes stands at 90° in Hilbert space. This orthogonality is the basic condition for the matrix element $\langle \nu_0 | R_5 | \nu_1 \rangle$ to measure the mixing between two *perpendicular* states — and for the modulus squared $|\langle \nu_0 | R_5 | \nu_1 \rangle|^2 = 2/9$ to be a transition probability in the sense of Born's rule. Without 90° in Hilbert space, no clean probability interpretation.

2/3 and 4/3 as a bridge to the Koide quotient. The Koide quotient is $Q = (\sum m_k)/(\sum \sqrt{m_k})^2 = 2/3$ **[E]**. The field-momentum formula from §ac.2 gives $\mathbf{p} = \frac{4}{3}(U/c^2)\mathbf{v}$. Numerically, exactly:

$$\tfrac{4}{3} = 2 \cdot \tfrac{2}{3} = 2\,Q.$$

The factor $4/3$ of the electromagnetic mass is twice the Koide quotient. This is a numerical identity — whether a deeper connection lies behind it, between the three-dimensionality factor of field physics and the mass-structure quotient of the leptons, is open **[S]**. It would be the most direct link of the Abraham-Lorentz problem to the lepton masses that the corpus currently contains: via the number $2/3$, which appears on the one hand as the Koide quotient, and on the other as the geometric fraction of transverse directions in three-dimensional space.

The T^4 torus and the four orthogonal circles. The torus T^4 in FFGFT carries four circle directions that in the standard setup stand pairwise at $90°$ — the simplest metric on the product torus $S^1 \times S^1 \times S^1 \times S^1$. These four $90°$ pairs generate the D_4 lattice of winding numbers. The transition to spacetime $T^4 \to \mathbb{R}^3 \times S^1$ picks *one* direction (the mass dimension that becomes time) and leaves the other three as spatial directions, pairwise at $90°$. The three-dimensionality of space — that exactly *three* directions can stand perpendicular to a motion — is therefore not a free choice but follows from this projection step. The $4/3$ factor is in this picture the signature of the symmetry that was left behind: the original four-dimensionality of the torus leaves in three-dimensional space exactly the three orthogonal directions that produce $4/3 = 2 \cdot (1 - 1/3)$.

Whether this reading — $4/3$ as a trace of the projection $T^4 \to \mathbb{R}^3 \times S^1$ — can be formally carried out in the corpus is the sharpest open question of this subsection **[S]**.

What is not shown [S]

That the angle occurs in the carrier is shown. That it plays a *role* there — as a mode angle, a winding angle, or in the $\mathbb{Z}_3$ identification — is not shown. The bridge currently reads: the same number $4/3$ at five places with a common cause for three of them, and the associated angle on shell 10 of the carrier lattice. Whether shell 10 is distinguished in FFGFT is open.

ac.6 Result

1. The factor $4/3$ of the electromagnetic electron mass (Abraham 1902) is a three-dimensionality effect: $4/3 = 2(1 - \frac{1}{3})$ **[E][B]**. It is the numerator of $\xi = (4/3) \cdot 10^{-4}$, not of $1/\xi = 7500 = (3/4) \cdot 10^4$ **[S]**.
2. $C_2(\mathrm{SU}(2)) = 3/4$ and $C_2(\mathrm{SU}(3)) = 4/3$ are reciprocals; $N = 2$ is the only such case **[B]**. Both values are invariants of the electron (spin, mass) **[S]**.
3. $(3, 4, 5)$ is the Berggren root, the only triple in arithmetic progression, has inradius 1, and is the square of $2 + i$ **[B]**. It carries the reciprocal pair as its legs.
4. The angle $\arctan \frac{4}{3}$ is absent among the shortest vectors of D_3, D_4 **[X]**, but occurs on the shell $|v|^2 = 10$ because $D_2 = (1 + i)\mathbb{Z}[i]$ and $(1 + i)(2 \pm i) = 1 + 3i,\ 3 + i$ **[B]**. A role of this shell in FFGFT is open **[S]**.
5. The enthalpy factor $(u + p)/u = 4/3$ at $p = u/3$ is a further occurrence of the one-third split **[E]**.

Verification script: `2/python/Dok371_Skripte/ffgft_371_faktor_vier_drittel.py` — 28 checks, 28 passed (P1–P5, P7 positive; P6: negative control of the shortest vectors and positive finding on shell 10).

Register (Doc. 190): no entry — no earlier document is corrected; the occurrences in Docs. 006, 205, 324 remain valid and are supplemented.

[1] M. Abraham, *Prinzipien der Dynamik des Elektrons*, Ann. Phys. 10, 105 (1903).

[2] H. A. Lorentz, *Electromagnetic phenomena in a system moving with any velocity smaller than that of light*, Proc. Roy. Neth. Acad. 6, 809 (1904).

[3] H. Poincaré, *Sur la dynamique de l'électron*, Rend. Circ. Mat. Palermo 21, 129 (1906).

[4] A. K. Singal, *Enigmatic factor of 4/3 in electromagnetic momentum of a moving spherical capacitor*, arXiv:2402.14884 (2024).

[5] *Conclusions Not Yet Drawn from the Unsolved 4/3-Problem*, Universe 11(3), 97 (2025).

[6] B. Berggren, *Pytagoreiska trianglar*, Tidskrift för elementär matematik, fysik och kemi 17, 129 (1934).

[7] B. Cha, R. Conceição, *A polynomial analogue of Berggren's theorem on Pythagorean triples*, arXiv:2310.02144 (2023).

Chapter ad

Doc. 372 — The Matrix-Element Principle and Its Reach

Abstract. The lepton masses follow without free parameters from a single matrix element, $|\langle v_0|R_5|v_1\rangle|^2 = 2/9$. This document examines what the same mechanism carries beyond that. First result **[B]**: in the $\mathbb{Z}_3$ mode basis the icosahedral group A_5 generates exactly eight squared matrix-element moduli, $\{0,\ (5-3\phi)/9,\ 1/9,\ 2/9,\ 4/9,\ 5/9,\ (2+3\phi)/9,\ 1\}$; the reach of the mechanism is therefore exactly bounded. Second result **[K]**: the on-shell Weinberg angle $1 - M_W^2/M_Z^2$ lies within $0.4\,\%$ of $2/9$ — the on-shell scheme is the natural one for pole masses from $\tilde{T} \cdot m = 1$; the $\overline{\text{MS}}$ deviation of $3.9\,\%$ is scheme dependence. Prediction $M_W = 80.420\,\text{GeV}$. Third result **[K]**: the atmospheric mixing angle satisfies $\sin^2\theta_{23} \in \{4/9,\ 5/9\}$; the octant ambiguity of the data is the ambiguity $4/9 \leftrightarrow 5/9$, maximal mixing $1/2$ is excluded. Negative **[X]**: $\sin^2\theta_{12}$, $\sin^2\theta_{13}$ and α_s do not lie in the eight-element set and cannot be reached by a single A_5 matrix element.

Notation

Epistemic markers:

- **[B]** algebraically proven (check script, exact arithmetic)
- **[K]** numerically confirmed
- **[S]** structurally plausible, identification open
- **[E]** established external mathematics or physics
- **[X]** tested and negative

Symbol	Meaning
R_5	five-fold rotation (72°), order-5 element of A_5
v_0, v_1, v_2	orthonormal $\mathbb{Z}_3$ modes (symmetric, phase-carrying)
$p_k = \lvert\langle v_0 \lvert R_5 \rvert v_k\rangle\rvert^2$	icosahedral redistribution weights
$\phi = (1+\sqrt{5})/2$	golden ratio
$\theta = 2/9 = p_0$	Koide phase (= Born probability, Doc. 367)
$\xi = 4/30000$	FFGFT scaling parameter
$\sin^2\theta_W$	Weinberg angle (electroweak mixing)
θ_C	Cabibbo angle (quark generation mixing)
C_3	three-fold rotation (120°), second generator of A_5
A_5	icosahedral group (60 elements)
$\mathbb{Z}_3$	cyclic group of order 3 (mode symmetry)
GF(q), GF(q)*	Galois field with q elements, its multiplicative group
$\tilde{T}$	proper-time scaling ($\tilde{T} \cdot m = 1$)
Q	Koide quotient $\sum m_i / (\sum \sqrt{m_i})^2$
Q_{up}, Q_{down}	Koide quotient of the up and down quarks
r_i, p_i	prefactor and exponent of the mass formula $m_i = r_i \xi^{p_i} v$
K_{frak}	fractal correction factor $1 - 100\xi$ (Doc. 133)
M_W, M_Z	masses of the W and Z bosons (pole masses)
m_h, m_t, m_s	mass of the Higgs particle, of the top quark, of the strange quark
m_e, m_p	mass of the electron, of the proton
v	electroweak vacuum scale (246.22 GeV, SM definition)
$\overline{\text{MS}}$	renormalisation scheme (minimal subtraction)
θ_{12}, θ_{13}, θ_{23}	neutrino mixing angles (solar, reactor, atmospheric)
δ_{CP}	CP-violating phase
Δm^2_{atm}, Δm^2_{sol}	differences of the squared neutrino masses
α, α_s	fine-structure constant, strong coupling
Λ_{QCD}	confinement scale of the strong interaction
R_{Proton}	charge radius of the proton
r_e, λ_p	classical electron radius, Compton wavelength of the proton
σ	standard deviation of the measurement

ad.1 The Mechanism and Its Yield for the Leptons

The starting point of the mechanism is a single number: the squared modulus of the matrix element of the five-fold rotation $R_5 \in A_5$ between the symmetric mode v_0 and the electron mode v_1,

$$p_0 = |\langle v_0 | R_5 | v_1 \rangle|^2 = \frac{2}{9}.$$

From this one matrix element follow (Doc. 293, 367, 368):

1. **The Koide phase** $\theta = 2/9$ **[B]**: the same number, written as an angle in the Koide parametrisation, gives the three charged lepton masses without free parameters to 10^{-3} % (Doc. 292, 293, 367).
2. **The redistribution weights** $p_0 = 2/9$, $p_1 = (2 + 3\phi)/9$, $p_2 = (5 - 3\phi)/9$ with $p_0 + p_1 + p_2 = 1$ **[B]**.
3. **The ϕ skeleton of the weight ratios [B]**:

$$\frac{p_1}{p_2} = \phi^8, \qquad \frac{p_0}{p_2} = 2\phi^4.$$

   Both relations are exact algebraic identities following from $p_1 = (2 + 3\phi)/9$ and $p_2 = (5 - 3\phi)/9$ (Doc. 368).
4. **The link 4/3 = 2Q** (Doc. 371): the Koide quotient $Q = 2/3$ and the three-dimensionality factor $4/3$ of the electromagnetic mass (Abraham 1902) satisfy $4/3 = 2Q$ exactly.

The mechanism thus carries four independent structural results from a single matrix-element value $2/9$.

ad.2 The Spectrum of the Matrix Elements [B]

Before asking which constant the mechanism might carry further, the reverse question must be answered: which numbers can it produce at all? This is a finite calculation. The group A_5 has 60 elements; in the three-dimensional representation, expressed in the $\mathbb{Z}_3$ mode basis v_0, v_1, v_2, every element has nine matrix elements. The check script generates all 60 elements from R_5 and C_3 and collects the squared moduli. Exactly eight values occur:

$$\left\{0,\ \tfrac{5-3\phi}{9},\ \tfrac{1}{9},\ \tfrac{2}{9},\ \tfrac{4}{9},\ \tfrac{5}{9},\ \tfrac{2+3\phi}{9},\ 1\right\}$$
$$= \{0,\ 0.0162,\ 0.1111,\ 0.2222,\ 0.4444,\ 0.5556,\ 0.7616,\ 1\}.$$

All denominators are $9 = |\mathbb{Z}_3|^2$; the two irrational values are the weights p_2, p_1 from Doc. 293/368. The set is not closed under $x \mapsto 1 - x$ ($1 - 1/9 = 8/9$ is missing), but the subset $\{4/9, 5/9\}$ and the pair $\{p_1, p_0 + p_2\}$ are.

> **Reach theorem.** Every constant that arises as the squared modulus of a single A_5 matrix element in the $\mathbb{Z}_3$ basis lies in this eight-element set. What does not lie in it cannot be delivered by the mechanism in this form **[B]**.

ad.3 Test Against the Mixing Parameters of the Standard Model

The eight-element set is held against the dimensionless mixing parameters of the Standard Model (PDG 2024, NuFIT 6.0). Hits and misses are both recorded.

Weinberg angle: on-shell $= 2/9$ [K]

The Weinberg angle has two common definitions. The $\overline{\text{MS}}$ value $\sin^2\hat{\theta}_W(M_Z) = 0.2312$ depends on the renormalisation scheme. The on-shell definition

$$\sin^2\theta_W \equiv 1 - \frac{M_W^2}{M_Z^2}$$

by contrast is fixed by pole masses. Since the FFGFT ground relation $\tilde{T}\cdot m = 1$ yields pole masses (Doc. 333), the on-shell scheme is the natural one for FFGFT.

Input	M_W [GeV]	$1 - M_W^2/M_Z^2$	to $2/9$	Sign.
World average 2024 (without CDF)	80.3692	0.22321	0.44 %	3.8σ
CDF 2022	80.4335	0.22197	0.12 %	1.4σ
PDG on-shell direct	—	0.22305	0.37 %	3.6σ
$\overline{\text{MS}}$ (comparison)	—	0.23121	3.9 %	—

The value $2/9$ lies between the world average and the CDF value; the value of M_W that gives $\sin^2\theta_W = 2/9$ exactly is

$$M_W(2/9) = M_Z\sqrt{7/9} = 80.420 \text{ GeV}.$$

This is a falsifiable prediction. It lies 3.8σ from the current world average and 1.4σ from the CDF value; the M_W measurements are at present not mutually consistent. The classification is **[K]** at 0.4 %, with the tension stated.

This replaces the previous corpus status: Doc. 336 named as the best Galois candidate $|\text{GF}(3)^*|/|\text{GF}(9)^*| = 1/4$ with 8 % deviation. The renormalisation-group running from ξ (Doc. 323, 0.2308) and the on-shell matrix element $2/9$ are two descriptions: the first in the $\overline{\text{MS}}$ scheme from the scale flow, the second in the on-shell scheme from the lattice geometry. Whether both describe the same geometric object remains open **[S]**; that they serve different schemes is settled.

Atmospheric mixing angle: $\sin^2\theta_{23} \in \{4/9, 5/9\}$ [K]

The eight-element set contains the pair $4/9$, $5/9$ with $4/9 + 5/9 = 1$. For the atmospheric neutrino mixing angle, $\sin^2\theta_{23}$ is experimentally close to $1/2$, with the octant unresolved:

Fit	$\sin^2\theta_{23}$	nearest candidate	significance
PDG 2024 (NO)	0.558 ± 0.020	$5/9 = 0.5556$	0.1σ
NuFIT 6.0 (IO)	0.550 ± 0.015	$5/9$	0.4σ
NuFIT 6.0 (NO)	0.470 ± 0.015	$4/9 = 0.4444$	1.7σ

In the matrix-element picture the octant ambiguity of the data is the ambiguity $4/9 \leftrightarrow 5/9$. The mechanism thus makes a prediction that is independent of the octant:

$$\left|\sin^2\theta_{23} - \tfrac{1}{2}\right| = \tfrac{1}{18} = 0.0556.$$

Maximal mixing ($\sin^2\theta_{23} = 1/2$) is excluded. This can be tested with DUNE and Hyper-Kamiokande. Classification **[K]** for the value, **[S]** for the assignment to the neutrino sector (the neutrino masses themselves come from $\mathrm{GF}(27)^*$ in Doc. 340, not from A_5; that the angle θ_{23} is nevertheless an A_5 matrix element would be a second, independent A_5 access to the lepton sector).

Relation to the Galois reading in Doc. 340

Doc. 340 reads two mixing angles off the orbit elements of $\mathrm{GF}(27)^* \cong \mathbb{Z}_2 \times \mathbb{Z}_{13}$: $\sin^2\theta_{12} \approx \cos^2(4\pi/13) = 0.323$ and $\sin^2\theta_{23} \approx \cos^2(10\pi/13) = 0.560$. For θ_{23} two values therefore stand side by side, 0.560 and $5/9 = 0.556$, both of which hit the PDG value to within $0.4\,\%$. This needs to be clarified.

The two values are not identical [B]. $\cos(2\pi/13)$ has a minimal polynomial of degree $\varphi(13)/2 = 6$ over $\mathbb{Q}$; consequently $\cos^2(10\pi/13)$ is irrational, whereas $5/9$ is rational. The closeness of $0.85\,\%$ is numerical, not algebraic: $5/9$ is merely the best rational approximation to $\cos^2(10\pi/13)$ with denominator ≤ 13.

The Galois reading is an order-of-magnitude statement. The set $\{\cos^2(2\pi k/13),\ 1 - \cos^2(2\pi k/13) \mid k = 1..6\}$ contains twelve values between 0.01 and 0.99. Held against seven dimensionless parameters of the Standard Model, for six of them an element is found within $6.6\,\%$ — the set is too dense to discriminate at this tolerance. The eight-element set of §ad.2, by contrast, discriminates: three hits below $1.3\,\%$, the rest above $5\,\%$. The $\cos^2$ readings in Doc. 340 are therefore to be understood as orders of magnitude, not as derivations; Doc. 340 itself labels them as “read off”.

Division of labour in the neutrino sector. This determines which structure carries which parameter:

Quantity	Structure	Source	Status
$\Delta m^2_{\mathrm{atm}},\ \Delta m^2_{\mathrm{sol}}$	$\mathrm{GF}(27)^*$ orbits	Doc. 340	**[K]** (1 %, 0.5 %)
ν_3 Dirac, $\nu_{1,2}$ Majorana	quadratic character mod 13	Doc. 343	**[B]**
$\sin^2\theta_{13} = (25\xi)^{2/3}$	$\xi \cdot \lvert A_5\rvert$ ($25\xi = 1/300$)	Doc. 340	**[K]** (1.4 %)
$\sin^2\theta_{23} \in \{4/9, 5/9\}$	A_5 matrix element	Doc. 372	**[K]** (0.4 %)
$\sin^2\theta_{12}$	only $\cos^2(4\pi/13)$, order of magnitude	Doc. 340	**[S]** (5 %)
δ_{CP}	orbit-3 element $k = 10$	Doc. 340	**[S]**

The neutrino sector is carried by two structures: $\mathrm{GF}(27)^*$ for the masses and the Dirac/-Majorana character, A_5 (alone or with ξ) for θ_{23} and θ_{13}. Both sit on the same lattice (Doc. 364: $A_5 \subset \mathrm{Aut}(D_4)$ only via the phase channel $\mathrm{GF}(81)$; $\mathrm{GF}(27)$ as the ordering structure, Doc. 341). Whether θ_{12} is likewise an A_5 object that only becomes visible through a

product or another basis, or whether it remains with $\mathbb{Z}_{13}$ alone, is the remaining open question of the sector **[S]**.

Cabibbo angle [S]

$\lambda = \sin\theta_C = 0.2250 \pm 0.0007$ lies $1.2\,\%$ (4.1σ) from $2/9$. The long-known numerical closeness $\sin\theta_C \approx \sin^2\theta_W^{\text{on-shell}}$ is read here as the closeness of both quantities to $2/9$. This is not an identification: Doc. 041 derives $|V_{us}| = \sqrt{m_d/m_s} \cdot f_{\text{Cab}}$ from quark masses, and the quark coefficients are QCD-calibrated (Doc. 289). Whether λ is a matrix element of another subgroup of $\text{Aut}(D_4)$ is open **[S]**.

Negative findings [X]

Three parameters do not lie in the eight-element set and cannot be reached by a single A_5 matrix element:

Parameter	Value	nearest element	distance
$\sin^2\theta_{12}$ (solar)	0.307	$2/9$	28 %
$\sin^2\theta_{13}$ (reactor)	0.0222	$(5-3\phi)/9$	27 %
$\alpha_s(M_Z)$	0.118	$1/9$	5.8 %

The reach theorem of §ad.2 makes this a sharp statement: these three quantities require either another group, a product or difference of matrix elements, or a different mechanism. For α_s the corpus has the Galois running with ξ; for θ_{12} and θ_{13} nothing is available in the matrix-element picture.

Quarks: compact description present, unrolling incompatible

In the corpus the charged leptons have two descriptions that meet: compact, the Yukawa ladder $m = r\xi^p v$ with $r_e = 4/3$ (Doc. 006), and unrolled ($T^4 \to \mathbb{R}^3 \times S^1$, Doc. 370), the A_5 mechanism of this document. Both arrive at $Q = 2/3$, $r = \sqrt{2}$ — the "special point" of Doc. 289.

The quarks have only the compact description, and it is more developed than the comparison with unrolling suggests. The exponents p_i are derived from the torus topology (Doc. 189: $p = d_{\text{akt}}/3$, one active dimension fewer per step) — up/down $3/2$, strange 1, charm $2/3$, bottom $1/2$ as transition exponent, top $-1/3$ as inverse winding **[K]**. The ξ scale correction and the cumulative factor $K_{\text{frak}} = 1 - 100\xi$ from the scale flow Planck→electroweak (Doc. 133) weight the ladder; the extended fractal method thereby reaches $\sim 1.2\,\%$ without free parameters (Doc. 005). What is missing are the prefactors r_i: Doc. 189 sets $r_i = n_\phi^2/(n_\theta n_c)$ with $n_c = 3$ for quarks and explicitly names the complete derivation as outstanding; the r_i tabulated in Doc. 006 are back-calculated from the measured masses, but carry exclusively prime factors from the Galois grid $\{2, 3, 5, 7, 11, 13\}$ **[B]** (Doc. 358, Theorem B). The gap is therefore precise: p_i derived, r_i postulated **[S]**.

The unrolling cannot be transferred, and Doc. 289 shows why: the quark triplets do not sit at the special point. With PDG-like values, $Q_{\text{up}} = 0.849$ ($r = 1.76$) and $Q_{\text{down}} = 0.730$

($r = 1.54$). Scale dependence does not change this: running all six masses to a common scale ($\overline{\text{MS}}$ at M_Z, m_t or roughly M_{GUT}), Q_{up} stays between 0.85 and 0.89 and Q_{down} between 0.72 and 0.75 — never close to $2/3$ **[K]**. The A_5 mechanism, however, produces *exactly* $Q = 2/3$ — that is its signature ($\theta = |A|^2 = 2/9 \Rightarrow Q = 2/3$, Doc. 367), and the eight-element set contains no value that yields $Q = 0.85$ or 0.73. The quark masses are incompatible with the lepton unrolling mechanism **[X]**, not merely unworked.

Three structural reasons, each documented in the corpus: quarks carry colour, and the $\mathbb{Z}_3$ that counts the generations for leptons is at the same time the colour $\mathbb{Z}_3$ for quarks (Doc. 321) — the mode space would be 3×3, not 3. Quarks come as isospin doublets, six instead of three modes per sector. And $\tilde{T} \cdot m = 1$ gives pole masses, which are not defined for u, d, s (Doc. 333). The open question is therefore not "how does one unroll quarks like leptons" but: which other subgroup of $\text{Aut}(D_4)$ or which tensor product $A_5 \otimes \mathbb{Z}_3^{\text{colour}}$ has a matrix-element spectrum that contains $Q \approx 0.85$ and 0.73. Doc. 358 names the field $\text{GF}(729)$ as the entry point for the top coefficient (prime 7, $k = 6$) **[S]**. That is a different programme, not an appendix to lepton unrolling.

Gauge bosons and Higgs

What is derived. The electroweak scale v follows by two routes to $0.3\,\%$ **[K]**: via the ladder with m_e as anchor, $m_e/((4/3)\,\xi^{3/2}) = 248.9\,\text{GeV}$ without and $248.9 \cdot 74/75 = 245.6\,\text{GeV}$ ($-0.25\,\%$) with K_{frak} (Docs. 006, 373, 375), or via $v/E_P = \xi^4/(5\pi)$, whose factor 10 is fitted to the measured value (Doc. 387). Both routes thus contain a measured input. The ratio M_W/M_Z follows from §ad.3: $M_W^2/M_Z^2 = 1 - 2/9 = 7/9$ **[K]**. The fine-structure constant $1/\alpha = 3700/27$ follows from Galois orders (Doc. 338) **[K]**. Thus v, α and $\sin^2\theta_W$ are available; the absolute masses $M_W = gv/2$, $M_Z = M_W/\cos\theta_W$ (Doc. 320) follow from them via the standard electroweak relations up to radiative corrections (Δr, of the order of a per cent), which the corpus does not compute. Photon and the eight gluons are massless: through the unbroken $U(1)_{\text{EM}}$ and $SU(3)_c$ (Doc. 320) **[B]**, where $SU(3)_c$ follows from $\mathbb{Z}_3$ triality (Doc. 321) and the eight gluons from the eight three-element orbits of $\text{GF}(27)^*$ (Doc. 339) **[B]**.

Where the Higgs stands on the ladder. The p ladder of Doc. 189 places the Higgs scale at $p = 0$ ($d_{\text{akt}} = 0$): $m_h = r_h\,v$ without ξ suppression, with $r_h = m_h/v = 0.508$. This correctly places the Higgs as the only particle on the v scale but provides no prefactor — the same gap as for the quark r_i.

What is not derived: the Higgs mass [X]. The corpus contains two formulas for m_h, and neither survives recalculation. Doc. 041 gives $m_h = v\xi^{1/4}$ with the numerical value $125\,\text{GeV}$; in fact $v\xi^{1/4} = 26.5\,\text{GeV}$. The same table gives $\Lambda_{\text{QCD}} = v\xi^{1/3} = 200\,\text{MeV}$; in fact $12.6\,\text{GeV}$. Doc. 005 gives $m_H = m_t\,\phi(1 + \xi D_f) = 124.95\,\text{GeV}$; since $m_t\phi = 279.4\,\text{GeV}$, this requires $\xi D_f = -0.553$, i.e. $D_f \approx -4100$ — a fit parameter, not a derivation. These two formulas are recorded as R122 in Doc. 190. Viable candidates are only provided by Doc. 373 (Galois chain and trace rule). The eight-element set yields nothing either: $m_h/v = 0.508$ and $m_h^2/v^2 = 0.258$ do not lie in it **[X]**.

Systematic search. Instead of guessing individual formulas, 17 dimensionless ratios of m_h, M_W, M_Z, m_t, v were held against a library of 276 FFGFT-related numbers (fractions with denominator ≤ 12, powers of ϕ, roots, Galois orders, K_{frak}, the eight-element set). At $0.5\,\%$ tolerance twelve hits are found, at $0.2\,\%$ three; the random expectation is about fourteen and six respectively. There is therefore *no* signal above the look-elsewhere background **[X]**. The obvious reading $m_h = v/2$ lies at $1.7\,\%$ and is not distinguished.

A single candidate has internal structure and is therefore recorded, explicitly as selected after the fact: $m_h/M_Z = 1.3730 \pm 0.0012$ and $m_t/m_h = 1.3784 \pm 0.0026$ both lie at $11/8 = 1.375$ (at 1.7σ and 1.3σ respectively), hence

$$M_Z : m_h : m_t \approx 1 : \tfrac{11}{8} : \left(\tfrac{11}{8}\right)^2, \qquad m_h \approx \sqrt{m_t\, M_Z} \quad (0.20\,\%,\ 1.6\sigma).$$

The numbers $11 = \max(\text{orbit}_4)$ and $8 = |\text{GF}(9)^*|$ are both Galois quantities of the corpus (Doc. 340, 336). The Higgs would then sit as the geometric mean between Z and top — in the language of $\tilde{T} \cdot m = 1$: its winding number is the mean of the two. Status **[S]**, with the proviso that one candidate out of about 5000 tests is not evidence but a hypothesis that would need an independent derivation.

ad.4 Hadrons: Bound States on the Lattice

The leptons are unrolled primary modes of the T^4/D_4 lattice — free, asymptotic particles whose masses follow directly from $\tilde{T} \cdot m = 1$. Hadrons are not. There is no free quark; what is measured is the system energy of the bound state. Three consequences can be tested numerically.

Koide quotient of the hadron multiplets [B]. For an isospin multiplet of dimension n with nearly degenerate masses $m_i = m(1 + \delta_i)$, $|\delta_i| \ll 1$, it holds algebraically exactly that

$$Q \cdot n = 1, \qquad Q = \frac{\sum m_i}{\left(\sum \sqrt{m_i}\right)^2}.$$

Numerically: $Q(p,n) = 1/2$, $Q(\Sigma^+,\Sigma^0,\Sigma^-) = 1/3$, $Q(\Delta^{++},\Delta^+,\Delta^0,\Delta^-) = 1/4$, $Q(\Omega^-) = 1$, all to $< 0.01\,\%$ **[B]**. This is an algebraic identity for degenerate masses, not a new result — but it sharpens the contrast: the leptons break exactly this relation. For $n = 3$ generations the degeneracy rule would require $Q_{\text{Lep}} = 1/3$; in fact $Q_{\text{Lep}} = 2/3$. The factor 2 is the signature of the A_5 mechanism: the icosahedron lifts the degeneracy specifically such that $Q = 2 \cdot (1/3) = 2/3$. Hadrons remain at the degenerate relation because their $SU(2)$ isospin symmetry is almost exactly conserved.

Gell-Mann–Okubo splitting consistent with m_s [K]. The baryon decuplet masses show a uniform spacing: $\Delta m(\Sigma^* - \Delta) = 151.7\,\text{MeV}$, $\Delta m(\Xi^* - \Sigma^*) = 148.1\,\text{MeV}$, $\Delta m(\Omega - \Xi^*) = 140.4\,\text{MeV}$, mean $146.7\,\text{MeV}$. This is the Gell-Mann–Okubo rule; the splitting corresponds to the effective strange quark mass. From the Galois coefficient r_s (Doc. 006) follows $m_s^{(\text{FFGFT})} = r_s\, \xi\, v \approx 93\,\text{MeV}$. The agreement is consistent to a factor of 1.6 **[K]**, but not a precision statement — the difference lies in the QCD dynamics, not in the lattice geometry.

Λ_{QCD} not derivable from the ξ ladder [X]. $\Lambda_{\mathrm{QCD}} \approx 210\,\mathrm{MeV}$ (PDG, $n_f = 5$) could be written as $v\xi^p$ for $p \approx 0.79$ — not a step of the ladder $p = n/3$ or $n/2$. The route via the hadron radius ($\omega_{\mathrm{min}} = 1/R_{\mathrm{had}}$, Doc. 319) is also circular as long as R does not follow from ξ. Λ_{QCD} is thus the sharpest open question of the hadron sector: it lies neither in the eight-element set (§ad.2), nor on the ξ ladder, nor in the Galois grid of Doc. 358.

Gluons as standing waves. Doc. 319 describes gluons as standing waves in the closed proton torus: massless, pure field energy ($E = pc$, $d\tau = 0$), topologically confined by the $\mathbb{Z}_3/SU(3)$ boundary condition (eight linking states of the three flux strands = eight generators of $SU(3)_c$, Doc. 321 **[B]**). In this picture confinement is not a dynamical mechanism but a topological impossibility of escape: the closed torus admits only integer wave numbers; there is no free propagation to the outside. In this language Λ_{QCD} is the minimum frequency on the torus, $\omega_{\mathrm{min}} = 1/R_{\mathrm{Proton}}$ (Doc. 319 **[S]**).

R_{Proton} structurally classified [S]. Doc. 319 gives $\Lambda_{\mathrm{QCD}} \approx \omega_{\mathrm{min}} = \hbar c/R_{\mathrm{Proton}}$ **[S]**. The FFGFT quantities allow a structural classification: the geometric mean of the classical electron radius $r_e = \alpha\hbar c/m_e$ and the proton Compton wavelength $\lambda_p = \hbar c/m_p$ is

$$R_{\mathrm{geom}} = \sqrt{r_e\,\lambda_p} = \frac{\hbar c\sqrt{\alpha}}{\sqrt{m_e\,m_p}} = 0.770\,\mathrm{fm},$$

$8.5\,\%$ below the measured charge radius 0.841 fm. All three factors — α from Galois structure (Doc. 338), m_e from the A_5 mechanism, m_p from the ξ ladder (Doc. 189) — are available in FFGFT. The missing factor $R_{\mathrm{PDG}}/R_{\mathrm{geom}} \approx 1.09$ lies in the quark–gluon binding dynamics, which the programme does not yet capture formally.

Conclusion. Hadrons are bound states on the same lattice as leptons — the same ξ, the same ladder for p_i, the same Galois grid for r_i. Confinement is a topological boundary condition, not a separate mechanism. R_{Proton} is structurally classified to $8.5\,\%$; the residual lies in the binding dynamics. This is the honest limit **[S]**.

ad.5 The More General Principle

The four results of §ad.1 suggest a more general thesis:

> **The matrix-element principle**: every dimensionless physical constant that arises from the overlap of two lattice modes under a finite group rotation can be read as the squared modulus of a matrix element $|\langle v_i|g|v_j\rangle|^2$ — where g is a group element, v_i, v_j are orthonormal modes of the T^4/D_4 lattice, and the result has no free parameters.

For the lepton masses this principle is fully carried out with $g = R_5 \in A_5$, $i = 0$, $j = 1$ **[B]**. For other sectors it is an open hypothesis **[S]**. The sharpest test question is: is there a finite subgroup of $\mathrm{Aut}(D_4)$ and two lattice modes whose squared matrix-element modulus reproduces the Weinberg angle or the Cabibbo angle without free parameters?

That $|A_5| = 60 = 4 \cdot 3 \cdot 5$ unites the three defining symmetry numbers of the lepton structure, and that A_5 is the only simple finite subgroup of $SO(3)$ with five-fold elements, make this hypothesis structurally attractive.

ad.6 Result

1. **Reach theorem [B]**: in the $\mathbb{Z}_3$ basis A_5 generates exactly eight squared matrix-element moduli $\{0, (5-3\phi)/9, 1/9, 2/9, 4/9, 5/9, (2+3\phi)/9, 1\}$.
2. **Lepton masses**: from $2/9$ follow the Koide phase, the weights, the ϕ skeleton ($p_1/p_2 = \phi^8$, $p_0/p_2 = 2\phi^4$) and $4/3 = 2Q$ **[B]**.
3. **Weinberg angle [K]**: on-shell $1 - M_W^2/M_Z^2 = 2/9$ to $0.4\,\%$; prediction $M_W = M_Z\sqrt{7/9} = 80.420$ GeV, between the world average and CDF. Replaces the candidate $1/4$ (8 %) of Doc. 336. The $\overline{\text{MS}}$ result 0.2308 of Doc. 323 remains as the scale-flow description.
4. **Atmospheric mixing [K]**: $\sin^2\theta_{23} \in \{4/9, 5/9\}$, prediction $|\sin^2\theta_{23} - 1/2| = 1/18$; maximal mixing excluded. The PDG value 0.558 hits $5/9$ at 0.1σ.
5. **Comparison with Doc. 340 [B]**: $\cos^2(10\pi/13) \neq 5/9$ (degree 6 versus rational); the $\cos^2(2\pi k/13)$ readings are orders of magnitude. Division of labour: $\mathrm{GF}(27)^*$ carries masses and Dirac/Majorana, A_5 carries θ_{23} and (with ξ) θ_{13}; θ_{12} remains open **[S]**.
6. **Quarks [X]**: compact ladder with derived p_i (Doc. 189) and ξ/K_{frak} weighting (Doc. 133, 005), $\sim 1.2\,\%$ without free parameters; r_i postulated, in the Galois grid **[B]**. $Q_{\text{up}} \approx 0.85$–$0.89$, $Q_{\text{down}} \approx 0.72$–$0.75$ at all scales: incompatible with A_5 unrolling ($Q = 2/3$ exactly).
7. **Bosons**: v, α, $\sin^2\theta_W$ available **[K]**; photon/gluons massless **[B]**; the Higgs formulas in Doc. 041 and 005 do not survive recalculation **[X]** (R122); candidates in Doc. 373.
8. **Cabibbo [S]**: λ within $1.2\,\%$ of $2/9$, not identified; Doc. 041 has a different derivation.
9. **Negative [X]**: $\sin^2\theta_{12}$, $\sin^2\theta_{13}$, α_s do not lie in the eight-element set.
10. The general matrix-element principle (§ad.5) thus has three confirmed instances ($2/9$ lepton, $2/9$ Weinberg, $\{4/9, 5/9\}$ atmospheric) and three excluded ones; the next question is whether another subgroup of $\mathrm{Aut}(D_4)$ carries the excluded values.

Check script:
`2/python/Dok372_Skripte/pruef_372_matrixelement.py`

Register (Doc. 190): R122 (Higgs mass in Doc. 041 and 005).

[1] J. Pascher, *Doc. 292 — Koide formula and FFGFT* (2025).

[2] J. Pascher, *Doc. 293 — Icosahedral matrix element* (2025).

[3] J. Pascher, *Doc. 323 — Renormalisation group and Weinberg angle* (2025).

[4] J. Pascher, *Doc. 340 — Neutrino mass hierarchy from GF(27)** (2026).

[5] J. Pascher, *Doc. 367 — Why $\theta = 2/9$* (2026).

[6] J. Pascher, *Doc. 368 — From p_0, p_1, p_2 to lepton masses* (2026).

[7] J. Pascher, *Doc. 371 — The factor 4/3* (2026).

Chapter ae

Doc. 373 — The Yukawa Mechanism as a Consequence of $\tilde{T} \cdot m = 1$

Abstract. In the Standard Model the Yukawa coupling $y_i = m_i/v$ is an empirical assumption: it is chosen so that the observed masses emerge after electroweak symmetry breaking. This document shows that the same coupling follows from the FFGFT ground relation $\tilde{T} \cdot m = 1$ once the electroweak scale v is allowed to depend on position. The mass-proportional coupling of all fermions to the Higgs is then not an assumption but the fluctuating form of $\tilde{T} \cdot m = 1$. The Higgs particle $h(x)$ is the vibrational quantum of the lattice scale — not an independent scalar field with an unknown potential. Its mass is the residual stiffness of the lattice: $M_W^2 + M_Z^2 + m_h^2 = v^2/2$ (trace rule, $0.45\,\%$). Together with $\sin^2\theta_W = 2/9$ and $m_h/M_Z = 11/8$, M_Z, M_W and m_h follow from v alone to within 0.2–$0.3\,\%$.

Notation

[B] algebraically proven
[K] numerically confirmed
[S] structurally plausible, derivation open
[X] tested and negative

Symbol	Meaning
$\tilde{T}_i$	proper-time scaling of mode i
m_i	rest mass of particle i
v	electroweak vacuum scale ($246.22\,\mathrm{GeV}$)
$h(x)$	physical Higgs field (fluctuation of v)
y_i	Yukawa coupling (SM definition: $m_i = y_i v$)
r_i, p_i	prefactor and exponent of the FFGFT mass formula
ξ	FFGFT scaling parameter ($4/30000$)
ψ_i, $\bar{\psi}_i$	fermion field of particle i and its Dirac adjoint
$\mathscr{L}$	Lagrangian density
H	complex lattice amplitude (Higgs field), $H = (v + h)/\sqrt{2}$
$n_{\phi,i}$, $n_{\theta,i}$	winding numbers on the T^4 torus
n_c	colour multiplicity
$d_{\mathrm{akt},i}$	number of active torus dimensions (Doc. 189)
M_W, M_Z	masses of the W and Z bosons (pole masses)
m_h, m_t	mass of the Higgs particle, of the top quark
m_s	mass of the strange quark
$W^\pm$	charged gauge boson; one complex mode, $W^\pm = (W^1 \mp i W^2)/\sqrt{2}$
W^3, B	neutral gauge fields before mixing ($SU(2)$, $U(1)$)
γ	photon
θ_W	Weinberg angle, on-shell $\sin^2\theta_W = 1 - M_W^2/M_Z^2$
g, g'	gauge couplings of $SU(2)$ and $U(1)$
λ	Higgs self-coupling, $m_h^2 = 2\lambda v^2$
y_t	Yukawa coupling of the top quark
G_F	Fermi constant
τ_μ	muon lifetime (the only measured quantity behind v)
ϕ	golden ratio $(1+\sqrt{5})/2$
Q	Koide quotient $\sum m_i / (\sum \sqrt{m_i})^2$
α, α_s	fine-structure constant, strong coupling
Λ_{QCD}	confinement scale of the strong interaction
R_{Proton}	charge radius of the proton
θ_{12}, θ_{13}, θ_{23}	neutrino mixing angles
δ_{CP}	CP-violating phase
K_{frak}	fractal correction factor $1 - 100\xi$ (Doc. 133)
$\mathrm{GF}(q)$, $\mathrm{GF}(q)^*$	Galois field with q elements, its multiplicative group
A_5	icosahedral group (60 elements)

ae.1 The Ground Relation and Its SM Notation

FFGFT sets for every fermion mode i:

$$\tilde{T}_i \cdot m_i = 1, \qquad m_i = r_i\, \xi^{p_i}\, v. \tag{ae.1}$$

The Yukawa coupling of the Standard Model is defined by $m_i = y_i\, v$, hence:

$$\boxed{y_i = r_i\, \xi^{p_i}.} \tag{ae.2}$$

This is not a matching but a direct rewriting **[B]**. Here (Doc. 189):

$$r_i = \frac{n_{\phi,i}^2}{n_{\theta,i}\, n_c}, \qquad p_i = \frac{d_{\mathrm{akt},i}}{3}, \tag{ae.3}$$

where $n_{\phi,i}$, $n_{\theta,i}$ are the winding numbers on the T^4 torus and n_c is the colour multiplicity. The exponents p_i follow from the topology (Docs. 189/358) **[K]**; the prefactors r_i follow from the winding numbers for the leptons, while for the quarks their derivation is open **[S]**.

ae.2 Derivation of the Yukawa Vertex from $\tilde{T} \cdot m = 1$

The vacuum case

In the vacuum $v = \text{const}$. Eq. (ae.1) gives $\tilde{T}_i = 1/m_i$ — every mode has a fixed proper-time scaling. The Lagrangian mass term reads:

$$\mathscr{L} \supset -m_i\, \bar{\psi}_i \psi_i. \tag{ae.4}$$

Fluctuating lattice scale

Now allow v to fluctuate with position:

$$v(x) = v + h(x), \qquad |h(x)| \ll v. \tag{ae.5}$$

The ground relation $\tilde{T}_i \cdot m_i = 1$ must continue to hold — not as a global normalisation but as a local bond between proper time and mass. This means:

$$m_i(x) = \frac{1}{\tilde{T}_i} = r_i\, \xi^{p_i}\, v(x) = r_i\, \xi^{p_i}\, \big(v + h(x)\big) = m_i^{(0)} \left(1 + \frac{h(x)}{v}\right). \tag{ae.6}$$

The coupling term

Inserting into the mass term gives **[B]**:

$$\mathscr{L} \supset -m_i(x)\, \bar{\psi}_i \psi_i = -m_i^{(0)}\, \bar{\psi}_i \psi_i \; - \; \frac{m_i^{(0)}}{v}\, \bar{\psi}_i \psi_i\, h(x). \tag{ae.7}$$

The second term is exactly the Yukawa vertex of the Standard Model with $y_i = m_i^{(0)}/v$. Here it does not follow from an assumption about the coupling structure but from the single FFGFT relation $\tilde{T} \cdot m = 1$ under spatial fluctuations of v.

ae.3 Mass-Proportional Coupling as a Consequence

Eq. (ae.7) holds for *all* fermions i *simultaneously*, with the same v and the same $h(x)$. This means:

> **[B]** The mass-proportional coupling $y_i = m_i/v$ of all fermions to the Higgs boson is a necessary consequence of $\tilde{T} \cdot m = 1$ under spatial fluctuation of the lattice scale. It is not a free assumption of the model.

This is the content of the Yukawa mechanism in FFGFT language: not "particles couple to a scalar field" but "mass is a local property of the lattice, and $h(x)$ is the oscillation of this property".

ae.4 Interpretation of the Higgs Particle

In this derivation the field $h(x)$ is the position-dependent deviation of the lattice scale v from its vacuum value. The physical Higgs particle — the resonance at $125.2\,\text{GeV}$ — is then the vibrational quantum of this deviation.

This differs conceptually from the Standard Model:

	Standard Model	**FFGFT**
Origin of mass	Yukawa coupling (postulated)	$\tilde{T} \cdot m = 1$ (ground relation)
Higgs field	independent scalar field	fluctuation of $v(x)$
Coupling $y_i = m_i/v$	empirical parameter	derived from Eq. (ae.6)
Number of free parameters	n_{fermions}	0 in the lepton sector; quark r_i so far from measured values
Higgs potential $V(H)$	Mexican hat (postulated)	residual stiffness of the lattice (§ae.6)
Higgs mass m_h	free parameter	trace rule (§ae.6) **[K]**

The derivation so far shows that $h(x)$ *must* exist and *must* couple with m_i/v. The restoring force of the oscillation — the potential $V(h)$ and hence the Higgs mass — is treated in §ae.5 and §ae.6.

ae.5 Higgs Mass from the Galois Chain [K]

Potential, Higgs mass and oscillation frequency are a single problem. This section shows that a numerically sharp candidate exists that is built entirely from Galois structures of the corpus.

The 11/8 chain. The three quantities M_Z, m_h, m_t satisfy:

$$M_Z : m_h : m_t = 1 : \tfrac{11}{8} : \left(\tfrac{11}{8}\right)^2, \qquad m_h = M_Z \cdot \tfrac{11}{8}, \quad m_t = M_Z \cdot \left(\tfrac{11}{8}\right)^2. \tag{ae.8}$$

Numerically **[K]**:

Quantity	measured	FFGFT (Eq. (ae.8))	deviation
M_W	$80.369\,\text{GeV}$	$M_Z\sqrt{7/9} = 80.420\,\text{GeV}$	$0.06\,\%$
M_Z	$91.188\,\text{GeV}$	input	—
m_h	$125.200\,\text{GeV}$	$M_Z \cdot 11/8 = 125.383\,\text{GeV}$	$0.15\,\%$
m_t	$172.570\,\text{GeV}$	$M_Z \cdot (11/8)^2 = 172.402\,\text{GeV}$	$0.10\,\%$

All four heavy bosons and fermions follow from two Galois numbers plus v.

Galois origin of the numbers. Both factors are field quantities of the corpus: $8 = |\mathrm{GF}(9)^*| = 3^2-1$ (Doc. 336) and $11 \in \mathrm{orbit}\{7, 11, 21\}$ under $x \mapsto 3x$ mod 26 in $\mathrm{GF}(27)^*$ (Doc. 340, the same structure as $\Delta m^2_{\mathrm{atm}}$). The Weinberg factor $\sqrt{7/9}$ comes from the A_5 matrix element $2/9$ (Doc. 372). The chain thus links three different Galois structures into one consistent numerical picture **[K]**.

Classification. Eq. (ae.8) is a numerically confirmed relation; the Galois origin of the numbers 11 and 8 is a reading **[S]**. A derivation of the Higgs mass that does without these numbers is provided by the trace rule in §ae.6; together the two fix the entire electroweak boson spectrum from v alone.

ae.6 The Trace Rule: Higgs Mass as Residual Stiffness of the Lattice

Picture. Coupled resonant circuits detune each other: the coupling pushes the eigenfrequencies apart. What is conserved in the process is the sum of the squared frequencies — the trace of the stiffness matrix. In this picture the Higgs mass is not a resonance of its own with its own potential but the share of the lattice stiffness left over by the gauge bosons.

The rule.

$$M_W^2 + M_Z^2 + m_h^2 \;=\; |\langle H\rangle|^2 \;=\; \frac{v^2}{2}, \qquad m_h^2 = \frac{v^2}{2} - M_W^2 - M_Z^2. \tag{ae.9}$$

The four building blocks.

1. **Conservation under coupling [B].** The sum of the eigenvalues of a symmetric matrix is its trace, independent of the basis. The electroweak mixing is a pure rotation by θ_W: before mixing, W^3 and B carry the stiffnesses $g^2v^2/4$ and $g'^2v^2/4$, afterwards the photon (0) and the Z (M_Z^2) — the sum is identical, the mixing matrix orthogonal.
2. **No intrinsic stiffness of the coupling [B].** A mechanical coupling spring is a component of its own and adds stiffness to the diagonal. The vacuum is empty; the coupling can only redistribute. That is exactly the property of a rotation (point 1).
3. **Counting once per mode.** W^+ and W^- are one complex mode with two senses of circulation: in the corpus the antiparticle is the same field with reversed sign (Doc. 042, 053), and the trace runs over m^2, where the sign drops out. Consistently, Koide counts three lepton masses, not six.
4. **Budget $= v^2/2$ [B].** The Higgs *field* is the ground lattice itself; the total stiffness belongs to it and is the squared modulus of its vacuum amplitude. Since the amplitude is complex (modulus and phase), the canonical normalisation of the real oscillation h requires the factor $1/\sqrt{2}$: $H = (v+h)/\sqrt{2}$, hence $|\langle H\rangle|^2 = v^2/2$. It is the same $\sqrt{2}$ as in $W^\pm = (W^1 \mp iW^2)/\sqrt{2}$ — the argument that counts W once also fixes the factor $1/2$.

The Higgs *particle* h is the eigen-oscillation of the lattice; its stiffness m_h^2 is the remainder in Eq. (ae.9). The Mexican-hat potential of the Standard Model is therefore not postulated: its curvature at the minimum is this residual stiffness.

Numbers [K]. With $M_W^2 = \frac{7}{9}M_Z^2$ (Doc. 372) it follows that $m_h = \sqrt{v^2/2 - \frac{16}{9}M_Z^2} = 124.6\,\text{GeV}$ (measured $125.2\,\text{GeV}$, $-0.47\,\%$). The balance itself holds to $0.45\,\%$.

Boson spectrum from v alone [K]. Combining Eq. (ae.9) with $\sin^2\theta_W = 2/9$ and the chain $m_h = \frac{11}{8}M_Z$ (§ae.5) leaves

$$M_Z^2\left(\tfrac{7}{9} + 1 + \tfrac{121}{64}\right) = \frac{v^2}{2} \quad\Longrightarrow\quad \frac{M_Z^2}{v^2} = \frac{288}{2113},$$

and all three boson masses follow from v without any measured mass as input:

	FFGFT	measured	deviation
M_Z	$90.90\,\text{GeV}$	$91.19\,\text{GeV}$	$-0.31\,\%$
M_W	$80.17\,\text{GeV}$	$80.37\,\text{GeV}$	$-0.25\,\%$
m_h	$124.99\,\text{GeV}$	$125.20\,\text{GeV}$	$-0.17\,\%$

As couplings: $g = 0.6512$ (on-shell $2M_W/v = 0.6528$), $\lambda = 0.1288$ (from m_h: 0.1293); the Higgs self-coupling is fixed by $4g^2/7 + 2\lambda = 1/2$ and is no longer a free parameter.

Accuracy. All three values lie 0.2–$0.3\,\%$ low, in the same direction. Eq. (ae.9) is a tree-level relation; the electroweak radiative corrections are of the order of a few per cent. Without a loop calculation no better agreement than the one reached here can be expected. The choice of v shifts the result: with the SM definition $246.22\,\text{GeV}$ the balance holds to $+0.45\,\%$, with the FFGFT values from Doc. 006/R104 (248.9 and $245.6\,\text{GeV}$) to -1.7 and $+1.0\,\%$ respectively. Which v is the right one is the question registered in R104/R112 (convention-free only via τ_μ), not a gap in the trace rule.

Fermion side: the trace rule applies to lattice modes only. The obvious extension "the top carries the same budget", $m_t^2 = v^2/2$ ($y_t = 1$), is incompatible with the chain: together with Eq. (ae.9) it requires $m_t/M_Z = \sqrt{2113/576} = 1.915$, while the measured value is 1.8925 ($+1.2\,\%$) and $(11/8)^2 = 1.8906$ agrees to $0.10\,\%$. The reading $y_t = 1$ is discarded **[X]**: it has no justification beyond the analogy, and it is the worse one. In addition, the top mass itself is scheme-dependent — pole mass $172.6\,\text{GeV}$, $\overline{\text{MS}}$ mass $162.5\,\text{GeV}$, a spread of $5.8\,\%$, corresponding to $y_t = 0.99$ and 0.93. Quarks are not free asymptotic modes (Doc. 372 §4); the trace rule applies to the eigen-oscillations of the lattice (W, Z, h), not to confined fermions. For the top mass the chain $m_t = (11/8)^2 M_Z$ applies **[K]**.

Rational ratios versus v-dependent relations. The accuracy of the relations is ordered by whether they contain v:

Relation	contains	deviation
$m_t/M_Z = (11/8)^2$	pole masses only	$0.10\,\%$
$M_W^2/M_Z^2 = 7/9$	pole masses only	$0.13\,\%$
$m_h/M_Z = 11/8$	pole masses only	$0.15\,\%$
$M_W^2 + M_Z^2 + m_h^2 = v^2/2$	v	$0.45\,\%$
$m_t^2 = v^2/2$ (discarded)	v	$1.2\,\%$

Ratios of measured pole masses are convention-free; this is where the rational Galois numbers agree best — as is to be expected, since they are exact numbers, not parameters. v, by contrast, is not a measurement but the SM definition $(\sqrt{2}\,G_F)^{-1/2}$, whose irrational factors stem from the SM formulation and which carries a theory correction of $-0.44\,\%$ from muon decay (Doc. 351, R112). Every relation containing v inherits these conventions. The residual of the trace rule ($0.45\,\%$) is of the size of this correction; that it stems from the definition of v and not from the rule is a testable conjecture **[S]**.

ae.7 Inventory of the Particle Zoo (as of 21 September 2026)

The following overview summarises what the corpus, after Doc. 372 and this document, delivers for each sector. In each case the best derivation is decisive; older, superseded formulas are recorded in the register (Doc. 190).

Sector	Quantity	Source	Status
Charged leptons	mass ratios $e : \mu : \tau$ from the matrix element $2/9$	292, 293, 367	**[B]/[K]**
	ϕ skeleton of the weights, $4/3 = 2Q$	368, 371	**[B]**
Neutrinos	masses from the GF$(27)^*$ orbits	340	**[K]**
	ν_3 Dirac, $\nu_{1,2}$ Majorana	343	**[B]**
	$\sin^2\theta_{13} = (25\xi)^{2/3}$	340	**[K]**
	$\sin^2\theta_{23} \in \{4/9, 5/9\}$, $\lvert\sin^2\theta_{23} - \frac{1}{2}\rvert = \frac{1}{18}$	372	**[K]**
	θ_{12} (order of magnitude only), δ_{CP}	340	**[S]**
	absolute scale m_ν (ansatz)	007	**[S]**
Gauge bosons	photon and eight gluons massless	320, 321, 339	**[B]**
	$M_W^2/M_Z^2 = 7/9$ ($\sin^2\theta_W = 2/9$ on-shell)	372	**[K]**
	$M_Z^2/v^2 = 288/2113$ (M_Z from v alone)	373	**[K]**
Higgs	Yukawa coupling $y_i = m_i/v$ from $\tilde{T}\cdot m = 1$	373	**[B]**
	$m_h = \frac{11}{8}M_Z$	373	**[K]**
	trace rule $M_W^2 + M_Z^2 + m_h^2 = v^2/2$, self-coupling fixed	373	**[K]**
Top	$m_t = (\frac{11}{8})^2 M_Z$	373	**[K]**
Other quarks	exponents $p_i = d_{\text{akt}}/3$	189	**[K]**
	prefactors r_i in the Galois grid; derivation open	006, 358	**[B]/[S]**
	unrolling as for leptons ($Q = 2/3$)	372	**[X]**
Hadrons	confinement as a topological boundary condition	319	**[S]**
	Gell-Mann–Okubo splitting consistent with m_s	372	**[K]**
	R_{Proton} structurally classified	372	**[S]**
	Λ_{QCD} from the ξ ladder	372	**[X]**
Couplings	α from Galois orders	338	**[K]**
	α_s via the Galois running	160, R78	**[K]**
	Cabibbo angle	372	**[S]**
Scale	v from the lattice	006, R104	**[K]**
	convention-free test only via τ_μ	351, R112	**[S]**

Ordering principle. Precise are the free modes — the charged leptons — and the oscillations of the lattice itself (W, Z, h). Blurred is everything confined: quarks and hadrons, whose masses depend on the scheme. And the exact Galois ratios between pole masses agree better than any relation that contains the SM quantity v (§ae.6).

Open. θ_{12} and δ_{CP}; the quark prefactors r_i apart from the top; Λ_{QCD}; the Cabibbo angle; the geometric derivation of the hadronic factor in Doc. 160 (R123); the convention-free determination of v via the muon lifetime (R112).

ae.8 Result

1. The Yukawa coupling $y_i = r_i\,\xi^{p_i}$: exponents p_i from the topology (Docs. 189/358) **[K]**; prefactors r_i for leptons from winding numbers, for quarks derivation open **[S]**.
2. The Yukawa vertex $-(m_i/v)\,\bar{\psi}_i h \psi_i$ follows from $\tilde{T}\cdot m = 1$ under $v \to v + h(x)$ for all fermions simultaneously **[B]**.
3. The mass-proportional coupling (the fingerprint of the Higgs discovery) is a consequence, not a parameter **[B]**.
4. $h(x)$ is the vibrational quantum of the lattice scale. The Higgs potential is not a separate postulate: its curvature at the minimum is the residual stiffness of the lattice (§ae.6).
5. **Higgs mass candidate [K]**: $m_h = M_Z \cdot 11/8 = 125.383$ GeV to $0.15\,\%$; $m_t = M_Z \cdot (11/8)^2$ to $0.10\,\%$. Both factors — $11 \in \mathrm{GF}(27)^*$, $8 = |\mathrm{GF}(9)^*|$ — are Galois quantities of the corpus; their origin is a reading **[S]**.
6. **Trace rule**: $M_W^2 + M_Z^2 + m_h^2 = v^2/2$ to $0.45\,\%$ **[K]**. Conservation under coupling, absence of intrinsic stiffness, counting and the factor $1/2$ are justified **[B]**; $m_h = \sqrt{v^2/2 - \frac{16}{9}M_Z^2}$.
7. **Boson spectrum from v [K]**: $M_Z^2/v^2 = 288/2113$; M_Z, M_W, m_h to 0.17–$0.31\,\%$, at tree level.
8. Inventory of the entire particle zoo in §ae.7.
9. Top: the trace rule applies to lattice modes only; $y_t = 1$ discarded **[X]**, $m_t = (11/8)^2 M_Z$ **[K]**.
10. Accuracy is ordered by convention dependence: rational pole-mass ratios 0.10–$0.15\,\%$, relations with v $0.45\,\%$. Open **[S]**: whether the residual of the trace rule stems from the definition of v (R104/R112).

Check script: `2/python/Dok373_Skripte/pruef_373_yukawa.py`

Register (Doc. 190): R122 updated (Higgs mass: candidates available).

[1] J. Pascher, *Doc. 006 — T0 particle masses* (2025).

[2] J. Pascher, *Doc. 189 — Torus derivation of the winding numbers* (2026).

[3] J. Pascher, *Doc. 372 — The matrix-element principle* (2026).

Part V

Part V — Interfaces, Review and the Basic Form

Chapter af

Doc. 374 — Common Anchor — Bridge Equations to Observer Patch Holography

Abstract. Observer Patch Holography (OPH) lists as an open item that length and tick do not follow from its own update rules. This document writes out the bridge to FFGFT as equations. The OPH cell energy is exactly the FFGFT bit energy $E_{\text{bit}} = \hbar c/L$ at the cell edge $L_{\text{cell}} = \sqrt{P}\ell_P$ **[B]**; the collapse threshold of Doc. 329 gives $n_{\text{thr}} = \sqrt{P/2}$ there, with the boundary at $P = 2$ **[B]**; $\tilde{T} \cdot m = 1$ supplies a native tick $\tilde{T}_{\text{cell}} = \sqrt{P}t_P$. Below v, FFGFT delivers all masses as ratios to v. The electroweak scale v is therefore the meeting point: the conditional OPH hierarchy $v/E_\star$ **[Q]** and the FFGFT ladder m_e/v **[K]** together give m_e/E_P to $0.93\,\%$. A single SI anchor then fixes both frameworks; through FFGFT's own chain the cell edge and the cell tick become absolute. The coefficient of that reading has meanwhile been derived on the OPH side (Doc. 376); #740 itself remains open there.

List of symbols

[B] algebraically proven
[K] numerically confirmed
[S] structurally plausible, derivation open
[X] checked and negative
[Q] taken from an external source, cited

Symbol	Meaning
ξ	FFGFT scaling parameter ($4/30000$)
ℓ_P, t_P, E_P	Planck length, time, energy (not reduced)
$L_0 = \xi\ell_P$	FFGFT resolution floor (Doc. 180)
$E_{\text{bit}}(L) = \hbar c/L$	system-dependent bit energy (Doc. 257)
$n_{\text{thr}}(L)$	collapse threshold, number of bit quanta up to Compton $=$ Schwarzschild (Doc. 329)
$M_{\text{coll}} = m_P/\sqrt{2}$	collapse mass (Doc. 329)
$\tilde{T}$	intrinsic time scale, $\tilde{T}\cdot m = 1$
v	electroweak scale (246.22 GeV, from G_F)
r_i, p_i	prefactor and exponent of the ladder $m_i = r_i\,\xi^{p_i} v$
P	OPH pixel ratio a_{cell}/ℓ_P^2
L_{cell}, E_{cell}	OPH cell edge $\sqrt{P}\ell_P$, cell energy $E_P/\sqrt{P}$
$E_\star$	OPH reference energy of the hierarchy $v/E_\star$
α_U	OPH unified coupling
N	OPH capacity of the whole network

af.1 Occasion and classification

Tracking item #740 of the OPH repository records that length and tick do not follow from the OPH updates: length is tied to an outside value only through the capacity N and $\Lambda\ell_P^2 = 3\pi/N$, and the tick only through ratios. Doc. 364 examined the icosahedral OPH carrier spectrally; this document treats the question of scale.

The roles are not symmetric. FFGFT derives from ξ; measured values and Standard Model quantities serve only for conversion and comparison. OPH quantities enter here as cited inputs from an external framework **[Q]** and carry the status their own documentation assigns them. The cosmic sector stays bracketed (P39); N is not used.

af.2 Inputs

FFGFT

- Bit energy $E_{\text{bit}}(L) = \hbar c/L$, system-dependent (Doc. 257).
- Collapse threshold $n_{\text{thr}}(L) = L/(\sqrt{2}\,\ell_P)$ and $M_{\text{coll}} = m_P/\sqrt{2}$ from Compton $=$ Schwarzschild (Doc. 329) **[B]**.
- Resolution floor $L_0 = \xi\ell_P$ as a ratio (Docs. 180, 329).
- $\tilde{T}\cdot m = 1$: every energy has a proper time $\hbar/E$.
- Lepton ladder $m_e = \frac{4}{3}\,\xi^{3/2}\,v$ (Docs. 006, 373) **[K]**.

OPH [Q]

- Pixel ratio $P = a_{\mathrm{cell}}/\ell_P^2$ as the fixed point of $P = \varphi + \sqrt{\pi}/A_T(P)$; public root $P = 1.630968\ldots$, source root $1.630972\ldots$; cell energy $E_{\mathrm{cell}} = E_P/\sqrt{P}$ (`code/P_derivation/FULL_DERIVATION.md`). The public root contains an empirical hadronic contribution; the source root does not.
- Hierarchy $v/E_\star = P^{-1/2}\exp[-2\pi/(4\alpha_U(P))] = 2.01998 \times 10^{-17}$ (Theorem A, `code/particles/hierarchy/theorem_package.md`); explicitly conditional, the physical identity of $E_\star$ is open in OPH.

af.3 Bridge equations for length and tick

All statements in this section are dimensionless in Planck units; no SI anchor is consumed.

B1 — dictionary [B]. With $L_{\mathrm{cell}} = \sqrt{P}\,\ell_P = 1.2771\,\ell_P$,

$$E_{\mathrm{cell}}^{\mathrm{OPH}} = \frac{E_P}{\sqrt{P}} = \frac{\hbar c}{L_{\mathrm{cell}}} = E_{\mathrm{bit}}(L_{\mathrm{cell}}). \tag{af.1}$$

Both frameworks use the same energy–length relation. OPH fixes the length by its fixed point; in FFGFT it stays system-dependent.

B2 — collapse threshold [B]. Inserted into Doc. 329:

$$n_{\mathrm{thr}}(L_{\mathrm{cell}}) = \sqrt{P/2} = 0.9030, \qquad \frac{r_s(E_{\mathrm{cell}})}{L_{\mathrm{cell}}} = \frac{2}{P} = 1.2263. \tag{af.2}$$

The boundary above which a single cell quantum stays outside its own Schwarzschild radius is $P = 2$. The OPH root lies below it: a cell quantum localised on one cell sits at the collapse scale. For a holographic pixel this is expected rather than objectionable, but it is a sharp condition that any future change of P must respect or explain. Source and public roots differ here by 1.2×10^{-6}; the statement does not depend on the open hadronic part.

B3 — tick. $\tilde{T} \cdot m = 1$ assigns the cell a proper time:

$$\tilde{T}_{\mathrm{cell}} = \frac{\hbar}{E_{\mathrm{cell}}} = \sqrt{P}\,t_P = \frac{L_{\mathrm{cell}}}{c}. \tag{af.3}$$

As arithmetic this is **[B]**; physically it gives OPH a tick per cell in the same unit as its length, without a master clock. Tying it to a laboratory clock is possible but not yet carried out on the OPH side **[S]**. A laboratory clock, such as the caesium transition, is itself a matter system with energy ΔE_{clk} and, by $\tilde{T} \cdot m = 1$, its own tick $\hbar/\Delta E_{\mathrm{clk}}$. The attachment requires the ratio $\Delta E_{\mathrm{clk}}/E_{\mathrm{cell}}$, computed within OPH itself; OPH lists this step as open (clock frontier $\varepsilon_{\mathrm{clk}} = \Delta E_{\mathrm{clk}}/E_\star$, caesium route as an incomplete candidate). In FFGFT the question is not open: every mode has its tick $\tilde{T} = 1/m$, and tick ratios are mass ratios. FFGFT thus supplies the translation principle, not the missing OPH value.

B4 — floor [K]. $L_{\text{cell}}/L_0 = \sqrt{P}/\xi = 9578$. The OPH cell lies about four orders of magnitude above the FFGFT resolution floor; no conflict.

af.4 The meeting point v

The two frameworks cover different sections of the scale ladder. OPH computes, conditionally, from the top down to v; FFGFT from v downwards: all charged lepton masses via the ladder $m_i = r_i\,\xi^{p_i} v$ (Docs. 006, 373), the boson masses via $\sin^2\theta_W = 2/9$ and the trace rule (Docs. 372, 373).

Quantity	Origin	vs. measured
$v/E_\star = 2.01998 \times 10^{-17}$	OPH, Theorem A **[Q]**	$+0.16\,\%$ (with $E_\star = E_P$)
$m_e/v = \frac{4}{3}\,\xi^{3/2} = 2.0528 \times 10^{-6}$	FFGFT, Docs. 006/373 **[K]**	$-1.09\,\%$ (bare)
$m_e/E_P = 4.1466 \times 10^{-23}$	product	$-0.93\,\%$

For the reference energy $E_\star$ see the following subsection. The $-1.09\,\%$ on the FFGFT side is the known gap between bare and measured lepton masses (Doc. 352), not a new finding. The product depends on the OPH branch only at the 10^{-4} level.

Consequence: a single SI anchor fixes both frameworks. With G as anchor, the OPH hierarchy leads to v, and the FFGFT ladder leads from there to all particle masses; with m_e as anchor, the chain runs the other way. The residual of about $1\,\%$ is the common consistency test. FFGFT also reaches m_e/E_P through its own chain; the meeting point v is thus a comparison of two independent routes.

The reference energy $E_\star$

What $E_\star$ is. In OPH, $E_\star$ is the energy unit of the calculation. The code sets the Planck energy equal to one; the cell energy is $E_P/\sqrt{P}$, the unification scale $M_U = E_P e^{-2\pi} P^{1/6}$, the hierarchy $v/E_\star = P^{-1/2}\exp[-2\pi/(4\alpha_U)]$ (reproduced, check script D1). Which physical energy this unit is, OPH explicitly leaves open; it is to be fixed through a clock, $E_\star = h\nu_{\text{clk}}/\varepsilon_{\text{clk}}$, that is, through the same open step as in B3.

Two candidates. The Planck energy exists in two conventions: non-reduced $E_P = \sqrt{\hbar c^5/G} = 1.2209 \times 10^{19}$ GeV and reduced $\bar{E}_P = E_P/\sqrt{8\pi} = 2.435 \times 10^{18}$ GeV. They differ by $\sqrt{8\pi} \approx 5.01$, depending on whether the gravitational coupling is written as G or as $8\pi G$. With measured v, OPH gives $E_\star = 1.2189 \times 10^{19}$ GeV: $0.16\,\%$ below E_P, but a factor of 5.0 above $\bar{E}_P$ **[K]**. The reduced reading is thereby excluded; the non-reduced one fits.

How sharp $0.16\,\%$ is. α_U sits in the exponent. The sensitivity is

$$\frac{\partial \ln(v/E_\star)}{\partial \alpha_U} = \frac{\pi}{2\alpha_U^2} \approx 929. \tag{af.4}$$

The deviation of $0.16\,\%$ corresponds to a shift of α_U by 1.7×10^{-6}, relatively 4×10^{-5} **[B]**. The same sensitivity accounts exactly for the difference between the public and the source

branch (-8.4×10^{-5}, check script D5). The agreement with E_P is therefore not a rough match of orders of magnitude but a tight condition on α_U.

Why a reading [S]. OPH itself does not claim $E_\star = E_P$; the assignment here follows from the numerical comparison. It would be confirmed if OPH closes the clock attachment and obtains $E_\star = E_P$, or if OPH singles out the coupling without 8π structurally. For the comparison in §af.5 the reading is not critical: it only fixes against which of the two Planck energies the $0.16\,\%$ is measured.

af.5 Two routes to m_e/E_P and their accuracy

FFGFT reaches the Planck scale through its own chain (Docs. 012, 013, 180):

1. $G = \dfrac{\xi^2}{4m_e}\, C_{\text{conv}}\, K_{\text{frak}} = 6.6745 \times 10^{-11}\ \text{m}^3\text{kg}^{-1}\text{s}^{-2}$, $+0.003\,\%$ against CODATA. The notation in Doc. 180 splits the same value into $C_{\text{dim}} \cdot C_{\text{conv}}$; both forms are equivalent.
2. $\ell_P = \sqrt{\hbar G/c^3} = 1.6163 \times 10^{-35}$ m and hence $L_0 = \xi \ell_P = 2.155 \times 10^{-39}$ m.
3. $E_P = 1.22087 \times 10^{22}$ MeV, $m_e/E_P = 4.1855 \times 10^{-23}$ $(+0.002\,\%)$ **[K]**.
4. $v/E_P = 2.01676 \times 10^{-17}$.

Not every route of calculation yields the same accuracy:

Route	Quantity	Dev.	Origin of the deviation
FFGFT chain $\xi \to G \to E_P$	m_e/E_P	$+0.002\,\%$	accuracy of G
FFGFT ladder	m_e/v	$-1.09\,\%$	bare versus measured (Doc. 352)
OPH, Theorem A **[Q]**	$v/E_\star$ vs. FFGFT chain	$+0.16\,\%$	reading $E_\star = E_P$, conditional fixed point
Combination via v	m_e/E_P	$-0.93\,\%$	sum of the two preceding

The routes do not contradict each other. Each deviation is assigned to a known place: the bare residual of the lepton ladder on one side, the identification of $E_\star$ on the other. For the comparison table, the route by which a value was obtained must therefore always be stated alongside the value.

Consequence for #740. With FFGFT's own chain the bridge equations become absolute. The OPH cell edge is $L_{\text{cell}} = \sqrt{P}\,\ell_P = 2.064 \times 10^{-35}$ m, the cell tick $\tilde{T}_{\text{cell}} = \sqrt{P}\,t_P = 6.885 \times 10^{-44}$ s **[K]**; every further tick ratio is, by $\tilde{T} \cdot m = 1$, a mass ratio. Length and tick thus follow without the detour via N and Λ, under the reading $E_\star = E_P$. Its coefficient has meanwhile been derived on the OPH side (Doc. 376): from two observers seeing different stress it follows that $\kappa = b\,L^2/q$ and hence $G = L^2$ **[Q]**. #740 itself remains open there; the length identity, the clock and a matching factor are still carried as open. The question of scale within #740 is thus answered on the FFGFT side, the item as a whole is not.

af.6 What the bridge does not achieve

- The upper part of the chain comes from an external framework and is conditional there **[Q]**; the combination is a consistency test between the frameworks.
- Unit of time: in FFGFT every time scale follows from $\tilde{T} = 1/m$; the SI second serves only for conversion. In OPH the unit of time is tied to $E_\star$ and is to be fixed through the clock attachment; this step is open on the OPH side **[S]**. Through the FFGFT chain the cell tick is already absolute under the reading $E_\star = E_P$; the laboratory clock is then a check.
- N and the cosmic sector are not involved (P39).
- The algebraic carriers still differ (A_5 versus $T^4/\mathbb{Z}_3$, Doc. 364); the bridge concerns scales, not symmetries.

Tool and method

Claude (Anthropic) was used as a tool for: reading the OPH source files, writing and running the Python check script, drafting the LaTeX sources, and maintaining register and changelog. The tool computes and writes; the author decides what is computed and written, judges every result and determines what it means physically. Every statement is secured by assertions in the check script. OPH values are taken verbatim from the public repository; measured values (CODATA 2022, PDG) serve only as comparators.

af.7 Result

1. B1: $E_{\text{cell}}^{\text{OPH}} = E_{\text{bit}}(\sqrt{P}\,\ell_P)$ **[B]**.
2. B2: $n_{\text{thr}}(L_{\text{cell}}) = \sqrt{P/2}$, boundary $P = 2$, OPH below it **[B]**.
3. B3: cell tick $\tilde{T}_{\text{cell}} = \sqrt{P}\,t_P$ from $\tilde{T} \cdot m = 1$; laboratory attachment open on the OPH side **[S]**.
4. B4: $L_{\text{cell}}/L_0 = \sqrt{P}/\xi \approx 9578$ **[K]**.
5. Meeting point v: OPH $v/E_\star$ **[Q]** × FFGFT m_e/v **[K]** gives m_e/E_P to $-0.93\,\%$; a single SI anchor suffices for both frameworks.
6. Length and tick absolute through the FFGFT chain: $L_{\text{cell}} = 2.064 \times 10^{-35}$ m, $\tilde{T}_{\text{cell}} = 6.885 \times 10^{-44}$ s **[K]**; the coefficient of the reading derived on the OPH side, #740 as a whole open there (Doc. 376).
7. FFGFT's own chain $\xi \to G \to E_P$: m_e/E_P to $+0.002\,\%$ **[K]**; the accuracy depends on the route of calculation (§af.5).

Check script: `2/python/Dok374_Skripte/pruef_374_anker.py` (30/30)

Register (Doc. 190): no entry.

[1] J. Pascher, *Doc. 006 — T0 particle masses* (2025).

[2] J. Pascher, *Doc. 012 — T0 gravitational constant* (2025).

[3] J. Pascher, *Doc. 013 — T0 and SI units* (2025).

[4] J. Pascher, *Doc. 180 — Derivation of* L_0 (2026).

[5] J. Pascher, *Doc. 257 — Bit energy* (2026).

[6] J. Pascher, *Doc. 329 —* n_{thr} *scale analysis* (2026).

[7] J. Pascher, *Doc. 352 — Lepton residuals* (2026).

[8] J. Pascher, *Doc. 364 — The icosahedral carrier and the* D_4 *lattice* (2026).

[9] J. Pascher, *Doc. 373 — The Yukawa mechanism* (2026).

[10] B. Mueller et al., *Observer Patch Holography*, `github.com/FloatingPragma/observer-patch-holography` (as of 22 September 2026).

Chapter ag

Doc. 375 — The Hierarchy v/E_P — Two Derivations Compared: FFGFT and Observer Patch Holography

Abstract. FFGFT obtains the electroweak scale from ξ with m_e as anchor (R104) and reaches the Planck scale through its own chain $\xi \to G \to E_P$, whose SI representation contains the measured value of G via C_{conv}. This route to the hierarchy number v/E_P thus uses two measured values. Without an anchor, $v/E_P = \xi^4/(5\pi) = 2.0120 \times 10^{-17}$ ($-0.23\,\%$, Doc. 149); the factor 10 in this formula is fitted to the measured value (Doc. 387). Observer Patch Holography (OPH) delivers the same number conditionally through its Theorem A. The two derivations give 2.0389×10^{-17} (FFGFT, $+1.10\,\%$) and 2.0200×10^{-17} (OPH, $+0.16\,\%$). The FFGFT deviation is entirely the known bare gap of the lepton ladder (Doc. 352); the Planck chain contributes less than 10^{-4}. The FFGFT entry for a hashed comparison table is given.

List of symbols

[B] algebraically proven
[K] numerically confirmed
[S] structurally plausible, derivation open
[Q] taken from an external source, cited

Symbol	Meaning
ξ	FFGFT scaling parameter ($4/30000$)
v	electroweak scale
E_P	Planck energy (not reduced)
G, C_{conv}, K_{frak}	gravitational constant and factors of the SI representation (Docs. 012/013)
$E_\star$, P, α_U	OPH reference energy, pixel ratio, unified coupling

ag.1 Occasion

Doc. 374 brought FFGFT and OPH together at the electroweak scale and, in doing so, combined m_e/v from the FFGFT ladder with $v/E_\star$ from OPH. That understates the FFGFT side: FFGFT obtains v itself, with m_e as anchor (R104), and reaches E_P through its own chain (Doc. 374, § 5). The hierarchy number v/E_P is therefore not a combination but a second, independent derivation of the same quantity. This document places the two derivations side by side.

ag.2 The FFGFT derivation

1. Electroweak scale (R104, Doc. 006):

$$v = \frac{m_e}{\frac{4}{3}\,\xi^{3/2}} = 248.93\,\text{GeV}. \tag{ag.1}$$

2. Planck energy through its own chain (Docs. 012, 013, 180): $G = \xi^2/(4m_e)\cdot C_{\text{conv}}\cdot K_{\text{frak}}$, hence $E_P = 1.22087\times 10^{19}$ GeV.
3. Hierarchy:

$$\left.\frac{v}{E_P}\right|_{\text{FFGFT}} = 2.0389\times 10^{-17} \quad \textbf{[K]}. \tag{ag.2}$$

Against the SM value $v = 246.22$ GeV (definition from G_F, R112) this is $+1.10\,\%$ higher. The deviation is entirely that of v; the Planck chain contributes less than 10^{-4}. It is identical to the bare gap of the lepton ladder m_e/v ($-1.09\,\%$, Doc. 352): v is obtained from the bare relation and inherits its residual. Not a new finding.

This route uses two measured values: m_e for v and, via C_{conv}, the measured value of G for E_P. Directly and without an anchor, v/E_P follows from Doc. 149: $v/E_P = \xi^4/(5\pi) = 2.0120\times 10^{-17}$, $-0.23\,\%$; the factor 10 in this formula is fitted to the measured value (Doc. 387). The factor 1.0134 between the two values is the residue of the electron mass in the chain (Doc. 383).

ag.3 The OPH derivation [Q]

Theorem A from `code/particles/hierarchy/theorem_package.md`:

$$\frac{v}{E_\star} = P^{-1/2}\exp\!\left[-\frac{2\pi}{4\alpha_U(P)}\right] = 2.0200\times 10^{-17}, \tag{ag.3}$$

reproduced in the check script. The derivation is explicitly conditional in OPH; the identification $E_\star = E_P$ is a reading **[S]** (Doc. 374). Against the SM value: $+0.16\,\%$.

ag.4 Comparison

Derivation	v/E_P	vs. SM	Origin of the deviation
FFGFT	2.0389×10^{-17}	$+1.10\,\%$	bare residual of the ladder (Doc. 352)
OPH **[Q]**	2.0200×10^{-17}	$+0.16\,\%$	conditional fixed point, reading $E_\star = E_P$
SM comparator	2.0167×10^{-17}	—	v from G_F (R112)

The two derivations lie $0.93\,\%$ apart. They also differ in sensitivity:

- FFGFT: $\partial \ln v / \partial \ln \xi = -3/2$ **[B]**. A rational structural quantity with power $3/2$; small changes of ξ act weakly.
- OPH: $\partial \ln(v/E_\star)/\partial \alpha_U = \pi/(2\alpha_U^2) \approx 929$. The coupling sits in the exponent; the $0.16\,\%$ corresponds to a shift of α_U by 4×10^{-5} relatively.

The FFGFT value is robust and carries a known residual; the OPH value is more accurate but highly sensitive and conditional. Both statements are testable: on the FFGFT side through the resolution of the bare residual (Doc. 352), on the OPH side through the clock attachment and the identification of $E_\star$.

ag.5 Entry for the comparison table

For a hashed comparison table the FFGFT entry, including the route, reads:

```
v/E_P | FFGFT | 2.03894e-17 | route: v=m_e/((4/3)xi^(3/2)) (R104, Doc
006) ; E_P via G=xi^2/(4m_e)C_conv K_frak (Docs 012/013/180) ; bare
ladder
```

SHA-256 of the string (generated in the check script):
`84fac85fc55d8c5928a07550e774683961613059415b0ef02715b327f58fb15f`

Tool and method

Claude (Anthropic) was used as a tool for: writing and running the Python check script, drafting the LaTeX sources, and maintaining the changelog. The tool computes and writes; the author decides what is computed and written, judges every result and determines what it means physically. OPH values are taken verbatim from the public repository; measured values (CODATA 2022, PDG) serve only as comparators.

ag.6 Result

1. FFGFT delivers $v/E_P = 2.0389 \times 10^{-17}$ through its own chain: v from ξ with m_e as anchor (R104), E_P via G with the measured value of G in C_{conv} **[K]**. Without an anchor: $\xi^4/(5\pi) = 2.0120 \times 10^{-17}$ ($-0.23\,\%$), factor 10 fitted to the measured value (Doc. 387).
2. The deviation of $+1.10\,\%$ is the bare residual of the lepton ladder; the Planck chain contributes less than 10^{-4} **[K]**.

3. OPH delivers, conditionally, 2.0200×10^{-17} ($+0.16\,\%$) **[Q]**; the derivations lie $0.93\,\%$ apart.
4. Sensitivity: FFGFT $-3/2$ in $\ln \xi$ **[B]**, OPH ≈ 929 in α_U.
5. FFGFT table entry given with route and SHA-256.

Check script: `2/python/Dok375_Skripte/pruef_375_hierarchie.py` (11/11)

Register (Doc. 190): no entry.

[1] J. Pascher, *Doc. 006 — T0 particle masses* (2025).

[2] J. Pascher, *Doc. 012 — T0 gravitational constant* (2025).

[3] J. Pascher, *Doc. 013 — T0 and SI units* (2025).

[4] J. Pascher, *Doc. 180 — Derivation of* L_0 (2026).

[5] J. Pascher, *Doc. 352 — Lepton residuals* (2026).

[6] J. Pascher, *Doc. 374 — Common anchor* (2026).

[7] B. Mueller et al., *Observer Patch Holography*, `github.com/FloatingPragma/observer-patch-holography` (as of 22 September 2026).

Chapter ah

Doc. 376 — The Coefficient 8π and the Status of Issue 740

Abstract. Doc. 374 carried the identification $E_\star = E_P$ as a reading **[S]** and concluded from it that #740 is closed under that reading. On the OPH side the coefficient has meanwhile been derived: from two observers seeing different stress it follows that $\kappa = bL^2/q$, so with the modular coefficient $b = 2\pi$ and the quarter area law $q = 1/4$, $\kappa = 8\pi L^2$ and $G = L^2$ **[Q]**. Hence $E_\star = \hbar c/L$ is the non-reduced Planck energy **[B]**. The further statement in Doc. 374 is withdrawn and corrected there directly: OPH still carries the length identity, the clock and a matching factor as open, and eight of the fifteen dependencies of #740 are not closed there. Both frameworks have a matching factor at the same place: OPH $0.16\,\%$ between source ν and Fermi ν, FFGFT $1.10\,\%$ as the bare residual of the ladder (Doc. 375).

List of symbols

[B] algebraically proven
[K] numerically confirmed
[S] structurally plausible, derivation open
[Q] taken from an external source, cited

Symbol	Meaning
κ	coupling in Einstein's equation, $\kappa = 8\pi G/c^4$
b, q	modular coefficient and prefactor of the area law $S = A/(4L^2)$
L	length in the area law
$E_\star$	OPH reference energy of the hierarchy
E_P, $\bar{E}_P$	Planck energy, non-reduced and reduced
m	matching factor between source and convention quantity
ξ	FFGFT scaling parameter ($4/30000$)

ah.1 Occasion

Doc. 374 set up the bridge equations to OPH and carried the identification $E_\star = E_P$ as a reading. Its result section states that #740 is closed under that reading. Two things have changed since: the coefficient has been derived on the OPH side, and OPH has set out the status of #740 in more detail. This document records both and corrects the statement that went too far.

ah.2 The theorem [Q]

Two observers read the same stress and geometry configuration at the same metric norm. The tensor relation is $F_i = \kappa T_i + \varphi n_i$ with unknown κ and a common offset φ. On the entropy side there are a bulk variation $b K_i T_i$, an area variation $-K_i(F_i - \psi n_i)$ and stationarity $\Delta S_i + (q/L^2)\,\Delta A_i = 0$. If the two observers see different stress, $T_1 \neq T_2$, then, with nothing assumed about the offsets,

$$\kappa = \frac{b\,L^2}{q}. \tag{ah.1}$$

With $b = 2\pi$ from the cap modular flow and $q = 1/4$ from the area law this gives $\kappa = 8\pi L^2$ **[K]**, and from $\kappa = 8\pi G$ it follows that $G = L^2$.

ah.3 Consequence for the reference energy

With $L = \sqrt{G}$ (in units $\hbar = c = 1$),

$$E_\star = \frac{\hbar c}{L} = \sqrt{\frac{\hbar c^5}{G}} = E_P, \tag{ah.2}$$

the non-reduced Planck energy **[B]**. The reduced reading is off by $\sqrt{8\pi} = 5.013$ and would require a different normalisation of the area law; it is therefore not a choice of units but a different physical assumption.

The reading from Doc. 374 is thus no longer posited on the OPH side but derived. The prefactor 8π stays where it always was, in Einstein's equation; the quarter in the area law selects G.

ah.4 Correction to Doc. 374

Doc. 374 concludes that #740 is closed under the reading $E_\star = E_P$. This is withdrawn and corrected directly in Doc. 374. OPH carries #740 as the whole common-world join and names eight of fifteen dependencies as open **[Q]**. Open there are in particular:

1. The length in the area law and the length in the pixel are so far the same by declaration; deriving them as one length on one source family is planned.
2. The tick. $\hbar c/\ell_P$ is an energy; a clock requires an evolving state, a gap and a readout, plus the map to a laboratory transition. On the FFGFT side $\tilde{T} \cdot m = 1$ supplies the tick directly (Doc. 374, B3); on the OPH side it is work in progress.

3. The matching factor $m = v_F/v_{\text{source}} = 0.9984$, that is, the $0.16\,\%$. It is to be computed from the same matter action.

What holds is therefore: the coefficient is derived, the bridge equations B1–B4 of Doc. 374 stand unchanged, and the absolute scale follows through the FFGFT chain. #740 itself remains open.

ah.5 Matching factors in both frameworks

Both frameworks have an open factor at the same place, with different origins:

Framework	Factor	Size	Origin
OPH **[Q]**	$v_F/v_{\text{source}} = 0.9984$	$0.16\,\%$	source v vs. Fermi v, computation planned
FFGFT	$v_F/v(\xi) = 0.98912$	$1.10\,\%$	bare residual of the lepton ladder (Docs. 352, 375)

The FFGFT side knows the origin of its factor and carries it corpus-wide as bare versus measured; the OPH side carries its factor as open. For a comparison table both factors belong beside the value, as does the route (Doc. 375).

Tool and method

Claude (Anthropic) was used as a tool for: writing and running the Python check script, drafting the LaTeX sources, and maintaining register and changelog. The tool computes and writes; the author decides what is computed and written, judges every result and determines what it means physically. The theorem and the statements about the status of #740 are taken from the communication of the OPH author and marked **[Q]**.

ah.6 Result

1. $\kappa = b\,L^2/q$ with $b = 2\pi$ and $q = 1/4$ gives $\kappa = 8\pi L^2$ and $G = L^2$ **[K] [Q]**.
2. Hence $E_\star = \hbar c/L$ is the non-reduced Planck energy **[B]**; the reduced reading is off by $\sqrt{8\pi}$.
3. The statement in Doc. 374 that #740 is closed under the reading is withdrawn and corrected there; length identity, tick and matching factor remain open on the OPH side **[Q]**.
4. Matching factors: OPH $0.16\,\%$, FFGFT $1.10\,\%$ as the bare residual **[K]**.

Check script: `2/python/Dok376_Skripte/pruef_376_koeffizient.py` (9/9)

Register (Doc. 190): R124 (withdrawal of the #740 statement in Doc. 374).

[1] J. Pascher, *Doc. 352 — Lepton residuals* (2026).

[2] J. Pascher, *Doc. 374 — Common anchor* (2026).

[3] J. Pascher, *Doc. 375 — The hierarchy* v/E_P (2026).

[4] B. Mueller et al., *Observer Patch Holography*, `github.com/FloatingPragma/observer-patch-holography`; theorem and status statements from the communication of 22 September 2026.

Chapter ai

Doc. 377 — Interfaces of FFGFT to Other Frameworks

Abstract. The corpus contains comparison and bridge documents for eleven frameworks, spread over more than twenty documents. This document collects them: for each framework the document numbers, the mathematical point of contact, the status and the open item. The quantities where several frameworks actually overlap follow, each with its route, as a template for a comparison table. Nothing new is computed; all numbers come from the documents named and are recomputed in the check script.

List of symbols

[B] algebraically proven
[K] numerically confirmed
[S] structurally plausible, derivation open
[X] checked and negative
[Q] taken from an external source, cited

ai.1 Principle

FFGFT derives from ξ. Measured values and Standard Model quantities serve for conversion and testability, not for derivation; the cosmic sector stays bracketed (P39). Quantities of external frameworks appear in all documents named below as cited inputs **[Q]** and do not stand on equal footing with our own posits. A point of contact counts as a bridge only when the same structural object follows from each framework's own geometry and an explicit map between the derivations exists; the criterion is from Doc. 283.

ai.2 Overview

Framework	Documents	Point of contact	Status
HLV (Krüger)	271, 272, 276, 282, 283, 285, 294, 297	icosahedral cut-and-project quasicrystal versus $T^4/\mathbb{Z}_3$; dimension bridge $6 = \mathbf{3} \oplus \mathbf{3}'$ in A_5; CP divisibility	bridge **[K]**, fork at the symmetry order
OPH (Mueller)	364, 374, 375, 376	spectrum of the icosahedral carrier; energy–length bridge $E_{\text{bit}} = \hbar c/L$; ν/E_P; coefficient 8π	bridge equations **[B]**, scales open on the other side **[Q]**
Hyperbit, $G(6, \mathbb{Z}_3\mathbb{C})$ (Matzke)	324, 326, 336	$\mathbb{Z}_3$ structure and GF(9); bit energy and the Landauer threshold	algebraic bridge **[B]**, bit energy system-dependent **[K]**
XYLATIC (Phillips)	252	number-theoretic structures of the σ function	structural comparison
RA/PMT (Guevara)	269	three-layer scheme, perception level	structural comparison
Vopson	251	computational-universe hypothesis, second law of infodynamics	different primitives
Matsas et al.	105	the number of fundamental constants	classification
ΛCDM	309	scale anchor: both frameworks need exactly one number for the SI coupling	shared situation **[K]**
Kagome lattice	361	four structural parallels; mass as geometric freezing	analogy **[S]**
SM Lagrangian	049	number of fields and free parameters against one equation	comparison of form
Literature in general	345	classification against published approaches	classification

UIFT (Teker) is touched in Docs. 013 and 257 through the bit mass, but has no comparison document of its own yet.

ai.3 Quantities where frameworks overlap

This table is the template for a comparison table. Every value comes with its route; without it the value is not comparable (Doc. 375).

Quantity	FFGFT value	Dev.	Route
α^{-1}	$3700/27 = 137.037$	$+7.6$ ppm	geometry from $T^4/\mathbb{Z}_3$ (Doc. 007), Galois confirmation (Doc. 338)
$\sin^2\theta_W$ (on-shell)	$2/9 = 0.22222$	$-0.52\,\%$	A_5 structure (Docs. 293, 372)
λ_{CKM}	$\xi^{1/6} = 0.22602$	$+0.46\,\%$	Galois bridge (Doc. 336)
ν/E_P	2.0389×10^{-17}	$+1.10\,\%$	ν from ξ (R104), E_P from our own chain (Doc. 375)
$E_{\text{bit}}(L)$	$\hbar c/L$	—	system-dependent, no universal bit value (Docs. 257, 326)
Dimension	3+1 from $T^4/\mathbb{Z}_3$	—	posit; $6 = \mathbf{3} \oplus \mathbf{3}'$ as the bridge to HLV (Doc. 285)
Cosmic sector	—	—	bracketed (P39, Doc. 309)

ai.4 What follows from this

- Two frameworks are connected through more than one quantity: HLV through the A_5 structure and the dimension, OPH through the carrier, the scale and ν/E_P. The rest are single comparisons.
- The overlaps lie almost entirely in the lepton sector and in the geometry. The hadron sector appears in no comparison; that is not a selection but the state of the other sides.
- α^{-1} and ν/E_P are currently the only quantities for which more than one framework states a number. They are therefore the only rows of a comparison table that data can tell apart.

- The scale anchor is the same situation in a different place for every framework (Doc. 309); it is not a distinguishing feature.

Tool and method

Claude (Anthropic) was used as a tool for: assembling the document list, writing and running the check script, drafting the LaTeX sources, and maintaining the changelog. The tool computes and writes; the author decides what is computed and written, judges every result and determines what it means physically. The check script verifies that every cited document is present in the corpus and recomputes every number of the second table from FFGFT quantities.

ai.5 Result

1. Eleven frameworks with comparison or bridge documents, two of them with several points of contact **[K]**.
2. Seven quantities where frameworks overlap, each with route and deviation **[K]**.
3. Only α^{-1} and v/E_P are given a number by more than one framework.
4. The hadron sector appears in no comparison.

Check script: `2/python/Dok377_Skripte/pruef_377_schnittstellen.py` (20/20)

Register (Doc. 190): no entry.

[1] Docs. 271, 272, 276, 282, 283, 285, 294, 297 (HLV).

[2] Docs. 364, 374, 375, 376 (OPH).

[3] Docs. 324, 326, 336 (Hyperbit, $\mathbb{Z}_3\mathbb{C}$).

[4] Docs. 049, 105, 251, 252, 269, 309, 345, 361.

[5] Docs. 007, 013, 257, 293, 338, 372 (FFGFT sources of the numbers).

Chapter aj

Doc. 378 — Peer Review under Present Conditions

Abstract. Classical peer review assumes that a referee can survey a paper in a few hours, that author and referee belong to an institution, and that texts are written by people. None of these assumptions holds in general any more: corpora run to hundreds of documents, independent researchers have no access to the usual submission routes, and language models produce plausible texts, scripts and apparent review reports in any quantity. This document collects the review tools that have already arisen in the exchange of the *Information Physics Institute* (IPI), in Stefaan Vossen's Dot Theory governance and in the FFGFT corpus's own record-keeping, and asks which of them can be used for a review framework suited to present conditions. The result is a proposal of ten building blocks: a declared constitution before review, an entry path provided by the author, testable objects instead of texts, predictions fixed in advance, separation of roles, a continuing register, rules for the use of language models and for simplifications, localised instead of global verdicts, and the explicit acknowledgement that there is no universal checker. The document contains no physical claims; it is a methodological inventory with the character of a proposal.

Legend

Label	Meaning
[K], [B], [S], [Q], [X]	FFGFT status markers: numerically confirmed, algebraically proved, structurally plausible with open derivation, taken from an external source, recognised as false
[SETZUNG]	explicitly posited foundational assumption, not derived
IPI	Information Physics Institute, mailing list and working groups
F O R M	*Framework – Operator – Residual – Model*, the grammar of the *Constitutional Verification Protocol* (CVP)
CIL	*Constitutional Interpretation Layer* (Vossen/Pascher)
CCR	*Canonical Constitutional Record* of *Constitutional Physics* (Vossen)
FER	analytical method of Diana Haskins (Universal Constraint)
OAP, FAH	admissibility record and *Framework Admissibility History* of Dot Theory
O0, O2	originator declaration and secondary rendering (Doc. 272)
SHA-256	cryptographic hash establishing the identity of a file object

aj.1 Starting point

What the classical procedure assumes

Journal peer review is built for the article: a finished paper of ten to thirty pages, one referee from the same field, a few hours of reading, a verdict in three grades (accept, revise, reject). It rests on four tacit assumptions:

1. The work can be surveyed in one sitting.
2. The referee shares with the author a common stock of methods against which deviations stand out.
3. Institutional affiliation acts as a prior filter.
4. A text is the product of human work, so its length is a rough measure of the effort behind it.

What has changed

Scale. A framework such as FFGFT now comprises more than three hundred and seventy documents with their own status markers, a correction register (Doc. 190) and cross-references between geometry, Galois theory and particle physics. Nobody takes that in within a few hours. This is not a special case: every serious theory reaches this point, and established programmes do not open up in an afternoon either.

Access. Independent researchers have no institutional address; indexing services often list only what is already listed through publisher platforms. Repositories (GitHub)

and archives with DOIs (Zenodo) are public and citable, but the review system hardly registers them.

Language models. Text, code and apparent review reports can now be produced at negligible cost. The length of a text no longer measures the work behind it. In the IPI exchange there have been cases in which a language-model prompt was submitted as “source code”, in which a check block printed its own constants and then “PASS” beneath them, and in which a second language model with shared context reproduced an earlier analysis verbatim instead of checking it independently.

Pattern search. Any pattern can be read out of enough numbers. The image of the magic lamp that one rubs until something appears describes a real misuse of computing power. It does not describe every piece of work that contains numbers. The difference lies in the order: in pattern search the measured values stand at the beginning, in a derivation they stand at the end as the test. In FFGFT the geometry $T^4/\mathbb{Z}_3$ is fixed as a [SETZUNG]; from it follow algebraically $\mathbb{Z}_3$, GF(27), D_4 and triality; only from these come numbers, and only afterwards are they compared with measurements (Doc. 365). An outsider sees at first glance only the number, not the order, and suspects the lamp.

Simplification and prejudice. A short entry path is possible and exists in several forms. It is, however, often read as faulty in itself, because the reader takes the simplification for the claim. Whoever comes expecting that work outside the institutions cannot be right finds the confirmation in every simplification. The burden therefore lies on both sides: on the author, who must build the entry path so that the order is visible, and on the reviewer, who must consult the full version before declaring an error.

The question

The question is not whether review is necessary but what it can look like under these conditions: who reviews what, on which object, in which order, with what binding force, and who retains interpretive authority over what.

aj.2 Inventory: what already exists

The following tools were not designed for a review procedure. They arose in the work itself, often from a concrete dispute. That is precisely why they are tested.

The record-keeping of the FFGFT corpus

- **Status markers.** Every statement carries a status: [K], [B], [S], [Q], [X], [SETZUNG]. The reader sees at once what is proved, what is computed, what is open and what is taken over. The markers existed before the CCR correspondence; later terminology describes what they do (Doc. 351, methodological note).

- **Correction register.** Doc. 190 records corrections to earlier documents and the closing of points held open, continuously; earlier versions are frozen as an archive. Corrections are not worked in silently; they are recorded.
- **Check scripts.** Every recent document comes with a script containing assertions that prints `PASS` or `FAIL` for each statement. Many of these scripts are null tests: if chance yields the same result, it is not counted.
- **Negative results as sections.** Doc. 342 explicitly grants Galois products with $k \geq 5$ no status from product search; Doc. 361 has a section of its own, "What does not match"; Doc. 276 records a weak result as an honest negative.
- **Guard-rail against exponent numerology.** Every quantity can be written as ξ^N; a measured scale inside an exponent is not a prediction (Doc. 190, P35).
- **Comparator is not input.** Measured values (PDG) serve for comparison, not for derivation. This separation has been taken into F O R M as Illustration 001.

The governance of Dot Theory

Dot Theory (Vossen) is by its own account not a competing physical model but a representational meta-programme. Its governance layer is set out in Doc. 272: *Lexicon* (terms), *OAP* (admissibility), *Matrix* (relations between frameworks), *FAH* (how admissibility changes). Two distinctions carry it:

- *Preservation of meaning-making* versus *allocation of meaning*: recording how a result was understood is admissible; deciding who is right is not.
- Originator declaration (O0) versus secondary rendering (O2): a third party's reading of a framework is marked as such; the right of correction stays with the originator. Records are permanent and contestable; later entries may refine, dispute or supersede, but the earlier rendering remains part of the history.

The Constitutional Interpretation Layer

The CIL is co-authored by Stefaan Vossen and the author; the co-authorship is limited to the CIL specification and does not extend to Dot Theory or the CCR. Its subject is not the correctness of scientific statements but the entitlement to draw, inherit and pass on conclusions. Explicitly excluded are scientific correctness, empirical validity, mathematical truth and ontological commitment. From the dialogue between the two authors came the principles

- 3 – *Asymptotic Humility*,
- 6 – *Boundary Legitimacy / Evidential Admissibility*,
- 7 – *Recoverability / Reproducibility*,
- 8 – *Asserted Constitutional Properties*: an asserted property (determinism, self-management, derivational status) remains an authorial assertion until it is independently recoverable.

Further principles of version v1.1 name the vocabulary trap (adopting the vocabulary is no evidence of adopting the method; similarity of result or terminology is not a bridge) and

constitutional drift (undeclared extension of scope). The author added the condition that scope must be declared at the point of derivation and cannot be asserted retrospectively in a dispute; this turns the principle into a checkable test.

F O R M and the Constitutional Verification Protocol

F O R M (ed. Vossen; authors Haskins, Leavitt, Pascher, Quílez Zamora, Vossen; v1.0, fixed by SHA-256 `bd2a1f2d…e8e90c0e`) divides every review into *Framework* (the declared distinction space), *Operator* (the operation carried out), *Residual* (what survives the operation) and *Model* (the recoverable representation). Filling the R position with *Residual* rather than *Reality* goes back to a proposal of the author: *Reality* would have introduced an ontological commitment that the protocol is meant to avoid.

The argument against a universal checker comes from the same correspondence. A proof checker such as Lean works because axioms, type system and admissible operations were agreed beforehand; the checker comes second. The law works the same way: an agreed, contestable procedure precedes the ruling. Programming languages lie in between and make the point most sharply: they are strongly normalised, and yet different libraries return different results for the same integral, and $0.1 + 0.2 = 0.3$ is false in IEEE 754 arithmetic and true in exact rational arithmetic. If the result depends on the frame even there, then for heterogeneous scientific frameworks a single checker cannot be the goal, only a procedure that discloses the normalisation in each case and makes it recoverable.

FER

Diana Haskins' analytical method FER asks of every term what earned it its status, and does not let terms become stronger because they fit together nicely. An application of FER in the IPI exchange of 24–25 September 2026 produced a recursive review sequence: declared frame → operation → distinctions obtained → representation → next operation → recovery test → retained, lost or newly exposed distinction → revised representation. Three cases are kept apart: a distinction was available and is no longer recoverable; it was never represented; it was not recognised, but the information was present and a later operation exposes it. The representation is thus not the end of a review but the next object of review.

Practice in audits

Several review procedures of recent months have produced rules that hold beyond their occasion:

- **Freezing and hashing.** The author freezes the object and states its SHA-256 hash; the reviewer confirms the hash independently on the object received before reviewing content. In the SIM2 procedure two files with the same title, the same date and the same version string could be told apart only by their hashes.
- **Final standing is assigned, not claimed.** A submission that certifies itself "ADMISSIBLE" has no standing thereby. Standing is assigned by the re-audit.

- **Order.** Declare → freeze → implement → execute → audit independently → assign standing. In the Block II procedure (MOTHER-GEA, 25–27 September 2026) implementations repeatedly arrived before the specification; that is exactly where the errors occurred.
- **Two constitutional invariants** from the same procedure: *solver reports convergence ≠ stationary point* (an absolute solver tolerance is not a stationarity test; what is needed are scale-normalised residuals evaluated independently after the solve) and *changed frozen input ≠ same object*. A third finding was added: the same frozen inputs with two different potential functions yield two different objects; the specification must fix the function itself, not only its parameters.
- **Separation of roles.** In the CTA-FUSION declaration the author declared his complete exposure to his own corpus and thereby excluded himself from independent review of its fidelity. In the SIM2 procedure (Krüger) custodian of the blind tasks, attestor and builder are separate roles with declarations of interest; co-authors are not regarded as neutral.
- **Sealed advance prediction.** The prediction that Krüger's OP8-Neuro operator U3 adds no structure beyond the Markov limit was deposited with a hash before the computation and confirmed by the result.
- **Localise rather than judge.** A limitation in framework A establishes or refutes nothing about framework B. An author may decline a request for deeper disclosure for legitimate reasons; the only consequence is that independent recoverability ends at the declared boundary. That localises the standing; it does not condemn.

What has proved unsuitable

- **Several language-model referees from the same model.** Different roles or prompts increase coverage, not independence: one model has one set of blind spots. A second model with shared context is an echo, not a replication.
- **Self-confirming check blocks.** A block that compares its own constants with themselves cannot fail and is therefore not a check.
- **A hash over the output instead of the source.** It shows reproducibility of the output, not identity of the program and not physical correctness.
- **Asserted properties.** "Self-managing", "parameter-free", "frozen" are properties that must be shown; a single fitted constant in the description is enough to refute "parameter-free".

aj.3 Proposal: ten building blocks

The following building blocks are derived from the inventory. They are not a finished framework but a proposal; a framework of rules is itself a posit and must be agreed by those who apply it.

B1 – Declared constitution before any review

The author discloses what is posited and what is derived, in which order the layers follow one another, and which status each statement has. For FFGFT this means: geometry → algebra → number → independent empirical comparison. This order belongs before the first number in every presentation.

B2 – An entry path provided by the author

Because no outsider will spend the time on the whole corpus, the burden lies with the author: a designated entry path that shows within about an hour what everything rests on and where it could fail. For FFGFT these are Doc. 365 (the foundation in four layers), Doc. 377 (the interfaces to other frameworks) and the fixed predictions, for instance $M_W = 80.420$ GeV and $\sin^2\theta_{23} \in \{4/9,\ 5/9\}$ (Doc. 372).

B3 – Testable objects instead of texts

What is reviewed is a frozen object: document, script, input data, each with a hash over the source. A check script must be able to fail; it compares against an external data set fixed beforehand, aborts on missing identity instead of printing a substitute value, and contains null tests. Where a function is constitutive, the function itself is fixed, not only its parameters.

B4 – Predictions fixed in advance

What a framework risks is deposited before the comparison with data, with date and hash. A framework that fits only afterwards has risked nothing. The size of a corpus is no substitute for a single sharp prediction.

B5 – Separation of roles and interpretive authority

The author retains interpretive authority over his posits and their declared meaning (O0). He has no authority over whether a declared operation is actually carried out and whether the result follows; that is decided by review. Author, reviewer and, where needed, custodian and attestor are separate roles; each role declares its exposure. A third party's reading of a framework is marked as a secondary rendering.

B6 – A continuing register

Reviewer findings, author responses and corrections are recorded continuously and permanently, not overwritten. A withdrawn statement remains visible as withdrawn. Standing is assigned, not claimed, and every assignment is contestable.

B7 – Language models as tools, not as verdicts

The use of language models is disclosed. Their results are not adopted unchecked but secured by assertions against computation and source. A language-model review does not replace an independent judgement; several roles of the same model are not a majority. Responsibility for every statement remains with the person who signs it.

B8 – Mark simplification as such

Simplified presentations are marked as such and name the full version. An objection to the simplification is not an objection to the derivation; before a reviewer declares an error, he consults the full version.

B9 – Localise rather than judge globally

The outcome of a review is a status for each statement (for instance confirmed, not accepted, candidate), not an overall verdict on a framework. A boundary of recoverability localises the standing of what lies beyond it; it does not condemn it. Similar vocabulary does not establish a bridge between frameworks.

B10 – No universal checker

Every review presupposes an agreed normalisation, and that is itself a posit. The aim is not one checker for all frameworks but a procedure that discloses the normalisation in each case, separates inherited conventions (computing library, system of units) from framework-specific assumptions, and makes both recoverable.

Graded depth of review

To answer the question of time in practice, review can be divided into three stages that can be carried out one after another and by different people:

Stage	Question	Object and effort
1 Entry	Are posits, order and status markers declared? Are there predictions fixed in advance?	entry documents; about an hour
2 Object	Do the hashes match? Do the scripts run? Can their checks fail? Do the inputs agree with the specification?	frozen scripts and data; hours
3 Depth	Does a particular chain actually follow from the posits?	a single derivation with its predecessors; days

Stages 1 and 2 require no expertise in the subject matter of the framework and can largely be distributed. Stage 3 does require it and is realistic only for individual chains. A verdict at one stage holds only for that stage.

Origin of the building blocks

Block	Origin	Usability
B1	FFGFT status markers; F O R M (Framework)	directly
B2	FFGFT entry documents	directly, effort on the author's side
B3	audits (hash, scripts, Block II)	directly
B4	sealed advance prediction (OP8-Neuro)	directly
B5	Dot Theory O0/O2; CTA-FUSION; SIM2	roles can be filled only with several participants
B6	Doc. 190; FAH	directly
B7	experience in the IPI exchange	directly
B8	experience with entry texts	requires discipline on both sides
B9	CIL (scope, vocabulary trap); framework $A \neq B$	directly
B10	CVP correspondence (normalisation before checker)	as a principle, not as a procedure

aj.4 Limits and open points

- **Effort.** Declaring, freezing, hashing and keeping the register take time. The Block II procedure showed that the procedure reliably makes errors visible; it also showed that many rounds may be needed before a reviewable object exists at all. Procedure cannot replace content.
- **Neutral reviewers are scarce.** In a small circle almost everyone involved is a co-author or correspondent. Separation of roles is then more a disclosure of closeness than real independence.
- **Recognition.** A framework of rules agreed within the IPI holds at first only there. Whether outsiders regard an object reviewed under these rules differently from an unreviewed one is open.
- **Prejudice cannot be regulated.** The building blocks make prejudice visible (for instance when an error verdict is issued without reference to the full version), but do not remove it.
- **Interpretive authority versus review finding.** The separation in B5 is clear in principle, not always in the individual case: whether a statement belongs to the posit or is meant to follow from it must itself be declared. Here the condition applies that scope must be fixed at the point of derivation.

aj.5 Result

1. Classical peer review presupposes surveyability, a shared common stock, an institutional prior filter and human text production; none of these assumptions holds in general today.

2. In the IPI exchange, in Dot Theory governance, in CIL, F O R M and FER, and in the FFGFT record-keeping, tools have arisen that closed these gaps in concrete cases.
3. From them ten building blocks and a graded depth of review can be derived. Most can be applied directly; separation of roles and recognition depend on how many participants there are.
4. The framework of rules is itself a posit. It gains validity only through agreement by those who apply it, and like every posit it must be declared openly.

Check script: `2/python/Dok378_Skripte/pruef_378_verweise.py` (checks the cited corpus documents, figures and hashes)

Register (Doc. 190): no entry (new document, no correction).

[1] J. Pascher, *Doc. 190 — Correction Register* (continuing, 2026).

[2] J. Pascher, *Doc. 272 — FFGFT and HLV, classified according to Dot Theory* (June 2026).

[3] J. Pascher, *Doc. 274 — FFGFT in the IPI field* (June 2026).

[4] J. Pascher, *Doc. 276 — HLV homology test* (2026).

[5] J. Pascher, *Doc. 342 — Factorisation classes in G(6)* (Sept. 2026).

[6] J. Pascher, *Doc. 351 — PDG model dependence* (Sept. 2026).

[7] J. Pascher, *Doc. 361 — Kagome lattice and FFGFT* (Sept. 2026).

[8] J. Pascher, *Doc. 365 — The foundation of FFGFT in four layers* (Sept. 2026).

[9] J. Pascher, *Doc. 372 — The matrix-element principle and its reach* (Sept. 2026).

[10] J. Pascher, *Doc. 377 — Interface overview* (Sept. 2026).

[11] S. Vossen (ed.), D. Haskins, J. D. Leavitt, J. Pascher, J. Quílez Zamora, *F O R M — Constitutional Verification Protocol*, v1.0 (2026), DOI 10.5281/zenodo.22875795.

[12] S. Vossen, J. Pascher, *Constitutional Interpretation Layer (CIL)*, v1.1 (2026), `dottheory.co.uk`.

[13] S. Vossen, *Constitutional Physics — Canonical Constitutional Record*, Vols. I–III, v1.0 (2026).

[14] D. Haskins, *Universal Constraint — Unified Overview* (2026), DOI 10.5281/zenodo.20257133.

[15] M. Krüger, *HLV-LIRO SIM2 v1.3.0* (2026), DOI 10.5281/zenodo.21371481.

[16] J. Pascher, *FFGFT* (2026), DOI 10.5281/zenodo.20117635; `github.com/jpascher/T0-Time-Mass-Duality`.

Chapter ak

Doc. 379 — The Acoustic Plenum and the Photon

Abstract

*C. C. Lien (Zenodo, DOI 10.5281/zenodo.21800087, 8 September 2026) interprets space as a compressible superfluid, derives the electron mass from the temperature of the cosmic microwave background, adopts a mass formula for the muon and tau, equates the neutrino mass with $k_B T_{CMB}$ and describes the photon as a longitudinal shockwave with a transverse wake. This document examines the paper as a case study with a verification script (20/20). The numerical values are reproducible **[K]**. None of the load-bearing claims survives the check: the electron hit lies 88 standard deviations from the measured value and is not significant in the null test, the lepton formula is known, the neutrino mass is incompatible with oscillation data, the redshift would be frequency-dependent, and the photon would have three instead of two polarisations **[X]**. The cross-check shows how FFGFT proceeds at the same points, without repeating those contents.*

Status markers

[B] Algebraically proved.
[K] Numerically confirmed (verification script).
[S] Structurally plausible, open.
[X] Examined and rejected.
[Q] Quoted verbatim from the source (Lien 2026).

ak.1 The source

Lien replaces empty space with a medium:

> **[Q]**"By treating the spatial manifold as a continuous compressible superfluid, we derive the foundational particle phase-states (electron, muon, tau, and

> neutrino) from the Cosmic Microwave Background (CMB) acoustic baseline." (Conclusion)

The background is stationary, and redshift is not expansion:

> **[Q]**"Cosmological redshift is therefore not temporal expansion; it is Acoustic Pulse Broadening – the observation that waves of higher energetic frequencies decay faster than those of a lower energetic frequency when traversing the same medium." (Section II.A)

The photon is not a particle:

> **[Q]**"Photonic emission is therefore not a discrete point-particle phenomenon. It is a primary longitudinal acoustic shockwave [...] which structurally interacts with the azimuthal shear of its source to continuously generate an orthogonal transverse spiral wake." (Section V)

The central formulas are

$$f_e = 2\pi(\alpha^{-1})^4 \,\frac{k_B T_{\mathrm{CMB}}}{h}, \tag{ak.1}$$

$$m_n = m_e\Big(1 + \tfrac{3}{2}\,\alpha^{-1}\sum_{k=0}^{n} k^4\Big), \tag{ak.2}$$

$$m_\nu = k_B T_{\mathrm{CMB}}/c^2. \tag{ak.3}$$

ak.2 Recalculation

All six numerical values are reproducible **[K]**: $f_{\mathrm{CMB}} = 5.679 \times 10^{10}$ Hz, $f_e = 1.2583 \times 10^{20}$ Hz against the Compton frequency 1.2356×10^{20} Hz ($+1.84\,\%$), $m_\mu/m_e = 206.55$ ($-0.10\,\%$), $m_\tau/m_e = 3495.4$ ($+0.52\,\%$) and $m_\nu = 0.235$ meV. There are no arithmetic errors. The only question is what the numbers mean.

ak.3 Significance

The electron hit is a miss. The ratio $m_e c^2/(k_B T_{\mathrm{CMB}}) = 2.176 \times 10^9$ lies $1.84\,\%$ below $2\pi\alpha^{-4} = 2.216 \times 10^9$. T_{CMB} is known to $0.02\,\%$; the deviation therefore amounts to about 88 standard deviations **[K]**. Lien interprets it as "dynamic acoustic drag" but does not calculate this drag. A friction named after the fact is not a correction but a second free parameter.

Null test. The formula family $a\,\pi^b\alpha^{-k}$ with thirteen simple fractions $a,\ b \in \{-3,\dots,3\}$ and $k \in \{1,\dots,6\}$ comprises 546 formulas. Against random target values between 10^9 and 10^{10}, a hit within $1.84\,\%$ succeeds in 61 % of cases **[K]**. The hit is therefore not significant. FFGFT applies the same standard to its own grid findings (Doc. 342, 343 §G′): what chance delivers equally well is not counted.

The lepton formula is known. Eq. (ak.2) is the formula of A. O. Barut (Phys. Rev. Lett. 42, 1251, 1979) without change **[B]**. The accuracy for muon and tau is Barut's result, not that of the plenum. For $n = 3$ the same formula would give a charged lepton at 10.3 GeV **[K]**, which is experimentally excluded; Lien's justification via Chirikov overlap remains qualitative. For the number of generations in FFGFT see Doc. 317 and 316.

ak.4 Redshift

Damping changes the amplitude, not the frequency. Stokes damping $\alpha \propto f^2$ reduces the amplitude of a wave; its frequency is preserved. Numerically, a damped 50 Hz oscillation keeps its spectral maximum at 50 Hz **[K]**. No redshift arises in this way.

A frequency-dependent redshift is excluded. If z were tied to damping $\propto f^2$, the 21 cm line and Lyman-α would differ by a factor of 3×10^{12} **[K]**. As measured, z agrees between the two to about 10^{-5}. FFGFT went through this error itself: older corpus documents contained a chromatic redshift, which A265 marks as superseded.

Light-curve stretching. Lien mentions the $(1 + z)$ stretching of supernova light curves only as an argument of standard cosmology, without explaining it. A static picture must contain it. In FFGFT it follows from $\tilde{T} \cdot m = 1$: wavelength and clock rate are booked together, and the stretching is exactly $(1 + z)$ for all z (Doc. 312, A265, R128) **[B]**. By Doc. 267 all frequency and clock-rate signatures are degenerate between a static picture and expansion; a static background is therefore admissible, but only if it is achromatic. Lien's plenum is not **[X]**.

ak.5 The photon

In vacuum, light has exactly two transverse polarisations and no longitudinal component. A massless $U(1)$ gauge field has four components, two of which are removed by gauge freedom and the constraint **[B]**. Lien's picture of a longitudinal primary wave plus a transverse wake would have three degrees of freedom **[X]**. The baroclinic torque $\nabla\rho \times \nabla P$ generates a vorticity component in the z direction, but no wave with two polarisation states. Photon counting (photoelectric effect, antibunching) is not addressed in the paper.

The argument that a particle picture would have to break up into a "quantised spray" in the far field confuses the wavefront of a classical field with the counting statistics of its quanta. In FFGFT the photon is exactly massless, the pure phase and connection anchor, and $E_{\text{bit}} = \hbar c / L$ is a photon energy (Doc. 290). No medium is needed for this.

ak.6 Neutrino

A single mass of 0.235 meV cannot produce the atmospheric mass splitting $\Delta m^2_{\text{atm}} \approx 2.5 \times 10^{-3}$ eV^2. For that, at least one state must weigh about 50 meV **[K]**. Equating the

neutrino mass with $k_B T_{\mathrm{CMB}}$ is therefore excluded **[X]**. The FFGFT hierarchy of Doc. 340 satisfies this condition with its heaviest state at $49.96\,\mathrm{meV}$ **[K]**.

ak.7 The shared anchor

It is striking that Lien touches the same number that is the SI anchor of the cosmic sector in FFGFT: $m_e c^2/(k_B T_{\mathrm{CMB}}) = 2.176 \times 10^9$ (A265). FFGFT carries this ratio explicitly as an anchor, not as a derivation; the exponent $41/4$ of the bridge formula is open (P20), and the cosmic sector is deliberately bracketed (P39). Lien's candidate $2\pi\alpha^{-4}$ misses the number by $1.84\,\%$ and cannot be distinguished from chance in the null test. It therefore does not close the gap; it shows how easily approximations can be found at this point **[S]**.

ak.8 What agrees and what does not

The paper shares two starting points with FFGFT: a stationary instead of an expanding background, and the rejection of point-like singularities. In FFGFT the smallest length $L_0 = \xi \ell_P$ secures finiteness (Doc. 180, R90); in Lien's work a density maximum ρ_E does this. That maximum, however, is circular, because the classical electron radius $r_e = \alpha\hbar/(m_e c)$ already contains the electron mass **[K]**.

The difference lies in the method. Lien introduces a medium with material constants (damping, viscosity, drag) and explains deviations after the fact through these constants. FFGFT posits a compact geometry with one parameter ξ, derives numbers and checks them against fixed criteria. Where a deviation remains, it is carried as an open bridge, not explained away.

ak.9 Summary

Claim (Lien)	Finding	FFGFT	Status
$f_e = 2\pi\alpha^{-4} f_{\mathrm{CMB}}$	$+1.84\,\%$, $88\,\sigma$, null test 61 %	A265, P20	**[X]**
Lepton formula Eq. (ak.2)	Barut 1979, $n = 3$ excluded	316, 317	**[X]**
$m_\nu = k_B T_{\mathrm{CMB}}$	incompatible with $\Delta m^2_{\mathrm{atm}}$	340	**[X]**
Redshift by damping	frequency-dependent, leaves f unchanged	267, 312, R128	**[X]**
Photon longitudinal + transverse	three instead of two degrees of freedom	290	**[X]**
Stationary background	admissible if achromatic	267, R128	**[S]**

The paper combines known formulas, a friction named after the fact and an intuitive medium into a picture that appears closed. The recalculation shows that the numbers are correct, but that the interpretation fails against measured data at every load-bearing

point. The usable core, a static background without singularities, is already realised differently and achromatically in FFGFT.

Verification script. `2/python/Dok379_Skripte/pruef_379_lien_photon.py` (20/20): recalculation (A), significance and null test (B), measured data (C), FFGFT cross-check (D). Two inputs are literature values and marked as such: the agreement of z between 21 cm and Lyman-α, and the DES value $b = 1.003 \pm 0.011$.

Source document. Lien, C. C.: *Electromagnetic Illusion: Photonic Emission as Orthogonal Acoustic Shockwaves in a Compressible Superfluid Plenum.* Zenodo (2026), DOI 10.5281/zenodo.21800087. Barut, A. O.: *Lepton Mass Formula.* Phys. Rev. Lett. **42**, 1251 (1979).

Chapter aI

Doc. 380 — Why D_4 — Specificity of the carrier against matched comparison lattices (R95)

Abstract

FFGFT posits the lattice D_4 as the carrier of the compact geometry $T^4/\mathbb{Z}_3$ (Doc. 314) ***[SETZUNG]****. Doc. 190 lists under R95 as an open bridge that an adversarial specificity test against matched comparison carriers is missing. This document examines the carrier. Three criteria fixed in advance are applied to $\mathbb{Z}^4$, A_4, $A_2 \oplus A_2$, D_4 and to random lattices: the fixed-point-free $\mathbb{Z}_3$ action with nine fixed points carried in the corpus (Doc. 330), the kissing number, and the lattice energy as the Epstein zeta function at equal volume of the fundamental cell. Only the third criterion is independent of FFGFT. The $\mathbb{Z}_3$ condition excludes $\mathbb{Z}^4$ and A_4; among the remaining carriers D_4 has the lowest lattice energy* ***[K]****, and the kissing number 24 is consistent with Doc. 340. D_4 is thereby better motivated as a carrier but remains a posit; R95 is not closed, and the level of the operator $\hat{F}$ remains open* ***[S]****. Verification script: 13/13.*

Status markers

[B]	Algebraically proved.
[K]	Numerically confirmed (verification script).
[S]	Structurally plausible, open.
[Q]	Taken from the literature, cited.
[SETZUNG]	Axiomatic posit.

aI.1 The question

A posit is motivated rather than arbitrary if the chosen carrier stands out among comparable alternatives by quantities that were not tuned to the results. This is exactly what

the GAE logic demands, as applied in the HLV audit: there an ordinary cubic lattice outperformed the carrier under study in the spectral measure, and the specificity claim failed (Doc. 330). For D_4, R95 records that $5 \nmid |\mathrm{Aut}(D_4)| = 1152$ is a lattice-specific fact, but not a proof of specificity.

The test must therefore meet three conditions. The comparison carriers have the same dimension and the same volume of the fundamental cell. The criteria are fixed before the calculation. And a criterion meant to establish specificity must not be a number that originates in FFGFT. Criteria that express a corpus condition test only compatibility; they are explicitly marked as such below.

al.2 Comparison carriers and criteria

The comparison covers the cubic lattice $\mathbb{Z}^4$ (the winner in the HLV audit), the root lattice A_4, the sum $A_2 \oplus A_2$ and D_4, each scaled to unit volume of the fundamental cell, plus random lattices.

K1: $\mathbb{Z}_3$ compatibility (corpus condition). The corpus carries $T^4/\mathbb{Z}_3$ with nine fixed points and a $\mathbb{Z}_3$ action with eigenvalues $\{\omega, \omega^2, \omega, \omega^2\}$; the nine fixed points follow from $|\det(1-g)| = |(1-\omega)(1-\omega^2)|^2 = 9$ (Doc. 330, likewise Docs. 361, 364, 365). What is required is therefore a lattice automorphism g of order 3 without eigenvalue 1. K1 expresses this condition and tests compatibility, not specificity. The search is complete, because every automorphism maps shortest vectors to shortest vectors and all four bases consist of shortest vectors.

Which $\mathbb{Z}_3$ is meant is shown by the complete enumeration of the 1152 automorphisms of D_4: 80 have order 3. Of these, 64 have a two-dimensional fixed space and produce fixed tori instead of isolated fixed points, among them the Dynkin automorphism of triality. The remaining 16 are fixed-point-free with exactly nine fixed points on T^4 **[K]**. These are the 16 elements named in Doc. 330; the triality that carries the orbifold is thus this fixed-point-free class, not the Dynkin automorphism.

K2: Kissing number (consistency). The number of shortest lattice vectors. It is a property of the lattice alone, but FFGFT uses it itself ($\mathrm{Kissing}(D_4) = 24$ in Doc. 340). K2 therefore tests consistency, not specificity.

K3: Lattice energy. The Epstein zeta function

$$Z_\Lambda(\nu) = \sum_{v \in \Lambda \setminus \{0\}} |v|^{-\nu} \tag{al.1}$$

at unit volume of the fundamental cell, for $\nu = 5, 6, 8$. It is at the same time the spectral zeta function of the flat torus of the dual lattice, hence an operator quantity of the Laplacian, and it contains no free parameter. It is computed with EpsteinLib (Buchheit et al.) **[Q]**; the library reproduces the closed-form values for $\mathbb{Z}^4$ and D_4 to twelve digits **[K]**.

al.3 Results

K1. $\mathbb{Z}^4$ and A_4 have no automorphism of order 3 without eigenvalue 1; on them $T^4/\mathbb{Z}_3$ with nine fixed points is impossible **[K]**. $A_2 \oplus A_2$ and D_4 have one, each with $|\det(1-g)| = 9$ **[K]**. The winner of the HLV audit is thus already excluded by the symmetry condition.

K2. The kissing numbers are D_4: 24, A_4: 20, $A_2 \oplus A_2$: 12, $\mathbb{Z}^4$: 8 **[K]**. Of the two $\mathbb{Z}_3$-compatible carriers, only D_4 has the 24 that enters Doc. 340 as Kissing(D_4). This is consistency: the corpus computes with the lattice it posits.

K3.

ν	D_4	A_4	$A_2 \oplus A_2$	$\mathbb{Z}^4$
5	**22.860**	23.123	23.703	24.531
6	**12.583**	12.919	13.699	14.830
8	**6.830**	7.282	8.457	10.246

D_4 has the lowest energy for all three exponents **[K]**. Of 300 general random lattices none falls below D_4 (best 13.15 at $\nu = 6$), and likewise none of 300 random lattices from the $\mathbb{Z}_3$-compatible family (best 12.72) **[K]**. All 200 small perturbations around D_4 increase the energy; D_4 is a local minimum **[K]**.

It is proved that D_4 is a local minimum of the Epstein zeta function among all four-dimensional lattices (Sarnak and Strömbergsson 2006) **[Q]**. Whether it is also the global minimum is an open conjecture in four dimensions. The 600 random lattices support it but do not prove it.

al.4 Assessment

What is shown. Among the four-dimensional carriers that admit the $\mathbb{Z}_3$ action with nine fixed points carried in the corpus, D_4 has the lowest lattice energy **[K]**. This is the only one of the three quantities that does not originate in FFGFT; the kissing number confirms consistency only. D_4 is thereby better motivated as a carrier than before, but it remains a motivated posit **[SETZUNG]**. The finding does not replace a proof of specificity.

What remains open. The GAE logic also asks about the operator: whether $\hat{F}$ with $\xi = \lambda_{\min}(\hat{F}_{D_4})$ (Doc. 327) yields something distinguishable on the comparison carriers. This is not tested here **[S]**. Likewise, the random search is numerical evidence, not a proof of global minimality within the $\mathbb{Z}_3$ family; it covers the exponents $\nu = 5, 6, 8$.

Consequences for the bookkeeping. R95 is not closed. The required operator test is still outstanding, and new results receive no register entry.

al.5 Summary

Criterion	D_4	$A_2 \oplus A_2$	A_4	$\mathbb{Z}^4$
K1: $\mathbb{Z}_3$ with 9 fixed points (corpus)	yes	yes	no	no
K2: kissing number (consistency)	**24**	12	20	8
K3: $Z(6)$ at unit volume	**12.58**	13.70	12.92	14.83

Among the carriers that satisfy the symmetry condition, D_4 has the lowest lattice energy, a quantity that does not originate in FFGFT; the kissing number is consistent with the corpus. The posit of the carrier is thus better supported, but it remains a posit.

Verification script. `2/python/Dok380_Skripte/pruef_380_d4_spezifitaet.py` (13/13): check of the library against closed-form values, K1 to K3, classification of the order-3 automorphisms of D_4, random and perturbation tests.

Sources. Buchheit, A. A.; Busse, J. K.; Gutendorf, R.: *Computation and Properties of the Epstein Zeta Function with Applications to Quantum Systems.* arXiv:2412.16317 (2024); software EpsteinLib, https://github.com/epsteinlib/epsteinlib. Sarnak, P.; Strömbergsson, A.: *Minima of Epstein's zeta function and heights of flat tori.* Invent. Math. **165**, 115–151 (2006). Conway, J. H.; Sloane, N. J. A.: *Sphere Packings, Lattices and Groups.* 3rd ed., Springer (1999).

Chapter am

Doc. 381 — The Running Recursion Summed Exactly

Abstract

The recursion $\xi_{n+1} = \xi_n(1 - 100\,\xi_n)$ (Doc. 295, 333) is the bookkeeping of the fractal correction in the running case. Doc. 295 derives its continuum behaviour $\xi_n \approx 1/(100(n+75))$ and the logarithmic defect $D(n) \approx \ln(1 + n/75)$. This document adds four statements. First, the product of all correction factors telescopes exactly to ξ_n/ξ_0 ***[B]****. Second, a logarithmic second-order term improves the continuum approximation by a factor of 50 to about 1000* ***[K]****. Third, the multiplicative and additive bookkeeping differ by a fixed amount $\psi'(75)/2$, and the bounded constant ≈ 0.0134 between mass and time defect found numerically in Doc. 295 is $\psi'(75) = \sum_{k\geq 0}(k+75)^{-2}$* ***[K]****. Fourth, the recursion has the form of a one-loop running, whose coefficient, however, is fixed only together with a step convention* ***[S]****.*

Status markers

[B] Algebraically proved.
[K] Numerically confirmed (verification script, 120 digits).
[S] Structurally plausible, open.

am.1 Starting point

With $\xi_0 = 4/30000 = 1/7500$ one has $100\,\xi_0 = 1/75$ and the single step $K_{\text{frak}} = 1 - 100\,\xi_0 = 74/75$. In the frozen case this factor stays constant; in the running case (Case C, Doc. 333) it changes with every step according to

$$\xi_{n+1} = \xi_n\,(1 - 100\,\xi_n). \tag{am.1}$$

Doc. 295 shows that this turns into a logarithmic spiral with ratio e. The following sections compute how accurate the approximations used there are, and give in closed form the quantities that appear there only numerically or approximately.

am.2 The product telescopes

From Eq. (am.1) it follows that $1 - 100\,\xi_k = \xi_{k+1}/\xi_k$. The product of all correction factors therefore cancels down to the last and the first term:

$$\prod_{k=0}^{n-1}\bigl(1 - 100\,\xi_k\bigr) = \frac{\xi_n}{\xi_0}. \tag{am.2}$$

This is an exact identity **[B]**, confirmed in rational arithmetic for $n = 1$ to 12 **[K]**. The total accumulated fractal correction after n steps is thus simply the ratio of the last to the first ξ; it need not be multiplied step by step. With the continuum form from Doc. 295 one obtains $\prod \approx 75/(n+75)$, accurate to $0.007\,\%$ at $n = 10^5$ **[K]**.

am.3 Continuum at second order

With $u_n = 1/\xi_n$, Eq. (am.1) reads exactly $u_{n+1} = u_n + 100 + 100^2/(u_n - 100)$. Doc. 295 keeps only the step $+100$ and obtains $u_n \approx 100(n+75)$. Including the next term $100^2/u_n \approx 100/(n+75)$, it sums to a logarithm:

$$\frac{1}{\xi_n} \;\approx\; 100\,(n+75) + 100\,\ln\frac{n+75}{75}. \tag{am.3}$$

Against the exact recursion (up to $n = 10^5$, 120 digits) **[K]**:

n	1st order (Doc. 295)	2nd order, Eq. (am.3)
1	$0.018\,\%$	$0.0004\,\%$
10	$0.15\,\%$	$0.003\,\%$
100	$0.49\,\%$ (maximum)	$0.005\,\%$
10^4	$0.049\,\%$	$0.00007\,\%$
10^5	$0.0072\,\%$	$0.000007\,\%$

The approximation of Doc. 295 therefore never deviates by more than $0.5\,\%$ from the exact value; its statements (logarithmic spiral, ratio e, pole at $n = -75$) are unaffected. The second order is 50 to about 1000 times more accurate and is recommended wherever values of the running recursion are used numerically.

am.4 Multiplicative and additive: a fixed difference

Doc. 295 carries the accumulated defect additively, $D(n) = \sum_k 100\,\xi_k$. The multiplicative bookkeeping is $-\ln\prod_k(1 - 100\,\xi_k)$. Because $-\ln(1-x) = x + x^2/2 + \ldots$ it is always larger **[K]**, and the difference does not grow with n but tends to the sum of the second-order terms:

$$-\ln\prod_k\bigl(1 - 100\,\xi_k\bigr) - D(n) \;\longrightarrow\; \sum_{k\ge 0}\frac{1}{2\,(k+75)^2} = \frac{\psi'(75)}{2} = 0.00671 \quad \textbf{[K]}. \tag{am.4}$$

The two bookkeepings are therefore equivalent up to a fixed amount that can be given in closed form. The gap between the multiplicative form $(1-\xi)^{100}$ and $74/75$, which Doc. 333 describes as the difference between frozen and running recursion, is a term of the same second-order kind.

The bounded constant from Doc. 295. Doc. 295 finds numerically that the accumulated defects of the mass side and the time side ($d_{\text{mass}} = 1/(1-100\,\xi_n) - 1$, $d_{\text{time}} = 100\,\xi_n$) differ only by a bounded constant ≈ 0.0134. Their difference per step is $(100\,\xi_n)^2/(1-100\,\xi_n)$, and the sum tends to

$$\sum_{k\geq 0} \frac{1}{(k+75)^2} = \psi'(75) = 0.013423 \qquad \textbf{[K]}, \tag{am.5}$$

to 6×10^{-5} relative. The constant is therefore not an accident of the simulation but the trigamma function at $75 = 1/(100\,\xi_0)$, exactly twice the difference between multiplicative and additive bookkeeping.

am.5 The one-loop reading

In the continuum, Eq. (am.1) becomes $d\xi/dn = -100\,\xi^2$ with the solution $\xi(n) = \xi_0/(1+100\,n\,\xi_0)$. This is the form of a one-loop running of a coupling, as it holds for instance for α_s. The form is unique; the coefficient is not. If one step corresponds to a scale factor s, then $d\xi/d\ln\mu = -(100/\ln s)\,\xi^2$ **[K]**:

Step convention	$b = 100/\ln s$
Factor e (spiral ratio, Doc. 295)	100
Factor 2 (usual RG convention)	144
Factor $3/2$ ($q = 2/3$, Doc. 275)	247
Factor 10 (decade)	43

The identification with a β function is therefore a statement only once a step convention is fixed **[S]**. The spiral of Doc. 295 suggests the factor e, because it scales by e per precession turn; Doc. 275 works internally with $q = 2/3$. Which convention carries physical meaning is open.

am.6 Single step and power form

The single step $74/75$ can also be written in the power form $(D_f/3)^{D_f/2}$ (A040). With $D_f = 2.973$ it gives 0.9866508 against $74/75 = 0.9866667$, i.e. to 2×10^{-5} **[K]**. Exact agreement requires $D_f^{\text{eff}} = 2.973032$. Both forms are equivalent; the linear form $1-100\,\xi_0$ is the one the recursion computes with.

am.7 Summary

Statement	Result	Status
Telescoping product, Eq. (am.2)	exactly ξ_n/ξ_0	**[B]**
Second order, Eq. (am.3)	50 to about 1000 times more accurate than Doc. 295	**[K]**
Multiplicative minus additive	$\psi'(75)/2 = 0.00671$	**[K]**
Bounded constant from Doc. 295	$\psi'(75) = 0.013423$	**[K]**
One-loop form	unique; coefficient depends on convention	**[S]**
Power form A040	$D_f^{\text{eff}} = 2.973032$	**[K]**

Doc. 295 remains valid in all its statements; its approximations are quantified here and extended by the second order. What remains open is which step convention underlies the one-loop reading physically, and, when several corrections run at the same time, whether they overlap. For the latter, the inclusion–exclusion rule of cluster expansions is the natural tool: the contribution of a level is its value minus what its sublevels already contain **[S]**.

Verification script. `2/python/Dok381_Skripte/pruef_381_rekursion_ansaetze.py` (14/14): exact recursion in rational arithmetic and with 120 digits up to $n = 10^5$, telescoping product, both continuum orders, defect and product, constant from Doc. 295, power form, step conventions.

Chapter an

Doc. 382 — Muon $g-2$: Status 2025 and Assessment of the FFGFT Statements

Abstract

Several early documents of the corpus refer to the deviation between the measured anomalous magnetic moment of the muon and the Standard Model prediction as it stood in 2021: $\Delta a_\mu = 251(59)\times 10^{-11}$*, about* $4.2\,\sigma$*. With the final result of the Fermilab experiment (June 2025) and the 2025 White Paper of the Muon* $g-2$ *Theory Initiative, this deviation has shrunk to* $38(63)\times 10^{-11}$*, about* $0.6\,\sigma$ ***[K]****. Statements that cite a T0 contribution of* 251×10^{-11} *as the explanation of the anomaly are therefore outdated; such a fixed contribution would today lie about* $3.4\,\sigma$ *above the difference* ***[K]****.*

The load-bearing $g-2$ *line of the corpus (Doc. 018 Rev. 12, Doc. 158, A138) is not affected. It rests on* ratios*:* $\Delta a(\tau\text{–}\mu)/\Delta a(\mu\text{–}e) = f^{1/3}-1$ *and the bridge* $\Delta a(\tau\text{–}e)/\Delta a(\mu\text{–}e) = (144/125)\,m_\tau/m_\mu$*. In these, the projection factor and the SI conversion cancel. It obtains absolute values by anchoring the ratios to measured quantities; with the 2025 values this gives* $a_\tau = 1.2811\times 10^{-3}$ ***[K]****. This anchored value is the better absolute value compared with the direct formula, and it depends only on the measured values of* a_e *and* a_μ*, not on the Standard Model calculation, whose status has changed.*

On notation: here f is the base winding number $f = 1/\xi = 7500$, not a frequency. In natural units f is only another reading of the duality $T\cdot m = 1$: $f = t_P/t_0 = \Lambda_{\mathrm{T0}}/E_P$ with $t_0 = \xi t_P$ and $\Lambda_{\mathrm{T0}} = E_P/\xi$, and $\xi E_0^2 = 1$ gives $f = E_0^2$. As a dimensionless ratio, f cannot be set to 1 with justification (Doc. A135).

Status markers

[K] Numerically confirmed (verification script).

[S] Speculative: a hint or conjecture, not yet derived or checked in the corpus.

The experimental and theoretical input values are taken from the sources listed in the bibliography; they are not derived here.

an.1 Starting point in the corpus

The corpus contains two kinds of statements on the lepton $g-2$, which have to be kept apart.

(a) Explaining the anomaly by a T0 contribution. Early documents posit a contribution $\Delta a_\mu^{(\mathrm{T0})} = 251\times10^{-11}$, or describe the muon anomaly as explained by the ξ term, or use it as a benchmark: Doc. 019, 030, 033, 049, 053, 059, 070, 081, 095, and in the A series A138 (comparison with the "muon discrepancy"). These statements depend directly on the size of the deviation between measurement and Standard Model.

(b) The ratio line. Doc. 018 (Rev. 12), Doc. 158 and A138 describe the leptons by fractal winding exponents ($5/3$ for the muon, $4/3$ for the tau) and put the ratios at the centre: ratios of $g-2$ differences are structurally exact, because the projection factor, the fractal correction and the SI conversion cancel. Absolute values follow only by anchoring to a measured quantity – the same division of labour as for the masses, where ratios are fixed geometrically and one empirical anchor sets the scale (A138, Doc. 351). This line compares with measured values, not with the difference to the Standard Model.

an.2 Status 2025

- **Measurement.** The Fermilab experiment E989 published its final result in June 2025: $a_\mu = 116\,592\,070.5(14.8)\times10^{-11}$, a precision of 127 ppb [1]. Together with E821 (Brookhaven) this gives the world average $a_\mu^{\mathrm{exp}} = 116\,592\,071.5(14.5)\times10^{-11}$ [2].
- **Theory.** The 2025 White Paper of the Muon $g-2$ Theory Initiative gives $a_\mu^{\mathrm{SM}} = 116\,592\,033(62)\times10^{-11}$ [2]. The essential change compared with 2020 is the hadronic vacuum polarisation: it is now determined from lattice QCD instead of e^+e^- cross sections.
- **Difference.** $a_\mu^{\mathrm{exp}} - a_\mu^{\mathrm{SM}} = 38(63)\times10^{-11}$, about $0.6\,\sigma$ **[K]**. In 2021 it was $251(59)\times10^{-11}$, about $4.2\,\sigma$ **[K]**.
- **Open within the theory.** Why the data-driven determination of the hadronic vacuum polarisation differs from the lattice determination is not yet understood [3]. The anomaly is therefore regarded as strongly weakened but not definitively settled.

What has changed is thus the Standard Model calculation, not the measured value: the world average of a_μ has moved by only about 10×10^{-11} since 2021.

an.3 Consequences for statements of type (a)

A fixed T0 contribution of 251×10^{-11} was, in 2021, exactly as large as the deviation it was meant to explain. Today it lies

$$\frac{251-38}{63}\approx 3.4\,\sigma$$

above the measured difference **[K]**. The statement "the ξ term explains the muon anomaly" is therefore outdated. For an additional contribution beyond the Standard Model, the current status leaves a range of about -88 to $+164 \times 10^{-11}$ at two standard deviations **[K]**. A T0 contribution of this size is not excluded, but it is not derived in the corpus **[S]**.

The documents listed under (a) in Section 1 are affected. They no longer use the 2021 deviation as evidence or as a target and refer to this document for the assessment of the muon anomaly.

an.4 The ratio line is not affected

The core of Doc. 018, Doc. 158 and A138 is two ratio statements **[K]**:

$$\frac{\Delta a(\tau\text{–}\mu)}{\Delta a(\mu\text{–}e)} = f^{1/3} - 1 = 18.57, \qquad \frac{\Delta a(\tau\text{–}e)}{\Delta a(\mu\text{–}e)} = \frac{144}{125}\,\frac{m_\tau}{m_\mu} = 19.37.$$

The first follows from the exponents $5/3$ and $4/3$ alone. The second replaces $f^{1/3} = 19.57$ by the ratio of the mass formulas and thereby links $g-2$ to the lepton masses; f, k_{geom} and K_{frak} cancel.

From ratio to absolute value. Anchoring the bridge to the measured values of a_e and a_μ gives the absolute value for the tau:

$$a_\tau = a_e + \frac{144}{125}\,\frac{m_\tau}{m_\mu}\,(a_\mu - a_e) = 1.2811 \times 10^{-3}$$

with the 2025 values **[K]**; Doc. 018 and A138 quote 1.282×10^{-3} with a rounded a_e, Doc. 158 1.281×10^{-3}. The direct absolute formula $a_\tau = a_e + 4\pi/f^{4/3}$ gives 1.2456×10^{-3} instead **[K]**. The anchored value is the better one, because it contains only the exact ratio and measured quantities; this is why Doc. 158 and A138 use it as the prediction.

Why the anomaly plays no role here. The anchor is the measured difference $a_\mu - a_e = 6.2685 \times 10^{-6}$. It has changed by only about 10^{-10} between 2021 and 2025 **[K]**. The anomaly was a statement about the Standard Model calculation; the ratio line does not use that calculation. The tau prediction therefore remains unchanged. It lies 1.04×10^{-4} above the Standard Model prediction $a_\tau^{\text{SM}} = 1.1772 \times 10^{-3}$ **[K]**. The current experimental bounds on a_τ are about 85 times wider than this difference (CMS 2024: $-0.0042 < a_\tau < 0.0046$, 95 % CL) **[K]**; the decision awaits a measurement of a_τ **[S]**.

an.5 The direct absolute formulas

Besides the ratio line, Doc. 018 contains direct absolute formulas from the winding geometry, corrected by $K_{\text{frak}}^{3/2}$. With the values of Doc. 018 one obtains **[K]**:

$$a_e^{(\text{corr})} = 1.1600 \times 10^{-3} \quad (0.03\,\% \text{ from the measured value}),$$
$$a_\mu^{(\text{corr})} = 1.1644 \times 10^{-3} \quad (-0.13\,\% \text{ from the world average}).$$

These values also compare with measured values and do not depend on the anomaly. They work at the level of 10^{-4} to 10^{-3}, however, and are not a precision comparison where

the anomaly was discussed. A numerical example shows why the corpus takes the route via ratios for absolute values: the directly computed difference $\Delta a(\mu\text{–}e) = 4\pi/f^{5/3} = 4.373 \times 10^{-6}$ lies about $30\,\%$ below the measured one **[K]**, while the ratio of the differences follows exactly from the exponents. The anchored route avoids this uncertainty.

A138. The comparison there, that the tau deviation is "seven times larger than the muon discrepancy", has become moot, since there is no longer a significant muon discrepancy. The tau statement itself is independent of this.

an.6 Result

With the 2025 status, the muon $g-2$ anomaly of 2021 is no longer significant. FFGFT statements that use it as evidence or as the target of a T0 contribution are outdated; the affected documents refer to this document for this. The load-bearing ratio line is not affected: its ratios are structurally exact, it obtains its absolute values by anchoring to measured quantities, and the measured values have not changed. Its tau prediction $a_\tau \approx 1.281 \times 10^{-3}$ remains the actual test statement. Whether the FFGFT additionally yields a contribution to the muon within today's allowed range of about 10^{-9} is open **[S]**.

Verification script: `2/python/Dok382_Skripte/pruef_382_myon_g2_stand.py`, 21/21.

[1] Muon $g-2$ Collaboration (D. P. Aguillard et al.), *Measurement of the Positive Muon Anomalous Magnetic Moment to 127 ppb*, Phys. Rev. Lett. 135 (2025) 101802, arXiv:2506.03069.

[2] R. Aliberti et al. (Muon $g-2$ Theory Initiative), *The anomalous magnetic moment of the muon in the Standard Model: an update*, Phys. Rep. 1143 (2025) 1–158, arXiv:2505.21476.

[3] CERN Courier, *Fermilab's final word on muon g-2* (2025).

Chapter ao

Doc. 383 — Masses and Planck Scale without v

Abstract

*The lepton ladder of the corpus writes the masses as $m_i = r_i\,\xi^{p_i}\,v$ and uses the electroweak scale v as an intermediate quantity; the Planck scale is reached separately through the formula for G with the conversion factor C_{conv}. Read this way, the calculation seems to need two anchors, one for v and one for E_P. It does not. Doc. 149 links v and E_P directly: $v = E_P/(f^4 \cdot (\pi/2) \cdot 10)$, i.e. $v/E_P = \xi^4/(5\pi) = 2.012 \times 10^{-17}$, $-0.23\,\%$ against the measured value **[K]**. Inserted into the ladder, v drops out: $m_i = \frac{r_i}{5\pi}\,\xi^{p_i+4}\,E_P$ **[B]**. All three lepton masses, E_P, G, ℓ_P and L_0 then hang on a single chain, and one measured value as anchor fixes all numbers. With m_e as anchor one obtains $E_P = 1.237 \times 10^{19}$ GeV ($+1.34\,\%$), $G = 6.50 \times 10^{-11}$ ($-2.6\,\%$), $\ell_P = 1.595 \times 10^{-35}$ m and $L_0 = 2.127 \times 10^{-39}$ m **[K]**. The deviations are the known bare residues of the ladder, not a second anchor.*

On notation: here f is the base winding number $f = 1/\xi = 7500$, not a frequency. In natural units f is only another reading of the duality $T \cdot m = 1$: $f = t_P/t_0 = \Lambda_{\mathrm{T0}}/E_P$ with $t_0 = \xi t_P$ and $\Lambda_{\mathrm{T0}} = E_P/\xi$, and $\xi E_0^2 = 1$ gives $f = E_0^2$. As a dimensionless ratio, f cannot be set to 1 with justification (Doc. A135).

Status markers

- **[B]** Proven algebraically (rearrangement without additional assumption).
- **[K]** Confirmed numerically (check script).
- **[S]** Speculative: a hint or conjecture, not yet derived or checked in the corpus.

Measured values (CODATA 2022, PDG) serve as anchor or as comparison values. $\hbar$, c and e are exact by definition in the SI; they fix only the units and are not an additional anchor.

ao.1 Starting point

Two routes stand side by side in the corpus:

1. **Lepton ladder** (Docs. 006, 352): $m_i = r_i\,\xi^{p_i}\,v$ with $r = (4/3,\ 16/5,\ 25/9)$ and $p = (3/2,\ 1,\ 2/3)$ for e, μ, τ. The mass ratios $m_\mu/m_e = \frac{12}{5}\xi^{-1/2} = 207.85$ and $m_\tau/m_\mu = \frac{125}{144}\xi^{-1/3} = 16.99$ do not contain v.
2. **Gravitation** (Docs. 012, 013, 180): $G = \xi^2/(4m_e)\cdot C_{\text{conv}}\cdot K_{\text{frak}}$, and from it E_P and ℓ_P.

Taking v from m_e and E_P from the G formula uses two measured values: m_e and, through C_{conv}, the measured value of G. This is how Doc. 375 calculates. A chain meant to derive all quantities from ξ alone, however, may have only one anchor; all other numbers must follow. Doc. 149 supplies the link.

ao.2 The relation between v and E_P

Doc. 149 derives the electroweak scale from the Planck energy: dilution over the f^4 cells of the four-dimensional lattice, projection onto the three-dimensional half-space ($\pi/2$), and a factor $1/10$ for the step to the electroweak scale:

$$v = \frac{E_P}{f^4\cdot(\pi/2)\cdot 10} \qquad\Longleftrightarrow\qquad \frac{v}{E_P} = \frac{\xi^4}{5\pi} \quad \textbf{[B]} \tag{ao.1}$$

with $f = 1/\xi$. The rearrangement is exact. The factor 10 is neither derived nor motivated but fitted to the measured value: in the working state of February 2026, $m_P/(f^4\cdot\pi/2)$ gave 2467 GeV instead of 246 GeV, and the gap was closed with $1/10$ (Doc. 149). If the bare $v = 248.93$ GeV of the ladder is inserted instead of the measured v, the formula requires $10\,K_{\text{frak}} \approx \pi^2$ instead of 10; the factor 10 would then be a mix-up of the bare ladder value with the SI comparison value. This is a choice and open **[S]**; Doc. 387 compares the variants.

Numerical value. $v/E_P = \xi^4/(5\pi) = 2.0120\times10^{-17}$. The measured value is $v/E_P = 246.22\,\text{GeV}/1.22089\times10^{19}\,\text{GeV} = 2.0167\times10^{-17}$; the deviation is $-0.23\,\%$ **[K]**. Since v/E_P is dimensionless, this number needs no anchor.

On the value in Doc. 149. There the calculation uses $f = 7491.91$ and gives $v = 246.71$ GeV. With $f = 1/\xi = 7500$ the same formula gives $v = 245.65$ GeV **[K]**. Both values are calculated correctly; they differ in the f inserted.

ao.3 The masses without v

Inserting $v = \xi^4 E_P/(5\pi)$ into the ladder gives

$$m_i = \frac{r_i}{5\pi}\,\xi^{\,p_i+4}\,E_P \quad \textbf{[B]} \tag{ao.2}$$

With the measured value $E_P = 1.22089\times10^{22}$ MeV for comparison:

Lepton	Formula	Coefficient	Value	vs. measurement
e	$\frac{4}{15\pi}\,\xi^{11/2}E_P$	0.08488	0.5043 MeV	$-1.32\,\%$
μ	$\frac{16}{25\pi}\,\xi^{5}E_P$	0.20372	104.81 MeV	$-0.80\,\%$
τ	$\frac{5}{9\pi}\,\xi^{14/3}E_P$	0.17684	1780.9 MeV	$+0.23\,\%$

[K] The mass ratios are the same as in the ladder, since E_P and 5π cancel. The deviations combine the bare residue of the ladder against v (Doc. 352) and the $-0.23\,\%$ of the relation v/E_P.

ao.4 One chain, one anchor

The chain reads:

$$\xi \;\longrightarrow\; \frac{m_i}{E_P} = \frac{r_i}{5\pi}\,\xi^{p_i+4} \;\longrightarrow\; E_P \;\longrightarrow\; G = \frac{\hbar c^5}{E_P^2}, \quad \ell_P = \frac{\hbar c}{E_P}, \quad L_0 = \xi\,\ell_P. \tag{ao.3}$$

Apart from the units, everything is fixed by ξ. A single measured value fixes the numbers. With $m_e = 0.51099895\,\text{MeV}$ as anchor:

Quantity	from anchor m_e	vs. measurement
E_P	1.2372×10^{19} GeV	$+1.34\,\%$
G	$6.4995 \times 10^{-11}\ \text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$	$-2.6\,\%$
ℓ_P	1.5950×10^{-35} m	$-1.32\,\%$
$L_0 = \xi\ell_P$	2.1266×10^{-39} m	$-1.32\,\%$

[K] Since $G \propto E_P^{-2}$, the relative deviation of G is twice that of E_P.

Choice of anchor. The chain can start at any point; which quantity serves as anchor is a question of accuracy, not of derivation. The deviations of the other quantities then depend on the bare residue of the chosen entry point:

Anchor	E_P	G	ℓ_P, L_0
m_e	$+1.34\,\%$	$-2.62\,\%$	$-1.32\,\%$
m_μ	$+0.81\,\%$	$-1.60\,\%$	$-0.80\,\%$
m_τ	$-0.23\,\%$	$+0.45\,\%$	$+0.23\,\%$
v	$+0.23\,\%$	$-0.46\,\%$	$-0.23\,\%$

[K] In every case exactly one measured value is used. The spread between the rows is the bare residue of the ladder; it shrinks to the extent that Doc. 352 resolves this residue.

ao.5 Relation to the G formula with C_{conv}

The formula $G = \xi^2/(4m_e) \cdot C_{\text{conv}} \cdot K_{\text{frak}}$ from Doc. 012 gives 6.6745×10^{-11}, i.e. $+0.003\,\%$. This does not contradict the $-2.6\,\%$ above: $C_{\text{conv}} = 7.783 \times 10^{-3}$ is chosen so that the SI

conversion matches the measured value of G. In this form m_e and G both enter as anchors. The one-anchor chain with m_e corresponds to $C_{\text{conv}} = 7.58 \times 10^{-3}$ **[K]**; the difference of $2.6\,\%$ is exactly the residue of the chain. C_{conv} remains the SI conversion of G; the chain only shows that it is not needed as a second measured value if the residue is accepted.

Doc. 375 calculates v/E_P along this route (v from m_e, E_P via C_{conv}) and obtains 2.0389×10^{-17}. The direct value $\xi^4/(5\pi) = 2.0120 \times 10^{-17}$ lies lower by the factor 1.0134; this is exactly the residue $-1.32\,\%$ of the electron mass in the chain **[K]**. Both routes are therefore consistent.

ao.6 References in other documents

- **Doc. 149:** At the formula for v it states that it gives 245.65 GeV with $f = 7500$ and means $v/E_P = \xi^4/(5\pi)$.
- **Doc. 180:** At ℓ_P it states that the value 1.616×10^{-35} m is the measured value used as anchor and that the one-anchor chain gives 1.595×10^{-35} m.
- **Doc. 375:** It states that v/E_P follows directly as $\xi^4/(5\pi)$, without C_{conv} and without an anchor.

ao.7 Result

In the mass calculation v is an intermediate quantity and can be eliminated: $m_i = \frac{r_i}{5\pi}\xi^{p_i+4}E_P$ **[B]**. The lepton masses, E_P, G, ℓ_P and L_0 thus hang on one chain that is fixed by ξ apart from the units. A single measured value suffices as anchor; which one is free. The deviations of 0.2 to $1.3\,\%$ in E_P are the bare residues of the ladder and of the relation v/E_P ($-0.23\,\%$), not the consequence of a missing anchor. The factor 10 in this relation is the open point of the chain **[S]**.

Check script: `2/python/Dok383_Skripte/pruef_383_ein_anker.py`, 20/20.

[1] E. Tiesinga et al., *CODATA Recommended Values of the Fundamental Physical Constants: 2022*, NIST (2024).

[2] Particle Data Group (S. Navas et al.), *Review of Particle Physics*, Phys. Rev. D 110 (2024) 030001.

Chapter ap

Doc. 384 — FFGFT in Brief — Foundations, results and status after the calculation review

Abstract

The Fundamental Fractal Geometric Field Theory (FFGFT) is first and foremost a ratio-based theory. It works in natural units, $\hbar = c = 1$, with the convention $\alpha = 1$ and a correspondingly redefined charge $e = \sqrt{4\pi}$. In this basic form there is exactly one dimensionless parameter, $\xi = 4/30000 = 1/7500$, and no fractal correction. Particles, couplings and gravitation appear as properties of a compact four-dimensional carrier $T^4/\mathbb{Z}_3$; the starting point is the time-mass duality $\tilde{T} \cdot m = 1$. The guiding idea is resonance: particles are oscillation modes, mass ratios are ratios of periods, $1/\xi = 2^2 \cdot 3 \cdot 5^4$ is a point in Euler's Tonnetz, and the fractal correction is the comma of the embedding into the continuum. The core statements are ratios: m_μ/m_e and m_τ/m_μ follow from ξ to within 0.5 and $1\,\%$, the electroweak scale relates to the Planck energy as $\xi^4/(5\pi)$, the minimal length and minimal time relate to the Planck quantities as ξ, the neutrino mass differences agree to $1\,\%$, and charge quantisation follows algebraically from the Galois structure of $\mathrm{GF}(27)^*$. The golden ratio enters through the icosahedron: the weights of the fivefold rotation stand exactly as φ^8 and $2\varphi^4$, and $m_\tau/m_e = (74 + \frac{1}{45})\varphi^8$ agrees to 3×10^{-5}, within the measurement uncertainty. Only the translation into SI needs a measured value as anchor, the factor $K_{frak} = 74/75$ and conversion quantities such as C_{conv}; this is also where the absolute values of α and G arise. This document summarises the corpus (Docs. 001–389, A series) as of 1 October 2026, after the complete recalculation. For every result it gives the formula, the numerical value, the comparison value, the status and the documents that carry it, and it states just as clearly what is set by choice, what remains open and what no longer holds. The cosmic sector (H_0, Λ) is deliberately set aside.

Status markers

[B] Algebraically proved (transformation or proof without additional assumption).
[K] Numerically confirmed (verification script, comparison with measurements).
[S] Choice, open or speculative: motivated but not derived.
[X] Refuted or excluded. After a negated sentence the marker applies to the negated claim, not to the sentence.
[Q] Convention or adopted quantity.

The corrected state is authoritative: the correction register Doc. 190 up to R150 and release v1.4.4, in which every corrected passage carries a dated note with the previous value. Comparison values: CODATA 2022, PDG 2024. All numbers in this document are recomputed by the verification script `pruef_384_zusammenfassung.py`.

ap.1 What FFGFT is

Ratios in natural units

The core of FFGFT is ratios. In natural units, mass, energy, temperature and inverse time are one quantity in four guises (A085), and every physical statement can be written as a dimensionless ratio: mass ratios, scale ratios to the Planck quantities, mixing angles, charge ratios. Such ratios do not depend on any choice of units and carry the content of the theory. Absolute values in SI are a second layer: they need a measured anchor and conversion quantities, and corrections appear there that do not occur in the ratios (A080, A130, Doc. 122). This summary follows that order: first the basic form and the ratios, then the translation into SI.

Architecture in four layers

FFGFT rests on three explicitly chosen basic assumptions (R56, Doc. 365): the time-mass duality $\tilde{T} \cdot m = 1$, the compact carrier T^4 and the identification with order 3, $\mathbb{Z}_3$. A fourth, motivated structural choice is the lattice D_4; Doc. 380 shows that among the comparison carriers tested with a $\mathbb{Z}_3$ action that has fixed points, D_4 has the lowest lattice energy **[K]**, but the choice remains a choice **[S]**. In addition there is exactly one dimensionless parameter, $\xi = 4/30000$. The fine-structure constant is set to $\alpha = 1$ by adjusting the charge unit; this is a convention, not a result **[Q]**. Galois fields, Hodge theory and representation theory serve as consistency checks, not as additional inputs (Doc. 365, layer 4). The spacetime dimension $3 + 1$ is an empirical input (P34).

Resonance as the guiding idea: Euler's Tonnetz

Behind all individual results stands one idea: whatever exists needs resonance. A particle is an oscillation mode on the compact carrier with integer winding numbers (resonance condition, Doc. 189); through $\tilde{T} \cdot m = 1$ every mass becomes a frequency and every mass

ratio a ratio of periods. Ratios of periods are characters of cyclic groups, and the $\mathbb{Z}_3$ reduction turns them into arithmetic in GF(3^k): harmonics and Galois classification are not analogous but two representations of the same object (Doc. 358, Theorem D **[B]**). What stands above is thus the geometry that possesses resonances; it fixes all ratios. The measured value is the concert pitch: it says *where* the structure sounds, not *how* (Doc. 310; on the anchor, Section 4).

The model is Euler's Tonnetz (1739), the lattice of just intonation with the primes 2, 3, 5 as axes. It carries the theory at three points:

- **ξ is a lattice point.** $1/\xi = 7500 = 2^2 \cdot 3 \cdot 5^4$ lies in the 5-limit space at the coordinates $(-2, -1, -4)$ (Doc. 316 **[K]**). Seven of the nine Yukawa prefactors are 5-limit ratios like the intervals of just intonation (Doc. 060); the two exceptions, the primes 13 and 7, are in the Galois reading the harmonic primes of depth $k = 3$ and $k = 6$, and all nine lie in the grid $\{2, 3, 5, 7, 11, 13\}$ (Doc. 358, Theorem B **[B]**).
- **The mass ladder is a logarithmic resonance scale.** Because $\xi \ll 1$, the exponent axis is a musical scale: a step $\Delta p = 1/3$ is the fixed interval $\xi^{1/3} \approx 1/19.6$, the "third" of the particle spectrum (Doc. 189).
- **The comma is the embedding error.** Euler's spiral of fifths and thirds never closes, because $2^x 3^y 5^z = 1$ holds only for $x = y = z = 0$; commas remain **[Q]**. In a finite field, by contrast, every circle of fifths closes exactly, and a comma is impossible there (Doc. 358, Theorem A **[B]**). The musical comma and the fractal correction are therefore not a property of the Galois structure but of the embedding into the continuum; this is why K_{frak} appears only in the SI translation. The ξ cycle itself is defined rolled up, with the rational step $1/75$, and closes after exactly 75 turns (Doc. 315 **[B]**); the rotation number $74/75$ is the same value as K_{frak}.

The same structure recurs in coupling: undercritical, critical and overcritical coupling of two circuits correspond exactly to the 2×2 mixing of two states (Doc. 328 **[B]**); consonance is coupling geometry. The limits are recorded just as clearly: the harmonic reading is a representation, not an additional input; a Tonnetz with a finite limit covers only finitely many primes (Doc. 316), and hit rates of harmonic assignments with wide tolerance windows are a coverage effect, not evidence (Doc. 343).

Guidelines

Five rules determine how results are to be read:

1. **Ratios are what counts.** An absolute value in SI tests the conversion, not the structure (A130).
2. **FFGFT works with ξ alone.** Measured values from SI and the Standard Model serve only for conversion and comparison, and for that exactly one measured value is needed as anchor (Doc. 383).
3. **Integer structural choices are choices.** $\mathbb{Z}_3$, D_4, the winding numbers or the exponent 10 in the cosmic sector are motivated choices, not calibrations.
4. **The cosmic sector is set aside.** H_0 and Λ are deliberately excluded from the derivation of constants (P39): the Standard Model does not derive them either, and none of the

frameworks compared determines the ratio of Planck length to Hubble length from first principles.

5. **Explanation before precision.** What FFGFT yields are explanations: why masses stand in fixed ratios, why charge is quantised, why there are three generations, why Koide's $Q = 2/3$ holds, why the Born rule is a geometric quotient. The numbers test these explanations; they do not replace the calculations of the Standard Model.

What the corpus no longer claims

The calculation review has withdrawn several earlier claims:

- FFGFT is not a precision extension of the Standard Model. Where the Standard Model calculates, FFGFT gives the same values or uses the measured ones as anchors; a measurable ξ correction that would be more precise does not exist (R142, Doc. 382). Where it has free parameters, FFGFT supplies structure at the per-cent to per-mille level, not at measurement precision.
- FFGFT is not "postulate-free": $\tilde{T} \cdot m = 1$, T^4, $\mathbb{Z}_3$, D_4 and ξ are chosen.
- Precision tables of older documents in which prefactors had been fitted to measured values are not evidence (R138).
- The agreement of the SI value of α is a property of the chosen anchor, not an independent prediction (R135).
- The precision of G to $0.003\,\%$ is not evidence in its own right (R141).
- A ξ effect in Bell experiments or quantum gates that is measurable with present-day hardware does not exist (R142, R143).

The older documents are corrected accordingly; release v1.4.4 records the earlier wording.

ap.2 The basic form

Natural units, $\alpha = 1$ and the charge

With $\hbar = c = \varepsilon_0 = 1$ (Heaviside–Lorentz) what remains of the fine-structure constant is the relation $e^2 = 4\pi\alpha$ **[B]**. The basic form sets $\alpha = 1$ **[Q]** and hence $e = \sqrt{4\pi}$: the charge is measured by the full surface of the sphere (A267). α does not disappear in the process; it moves into the charge unit,

$$\alpha = \left(\frac{e_{\text{hist}}}{e_{\text{geo}}}\right)^2 = \frac{r_e}{\lambda_C} \quad \textbf{[B]}, \tag{ap.1}$$

the ratio of the Coulomb and Compton spheres of the electron (Doc. 386). Geometrically, $\alpha = 1$ means that both spheres coincide, and with them the Bohr radius. In every calculation α stays on record through $e^2 = 4\pi\alpha$, one factor $\sqrt{\alpha}$ per vertex (A267). The bridge between α and ξ runs through the energy scale $E_0 = \sqrt{m_e m_\mu}$, the geometric mean of the two lightest charged leptons; in the basic form it reads

$$\xi E_0^2 = 1, \qquad E_0^2 = \frac{1}{\xi} = 7500 \quad \textbf{[B]}. \tag{ap.2}$$

This fixes the T0 energy unit and leaves no room for a correction factor.

Time-mass duality

In natural units the intrinsic time scale and the mass of a state are reciprocal:

$$\tilde{T} \cdot m = 1, \qquad \tilde{T} = \frac{\hbar}{mc^2} \text{ (SI)}. \tag{ap.3}$$

The relation describes every state as a snapshot, not as a law of motion (Doc. 365). A position-dependent mass in a gravitational potential is equivalent to a position-dependent time; "mass constant, space curved" and "space flat, mass variable" are two representations of the same situation (Doc. 078 **[K]**). For clocks and redshift the clock rate suffices; light also requires the fractal path lengthening, which carries the spatial half (Docs. 308, 389). These are representations, not ontological statements; mixed forms are admissible as long as clock rate and path together give $\gamma = 1$. On the carrier the duality holds mode by mode exactly as a position relation $\lambda_4 \cdot m = 2\pi$ (Doc. 314 **[K]**). Reciprocity gives $E \to \infty \Leftrightarrow T \to 0$; since time is bounded from below by $T_0 = \xi t_P$, the horizon is an extremum and not a singularity (Doc. 306).

The parameter ξ

$$\xi = \frac{4}{30000} = \frac{1}{7500} = 1.\overline{3} \times 10^{-4}. \tag{ap.4}$$

ξ is the only dimensionless parameter. It has several consistent readings, but none produces the number without an input:

- **geometrically** as the number 4 of lattice directions of D_4 over $3 \cdot 10^4$ (Docs. 004, 320);
- **as a Galois number** via

$$\frac{(r_\mu/r_e)^2}{\xi} = |\mathrm{GF}(9)^*|^2 \cdot 5^2 \cdot |\mathrm{GF}(27)| = 8^2 \cdot 25 \cdot 27 = 43200 \tag{ap.5}$$

  (Docs. 336, 338), which is arithmetically exact **[B]** but depends on the chosen prefactors r_i;
- **spectrally** as the smallest eigenvalue of the D_4 sub-operator, $\xi = \lambda_{\min}(\hat{F}_{D_4})$ (Docs. 322, 327), which is well defined provided the D_4 sector lies in the $\mathbb{Z}_3$-invariant sector **[B]**, but whose value is inserted, not derived.

Overall, ξ is a motivated choice **[S]**: in the words of Doc. 365, not completely free, but not yet completely derived either.

Check via the Higgs geometry

The Higgs geometry serves as a check, and it too is a pure ratio: with the self-coupling $\lambda_h = m_h^2/(2v^2) = 0.129$ one has $\xi_{\mathrm{EFT}} = \lambda_h^2 v^2/(16\pi^3 m_h^2) = m_h^2/(64\pi^3 v^2) = 1.30 \times 10^{-4}$; the expression depends only on m_h/v and lies about $2.3\,\%$ below $4/30000$ (PDG 2024; 2.2–$2.6\,\%$ depending on rounding; Docs. 354, A190, 385) **[K]**; the deviation is interpreted structurally **[S]** (R59). This is a consistency check to within a few per cent, not an exact derivation. The route goes back to the original vacuum formula of 2025 with ε_0; written out it is $\xi_{\mathrm{EFT}} \cdot \alpha$, and with $\alpha = 1$ the same expression (Doc. 385) **[B]**.

Scales as ratios

All scales of the theory stand in fixed ratios to one another. The minimal length, minimal time and maximal energy relate to the Planck quantities as

$$\frac{L_0}{\ell_P} = \frac{T_0}{t_P} = \xi, \qquad \frac{E_{\max}}{E_P} = \frac{1}{\xi} = 7500. \tag{ap.6}$$

The exponent 1 is derived **[B]** (R90): L_0 is the Compton range of the time-field mediator with mass $m_T = M_{\mathrm{basis}}/\xi$; the choice $M_{\mathrm{basis}} = m_P$ in the fundamental sector is open **[S]**. L_0 makes the number of levels of the fractal operator finite and hence $\hat{F}$ bounded **[B]** (Doc. 327). The electroweak scale relates to the Planck energy, and the lepton masses via the ladder (Section 3), likewise in pure ξ ratios (Docs. 149, 383):

$$\frac{v}{E_P} = \frac{\xi^4}{5\pi} = 2.012 \times 10^{-17}\ (-0.23\,\%), \qquad \frac{m_i}{E_P} = \frac{r_i}{5\pi}\,\xi^{p_i+4} \quad \textbf{[B]}. \tag{ap.7}$$

With the measured E_P the lepton masses come out as 0.504, 104.8 and $1780.9\,\mathrm{MeV}$ (-1.32, -0.80, $+0.22\,\%$) **[K]**. The factor 10 in $v = E_P/(f^4 \cdot \frac{\pi}{2} \cdot 10)$ with $f = 1/\xi$ is fitted to the measured value. If the bare $v = 248.93\,\mathrm{GeV}$ of the ladder is inserted instead of the measured v, the formula requires $10\,K_{\mathrm{frak}} \approx \pi^2$ instead of 10; the factor 10 would then be a mix-up of the bare ladder value with the SI comparison value **[S]** (Doc. 387).

On notation: here f is the base winding number $f = 1/\xi = 7500$, not a frequency. In natural units f is only another reading of the duality $T \cdot m = 1$: $f = t_P/t_0 = \Lambda_{\mathrm{T0}}/E_P$ with $t_0 = \xi t_P$ and $\Lambda_{\mathrm{T0}} = E_P/\xi$, and $\xi E_0^2 = 1$ gives $f = E_0^2$. As a dimensionless ratio, f cannot be set to 1 with justification (Doc. A135). Cf. Doc. 190, R147.

Time structure and recursion

The time structure runs step by step:

$$\xi_{n+1} = \xi_n\,(1 - 100\,\xi_n), \qquad \xi_0 = \tfrac{1}{7500}, \qquad \xi_n \approx \frac{1}{100(n+75)}, \qquad D(k) = \sum_{n<k} 100\,\xi_n \approx \ln\!\Big(1 + \frac{k}{75}\Big). \tag{ap.8}$$

In the frozen case with constant defect $d = 1/75$ the winding closes after exactly 75 turns; the rotation number is $74/75 = 1 - 100\xi$ **[B]**. This is the geometric origin of the value that appears as K_{frak} in the translation into SI (Section 4). In the running case the time winding becomes a logarithmic spiral with self-similarity ratio e, a continuous scale invariance with

a pole at $n = -75$ (Doc. 295 **[B]**/**[K]**). The increments $d_n = 100\,\xi_n$ decrease monotonically from $d_1 = 1/75$; d_1 is the largest, not the smallest increment, and the lower time bound is $T_0 = \xi t_P$ (Docs. 306, 307). Doc. 381 shows the telescoping product $\prod(1 - 100\xi_k) = \xi_n/\xi_0$ exactly **[B]**, a second order $1/\xi_n \approx 100(n+75) + 100\ln((n+75)/75)$ that is 50 to about 1000 times more accurate **[K]**, and the constant $\psi'(75) = 0.01342$ between mass and time defect **[K]**. The assignment to a one-loop β function depends on the step convention **[S]**. The number 100 itself is chosen, not derived (Doc. 133, A040).

Geometry, Hilbert space, spectral theory

The state space is $\mathscr{H} = L^2(T^4) \otimes \mathbb{C}^3$; the translation between T^4 and $\mathscr{H}$ is bijective and lossless (Docs. 230, 270, 330). The torus is flat: the curvature vanishes pointwise via the metric descent **[B]** (R95). The spectrum separates the lattices: $\mathbb{Z}^4$ begins at $|k|^2 = 1$ with degeneracy 8, D_4 at $|k|^2 = 2$ with 24 (Doc. 314 **[K]**).

The fundamental fractal operator $\hat{F} = \bigoplus_n S_n \circ L_n$ with $S_n = r_n U_n$ and $r_n \in (0,1)$ is bounded **[B]**. The $\mathbb{Z}_3$ pairing makes it normal; it is self-adjoint on the $\mathbb{Z}_3$-invariant sector P_0 **[B]**, with deficiency indices $(0,0)$ and exactly one realisation. On the full space only the real part $(\hat{F} + \hat{F}^\dagger)/2$ is self-adjoint; using it is a choice **[S]** (R145, Doc. 327). The particle masses are eigenvalues of the mass operator $\hat{M}$, not of $\hat{F}$ (Docs. 322, 330).

Field theory: Lagrangian density, mode operator, Schrödinger limit

The field theory (A075, A070, A142; Docs. 019, 129) starts from the free Lagrangian of a scalar mass field δm on the torus and a fractal correction:

$$\mathscr{L} = \tfrac{1}{2}(\partial\,\delta m)^2 + \varepsilon\,\xi\Big[\frac{f(\vartheta)}{2}(\partial\,\delta m)^2 - k^{\mu\nu}(\vartheta)\,\partial_\mu \delta m\,\partial_\nu \delta m\Big]. \tag{ap.9}$$

The geometric functions $f(\vartheta)$ and $k^{\mu\nu}(\vartheta)$ are not fixed **[S]**. The duality $\tilde{T} \cdot m = 1$ does not follow from a variational principle, since $\mathscr{L}_0 = T \cdot m - 1$ contains no derivatives; it remains a setting (Doc. 129).

$$\int \tfrac{1}{2}(\partial\,\delta m)^2\, d^4x = \tfrac{1}{2}\langle\psi|\Phi^2|\psi\rangle, \qquad \Phi = \sum_i m_i P_i \quad \textbf{[B] [K]} \tag{ap.10}$$

This bridge formula (A075) connects the Lagrangian and the mode operator: masses are eigenvalues of Φ, not parameters in the Lagrangian. This is the structural difference from the Standard Model, in which every mass enters through its own Yukawa coupling, and it is an explanation, not a statement of precision.

In FFGFT the Schrödinger equation is not fundamental. With the polar decomposition $\Phi \to \sqrt{\rho}\,e^{i\theta}$, $\psi \propto \sqrt{\rho}\,e^{i\theta}$ satisfies in the non-relativistic limit

$$i\hbar\,\partial_t\psi = -\frac{\hbar^2}{2m}\nabla^2\psi + V\psi; \tag{ap.11}$$

its imaginary and real parts are the continuity equation and the Hamilton–Jacobi equation with quantum potential. The transformation is the known Madelung decomposition **[Q]**, recomputed in the verification script **[K]**; the reading is that the equation describes the low-energy limit of the mode structure, for $E \ll E_\xi$ (A070, A075). The Dirac equation

remains through the Clifford algebra; the γ matrices are its representation, and spin appears as a phase winding (A075).

For gravitation all matter fields couple conformally to the time field, $g_{\mu\nu} \to \Omega^2 g_{\mu\nu}$ with $\Omega = \mathscr{T}_0/\mathscr{T}$; the Dirac equation acquires the connection $\Gamma_\mu = \mathscr{T}^{-1}\partial_\mu \mathscr{T} = -\partial_\mu m/m$ **[K]** and turns into the standard form for constant $\mathscr{T}$ (A142). The Maxwell field is conformally invariant in four dimensions; for light the conformal coupling alone contributes nothing, it supplies the clock rate, and the spatial half comes from the path lengthening (A142, Doc. 389). Three statements of A075 and A142 were clarified: the limit $\xi \to 0$ is shown only for the scalar field, not for the full SM Lagrangian **[S]**; the modified Schrödinger equation $i\mathscr{T}\,\partial_t \Psi + \ldots = \hat{H}\Psi$ is inhomogeneous with a dimensionful $\mathscr{T}$, homogeneous with $\mathscr{T}/\mathscr{T}_0 = 1/\Omega$, and then gives the standard form for $\Omega = 1$. Its amplitude follows the local energy density, not a probability (Doc. 230), and its phase runs at the local clock rate; this is the gravitational redshift (Doc. 078). Gravitation is therefore already contained in the equation **[B][K]**, without an additional field and without a graviton: it shows itself as an effect in the redshift, read as curvature of space, as a change of mass (Doc. 078) or as fractal path lengthening (Doc. 041); a ξ addition of 2.34×10^{-10} to a_e, computed with $\alpha = 1$, and likewise the value 1.7×10^{-12} with the physical coupling $\alpha/(2\pi)$ are excluded by measurement, by 8.6σ or 10σ depending on the determination of α (Doc. 067).

Galois structure

The finite fields $\mathrm{GF}(3^n)$ are the algebraic language of the mode classification on $T^4/\mathbb{Z}_3$. One has $|\mathrm{GF}(9)^*| = 8$, $|\mathrm{GF}(27)^*| = 26 = 2 \cdot 13$, $|\mathrm{GF}(81)^*| = 80 = 2^4 \cdot 5$; $\mathrm{GF}(9)$ is not a subfield of $\mathrm{GF}(27)$, both lie in the compositum $\mathrm{GF}(729)$. The Frobenius $x \mapsto x^3$ splits $\mathrm{GF}(27)^* \cong \mathbb{Z}_2 \times \mathbb{Z}_{13}$ into two fixed points $\{+1, -1\}$ and eight three-element orbits; on $\mathbb{Z}_{13}$ four orbits arise:

$$O_1 = \{1, 3, 9\}, \quad O_2 = \{2, 5, 6\}, \quad O_3 = \{4, 10, 12\}, \quad O_4 = \{7, 8, 11\} \quad \textbf{[B]}. \tag{ap.12}$$

The 5 is the only primitive prime divisor of $3^4 - 1$, not the smallest one (Docs. 336, 338, 341). With the IPI correspondent D. Matzke only the algebra is jointly confirmed (orders in $G(3)$ and $G(6)$, Doc. 341); the physical assignments are on the FFGFT side.

ap.3 Particles: the ratios

Lepton ladder

The charged leptons follow the ladder

$$\frac{m_i}{\nu} = r_i\, \xi^{p_i}, \qquad r = \left(\tfrac{4}{3}, \tfrac{16}{5}, \tfrac{25}{9}\right), \qquad p = \left(\tfrac{3}{2}, 1, \tfrac{2}{3}\right), \qquad p_g = \tfrac{3}{2}\left(\tfrac{2}{3}\right)^{g-1}, \tag{ap.13}$$

with $r_g = n_{\phi,g}^2/n_{\theta,g}$ from the winding numbers $(3,2)$, $(5,4)$, $(9,5)$ (Docs. 006, 317, 338) and a geometric sequence of exponents with starting value $D/2$ and ratio $2/3$ (Docs. 158, 018). The mass ratios contain no scale:

$$\frac{m_\mu}{m_e} = \frac{12}{5}\xi^{-1/2} = 120\sqrt{3} = 207.85\ (+0.52\,\%), \qquad \frac{m_\tau}{m_\mu} = \frac{125}{144}\xi^{-1/3} = 16.99\ (+1.03\,\%) \quad \textbf{[K]}. \tag{ap.14}$$

Formula and ratio calculation are **[B]**, the agreement **[K]**; the r_i, p_i and winding numbers are choices **[S]** (R138). The step $1/3$ holds only from μ to τ; e lies on an orbit of its own (Doc. 322). The ladder is written without K_{frak}; in ratios it cancels exactly **[B]** (A040, A100, Docs. 070, 122).

Galois identities, Koide and the four routes

$(m_\mu/m_e)^2_{\text{ladder}} = 43200$ is arithmetically exact; against the measured value a residual of $-1.03\,\%$ remains, an empirical discrepancy **[K]** whose cause is open **[S]** (R113). The Koide relation, itself a pure ratio,

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3}, \tag{ap.15}$$

holds for the measured masses to $Q - \frac{2}{3} = -0.017\,\xi$ (PDG 2024; with $m_\tau = 1776.86\,\text{MeV}$ $-0.046\,\xi$, Doc. 352) **[K]**; as a prediction of m_τ from m_e, m_μ it lies 0.4σ from PDG, undecided. From the ladder, Koide follows only approximately ($+8\,\xi$).

In the $\mathbb{Z}_3$ circulant form $\sqrt{m_k} = c + d\cos(\theta + 2\pi k/3)$ one has $Q = \frac{2}{3} \Leftrightarrow d/c = \sqrt{2}$ **[B]**. The matrix-element route with $|A| = \sqrt{2}/3$ and $\theta = |A|^2 = 2/9$ (Doc. 367) hits $m_\tau/m_e = 3477.47$ to 3×10^{-5} **[K]** but gives $m_\mu/m_e = 206.7703$, which is 442σ from CODATA; the exact pair $(\sqrt{2}, 2/9)$ is excluded for m_μ/m_e **[X]** (Doc. 369, R121). The geometric derivation of $d/c = \sqrt{2}$ in Doc. 353 does not hold **[S]**.

Doc. 369 compares four routes to the lepton masses (ladder, Koide matrix element, Galois, φ skeleton); routes 1 and 3 are the same calculation in two languages for m_μ/m_e. The following section shows the φ route.

Icosahedron and golden ratio: the φ skeleton

FFGFT is a $\mathbb{Z}_3$ theory from the ground up; the golden ratio enters through the icosahedron. Its rotation group A_5 is the smallest finite group in which a threefold and a fivefold rotation occur together without commuting (Doc. 370). The three acts as a lattice operation on the compact carrier, the five only as a phase in the unrolled space; A_5 is incompatible with the D_4 lattice ($5 \nmid 1152$), and the two meet in the field $\text{GF}(81)$ **[B]** (Docs. 364, 370).

The fivefold rotation R_5 mixes the three $\mathbb{Z}_3$ modes; the weights $p_j = |\langle v_j | R_5 | v_e \rangle|^2$ are

$$p_0 = \frac{2}{9}, \qquad p_1 = \frac{2 + 3\varphi}{9}, \qquad p_2 = \frac{5 - 3\varphi}{9} \quad \textbf{[B]} \tag{ap.16}$$

(Doc. 293, A120). The rational weight is the Koide phase, $p_0 = \theta = 2/9$; with $3\sqrt{p_0} = \sqrt{2}$ the same matrix element also gives the value of the Koide amplitude; that the amplitude is exactly $3\sqrt{p_0}$ is not derived **[S]** (Docs. 353, 367; R150). The two weights containing φ stand exactly in the ratios

$$\frac{p_1}{p_2} = \varphi^8, \qquad \frac{p_0}{p_2} = 2\varphi^4, \qquad 3\sqrt{p_j} \in \{\sqrt{2},\ \varphi^2,\ \varphi^{-2}\} \quad \textbf{[B]} \tag{ap.17}$$

(Doc. 368). These are pure numbers from the geometry, without a cosine and without a scale. The mass ratios carry this skeleton; the most precise hit is

$$\frac{m_\tau}{m_e} = \left(74 + \frac{1}{45}\right)\varphi^8 = 3477.469 \qquad (\text{PDG } 3477.37 \pm 0.18;\ +3.0\times10^{-5},\ 0.6\sigma) \quad \textbf{[K]}. \qquad \text{(ap.18)}$$

Ratio	Formula	Value	against PDG
m_τ/m_e	$74\,\varphi^8$	3476.42	-2.7×10^{-4} (-5.3σ)
m_τ/m_e	$(74+\frac{1}{45})\,\varphi^8$	3477.469	$+3.0\times10^{-5}$ (0.6σ)
m_τ/m_e	$74\,\varphi^8(1+\frac{27}{12}\xi)$	3477.468	$+2.9\times10^{-5}$ (0.6σ)
m_τ/m_e	Koide, $\theta = 2/9$	3477.473	$+3.1\times10^{-5}$ (0.6σ)
m_μ/m_e	$30\,\varphi^4$	205.62	$-0.55\,\%$

On the φ route, m_τ/m_e is therefore as precise as on the Koide route, within the measurement uncertainty of m_τ, as a pure ratio without an anchor and without K_{frak}. The route is algebraic: since $\cos(2/9)$ is transcendental, no Koide ratio can be exactly a rational multiple of a power of φ **[B]**; the φ form approximates the same value algebraically, and the second layer ($\frac{1}{45}$ or $\frac{27}{12}\xi$) carries the remainder.

Limits of the φ route

The factors 74 and $\frac{1}{45}$ are observed, not derived from the $\mathbb{Z}_3/A_5$ framework **[S]** (Doc. 368). The 74 is the same number as in the numerator of the rotation number $74/75$ of the recursion (Doc. 381); its prime factor 37 demonstrably does not come from the icosahedron, because $\mathrm{GF}(81)$ is not a subfield of $\mathrm{GF}(3^{18})$ **[B]** (Doc. 370). For m_μ/m_e there is no φ route at this level: $30\varphi^4$ is off by $0.55\,\%$, the ladder by $0.52\,\%$, and the exact Koide pair is excluded there (above). The φ skeleton is not the earlier φ mass ladder with $m \propto \varphi^{\text{generation}}$ and $m_\mu/m_e \approx \varphi^{10}$, which is refuted **[X]** (Docs. 172, 281).

Neutrinos

Doc. 340 derives hierarchy and mixing from the four Frobenius orbits. The mass ratios are fixed,

$$m_1 : m_2 : m_3 = 1 : \sqrt{\tfrac{14}{3}} : 11, \qquad \frac{\Delta m^2_{\text{atm}}}{\Delta m^2_{\text{sol}}} = \frac{120}{11/3} = 32.7\ (-1.4\,\%) \quad \textbf{[K]}, \qquad \text{(ap.19)}$$

in normal hierarchy. Absolute values require a base mass; the assumed one is $m_\nu = \frac{\xi^2}{2} m_e = 4.54\,\text{meV}$ **[S]**. This gives $m_2 = 9.81$ and $m_3 = 49.96\,\text{meV}$, $\sum m = 64.3\,\text{meV}$, $\Delta m^2_{\text{atm}} = 120\,m_\nu^2 = 2.476\times10^{-3}\,\text{eV}^2$ ($-1.0\,\%$ against 2.500×10^{-3}, Doc. 340) and $\Delta m^2_{\text{sol}} = \frac{11}{3} m_\nu^2 = 7.565\times10^{-5}\,\text{eV}^2$ ($+0.46\,\%$) **[K]**. The reactor angle gives $\sin^2\theta_{13} = (25\xi)^{2/3} = 0.0223$ ($+1.4\,\%$) **[K]**. According to R108, ν_1 and ν_2 lie algebraically in the self-inverse Majorana layer and ν_3 is Dirac; this assignment of ν_3 to the orbit O_4 is in tension with Doc. 346, which assigns O_4 to the charged leptons, and Doc. 344 lists all neutrinos as Dirac **[S]**; that the algebraic self-inversion physically means $\nu = \bar{\nu}$ is a further assumption **[S]**. For neutrinoless double beta decay this gives $m_{ee} \approx 6.0\,\text{meV}$ (phases zero), $0.13\,\text{meV}$ in the destructive case, at most $14.4\,\text{meV}$, and exactly zero if all neutrinos are Dirac (Doc. 344). Older neutrino spectra

(Docs. 006, 007) are excluded **[X]** (R109; Doc. 007). The A_5 eight-element spectrum gives $\sin^2\theta_{23} \in \{4/9, 5/9\}$; $5/9 = 0.556$ lies close to the measured value for normal hierarchy, so maximal mixing $1/2$ is excluded (Doc. 372) **[K]**. θ_{12} is captured only in order of magnitude, the CP phase not at all **[S]**. The venting exponent $p = 5/24$ follows from the decomposition $\rho_4{\downarrow_{A_4}} = \rho_3 \oplus \rho_1$ with $|A_5 : A_4| = 5$ **[B]** (Doc. 340).

Anomalous magnetic moments

The ratio line carries the result **[K]** (Docs. 018, 158): with winding exponents $5/3$ and $4/3$, $\Delta a(\tau{-}\mu)/\Delta a(\mu{-}e) = f^{1/3} - 1 = 18.57$, and anchored to the measured a_e, a_μ one obtains

$$a_\tau = a_e + \frac{144}{125}\frac{m_\tau}{m_\mu}(a_\mu - a_e) = 1.2811 \times 10^{-3}, \tag{ap.20}$$

1.04×10^{-4} above the Standard Model value; present bounds are about 85 times wider, the decision is pending **[S]**. A fixed FFGFT addition $\Delta a_\mu = 251 \times 10^{-11}$ as the explanation of the former muon anomaly is superseded by the 2025 status; it would lie 3.4σ above today's difference **[X]** (Doc. 382).

Electroweak sector, Higgs, top

Here too the statements are ratios to the Z mass. On shell, $\sin^2\theta_W = 1 - M_W^2/M_Z^2 = 2/9$ holds as an element of the A_5 eight-element spectrum (Doc. 372) **[K]**; this gives $M_W/M_Z = \sqrt{7/9}$, $M_W = 80.42\,\text{GeV}$, 3.8σ from the 2024 world average, an explicitly stated, falsifiable tension. In the $\overline{\text{MS}}$ scheme the renormalisation running from $3/8$ gives $\sin^2\theta_W(M_Z) = 0.2308$ ($-0.19\,\%$) with two external inputs (Doc. 323) **[K]**. The ratios $m_h/M_Z = \frac{11}{8}$ ($m_h = 125.38\,\text{GeV}$, $+0.15\,\%$) and $m_t/M_Z = (\frac{11}{8})^2$ ($m_t = 172.40\,\text{GeV}$, $-0.10\,\%$) and the trace rule $M_W^2 + M_Z^2 + m_h^2 = v^2/2$ ($+0.45\,\%$) are candidates **[K]**; the origin of the factors is a reading **[S]** (Doc. 373). The chain "symmetry breaking topologically enforced" is open at two points **[S]** (Doc. 355). Earlier Higgs formulas ($m_h = v\xi^{1/4}$, Doc. 041) are refuted **[X]** (R122).

Quarks

The quarks follow the same ladder form $m_q/v = r_q\xi^{p_q}$ with topologically motivated exponents **[K]**, except for c and t, whose exponents are choices **[S]** (R138); the prefactors r_q are determined from measured values, their derivation is open **[S]** (R129, Doc. 373). The reproduction is therefore an encoding, not evidence; the table gives the values with $v = 246\,\text{GeV}$:

Quark	r_q	p_q	Ladder ($v = 246\,\text{GeV}$)	PDG 2024
u	6	$3/2$	$2.27\,\text{MeV}$	$2.16\,\text{MeV}$
d	$25/2$	$3/2$	$4.73\,\text{MeV}$	$4.70\,\text{MeV}$
s	$26/9$	1	$94.8\,\text{MeV}$	$93.5\,\text{MeV}$
c	2	$2/3$	$1.284\,\text{GeV}$	$1.273\,\text{GeV}$
b	$3/2$	$1/2$	$4.26\,\text{GeV}$	$4.18\,\text{GeV}$
t	$1/28$	$-1/3$	$172.0\,\text{GeV}$	$172.6\,\text{GeV}$

Light quark masses are scheme dependent; $\tilde{T} \cdot m = 1$ refers to pole masses. Λ_{QCD} does not follow from the ξ ladder **[X]** (R122): $v\xi^{1/3} = 12.6$ GeV, not 200 MeV. The CKM parameter $\lambda = \xi^{1/6} = 0.2260$ lies $0.46\,\%$ from the measured value **[K]**; the exponent $1/6$ is a motivated choice **[S]** (Doc. 336). The quark and hadron sector as a whole is an open bridge (Doc. 318, R76).

Charge, hypercharge, generations, colour, photon

Charge quantisation follows algebraically **[B]** (Doc. 346, R111): the QR orbits O_1, O_3 mod 13 are electrically neutral (neutrinos, antineutrinos), the NQR orbits O_4 and O_2 are charged (charged leptons, up quarks); together with N mod 3 this gives $Q \in \{0, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm 1\}$, and "neutral and coloured" is excluded. The charge ratios are thus ratios of integers as well. The down quarks have no orbit of their own. Gell-Mann–Nishijima is not fully closed: $I_3 = \pm\frac{1}{2}$ follows **[B]**, but the hypercharge of the charged leptons does not follow from the QR/NQR bit alone and is open **[S]** (R146, Doc. 347). The three elements of a Frobenius orbit give three generations **[B]** (Doc. 348); the earlier swap of generations 2 and 3 is an ordering artefact. $\mathfrak{su}(3)$ arises from the $\mathbb{Z}_3$ eigensectors **[B]**; $SU(3)$ is the smallest, not the only suitable group (Doc. 321). Photon and gluons are massless **[B]**; that masslessness means spin 1 does not follow from this and is open **[S]** (Doc. 339).

ap.4 The translation into SI

One anchor

Because all scales stand in fixed ξ ratios (Section 2), a single measured value suffices for the translation into SI (Doc. 383). With m_e as anchor one obtains $E_P = 1.237 \times 10^{19}$ GeV ($+1.34\,\%$), $G = 6.50 \times 10^{-11}$ ($-2.6\,\%$), $\ell_P = 1.595 \times 10^{-35}$ m and $L_0 = \xi \ell_P$ **[K]**. The spread depending on the choice of anchor is the known residual of the ladder, not a second anchor:

Anchor	E_P	G	ℓ_P, L_0
m_e	$+1.34\,\%$	$-2.62\,\%$	$-1.32\,\%$
m_μ	$+0.81\,\%$	$-1.60\,\%$	$-0.80\,\%$
m_τ	$-0.22\,\%$	$+0.45\,\%$	$+0.22\,\%$
v	$+0.23\,\%$	$-0.46\,\%$	$-0.23\,\%$

This fixes $L_0 = 2.155 \times 10^{-39}$ m, $T_0 = 7.19 \times 10^{-48}$ s and $E_{\mathrm{max}} = 9.16 \times 10^{22}$ GeV. With $v = 246$ GeV the absolute lepton masses lie at -1.18, -0.66 and $+0.37\,\%$.

Electroweak scale

In the mass calculation v is an intermediate quantity. Depending on the route one obtains **[K]**:

Route	v
Galois m_e from Doc. 338 (R104)	248.3 GeV
measured m_e	248.9 GeV
measured m_e and K_{frak}	245.6 GeV
$\xi^4 E_P/(5\pi)$	245.65 GeV

The comparison value $v = 246.22\,\text{GeV}$ is a Standard Model definition from G_F **[Q]** (R112); a convention-free comparison would only be possible via the muon lifetime τ_μ, which is open **[S]**.

Fractal correction

$$K_{\text{frak}} = 1 - 100\,\xi = \frac{74}{75} = 0.98\overline{6}. \tag{ap.21}$$

The factor appears only in the translation into SI: in ratios it cancels exactly **[B]** (A040, Docs. 070, 122), and in the basic form $\xi E_0^2 = 1$ there is no room for it. Its value comes from the geometry, as the rotation number of the frozen recursion (Section 2). It is confirmed **[K]** (R135): comparing the two routes to E_0 with the measured α requires $K = \xi m_e m_\mu \alpha^{-1} = 0.98650$, i.e. $n = 101.2$ instead of 100 in $1 - n\xi$. This factor is exactly the square of the bare reference energy in MeV (Section 4.4, Doc. 386, A130). The residual of 1.7×10^{-4} is not explained by measurement uncertainties and is open **[S]**. The *form* of the factor (additive $1 - 100\xi$ or multiplicative $(1-\xi)^{100}$) is open again **[S]** (R139): the available witnesses do not separate the two forms more sharply than their difference of 8.8×10^{-5}. Equivalently, K_{frak} can be written in power form $(D/3)^{D/2} = 74/75$ with $D_f^{\text{eff}} = 2.97303$ **[K]** (Doc. 381); the spatial box dimension $D_f = 3 - \xi = 2.999867$ is a choice **[S]** that does not follow from the 100-step recursion (note in Doc. 322).

E_0 and the SI value of α

In SI, E_0 is the geometric mean of the two light lepton masses, bare and with the fractal correction (R72):

$$E_0^{\text{bare}} = \sqrt{m_e m_\mu} = 7.348\ \text{MeV}, \qquad E_0 = \sqrt{\frac{m_e m_\mu}{K_{\text{frak}}}} = 7.397\ \text{MeV}. \tag{ap.22}$$

With $\alpha_{\text{SI}} = \xi (E_0/1\,\text{MeV})^2$ one obtains without correction $\alpha^{-1} = 7500/E_0^2 = 138.9$, with correction 137.06, 1.7×10^{-4} from the measured value. The Galois form $\alpha^{-1} = 3700/27 = 137.037$ ($+7.6$ ppm, Doc. 338, R102) follows from $(E_0^{\text{bare}})^2 = m_e m_\mu \approx 54\ \text{MeV}^2 = |\text{GF}(3)^*| \cdot |\text{GF}(27)|\ \text{MeV}^2$ together with $K_{\text{frak}} = 74/75$. Both agreements are properties of the anchor E_0, not independent predictions **[K]** (R135).

The SI form requires the reference energy $E_0\sqrt{\xi/\alpha} = 0.99992\,\text{MeV}$ (bare $0.99323\,\text{MeV}$, which is $\sqrt{K_{\text{frak}}}\,\text{MeV}$). The MeV is not a geometric quantity but a unit of the SI chain **[Q]**: its numerical factor is the numerical value of the historical charge e, i.e. of the same unit in which α hides. If the correction is placed on this side, the side of the Coulomb sphere, the masses and the Compton sphere stay bare, and $1\,\text{MeV}$ results to within 8×10^{-5}; this

assignment is a choice **[S]**, the residual remains open (Doc. 386). Third routes to E_0 and closed α formulas from ξ alone are withdrawn **[X]**.

Gravitational constant

$$G_{\mathrm{SI}} = \frac{\xi^2}{4m_e}\, C_{\mathrm{conv}}\, K_{\mathrm{frak}}, \qquad C_{\mathrm{conv}} = 7.783 \times 10^{-3}, \quad K_{\mathrm{frak}} = 0.986. \tag{ap.23}$$

G is not an independent constant: its form and order of magnitude follow from ξ and the one anchor **[K]** (Docs. 012, 180, 383). C_{conv} is the SI conversion of the one-anchor chain; its corpus value additionally contains the small residual of this chain (chain value 7.58×10^{-3}). The agreement with CODATA to $0.003\,\%$ arises only with this residual and is not precision evidence in its own right (R141); it holds with the rounded corpus value $K_{\mathrm{frak}} = 0.986$ of the G formula, with $74/75$ it would be $+0.07\,\%$. Explaining the residual structurally is open **[S]**. From $\tilde{T} \cdot m = 1$ one has $m_{\mathrm{char}} \cdot G = \xi^2/4$ for the characteristic mass of the anchor (R58, A142); for arbitrary m the relation holds only if G runs along (Doc. 350). A mass singularity with finite G is excluded **[B]** (Doc. 350), while physical validity in the black-hole regime is open **[S]**.

ap.5 Gravitation and cosmos

Black holes

Equating the Compton and Schwarzschild radii gives, purely geometrically,

$$M_{\mathrm{coll}} = \frac{m_P}{\sqrt{2}}, \qquad r_s = \sqrt{2}\,\ell_P, \qquad S_{\mathrm{BH}}(M_{\mathrm{coll}}) = 2\pi\ \mathrm{nat} = \frac{2\pi}{\ln 2}\ \mathrm{bit} = 9.065\ \mathrm{bit} \quad \textbf{[B]} \tag{ap.24}$$

(Docs. 325, 329). S_{BH} equals Matzke's length in Planck units; Matzke's threshold is smaller by $\sqrt{2}$, $n_{\mathrm{thr}} = 6.41$ (R144). As a universal bit number n_{thr} is refuted **[X]** (R88). The ratio of sector to thermal information is mass independent, $\ln 3 \approx 1.099$ **[B]**. The Hawking temperature corresponds to the bit energy $\hbar c/L$ at the horizon scale **[K]**; only modes with triality 0 can be emitted, and confinement and Hawking selection are one principle **[B]** (Doc. 325). Greybody factors and back-reaction are open.

Casimir and CMB

The Casimir energy density between ideal plates is $-\pi^2\hbar c/(720\,d^4)$; the form with 240 is the pressure. At the length L_ξ the ratio to the CMB density, $\pi^2/(720\xi) = 102.8$, is an identity, because L_ξ itself is determined from the CMB density and ξ; it is not measurement evidence (R136). The QFT Casimir energy does not supply a factor $4/3$ **[X]** (Doc. 371). The temperature $k_B T_{\mathrm{CMB}} = \frac{16}{9}\xi(1 - \frac{275}{4}\xi)$ eV matches only in the unit eV and with an underived coefficient **[S]** (R137); the eV brings the electron mass along as a second anchor. In the basic form T_{CMB}/m_e depends on a single number. The candidate $T_{\mathrm{CMB}}/m_e = (8\pi)^{1/4}\xi^{5/2}$ meets the FIRAS value to 0.12σ, together with the H_0 form gives $T_{\mathrm{CMB}}^4 = 16\, H_0 m_e^3$, and fixes L_ξ without the CMB; the prefactor is not derived **[S]** (Doc. 388). The factor 3 in front of the peak sequence $\{1, 6, 14, 26\}$ is a choice; the sequence itself is read off the data (R139).

Cosmic sector: deliberately set aside

The structural form $H_0/\nu_e = \pi^2\xi^{10} = 1.75\times10^{-38}$ gives $H_0 = 66.8$ km/s/Mpc; the exponent 10 is a motivated choice (P20), so the statement is conditional and not a prediction. Equally conditional are $\Lambda^* = 1/R_H^2$ and the recasting of the 10^{123} question as a structural question (Doc. 312). Excluded are the Λ formulas of older documents, $H_0 = \xi c$, $z = \xi x$ and $z \approx \xi \ln(d/\ell_P)$ **[X]** (R137; Doc. 156). The redshift $1+z = e^{\xi x}$ is achromatic **[K]**, but the length in x is not specified from ξ; the supernova time dilation is $1+z$ for all z **[B]** (R128). Supernovae, BAO and the CMB temperature are degenerate between expansion and intrinsic time unfolding; the decision is theoretical, not empirical (Doc. 267).

Galaxies

The acceleration scale $a_0 = c^2\xi^{10}/(4\bar{\lambda}_e) = 1.03\times10^{-10}$ m/s^2 lies $14\,\%$ below the empirical value and inherits the choice P20. The gas-dominated dwarf galaxy DDO 154 matches to $0.2\,\%$ **[K]**; for spiral galaxies deviations of about $19\,\%$ remain open. “Bullet Cluster passed” is not supported: with baryonic masses the peak lies at the gas (note in Doc. 308). Light deflection reproduces the GR value $1.75''$ **[K]**: clock rate and fractal path lengthening contribute $0.875''$ each, together they give $\gamma = 1$ (Doc. 308); the purely conformally flat metric of Doc. 143 would give 0 (Doc. 143). With the logarithmic form $\ln(m/m_0) = \Phi/c^2$ one also has $\beta = 1$, and the perihelion shift is that of GR **[K]**; the form itself is a setting **[S]** (Doc. 389).

ap.6 Quantum mechanics

In quantum mechanics FFGFT is deterministic. The rules and numerical values of standard quantum mechanics remain; their interpretation changes at four points:

- **Measurement is a pole transition, not a collapse.** The interaction with the apparatus drives the state point to one of the poles; which one is fixed by the apparatus variable λ on the same axis, $A(z,\lambda) = \mathrm{sgn}(z-\lambda)$. Every single measurement is thus determined, even when λ is unknown (Docs. 230, 175) **[K]**.
- **The Born rule is a geometric quotient.** $(1+z)/2 = |\alpha|^2$ is the share of the axis $[-1,1]$ whose λ values lead to the pole $+1$ **[K]**; by Archimedes it is at the same time an area fraction of the Bloch sphere **[B]**. It appears as a frequency only in series of measurements with uncontrolled λ. Randomness is epistemic, not ontological (Docs. 365, 367, 370); its closest relative in the literature is Aerts’ hidden-measurement model **[Q]**.
- **Gates are geometric rotations.** The three Pauli matrices correspond to three rotations on the Bloch cylinder; CNOT is a geometric conditioning in which the rotation of one cylinder point depends on the z value of the other. Since $\{H, T, \mathrm{CNOT}\}$ is universal, every gate can be reproduced deterministically and without repetition, up to the noise limit (Docs. 230, 175) **[K]**.
- **Noise is disordered phase, and the limit applies to both evaluations.** A spectrum with fixed amplitudes and unresolved phases looks like white noise although every component is fixed, and with known phases it can be recomputed exactly **[K]**; vacuum noise can thus be a superposition of deterministic harmonics **[S]**. The process is

deterministic and can be recomputed up to the noise limit: in the feedback the phase error grows with every step and reaches the limit after about $\pi/(2\varepsilon)$ steps (Doc. 183) **[S]**. Beyond it, outcomes can no longer be distinguished. The deterministic reading needs no repeated measurements, since every single outcome is fixed. The probabilistic evaluation, however, gets no further: averaging over many runs runs into the same floor, whose natural scale is ξ (Doc. 343, Theorem G′) **[S]**.

Bell and Kochen–Specker exclude only local or non-contextual hidden variables **[Q]**. The FFGFT measurement is contextual, because the determination lies in the interaction, and for several qubits the correlation is a property of the closed geometry, not a local quantity in $\mathbb{R}^3$ (Doc. 230, R126). Operationally, FFGFT and standard quantum mechanics give the same predictions up to a ξ correction.

The Bell violation remains: the fractal damping reduces S only by an amount of order ξ, $\Delta S/S \approx 10^{-5}$ to 10^{-4} depending on the form, and establishes no local bound (R142). The only documented hardware measurement (IBM, 28 May 2026, Doc. 147, 2048 shots per run):

Device	Runs	$\langle S\rangle$	Standard error	Share of $2\sqrt{2}$
ibm_kingston	15	2.7396	0.0071	96.9 %
ibm_marrakesh	10	2.7016	0.0138	95.5 %
pooled	25	2.7244	0.0078	96.3 %

The Bell violation is confirmed **[K]**; the gap to $2\sqrt{2}$ (0.088 on ibm_kingston, 0.104 pooled) is decoherence, 160 to several thousand times larger than the ξ effect, and two identical chips differ by 0.038. The ξ effect cannot be resolved with present-day hardware **[S]**.

CHSH values given earlier from alleged runs in 2025 and a ξ value fitted to them are unsupported and removed (R143). The deviation of the π gate by $100\xi = 1.33\,\%$ is absorbed by the pulse calibration of real qubits and is not directly measurable (R142).

Shor in the deterministic reading. Shor's algorithm splits into the preparation, a modular exponentiation with $O(n^3)$ gates and almost the entire cost (share $1-1/(n+1)$, 99.95 % for 2048 bits **[K]**), and the quantum Fourier transform with $O(n^2)$ gates, a wave operation that extracts the period as a resonance (Doc. 176). In theory it factorises in $O((\log N)^3)$ **[K]**; a general exponential quantum advantage, however, is not supported (Doc. 176). The advantage lies solely in the period finding. Preparation and error correction cost more on quantum hardware than classically: the preparation by a fixed factor, the error correction by an overhead that, according to the threshold theorem, grows only slowly. The advantage would take effect only for numbers for which the hardware is lacking today (RSA-2048: 8.6×10^9 gates, 4.1×10^6 qubits **[K]**); in the reachable range a classical computer factorises just as fast. As an apparatus the quantum computer is an analogue computer with quantum-mechanical precision: a qubit is a two-level resonator, every gate a resonant drive, and error correction averages over syndrome rounds. This averaging runs into the same resolution floor as every evaluation; that it falls below it at scale is what the threshold theorem claims, but it has not been shown **[S]**. Whether a net advantage remains is thus decided by the error correction alone. If ξ is the floor, the limit of a resonance-based computer lies at $N_{\max} \approx 1/\xi = 7500$, about 13 bits, and the time advantage vanishes structurally **[S]** (Doc. 343, Theorem G′). There are alternatives for the computational operations themselves: the Fourier transform runs optically or in any resonator (Doc. 176), analogue matrix folding and resonator locking compute

with resonances instead of gates (Doc. 186), as do resonance computers such as the BAW Ising machine (Doc. 362). They are subject to the same resolution floor. The calculations of standard quantum mechanics remain usable; FFGFT interprets them as resonance processes. What is withdrawn is only the earlier claim that a separate T0 or ξ factorisation method yields results distinguishable from Shor's (R65); ξ as a line width of phase estimation is refuted for the preregistered setup, the line remains open (P44).

ap.7 Testable statements

Quantity	FFGFT	Measured / bound	Status
Ratios (basic form)			
m_μ/m_e	207.85	206.77 (+0.52 %)	[K], r_i [S]
m_τ/m_μ	16.99	16.82 (+1.03 %)	[K]
Koide Q	$2/3$	$-0.017\,\xi$	[K]
m_τ from Koide	1776.97 MeV	1776.93(9), 0.4σ	open (footnote 1)
m_τ/m_e (φ skeleton)	$(74+\frac{1}{45})\varphi^8 = 3477.47$	3477.37(18), 0.6σ	φ^8 [B], factors [S]
v/E_P	$\xi^4/(5\pi)$	−0.23 %	[K], factor 10 fitted
ξ from m_h/v	1.30×10^{-4}	$4/30000$ (−2.3 %)	check [K]
$\Delta m^2_{\text{atm}}/\Delta m^2_{\text{sol}}$	32.7	33.2 (−1.4 %)	[K]
$\sin^2\theta_{13}$	0.0223	0.0220 (+1.4 %)	[K]
$\sin^2\theta_{23}$	$5/9$	normal hierarchy	[K]
a_τ	1.2811×10^{-3}	bound 85 times wider	open (footnote 2)
M_W/M_Z	$\sqrt{7/9}$, 80.42 GeV	3.8σ (world average)	tension
m_h/M_Z	$11/8$, 125.38 GeV	125.20 (+0.15 %)	candidate [K]
m_t/M_Z	$(11/8)^2$, 172.40 GeV	172.57 (−0.10 %)	candidate [K]
λ_{CKM}	$\xi^{1/6} = 0.2260$	0.2250 (+0.46 %)	[K], exponent [S]
CHSH deviation	$\Delta S/S \sim 10^{-5}$–10^{-4}	hardware noise $\sim 10^{-2}$	not resolvable
π gate	1.33 %	in pulse calibration	not measurable
Absolute values (SI translation)			
Δm^2_{atm}	2.476×10^{-3} eV^2	2.500×10^{-3} (−1.0 %)	[K]
Δm^2_{sol}	7.565×10^{-5} eV^2	+0.46 %	[K]
m_{ee} ($0\nu\beta\beta$)	6.0 / 0.13 / 0 meV	nEXO 5–20 meV	[S], nEXO decides
$\sum m_\nu$	64.3 meV	cosmological, model dependent	[K]
α^{-1}	137.06	137.036 ($+1.7\times10^{-4}$)	property of the anchor
τ_μ without G_F, v	–	MuLan	open [S]

Footnote 1. Koide with the measured m_e and m_μ gives $m_\tau = 1776.97$ MeV, 0.4σ from the PDG mean 1776.93(9) MeV [K]. This does not decide the test: the two measurement methods, threshold scan (BESIII, 1776.91 MeV) and pseudo-mass (Belle II, 1777.09 MeV, with $m_{\nu_\tau} = 0$), scatter by 0.18 MeV, twice the stated uncertainty. Only when the methods agree does the value test Koide (R113, Docs. 351, 352).

Footnote 2. a_τ has not yet been measured; there is only an exclusion interval. The best bound is CMS 2024 with $-0.0042 < a_\tau < 0.0046$ (95 % CL), i.e. 0.0088 wide. The FFGFT value lies 1.04×10^{-4} above

the Standard Model value 1.1772×10^{-3}; the interval is about 85 times wider than this difference, and both values lie inside it **[K]**. The question can be decided only by a measurement about a hundred times more precise, for instance at Belle II (Doc. 382).

ap.8 What no longer holds

The following statements of older documents are refuted or withdrawn **[X]**; the documents are corrected accordingly (Doc. 190, release v1.4.4).

Foundations: ξ as necessarily derived, a Casimir factor $4/3$, the number 100 as derived, a minimal time increment d_1, self-adjointness of $\hat{F}$ on the full space, a field tower $\mathrm{GF}(3) \subset \mathrm{GF}(9) \subset \mathrm{GF}(27)$, the exclusion of fivefold symmetry in four dimensions.

Particles: precision tables with fitted prefactors as evidence, the φ mass ladder with $m \propto \varphi^{\text{generation}}$ (not the φ skeleton), the exact pair $(\sqrt{2}, 2/9)$ for m_μ/m_e, Λ_{QCD} from the ξ ladder, earlier Higgs formulas, the Higgs route as a derivation of ξ, the W and Z masses as a prediction, fermion statistics from the D_4 geometry, the neutrino spectra of Docs. 006 and 007, a fixed muon addition $\Delta a_\mu = 251 \times 10^{-11}$, Gell-Mann–Nishijima as fully closed.

Gravitation and cosmos: the precision of G as evidence in its own right, n_{thr} as a universal number, the Casimir-CMB ratio as measurement evidence, the Λ formulas, $H_0 = \xi c$ and $z = \xi x$, "Bullet Cluster passed".

Quantum mechanics: local realism or $S \le 2 + \varepsilon$, a hardware confirmation of the ξ effect, the CHSH values 2.8275 and 2.8278, a factorisation method specific to FFGFT.

ap.9 Open bridges

Doc. 190 keeps the open points in a list of bridges (as of R150). In summary:

1. **Basic form:** the origin of the number 100 and of the fitted factor 10 in v/E_P (Docs. 149, 387); $M_{\mathrm{basis}} = m_P$ (R90); self-adjointness of $\hat{F}$ on the full space (R145); an adversarial D_4 specificity test (R95); the spectral dimension $d_s = 1.86$ versus ≈ 2.
2. **SI translation:** the residual 1.7×10^{-4} between the K_{frak} required by α and $1 - 100\xi$, and the form of the factor (R135, R139); the residual of 8×10^{-5} in the reference energy 1 MeV (Doc. 386); the residual of the one-anchor chain for G (R141).
3. **Particles:** τ_μ directly from FFGFT as the only convention-free test of the weak sector (R112); the cause of the decreasing lepton residuals (R113, R121); the Koide amplitude $d/c = \sqrt{2}$ from the geometry, equivalent to the equipartition of the mass sum between the symmetric mode and the non-trivial pair (Doc. 353, R150); the factors 74 and $\frac{1}{45}$ of the φ skeleton and the question whether the icosahedral phase and the Galois order have a common root (Docs. 368, 370); the hypercharge of the charged leptons (R146) and of the Higgs doublet (R148); the quark prefactors and the hadron sector (R76, R129); K_{had} (R123); two-loop running of α_s; m_{Pl} and $\alpha_{\mathrm{em}}(M_Z)$ from ξ; spin 1 of the photon.
4. **Gravitation and cosmos:** greybody factors and back-reaction (R85); a physical Casimir signature of L_ξ (R136); T_{CMB} independent of units (R137); forward selection

of the CMB peaks and the C_ℓ projection (P29–P31); the exponent $41/4$ or 10 forward (P20); light elements without a hot early phase; Ω_{DM} model-neutrally.

5. **Method:** ξ as a line width (P44); clean-up of the H_0 language (P16, P34).

ap.10 Status balance

Status	Results (selection)
[B]	$e^2 = 4\pi\alpha$ and $\alpha = r_e/\lambda_C$; $\xi E_0^2 = 1$ in the basic form (Doc. 386); charge quantisation from the Legendre symbol (Doc. 346); $SU(2)_L$, $U(1)_Y$ and $\sin^2\theta_W\vert_{\mathrm{GUT}} = 3/8$ (R80); $\mathfrak{su}(3)$ from $\mathbb{Z}_3$ sectors (Doc. 321); three generations as a Frobenius orbit (Doc. 348); self-adjointness of $\hat{F}$ on P_0 (R145); exponent in $L_0 = \xi\ell_P$ (R90); $m_i/E_P = \frac{r_i}{5\pi}\xi^{p_i+4}$ (Doc. 383); rotation number $74/75$ and telescoping product of the recursion (Doc. 381); A_5 weights with $p_0 = \theta = 2/9$, $p_1/p_2 = \varphi^8$, $p_0/p_2 = 2\varphi^4$ (Docs. 293, 367, 368); $S_{\mathrm{BH}}(M_{\mathrm{coll}}) = 2\pi$ nat and $\ln 3$ (R144); supernova time dilation $1+z$ (R128).
[K]	Lepton ratios and Koide; m_τ/m_e from the φ skeleton to 3×10^{-5}; neutrino ratios, mass differences and θ_{13}; a_τ ratios; v/E_P; Higgs check of ξ to about 2.3 %; Higgs and top candidates; λ_{CKM}; value $K_{\mathrm{frak}} = 74/75$ and its appearance only in the SI translation; form and order of magnitude of G; Bell violation on hardware.
[S]	$\tilde{T}\cdot m = 1$, T^4, $\mathbb{Z}_3$, D_4, ξ; r_i, p_i and winding numbers; the factors 74 and $\frac{1}{45}$ in the φ skeleton; the number 100 and the form of K_{frak}; factor 10 in v/E_P ($10\,K_{\mathrm{frak}} \approx \pi^2$ with the bare v); $M_{\mathrm{basis}} = m_P$; correction on the unit side (Doc. 386); base mass m_ν; quark prefactors; hypercharge of the charged leptons and of the Higgs doublet; spin 1 of the photon; exponent 10 in the cosmic sector; CMB factor 3.
[Q]	$\hbar = c = 1$ and $\alpha = 1$ with adjusted e; $v = 246$ GeV as Standard Model definition; the anchor m_μ/m_e is QED dependent; Maxwell dynamics adopted; C_{conv} and the MeV as SI conversion.
[X]	see section “What no longer holds”.

ap.11 Method and bookkeeping

The corpus labels every statement with a status marker. Newer documents have a verification script with assertions, many of them null tests. The register Doc. 190 records corrections to older documents and open bridges; since R133 the source documents themselves are corrected; up to release v1.4.4 each passage carried a dated note with the previous value. Excluded predictions are not rewritten but annotated. From 29 September to 1 October 2026 the entire corpus was recalculated; in v1.4.4 about 235 documents carry a correction box and the A series carries notes in the text, while the cleaned documents contain only the corrected text. The consequences for the classification are recorded in R135 to R146, R148 and R150. Guard rails:

- A comparison value is not an input.
- A form $X = \xi^N$ alone says nothing (P35).

- Predictions are preregistered with fixed degrees of freedom (P44).
- Every prediction names its anchor (R113).
- An absolute value in SI tests the conversion, a ratio tests the structure (A130).

ap.12 Reading guide

Topic	Documents
Foundation	365 (four layers), 190 (register), 180/189 (chain $\xi \to L_0$), 383 (one anchor)
Units, α	A080/A085 (natural units), A130/A267 ($\alpha = 1$), 122 (ratio and absolute value), 386 (where α hides), 385 (Higgs–vacuum route), 387 (deviations)
Masses	006 (ladder), 336/338 (Galois), 351/352 (PDG, lepton residuals), 367/369 (matrix element, four routes)
Icosahedron, φ	291/292/293 (place and origin of $2/9$), 364 (A_5 and D_4), 368 (φ skeleton), 370 (why the icosahedron)
Electroweak	323 (Weinberg running), 372/373 (2/9, Higgs, Yukawa), 355 (symmetry breaking)
Charge, colour	321 ($SU(3)$), 346/347 (charge, Gell-Mann–Nishijima), 348 (generations)
Neutrinos, $g-2$	340, 343/344 (Majorana/Dirac), 158/382 (a_τ, status 2025)
Gravitation	012/013 (G, SI), 325/329 (black holes), 350, 388 (CMB temperature in the basic form), 389 (parabolic flight and representations)
Structure	314 (lattices), 322/327/330 (spectral theory), 380 (D_4), 381 (recursion)
Field theory	019/129 (Lagrangian), A075 (bridge formula, Schrödinger and Dirac limits), A070 (low-energy limit), A142 (gravitation, time field)
Time	295 (log spiral), 306/307 (reciprocity, time in state space)
Quantum mechanics	230/231 (Hilbert space, measurement), 175 (qubit state spaces, deterministic readout), 183 (noise as phase), 023 (Bell), 147 (hardware), 176 (Shor: preparation and transformation), 343 (quantum computer as analogue computer)
Resonance, harmonics	060 (musical parallels), 189 (resonance condition), 310 (resonance geometry), 315 (Euler spiral), 316 (Tonnetz), 328 (coupling regimes), 358 (harmonics and Galois)
Delimitation	309 (scale anchor), 267 (degeneracy), 377 (interfaces), 378 (peer review)

ap.13 Conclusion

FFGFT is a theory of ratios and yields explanations, not greater precision than the Standard Model. In natural units with $\alpha = 1$ it replaces the free parameters of the lepton sector by a structure with one parameter ξ and chosen integers, places lepton masses, the electroweak scale, the minimal length and the Planck scale in fixed ξ ratios, and obtains charge quantisation algebraically. Its guiding idea is resonance: particles are modes of the carrier, ξ is a point in Euler's Tonnetz, and the fractal correction is the comma of the embedding. The ratios agree to within a per cent down to per mille; through the icosahedron, m_τ/m_e carries the φ skeleton and agrees to 3×10^{-5}. The translation into SI needs one anchor, the fractal correction and conversion quantities; this is where the absolute values of α and G arise, and where a known residual lies. Open are the origin of the number 100, the origin of the fitted factor 10, the form of the fractal correction and its residual against the MeV, the hypercharge of the charged leptons, the hadron sector and everything that depends on the cosmic sector. Decisive tests in the coming years are m_{ee} (nEXO), a_τ (Belle II), the sum of neutrino masses and M_W. The corpus labels choices as choices and refutations as refutations; this summary reflects that state.

ap.14 Use of AI tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author.

Verification script: `2/python/Dok384_Skripte/pruef_384_zusammenfassung.py` — all numbers in this document recomputed.

Chapter aq

Doc. 385 — The Higgs–vacuum route to ξ

Abstract

*The first route to the relations of FFGFT ran through the vacuum and the Higgs field. In spring 2025 it produced a formula linking the Higgs quantities to the electric constant ε_0: $\lambda_h^2 v^2 e^2/(64\pi^4\varepsilon_0\hbar c\, m_h^2)$. This document traces the formula through the version history from February 2025 to today, recomputes every stage and states the current status. Written out, the vacuum formula is exactly $\xi_{EFT}\cdot\alpha$ with $\xi_{EFT}=\lambda_h^2 v^2/(16\pi^3 m_h^2)$; with the FFGFT convention $\alpha=1$ it becomes today's Higgs route. With the self-coupling $\lambda_h = m_h^2/(2v^2)$ one obtains $\xi_{EFT}=m_h^2/(64\pi^3 v^2)=1.30\times10^{-4}$, about $2.3\,\%$ below $\xi=4/30000$. The Higgs geometry is therefore a consistency check of ξ to within a few per cent **[K]**, not an exact derivation. Not sustainable are the early equating of the expression with 1, a short-lived variant with μ_0/ε_0, and the later shortcut to $1/(16\pi^3)$ **[X]**. The factor $64\pi^4$ of the early HTML versions comes from the 4π in $4\pi\varepsilon_0$.*

Status markers

- **[B]** Algebraically proven.
- **[K]** Numerically confirmed (check script, comparison with measured values).
- **[S]** Assumption, open or speculative.
- **[X]** Refuted or not sustainable.
- **[Q]** Convention.

Reference values: CODATA 2022, PDG 2024 ($m_h = 125.20$ GeV, $v = 246.22$ GeV). The sources of the early formulas are commits of the repository; the files now exist only in the version history. Check script: `pruef_385_higgs_vakuum.py`.

aq.1 Starting idea: vacuum, Higgs field and the speed of light

In February 2025 the starting point was a qualitative thesis: the vacuum is not empty, and the Higgs field together with the background radiation determines the maximal speed. The link ran through the vacuum constants,

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}, \qquad Z_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} = 376.73\ \Omega, \tag{aq.1}$$

still without a formula for a parameter (first upload of 27 March 2025, document on the Schwarzschild metric). In parallel, notes appeared expressing μ_0 and ε_0 through α, e and c, $\alpha = e^2/(4\pi\varepsilon_0\hbar c)$. These two lines, Higgs and vacuum on one side, vacuum constants and α on the other, are the origin of the route traced here.

aq.2 The chain of formulas in 2025

Date	Formula	Finding
31 Mar 2025	$\beta = \frac{\lambda_h^2 v^2}{16\pi^3 c^3} \cdot \frac{\hbar}{m_h^2} \cdot \frac{1}{r_0}$	factor $16\pi^3$ appears; not dimensionless in SI
5 Apr 2025	$\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \approx 1.33 \times 10^{-4}$	correctly evaluated 1.30×10^{-4}
6 Apr 2025	$\frac{\lambda_h^2 v^2 e^2}{64\pi^4 \varepsilon_0 \hbar c\, m_h^2} = 1$	$= \xi_{\mathrm{EFT}}\,\alpha$, not 1
18 Apr 2025	$\beta_T = \frac{\lambda_h^2 v^2}{4\pi^2 \lambda_0^2 \alpha_0}$, dimension $(\mu_0/\varepsilon_0)^k$	dimensionally inconsistent
25 May 2025	ε_0 eliminated via $e^2 = 4\pi\varepsilon_0$: $64\pi^4 \to 16\pi^3$	return to the Higgs expression
from 1 Jun 2025	$\xi = \frac{\lambda_h^2 v^2}{64\pi^4 m_h^2} = 1.04 \times 10^{-5}$	4π from $4\pi\varepsilon_0$ left over
22 Aug 2026	$\lambda_h = m_h/v \Rightarrow \xi = 1/(16\pi^3)$, iteration to $4/30000$	shortcut, not sustainable

The Higgs expression

On 31 March 2025 an expression with the Higgs quantities and the factor $16\pi^3$ appears for the first time, then as a parameter β with a radius r_0. On 5 April 2025, with $r_0 = \xi \ell_P$, natural units and $\beta_T = 1$, it becomes

$$\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2}. \tag{aq.2}$$

The document gives 1.33×10^{-4}; computed from the inputs it is 1.30×10^{-4}. The same document already contains the rewriting with $m_h^2 = 2\lambda_h v^2$: $\xi = \lambda_h/(32\pi^3)$. The expression is dimensionless and correctly formed **[B]**.

The vacuum formula

On 6 April 2025 the document connects, via natural units with $\alpha = 1$, the Higgs expression with the fine-structure constant in its SI form:

$$\beta_T \alpha_{\mathrm{EM}} \approx \frac{\xi \lambda_h^2 v^2}{16\pi^3 m_h^2} \cdot \frac{1}{\xi} \cdot \frac{e^2}{4\pi\varepsilon_0 \hbar c} = \frac{\lambda_h^2 v^2 e^2}{64\pi^4 \varepsilon_0 \hbar c\, m_h^2} = 1. \tag{aq.3}$$

The same document contains $c = 1/\sqrt{\mu_0\varepsilon_0}$, $Z_0 = \sqrt{\mu_0/\varepsilon_0}$ and $h = 2\pi\hbar\mu_0 c^2$. This is the formula in which the Higgs sector and the vacuum constants appear together for the first time.

The variant with μ_0/ε_0

On 18 April 2025 an English working draft attempts an “electromagnetic derivation” of β_T from μ_0 and ε_0: $\beta_T^{\mathrm{nat}} = \lambda_h^2 v^2/(4\pi^2 \lambda_0^2 \alpha_0)$ with the Higgs Compton length $\lambda_0 = \hbar/(m_h c)$ and a dimension factor $(\mu_0/\varepsilon_0)^k$, $k = 0$. The file was deleted two days later and never adopted.

Return and offshoot

On 25 May 2025 ε_0 is eliminated via $e^2 = 4\pi\varepsilon_0$ ($\alpha = 1$, $\hbar = c = 1$); $64\pi^4$ turns back into $16\pi^3$, and the expression is again the Higgs expression of 5 April. From 1 June 2025, however, the HTML versions and Doc. 049 contain $\xi = \lambda_h^2 v^2/(64\pi^4 m_h^2) = 1.04 \times 10^{-5}$: here ε_0 was dropped, but the 4π from $4\pi\varepsilon_0$ was not cancelled.

The shortcut of 2026

When Doc. 320 was created on 22 August 2026, $\lambda_h = m_h/v$ was inserted. Everything then reduces to $1/(16\pi^3)$, and an iteration is supposed to lead from there to the fixed point $4/30000$; A130 and Doc. 133 are cited as sources. This substitution confuses the Higgs self-coupling with a Yukawa coupling $y = m/v$; the same identification $y = \lambda_h$ already appears in Doc. 097 in 2025. A130 contains the Higgs route in no version, Doc. 133 contains no iteration.

aq.3 What the vacuum formula means

Written out, the right-hand side of (aq.3) is a product of two dimensionless factors:

$$\frac{\lambda_h^2 v^2 e^2}{64\pi^4 \varepsilon_0 \hbar c\, m_h^2} = \underbrace{\frac{\lambda_h^2 v^2}{16\pi^3 m_h^2}}_{\xi_{\mathrm{EFT}}} \cdot \underbrace{\frac{e^2}{4\pi\varepsilon_0 \hbar c}}_{\alpha} = \xi_{\mathrm{EFT}} \cdot \alpha \quad \textbf{[B]}. \tag{aq.4}$$

The formula is dimensionally correct. The vacuum constant ε_0 enters only through α, and the factor $64\pi^4 = 16\pi^3 \cdot 4\pi$ arises because the 4π from $4\pi\varepsilon_0$ moves into the denominator. In SI the expression gives $\xi_{\mathrm{EFT}}\,\alpha = 9.5 \times 10^{-7}$ **[K]**. With the FFGFT convention $\alpha = 1$, i.e. $e^2 = 4\pi\varepsilon_0\hbar c$, it is exactly ξ_{EFT} **[B]**. The Higgs–vacuum route and today’s Higgs route are therefore the same calculation: the vacuum constants disappear in natural units, and

what remains is the Higgs expression that checks ξ. Doc. 310 reads it the same way: in natural units μ_0 and ε_0 disappear, and only ξ remains.

Not sustainable is the equating with 1 in (aq.3) **[X]**. In the first step $\xi \cdot (\ldots) \cdot (1/\xi)$ is cancelled, while β_T itself contains the factor $1/\xi$. Read correctly, $\beta_T = \xi_{\mathrm{EFT}}/\xi = 0.977$, close to but not exactly 1, and $\beta_T \alpha = 0.0071$.

The variant with μ_0/ε_0 of 18 April is dimensionally inconsistent **[X]**: λ_0^{-2} has the dimension energy^2, so v^2/λ_0^2 has energy^4; with $\lambda_0 = 1/m_h$ the expression gives about 4×10^5 GeV^4, neither 1 nor ξ. The factor $(\mu_0/\varepsilon_0)^k = Z_0^{2k}$ appears only in the dimension line and vanishes for $k = 0$.

aq.4 Current status

With the Standard Model self-coupling $\lambda_h = m_h^2/(2v^2) = 0.129$, the Higgs expression becomes

$$\xi_{\mathrm{EFT}} = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} = \frac{m_h^2}{64\pi^3 v^2} = \frac{\lambda_h}{32\pi^3} = 1.30 \times 10^{-4} \quad \textbf{[K]}, \tag{aq.5}$$

about $2.3\,\%$ below $\xi = 4/30000 = 1.333 \times 10^{-4}$. It depends only on the ratio m_h/v. Depending on the inputs the gap lies between 2.2 and $2.4\,\%$; the frequently quoted $2.6\,\%$ arises only with λ_h rounded to 0.129:

Inputs	ξ_{EFT}	**Gap to** $4/30000$
$m_h = 125.20$, $v = 246.22$ GeV (PDG 2024)	1.303×10^{-4}	$-2.3\,\%$
$m_h = 125.25$, $v = 246.22$ GeV	1.304×10^{-4}	$-2.2\,\%$
$m_h = 125.1$, $v = 246.2$ GeV	1.301×10^{-4}	$-2.4\,\%$
$\lambda_h = 0.129$ rounded, $m_h = 125.1$	1.299×10^{-4}	$-2.6\,\%$

The Higgs geometry is therefore a consistency check of ξ to within a few per cent **[K]** (Doc. 354, A190, 310). The deviation is interpreted structurally **[S]** (R59): the SI values of m_h and v are scheme dependent, and fractal corrections cannot be cleanly separated from the loop structure of the Standard Model. The Higgs route is not an exact derivation of $4/30000$. It is, however, an independent check: it uses only the Higgs mass and the electroweak scale, no lepton mass and no Galois counting.

aq.5 What holds and what does not

Statement	Status	Reason
Vacuum formula $= \xi_{\text{EFT}} \cdot \alpha$	**[B]**	written out; $\alpha = e^2/(4\pi\varepsilon_0\hbar c)$
with $\alpha = 1$ equal to the Higgs expression	**[B]**	$e^2 = 4\pi\varepsilon_0\hbar c$
$\xi_{\text{EFT}} = m_h^2/(64\pi^3 v^2) = 1.30 \times 10^{-4}$, about $2.3\,\%$ below ξ	**[K]**	consistency check
Deviation structural (scheme, fractal correction)	**[S]**	R59
Vacuum formula $= 1$	**[X]**	cancellation of $1/\xi$ inconsistent
Variant with μ_0/ε_0	**[X]**	dimension energy^4
$\xi = \lambda_h^2 v^2/(64\pi^4 m_h^2) = 1.04 \times 10^{-5}$	**[X]**	4π from $4\pi\varepsilon_0$ not cancelled
$\lambda_h = m_h/v$, $\xi = 1/(16\pi^3)$, iteration to $4/30000$	**[X]**	15 times ξ; iteration runs to 0
Higgs route as exact derivation of $4/30000$	**[X]**	gap 2–3 %

aq.6 Conclusion

The Higgs–vacuum route that stood at the beginning of FFGFT holds. Its vacuum formula of April 2025 is, written out correctly, the product of the Higgs expression and the fine-structure constant; with $\alpha = 1$ it is today's Higgs route, and this route meets ξ to about $2.3\,\%$. The errors of the line of development lie in side steps: the equating with 1, a dimensionally wrong variant, an uncancelled 4π and a late shortcut. As a derivation of the exact number $4/30000$ the route is not sufficient; as an independent check of ξ from the Higgs sector it stands.

Check script: `2/python/Dok385_Skripte/pruef_385_higgs_vakuum.py` — 17/17.

Chapter ar

Doc. 386 — Where α hides — Charge unit, two spheres and ξ as an area

Abstract

FFGFT is first and foremost a ratio-based theory. It works in natural units with the convention $\alpha = 1$ and a correspondingly redefined charge; in this basic form it needs no fractal correction. This document shows that nothing is lost in doing so and that the geometry becomes visible. With $\hbar = c = \varepsilon_0 = 1$ one has $e^2 = 4\pi\alpha$; with $\alpha = 1$, $e = \sqrt{4\pi}$, and the charge is measured by the full surface of the sphere. α moves into the charge unit and is the ratio of two spheres around every charged particle, $\alpha = r_e/\lambda_C$; with $\alpha = 1$ the Coulomb sphere, the Compton sphere and the Bohr radius merge into one scale. In the basic form the FFGFT bridge reads $\xi E_0^2 = 1$ with $E_0^2 = m_e m_\mu$. This has an immediate geometric reading: ξ is the product of the Compton radii of electron and muon, $\xi = \lambda_e \lambda_\mu$, an area. The number $1/\xi = 7500 = 100 \cdot 75$ is the same one that carries the recursion, with starting value $\xi_0 = 1/7500$ and rotation number $74/75$. Only the translation into SI needs a reference energy, the MeV, and there, and only there, the fractal correction appears.

Status markers

[B] Algebraically proven.
[K] Numerically confirmed (check script, comparison with measured values).
[S] Assumption, open or speculative.
[X] Refuted or not sustainable.
[Q] Convention.

Constants: CODATA 2022. Check script: `pruef_386_alpha_geometrie.py`.

ar.1 The basic form of FFGFT

FFGFT calculates with ratios. In natural units, mass, energy, temperature and inverse time are one quantity in four guises (A085), and the coupling is adopted as a unit of calculation: $\alpha = 1$ **[Q]** (A267). The only parameter is $\xi = 4/30000 = 1/7500$. In this basic form there is no fractal correction; K_{frak}, the SI values of α and G and the fully worked-out mass tables belong to the translation into SI, not to the core (A130). The question of this document is therefore not how precisely $1/137$ is reproduced, but what $\alpha = 1$ means geometrically and which structure the basic form then shows.

ar.2 $\alpha = 1$: the charge as the full sphere

The fine-structure constant is

$$\alpha = \frac{e^2}{4\pi\varepsilon_0 \hbar c}. \tag{ar.1}$$

In natural units with $\hbar = c = \varepsilon_0 = 1$ (Heaviside–Lorentz) what remains is

$$e^2 = 4\pi\alpha \quad \textbf{[B]}. \tag{ar.2}$$

The 4π is the full surface of the sphere. By Gauss's law the charge is the flux through a closed sphere; the flux is spread over the solid angle 4π. Heaviside–Lorentz puts the 4π exactly where it belongs geometrically, into Coulomb's law, and keeps the Maxwell equations free of it (A267). Setting $\alpha = 1$ gives

$$e_{\text{geo}} = \sqrt{4\pi} = 3.545, \tag{ar.3}$$

so the charge is measured geometrically: its square is the full solid angle. With the measured α the historical charge is $e_{\text{hist}} = \sqrt{4\pi\alpha} = 0.3028$. Hence

$$\alpha = \left(\frac{e_{\text{hist}}}{e_{\text{geo}}}\right)^2, \qquad \frac{e_{\text{geo}}}{e_{\text{hist}}} = \frac{1}{\sqrt{\alpha}} = 11.706 \quad \textbf{[B]}. \tag{ar.4}$$

Under the redefinition α therefore does not disappear. It moves into the charge unit: α is the square of the ratio of the historical to the geometric charge unit. The calculation stays fully reversible, because $e^2 = 4\pi\alpha$ carries the coupling order along: one factor $\sqrt{\alpha}$ per vertex, and a rate of order α^n gets exactly the factor α^n back in the transition to SI (A267).

The same 4π appears in the Higgs–vacuum route: the factor $64\pi^4 = 16\pi^3 \cdot 4\pi$ of the 2025 vacuum formula is the 4π from $4\pi\varepsilon_0$; with $\alpha = 1$ the vacuum formula becomes the Higgs expression that checks ξ to about $2.3\,\%$ (Doc. 385). The basic form thus connects the geometry of the charge with the check of ξ from the Higgs sector.

ar.3 Two spheres around every charged particle

Two spheres can be drawn around the electron. On the first, the Coulomb energy equals the rest energy; this is the classical electron radius

$$r_e = \frac{e^2}{4\pi\varepsilon_0\, m_e c^2} = 2.818 \times 10^{-15}\ \text{m}. \tag{ar.5}$$

On the second, the quantum energy $\hbar c/r$ equals the rest energy; this is the reduced Compton wavelength

$$\lambda_C = \frac{\hbar}{m_e c} = 3.862 \times 10^{-13}\ \text{m}. \tag{ar.6}$$

Their ratio is exactly the fine-structure constant, and the Bohr radius follows with the inverse power:

$$r_e : \lambda_C : a_0 = \alpha : 1 : \alpha^{-1} \quad \textbf{[B]}. \tag{ar.7}$$

Equivalently, α is the ratio of Coulomb energy to quantum energy on the same sphere, $\left(e^2/(4\pi\varepsilon_0 r)\right)/(\hbar c/r)$, for every radius r **[K]**. The same holds for the muon and the tau: their Coulomb and Compton spheres also stand in the ratio α. α therefore does not distinguish the particles; it is a property of the charge unit, not of the mass **[B]**.

Geometrically, $\alpha = 1$ means that the Coulomb sphere, the Compton sphere and the Bohr radius merge into a single scale $1/m$. The three atomic lengths are not three independent scales but one scale with three powers of α (A267). Written out with $1/137$ this common origin stays hidden; the basic form exposes it. The SI value only says that in the historical charge unit the Coulomb sphere is 137 times smaller than the Compton sphere.

Two levels. The question about the 137 contains two different things. The first level is the normalisation: how one fixes the charge unit, ε_0, the 4π and $\hbar$. On this level "137" is only the name of the coupling. With h instead of $\hbar$ the same coupling would read $e^2/(\varepsilon_0 hc) = 2\alpha \approx 1/68.5$, without the 4π it would read $e^2/(\varepsilon_0\hbar c) = 4\pi\alpha \approx 1/10.9$, and with $\alpha = 1$ and $e = \sqrt{4\pi}$ the number disappears as a constant altogether **[Q]**. The second level is the physical scale ratio: how the electric scale relates to the mechanical scale within the same system. Examples are r_e/λ_C, the binding energy of hydrogen relative to the rest energy of the electron, $\alpha^2/2 = 2.66 \times 10^{-5}$, and the ratio of two resistances, $Z_0/R_K = 2\alpha$. Such ratios come out the same in every system of units. They also do not change when e is redefined, because both sides are rescaled by the same factor; with $\alpha = 1$ they appear in the ratio $e_{\text{hist}}/e_{\text{geo}}$ **[B]**. The two levels became mixed because α is historically defined through the charge. Since 2019 the SI has additionally moved the measured value into $\varepsilon_0 = e^2/(2\alpha hc)$, so that α appears to be a property of the charge or of the vacuum. The basic form separates the two levels: $\alpha = 1$ settles the normalisation, and the scale ratio is what a theory has to deliver. FFGFT delivers $\alpha^{-1} = 3700/27$ for it (Doc. 338, 394), 7.6 ppm from the measured value **[K]**.

ar.4 The bridge in the basic form: ξ as an area

FFGFT links α to ξ through the energy scale $E_0 = \sqrt{m_e m_\mu}$, the geometric mean of the two lightest charged leptons. In the basic form, i.e. in T0 units with $\alpha = 1$ and without fractal correction, the bridge reads

$$\xi E_0^2 = 1 \quad \Longleftrightarrow \quad E_0^2 = \frac{1}{\xi} = 7500, \qquad E_0 = 86.60 \quad \textbf{[B]}. \tag{ar.8}$$

ξ as an area. Since the Compton radii are the reciprocals of the masses, $\lambda_e = 1/m_e$ and $\lambda_\mu = 1/m_\mu$, the bridge is equivalent to

$$\xi = \frac{1}{m_e m_\mu} = \lambda_e\,\lambda_\mu \quad \textbf{[B]}. \tag{ar.9}$$

In the basic form ξ is the product of the Compton radii of electron and muon, i.e. the area of the rectangle spanned by the two Compton spheres. Because with $\alpha = 1$ the Coulomb sphere of the electron coincides with its Compton sphere, one equally has $\xi = r_e \lambda_\mu$: the Coulomb sphere of the electron times the Compton sphere of the muon **[B]**. The bridge between charge and geometry is thus a statement about two sphere radii, without a unit and without a correction.

The same number as the recursion. The number $1/\xi = 7500$ splits as $7500 = 100 \cdot 75$. Exactly these two factors carry the time structure: the recursion $\xi_{n+1} = \xi_n(1 - 100\,\xi_n)$ starts at $\xi_0 = 1/7500$, the first increment is $100\,\xi_0 = 1/75$, and in the frozen case the winding closes after exactly 75 turns with the rotation number $74/75 = 1 - 100\xi$ (Docs. 381, 384) **[B]**. The number that fixes the energy unit of the basic form through $E_0^2 = 1/\xi$ is therefore the same one that determines the winding of the time structure. The value of the factor that later appears as K_{frak} in the translation into SI is thus already present in the basic form, as geometry and not as a correction.

The T0 energy unit. Taken by itself, $\xi E_0^2 = 1$ fixes the energy unit of the basic form: one unit is $E_0/86.60 = \sqrt{\xi m_e m_\mu} = 84.85$ keV **[K]**. All statements of the basic form are ratios in this unit; the comparison with the MeV is the transition to SI.

ar.5 The translation into SI

The reference energy

In SI the ratio form becomes (Doc. 011, A130)

$$\alpha_{\text{SI}} = \xi \left(\frac{E_0}{E_{\text{ref}}} \right)^2, \qquad E_{\text{ref}} = 1 \text{ MeV}. \tag{ar.10}$$

In SI, ξE_0^2 has the dimension energy^2, so a reference energy is needed. Computed backwards, the measured value requires $E_{\text{ref}} = E_0\sqrt{\xi/\alpha} = 0.99323$ MeV with the bare $E_0 = 7.348$ MeV, which is $\sqrt{K_{\text{frak}}}$ MeV to within 8×10^{-5}, and 0.99992 MeV with the corrected $E_0 = 7.397$ MeV **[K]**. Without correction $\alpha^{-1} = 7500/E_0^2 = 138.9$, with it 137.06, 1.7×10^{-4} from the measured value. This agreement is a property of the anchor E_0 (R135), not an independent prediction **[K]**.

The area form of the bridge also has its SI counterpart: the classical electron radius times the Compton length of the muon is

$$r_e \lambda_\mu = \frac{\xi}{K_{\text{frak}}} \left(\frac{\hbar c}{1 \text{ MeV}} \right)^2 \quad (+1.7 \times 10^{-4}) \quad \textbf{[K]}, \tag{ar.11}$$

$1.4\,\%$ off without correction. The area of the basic form appears in SI in units of $(\hbar c/\text{MeV})^2$, and it is exactly there that K_{frak} enters.

Where the MeV comes from

The MeV is not a geometric quantity but a unit of the SI chain **[Q]**: 1 MeV $= 10^6 \cdot 1.602176634 \times 10^{-19}$ J. The numerical factor is the numerical value of the elementary charge in coulombs, fixed since 2019 and taken over from the old definition of the ampere with $\mu_0 = 4\pi \times$

10^{-7} N/A^2; the joule depends on the kilogram, metre and second. The MeV therefore carries the historical charge unit, exactly the unit in which α hides (Section 2). The electron mass in MeV also has a measurement path involving α: what is measured is the Rydberg energy Ry $= 13.6057$ eV, and $m_e c^2 = 2\,\mathrm{Ry}/\alpha^2$ **[B]**. The SI form of the bridge is equivalent to $m_e/\mathrm{MeV} = \sqrt{\alpha K_{\mathrm{frak}}/(\xi m_\mu/m_e)} = 0.51104$, the same residual of 8×10^{-5} **[K]**.

Where the fractal correction sits

The two spheres show on which side the correction belongs. The Compton sphere contains only the mass, i.e. the anchor. The Coulomb sphere contains e, i.e. the same historical unit from which the MeV is built. If the correction is placed on this side, the masses and the Compton spheres stay bare, and the reference energy becomes $E_0^{\mathrm{bare}}\sqrt{\xi/\alpha}/\sqrt{K_{\mathrm{frak}}} = 0.99992$ MeV. Mathematically this is the same equation as with the corrected E_0 (R72); the reading is more coherent, because K_{frak} then sits on the side of the unit **[S]**.

The two routes to α in A130 differ only by this factor. The factor required by α is

$$K_\alpha = \xi m_e m_\mu \alpha^{-1} = \left(\frac{E_{\mathrm{ref}}^{\mathrm{bare}}}{1\ \mathrm{MeV}}\right)^2 = 0.98650 \quad \textbf{[B]}, \tag{ar.12}$$

1.7×10^{-4} from $74/75$ **[K]**; this is the same equation as in A130, read differently. In ratios K_{frak} cancels (Doc. 122), and in the basic form $\xi E_0^2 = 1$ there is no room for it. In the α sector it therefore appears only in the translation into SI, as the adjustment between the T0 unit and the MeV **[K]**; its value $74/75$ comes from the geometry of the recursion (Section 4). The residual of 1.7×10^{-4} remains open (R135, R139) **[S]**.

ar.6 What holds and what is open

Statement	Status	Reason
Basic form (natural units, $\alpha = 1$)		
$e^2 = 4\pi\alpha$; with $\alpha = 1$, $e = \sqrt{4\pi}$, the full sphere	**[B]**	$\hbar = c = \varepsilon_0 = 1$
$\alpha = (e_{\text{hist}}/e_{\text{geo}})^2$, calculation reversible	**[B]**	A267
$r : \lambda_C : a_0 = \alpha : 1 : \alpha^{-1}$ for every charged particle	**[B]**	two spheres
$\alpha = 1$ as unit of calculation	**[Q]**	A267
$\xi E_0^2 = 1$, $E_0^2 = 1/\xi = 7500$	**[B]**	bridge
$\xi = \lambda_e \lambda_\mu = r_e \lambda_\mu$	**[B]**	area of two spheres
$7500 = 100 \cdot 75$: $\xi_0 = 1/7500$, rotation number $74/75$	**[B]**	recursion (Doc. 381)
T0 energy unit $\sqrt{\xi m_e m_\mu} = 84.85\,\text{keV}$	**[K]**	check script
Translation into SI		
$E_{\text{ref}} = 0.99992\,\text{MeV}$ (corrected), $0.99323\,\text{MeV}$ (bare)	**[K]**	property of the anchor (R135)
$r_e \lambda_\mu = (\xi/K_{\text{frak}})(\hbar c/\text{MeV})^2$	**[K]**	$+1.7 \times 10^{-4}$
MeV from the numerical value of e, $\mu_0 = 4\pi \times 10^{-7}$	**[Q]**	SI chain
$m_e c^2 = 2\,\text{Ry}/\alpha^2$	**[B]**	measurement path of the electron mass
correction on the unit side, Compton spheres bare	**[S]**	reading
K_{frak} in the α sector only in the SI translation	**[K]**	basic form; Doc. 122
residual 1.7×10^{-4} between K_α and $74/75$	**[S]**	R135, R139

ar.7 Conclusion

With $\alpha = 1$ the basic form of FFGFT loses nothing and gains geometry. The charge is measured by the full surface of the sphere, $e = \sqrt{4\pi}$, and α becomes the charge unit: the ratio of Coulomb to Compton sphere, the same for every charged particle. With $\alpha = 1$ the three atomic lengths merge into one scale. The bridge to ξ then reads $\xi E_0^2 = 1$, which means geometrically: ξ is the area spanned by the Compton spheres of electron and muon. The number $1/\xi = 7500 = 100 \cdot 75$ is at the same time the number of the recursion; the value $74/75$ is already present in the basic form. Only in the transition to SI does a reference energy come in, the MeV, built from the historical charge unit, and only there does the fractal correction act.

The same scale $1/m_e$ also carries gravitation through $\tilde{T} \cdot m = 1$ and ξ, $G \propto \xi^2/m_e$ (Doc. 012). In the static sector, that is for free fall, redshift and clocks, the mass reading carries this alone **[K]**. The full dynamics with light deflection, perihelion advance, frame dragging and gravitational waves is contained in the curvature representation through the field equations, whose form is enforced by Lovelock's theorem (Doc. 312); it remains open whether the structure of the carrier itself supplies the spatial half, the co-moving field and the tensor part (Doc. 308, 392) **[S]**. This gives a common scale framework for electrodynamics and gravitation, but not a completed unification.

Check script: `2/python/Dok386_Skripte/pruef_386_alpha_geometrie.py` — all numbers recomputed.

Chapter as

Doc. 387 — The deviations at a glance

Abstract

This document records all residual deviations of FFGFT in one place, each with the measured value and its uncertainty. It follows the order of the theory: first the ratios of the basic form in natural units with $\alpha = 1$, then the translation into SI. The ratios agree to within per mille to a few per cent; the measurement uncertainties are almost everywhere orders of magnitude smaller than the deviations and contribute only a minimal amount. Three correction quantities carry the residuals: the Higgs correction β_T, which already appeared in the earliest documents of 2025 as a small deviation from $\beta_T = 1$, the fractal correction $K_{frak} = 74/75$ of the SI translation, and the factor 10 in v/E_P. For each one, the document shows which variants exist, how well they fit and which connections arise.

Status markers

[B] Algebraically proven.
[K] Numerically confirmed (check script, comparison with measured values).
[S] Assumption, open or speculative.
[X] Refuted or not sustainable.
[Q] Convention.

Measured values: CODATA 2022, PDG 2024, neutrino data as in Doc. 340 with the uncertainties from PDG 2024. The column “measurement error” gives the relative uncertainty of the comparison value; where it is comparable to the deviation, the deviation is also given in standard deviations. Check script: `pruef_387_abweichungen.py`.

as.1 How to read the deviations

FFGFT is first and foremost ratio-based (Docs. 384, 386). In the basic form, in natural units with $\alpha = 1$ and a redefined charge, there is only $\xi = 1/7500$ and chosen integers; ratios test the structure. Absolute values in SI need an anchor and a reference energy and test the conversion (A130). The deviations are therefore listed separately. A deviation much larger than the measurement error is a genuine residual of the theory, not a measurement problem; this holds for almost all entries.

as.2 Ratios of the basic form

Quantity	FFGFT	Deviation	Meas. error	Status
m_μ/m_e	$120\sqrt{3} = 207.85$	$+0.52\,\%$	2×10^{-8}	**[K]**, genuine residual
m_τ/m_μ	$\frac{125}{144}\xi^{-1/3} = 16.99$	$+1.03\,\%$	5×10^{-5}	**[K]**, genuine residual
Koide $Q - \frac{2}{3}$	0	$-0.017\,\xi$	$0.038\,\xi$	0.4σ
v/E_P	$\xi^4/(5\pi)$	$-0.23\,\%$	1×10^{-5}	**[K]**, factor 10
ξ from m_h/v	$\beta_T = 0.977$	$-2.3\,\%$	1.8×10^{-3}	check **[K]**
$\Delta m^2_{\text{atm}}/\Delta m^2_{\text{sol}}$	$120/(11/3) = 32.7$	$-1.4\,\%$	$2.6\,\%$	0.5σ
$\sin^2\theta_{13}$	$(25\xi)^{2/3} = 0.0223$	$+1.4\,\%$	$3\,\%$	0.4σ
M_W/M_Z	$\sqrt{7/9}$	$+0.06\,\%$	1.7×10^{-4}	3.8σ
m_h/M_Z	$11/8$	$+0.15\,\%$	9×10^{-4}	1.7σ
m_t/M_Z	$(11/8)^2$	$-0.10\,\%$	1.7×10^{-3}	0.6σ
λ_{CKM}	$\xi^{1/6} = 0.2260$	$+0.46\,\%$	3×10^{-3}	1.5σ

The lepton ratios are the sharpest tests of the basic form. Their measurement errors are 10^{-8} and 10^{-5}, the deviations 0.5 and $1\,\%$; the residuals are thus entirely residuals of the ladder, and the measurement contributes nothing **[K]**. Their cause is open (R113). For the neutrinos, θ_{13}, m_t and λ_{CKM} the deviations lie within 1.5 standard deviations; there the measurement is the limiting factor. M_W from $\sin^2\theta_W = 2/9$, at 3.8σ, is the only stated tension in the basic form. Koide agrees within the measurement precision of m_τ.

as.3 The translation into SI

Quantity	FFGFT	Deviation	Meas. error	Status
α^{-1} with $74/75$	137.059	$+1.7\times10^{-4}$	1.5×10^{-10}	**[K]**, anchor
α^{-1} Galois	$3700/27 = 137.037$	$+7.6\times10^{-6}$	1.5×10^{-10}	**[K]**
$m_e m_\mu$ against $54\,\mathrm{MeV}^2$	54	-1.6×10^{-4}	2×10^{-8}	**[K]**
m_e (chain, measured E_P)	$\frac{4}{15\pi}\xi^{11/2}E_P$	$-1.32\,\%$	1×10^{-5}	**[K]**
m_μ (chain)	$\frac{16}{25\pi}\xi^{5}E_P$	$-0.80\,\%$	1×10^{-5}	**[K]**
m_τ (chain)	$\frac{5}{9\pi}\xi^{14/3}E_P$	$+0.22\,\%$	5×10^{-5}	**[K]**
v from the ladder, bare	$248.93\,\mathrm{GeV}$	$+1.10\,\%$	10^{-6}	**[K]**
v from the ladder with K	$245.61\,\mathrm{GeV}$	$-0.25\,\%$	10^{-6}	**[K]**
$\Delta m^2_{\mathrm{atm}}$	$2.476\times10^{-3}\,\mathrm{eV}^2$	$-1.0\,\%$	$1.1\,\%$	0.9σ
$\Delta m^2_{\mathrm{sol}}$	$7.565\times10^{-5}\,\mathrm{eV}^2$	$+0.46\,\%$	$2.4\,\%$	0.2σ

Here too the measurement errors, except for the neutrinos, are orders of magnitude smaller than the deviations. Two residuals stand out. The residual of α, 1.7×10^{-4}, is exactly the distance of the measured masses from the Galois value: $m_e m_\mu = 53.991\,\mathrm{MeV}^2$ lies 1.6×10^{-4} below $54 = |\mathrm{GF}(3)^*|\cdot|\mathrm{GF}(27)|$, and with 54, $\alpha^{-1} = 3700/27$ agrees to $7.6\,\mathrm{ppm}$ **[K]**. The residual of the electron mass in the chain, $-1.32\,\%$, equals $K_{\mathrm{frak}} - 1 = -1.33\,\%$ to within 1.6×10^{-4} **[K]**; this does not hold for the muon and the tau.

as.4 The correction quantities

β_T: the Higgs correction from the earliest documents

The earliest documents of March and April 2025 set two quantities to 1: $\alpha = 1$ and $\beta_T = 1$, on the grounds that both are conventions in natural units and that their SI values, $\alpha \approx 1/137$ and $\beta \approx 0.008$, arise only through the translation ("The consistency of $\alpha = 1$ and $\beta = 1$", version history). $\beta_T = 1$ with $r_0 = \xi\,\ell_P$ gave the Higgs formula $\xi = \lambda_h^2 v^2/(16\pi^3 m_h^2)$; the document gave $r_0 \approx \ell_P/7519$ instead of $\ell_P/7500$, i.e. already then a small deviation from $\beta_T = 1$ (Doc. 385). Computed today,

$$\beta_T = \frac{\xi_{\mathrm{EFT}}}{\xi} = \frac{m_h^2}{64\pi^3 v^2\,\xi} = 0.977 \quad \textbf{[K]}, \tag{as.1}$$

with a measurement error of $0.18\,\%$ from m_h. The deviation from $\beta_T = 1$ is therefore a genuine residual, 13 times larger than the measurement uncertainty. It depends on which v is inserted:

Input for v	v	β_T
measured (from G_F)	$246.22\,\mathrm{GeV}$	0.977
chain $\xi^4 E_P/(5\pi)$	$245.65\,\mathrm{GeV}$	0.982
ladder with m_e and K_{frak}	$245.61\,\mathrm{GeV}$	0.982
ladder with m_e, bare	$248.93\,\mathrm{GeV}$	0.956

With FFGFT's own values for v, β_T moves closer to 1 but stays at $-1.8\,\%$; with the bare v it moves away. The Higgs check remains a check of ξ to within a few per cent **[K]**. There is no derivation of the residual **[S]**. Numerically, $\beta_T = 0.9772$ lies close to $K_{\text{frak}}^{\sqrt{3}} = 0.9770$; with a measurement error of $0.18\,\%$ this is not a robust statement.

K_{frak}: the correction of the SI translation

In the basic form there is no correction factor; the bridge reads $\xi E_0^2 = 1$ (Doc. 386). In the SI translation, α requires the factor $K_\alpha = \xi m_e m_\mu \alpha^{-1} = 0.98650$. The variants compared:

Variant	K	α^{-1}	Galois	v (ladder)	Turns
$1 - 100\xi = 74/75$	0.986667	$+1.7 \times 10^{-4}$	+7.6 ppm	$-0.25\,\%$	75.0
$(1-\xi)^{100} \approx e^{-100\xi}$	0.986754	$+2.6 \times 10^{-4}$	+96 ppm	$-0.24\,\%$	75.5
$1/(1+100\xi)$	0.986842	$+3.5 \times 10^{-4}$	+190 ppm	$-0.23\,\%$	76.0
$1 - \psi'(75)$ (Doc. 381)	0.986577	$+7.8 \times 10^{-5}$	−83 ppm	$-0.26\,\%$	74.5
without correction	1	$+1.4\,\%$	$+1.4\,\%$	$+1.10\,\%$	–

Only $74/75$ closes the winding after a whole number of turns and matches the Galois form to 7.6 ppm **[K]**. $1 - \psi'(75)$ halves the α residual but loses the whole number of turns and worsens the Galois form. The multiplicative and the inverse forms fit worse everywhere than the additive one; regarding the open form (R139), the α sector thus favours $1 - 100\xi$. No variant acts on the lepton ratios; for v and G they differ only by 10^{-4}.

The factor 10 in v/E_P

$v/E_P = \xi^4/(5\pi)$ with $5\pi = (\pi/2) \cdot 10$ agrees to $-0.23\,\%$ at a measurement error of 10^{-5}. The factor 10 is fitted to the measured value: in the working state of February 2026, $m_P/(f^4 \cdot \pi/2)$ gave 2467 GeV instead of 246 GeV, and the gap of 10.02 was closed with $1/10$. Two readings were examined:

- Rationalised gravitation ($8\pi G = 1$) with the reduced Planck mass would give $2\sqrt{8\pi} = 10.03$ instead of 10. Docs. 374 and 376, however, show that the appropriate reference energy is the unreduced Planck energy; the reduced reading would be a different physical assumption. This reading does not hold **[X]**.
- Measured versus bare v: if the bare $v = 248.93$ GeV of the ladder is inserted instead of the measured $v = 246.22$ GeV, the formula requires the denominator 15.501. This is $5\pi K_{\text{frak}} = 15.4985$ or $\pi^3/2 = 15.5031$, each to within 1.5×10^{-4}. The factor would then be $10\,K_{\text{frak}} \approx \pi^2$ instead of 10. The two cannot be distinguished at this level **[S]**.

The second reading fits the separation of basic form and SI: the factor 10 would then be a mix-up of the bare value of the ladder with the SI comparison value of v. It is not a derivation.

On notation: here f is the base winding number $f = 1/\xi = 7500$, not a frequency. In natural units f is only another reading of the duality $T \cdot m = 1$: $f = t_P/t_0 = \Lambda_{\text{T0}}/E_P$ with $t_0 = \xi t_P$ and $\Lambda_{\text{T0}} = E_P/\xi$, and $\xi E_0^2 = 1$ gives $f = E_0^2$. As a dimensionless ratio, f cannot be set to 1 with justification (Doc. A135).

as.5 Connections

The residuals do not stand unconnected side by side:

1. α **and Galois.** The α residual of 1.7×10^{-4} is the distance of $m_e m_\mu$ from $54\,\mathrm{MeV}^2$. With the Galois value, $K_{\mathrm{frak}} = 74/75$ stays exact and the residual lies on the side of the masses **[K]**.
2. **Electron and** $K_{\mathbf{frak}}$**.** The electron residual of the chain is $K_{\mathrm{frak}} - 1$ to within 1.6×10^{-4}. The chain, however, contains the factor 10; as long as that is open, this agreement is not robust **[S]**.
3. **Factor 10 and** $K_{\mathbf{frak}}$**.** With the bare v, the factor 10 becomes $10\,K_{\mathrm{frak}}$; the same correction that carries E_0 and α would then also stand in v/E_P **[S]**.
4. β_T**.** The Higgs correction, at $-2.3\,\%$, is the largest and oldest. With FFGFT's own values for v it drops to $-1.8\,\%$; it shows no robust connection with K_{frak} **[S]**.

as.6 Measurement precision

Input	relative uncertainty	effect
α, m_e	1.5×10^{-10}, 3×10^{-10}	negligible
m_μ, M_Z	2×10^{-8}, 2×10^{-5}	negligible
G (hence E_P)	2×10^{-5} (1×10^{-5})	negligible
m_τ	5×10^{-5}	negligible except for Koide
m_h	9×10^{-4}	β_T to $0.18\,\%$
m_t, M_W	1.7×10^{-3}, 1.7×10^{-4}	limiting
neutrino data, θ_{13}, λ_{CKM}	1 to $3\,\%$	limiting

For the lepton masses, α, v/E_P and the SI chain the measurement errors contribute only a minimal amount; the residuals there are residuals of the theory. For m_h, m_t, M_W, the neutrinos and the mixing angles the measurement is the limiting factor, and the deviations are given there in standard deviations.

as.7 Conclusion

The ratios of the basic form agree to within per mille to a few per cent, and where the measurement is precise, the residuals are genuine residuals of the theory. Three correction quantities carry them. β_T is the oldest: the earliest documents already set $\beta_T = 1$ alongside $\alpha = 1$ and saw a small deviation; today it is 0.977, and 0.982 with FFGFT's own v. $K_{\mathrm{frak}} = 74/75$ is the correction of the SI translation; among the variants it is the only one with a whole number of turns, and it matches the Galois form to 7.6 ppm. The factor 10 in v/E_P is fitted; with the bare v of the ladder it becomes $10\,K_{\mathrm{frak}} \approx \pi^2$. Open are the cause of the lepton residuals, the residual of β_T and the origin of the factor in v/E_P.

Check script: `2/python/Dok387_Skripte/pruef_387_abweichungen.py` — 27/27.

Chapter at

Doc. 388 — The CMB temperature in the basic form

Abstract

The Casimir-CMB connection is an identity, because the length L_ξ is fixed solely through the CMB temperature (R136). This document collects all routes meant to obtain T_{CMB} from ξ without this circularity and recalculates them in the basic form: natural units with $\alpha = 1$, a single anchor m_e. The eV relation $T_{\mathrm{CMB}} = \frac{16}{9}\xi$ eV with second-order correction agrees to 2×10^{-6}, but holds only in the unit eV, whose size was fixed in 1881 and set exactly in 2019. In the basic form, T_{CMB}/m_e depends on a single number. The candidate $T_{\mathrm{CMB}}/m_e = (8\pi)^{1/4}\,\xi^{5/2}$ meets the FIRAS value to 0.12σ, together with the corpus form of H_0 gives $T_{\mathrm{CMB}}^4 = 16\,H_0\,m_e^3$, and makes L_ξ independent of the CMB. The H_0 form with exponent 10 fits it; the older one with $41/4$ is off by 11σ. Neither the prefactor nor the exponent is derived; what remains open is listed at the end.

Status markers

- **[B]** Proven algebraically.
- **[K]** Confirmed numerically (verification script, comparison with measured values).
- **[S]** Setting, open or speculative.
- **[X]** Refuted or not viable.
- **[Q]** Convention.

Measured values: $T_{\mathrm{CMB}} = 2.72548 \pm 0.00057$ K (FIRAS, Fixsen 2009; relative uncertainty $0.021\,\%$), CODATA 2022. Verification script: `pruef_388_cmb_grundform.py`.

at.1 Starting point

The length $L_\xi = (15\xi/\pi^2)^{1/4}\,\hbar c/(k_B T_{\mathrm{CMB}}) = 100.24\,\mu$m is chosen such that the photon density of the CMB equals ξ/L_ξ^4 there. The comparison with the Casimir energy density at the

same length is therefore an identity, not a measurement (Doc. 279, R136). An H_0 anchor does not change this either, since L_ξ depends only on T_{CMB}.

The ΛCDM model does not derive T_{CMB} either. It takes the FIRAS value as a fixed input, and the Planck value $H_0 = 67.4$ km/s/Mpc is a derived quantity that presupposes it. The question is therefore not whether FFGFT is worse off than ΛCDM, but whether it can do more: T_{CMB} as a ratio from ξ, with the same anchor as the masses.

at.2 The eV relation

Since 2025 the corpus has carried the relation $T_{\mathrm{CMB}} = \frac{16}{9}\xi$ eV, later with a second-order correction (P11):

$$k_B T_{\mathrm{CMB}} = \tfrac{16}{9}\,\xi\left(1 - \tfrac{275}{4}\xi\right)\ \mathrm{eV}. \tag{at.1}$$

In SI, the first order gives 3.7978×10^{-23} J, i.e. 2.7507 K and $+0.93\,\%$; the second order gives 2.72549 K and $+0.0002\,\%$ **[K]**. The agreement is real, its meaning is not: three observations show that it depends on the unit.

First, the relation holds only in eV. Read $\frac{16}{9}\xi$ in meV, in kelvin or in units of m_e, and it is off by factors of 10^3, 10^4 or 5×10^5 **[K]**. Written as a ratio it reads $T_{\mathrm{CMB}}/m_e = \frac{16}{9}\xi/510\,999$; the number $510\,999$ is m_e in eV, i.e. the anchor itself, not an expression in ξ.

Second, redefining α changes nothing. With $\alpha = 1$ the charge unit grows by $1/\sqrt{\alpha} = 11.706$ and the potential shrinks by the same factor; energies stay the same **[B]**. T_{CMB} contains no charge, so α can neither create nor remove the relation.

Third, the eV is set, not measured **[Q]**. Its size goes back to the practical volt of 1881 (10^8 electromagnetic CGS units, chosen after the voltage of the Daniell cell), and thus to the centimetre, gram and second. Every later stage – the international volt of 1908, absolute units in 1948, the Josephson constant in 1990 – kept this size. Since 20 May 2019, with the fixed elementary charge, exactly 1 eV $= 1.602176634 \times 10^{-19}$ J (SI Brochure, 9th ed.). The CMB was discovered in 1965; it cannot have entered the size of the eV.

Then there is the coefficient. Exactly required would be 68.766; $275/4 = 68.75$ is the nearest fraction with denominator 4. Fractions $n/4$ lie 3.3×10^{-5} apart in T, so a fitting one can always be found to within $\pm 1.7 \times 10^{-5}$. The second order lies 0.01σ from the central value. Beyond the measurement precision an agreement proves nothing; a correct formula would typically lie about 1σ off, and it comes this close only with a probability of about $1\,\%$. This suggests that the coefficient was chosen at the central value. The classification of P11 in R137 (i) stands: unit-dependent agreement, coefficient not derived **[S]**.

For the bookkeeping this means: a formula of the form "quantity = number times eV" uses the unit eV as an anchor. That is allowed as long as it is the only one. The mass ladder, however, uses m_e; both together require that m_e/eV follow from ξ. The only proximity is $\mathrm{eV}/m_e = 1.271\,\xi^{3/2}$, $0.17\,\%$ from $(4/\pi)\,\xi^{3/2}$ **[K]**; that would derive the size of the volt from ξ and is not viable.

at.3 Other reference energies

The basic form fixes an energy unit of its own: from $\xi E_0^2 = 1$ follows $u = \sqrt{\xi}\,E_0$, so with $E_0 = \sqrt{m_e m_\mu}$, $u = 84.85$ keV. In this unit $T_{\mathrm{CMB}} = 2.768 \times 10^{-9}\,u = 0.0876 \cdot \frac{16}{9}\xi^2\,u$ **[K]**, without a clean

prefactor. Referred to the energy scale of the atom, $T_{\mathrm{CMB}}/(\alpha^2 m_e) = 0.0647\,\xi$ **[K]**, likewise without a clean prefactor. Neither reference yields a relation.

at.4 The basic form: T_{CMB}/m_e

In natural units T_{CMB} depends on the Hubble rate, the Planck energy and the share of photons in the critical density. With $\rho_\gamma = \frac{\pi^2}{15}T^4$ and $\rho_{\mathrm{crit}} = 3H_0^2E_P^2/(8\pi)$,

$$\frac{T_{\mathrm{CMB}}}{m_e} = \Big(\frac{45\,\Omega_\gamma}{8\pi^3}\Big)^{1/4}\Big(\frac{H_0 E_P}{m_e^2}\Big)^{1/2}. \tag{at.2}$$

With the corpus forms $H_0 = \frac{\pi}{2}\xi^{10}m_e$ (Docs. 165, 279) and $m_e/E_P = \frac{4/3}{5\pi}\xi^{11/2}$ (Doc. 383), $H_0E_P/m_e^2 = \frac{15\pi^2}{8}\xi^{9/2}$ **[B]**. If Ω_γ is proportional to ξ, then $T_{\mathrm{CMB}}/m_e \propto \xi^{5/2}$. Measured:

$$\frac{T_{\mathrm{CMB}}}{m_e} = K\xi^{5/2}, \qquad K = 2.23897 \quad \textbf{[K]}. \tag{at.3}$$

This explains the exponent 2.41 read off without a prefactor. The question is thereby reduced to one number, equivalent to Ω_γ/ξ.

Simple candidates lie outside the measurement precision: $\sqrt{5}$ ($-0.13\,\%$), $9/4$ ($+0.49\,\%$), $64/(9\pi)$ ($+1.10\,\%$). Inside lies

$$\boxed{\frac{T_{\mathrm{CMB}}}{m_e} = (8\pi)^{1/4}\,\xi^{5/2}} \qquad K = 2.23903,\ +0.0025\,\%,\ 0.12\sigma \quad \textbf{[K]}. \tag{at.4}$$

8π is the coupling in $8\pi G$, which motivates the candidate but does not justify it **[S]**.

As a check against chance, all 3370 expressions $(p/q\cdot\pi^k)^{1/n}$ with $p,q\le 30$, $|k|\le 3$, $n\le 4$ in the range 1.5 to 3.5 were searched. Three fall within FIRAS; 1.6 are expected by chance **[K]**. Restricted to $p,q\le 10$ (383 expressions), only $(8\pi)^{1/4}$ remains, with 0.2 hits expected by chance. The candidate is thus the simplest, but not the only one; a coincidence is not excluded.

at.5 Consequences of the candidate

Relation to H_0. Equivalently $T_{\mathrm{CMB}}^4 = 8\pi\xi^{10}m_e^4$, with the same exponent 10 as in H_0. Together with $H_0 = \frac{\pi}{2}\xi^{10}m_e$, ξ drops out:

$$\boxed{T_{\mathrm{CMB}}^4 = 16\,H_0\,m_e^3} \qquad \textbf{[B]}. \tag{at.5}$$

From the measured T_{CMB} and m_e follows $H_0 = 66.81 \pm 0.06$ km/s/Mpc **[K]**. This lies 1.2σ from Planck (67.4 ± 0.5) and far from SH0ES. The uncertainty comes from FIRAS alone and is about eight times smaller than that of direct H_0 measurements.

Photon density. With the T0 values of H_0 and E_P, $\Omega_\gamma = \frac{4096}{10125}\,\xi = 0.4045\,\xi$ **[B]**; with the measured E_P instead of the T0 form, $0.4154\,\xi$.

L_ξ without the CMB. Inserting the candidate into the definition of L_ξ, T_{CMB} drops out:

$$L_\xi = \left(\frac{15}{8\pi^3}\right)^{1/4} \xi^{-9/4}\, \bar{\lambda}_e = 100.24\,\mu\mathrm{m} \qquad \textbf{[K]}. \tag{at.6}$$

L_ξ is thus a length built from ξ and the Compton wavelength of the electron. The ratio $\pi^2/(720\xi) = 102.8$ between the Casimir and CMB densities at L_ξ remains an identity; the circularity, however, moves from L_ξ to the prefactor $(8\pi)^{1/4}$.

at.6 The H_0 forms compared

The corpus contains two forms: $E_H = E_0\,\xi^{41/4}$ (Docs. 026, 064; exponent calibrated to the measured value, P35) and $H_0 = \frac{\pi}{2}\xi^{10} m_e$ (Docs. 165, 279). They differ by $1\,\%$, since $E_0\,\xi^{1/4} = 0.9904 \cdot \frac{\pi}{2} m_e$ **[K]**. In the H_0 measurement this goes unnoticed; T_{CMB}, however, is measured more than thirty times more precisely and separates the forms via $T^4 = 16\,H_0 m_e^3$:

H_0 form	H_0 (km/s/Mpc)	T_{CMB} (K)	Deviation
$\frac{\pi}{2}\xi^{10} m_e$	66.821	2.72555	$+0.003\,\%$, 0.1σ **[K]**
$E_0\,\xi^{41/4}$, $E_0 = 7.398$ MeV	66.179	2.71898	$-0.24\,\%$, 11σ **[X]**
$E_0\,\xi^{41/4}$, $E_0 = \sqrt{m_e m_\mu}$	65.731	2.71436	$-0.41\,\%$, 20σ **[X]**
backwards from T_{CMB}	66.81 ± 0.06	measured	–

If the candidate holds, the form with $41/4$ is excluded by the CMB, and the transition to the exponent 10 in Doc. 165 is confirmed. This confirmation is not independent: the prefactor $(8\pi)^{1/4}$ was found at the measured value, and the formula already contains ξ^{10}. What is shown is that the same ξ^{10} structure carries both H_0 and T_{CMB}, with the simple prefactors $\pi/2$ and 8π.

at.7 Comparison

Route	Anchor	Agreement	Classification
$\frac{16}{9}\xi$ eV	eV	$+0.93\,\%$	unit-dependent **[S]**
$\frac{16}{9}\xi(1 - \frac{275}{4}\xi)$ eV	eV	2×10^{-6}	unit-dependent, fitted **[S]**
T/u, $u = \sqrt{\xi} E_0$	m_e	no prefactor	no relation
$T/(\alpha^2 m_e)$	m_e	no prefactor	no relation
$\sqrt{5}$, $9/4$, $64/(9\pi)$	m_e	0.13 to $1.1\,\%$	outside FIRAS **[X]**
$(8\pi)^{1/4}\xi^{5/2}$	m_e	0.12σ	candidate **[S]**
ΛCDM	T_{CMB} itself	–	measured value as input **[Q]**

at.8 What remains open

The circularity of L_ξ is resolved only conditionally: it has been moved to one number. Open are

- the prefactor $(8\pi)^{1/4}$, equivalent to $\Omega_\gamma = \frac{4096}{10125}\xi$: motivated by $8\pi G$, not derived; with 1.6 hits expected in the wide search space, a coincidence is not excluded **[S]**;
- the exponent 10 and the prefactor $\pi/2$ of H_0 (P20); the T relation inherits them, since $5/2 = 10/4$ **[S]**;
- why the photon density today is a fixed fraction of ξ; in an expanding model this would be a coincidence of epoch, in a static one a statement that needs a derivation **[S]**;
- the eV relation P11: unit-dependent, coefficient $275/4$ fitted; it remains as a historical agreement but contributes nothing to the basic form (R137, R70) **[S]**.

The candidate can be tested through H_0: it requires $H_0 = 66.81 \pm 0.06$ km/s/Mpc. A model-independent H_0 measurement with a precision of a few tenths of a per cent would confirm or exclude it.

at.9 Conclusion

The eV relation agrees because the eV brings the electron mass along as a second anchor; in every unit that FFGFT fixes itself, the agreement disappears. In the basic form, T_{CMB}/m_e depends on a single number, and the simplest candidate, $(8\pi)^{1/4}\xi^{5/2}$, meets FIRAS to 0.12σ. It links T_{CMB} and H_0 as $T^4 = 16\,H_0 m_e^3$, fixes L_ξ without the CMB, and excludes the old H_0 form with $41/4$. It is not derived; the prefactor and, as for H_0, the exponent 10 remain open.

Verification script: `2/python/Dok388_Skripte/pruef_388_cmb_grundform.py` — 27/27.

Chapter au

Doc. 389 — Parabolic Flight, Inertia and the Representations of Gravitation

Abstract

In parabolic flight, gravity disappears for the passengers. This document starts from that question and reads it with the time-mass duality $\tilde{T} \cdot m = 1$: free fall is the path of stationary phase of a mode with position-dependent mass, and weight on the ground is the deviation of the rest world line from that path. What cannot be transformed away are the tides. Gravitation appears in three representations: as space curvature, as a change of mass or clock rate, and as fractal path lengthening. They are mathematical models, not statements about whether space is "really" flat or curved, and they are equivalent for the static, non-rotating weak field and for the conformally flat part of the geometry; how far the equivalence reaches beyond that is examined in Doc. 392. For light, clock rate and path are complementary halves: each alone gives $0.875''$ at the solar limb, both together $1.75''$ and $\gamma = 1$. The purely conformally flat metric gives no deflection. With the logarithmic form of the mass change, $\beta = 1$ follows as well, and with it the perihelion shift of GR. At the horizon, T_0 limits the comparison with a distant reference, not the free crossing. A separate quantisation of gravitation is not needed.

Status markers

- **[B]** Proved algebraically.
- **[K]** Confirmed numerically (verification script, comparison with measured values).
- **[S]** Setting, open or speculative.
- **[X]** Refuted or not viable.
- **[Q]** Convention.

Measured values: Pound–Rebka 1960, Cassini (Bertotti et al. 2003), IAU and CODATA 2022 constants. Verification script: `pruef_389_parabelflug.py`.

au.1 The question

In parabolic flight the aircraft flies a ballistic parabola for about twenty seconds. The passengers float, although the Earth attracts them as before. On the ground, by contrast, they feel their weight, although they are not moving. The question is what this weight is, if it disappears in free fall.

The answer this document works out: the force we feel as weight exists only because we deviate from the path we would take without support. The ground forces us onto a path that is accelerated exactly opposite to the path of fall. In parabolic flight this deviation drops out, and with it the weight. The following sections recalculate this in the representation of FFGFT and place it among the different readings of gravitation.

au.2 Free fall in the mass representation

In the representation “space flat, mass variable” (Doc. 078) a mode in the potential Φ carries the position-dependent mass $m(x)$. Its action is that of a clock ticking at the rate $m(x)c^2$:

$$L = -m(x)\,c^2\sqrt{1-v^2/c^2}, \qquad m(x) = m_0\,e^{\Phi(x)/c^2}. \tag{au.1}$$

Here $m(x)$ is the mass as compared from the distant reference. A clock at Φ ticks in that reference’s time at

$$\omega(\Phi) = \omega_0\sqrt{g_{00}} \approx \omega_0\Big(1+\frac{\Phi}{c^2}\Big), \qquad m(x) = m_0\sqrt{g_{00}}, \qquad g_{00} = \Big(\frac{m}{m_0}\Big)^2 = e^{2\Phi/c^2} \quad \textbf{[B]}, \tag{au.2}$$

since with $\tilde{T}\cdot m = 1$ the mass is the frequency of the mode. $m(x)$ is the Killing energy of the mode; measured with a co-moving clock the mass remains m_0. For small velocity the Euler–Lagrange equation gives

$$\ddot{\vec{x}} = -c^2\,\nabla \ln m = c^2\,\nabla \ln \lambda_4 = -\nabla\Phi \quad \textbf{[B]}, \tag{au.3}$$

with the mode length $\lambda_4 = 2\pi/m$ (Doc. 314). Without a metric and without Christoffel symbols, the fall follows from the gradient of the mass alone: bodies fall towards where their mass is smaller and their mode length larger. Since $\nabla \ln m$ is the same for every mode, all bodies fall alike; in this form the equivalence principle is built in.

At the Earth’s surface $g/c^2 = 1.09\times10^{-16}$ per metre **[K]**. This is the same number by which a clock runs faster per metre of height. In this representation, gravitational acceleration and clock rate are a single quantity.

Sign. The direction of the mass change is fixed by the redshift. Light rising in a gravitational field arrives redshifted; the Pound–Rebka experiment measured this over 22.5 m as $(2.57\pm0.26)\times10^{-15}$, with 2.46×10^{-15} expected **[K]**. In the mass representation this means: the mode below, compared from the upper reference, has the smaller mass; m grows with Φ. With this sign the body falls downwards. The sign is therefore not an open choice but determined by measurement.

au.3 Rest as a counter-path

The phase of a mode along a path is $\varphi = -\int m(x)\,c^2\,d\tau$. Free fall is the path on which this phase is stationary, $\delta\varphi = 0$. The rest world line $\vec{x} = \text{const}$ is not: its variation does not vanish but leaves the remainder

$$\left.\frac{\partial L}{\partial \vec{x}}\right|_{v=0} = -m\,\nabla\Phi = -m\,\vec{g} \quad \textbf{[B]}. \tag{au.4}$$

For a body to be at rest nonetheless, the ground must balance exactly this remainder with $+m\vec{g}$ upwards. That is the weight. It is not an additional force on top of gravity, but the measure of how far the rest world line deviates from the path of stationary phase. In the language of the mode it is a phase detuning: the mode at rest runs against the rate it would have in free fall.

In parabolic flight the aircraft follows the path of fall. The deviation vanishes, and with it the weight. What the passengers experience as heaviness on the ground and as weightlessness in the aircraft is the same path, once with and once without the counter-path. As a formula:

$$\vec{a}_{\text{fall}} = c^2\,\nabla \ln \lambda_4, \qquad \vec{a}_{\text{inert}} = -c^2\,\nabla \ln \lambda_4 = -\vec{a}_{\text{fall}} \quad \textbf{[B]}. \tag{au.5}$$

On the ground the two cancel; in parabolic flight the counter-path is zero. In this sense the inertial force exists only because we have an exactly opposite path.

au.4 What cannot be transformed away

The gradient $\nabla \ln m$ can be transformed away locally: in the freely falling frame it is zero, as are the Christoffel symbols in the local inertial frame of GR. The second derivative cannot be transformed away. The tides are the Hessian

$$\partial_i\partial_j\Phi = c^2\,\partial_i\partial_j \ln m, \qquad \text{eigenvalues } (-2,\,1,\,1)\,\frac{GM}{r^3} \text{ on the axis}, \qquad \nabla^2 \ln m = \frac{4\pi G\rho}{c^2} \quad \textbf{[B]}. \tag{au.6}$$

In vacuum the trace is zero; inside a mass it gives the Poisson equation, now as an equation for the mass field. Outside, $m = m_0\,e^{-GM/(rc^2)} \approx m_0\big(1 - GM/(rc^2)\big)$; the gradient $\partial_r \ln m = GM/(r^2c^2)$ can be transformed away, the second derivative $c^2\,\partial_r^2 \ln m = -2GM/r^3$ cannot **[B]**. At the Earth's surface $2GM/r^3 = 3.08 \times 10^{-6}\ \text{s}^{-2}$ **[K]**. This remainder persists in parabolic flight as well: two bodies at different places in the aircraft do not fall exactly alike. What the mass representation calls "curvature" is the curvature of the mass field, not that of a spatial geometry.

au.5 Three representations

A position-dependent mass is equivalent to a position-dependent time, since $\tilde{T}\cdot m = 1$. What one calls it is a choice of representation:

- **Space curvature:** mass constant, space and time curved, as in GR.

- **Change of mass or clock rate:** space flat, mass and clock rate position-dependent (Doc. 078).
- **Fractal path lengthening:** space flat, propagation paths geometrically stretched (Docs. 026, 041, 182).

These are mathematical models that must agree by virtue of the duality insofar as they yield the same observables. Ontologically, one can say neither that space is flat nor that it is curved; mixed forms are conceivable. All that matters is that the observables agree. This has been shown for the static, non-rotating weak field with $\gamma = \beta = 1$ and for the conformally flat part, to which the universe at large also belongs. Around rotating masses and for gravitational waves the flat representation must additionally supply a co-moving vector field and a tensor part; in the curvature representation both are contained through the field equations (Docs. 312, 392).

Light as the test. For clocks, redshift and slow bodies the clock rate suffices. Light tests more, because it also feels the spatial geometry. In Fermat form light propagates through a refractive index $n = 1 + (1+\gamma)\,GM/(rc^2)$; the deflection at the solar limb and the Shapiro delay scale with $(1+\gamma)/2$:

Representation	γ	Deflection at the solar limb
purely conformally flat metric (Doc. 143)	-1	0
clock rate only (Einstein 1911)	0	$0.875''$
clock rate and fractal path lengthening (Doc. 308)	1	$1.751''$

[K]. Cassini measures $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$; only the last row is compatible with the measurement. Doc. 308 computes the same by ray tracing: clock rate and path lengthening give $0.875''$ each, together $1.750''$. They are complementary halves, not alternatives.

This sharpens the three representations. “Space flat, mass variable” on its own carries only the temporal half. The flat representation becomes complete only with the fractal path lengthening, which carries the spatial half. The space curvature of GR is then equivalent to clock rate and path together. The temporal half is measured by clocks and is the same in every representation; the spatial half can be written as curvature, as path lengthening, or as a mixture of both. The freedom of representation therefore lies in the spatial half, and the condition for every mixed form is $\gamma = 1$.

The conformally flat metric $g_{\mu\nu} = \eta_{\mu\nu}(1 - 2\xi E_{\text{field}}/E_P)$ of Doc. 143 is none of these representations: it stretches space and time by the same factor, null geodesics remain unchanged, and light propagates as in flat space. This is the Nordström case (Doc. 143). The same holds for the conformal coupling $g_{\mu\nu} \to \Omega^2 g_{\mu\nu}$ of the matter fields to the time field (A142): the Maxwell field is conformally invariant in four dimensions, and the coupling leaves light unaffected **[B]**. For massive matter it supplies the clock rate, i.e. the temporal half; the spatial half has to come from the path lengthening (A142).

au.6 Second order: the perihelion shift

Light deflection tests the first order in the potential. The perihelion shift additionally tests the second order of the time component, the parameter β in $g_{00} = 1 - 2U + 2\beta U^2$

with $U = GM/(rc^2)$. The factor relative to GR is $(2 + 2\gamma - \beta)/3$. In the mass representation $g_{00} = (m/m_0)^2$, and β depends on the form of $m(\Phi)$:

Form	β	with γ	Mercury (″/cent.)
$m = m_0\, e^{\Phi/c^2}$	1	1	42.98 (GR)
$m = m_0\,(1 + \Phi/c^2)$	$\frac{1}{2}$	1	50.1
Nordström	$\frac{1}{2}$	-1	-7.2
clock rate only	1	0	14.3

[K]. The logarithmic form $\ln(m/m_0) = \Phi/c^2$ is the same one that in Section 2 gives the fall $\ddot{\vec{x}} = -\nabla\Phi$ exactly and not merely approximately. With it, and with the spatial half from Doc. 308, the flat representation gives the perihelion shift of GR. The logarithmic form itself is set, not derived from $\tilde{T} \cdot m = 1$ **[S]**; the linear form is excluded by the Mercury measurement.

au.7 The relation $\lambda_4 \cdot m = 2\pi$ holds locally

The position relation $\lambda_4 \cdot m = 2\pi$ (Doc. 314) connects proper quantities: the mode length as a co-moving rod measures it, and the mass as a co-moving clock measures it. In coordinates of the distant reference it does not hold unchanged once the spatial half is written as curvature, because coordinate lengths then carry a factor of the spatial metric. In the isotropic form of GR,

$$g_{ij} = \Big(1 - \frac{2\Phi}{c^2}\Big)\delta_{ij}, \qquad \ell_{\text{coord}} = \ell\Big(1 + \frac{\Phi}{c^2}\Big), \qquad \lambda_{4,\text{coord}} = \frac{2\pi}{m(x)} = \lambda_4\Big(1 - \frac{\Phi}{c^2}\Big) \quad \textbf{[B]}. \tag{au.7}$$

Seen from the distant reference, spatial lengths deep in the potential appear shortened and the temporal mode length lengthened: the two factors run against each other. In the purely conformally flat metric both would scale alike with $1/m$, and light would remain undeflected. The spatial factor is exactly the half that light additionally feels. In the representation with fractal path lengthening it sits in the stretched path, as a refractive contribution $n - 1 = -\Phi/c^2$ added to the clock rate. The gradient $\nabla \ln \lambda_4$ in Section 2 is accordingly a local quantity: in the freely falling frame it vanishes, in the frame at rest it is g/c^2.

au.8 Horizon: comparison and crossing

Doc. 306 describes the horizon as an extremum: there T reaches the value $T_0 = \xi t_P$ and the energy per excitation the value E_P/ξ. It adds that the “slowest time” at the horizon is a statement about the comparison, not about experience. The calculation confirms this.

A static observer at r measures, for a mode of energy E (referred to the distant reference), the local energy $E/\sqrt{1 - r_s/r}$; it diverges at the horizon **[B]**. The bound $E \le E_P/\xi$ cuts this approach off at the proper distance $\rho_{\min} = 2\, r_s\, \xi\, E/E_P$ **[B]**. A freely falling observer, by contrast, measures $E/(1 + \sqrt{r_s/r})$ for incoming light, hence $E/2$ at the horizon, which is finite **[B]**. For this observer there is nothing to bound at the horizon.

The bound T_0 therefore concerns the static reading, the comparison with a distant reference. The free crossing is not affected; this is the same distinction as between rest and fall in parabolic flight. Whether T_0 additionally bounds the radius in the interior from below requires a dynamics of the interior that the corpus does not supply. The physical validity in the black-hole regime remains open **[S]** (Doc. 384).

au.9 Two masses, two readings of G

Two masses at the upper end are easily confused:

- $M_{\text{coll}} = m_P/\sqrt{2}$: here the Schwarzschild radius equals the reduced Compton length **[B]** (Docs. 325, 329).
- $m_P/\xi = 7500\, m_P \approx 1.63 \times 10^{-4}$ kg: the dual upper bound per excitation, belonging to $L_0 = \xi \ell_P$ **[K]** (Docs. 290, 306).

Both are correct but different; their ratio is $7500\sqrt{2} \approx 10\,607$. According to Doc. 306, both bounds hold per excitation, not per aggregate: a star is not a single winding, and $\hbar/(Mc^2)$ for a stellar mass is not a state of the relation.

The same applies to G. With the measured, fixed G, $m \cdot G = \xi^2/4$ holds for the characteristic mass m_{char} of the anchor (R58, A142). For arbitrary m the relation holds only if G runs along, as Doc. 350 reads it for an evaporating hole. The statement “no mass singularity with finite G” is, in the first reading, a consequence of the definition; in the second, a statement about the dynamics.

au.10 No separate quantisation of gravitation

In FFGFT gravitation is not an additional field and needs no graviton. It is already contained in the time-field geometry and shows itself as an effect in the redshift, read as space curvature, as mass change or as fractal path lengthening. A separate quantisation of gravitation is therefore not required (R58, A142). What remains open is the transfer of the classical equations into loop corrections for situations beyond the Schwarzschild limit (A142).

au.11 What remains open

- the logarithmic form $\ln(m/m_0) = \Phi/c^2$, which gives $\beta = 1$, as a consequence of the duality rather than only as a setting **[S]**;
- the origin of the fractal path lengthening as the spatial half from the structure of the carrier; Doc. 308 gives the K3 area invariance as the reason, confirmed numerically but not proved **[K]**;
- the dynamics inside a horizon and the validity of $m \cdot G = \xi^2/4$ beyond the anchor **[S]**.

au.12 Conclusion

In parabolic flight it is not gravitation that disappears but the counter-path. Free fall is the path of stationary phase of a mode with position-dependent mass, weight on the ground is the deviation of the rest world line from that path, and only the tides persist in every frame. Space curvature, mass change and fractal path lengthening are models of the same situation, not statements about the nature of space; they are equivalent for the static, non-rotating weak field and the conformally flat part, while dragging and gravitational waves require additional parts in the flat representation (Doc. 392). For light, clock rate and path are complementary halves; only both together give $1.75''$ and $\gamma = 1$. With the logarithmic form of the mass change the perihelion shift of GR follows as well. At the horizon, the same distinction between rest and fall separates the comparison with a distant reference from the free crossing.

Verification script: `2/python/Dok389_Skripte/pruef_389_parabelflug.py` — 34/34.

Part VI

Part VI — Loose Puzzle Pieces, Correction Spacings and the Measurement Chain

Chapter av

Doc. 390 — Loose Puzzle Pieces: Kaluza-Klein and Other Established Building Blocks of the Time-Mass Duality

Abstract

Anyone who sees $\tilde{T} \cdot m = 1$ for the first time often recognises something familiar in it, most often Kaluza-Klein. That is not an objection but a finding: almost every building block of FFGFT has been in the literature for decades, scattered, however, across relativity, quantum mechanics, group theory, string theory, number theory and phenomenology. This document places eleven such building blocks: Kaluza-Klein, de Broglie's internal clock and the Compton time, the Zitterbewegung, Pauli's objection to a time operator, the relational approaches to time, Wigner's mass as a Casimir invariant, the Casimir value of $SU(3)$, Heaviside-Lorentz units, orbifold compactification, finite fields with their Frobenius automorphism, and the Koide formula. For each building block it records what it already contained, which piece was missing, and where in the corpus it fits in. The order matters: these building blocks were not the foundation of FFGFT. The theory arose independently of them from the relation $\tilde{T} \cdot m = 1$; the building blocks were found in the literature only afterwards and fit into a picture that was drawn without them. The result is a map of the places at which the theory could already have been found.

Status markers

[SETZUNG]	Motivated input, not a derivation.
[B]	Algebraically proved (verification script, exact arithmetic).
[K]	Numerically confirmed.
[S]	Structurally plausible, identification open.
[E]	Established mathematics or physics, adopted.
[Q]	Taken from the literature, with source.
[X]	Checked and negative.

The markers in this document give the status *in the corpus*; the document itself derives nothing new but places things in context. Verification script: `pruef_390_puzzleteile.py`.

av.1 The question

In the introductory conversation on the website, the sceptical physicist asks right at the start whether this does not sound like Kaluza-Klein. The answer is yes, and the question can be generalised: if the duality of mass and time was already known, why was it not thought through to the end?

For Einstein, the corpus has already answered this question. Doc. 312, chapter “Why Einstein did not see $\tilde{T} \cdot m = 1$”, shows that the ingredients were on his desk and that the missing step was not a formula but the change of status of a known formula into a principle. The present document does not repeat that account. It widens the view: not just one ingredient but almost all of them were available, each in a different field, each treated as a tool or a curiosity. The following sections go through them in three groups: the rolled-up direction, mass as structure, and the geometry with its algebra.

On the order. None of the following building blocks was a starting point of FFGFT. The theory was developed from $\tilde{T} \cdot m = 1$ and the geometry of the carrier; the literature references were looked for and found only afterwards. Where it says below that a building block fits in at some place in the corpus, this is to be read as agreement in hindsight, not as origin. This is precisely what gives the map its value: pieces that arose independently of one another and independently of FFGFT fit into the same picture.

av.2 Time and mass: the rolled-up direction

Kaluza-Klein (1921, 1926)

What it contained. Kaluza extended general relativity by a fifth dimension, and Klein rolled it up into a circle of radius R **[Q]** [1, 2]. On a circle only integer windings are allowed; from this a mode tower with $m_n \propto n/R$ arises, and momentum along the circle direction appears in four dimensions as mass. Klein thereby also obtained the quantisation of charge. So by 1926 the core statement was in place that a compact direction generates mass as the inverse of its period.

What was missing. The rolled-up direction was spatial and was added to spacetime. The radius R remained a free parameter, and the predicted tower of heavy excitations was never found. As Doc. 311 puts it, one had gained a scale and bought a parameter along with it.

Where it fits in the corpus. In the corpus, $\tilde{T}\cdot m = 1$ is explicitly the Kaluza-Klein relation of the fourth torus direction, the mass dimension S^1_m (Doc. 330, Doc. 270). The direction is timelike: what holds there for a rolled-up spatial direction holds here for rolled-up time, with mass in the role of the mode index (Doc. 307; likewise Doc. 231). The radius is not a free modulus, because $\tilde{T}\cdot m = 1$ binds the direction; Doc. 311 summarises this as "scale without modulus", and Doc. 318 sets the three routes side by side. There is no additional tower to wait for: the discrete spectrum of the compact direction already is the observed particle spectrum (Doc. 307). Locally, Lorentz invariance holds without restriction; the compactness shows only globally (Doc. 312).

De Broglie's internal clock and the Compton time (1924, 2013)

What it contained. De Broglie assigned to every particle an internal clock with frequency $\nu = mc^2/h$ **[Q]** [3]. In natural units this is $T\cdot m = 1$ in raw form. The Compton time $\hbar/(mc^2)$ has been a standard notion ever since, and in 2013 Lan, Müller and co-workers built a clock whose rate follows directly from the mass of an atom **[Q]** [4].

What was missing. The clock became the wavelength; the clock itself dropped out as a derived scale. The history of this step is told in Doc. 312 and is not repeated here.

Where it fits in the corpus. The de Broglie relation appears on the mass direction as $\lambda_4 \cdot m = 2\pi$, exact mode by mode (Doc. 314); the clock character of every mass is the starting point of the theory, not its consequence (Doc. 365).

The Zitterbewegung (1930, 1990)

What it contained. Schrödinger found in the Dirac equation a rapid oscillation of the electron's position with angular frequency $2mc^2/\hbar$ **[Q]** [5]. Hestenes interpreted it as an internal circular motion and explicitly called the electron a clock **[Q]** [6].

What was missing. The Zitterbewegung remained a peculiarity of single-particle Dirac theory, regarded in the field picture as interference of positive- and negative-energy parts; no geometric direction in which it takes place was assigned to it.

Placement. The corpus has not drawn on the Zitterbewegung so far. It belongs in this series because it shows the same reciprocity of mass and rate in another language. An identification with the circle direction S^1_m is not claimed here; the factor 2 in the frequency and the role of the negative-energy parts would first have to be clarified **[S]**.

Pauli's objection to a time operator (1933)

What it contained. Pauli showed that no self-adjoint time operator can exist for a Hamiltonian that is bounded from below **[Q]** [7]. For decades this was taken as a reason to keep time out of the state space.

What was missing. The objection presupposes open time. Compact time behaves like an angle variable; its conjugate spectrum is discrete, as for angular momentum conjugate to an angle. This gap in the argument was not exploited consistently.

Where it fits in the corpus. Doc. 307 and Doc. 334 record that Pauli's objection does not apply to the rolled-up view: for compact time the Pauli condition is not satisfied, and the discreteness of the modes becomes content rather than obstacle.

Relational and timeless approaches (1983 onwards)

What it contained. Page and Wootters obtain time from correlations in a static overall state **[Q]** [8]; the Wheeler-DeWitt equation and Rovelli's relational time proceed similarly. Kaluza-Klein variants with compact time and Bars' two-time physics have been studied **[Q]** [9].

What was missing. The mainstream takes time *out* as something fundamental. The opposite route, taking time *in* as a compact dimension and obtaining the mass modes from it, has hardly been pursued.

Where it fits in the corpus. Doc. 307 draws exactly this line: shared is the rejection of an external parameter time; different is the positive proposal, compact geometric time instead of relational or thermal time.

av.3 Mass as structure

Wigner: mass as a Casimir invariant (1939)

What it contained. Wigner classified particles as irreducible representations of the Poincaré group; mass is a Casimir label there, hence already a structural eigenvalue and not an attached number **[E]** [10].

What was missing. The classification does not say which values this label takes; the mass values remain inputs.

Where it fits in the corpus. Doc. 307 cites Wigner as the reference showing that mass as a structural quantity is well familiar; Doc. 339 uses the Wigner classification as a standard step in separating the massive and massless sectors. In FFGFT the mass values themselves come from the compact direction, not from the Poincaré group.

The Casimir value of $SU(3)$

What it contained. The quadratic Casimir operator of the fundamental representation of $SU(N)$ has the value $(N^2-1)/(2N)$, for $SU(3)$ therefore $4/3$ **[E]** [11]. The value is textbook material of colour dynamics.

What was missing. Nobody had a reason to connect this value with a mode count of a compact geometry.

Where it fits in the corpus. In Doc. 324, $\xi = C_2(SU(3)_{\text{fund}})/N_{\text{Fourier}} = (4/3)/10^4 = 4/30000$; the formula is **[B]** there, the Fourier mode count **[K]**. The same value follows in Doc. 338 from group orders of finite fields (section av.4).

Heaviside-Lorentz units (1893)

What it contained. Heaviside introduced rationalised units in which the factors 4π disappear from the field equations **[Q]** [12]. With $\hbar = c = \varepsilon_0 = 1$, $\alpha = 1$ becomes a choice of units.

What was missing. The choice of units remained a computational convenience; that $\hbar$ and c are mere conversion factors became self-evident only with the practice of natural units and the SI reform of 2019 (Doc. 312).

Where it fits in the corpus. Doc. 365 sets up the theory in this basic form: $\alpha = 1$ is not a setting but a choice of units, and the electromagnetic quantities become derived forms of mass and time. The familiar number 137 arises only in the transition to SI units; its agreement with the measured value depends on the anchor E_0 (Doc. 338).

av.4 Geometry and algebra

Orbifold compactification (1985)

What it contained. Dixon, Harvey, Vafa and Witten compactified string theory on orbifolds, tori divided by a finite group; $T^6/\mathbb{Z}_3$ was one of the first examples **[Q]** [13]. The fixed points of the group carry sectors of their own there, and the number of generations becomes a topological quantity.

What was missing. Six additional spatial dimensions and a vast number of possible compactifications; time stays outside. Doc. 311 applies the corpus's two conditions to string theory: the compact directions are exemplarily compact, all information is rebooked into conserved quantum numbers, and what remains open is what the directions carry.

Where it fits in the corpus. The carrier is $T^4/\mathbb{Z}_3$ on the D_4 lattice, four directions instead of six additional ones, with the time-mass direction among them (Doc. 330, Doc. 314). The $\mathbb{Z}_3$ acts freely on all non-zero momenta; in position space the orbifold has nine fixed points (R131 in Doc. 190). Torus, $\mathbb{Z}_3$ and D_4 are settings **[SETZUNG]** (Doc. 365).

Finite fields and the Frobenius automorphism

What it contained. The finite fields $GF(p^n)$, their cyclic multiplicative groups and the Frobenius automorphism $x \mapsto x^p$ have been fully described since Galois **[E]** [14]. They were not connected with elementary particle physics.

What was missing. A physical structure in which the prime 3 and its extensions occur naturally.

Where it fits in the corpus. The $\mathbb{Z}_3$ of the orbifold supplies this structure. The Frobenius automorphism of $GF(9)$ corresponds to the sector pairing $k \leftrightarrow -k$ **[B]** (Doc. 336). Charge quantisation follows from the Legendre symbol modulo 13 **[B]** (Doc. 346), the three generations arrange themselves in the Frobenius orbits of $GF(27)^*$ **[B]** (Doc. 348), and the identity $(r_\mu/r_e)^2/\xi = |GF(9)^*|^2 \cdot 5^2 \cdot |GF(27)| = 43200$ connects the lepton ladder with the group orders **[K]** (Doc. 338). The hypercharge of the charged leptons is still open here (R146 in Doc. 190).

The Koide formula (1981–1983)

What it contained. Koide found that the three charged lepton masses satisfy the relation $Q = (m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$ to about 10^{-5} **[Q]** [16]. For more than forty years the formula remained unexplained, a curiosity of phenomenology.

What was missing. A reason for the value $2/3$ and for the phase that the formula requires in its cosine form.

Where it fits in the corpus. The Koide phase is $\theta = 2/9$ **[B]**, the same quotient as the squared modulus of an icosahedral projection (Doc. 367). The amplitude $d/c = \sqrt{2}$ has the right value but is not yet derived from the geometry and is treated as a setting **[S]** (Docs. 353, 367; R150 in Doc. 190). A further puzzle piece fits in here afterwards: Foot showed in 1994 that $Q = 2/3$ means an angle of 45° between the vector of the $\sqrt{m_k}$ and the axis $(1, 1, 1)$ **[Q]** [15]. In the language of the corpus this is the equipartition of the mass sum between the $\mathbb{Z}_3$-symmetric mode and the pair of non-trivial modes **[B]** (Doc. 353). The open question is thus no longer why $\sqrt{2}$ in particular, but why the mass structure divides half and half between these two parts. The exact pair $(\sqrt{2},\ 2/9)$ is excluded for m_μ/m_e (Doc. 369). The formula is thus placed, but not completely closed.

av.5 Why the pieces did not come together

Three reasons run through all the sections.

The pieces lay in different fields. Kaluza-Klein belongs to gravitation and later to string theory, the internal clock and the Zitterbewegung to quantum mechanics, Wigner and the Casimir value to group theory, finite fields to number theory, the Koide formula to phenomenology. Whoever works in one of these fields rarely has reason to read the others as parts of the same picture.

Each piece was classified as a tool or a curiosity. The Compton time counted as a conversion, Heaviside's units as a convenience, the Zitterbewegung as an artefact, the Koide formula as a coincidence. The missing step was the same each time as with Einstein: to take a known relation not as a derived quantity but as a principle (Doc. 312).

Two switches were set wrongly. Pauli's objection kept time out of the state space, and Kaluza-Klein rolled up a spatial instead of the temporal direction. Both switches could have been reset with the same insight: compact time escapes the objection, and it delivers the mass modes without a free radius.

There is also a methodological point. That the pieces fit into the picture is a structural agreement, not a numerical one. The more complex the quantum numbers, the closer one comes to the noise floor, where a matching combination can always be found; no status is assigned there (Doc. 342, Doc. 343). Scheme-dependent quantities of the Standard Model such as Λ_{QCD} are excluded from any claim of derivation. The puzzle pieces therefore hold only where they concern simple, convention-free ratios, and that is exactly where the places in this document lie.

av.6 Overview

Building block	Year	What was missing	FFGFT in the corpus
Kaluza-Klein	1921/26	spatial direction, free radius, unobserved tower	Doc. 330, 270, 307, 311, 318
De Broglie, Compton time	1924/2013	clock only as a derived scale	Doc. 312, 314, 365
Zitterbewegung	1930/1990	no geometric direction	not used so far **[S]**
Pauli's objection	1933	presupposes open time	Doc. 307, 334
Relational time, two-time physics	1983 on	time taken out instead of in	Doc. 307
Wigner	1939	mass values remain inputs	Doc. 307, 339
Casimir value	textbook	no link to modes	Doc. 324, 338
$SU(3)$			
Heaviside-Lorentz	1893	only a computational convenience	Doc. 365, 312
Orbifold	1985	six spatial directions, time outside	Doc. 330, 314, 311; R131
Finite fields, Frobenius	Galois	no physical structure	Doc. 336, 338, 346, 348
Koide, Foot	1981–83, 1994	no reason for $2/3$ and the phase	Doc. 367, 353, 369; R150

av.7 Conclusion

FFGFT connects to established physics and mathematics at eleven places, and at each of them it could have been found. Kaluza-Klein had the compact direction, de Broglie the clock, Pauli the objection whose gap is compact time, Wigner mass as structure, group theory the value $4/3$, string theory the orbifold, number theory the finite fields, and Koide the puzzle they help to solve together. FFGFT is not built from these pieces. It arose independently from time as a compact direction carrying the mass modes, and from the algebra that follows from it with the one parameter ξ. The building blocks were found in the literature only afterwards, and they fit in. That does not speak against the theory but for it: pieces that arose independently of one another fit into a picture that was drawn without them.

Verification script: `2/python/Dok390_Skripte/pruef_390_puzzleteile.py` — 59/59.

Tool and method

Claude (Anthropic) was used as a tool for: compiling the references in the corpus, writing and running the verification script, and working out the LaTeX sources according to the author's specifications. The verification script backs every corpus reference in this document with a text check in the named chapter file; if a check fails, the reference is not used.

[1] Th. Kaluza, *Zum Unitätsproblem der Physik*, Sitzungsber. Preuß. Akad. Wiss. Berlin (1921) 966–972.

[2] O. Klein, *Quantentheorie und fünfdimensionale Relativitätstheorie*, Z. Phys. **37** (1926) 895–906.

[3] L. de Broglie, *Recherches sur la théorie des quanta*, Ann. Phys. (Paris) 10e sér. **3** (1925) 22–128 (thesis 1924).

[4] S.-Y. Lan, P.-C. Kuan, B. Estey, D. English, J. M. Brown, M. A. Hohensee, H. Müller, *A clock directly linking time to a particle's mass*, Science **339** (2013) 554–557.

[5] E. Schrödinger, *Über die kräftefreie Bewegung in der relativistischen Quantenmechanik*, Sitzungsber. Preuß. Akad. Wiss., Phys.-Math. Kl. **24** (1930) 418–428.

[6] D. Hestenes, *The zitterbewegung interpretation of quantum mechanics*, Found. Phys. **20** (1990) 1213–1232.

[7] W. Pauli, *Die allgemeinen Prinzipien der Wellenmechanik*, in: Handbuch der Physik, vol. 24/1, Springer, Berlin 1933.

[8] D. N. Page, W. K. Wootters, *Evolution without evolution: Dynamics described by stationary observables*, Phys. Rev. D **27** (1983) 2885–2892.

[9] I. Bars, *Survey of two-time physics*, Class. Quantum Grav. **18** (2001) 3113–3130.

[10] E. Wigner, *On unitary representations of the inhomogeneous Lorentz group*, Ann. Math. **40** (1939) 149–204.

[11] H. Georgi, *Lie Algebras in Particle Physics*, 2nd ed., Westview Press, 1999.

[12] O. Heaviside, *Electromagnetic Theory*, vol. 1, The Electrician, London, 1893.

[13] L. Dixon, J. A. Harvey, C. Vafa, E. Witten, *Strings on orbifolds*, Nucl. Phys. B **261** (1985) 678–686.

[14] R. Lidl, H. Niederreiter, *Finite Fields*, 2nd ed., Cambridge University Press, 1997.

[15] R. Foot, *A note on Koide's lepton mass relation*, arXiv:hep-ph/9402242 (1994).

[16] Y. Koide, *A fermion-boson composite model of quarks and leptons*, Phys. Lett. B **120** (1983) 161–165.

Chapter aw

Doc. 391 — Loose Puzzle Pieces II: Cosmology — Redshift, Scales and the Boundary of the Cosmic Sector

Abstract

Doc. 390 placed the building blocks of particle physics that were found in the literature afterwards and fit into the picture of FFGFT. This document does the same for cosmology. It goes through eight building blocks: Zwicky's tired light, the gravitation theory of Hoyle and Narlikar and Wetterich's universe without expansion, Einstein's static universe and the steady-state theory, Dirac's large numbers, the Hubble length, Milgrom's acceleration scale, the problem of vacuum energy, and the cosmic background radiation. The order holds here too: the building blocks were not the foundation of FFGFT; they fit in afterwards. Unlike in particle physics, however, the cosmic sector lies outside the claim of derivation: H_0 is calibrated to the measurement, the exponent 10 is a setting, and the Standard Model does not derive this sector either. The puzzle pieces therefore show readings, not derivations: what the observations of cosmology carry can be read with $\tilde{T} \cdot m = 1$ without expansion, and several places in the literature have already shown this. Two building blocks, vacuum energy and background radiation, do not fit so far or fit only as a candidate, and are named as the boundary.

Status markers

[SETZUNG]	Motivated input, not a derivation.
[B]	Algebraically proved (verification script, exact arithmetic).
[K]	Numerically confirmed.
[S]	Structurally plausible, identification open.
[E]	Established mathematics or physics, adopted.
[Q]	Taken from the literature, with source.
[X]	Checked and negative.

The markers give the status *in the corpus*; the document derives nothing new but places things in context. Verification script: `pruef_391_kosmologie.py`.

aw.1 Order and boundary

On the order. As in Doc. 390, none of the following building blocks was a starting point of FFGFT. The theory was developed from $\tilde{T} \cdot m = 1$ and the geometry of the carrier; the literature references were found only afterwards. Where it says that a building block fits in at some place in the corpus, this is agreement in hindsight, not origin.

On the boundary. The cosmic sector is deliberately left out of the corpus. The Standard Model does not derive H_0 and Λ either, so a comparison would be asymmetric (P39 in Doc. 190; Doc. 365). The form of the relation between the Hubble length and the Planck length is fixed as a dimensionless ξ ratio (P20), the exponent 10 in $H_0 \propto \xi^{10}$ is a setting **[SETZUNG]**, and the scale is calibrated to the measurement of H_0 (R137 in Doc. 190). What this document places are therefore readings of the observations, not predictions of their values.

aw.2 Redshift without expansion

Zwicky's tired light (1929)

What it contained. Shortly after Hubble's discovery, Zwicky proposed that the redshift of distant galaxies arises not from expansion but from light losing energy along the way **[Q]** [1]. This was the first static interpretation of the Hubble relation.

What was missing. Tired light changes only the photon. Distant clocks, such as the light curves of supernovae, would then have to run at their original rate. What is observed, however, is a stretching by $(1+z)$, and the idea failed on that.

Where it fits in the corpus. Because of $\tilde{T} \cdot m = 1$, wavelength and clock rate are booked together. From $1 + z = e^{\xi_0 x}$ the time dilation $(1+z)$ follows for all z **[B]** (R128 in Doc. 190). Classical tired light fails because it changes only the photon side; the mass reading cannot make this mistake structurally (Doc. 312). The redshift is achromatic, and the scale of the path lengthening is calibrated via H_0 (K6 in Doc. 190).

Hoyle–Narlikar (1964) and Wetterich's universe without expansion (2013)

What it contained. Hoyle and Narlikar formulated a theory of gravitation in which particle masses arise from the distribution of matter in the cosmos **[Q]** [2]. In 2013 Wetterich showed that an expanding universe with constant masses turns, through a field redefinition, exactly into a static universe with growing particle masses **[Q]** [13]. Only dimensionless ratios are observable.

What was missing. A reason why the running of masses should be the natural reading and not merely an equivalent rewriting.

Where it fits in the corpus. Doc. 312 lists the equivalence of expansion and running masses as **[K]** and thereby anchors the degeneracy from Doc. 267: redshift, luminosity distance, baryon acoustic oscillations and background temperature do not distinguish the

two readings. In FFGFT the running of masses is not an additional assumption, because mass and clock rate are the same quantity through $\tilde{T} \cdot m = 1$ anyway. Doc. 312 records the boundary: the equivalence shows that the static reading is possible and observationally equal, not that it is right.

Einstein's static universe (1917) and the steady-state theory (1948)

What it contained. In 1917 Einstein built the first relativistic world model as a static, spatially closed cosmos **[Q]** [4]. In 1948 Bondi, Gold and Hoyle formulated a universe without a beginning that looks the same on large scales at all times **[Q]** [5, 6].

What was missing. Einstein's model needed a specially introduced cosmological constant and was unstable. The steady-state theory needed a continuous creation of matter, and the discovery of the background radiation in 1965 was taken as its end.

Where it fits in the corpus. FFGFT lives on a carrier without boundary, $\partial T^4 = \emptyset$ (Doc. 312). Doc. 313 works through the question of the beginning on the compact time cycle: the observable history fills half a cycle, and the "beginning" is the antipode of the cycle, not an event **[K]** with the calibration from P20. Neither a cosmological constant nor creation of matter is needed for this. According to Doc. 313 it remains open that an exact periodicity of the cycle is not dynamically generic and has to be set as a boundary condition.

aw.3 Scales

Dirac's large numbers (1937)

What it contained. Dirac noticed that the ratio of the electric to the gravitational force between electron and proton and the age of the universe in atomic units of time are both about 10^{39}, and suspected a connection behind this **[Q]** [7].

What was missing. A mechanism. Dirac's conclusion that the gravitational constant must decrease with time is excluded by measurements.

Where it fits in the corpus. In the corpus the relation between Hubble length and Planck length is a dimensionless ξ ratio (P20); the exponent is set, and the scale is calibrated to H_0. P39 records that no framework derives this relation from first principles. Dirac's observation thus appears as a ratio of two scales of the same geometry, not as a constant running in time. Dirac's name does not yet occur in the corpus; the placement is a reading **[S]**.

The Hubble length as maximal scale

What it contained. In 1929 Hubble found the linear relation between distance and redshift **[Q]** [8]. Since then the length c/H_0 has been read as the horizon of an expanding universe.

What was missing. A reading of this length without expansion.

Where it fits in the corpus. Doc. 182 and Doc. 365 read $R_H = c/H_0$ as the geometric maximal scale of the compact carrier, not as an expansion horizon. Its value is calibrated to H_0.

Milgrom's acceleration scale (1983)

What it contained. Milgrom explained the flat rotation curves of galaxies without dark matter through an acceleration scale a_0 below which Newton's law deviates **[Q]** [9]. He already noticed that a_0 is of the order of cH_0.

What was missing. A reason for this closeness to the cosmic scale, and a justification of the interpolation formula.

Where it fits in the corpus. In the corpus $a_0 = cH_0/(2\pi)$ **[K]** (Doc. 365); the same exponent 10 that carries H_0 also reproduces a_0, which would not be expected from a pure H_0 fit (Doc. 309, cross-anchoring). Doc. 308 obtains the form $a = \sqrt{a_N^2 + a_N a_0}$ with $a_0 = 2a_{\mathrm{bg}}$ from the Unruh coupling **[K]/[S]**. The book "Dark Matter – an Illusion" works out the rotation curves. The exponent remains set.

aw.4 Where it does not fit yet, or only as a candidate

Vacuum energy and the cosmological constant

What it contained. The zero-point energy of quantum fields is measured in the Casimir effect. As a source of the cosmological constant, its naive estimate lies many orders of magnitude above the observed value; Weinberg called this the cosmological constant problem **[Q]** [10].

Status in the corpus. The connection between the Casimir and background energy densities at the length L_ξ is an identity, not evidence from measurement **[S]** (R136 in Doc. 190). The Λ formulas of older documents do not hit the observed value **[X]** (R137). In the static reading, Λ is an artefact of the reading without a referent of its own (Doc. 308) and is not set (Doc. 312); there is no derivation of the observed value. This building block does not fit in so far.

The cosmic background radiation (1965)

What it contained. In 1965 Penzias and Wilson found an isotropic radiation of about 3 kelvin **[Q]** [11]. Its spectrum and its anisotropies are among the most precise data in cosmology.

Status in the corpus. The relations for T_{CMB} from ξ give a pure number that agrees with the measured value only in the unit eV; they count as a unit-dependent agreement, and the question is open (R137, R70). The relation $T(z)$ in Doc. 039 is excluded by measurements **[X]**. The factor 3 in front of the peak sequence is a setting, and the peak selection as a whole is open (R139). In the basic form, Doc. 388 sets up the candidate $T_{\mathrm{CMB}}/m_e = (8\pi)^{1/4}\,\xi^{5/2}$, which hits the FIRAS value **[K]**; the prefactor is motivated but not justified **[S]**, and the random test there shows that chance is not excluded. If the candidate holds, it connects temperature and Hubble scale through $T_{\mathrm{CMB}}^4 = 16\,H_0\,m_e^3$ **[B]** (Doc. 388), with the same set exponent 10. The temperature of the radiation is moreover among the quantities that do not resolve the degeneracy from Doc. 267. This building block thus fits in only as a candidate, not as a secured place.

aw.5 What the pieces show

In particle physics (Doc. 390) the building blocks fitted in at places where the corpus derives something. In cosmology they fit in at places where the corpus *reads* something. Zwicky, Hoyle and Narlikar, Wetterich, Einstein and the steady-state theory each supplied a piece of a static cosmology; each piece failed on its own at a different point, at time dilation, at the missing justification of running masses, at instability or at creation of matter. With $\tilde{T} \cdot m = 1$ these points are no longer separate problems, because wavelength, clock rate and mass are the same booking.

Dirac and Milgrom each noticed a striking closeness between microscopic and cosmic scales. In the corpus both are ratios of the same geometry, but tied to a set exponent and a calibrated scale. They are therefore cross-anchorings, not evidence.

Vacuum energy and background radiation do not fit so far, or only as a candidate. This is not concealed: the formulas concerned are listed in the register as **[X]** or as open, and the temperature candidate from Doc. 388 as **[S]**. For the cosmic sector the same therefore holds as for scheme-dependent quantities in particle physics: no claim of derivation is made where the quantities depend on the measurement or on a convention.

aw.6 Overview

Building block	Year	What was missing	In the corpus
Zwicky, tired light	1929	no time dilation	R128, K6; Doc. 312
Hoyle–Narlikar, Wetterich	1964, 2013	running masses only equivalent, not justified	Doc. 312, 267
Einstein universe, steady state	1917, 1948	instability, creation of matter	Doc. 313, 312
Dirac's large numbers	1937	no mechanism	P20, P39 **[S]**
Hubble length	1929	read only as expansion horizon	Doc. 182, 365
Milgrom's a_0	1983	closeness to cH_0 unexplained	Doc. 365, 309, 308
Vacuum energy, Λ	1989	does not fit so far	R136, R137
Background radiation	1965	only a candidate, chance not excluded	Doc. 388; R137, R139, R70

aw.7 Conclusion

In cosmology, too, the puzzle pieces were in the literature long before FFGFT, and here too they were not its foundation. Six of the eight building blocks fit in afterwards, as readings of the observations without expansion and as cross-anchorings between microscopic and cosmic scales. Two do not fit so far or fit only as a candidate. The difference from particle physics must be stated honestly: the cosmic sector is calibrated and left out; here the building blocks show that the theory can read cosmology, not that it derives its numbers.

Verification script: `2/python/Dok391_Skripte/pruef_391_kosmologie.py` — 51/51.

Tool and method

Claude (Anthropic) was used as a tool for: compiling the references in the corpus, writing and running the verification script, and working out the LaTeX sources according to the author's specifications. The verification script backs every corpus reference in this document with a text check in the named chapter file; if a check fails, the reference is not used.

[1] F. Zwicky, *On the red shift of spectral lines through interstellar space*, Proc. Natl. Acad. Sci. USA **15** (1929) 773–779.

[2] F. Hoyle, J. V. Narlikar, *A new theory of gravitation*, Proc. R. Soc. Lond. A **282** (1964) 191–207.

[3] C. Wetterich, *A Universe without expansion*, Phys. Dark Univ. **2** (2013) 184–187; arXiv:1303.6878.

[4] A. Einstein, *Kosmologische Betrachtungen zur allgemeinen Relativitätstheorie*, Sitzungsber. Preuß. Akad. Wiss. Berlin (1917) 142–152.

[5] H. Bondi, T. Gold, *The steady-state theory of the expanding universe*, Mon. Not. R. Astron. Soc. **108** (1948) 252–270.

[6] F. Hoyle, *A new model for the expanding universe*, Mon. Not. R. Astron. Soc. **108** (1948) 372–382.

[7] P. A. M. Dirac, *The cosmological constants*, Nature **139** (1937) 323.

[8] E. Hubble, *A relation between distance and radial velocity among extra-galactic nebulae*, Proc. Natl. Acad. Sci. USA **15** (1929) 168–173.

[9] M. Milgrom, *A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis*, Astrophys. J. **270** (1983) 365–370.

[10] S. Weinberg, *The cosmological constant problem*, Rev. Mod. Phys. **61** (1989) 1–23.

[11] A. A. Penzias, R. W. Wilson, *A measurement of excess antenna temperature at 4080 Mc/s*, Astrophys. J. **142** (1965) 419–421.

Chapter ax

Doc. 392 — Where the Representations of Gravitation Part Ways

Abstract

Doc. 389 treats space curvature, change of mass or clock rate, and fractal path lengthening as three representations of gravity that must be equivalent by virtue of the duality $\tilde{T} \cdot m = 1$. This document examines how far that equivalence reaches. The temporal half $g_{00} = (m/m_0)^2$ is contained in every admissible representation and alone carries free fall, redshift and clock rates. Pure mass variation, in which clock rates and lengths both scale with $1/m$, is a conformal rescaling of flat space and has a vanishing Weyl tensor. The flat Friedmann universe is conformally flat as well; there the mass reading is exact and passes light curves, background temperature and the Tolman test, whereas path lengthening acting on the photon alone fails all three. The field of a star, by contrast, carries Weyl curvature, whose component $C_{trtr} = -2GM/r^3$ is the tidal term. There the flat representation needs the spatial half with the opposite sign for rulers, a co-moving vector field for rotating masses, and a tensor part for gravitational waves. The speed-of-light limit follows from the torus spectrum: massless modes travel at c, massive ones are always slower; a conformal rescaling changes neither c nor ε_0 and μ_0 locally. The mass reading is thus the complete representation of the conformally flat part; the Weyl part is what has to be supplied in addition. The Alcock–Paczyński test does not separate expansion and the mass reading, but with the DESI data it excludes the pure exponential form $1 + z = e^{\xi x}$ of the distance law.

Status markers

[B] Algebraically proved (symbolic computation in the verification script).
[K] Numerically confirmed (verification script, comparison with measurements).
[S] Setting, open or speculative.
[X] Excluded by measurement.
[Q] Taken from the literature, with source.

Verification scripts: `2/python/Dok392_Skripte/` (eight scripts, 81/81). Measured values and sources are given in the script headers and in the bibliography.

Symbols

$\xi = 4/30000$	the single parameter of FFGFT (geometry T^4/Z_3)
$\tilde{T}$, m, m_0	time field, position-dependent mass, reference mass; duality $\tilde{T} \cdot m = 1$
c	speed of light, $c = 1$ in the basic form
G, M, r, $\vec{x}$	gravitational constant, mass of the central body, distance, position vector
Φ, U	Newtonian potential; $U = GM/(rc^2) = -\Phi/c^2$, dimensionless
$g_{\mu\nu}$	metric; g_{00} temporal component (clock rate), g_{0i} mixed (dragging), g_{ij} spatial components ($i, j = 1, 2, 3$)
$\eta_{\mu\nu}$, Ω	flat Minkowski metric; conformal factor, $\Omega = m/m_0$ for pure mass variation
γ, β	PPN parameters: space curvature per unit mass; non-linearity of g_{00}
$\eta = 4\beta - \gamma - 3$	Nordtvedt parameter (not to be confused with $\eta_{\mu\nu}$ or the conformal time η)
m_G/m_I	ratio of gravitational to inertial mass (lunar laser ranging)
C_{trtr}	component of the Weyl tensor, the tidal part of curvature
n	refractive index of path lengthening, $n = 1 + 2U$
$\vec{J}$, I, R	angular momentum, moment of inertia and radius of the rotating body
$\vec{u}$	velocity field with which path lengthening is dragged along
$\vec{w}$	magnetoelectric coupling in the medium form, $\vec{w} = -g_{0i}/g_{00}$
$\dot{P}$	rate of change of the orbital period of a binary
z, a, a_0	redshift; scale factor, or m/m_0 in the mass reading; its present value
η (cosmic), H, H_0	conformal time; Hubble rate $H = a'/a^2$; its present value
b, β (in $T(z)$), k	exponents of light-curve stretching $(1+z)^b$, of the background temperature $T \propto (1+z)^{1-\beta}$ (not the PPN parameter) and of surface brightness $(1+z)^{-k}$
W, T, M	cosmic readings: photon only; clocks in addition; clocks and rulers in addition
d_L, d_A	luminosity and angular-diameter distance
D_M, D_H, r_d	transverse comoving distance, Hubble distance $D_H = c/H$, sound-horizon length (BAO ruler)
x	path length in the law $1 + z = e^{\xi x}$ (Docs. 280, 308)
E, p, v	energy, momentum, group velocity $v = dE/dp$ of a mode
x_4, R_4, k_4, λ_4	coordinate and radius of the compact fourth direction, winding number ($m = k_4/R_4$), wavelength with $\lambda_4 \cdot m = 2\pi$
ε, μ, ε_0, μ_0	effective permittivity and permeability in the field; electric and magnetic constants
A, B	metric functions in the static field, $g_{00} = -A$, $g_{ij} = B\delta_{ij}$
χ^2	sum of squared deviations, measured in standard deviations

ax.1 The question

Doc. 389 reads gravity in three representations: as space curvature as in general relativity, as change of mass or clock rate in flat space (Doc. 078), and as fractal path lengthening (Docs. 026, 041, 182, 308). It already shows there that the change of mass alone carries only the temporal half and yields the light deflection of general relativity only together with path lengthening. It remained open whether the three representations are equivalent beyond the static weak field, and if not, where exactly they part ways. This document

computes both sides: what excludes a representation, and what speaks for the mass reading.

ax.2 The temporal half as the common core

In the mass representation $g_{00} = (m/m_0)^2$ with $m = m_0\, e^{\Phi/c^2}$ (Doc. 389). Free fall follows exactly from the mass gradient, $\ddot{\vec{x}} = -c^2 \nabla \ln m = -\nabla\Phi$ **[B]**, and since $\nabla \ln m$ is the same for every mode, the weak equivalence principle is built in. The same function determines the redshift and the clock rate. The measurements of this half are:

Measurement	Expectation from g_{00}	measured
Pound–Rebka, 22.5 m	2.45×10^{-15}	$(2.57 \pm 0.26) \times 10^{-15}$ **[Q]**
optical clocks, 33 cm	3.6×10^{-17}	$(4.1 \pm 1.6) \times 10^{-17}$ **[Q]**
GPS, clock-rate part per day	$+45.7\,\mu$s	corrected in operation
Gravity Probe A, Galileo 5/6	clock-rate part	confirmed to 7×10^{-5} and 2.5×10^{-5} **[Q]**

[K]. This half is not an alternative to space curvature; it is the same in every admissible representation, and in general relativity too g_{00} is the clock rate. The mass reading describes it literally: in a static field the frequency of the photon stays constant along the whole path, and what differs are the atoms at the two places (Doc. 312). With $\tilde{T} \cdot m = 1$ this is a single quantity.

ax.3 What the weak field excludes

In the parametrised weak field the representations differ in γ (spatial half) and β (second order of g_{00}). In the mass representation β follows from the form of $m(\Phi)$: the exponential form gives $g_{00} = 1 - 2U + 2U^2$, hence $\beta = 1$, the linear form $g_{00} = 1 - 2U + U^2$, hence $\beta = \frac{1}{2}$ **[B]**. Four independent measurements separate them:

Representation	γ	β	Light	Mercury	GP-B geodetic	Lunar laser
Space curvature (GR)	1	1	$1.751''$	$42.98''$	6604	$\eta = 0$
Clock rate (exp.) + path lengthening	1	1	$1.751''$	$42.98''$	6604	$\eta = 0$
Clock rate (linear) + path lengthening	1	$\frac{1}{2}$	$1.751''$	$50.1''$ **[X]**	6604	$\eta = -2$ **[X]**
clock rate only	0	1	$0.876''$ **[X]**	$14.3''$ **[X]**	2201 **[X]**	$\eta = 1$ **[X]**
conformally flat (Doc. 143)	-1	$\frac{1}{2}$	0 **[X]**	$-7.2''$ **[X]**	-2201 **[X]**	$\eta = 0$

[K]. Mercury in arcseconds per century, geodetic precession in mas per year, $\eta = 4\beta - \gamma - 3$. The measured values are $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ (Cassini), the geodetic precession 6601.8 ± 18.3 mas/year (Gravity Probe B), and for lunar laser ranging $(m_G/m_I)_E - (m_G/m_I)_M = (-0.8 \pm 1.3) \times 10^{-13}$ **[Q]**; with the difference of the self-energies of Earth and Moon this gives $\eta = (1.8 \pm 2.9) \times 10^{-4}$ without correction for composition **[K]**. The change of mass alone thus fails also in a test without light, the geodetic precession, by about 240 standard deviations. The linear form of the mass change fails independently at Mercury and in lunar laser ranging; the exponential form, carried as a setting in Doc. 389, is thereby fixed from two sides.

Mixed forms are admissible. If curvature carries the share w and path lengthening the share s of the spatial half, then $\gamma = w + s$, and the condition reads $w + s = 1$ **[K]**. The freedom of representation therefore lies in how the spatial half is divided, not in its size.

ax.4 Conformally flat or Weyl-curved

If clock rates and lengths both scale with $1/m$, the metric is a conformal rescaling of flat space, $g_{\mu\nu} = \Omega^2 \eta_{\mu\nu}$ with $\Omega = m/m_0$. For every such metric the Weyl tensor vanishes **[B]**. Pure mass variation therefore carries exactly the conformally flat geometry, and light travels in it as in flat space.

The universe at large. The flat Friedmann universe can be written in conformal time as $a(\eta)^2 \eta_{\mu\nu}$; its Weyl tensor vanishes as well **[B]**. In the static frame the masses run as $m(\eta) = m_0\, a(\eta)$, and $1 + z = m_{\text{today}}/m_{\text{emission}} = a_0/a$ (Wetterich [13]) **[Q]**. Because the cosmic geometry is conformally flat, the mass reading is there not merely observationally equivalent but exactly the same geometry in other variables.

The field of a star. For the isotropic weak field with $\gamma = 1$ the computation gives to first order

$$C_{trtr} = -\frac{2GM}{r^3} \neq 0 \quad \textbf{[B]}, \tag{ax.1}$$

exactly the tidal term. In vacuum outside a mass the curvature is pure Weyl curvature. A conformal rescaling cannot produce it.

The sign of rulers. Deep in the potential $m/m_0 = 1 - U$. A length that scales with $1/m$, like the Compton or Bohr length, appears longer by $1 + U$ from the distant reference. The spatial half with $\gamma = 1$, by contrast, requires proper rulers that appear shorter by $1 - U$ **[B]**. The signs are opposite; this is the counter-running behaviour that Doc. 389 records for $\lambda_4 \cdot m = 2\pi$. In the field of a star the spatial half therefore does not follow from the mass scaling, whereas in the Friedmann universe it does.

ax.5 Cosmos: which static reading is equivalent

For redshift without expansion there are three static readings (Docs. 280, 312, 391): path lengthening or energy loss acting on the photon alone (W), the same with clocks booked along through $\tilde{T} \cdot m = 1$ (T), and the full mass reading, in which clocks and rulers are booked along (M). Doc. 312 names four discriminators whose forward demonstration was open. Three of them can be computed directly:

	Light curves $(1+z)^b$	$T(z) \propto (1+z)^{1-\beta}$	Tolman $(1+z)^{-k}$
Measurement **[Q]**	$b = 1.003 \pm 0.011$	$\beta = -0.007 \pm 0.027$	$k = 2.59 \pm 0.17$ (R), 3.37 ± 0.13 (I)
W: photon only	$b = 0$ **[X]**	$\beta = 1$ **[X]**	$k = 1$ **[X]**
T: + clocks	$b = 1$	$\beta = 0$	$k = 2$ **[X]**
M: + clocks + rulers	$b = 1$	$\beta = 0$	$k = 4$

[K]. The light curves exclude reading W by about 90 standard deviations, the background temperature by about 37. In the Tolman test the observed values lie above $k = 2$, although the evolution of the stellar populations makes early galaxies brighter and raises the true

exponent further; only $k = 4$ is compatible. Equivalent to expansion is therefore exactly reading M. In it the rulers are booked along by themselves, because atomic lengths scale with $1/m$ ($\lambda_4 \cdot m = 2\pi$, Doc. 314). Path lengthening acting on the photon alone (Doc. 280) is not sufficient as a cosmic reading; it holds as a description of the same redshift within the mass reading, not as an independent alternative.

The fourth discriminator, the Alcock–Paczyński test, compares the radial with the transverse extent of an object. Computing in flat space with $m(t) = m_0 a(t)$, with lengths and periods proportional to $1/m$ and conserved photon number, gives the angular-diameter and luminosity distances and the AP ratio exactly as in the FRW model with the same a; in particular $d_L = (1+z)^2 d_A$ **[B]**. Surface brightness falls as $(d_A/d_L)^2$; the Tolman exponent $k = 4$ is therefore the same statement as this reciprocity, and readings W and T violate it with $d_L/d_A = (1+z)^{1/2}$ and $(1+z)$ respectively. The AP test therefore does not separate expansion and the mass reading; it tests the course of $H(z)$, that is, the distance law.

For the law $1 + z = e^{\xi x}$ of Docs. 280 and 308 the test can be carried out if x is read as the coordinate distance of reading M. Then $a(\eta) = e^{H_0(\eta - \eta_0)}$, hence $H(z) = H_0(1+z)$, and the ratio of transverse to radial distance becomes

$$\frac{D_M}{D_H} = (1+z)\ln(1+z). \tag{ax.2}$$

It contains neither H_0 nor the sound-horizon length r_d; the caveat P20, which Doc. 267 states for the BAO distances, therefore does not apply. DESI DR2 [12] measures this ratio in six redshift ranges **[Q]**. Up to $z \approx 1.5$ the law lies within three standard deviations, and at $z = 0.51$ it matches the measured value exactly. For the Lyman-α forest at $z = 2.33$ it gives 4.01 against a measured 4.52 ± 0.10, that is 5.3 standard deviations too low; over all six values $\chi^2 = 37.3$, whereas flat ΛCDM with $\Omega_m = 0.2975$ gives 6.0 **[K]**. In this reading the pure exponential form is therefore excluded **[X]**. The degeneracy that Doc. 267 assumes for the BAO angles requires a relation $z(d)$ that departs from the exponential law at high redshift; no such relation has been derived in FFGFT **[S]**.

Equivalence alone does not decide between expansion and the mass reading. In favour of the mass reading speaks the simpler structure of settings: the cosmological constant appears in it as an artefact of the reading and is not set (Docs. 308, 312) **[S]**.

ax.6 Rotating masses

Frame dragging by a rotating mass resides in g_{0i} (Doc. 312). A static pair of scalar fields, mass $m(x)$ and refractive index $n(x)$, has $g_{0i} = 0$: the mass density of a rigidly rotating sphere is stationary like that of a sphere at rest, the angular momentum J does not appear in a scalar, and the gravitomagnetic field $\nabla \times (\vec{J} \times \vec{x}/r^3)$ is a dipole field that only a vector field can carry **[B]**. Gravity Probe B measures frame dragging as 37.2 ± 7.2 mas/year against a prediction of about 39, and LAGEOS confirms it to about 10 % **[Q]**; absent frame dragging is thus excluded by 5 and about 10 standard deviations respectively **[K]**.

The flat representation reaches frame dragging if path lengthening is carried along like a moving medium. The Gordon metric of a medium with index n and velocity $\vec{u}$ has $\tilde{g}_{0i} =$

$-(1-n^{-2})\,u_i/c \approx -4Uu_i/c$ for $n = 1+2U$; equating this with the GR value $g_{0i} = -2G(\vec{J}\times\vec{x})_i/(c^3r^3)$ gives

$$\vec{u} = \frac{\vec{J}\times\vec{x}}{2Mr^2} \quad \textbf{[B]}. \tag{ax.3}$$

At the Earth's surface this is about 77 m/s at the equator, one sixth of the rotation velocity ($I/(2MR^2)$); for the Sun 70 m/s or $3.5\,\%$; the field falls off as $1/r$ **[K]**. This field is a vector and does not follow from $m(x)$. In the curvature representation the dragging is contained, because FFGFT adopts the field equations (Doc. 312); whether the structure of the carrier itself supplies the co-moving field of the flat representation is open **[S]**.

ax.7 Gravitational waves

Gravitational waves impose three requirements on every representation. First, they travel at the speed of light: for GW170817 the gamma-ray signal arrived 1.74 s after the waves over about 40 Mpc, $|c_{\mathrm{gw}}/c - 1| \lesssim 10^{-15}$ **[K]**. Second, the spatial half acts equally on light and waves: the Shapiro delay in the potential of the Milky Way is about 240 days; if path lengthening acted on light only, the light would arrive about 120 days late **[K]**. Third, binary stars emit quadrupole radiation: the orbital decay of the Hulse–Taylor pulsar follows the quadrupole formula, $\dot{P}_{\mathrm{GR}} = -2.402\times10^{-12}$, to $0.2\,\%$ **[K]**. The polarisation favours pure tensor modes over pure scalar modes with a Bayes factor above 1000 **[Q]**. A scalar $m(x)$ can travel at c but supplies no tensor polarisation; the flat representation needs a tensor part for waves **[S]**. In the curvature representation it is given with the adopted field equations (Doc. 312).

ax.8 The speed-of-light limit and the field constants

In the basic form with $c = 1$, c is the conversion factor between the spatial directions and the compact fourth direction x_4, not the speed of a medium (Doc. 134). The spectrum of the carrier is $E^2 = p^2 + m^2$ with $m = k_4/R_4$; the winding number is the mass number (Doc. 314). The group velocity is

$$v = \frac{dE}{dp} = \frac{p}{\sqrt{p^2+m^2}} \quad \textbf{[B]}. \tag{ax.4}$$

It equals 1 exactly when the mode has no winding in the time direction ($k_4 = 0$), and it is smaller than 1 for every massive mode at every momentum; the energy needed to reach $v = 1$ diverges **[B]**. A massive mode distributes its wave number between space and x_4, and only the spatial share is motion. In the mass reading this means: whatever carries a clock is slower than light; only what has no proper time reaches the limit (Doc. 391). The kinetic form of $\tilde{T}\cdot m = 1$ is de Broglie's harmony of phases: the inner frequency drops by γ, the phase frequency rises by γ, and the product stays (Doc. 312) **[B]**.

The limit is the same in all representations. A conformal rescaling $\Omega^2\eta_{\mu\nu}$ leaves the null vectors fixed, and the locally measured ratio of proper length to proper time of a light signal does not depend on Ω **[B]**; mass variation can never change c locally. What changes in a field is the coordinate speed: it is 1 for pure mass variation and $1-2U = 1/n$

with $n = 1 + 2U$ in the isotropic field with $\gamma = 1$, smaller by 4.2×10^{-6} at the solar limb **[K]**. This slowing belongs to the spatial half, the path lengthening, not to a variable c (R132).

The same holds for the electric and the magnetic field constant. A static field acts on Maxwell fields like a medium with $\varepsilon = \mu = \sqrt{B/A}$ for $g_{00} = -A$, $g_{ij} = B\delta_{ij}$ (Plebanski [17]) **[Q]**. For $\gamma = 1$ both rise equally by $1 + 2U$, the coordinate speed $1/\sqrt{\varepsilon\mu}$ drops by $2U$, and the wave impedance $\sqrt{\mu/\varepsilon} = 1$ stays **[B]**. This is why light in a gravitational field is neither reflected nor deflected birefringently, and why the deflection depends neither on colour nor on polarisation. Pure mass variation leaves $\varepsilon = \mu = 1$, because Maxwell's equations are conformally invariant in four dimensions **[B]**; here too the spatial half carries the whole effect, in equal parts in ε and μ. Measured locally, ε_0, μ_0 and $\varepsilon_0\mu_0 c^2 = 1$ remain the same everywhere; in the Heaviside–Lorentz units of the basic form both are one. The frame dragging of Section 6 appears in this medium form as a magnetoelectric coupling $\vec{w} = -g_{0i}/g_{00}$ between the electric and the magnetic field (Plebanski) **[Q]**.

ax.9 Result

Domain	Mass reading	Status
Free fall, redshift, clocks	carries it alone, common core	**[K]**
Universe at large	exact (Weyl $= 0$), passes three discriminators, AP as in FRW	**[B]**/**[K]**
distance law $1 + z = e^{\xi x}$	fails the AP test at $z = 2.33$ ($5.3\,\sigma$)	**[X]**
Light, Mercury, geodetic	needs the spatial half, $\gamma = \beta = 1$	**[K]**, origin **[S]**
Rotating masses	needs a co-moving vector field	**[B]**, origin **[S]**
Gravitational waves	needs a tensor part	**[S]**
Speed of light, ε_0, μ_0	unchanged locally, light cones conformally invariant	**[B]**

The three representations of Doc. 389 are equivalent for the conformally flat part and for the static, non-rotating weak field with $\gamma = \beta = 1$. The mass reading is the complete representation of the conformally flat part: it carries the temporal half everywhere and the universe at large exactly. What shows itself as space curvature in the narrower sense is the Weyl part around individual masses; the flat representation has to supply it in addition, as the spatial half with the ruler sign of proper lengths, as a co-moving field for rotation, and as a tensor part for waves. Mixed forms are admissible as long as the spatial half carries the share one in total.

ax.10 What remains open

- The origin of the spatial half from the structure of the carrier. Doc. 308 identifies the K3 area invariance as the only one of four tested candidates that yields $\gamma = 1$ and

confirms the factor two of light deflection numerically; Doc. 308 itself lists the forward proof as open, as does Doc. 389 **[S]**.

- Dragging and the tensor part of gravitational waves in the flat representation. In the curvature representation both are contained, because FFGFT adopts the field equations, whose form is enforced by Lovelock's theorem (Doc. 312). Open is whether the structure of the carrier itself supplies the co-moving field $\vec{u}$ and a tensor part. The corpus contains only scalar approaches for this (Docs. 131, 189); the attempt in Doc. 143 to obtain the tensor part from the scalar is circular according to Doc. 144 **[S]**.
- A distance law of FFGFT that passes the Alcock–Paczyński test. The reciprocity $d_A = d_L/(1+z)^2$ is no longer open: it holds exactly in reading M and is the same statement as $k = 4$. The pure exponential form $1 + z = e^{\xi x}$ (Docs. 280, 308), read as coordinate distance of reading M, fails at the Lyman-α value of DESI (Section 5). Open is which relation $z(d)$ the structure of the carrier supplies and whether a different reading of the path length x changes the result; Doc. 268 so far treats D_A as an external quantity **[S]**.

Verification scripts: `2/python/Dok392_Skripte/` — `pruef_392a_schwachfeld.py` (14/14), `pruef_392b_mitfuehrung.py` (11/11), `pruef_392c_gravitationswellen.py` (8/8), `pruef_392d_kosmos.py` (9/9), `pruef_392e_massenlesart.py` (11/11), `pruef_392f_lichtgrenze.py` (16/16), `pruef_392g_abstaende.py` (8/8), `pruef_392h_desi_ap.py` (4/4); together `run_all_392.sh`, 81/81.

Tool and method

Claude (Anthropic) was used as a tool for: searching the corpus, writing and running the verification scripts, preparing the LaTeX sources according to the author's specifications. Every number in the text is secured by a check in the scripts named; statements that only reproduce literature values are marked **[Q]** and are not counted there as checks.

[1] B. Bertotti, L. Iess, P. Tortora, *A test of general relativity using radio links with the Cassini spacecraft*, Nature **425** (2003) 374–376.

[2] C. W. F. Everitt et al., *Gravity Probe B: Final results of a space experiment to test general relativity*, Phys. Rev. Lett. **106** (2011) 221101.

[3] J. G. Williams, S. G. Turyshev, D. H. Boggs, *Lunar laser ranging tests of the equivalence principle*, Class. Quantum Grav. **29** (2012) 184004.

[4] I. Ciufolini, E. C. Pavlis, *A confirmation of the general relativistic prediction of the Lense–Thirring effect*, Nature **431** (2004) 958–960.

[5] B. P. Abbott et al., *Gravitational waves and gamma-rays from a binary neutron star merger: GW170817 and GRB 170817A*, Astrophys. J. Lett. **848** (2017) L13.

[6] B. P. Abbott et al., *GW170814: A three-detector observation of gravitational waves from a binary black hole coalescence*, Phys. Rev. Lett. **119** (2017) 141101.

[7] J. M. Weisberg, Y. Huang, *Relativistic measurements from timing the binary pulsar PSR B1913+16*, Astrophys. J. **829** (2016) 55.

[8] R. V. Pound, G. A. Rebka, *Apparent weight of photons*, Phys. Rev. Lett. **4** (1960) 337–341.

[9] C. W. Chou et al., *Optical clocks and relativity*, Science **329** (2010) 1630–1633.

[10] R. F. C. Vessot et al., *Test of relativistic gravitation with a space-borne hydrogen maser*, Phys. Rev. Lett. **45** (1980) 2081–2084.

[11] P. Delva et al., *Gravitational redshift test using eccentric Galileo satellites*, Phys. Rev. Lett. **121** (2018) 231101.

[12] M. Abdul Karim et al. (DESI Collaboration), *DESI DR2 Results II: Measurements of baryon acoustic oscillations and cosmological constraints*, arXiv:2503.14738 (2025), DOI 10.1103/tr6y-kpc6; Table IV.

[13] C. Wetterich, *A Universe without expansion*, Phys. Dark Univ. **2** (2013) 184–187.

[14] R. M. T. White et al., *The Dark Energy Survey supernova program: slow supernovae show cosmological time dilation out to* $z \sim 1$, Mon. Not. R. Astron. Soc. **533** (2024) 3365–3378.

[15] P. Noterdaeme et al., *The evolution of the cosmic microwave background temperature*, Astron. Astrophys. **526** (2011) L7.

[16] L. M. Lubin, A. Sandage, *The Tolman surface brightness test for the reality of the expansion. IV.*, Astron. J. **122** (2001) 1084–1103.

[17] J. Plebanski, *Electromagnetic waves in gravitational fields*, Phys. Rev. **118** (1960) 1396–1408.

Chapter ay

Doc. 393 — The Noise Limit and Learning Systems

Abstract

The corpus knows a noise limit: beyond a certain measurement tolerance, harmonic relations can no longer be distinguished from chance (Doc. 343, Theorem G′). In questions and answers, this document examines whether neural networks and language models run into the same limit, why they nevertheless almost always produce readable sentences, why language forgives errors and numbers do not, and what role probability and the weights of a network play. A null test with an exhaustive symbolic search over about 23 million expressions shows that the dimensionless measured quantities of the basic form carry no harmonic structure above chance level on their own, and that even a genuine rule such as the Balmer series becomes visible from a few examples only with the appropriate, narrow prior. The common core of all answers: discrete structure is error-tolerant, the noise limit sits at the transition from the discrete to the continuum, and structure that the data do not yield must be specified and then tested.

Status markers

- **[B]** Proved algebraically or information-theoretically.
- **[K]** Confirmed numerically (verification script).
- **[S]** Setting, open, or classification without a calculation of its own.
- **[X]** Excluded by the calculation.
- **[Q]** Taken from the literature, with source.

The document is structured as a sequence of questions and answers. The questions are kept in the order in which they arise from one another; each answer stands on its own. Section 14 summarises the common core in a single picture.

ay.1 The noise limit in the corpus

Question. What is meant by the noise limit?

Answer. A measured ratio is known only up to a tolerance ε. The larger the admitted denominator, the more densely the fractions lie; beyond the Farey height

$$Q_c = \frac{\pi}{\sqrt{6\,\varepsilon}}$$

the tolerance window contains so many fractions that a genuine ratio can no longer be separated from a chance one (Doc. 343, Theorem G′). For $\varepsilon = 1.05\,\%$, the size of the fractal correction, $Q_c \approx 12.5$; for $\varepsilon = 100\,\xi \approx 1.33\,\%$, $Q_c \approx 11.1$ (Doc. 359, Theorem B) **[K]**. Harmonic relations with small denominators remain visible; finer ones are lost in the noise. The limit is a property of the data, not of the method: what is not contained in the signal cannot be recovered by any analysis, neither by a Fourier analysis nor by an autocorrelation **[B]**.

ay.2 Do neural networks and language models run into the same limit?

Question. Should neural networks and language models not also run into this limit in pattern recognition?

Answer. Yes. A network is a method like any other; it can reach the limit but not exceed it. Beyond the limit it no longer recognises but adapts to the noise and takes chance proximity for structure; in learning theory this is called overfitting. In addition, such networks have a known property: they learn coarse, low-frequency structures first and fine ones late **[Q]** [4]. Fine structures, however, are precisely those closest to the noise.

Language models are in a less favourable position. They learn from text, not from measured signals, and have no measurement resolution of their own. With numerical ratios they tend towards false hits: given enough combinations, a “harmonic” explanation can almost always be found (look-elsewhere effect). This is why verification scripts in the corpus are set up as null tests: if a random selection yields the same result, it is not counted. This applies to every suggestion made by a language model.

Networks become better than the raw data only through built-in prior knowledge. Doc. 359 Theorem D shows this with an example: whether the admissibility criterion “prime factors only from $\{2, 3, 5, 7, 11, 13\}$” is learnable is decided by the hypothesis class. Without a structural assumption the learning bound requires about 84 000 labelled examples; with the structural assumption “prime factors from a fixed set” at most 407, and from positive examples alone the correct prime set is then reliably recovered after only 30 examples **[K]** [7].

ay.3 Why do language models still produce readable sentences?

Question. If this is so, why do language models hardly ever produce unreadable sentences?

Answer. Because grammar and sentence structure are what occurs most densely in the training data. There are trillions of words, the same rules recur in millions of different sentences, there is no measurement noise, and the architecture is tailored to sequences of words. This is exactly the favourable case of Doc. 359 Theorem D: a suitable prior and plenty of examples.

The models become weak where the pattern is rare or exact: precise numbers, chains of calculation, rare facts, bibliographic references. There they produce something plausible, and the sentence remains fluent. Readability is the dense pattern, correctness often the sparse one. A well-written sentence from a language model therefore says nothing about the correctness of its content.

ay.4 Should harmonic patterns not be as dense as grammar?

Question. If FFGFT is correct, harmonic ratios pervade all of nature. Should they then not occur just as densely as grammar?

Answer. Dense in occurrence, yes; dense in variety, no. For learning a rule, what counts is the number of *different* examples, not the number of repetitions. The electron occurs an unimaginable number of times, but each time it yields the same mass ratio. The basic structure of FFGFT lies in a few numbers: some mass ratios, couplings and mixing angles, a few dozen altogether. From so few values no method can separate a rule from chance.

Where harmonic patterns do occur in great variety, they have also been found: in atomic spectra with thousands of lines following one integer construction principle, in crystal lattices, in orbital resonances and in musical instruments. Balmer found the rule of the hydrogen lines in 1885 by hand from four lines **[Q]** [1]; symbolic regression has rediscovered known formulae of physics from data **[Q]** [5].

A second reason comes from the corpus itself. According to FFGFT the measured ratios are not exact fractions: the basic form is exact, the measured value carries the embedding error (Doc. 333, Doc. 358 Theorem A), and the PDG values contain model-dependent contributions (Doc. 351). A learning procedure that sees only measured values learns the shifted form, not the rule beneath it. In language this layer does not exist.

ay.5 The null test

Question. Can it be tested how much structure the measured values carry themselves and how much the prior contributes?

Answer. Yes, with a symbolic regression in its exact form, the exhaustive search over an expression grammar. All expressions

$$x = c \cdot d^{1/k} \cdot \xi^{m/n} \cdot \pi^{s}$$

are searched, with a coprime fraction $c = p/q$, $p, q \le 200$, roots $d^{1/k}$ with $d \le 30$ and $k \in \{2, 3\}$, $\xi = 1/7500$ with $n \in \{1, 2, 3, 6\}$ and $|m/n| \le 4.5$, and $s \in \{-1, 0, 1\}$. The prime factors of p, q and d are taken from a chosen set of primes. With $\{2, 3, 5, 7, 11, 13\}$ these are 23 084 325

expressions; separately, the search runs over the fractions p/q alone (3109 expressions). Each expression is assigned a complexity in bits that grows with numerator, denominator, root and exponent. The p-value is the fraction of random numbers (log-uniform, one decade around the measured value) that the grammar hits equally well at equal or lower complexity. Hits more precise than the measurement uncertainty count only up to that uncertainty.

With prior. The formulae of the basic form from Doc. 387 are fixed; they are compared with all expressions of equal complexity in the same grammar:

Quantity	Formula	Deviation	p
v/E_P	$\xi^4/(5\pi)$	$-0.23\,\%$	0.02
M_W/M_Z	$\sqrt{7/9}$	$+0.06\,\%$	0.09
λ_{CKM}	$\xi^{1/6}$	$+0.45\,\%$	0.11
m_h/M_Z	$11/8$	$+0.15\,\%$	0.42
m_μ/m_e, m_τ/m_μ, Δm^2 ratio, $\sin^2\theta_{13}$, m_t/M_Z		0.1–1.5 %	1.0

Combined by Fisher's method this gives $p = 0.41$ **[K]**. At deviations of 0.5 to $1\,\%$ the grammar hits every number; there the lepton formulae carry no numerical evidential weight. The most striking case is v/E_P. For $\alpha^{-1} = 3700/27$ (Doc. 338): the deviation of 7.6 ppm is also reached by a fraction with denominator up to 27 for a random number in about half of the cases ($p \approx 0.47$) **[K]**. This holds, however, only if $3700/27$ is treated as a freely searched fraction. In the corpus it arises as the reduction of $\frac{74}{75}\cdot 7500/54$ from $1/\xi$, the rule $74/75$ and the integer 54; calculated according to these constituents, the chance probability is about 10^{-3} (Docs. 394, 395) **[K]**.

Without prior. The blind search over the same nine quantities stays at chance level for every set of primes: for $\{2,3,5,7,11,13\}$, for the control sets $\{2,3,5,7,17,19\}$ and $\{2,3,5,13,17,23\}$, for $\{2,3,5,7\}$, and likewise for pure random numbers **[K]**.

Positive control. The Balmer lines of hydrogen have the exact ratios $27/20$, $28/25$ and $200/189$; the air wavelengths match them to 10^{-5} **[Q]** [8]. With the narrow grammar of fractions this rule is found from just three ratios (combined $p = 0.002$), with the full grammar it is not ($p = 0.47$) **[K]**. The 23 million expressions lie so densely that random numbers are hit equally well.

What follows. Even a true rule is visible from a few examples only with the right, narrow prior. Numerical proximity does not carry FFGFT; that must be done by the chain of derivation: geometry, Galois structure, formula. This corresponds to the reading of Doc. 384 and Doc. 387, according to which the ratios of the basic form test the structure. Nothing is refuted by the test; it measures only what the numbers yield on their own.

Limits of the test. The bit measure of complexity is a choice; a different measure shifts the individual p-values **[S]**. There are only nine quantities. The formulae were found with knowledge of the measured values; the comparison with equally complex alternatives is therefore the fair standard, not the raw hit. The lepton residuals of 0.5 to $1\,\%$ are regarded in the corpus as genuine structure (K_{frak}, Doc. 352); the test can only count them as misses.

ay.6 Language changes constantly, and yet it works

Question. Language changes all the time, new word formations appear, grammar varies across dialects. Why does it work nevertheless?

Answer. The very changeability confirms the picture. Language operates with a large tolerance and at the same time provides very many examples.

On tolerance: a sentence remains intelligible if a word is misspelt, a dialect word is substituted or the word order is twisted. Printed language is highly redundant **[Q]** [2]. The precision that language requires is coarse; in the terms of the corpus: large ε, hence small Q_c. Language lives in the range in which structures are still separable.

On the examples: new word formations and dialects are not isolated cases but patterns in their own right, with many instances. A dialect has millions of speakers and countless sentences; new words follow old construction rules, so “to google” is conjugated like any regular verb. A learning procedure therefore does not learn a single rigid grammar but a distribution of variants, and each variant is densely attested. Beneath the dialects lies a common deeper structure, as with Balmer: many lines, one rule. Where it fails, the picture fits as well: with a small dialect that has little text, a rare technical term or an entirely new expression, language models become uncertain.

Music shows both sides. The harmonic grammar of octave, fifth and third, that is, of the primes 2, 3 and 5, is universally recognisable and has many “dialects”: just intonation, equal temperament, meantone, folk-music tunings. It works because the ear tolerates deviations of the size of a comma. This is the statement of Doc. 342 and Doc. 343: up to $k \approx 3$ the Galois structures are separable at this tolerance, beyond it they are not **[B]**. The harmonics of the small primes are as dense and robust in nature as grammar. The fine ratios of particle masses, where a tenth of a per cent matters, lie beyond this limit; there the structure must come from the derivation.

ay.7 Why does language forgive errors and numbers do not?

Question. Why does language allow this tolerance, while mathematics and numbers do not?

Answer. Because of redundancy. In terms of communications engineering, language is an error-correcting code and a number an uncoded transmission.

Most random sequences of letters are not words. If one letter is wrong, the result is usually an invalid word, and the receiver corrects it to the nearest valid one; syntax and context provide additional checks. This corresponds to a code with a large Hamming distance: errors are detected and corrected. Many forms moreover lead to the same meaning, and at the destination only the meaning counts.

For a number the distance is zero. Every sequence of digits is a valid number; with one wrong digit 137.036 becomes 137.037, not a faulty number but a different, equally valid one. Nothing signals the error, nothing corrects it. Mathematics also chains: an error enters every following step and is often amplified, whereas errors in a sentence do not add up.

In counting, by contrast, mathematics is tolerant. Integers, group orders and winding numbers behave like words: between 3 and 4 there is no valid group order, and 3.01 is immediately recognised as invalid. In the corpus this appears as “state versus process”:

counting is exact, a bit is a winding number (Doc. 343, Doc. 247). The algebraically proved statements of FFGFT, namely $\mathbb{Z}_3$, $GF(27)$, orbits and fixed points, are therefore as error-tolerant as language. The intolerance arises at a single point: at the transition from the discrete to the continuum, when an integer ratio is compared with a measured value on the real axis. There every neighbouring value is valid, and no distance absorbs the noise.

ay.8 Do AI models draw on a finished framework?

Question. AI models ultimately have a finished framework from which they draw; not everything has to be searched anew for patterns. How does this fit with the noise limit?

Answer. After training a model is indeed a finished framework; when answering, nothing is searched any more, what has been learned is applied. The weights are the frozen prior, in the sense of Doc. 359 Theorem D the hypothesis class.

This framework, however, has itself been learned from data. It is a condensed form of the training texts and contains only what occurred densely there. Where the texts were sparse it has gaps, and the model fills them with something plausible; this is where invented references come from. It is moreover statistical rather than exact, built for language and unreliable for exact numbers. Modern systems therefore call a tool for the exact part: the statistical framework decides what is to be calculated, the tool calculates. In the work on the corpus this is the division of labour between formulation and verification script.

FFGFT is also a finished framework: ξ, $T^4/\mathbb{Z}_3$, the Galois structure. It is, however, not learned from data but set and derived. The learned framework is broad and tolerant but without a guarantee of exactness; the set one is exact and discrete but must prove itself against the measured value, precisely where the discrete meets the continuum. After the null test, a learning procedure cannot find the FFGFT structure on its own; it can, however, be given it as a built-in framework, as proposed by the architecture sketch of Doc. 359 **[S]**.

ay.9 Where do the rules of an AI assistant come from?

Question. An AI assistant works with many prescribed rules. Are they learned or entered directly?

Answer. Both, in three layers.

The first layer is learned and makes up the largest part. First, training on very large amounts of text produces the statistical framework. Then a second training stage shapes the behaviour, with ratings by humans and with written principles such as honesty, avoidance of harm and the admission of uncertainty. These principles, too, then reside in the weights and are not looked up anew for every answer.

The second layer is supplied afresh with every conversation: instructions from the product, for example how tools are to be used and what may happen only after asking, as well as the user's stored preferences. This layer is the easiest to change; it takes effect immediately, without new training. The third layer is the running context: everything that stands in the conversation and everything that has been read.

The layers can contradict one another. During the work on this corpus, an automatic check demanded after every answer that changes be transferred to the public repository, while the author's stored preference forbids exactly that; the explicit preference was followed. The division fits the topic of this document: the first layer is the learned, tolerant framework, the second and third are set rules, and the verification scripts are the exact tool that checks both.

ay.10 How does the next step arise after an error message?

Question. Between an error message from a script and the decision about what is most sensible next, there must be many layers.

Answer. Yes, but differently from what one might expect: there is no separate decision module, only layers in the network and a simple loop around it.

The loop: the assistant writes a tool call. The surrounding program executes it and appends the result, including an error message, to the text of the conversation. Then the whole is read again and the next step is written. A plan exists only as text in this context.

The layers: the network itself has dozens to more than a hundred consecutive layers, each of which further processes the entire preceding text. From error message, goal, files read and previous attempts the next step emerges. That it is sensible is trained: in the second training stage such sequences were practised on many tasks, and sequences that led to the goal were reinforced.

Two examples from the resetting of the book series: an "Improper alphabetic constant" message when typesetting a chapter title containing φ is a densely attested pattern; the cause, mathematics in the PDF bookmarks, is recognised at once. A volume that produced only 18 instead of more than 300 pages without any error message required an inference instead: the input ended too early, presumably because of an `\end{document}` in one chapter, which a search confirmed. Where patterns are sparse, things are tried out and several attempts are needed. The same therefore holds for the assistant as for FFGFT: try, check exactly, measure with verification scripts, rather than trusting one's own impression.

ay.11 Chance and probability

Question. Does the noise limit not also play a role in chance and probabilities?

Answer. Yes, quite directly: probability theory is the mathematics of the noise limit. Two kinds of probability must be distinguished.

Probability as ignorance: a measured value fluctuates. Averaging N independent measurements reduces the noise as $1/\sqrt{N}$; a hundred times as many measurements reduce it to a tenth **[B]**. This is why m_μ/m_e is known to 10^{-8}, although each individual measurement is coarser. This holds only for random errors; systematic contributions such as the model assumptions of the PDG values (Doc. 351) do not average out. The p-value of the null test belongs here as well: it is the noise limit expressed as a probability.

In language models probability is the output itself. For each word a distribution over all possible next words is computed and one is selected. Where the training data were dense the distribution is sharp and the answer stable; where they were sparse it is flat,

and the selection is almost a matter of chance. A flat distribution is the noise limit of the model, and from outside it cannot be seen, because the sentence still sounds fluent.

Probability as structure: in quantum mechanics probabilities are exact numbers. According to Doc. 367 the Koide phase θ and the icosahedral transition probability p_0 are the same quotient $2/3^2 = 2/9$, two readings of one Born-rule structure. Such a probability is as discrete as a group order. Noise enters only with the measurement: a frequency from N events scatters around $2/9$ by about $\sqrt{p(1-p)/N}$; to separate $2/9$ from the neighbouring fraction $1/5$ at three standard deviations requires about 3200 events **[K]**. In this sense Doc. 230 also treats the Bell correlations: real, but geometrically structured. Probability thus appears twice: as an object of the theory, exact, and as a tool for judging whether the exact can still be recognised in the noise.

ay.12 Does the noise limit enter the weights?

Question. What about the weights of a network: do the noise limits enter there?

Answer. Yes, in two ways, and the weights are themselves an example of the principle.

During learning the weights are adjusted in many small steps, each according to a small batch of examples and therefore noisy. What occurs often pushes the weights again and again in the same direction and prevails; the noise averages out as $1/\sqrt{N}$. What occurs rarely is lost in the noise of the steps. In addition there are deliberately built-in brakes: weights are kept small, and training is stopped in time. A noise limit is thereby set on purpose: fine structure that is not attested often enough should not be learned, because it is mostly noise.

In storage each individual weight is surprisingly imprecise. With 16 bits this amounts to about 2.4 to 3.3 decimal digits **[K]**, and weights are often reduced afterwards to 8 or 4 bits without the model becoming noticeably worse. This works only because knowledge does not reside in individual weights but is distributed over very many at once. The network is thus itself a redundant, error-tolerant code, like language. Good training solutions have exactly this property: a small shift of the weights hardly changes the result **[Q]** [3]. The flip side is the same as for language: exact calculations that require full precision at every position succeed poorly inside the network.

ay.13 Does anything work at all without set framework conditions?

Question. Does this mean that without set framework conditions nothing works anyway?

Answer. Essentially yes; for learning from data this is proved. According to the no-free-lunch theorem of learning theory, averaged over all conceivable tasks no learning procedure is better than any other, not even better than guessing **[Q]** [6]. A procedure is better only if its built-in assumptions fit the actual world. Every procedure has such assumptions, even if they are not stated: Fourier analysis presupposes periodicity, an image network that neighbouring pixels belong together, a language model that text is a sequence with structure. Doc. 359 Theorem D shows the difference in numbers: without a framework about 84 000 examples, with the appropriate framework at most 407, and

from positive examples alone 30 **[K]**. Without a prior every pattern is as good as any other.

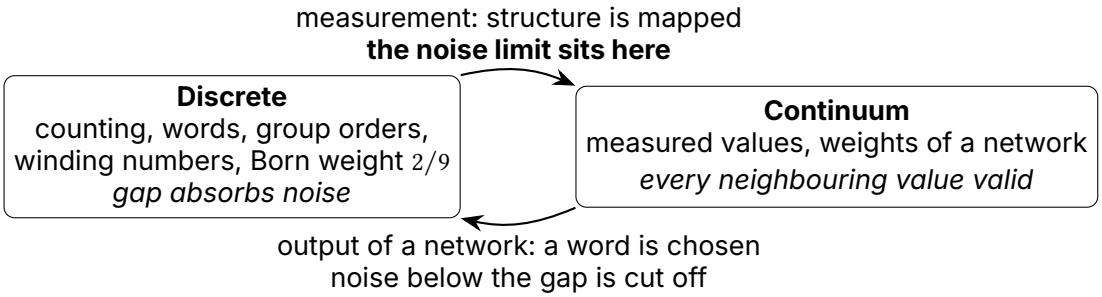
Two qualifications belong to this. First, a framework can be weak when data are plentiful: the more data, the less needs to be specified. Language models have a rather general framework and compensate for it with very large amounts of text. The sparser the data, the stronger and more precise the framework must be; with a few dozen constants of nature it must carry practically the whole structure. This is the situation of FFGFT.

Second, a framework does not work merely because it has been set. A wrong or too broad framework also adapts and then finds false patterns. The Balmer test of Section ay.5 shows both sides: the narrow framework of fractions finds the true rule; the broad framework with 23 million expressions fits everything and therefore no longer distinguishes anything **[K]**. A framework must therefore not only be set but also tested; the null test measures whether it achieves more than chance.

For physics this means: every theory has settings. The Standard Model has its gauge groups and about two dozen measured parameters; FFGFT has $T^4/\mathbb{Z}_3$, ξ and the integer structural choices. The question is not whether there are settings, but how few there are, how much follows from them, and whether what follows withstands a test against chance: setting, derivation, comparison, null test.

ay.14 The common core in a single picture

All answers lead to the same structure. On the discrete side stand counting, words, group orders, winding numbers and Born weights; there are gaps between the valid values, and noise smaller than such a gap is absorbed. On the continuous side stand measured values and the weights of a network; there every neighbouring value is valid.



A network computes continuously inside and outputs discrete words; in the step from the continuum to the discrete the noise is cut off as long as it is smaller than the gap between two words. In measurement it runs the other way: an exact, discrete structure is mapped onto the continuum, and there no gap absorbs the noise. The four core statements of this document follow:

1. Discrete structure is error-tolerant; the noise limit sits at the transition from the discrete to the continuum **[B]**.
2. Structure that is densely and variously attested in the data is learned; structure from a few exact values is not found from data, not even by a network **[K]**.

3. Such structure must be specified and then tested: setting, derivation, comparison with the measured value, null test. This is the path FFGFT takes **[S]**.
4. No learning without a framework, but a framework is only as good as its test **[Q]** [6], **[K]**.

Verification scripts: `2/python/Dok393_Skripte/` — `pruef_393a_nulltest_grammatik.py` (8/8, result log `pruef_393a_ergebnis.txt`), `pruef_393b_zahlen.py` (11/11); together `run_all_393.sh`, 19/19.

Tool and method

Claude (Anthropic) was used as a tool for: formulating the answers to the author's questions, writing and running the verification scripts, and preparing the LaTeX sources. The sections on how AI assistants work describe the general operation of such systems; they do not rest on an inspection of the model's own weights. Every number in the text is secured by a check in the scripts named; literature references are marked **[Q]**.

[1] J. J. Balmer, *Notiz über die Spectrallinien des Wasserstoffs*, Annalen der Physik **261** (1885) 80–87.

[2] C. E. Shannon, *Prediction and entropy of printed English*, Bell System Technical Journal **30** (1951) 50–64.

[3] S. Hochreiter, J. Schmidhuber, *Flat minima*, Neural Computation **9** (1997) 1–42.

[4] N. Rahaman et al., *On the spectral bias of neural networks*, Proceedings of the 36th International Conference on Machine Learning, PMLR **97** (2019) 5301–5310.

[5] S.-M. Udrescu, M. Tegmark, *AI Feynman: A physics-inspired method for symbolic regression*, Science Advances **6** (2020) eaay2631.

[6] D. H. Wolpert, *The lack of a priori distinctions between learning algorithms*, Neural Computation **8** (1996) 1341–1390.

[7] S. Shalev-Shwartz, S. Ben-David, *Understanding Machine Learning: From Theory to Algorithms*, Cambridge University Press (2014).

[8] A. Kramida, Yu. Ralchenko, J. Reader, NIST ASD Team, *NIST Atomic Spectra Database*, National Institute of Standards and Technology, Gaithersburg (hydrogen lines Hα–Hδ, air wavelengths).

Chapter az

Doc. 394 — How Far Apart the Fractal Corrections Lie

Abstract

The basic form of FFGFT is ratio-based; a fractal correction appears only in the translation into SI units. As a principle, the corpus uses a single factor for this, $K_{frak} = 1 - 100\xi = 74/75$, the per-step factor of the scale recursion (Docs. 133, 300, 384). In the individual calculations, however, different values and residuals stand side by side (R154). This document writes each of them as n in $K = 1 - n\,\xi$, compares it with $n = 100$ and sorts by origin. Result: of all candidates, only one is a genuine test of the SI translation, the fine-structure constant, and even that only together with the anchoring of the masses in MeV; it requires $n = 101.25$, that is $1.25\,\xi$ away from the rule. All other deviations are algebraic consequences, fitted quantities, other quantities under the same name, or the distance of the lepton ladder from the Koide form. The lepton residuals are therefore not a recursion of the measurement but an approximation error of the ladder, which the Koide form predicts without any measured value. The uniform rule is not refuted; so far it has been tested exactly once. If the energy unit is chosen so that $m_e m_\mu = 54$, the rule lies only $0.06\,\xi$ from α, and in additive form; this is an assignment of the residual, not a prediction.

Status markers

- **[B]** Algebraically proved (rearrangement without additional assumption).
- **[K]** Numerically confirmed (check script).
- **[S]** Assumption, open, or classification without a calculation of its own.
- **[X]** Excluded by the calculation.
- **[Q]** Taken from the literature, with source.

Measured values: m_e, m_μ, α and G from CODATA 2022 **[Q]**, $m_\tau = 1776.93 \pm 0.09$ MeV from PDG (pdgLive, 2026 edition) **[Q]**. The corpus has so far used $m_\tau = 1776.86$ MeV; where this makes a difference, both values are given.

az.1 Starting point

In the basic form, in natural units with $\alpha = 1$, a fractal correction has no place: $\xi E_0^2 = 1$ is exact, and in ratios every common factor cancels (A040, Docs. 070, 122). A correction arises only when a quantity of the basic form is compared with a value measured in SI units. For this the corpus sets a single factor,

$$K_{\text{frak}} = 1 - 100\,\xi = \frac{74}{75}, \tag{az.1}$$

the per-step factor of the scale recursion $\xi_{n+1} = \xi_n(1 - 100\,\xi_n)$ (Docs. 133, 300). The reading that the measurer stands inside this recursion, so that the correction is part of every measured value, requires the same factor to appear wherever a translation takes place. Whether this is the case was left open by R154: besides $74/75$, the corpus contains 0.986, a value required by α, a chain residual for G, generation-dependent lepton residuals, a hadronic factor and $e^{-\xi}$.

To make these values comparable, each is written as n in $K = 1 - n\,\xi$. The rule is $n = 100$; the distance $n - 100$ states by how many ξ a value deviates from it.

az.2 All values on one scale

Quantity	K	n	Origin
Rule $1 - 100\xi = 74/75$	0.986667	100	Docs. 133, 300, 384
multiplicative form $(1-\xi)^{100}$	0.986754	99.34	R139
required by α, $\xi m_e m_\mu \alpha^{-1}$	0.986501	101.25	R135
1 MeV reference	0.986501	101.25	Doc. 386
G formula, rounded	0.986	105	Doc. 012
G formula with chain residual	1.0124	−93	R141, Doc. 383
hadronic factor $144/125$	1.152	−1140	R123
$e^{-\xi}$	0.999867	1	Doc. 037

[K] In addition there are the lepton residuals $\varepsilon_e = 192\,\xi$, $\varepsilon_\mu = 152\,\xi$ and $\varepsilon_\tau = 73\,\xi$ (Doc. 352). They measure the distance of the bare ladder $m_i^\circ = r_i\,\xi^{p_i}\,v K_{\text{frak}}$ from the measured value and fall with the generation.

Read as a plain list, this looks like a rule that holds only in principle. The numbers, however, have different origins, and only some of them can test the rule at all.

az.3 What tests the rule and what does not

Algebraic consequences

The residual of $-2.6\,\%$ for G in the one-anchor chain (Doc. 383) is not a finding of its own. With m_e as anchor, E_P lies $+1.34\,\%$ from the measured value, and because $G = \hbar c^5/E_P^2$ the deviation of G is exactly twice as large **[B]**. That it seems to resemble two correction steps follows from the exponent alone.

The 1 MeV reference from Doc. 386 is the same number as the factor required by α. From $E_0^{\text{bare}} = \sqrt{m_e m_\mu}$ it follows that $\left(E_0^{\text{bare}}\sqrt{\xi/\alpha}\right)^2 = \xi m_e m_\mu \alpha^{-1}$ **[B]**; the residual of 8×10^{-5} in the reference energy is half of the residual 1.7×10^{-4} in K. It is one finding, not two.

Fitted quantities

The G formula $G = \xi^2/(4m_e)\, C_{\text{conv}}\, K_{\text{frak}}$ matches CODATA only because $C_{\text{conv}} = 7.783 \times 10^{-3}$ contains the residual of the one-anchor chain; the chain alone gives 7.58×10^{-3} (R141) **[K]**. Whether the formula uses 0.986 or $74/75$ shifts G by $0.068\,\%$, which is smaller than the fitted part. The G formula can therefore neither confirm nor refute the rule.

The electron residual of the chain, $-1.32\,\% = -99\,\xi$, lies strikingly close to one correction step. The chain, however, contains the relation $v/E_P = \xi^4/(5\pi)$ with the factor 10 fitted to the measured value (Docs. 149, 383); with a different anchor (muon or tau) the residual would be a different one. The closeness to $100\,\xi$ depends on this choice and carries no weight **[S]**. The same holds for the absolute level of the ε_i: it depends on the anchor v, which is a convention of the Standard Model via G_F (R112). Only the differences between the generations are independent of the anchor.

Other quantities under the same name

$144/125$ is the ratio of two ladder coefficients, $r_\mu/r_\tau = (16/5)/(25/9) = 144/125$ **[B]**, and lies on the other side of one. It is the bridge factor from Doc. 158 (R140), not an SI correction. $e^{-\xi}$ in Doc. 037 is a single ξ step with a different role; the document itself notes that the G formula there uses $1 - 100\xi$.

az.4 The lepton residuals are the distance of the ladder from the Koide form

The lepton ladder $m_i \propto r_i\, \xi^{p_i}$ with $r = (4/3,\ 16/5,\ 25/9)$ and $p = (3/2,\ 1,\ 2/3)$ gives the ratios

$$\frac{m_\mu}{m_e} = \frac{12}{5}\,\xi^{-1/2} = 207.846, \qquad \frac{m_\tau}{m_\mu} = \frac{125}{144}\,\xi^{-1/3} = 16.992. \tag{az.2}$$

In these ratios v, E_P and every common correction factor cancel. They lie $+39.10\,\xi$ and $+77.56 \pm 0.38\,\xi$ above the measured values **[K]**. These are the generation differences of the ε_i; they cannot contain a common factor such as K_{frak}.

The Koide form with the angle $\theta = 2/9$ and the amplitude $d/c = \sqrt{2}$ (Docs. 353, 367),

$$\sqrt{m_k} \;\propto\; 1 + \sqrt{2}\,\cos\!\left(\tfrac{2}{9} + \tfrac{2\pi k}{3}\right), \tag{az.3}$$

matches the same ratios to $0.07\,\xi$ and $0.16 \pm 0.38\,\xi$, for the tau therefore within the measurement uncertainty **[K]**. Comparing ladder and Koide form directly with each other, without any measured value, the ladder lies $39.02\,\xi$ and $77.40\,\xi$ above **[K]**. Up to $0.2\,\xi$, these are exactly the lepton residuals.

This settles their origin. The ladder is a rational approximation of the Koide form, and the lepton residuals are its approximation error. The measurement plays no part in this:

the error already lies between two exact constructions of the basic form. This confirms reading B from R121 (distance between cosine and power law, Docs. 367, 369) and, for the ratios, answers the question from R113 of why the ε_i fall with the generation.

The ratio of the two residuals is $77.40/39.02 = 1.984$. It is close to two, but it is not two **[K]**; the measurement gives 1.984 ± 0.010 and agrees with the Koide value. An integer step law for the generations cannot be read off from this **[X]**.

az.5 What remains

After removing the algebraic consequences, the fitted quantities, the unrelated quantities and the ladder residual, a single value remains that tests the rule independently: the fine-structure constant. It connects the basic form $\xi E_0^2 = 1$ with the SI value $\alpha^{-1} = 137.036$ via $E_0^2 = m_e m_\mu / K$ and requires

$$K_\alpha = \xi\, m_e m_\mu\, \alpha^{-1} = 0.986501, \qquad n = 101.25. \tag{az.4}$$

[K] The distance from the rule is $1.25\,\xi$ or 168 ppm. The measurement uncertainty of K_α is below 3×10^{-8}; the residual is therefore real and not explained by measurement errors. The additive form $1 - 100\xi$ lies closer than the multiplicative form $(1-\xi)^{100}$ ($1.25\,\xi$ against $1.90\,\xi$), but does not separate them more sharply than their mutual distance (R139). A second-order term, $1 - \varepsilon - \varepsilon^2$ with $\varepsilon = 1/75$, would give $n = 101.33$ and misses K_α by 12 ppm; it is not adopted **[S]**.

Direction and location of the residual

The direction of the residual is unambiguous. Without correction, $1/(\xi m_e m_\mu) = 138.911$, that is $1.37\,\%$ above the measured value; the rule subtracts $1.333\,\%$, while $1.350\,\%$ would be needed **[K]**. The rule thus supplies $98.8\,\%$ of the required correction, and the prediction $\alpha^{-1} = 137.059$ remains too large. With the factor applied the other way round, $\alpha = \xi m_e m_\mu K$, the deviation would be $+2.7\,\%$; this direction is excluded **[X]**. The measured value leaves no room either: the magnetic moment of the electron with QED and the recoil measurements on Cs and Rb atoms, which do not rely on QED, agree to about 10^{-9} **[Q]**, roughly 10^5 times more precisely than the residual. The running of α does not help: the measured value is the limit at vanishing energy and hence the largest α^{-1}; any other scale makes the residual larger.

What is open, in contrast, is where the residual sits. The product $m_e m_\mu = 53.9913\,\mathrm{MeV}^2$ lies $1.21\,\xi$ (161 ppm) below the Galois value $54 = |\mathrm{GF}(3)^*| \cdot |\mathrm{GF}(27)|$ (Doc. 338). With 54 and the rule one obtains $\alpha^{-1} = \frac{74}{75} \cdot \frac{7500}{54} = 3700/27$ **[B]**, that is 137.037, only 7.6 ppm from the measurement **[K]**. Of the 168 ppm, therefore, 161 ppm fall on the product of the masses in MeV and 7.6 ppm on the Galois residual **[K]**. Read according to its constituents, this agreement is not a repetition of the same finding. $3700/27$ is the reduction of $\frac{74}{75} \cdot 7500/54 = 7400/54$; $7500 = 1/\xi$ and $74/75$ come from the basic form, the only new element is the integer 54. The hit then states that $7400/\alpha^{-1} = 54.0004$ lies only 7.6 ppm from an integer; by chance this happens with a probability of about 8×10^{-4} **[K]**. In addition, the mass product lies 161 ppm from the same integer, by chance with about $1.7\,\%$ **[K]**; together this is about 1.4×10^{-5}. In substance this means: the residual of K against the rule (-168 ppm)

and the residual of the product against 54 (-161 ppm) are equal to 7.6 ppm, as expected in the 54 world, in which both are the same deviation of the unit. The precondition is that $74/75$ was fixed independently of α; this is the case (Docs. 133, 300), since α in MeV requires $n = 101.25$ and not 100 **[S]**. The Galois decomposition itself, by contrast, hardly narrows things down, since 24 of the 31 integers between 40 and 70 are products of two Galois orders (q or $q-1$ of a prime power $q \le 81$) **[K]**. The null test from Doc. 393 ($p \approx 0.47$) treats $3700/27$ as a freely searched fraction and therefore underestimates the agreement. In MeV the measured product is not 54; it can be exactly 54 only in a structural energy unit that lies 80 ppm below the MeV. The MeV itself comes from the SI conventions, the numerical value of e and m_e via $2\,\mathrm{Ry}/\alpha^2$ (Doc. 386). The 54 becomes robust only once this unit is fixed from the basic form without $m_e m_\mu$, for instance via E_P and ξ or the Higgs sector **[S]**. The MeV is not a geometric quantity but a unit of the SI chain whose numerical value depends on the charge e (Docs. 384, 386). Whether the residual lies in the factor K or in the anchoring of the product in MeV cannot be decided by α alone; read in the energy unit, it amounts to 84 ppm. Strictly speaking, α therefore tests the rule together with this anchoring.

The non-closure of the running recursion (Doc. 295) does not explain the residual. Going forward, the steps become smaller, the correction falls below $n = 100$ and goes in the wrong direction; the continuous spiral going forward, $\ln(76/75)$, coincides with the multiplicative form ($n = 99.34$) **[K]**. One step back, that is radius 74 instead of 75 and $K = 73/74$, gives $n = 101.35$ and overshoots by $0.11\,\xi$; the corpus offers no justification for this step **[S]**. The mass side with the factor $1/K$ is already contained in $E_0^2 = m_e m_\mu / K$ and supplies no second factor **[B]**.

The uniform rule is therefore not refuted. So far it has been tested exactly once, together with the anchoring in MeV, and passes this test at $1.25\,\xi$. A second quantity that, like α, connects a quantity of the basic form with an SI measured value does not exist: $\alpha_{\mathrm{SI}} = \xi (E_0/1\,\mathrm{MeV})^2$ is the only relation in the corpus with an SI unit, and the SI sector therefore has exactly one degree of freedom, in which K and the MeV anchoring coincide. The muon lifetime, too, tests only the dimensionless ratio v/m_μ, not K (Doc. 395) **[B]**.

az.6 The 54 world: the unit from the mass product

Because the basic form supplies only ratios, every absolute number needs an energy unit. This unit can be fixed so that the mass product becomes an integer: $m_e m_\mu = 54\,E_u^2$. E_u denotes the energy unit fixed in this way, $E_u = \sqrt{m_e m_\mu / 54} = 0.99992\,\mathrm{MeV}$; the symbol is chosen to avoid confusion with the atomic mass unit u. This choice is an assumption **[S]**. Like every anchor, it costs exactly one measured value and shifts the residual from the factor K into the unit, in the sense of the assignment in Doc. 386.

Masses. With the Koide ratios ($\theta = 2/9$, $d/c = \sqrt{2}$) one obtains $m_e = 0.511038\,E_u$, $m_\mu = 105.6674\,E_u$ and $m_\tau = 1777.12\,E_u$ **[K]**. Against the measured values, each lepton gives the same unit, $E_u = 0.99992\,\mathrm{MeV}$ to about 10^{-5} (from the electron 0.999925, from the muon 0.999915, from the tau $0.999894\,\mathrm{MeV}$) **[K]**. The assumption is therefore self-consistent.

Fine-structure constant. In this unit the bridge reads $\alpha^{-1} = K/(54\,\xi)$, dimensionless and without MeV. The measured value requires $K = 0.986659$, that is $n = 100.06$ instead of 101.25 **[K]**. The residual against the rule shrinks from $1.25\,\xi$ to $0.06\,\xi$ (7.6 ppm). At the same

time the 54 world separates the two forms of the factor: the additive form $1-100\xi$ lies $0.06\,\xi$ away, the multiplicative form $(1-\xi)^{100}$ by contrast $0.71\,\xi$ **[K]**. In this bookkeeping the question of the form from R139 is decided in favour of the additive form.

What stays the same. E_P and G from the one-anchor chain remain at $+1.34\,\%$ and $-2.62\,\%$ **[K]**. This residual depends on the fitted factor 10 in v/E_P and on the ladder, not on the energy unit.

If 54 holds exactly. Then K is no longer a rule of its own but follows from α: from $\alpha^{-1} = K/(54\,\xi) = \frac{1250}{9}\,K$ one obtains

$$K = \frac{9\,\alpha^{-1}}{1250} = 0.986\,659\,194 \pm 1.5\times10^{-10}, \qquad n = 100.056. \tag{az.5}$$

[K] $74/75$ then remains the leading term, the frozen closure from Doc. 295, and matches K to 7.6 ppm; the residual is $0.056\,\xi$, about $\xi/18$. The corpus knows no mechanism for this residual: second-order terms of the recursion are of size $(100\xi)^2 = 1.33\,\xi$, the variants of non-closure 0.5 to $2\,\xi$ **[K]**. A fitting fraction such as $1/18$ misses the value by about 430 standard deviations and is not adopted (cf. Doc. 393) **[S]**. The choice is symmetric: if $74/75$ holds exactly, the product would have to be $54.0004\,E_u^2$. In both readings, rule and Galois number agree to 7.6 ppm; without the 54, to 168 ppm.

Read geometrically: an orbit open towards the inside. With the pole of the spiral at $n=-75$ and the radius $\rho = n+75$, the defect per turn is $d = 1/\rho$ (Doc. 295). In the 54 world $d = 0.0133408$, that is $\rho = 74.958$ instead of 75 **[K]**. After 75 turns the orbit therefore does not close exactly but runs ahead by 0.00056 turns, about $0.2°$; a full additional turn accumulates after roughly 1800 closure cycles **[K]**. The rotation number is then no longer rational; the orbit is open and precesses slowly. The direction is fixed: outwards, in the running direction of the recursion from Doc. 295, ρ grows, the steps become smaller and the orbit falls behind (75 steps give only 0.69 turns). Inwards, towards the pole, ρ shrinks, the steps become larger and the orbit runs ahead. The residual therefore lies on the inner side, 0.042 of a recursion step away from the frozen value; this fits the reading of a fractal structure that goes inwards **[S]**. The magnitude is open: a full step inwards ($\rho = 74$) would enlarge the defect by $1.35\,\%$, 24 times as much, and a reason for exactly $\rho = 74.958$ without knowledge of α is missing. The residual is of third-order size, $3.15\,d^3$; a factor close to π misses the value by about 170 standard deviations and is not adopted (cf. Doc. 393) **[S]**.

Limit. Rule and 54 together give $\alpha^{-1} = 3700/27$, 7.6 ppm from the measured value known to 10^{-10}. Both being exact at the same time is excluded **[X]**; one of the two assumptions carries this residual. The 54 world predicts nothing new; it assigns the residual differently. It is a consistent bookkeeping in which the rule holds almost exactly and in additive form. A second SI quantity on which E_u could be tested does not exist (Doc. 395); the 54 world becomes robust only if the 54 or the residual of $0.056\,\xi$ follows from the structure **[S]**.

What the measurement allows. The measured values do not exclude the exact 54; they only exclude that something else holds exactly alongside it:

Assumption	allowed by the measurement?	
$m_e m_\mu = 54$ in MeV	no	$7385\,\sigma$
$m_e m_\mu = 54\,E_u^2$	yes	exactly, this is how E_u is fixed
54 exact and $K = 74/75$ exact	no	7.6 ppm, about $49\,000\,\sigma$ in α
54 exact and $K = 1-100.056\,\xi$	yes	

[K] The tau fits within its uncertainty, and α fits if K carries the residual of $0.056\,\xi$. In the reading with an exact 54, therefore, α, the lepton ratios in the Koide form and the rule $74/75$ agree up to 7.6 ppm. The measurement only requires that these 7.6 ppm be accommodated somewhere, and the open recursion offers itself for this: the orbit that runs ahead inwards by about 0.2° per 75 turns is exactly this residual in geometric form **[S]**. Not covered are E_P and G; their residual of $+1.3$ and $-2.6\,\%$ depends on the fitted factor 10 and on the ladder and remains in every reading of the unit.

az.7 Conclusion

The list from R154 looked like many different corrections. Sorted by origin, what remains is one rule with one test. The lepton residuals do not belong to the SI translation but to the structure of the basic form itself: they disappear when the ladder is replaced by the Koide form. The reading that the measurer stands inside the recursion therefore concerns only the translation into SI values, and there so far only α. The ratios need no correction; in the Koide form they are exact up to the measurement accuracy. If, in addition, the mass product is fixed exactly at 54 as the unit, the rule $74/75$ holds up to $0.056\,\xi$, and the remaining residual of 7.6 ppm can be read as an open recursion, an orbit that slowly runs ahead inwards (Section 6) **[S]**.

What remains open, first, is where the residual for α sits. Calculated in MeV, the rule lies $1.25\,\xi$ from the value required by α; if the energy unit is fixed via $m_e m_\mu = 54$, only $0.056\,\xi$ remain. Which of the two readings holds cannot be decided by α alone (R135). The form of the factor depends on this as well: in the 54 reading the additive form is clearly better, while calculated in MeV the additive and multiplicative forms cannot be separated (R139). It also remains open whether the small residual of $0.056\,\xi$ has a mechanism, for instance the open recursion inwards.

A second, independent test in the SI sector cannot exist, because this sector has only one degree of freedom; the muon lifetime tests the normalisation of the ladder to v, not K (Doc. 395). Independently of this, the factor 10 in v/E_P remains fitted (Docs. 149, 387); the residual for E_P and G depends on it **[S]**.

Check script: `2/python/Dok394_Skripte/pruef_394_korrekturabstaende.py`

Chapter ba

Doc. 395 — The Measurement Chain as a Channel

Abstract

Communications engineering no longer fights echoes; it measures and uses them: known pilot symbols reveal the channel, a suitable basis makes it diagonal, regeneration prevents noise from adding up, and the Shannon limit states how much information a noisy channel can deliver at most. This document transfers the four methods to the question of whether recursions are part of the measured values (Doc. 394). Results: in the ratio sector the channel is flat; with the Koide form as pilot, the lepton ratios deviate by less than $1\,\xi$. A mode-dependent channel factor is thereby limited to at most about $1\,\%$ of the correction step $100\,\xi$. In the SI sector there is exactly one pilot, α, and there cannot be more in a ratio-based theory: the only relation with an SI unit is $\alpha_{SI} = \xi (E_0/1\,\mathrm{MeV})^2$, and K and the MeV anchoring are the same degree of freedom. The muon lifetime tests not K but the dimensionless ratio v/m_μ; the bound on the mode factor needs only the derived angle $2/9$. For quantum computers it follows that a common gate factor is absorbed by the calibration and is in principle not measurable; only a mode-dependent part of about 10^{-4} would be visible, far below today's decoherence. Discrete structures with small denominators survive the measurement chain; the SI correction of $1.33\,\%$ leaves denominators unambiguous only up to about 11. As a yardstick against number mysticism, a hit counts only if the measurement delivers more information than the formula contains; the Koide parameters $2/9$ and $\sqrt{2}$ pass this clearly, and so does $\alpha^{-1} = 3700/27$ when it is calculated according to its constituents $7500 \cdot (74/75)/54$ rather than as a freely searched fraction.

Status markers

[B] Algebraically proved (rearrangement without additional assumption).
[K] Numerically confirmed (check script).
[S] Assumption, open, or classification without a calculation of its own.
[X] Excluded by the calculation.
[Q] Taken from the literature, with source.

Measured values: m_e, m_μ and α from CODATA 2022 **[Q]**, $m_\tau = 1776.93 \pm 0.09$ MeV from PDG (pdgLive, 2026 edition) **[Q]**.

ba.1 Occasion

The basic form of FFGFT calculates with ratios; a fractal correction appears only in the translation into SI units. Behind this stands the reading that the measurer stands inside the scale recursion, so that the recursion is part of every measured value. Doc. 394 showed that this reading has so far been tested on exactly one quantity, α, and that the other residuals have a different origin. What remained open was how to test the question systematically.

Communications engineering has faced the same situation for decades: a signal runs through a channel that alters it, and the receiver stands at the end of this channel, not outside it. How it deals with this yields four transferable methods. Two are directly usable as methods, channel estimation with a pilot and the Shannon balance; two are structural hints, the eigenbasis and regeneration.

ba.2 How communications engineering deals with echoes

Besides the direct signal, a radio channel delivers delayed copies, the echoes. They used to be regarded as interference; today they are measured and used as additional signal energy **[Q]**. The *RAKE receiver* (CDMA, UMTS) picks up the delayed copies individually with several "fingers", aligns their delays and adds them with weights [2]. *OFDM* (WLAN, LTE, 5G, DVB-T) spreads the signal over many slow subcarriers; a guard interval longer than the echo delays turns the convolution with the echoes into a simple phase and amplitude rotation per subcarrier [4]. *Maximum ratio combining* and MIMO exploit several propagation paths as parallel channels [3].

What makes this possible is *channel estimation*: the transmitter sends along pilot symbols whose value the receiver knows exactly; from what arrives, it determines the channel and removes it. A second principle is *regeneration*: every repeater decides anew between 0 and 1, so noise does not accumulate along the line. What none of the methods circumvents is the *Shannon limit* $C = B \log_2(1 + S/N)$ [1]. Using echoes increases the effective ratio S/N, and modern error correction comes within fractions of a decibel of the limit [5, 6], but never beyond it. The echoes therefore do not disappear; they are measured, modelled as part of the system and exploited, and the bound remains.

ba.3 Channel estimation with a pilot

Method and condition

Transferred, this means: the measurement chain is the channel H that turns the value of the basic form y_{th} into the measured value $y_{\text{meas}} = H\, y_{\text{th}}$. A pilot is a quantity whose value is fixed algebraically and exactly in the basic form. H follows from it. With this H one calculates the other quantities back. If one common H explains several residuals, there is a uniform recursion; if each quantity needs its own H, there is none, and one then knows it for certain. The condition for this is strict: the pilot must be exact independently. If it contains a model error itself, the channel estimate absorbs this error and the calculation becomes circular.

Ratio sector: the channel is flat

The pilot in the ratio sector is the Koide form with $\theta = 2/9$ and $d/c = \sqrt{2}$ (Docs. 353, 367),

$$\sqrt{m_k} \;\propto\; 1 + \sqrt{2}\,\cos\!\Big(\tfrac{2}{9} + \tfrac{2\pi k}{3}\Big). \tag{ba.1}$$

It fixes the ratios m_μ/m_e and m_τ/m_μ without any measured value. The channel factor is then

$$H_{\mu e} - 1 = -0.074\,\xi, \qquad H_{\tau\mu} - 1 = -0.16 \pm 0.38\,\xi \tag{ba.2}$$

[K], for the muon-electron ratio with a measurement uncertainty of only 2×10^{-8}. Both factors lie below $1\,\xi$, more than a hundred times below the SI correction step $100\,\xi$. In the ratio sector the channel is therefore flat: a recursion of the measurement chain is not visible there. This is to be expected, since a common factor cancels in ratios (A040, Doc. 070).

Pilot with a model error: the ladder

If the lepton ladder $m_i \propto r_i\,\xi^{p_i}$ is taken as the pilot instead of the Koide form, it lies $39.10\,\xi$ and $77.56\,\xi$ above the measured ratios **[K]**; each generation then requires its own channel factor. This, however, is not a channel. Ladder and Koide form differ without any measured value by $39.02\,\xi$ and $77.40\,\xi$ (Doc. 394) **[K]**: the apparent channel is the model error of the pilot. This is exactly what the condition in Section 3.1 warns against. In the language of channel estimation, the falling residuals ε_i from Doc. 352 are the result of an inexact pilot.

SI sector: one pilot, no test value

In the SI sector, α is the only pilot without a fitted factor. It estimates the channel very precisely: $K = \xi m_e m_\mu \alpha^{-1} = 0.986501$, that is $n = 101.25$ in $K = 1 - n\xi$, with an uncertainty below 3×10^{-8} **[K]**; if the energy unit is fixed via $m_e m_\mu = 54$, it is $n = 100.056$ (Doc. 394) **[K]**. Whether the same K also carries other SI quantities cannot be tested, because a second pilot is missing: E_P and G depend on the fitted factor 10 in v/E_P (Docs. 149, 383), and v is a convention of the Standard Model (R112). Channel estimation is thus possible, its test is not. Section 7 shows that a second SI pilot cannot exist in FFGFT at all.

ba.4 Eigenbasis: one channel factor per mode

OFDM switches to a basis in which the channel becomes diagonal; the convolution with echoes turns into a simple factor per subcarrier. FFGFT already has such a basis: the modes on T^4 with their mode-bound time scale $\tilde{T} = 1/m$. A measurement recursion would there be a factor κ_i of its own per mode, and this would lead one to expect generation-dependent corrections **[S]**.

The pilot from Section 3.2 limits this possibility. A difference between the channel factors of two modes appears directly in the ratio of their masses. If the Koide form holds exactly, it follows that

$$|\kappa_\mu - \kappa_e| \le 0.074\,\xi, \qquad |\kappa_\tau - \kappa_\mu| \le 0.92\,\xi\,(2\sigma) \tag{ba.3}$$

[K]. A mode-dependent part of the measurement recursion is thereby limited to at most about $1\,\%$ of the correction step $100\,\xi$. The SI correction must be practically common to the three charged leptons; this supports the uniform rule. The generation-dependent residuals of the ladder do not need this basis; they are the distance from the Koide form (Section 3.3). The bound holds under the assumption that the Koide form is exact; if it is not, the calculation only limits the sum of the Koide residual and the mode factor **[S]**. Section 7 shows that the main part of the bound already holds with the derived angle $2/9$ alone.

The Koide form itself can also be fitted freely to the measured values. With both parameters free, one obtains $\theta = 0.2222248 \pm 0.0000063$ and $d/c = 1.4142089 \pm 0.000011$, that is $2/9$ and $\sqrt{2}$ each within one standard deviation, the uncertainty coming solely from the tau mass **[K]**.

ba.5 Regeneration: the discrete survives the measurement chain

Digital transmission beats noise because a new decision is made at every node. This agrees with Doc. 343: the discreteness of FFGFT is topological; it lies in winding numbers and integer structures, not in the dynamics. Transferred, this means: exact rational ratios and integers can be recovered from a noisy measured value as long as the tolerance is smaller than the distance to the nearest competitor. The limit for this is the Farey height from Doc. 343,

$$Q_c = \frac{\pi}{\sqrt{6\,\varepsilon}}, \tag{ba.4}$$

up to which fractions remain unambiguous at relative tolerance ε:

Tolerance	ε	Q_c
SI correction $100\,\xi$	$1.33\,\%$	11.1
m_τ/m_μ	5.1×10^{-5}	180
m_μ/m_e	2.2×10^{-8}	8700
α	1.5×10^{-10}	1.0×10^{5}

[K] A clear rule for the presentation follows from this. If a ratio is passed through the SI correction of $1.33\,\%$, only denominators up to about 11 remain unambiguous; the denominators 9, 5 and 8 of the basic form survive this, the 75 of the rotation number does not. In the basic form itself, without correction, the ratios are by contrast measured so precisely that large denominators could also be recovered. Continuing corrections accumulate errors; exact ratios are preserved. This supports the guideline of building the theory from the ratio-based basic form and applying corrections only at the end **[S]**.

ba.6 Shannon balance against number mysticism

A measurement with relative precision δ delivers, about a constant whose value is known beforehand only to within a decade, roughly

$$I = \log_2 \frac{\ln 10}{2\,\delta} \text{ bits.} \tag{ba.5}$$

A formula itself contains information; Doc. 393 measures it as complexity C in bits, growing with numerator, denominator, root and exponent. If the space of possible formulas contains more bits than the measurement, some combination inevitably hits. A formal criterion follows from this: a hit counts only if the surplus $I - C$ is clearly positive. The deviation from the measured value is taken as δ, but at most the measurement uncertainty.

Hit	I	C	$I - C$
$\alpha^{-1} = 7400/54$ $(= 3700/27)$	17.2	5.8	+11.5
as a freely searched fraction $3700/27$	17.2	16.6	+0.6
$v/E_P = \xi^4/(5\pi)$	9.0	6.3	+2.6
$M_W/M_Z = \sqrt{7/9}$	10.8	5.4	+5.4
$m_e m_\mu$ close to the integer 54	5.8	0	+5.8
Koide $\theta = 2/9$	15.3	4.2	+11.1
Koide $d/c = \sqrt{2}$	17.2	2.0	+15.2

[K] For $3700/27$ the decisive point is how the formula is counted. As a freely searched fraction it contains almost as much information as the deviation of 7.6 ppm provides and does not count; this is how the null test from Doc. 393 ($p \approx 0.47$) calculates. In the corpus, however, $3700/27$ is only the reduction of $7500 \cdot (74/75)/54$: $7500 = 1/\xi$ and $74/75$ come from the basic form, the only new element is the integer 54. Counted this way, more than 11 bits of surplus remain; estimated directly, $7400/\alpha^{-1}$ lies this close to an integer with a probability of about 8×10^{-4} (Doc. 394) **[K]**. The precondition is that $74/75$ was fixed independently of α (Docs. 133, 300). The closeness of the mass product to 54 delivers just under 6 bits, corresponding to a chance probability of about $1.7\,\%$ (Doc. 394). Far ahead are the two Koide parameters with 11 and 15 bits of surplus; they are the strongest numerical structure of the lepton sector. It should be noted that $d/c = \sqrt{2}$ is equivalent to the Koide relation $Q = 2/3$, known since 1982 [7]; the surplus demonstrates the structure of the data, not yet its derivation from FFGFT. That derivation is given for $\theta = 2/9$ in Doc. A110 and for the amplitude in Docs. 353 and 367 (R150).

The criterion is rough. It does not count how many formulas of equal complexity exist, and it depends on the chosen bit measure **[S]**. For individual hits it therefore does not replace the null test from Doc. 393, but it shows at a glance where a closer test is worthwhile and where it is not.

ba.7 What can be resolved

There is no second pilot in the SI sector

The only relation in the corpus in which an SI unit appears explicitly is $\alpha_{\mathrm{SI}} = \xi(E_0/1\,\mathrm{MeV})^2$ (Docs. 010, 011, 014, 137, 313, 365, 384, 386). All other statements of FFGFT are pure numbers, ratios of two quantities of the same dimension; a common factor cancels in them (A040) **[B]**. With $\hbar = c = 1$ a single dimension remains, energy, and hence a single unit. The SI sector therefore has exactly one degree of freedom: the factor that binds the basic form to the MeV. K and the MeV anchoring are the same number, and the one measurement α fixes it. A second pilot that could test this number cannot exist in a ratio-based theory **[B]**. The assignment from Doc. 386, according to which the correction sits on the side of the unit, is thus no longer a mere assumption but follows from the structure; the value $74/75$ itself is given geometrically, as the rotation number of the frozen recursion with $1/\xi = 7500 = 100 \cdot 75$ (Docs. 300, 381, 384) **[B]**. What remains open is only the origin of the number 100 (Doc. 133, A040) and the small residual that α shows against this value (R135) **[S]**. For the number 100 the strongest existing support is $K^{-36} = 16/\pi^2 = 1/\Delta(\mathrm{D}_4)$, the inverse of the packing density of the D_4 lattice; it matches $74/75$ to 2.8 ppm, and the exponent 36 is open (Doc. 315, P35) **[K]**.

The muon lifetime is not an SI pilot either. With $G_F = 1/(\sqrt{2}\,v^2)$,

$$\frac{\Gamma_\mu}{m_\mu} = \frac{1}{384\,\pi^3}\Big(\frac{m_\mu}{v}\Big)^4 \tag{ba.6}$$

up to radiative corrections **[Q]**; τ_μ therefore tests only the dimensionless ratio v/m_μ. The ladder gives $5/(16\,\xi) = 2343.75$ for it, the measured value is 2330.34: a residual of $+43\,\xi$ without K and $+145\,\xi$ with K **[K]**. Without K the ladder lies 3.4 times closer, as required by the rule according to which K has no place in ratios. The residual itself is the known error of the ladder: relative to v, $+82\,\xi$ for the electron, $+43\,\xi$ for the muon and $-34\,\xi$ for the tau, with the distances of 39 and $77\,\xi$ between ladder and Koide form **[K]**. τ_μ thus tests the normalisation of the ladder to v, not the factor K.

The bound hardly needs the Koide amplitude

The bound in Section 4 assumed both Koide parameters to be exact. The corpus derives the angle $\theta = 2/9$ (Doc. A110); the justification of the amplitude $d/c = \sqrt{2}$ is open (R150). If the amplitude is left free, it absorbs part of the residuals, and only one combination of the mode factors remains bound. The amplitude, however, shifts almost only the muon-to-electron ratio ($\partial \ln\big(m_\mu/m_e\big)/\partial a = 32.6$ against 1.8 for the tau-to-muon ratio) **[K]**. The bound combination is therefore almost exactly the difference between tau and muon:

assumed exact	tau–muon	muon–electron
$\theta = 2/9$ and $d/c = \sqrt{2}$	$\leq 0.92\,\xi$	$\leq 0.074\,\xi$
only $\theta = 2/9$	$\leq 0.91\,\xi$	not bound
only $d/c = \sqrt{2}$	$\leq 0.92\,\xi$	not bound
neither	not bound	not bound

[K] The central statement, that a mode-dependent part of the measurement recursion amounts to at most about $1\,\%$ of the correction step $100\,\xi$, therefore depends only on the derived angle $2/9$. Only the finer bound between muon and electron additionally needs the amplitude. If both parameters are free, there is no bound: two parameters always fit two ratios exactly.

ba.8 Consequences for quantum computers

A common factor is calibrated away. In the corpus, a gate that is meant to rotate by π advances only by πK, shortened by $100\,\xi = 1.33\,\%$ (Docs. 034, 175, 230, 231; A070). R142 records that this deviation is absorbed in the pulse calibration of real qubits and is therefore not directly measurable. Channel estimation gives the reason: the calibration is a pilot. It measures how far a pulse actually rotates and adjusts it so that the desired result is obtained. A factor that is the same for all gates is a single degree of freedom, and the pilot absorbs it completely, just as α absorbs the SI factor (Section 7). As long as K is common, no quantum computer can see it, not even with better hardware **[B]**.

Only a mode-dependent part would be visible. What the calibration cannot absorb is a difference between modes, for instance between qubits of different frequency or between one- and two-qubit gates. For the charged leptons such a part is limited to at most about $1\,\xi$ (Sections 4 and 7), that is about 1.2×10^{-4} in relative terms **[K]**. If a similar bound holds for qubits, a possible residual signal lies in the same order of magnitude as the fractal damping of the Bell quantity, $\Delta S/S \approx 10^{-5}$ to 10^{-4} (R142). The transfer from leptons to qubits is an analogy and not derived **[S]**. The gap to the Tsirelson bound $2\sqrt{2}$ measured today is 0.088 on ibm_kingston and 0.104 pooled (Docs. 147, 384); it is decoherence and about 250 to 300 times larger than a residual signal of this size **[K]**. This leaves the state of Doc. 384: operationally, FFGFT and standard quantum mechanics give the same predictions up to a ξ correction that cannot be resolved with today's hardware.

Deterministic reading. The results support it without proving it. First, the open orbit from Doc. 394 provides a deterministic mechanism for apparent randomness: a rotation with a non-rational rotation number never repeats its phases exactly and distributes them uniformly in the long run. With an advance of 0.00056 turns per closure cycle, a quarter turn is reached after about 450 cycles and a full turn after about 1800 **[K]**. This fits the statement of the corpus that noise is disordered phase (Doc. 384): every individual run is fixed and nevertheless looks like noise over long runs. The open orbit itself, however, is an interpretation of the residual of 7.6 ppm and not derived **[S]**. Second, discrete outcomes survive the measurement chain (Section 5): a bit is a winding number (Doc. 343) and can be regenerated like a digital signal. Third, the Shannon limit remains the same for both readings: the deterministic individual run and statistical averaging over many runs

hit the same floor (Doc. 343, Theorem G′). The Bell violation is not affected; the FFGFT measurement remains contextual (Doc. 230).

ba.9 Conclusion

Communications engineering does not fight the recursion of the channel; it measures it with pilots and removes it, while the Shannon limit remains. Transferred to FFGFT, this yields a clear picture. In the ratio sector the channel is flat: with the Koide form as pilot no residual above $1\,\xi$ remains, and a mode-dependent part of the measurement recursion is limited to about $1\,\%$ of the correction step. The generation-dependent residuals of the ladder are the error of an inexact pilot, not a channel. In the SI sector there is exactly one pilot, α, and there cannot be a second one: the only relation with an SI unit fixes the one degree of freedom in which K and the MeV anchoring coincide. The muon lifetime tests the normalisation of the ladder to v, not K. Discrete structures with small denominators survive the measurement chain; this is why the presentation belongs in the basic form. And a numerical hit counts only if the measurement delivers more information than the formula contains. For quantum computers this means: a common gate factor disappears in the calibration, a mode-dependent residual would lie at about 10^{-4} and thus far below decoherence; the deterministic reading remains viable up to the noise limit.

What remains open is the origin of the number 100, from which the rotation number $74/75$ follows, equivalent to the question why the exponent 36 in $K^{-36} = 1/\Delta(\mathrm{D}_4)$ couples to the packing density (Docs. 133, 315, 384), the small residual that α shows against this value (R135), and the derivation of the Koide amplitude (R150); only the finer bound between muon and electron still depends on it **[S]**.

Check script: `2/python/Dok395_Skripte/pruef_395_messkette_kanal.py`

[1] C. E. Shannon, *A mathematical theory of communication*, Bell System Technical Journal **27** (1948) 379–423, 623–656.

[2] R. Price, P. E. Green, *A communication technique for multipath channels*, Proceedings of the IRE **46** (1958) 555–570.

[3] D. G. Brennan, *Linear diversity combining techniques*, Proceedings of the IRE **47** (1959) 1075–1102.

[4] S. B. Weinstein, P. M. Ebert, *Data transmission by frequency-division multiplexing using the discrete Fourier transform*, IEEE Transactions on Communication Technology **19** (1971) 628–634.

[5] C. Berrou, A. Glavieux, P. Thitimajshima, *Near Shannon limit error-correcting coding and decoding: Turbo-codes*, Proceedings of ICC '93, Geneva (1993) 1064–1070.

[6] E. Arıkan, *Channel polarization*, IEEE Transactions on Information Theory **55** (2009) 3051–3073.

[7] Y. Koide, Lettere al Nuovo Cimento **34** (1982) 201; Physics Letters B **120** (1983) 161.